INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

# Banking

*What the industry is, how it runs, and how to model it — for architects who have never worked in it*

A working guide for an enterprise architect walking into a bank for the first time. The balance sheet as the product, and why that changes everything. The payment rails, which in South Africa are their own country. The credit cycle, the capital rules and the regulator who reads your returns. BIAN and ISO 20022, and what a reference architecture is actually for. Then the part that matters most: how to turn all of it into a model somebody on the executive committee would recognise as true.

**South African lens. Sourced throughout. October 2026.**

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Thotnr Consulting

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CHAPTER 1

# The industry in one chapter

*A bank sells the use of other people's money and finds out later what the credit cost. This chapter is the whole book compressed: enough to follow the conversation in a bank this afternoon, and a map of where the rest of the detail lives.*

## What a bank actually is

Two ordinary mornings.

A delivery driver in Tembisa is paid on the 25th. By the time she wakes up, R420 has already left her account by debit order to a furniture retailer, because that collection ran in a window that opens before most people are awake. She draws R600 in cash at a supermarket till on her way to work, pays R7.50 a month for the account, and has never been inside a branch.

A property developer in Umhlanga draws the fourth tranche of a R180 million construction facility. The money moves to a conveyancing attorney's trust account. A mortgage bond is registered at the Deeds Office, which takes weeks and over which the bank has no control. The rate on the facility is quoted over an interbank benchmark that is itself being withdrawn, which means somebody has to find every contract that references it and repaper them against a deadline.

Different customers. Different systems. Different regulators watching. The same institution, doing the same four things.

A bank exists because four mismatches in the economy are expensive for people to resolve with each other directly. Learn the four and most of what follows stops being arbitrary.

**Maturity transformation.** Savers want their money back on demand, or in weeks. Borrowers want twenty-year home loans and five-year plant finance. The bank stands between the two, funding long assets with short liabilities, and absorbs the mismatch. South African banks held R1.96 trillion of mortgage advances against R6.67 trillion of total deposits at end-December 2025 [2].

**Liquidity transformation.** The bank promises every depositor instant access to money it has simultaneously lent out for twenty years. That promise is only ever partly true. It works because withdrawals are statistically predictable in aggregate and the bank holds a buffer of assets it can sell fast — liquid assets were 16.0% of total assets and 28.4% of short-term liabilities on 2025 averages [3]. When the statistical assumption fails, the bank fails in days rather than quarters, which is why daily liquidity discipline is not a reporting exercise.

**Credit intermediation.** The bank pools credit risk, assesses it at a scale no individual saver could manage, prices it, and takes the loss when it is wrong. Impaired advances were R307.6bn, 5.20% of gross loans and advances, on 2025 averages [3].

**Payments.** The bank operates the account from which value moves, and provides finality, availability and interoperability. This is the only one of the four that earns a fee rather than a spread, and the only one where a non-bank can get between the bank and the customer at the layer the customer can see.

### Why a bank is not a shop with money

The shop analogy is the one newcomers reach for, and it misleads in five specific ways, each of which turns into an architectural constraint.

The inventory is other people's money, repayable on demand. A shop that loses stock loses a quarter's profit. A bank that loses depositor confidence ceases to exist, because liabilities run faster than assets liquidate. "Who are our largest depositors, and are they correlated?" is therefore a question somebody has to be able to answer today, not next month.

Revenue is the *difference* between two prices, not a markup on one. A bank can grow its loan book and shrink its earnings at the same time. Volume dashboards without margin dashboards are not incomplete; they are actively misleading.

Cost of goods sold is unknown at the point of sale. The true cost of a loan is the credit loss that emerges later, estimated at origination by a model. So the largest cost line in a lending business is a statistical estimate posted as an accounting entry, with a full reconciliation and audit obligation attached.

The balance sheet is a scarce, capital-constrained resource. A bank cannot lend more without capital and stable funding. Every product decision is therefore an allocation decision, and "good for the customer" and "we can afford to write it" are separate tests.

And the bank is a node in shared infrastructure it does not control. It cannot change its payment message format on its own. The format changes when the clearing operator and the central bank change it. South African payments settle through SAMOS, operated by the South African Reserve Bank, with retail clearing through PayInc — the renamed BankservAfrica, in which the Reserve Bank took a 50% stake in late 2025 [16][17].

> **The balance sheet is the product, not a by-product.** For a manufacturer the financial statements report on the business. For a bank they *are* the business, written down. Every product the bank sells becomes a line on one side of that statement, and every line carries a capital charge, a liquidity classification and a maturity profile. An architect who does not read the balance sheet is working on a model of something else.

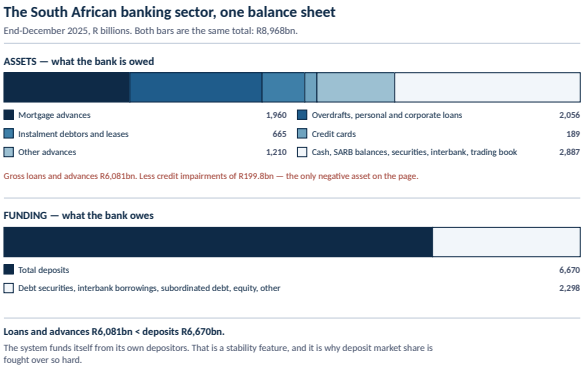



**Figure 1.1** The South African banking sector as one balance sheet, end-December 2025, drawn to scale [2]. Assets are what the bank is owed; deposits are what it owes. The impairment allowance is the only negative asset on the page. Chapter 6 does the arithmetic properly.

Three readings of that picture are worth doing slowly, because newcomers get all three wrong.

First, assets are what the bank is *owed* and deposits are what it *owes*. Your bank account is the bank's liability. The loan on your house is the bank's asset. Every data model that treats a customer balance as something the bank holds *for* someone has already lost the accounting.

Second, loans of R6.08 trillion are less than deposits of R6.67 trillion [2], so the South African system funds itself from its own depositors in aggregate. That is a stability feature and it is also why deposit market share is fought over so hard: the cheap deposit is the margin.

Third, impairments of R199.8bn are about 3.3% of loans while impaired advances are 5.20% [2][3]. The gap is not an error. It is collateral the bank expects to realise. Which is why a mortgage book and an unsecured book at identical arrears levels carry completely different provisions, and why a single coverage-ratio key performance indicator across a group that holds both is a number with no meaning.

## The seven things an architect must understand

Each of these has a chapter of its own later. What is here is the mechanism and the consequence — enough to follow a conversation this afternoon, ask one question that lands, and know which answer should worry you. They build on each other, so read them in order.

### 1. The lines of business decide nearly everything

A bank is not one business. It is a dozen businesses sharing a licence, a general ledger and a customer master, and the line decides the capital charge, the regulator, the system, the kind of person who makes the decision and how fast anything can change.

Retail transactional banking is millions of small, standardised, straight-through events priced by a fee schedule. Home loans are half of all consumer credit in South Africa by value — R1,265bn of gross debtors book, 52.16% of the total — granted almost entirely to the top income band, with 99.75% of mortgage rand value going to consumers earning above R15,000 a month [8]. Corporate and investment banking is a few thousand named clients, bespoke structures, and an architecture dominated by limits, collateral, confirmations and settlement finality. Treasury is not customer-facing at all and is where the whole balance sheet is managed.

The failure is predictable. A group buys one platform intended to serve both high-volume retail and negotiated corporate lending, and it serves neither. **Chapter 2** takes the twelve lines one at a time.

### 2. There are two revenue engines and they are computed differently

**Net interest income** is interest earned on assets less interest paid on liabilities. **Net interest margin** is that figure over average interest-earning assets. The big-bank South African margin, excluding Capitec and Investec, was 443 basis points for FY2025, down from 451 [1].

**Non-interest revenue** is fees, commissions, trading income, the insurance result and investment income. It grew 9.3% in FY2025, faster than net interest income [1]. At Capitec, net non-interest income of R16.1bn exceeded net interest income of R12.7bn in the six months to 31 August 2026 [7].

Here is the part that matters to you, and it is a data statement rather than a finance one. Net interest income is computed from the balance sheet and the rate environment: balances, rates, repricing dates, average balances over a period. Non-interest revenue is computed from events: a transaction happened, a fee applied, a counter incremented. Different grain, different latency, different reconciliation. One management-information platform asked to serve both usually produces a product-profitability layer nobody trusts. **Chapter 6** does the arithmetic.

### 3. The policy rate flows through the book faster than anyone expects

In South Africa, prime is a real, single, national reference rate pegged mechanically at the repo rate plus 3.5 percentage points — 7.25% and 10.75% respectively from 25 September 2026 [11]. Most retail lending is quoted as "prime plus or minus x". There is no South African equivalent of the United States thirty-year fixed mortgage market at scale.

So a Monetary Policy Committee announcement reprices an entire loan book overnight, automatically, including loans that are mid-registration. A home-loan servicing system that cannot do that is not fit for this market. At the same time a large part of the liability base — current accounts, low balances, money in transit — does not reprice at all, so the margin widens when rates rise with no management action. That is the **endowment effect**, South African banks disclose it as its own line, and it is why deposit market share is margin share.

> The inflation target changed on 12 November 2025, from a 3–6% range to a 3% point target with a one percentage point tolerance band, to be implemented over two years [10]. A lower nominal anchor means a lower steady-state policy rate, a smaller endowment benefit and a structurally thinner deposit-franchise margin. Anybody modelling a South African bank's net interest income on the 2015–2024 rate distribution is modelling a regime that no longer exists.

#### 4. Credit cost is an estimate that must be reproducible for twenty years

IFRS 9 puts every credit exposure in one of three stages: performing, with a twelve-month expected credit loss; a significant increase in credit risk since origination, with a lifetime loss; and credit-impaired, also lifetime. The trigger for the middle one is a comparison *to origination*.

Read that again, because it is the single most expensive sentence in banking data architecture. To know whether risk has increased significantly, the bank needs the risk it assessed on the day the loan was written — the origination-date probability of default or risk grade — stored immutably, per contract, for the twenty years of a mortgage. Most core systems store the *current* risk grade and overwrite it monthly.

Four more consequences follow, each of which breaks a conventional loan system: every stage transition and every cure has to be a dated, auditable event rather than a status field; the macroeconomic scenarios and their weights have to be versioned, because the auditor will ask whose forecast this was and who approved it; collateral valuations have to be dated and linked to exposures; and the loss allowance needs a movement-complete reconciliation — originations, derecognitions, stage transfers, remeasurement, write-offs, recoveries, model changes, each separately derivable. **Chapters 9 and 10** build the data model that makes this possible, and **Chapter 15** covers what the supervisor does with it.

#### 5. Payments is shared infrastructure, and in South Africa it is its own country

Most bank integration work is payments work, and almost every assumption a reader brings from the United Kingdom, the United States or the European Union is wrong here.

Cheques were withdrawn entirely on 31 December 2020 — not reduced, withdrawn [15]. Debit-order mandates for authenticated collections are held at the *payer's* bank, which validates every collection against the agreed terms before processing it; there is no SEPA or NACHA analogue. Interchange is determined by the central bank for several streams, not set by the card schemes. AEDO and NAEDO, the early-debit-order rails, were switched off on 31 October 2021 [14]. And the governance of the whole system changed in 2026: the Payments Association of South Africa's recognition as the industry's management body was withdrawn, with its functions split between the Reserve Bank itself and PayInc over August and September 2026 [16].

> South Africa does not have a regulated open banking regime. Never write that it does. What exists is bilateral commercial APIs on each bank's own terms, plus SARB Directive No. 2 of 2024, which regulates — rather than bans — the issuing of EFT credit payment instructions on a payer's behalf by screen scraping, and requires those providers to register with the Reserve Bank. Anybody importing a UK Open Banking or PSD2 reference architecture is designing for obligations that do not exist here and omitting the one that does.

**Chapter 12** maps the rails and the systems that connect to them.

#### 6. Time is a first-class dimension, and the regulator enforces it

A twenty-five-year mortgage commits a bank to administering a contract under rules that have not been written yet. A reckless-lending challenge brought in 2031 is answered against the affordability rules as they stood on the day of the decision. A restated impairment number has to be reproducible with the model version and scenario set that produced it.

That makes three dates load-bearing on almost every record: when the thing happened, when the change takes effect, and when the bank came to know it. Effective dating is not a nicety here, and "we overwrite and keep an audit log" is not the same capability. Nothing in the origination chain — application, assessment inputs, bureau data, affordability calculation, decision, disclosure document, agreement — may be overwritten. **Chapter 10** treats this properly.

#### 7. The supervisor reads your data, not your architecture diagram

South Africa runs a Twin Peaks model. The Prudential Authority, housed inside the Reserve Bank, supervises safety and soundness. The Financial Sector Conduct Authority supervises conduct. The National Credit Regulator — and in a South African banking sentence "NCR" means the National Credit Regulator, not the hardware company — registers credit providers under the National Credit Act. The Financial Intelligence Centre enforces the anti-money-laundering regime. The Corporation for Deposit Insurance pays depositors when a bank fails.

Three of those obligations are, in substance, data specifications rather than policies.

The BA 900 return requires every balance classifiable by counterparty institutional sector, instrument type, currency and residual maturity, simultaneously, monthly, reconciled to the general ledger. If those four are not governed attributes in the product and party masters, the return is assembled in spreadsheets — and because it feeds national monetary statistics, the errors are publicly visible. Note also that Government Notice 5802 of 31 January 2025, effective 1 February 2025, removed the BA return *forms* from the Regulations: returns are now determined by the Authority in writing [27]. The specification can change without a gazette cycle, so a reporting architecture has to be able to absorb that.

Deposit insurance cover is R100,000 per qualifying depositor per bank, and the Corporation for Deposit Insurance has specified the single customer view as an actual data structure — aggregation rules, a sharing ratio per joint account holder, beneficiary look-through for trust and stokvel accounts, a prescribed field set, and monthly submission within ten working days of month-end [20]. A bank with five customer masters cannot produce it. This is the most under-appreciated data requirement in South African retail banking.

Cross-border payments are each a reporting event to the Reserve Bank's Financial Surveillance department, with a balance-of-payments category captured *from the customer at the point of transacting*, because it encodes economic purpose and cannot be derived afterwards. That makes it a mandatory field in every channel that can originate a cross-border payment, APIs included.

**Chapter 15** takes the instruments one at a time, with commencement dates.

## The money, in one page

Four numbers run every management conversation in a bank. Learn them in this order.

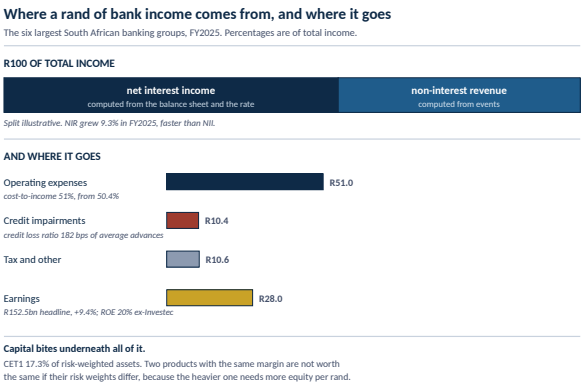



**Figure 1.2** Where a rand of bank income comes from and where it goes, for the six largest South African banking groups in FY2025 [1]. The income split is illustrative; the cost, impairment and earnings figures are reported. Capital sits underneath all of it and is not an expense line.

**Net interest income** is the first and largest. Decompose it four ways, because management will: volume, meaning more book at the same spread; rate or margin, the same book at a different spread; mix, a reallocation between differently-spread products; and endowment, the policy rate flowing through assets faster than liabilities. A rand of unsecured lending earns multiples of a rand of mortgage, which is why FirstRand's 5.29% group margin against Nedbank's 3.75% is mostly mix rather than skill [5][6]. Any analytics layer that cannot produce a volume, rate, mix and endowment attribution will be rebuilt within two years. It is the most-requested and least-often-delivered capability in bank management information.

**Non-interest revenue** is the second, and in South Africa it is growing faster. It moves on transaction volumes and the fee schedule, on card interchange, on trading and markets, and on attached insurance and asset management. Capitec's digital transaction volumes, excluding value-added services, were 662 million in six months, up 26% [7].

**Impairments** are the third, expressed as a credit loss ratio: the impairment charge over average gross advances, in basis points. The FY2025 aggregate across the six majors was 182 basis points [1]. Individual banks reported very different numbers in the following period — Standard Bank 73, Absa 94, Nedbank 95 against a 60–100 through-the-cycle range, FirstRand 105 [4][6]. African Bank's personal banking book ran at 11.5% [28]. None of those is inconsistent with the others. They are different book mixes. A single credit-loss key performance indicator across a group holding both mortgages and unsecured lending is meaningless without segmentation, and building one is a reliable way to be disbelieved.

**Cost-to-income** is the fourth: operating expenses over total income. The South African majors ran 51% in FY2025, from 50.4% [1]. Capitec ran 36% [7]. African Bank ran 70.7% [28]. That spread is the argument for architectural simplicity, stated numerically, and it is worth knowing why it exists: branch and ATM estates in managed decline but still large; cash, which is expensive to handle, move, insure and reconcile in a way Nordic banks have forgotten; core estate age and count, because decades of mergers leave multiple cores and multiple customer masters with a permanent integration run cost; mandatory regulatory programme load; and fraud and dispute handling, where R2.4bn of digital banking crime losses and R1.7bn of card fraud in 2025 generated operational cost many multiples of the loss itself [21].

Underneath all four sits capital. Each exposure is multiplied by a risk weight to produce risk-weighted assets, and capital ratios are capital divided by that. The South African minimum is built in layers — the Basel minima, a Pillar 2A systemic add-on, a bank-specific Pillar 2B individual capital requirement, a domestic systemically important bank add-on, a capital conservation buffer and a countercyclical buffer, which South Africa set at a positive 1% from 1 January 2026 [12]. Note one disclosure asymmetry an architect building external reporting will trip over: a bank must disclose its systemic add-on and must *not* publicly disclose its bank-specific Pillar 2B requirement [12].

> "Capital-efficient" has a precise meaning. Two products with identical net interest margins are not worth the same if their risk weights differ, because the heavier one needs more equity per rand of balance. A capital-efficient version of a product means one of: more or better collateral, a shorter maturity, a better-rated counterparty, a risk-transfer structure, or distribution off the balance sheet. Which means risk weight and capital charge have to be deal-level attributes in the pricing engine, and the pricing calculator has to produce a risk-adjusted return on capital rather than a margin. Nedbank states its cost of equity as 14.0% [6]; a line earning 12% destroys value while growing.

## The South African market in a page

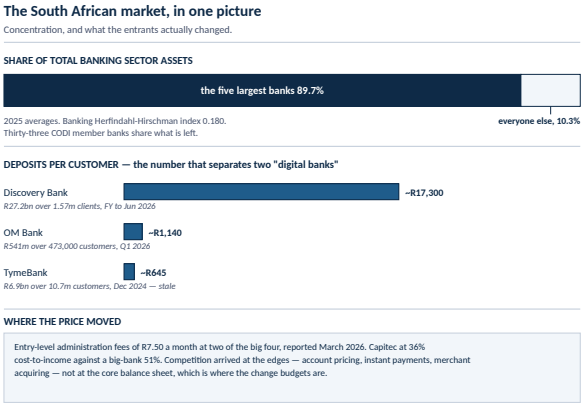



**Figure 1.3** Concentration, the two-tier customer economics, and where the entrants actually moved the price [3][7][23][24]. The TymeBank deposit figure is as at December 2024 and materially stale; it is shown because the order of magnitude is the point.

Total banking sector assets were R8.97 trillion at end-December 2025 [2]. Concentration is extreme: the five largest banks held 89.69% of total assets on 2025 averages, with a banking Herfindahl-Hirschman index of 0.180, and the six largest have held above 90% for a decade [3]. PwC's Major Banks Analysis covers Absa, Capitec, FirstRand, Investec, Nedbank and Standard Bank, whose South African operations are 93% of sector assets on December 2025 BA 900 data [1].

That concentration has four architectural consequences, and they are not obvious.

Four or five institutions set every de facto integration standard in the country. "Integrating to the banks" means four API estates plus a long tail, and none of the four will change for you.

Supervision of a few large institutions is intensive, so the per-bank compliance surface is heavy and systemic designation brings extra capital.

Competition arrives at the edges — entry-level account pricing, instant payments, merchant acquiring — and not at the core balance sheet, which is where the change budgets cluster. That mismatch is the central tension in most South African bank portfolios.

And incumbent cores are decades old and very large, so coexistence architecture is the normal state rather than a transitional one. Plan for it as a permanent condition.

### The big five, and the two numbers that matter about everyone else

Standard Bank is the largest by assets, with twenty million active customers across twenty-one markets and Africa Regions contributing 40% of group headline earnings at H1 2026 [4]. FirstRand runs distinct franchises rather than segments — FNB for retail and commercial, RMB for corporate and investment, WesBank for vehicle and asset finance — and posted the highest margin and return on equity of the five [5]. Absa and Nedbank are of similar size, both reporting a 15.0% return on equity at H1 2026, Nedbank against an explicitly stated 14.0% cost of equity [6]. Investec is about 7% of sector assets and its engine is private banking, where it holds a 41% share of clients earning above R5 million a year [9].

Capitec is the structural outlier and the most useful teaching case in the market: 26.6 million active clients of whom only 10.4 million are "fully banked", a 36% cost-to-income ratio, a 31% return on equity, and an earnings mix in which fintech — value-added services plus a mobile network operator — contributed 29% [7]. Three per cent of sector assets producing R9.5bn of half-year earnings is what one modern core plus one app produces when it is not carrying four decades of acquisitions.

Then the entrants, where two numbers tell the whole story. Deposits per customer at TymeBank were roughly R645 at the end of 2024; at Discovery Bank they were roughly R17,300 in the year to June 2026 [23][24]. Ten million customers producing under R7bn of deposits is a distribution success and a balance-sheet problem, which is why that bank's strategy is merchant and payroll lending rather than mortgages. The same two words, "digital bank", cover two entirely different balance sheets and therefore two entirely different architectures.

> Bank results figures go stale within six months and sector aggregates within a quarter. Read every banking number with four qualifiers attached: gross or net of impairments; group or registered bank, because a listed group includes insurance, asset management and offshore operations; headline, normalised or IFRS, because the JSE's primary measure is headline earnings; and point-in-time or average, because margins and ratios are usually struck on average balances while capital is period-end. Architects who drop those qualifiers ship reconciliation defects into every management-information layer they build.

## What breaks, predictably

You will walk into some version of all four of these. None is a surprise to anyone who has worked in a bank, and none of them is on the architecture diagram.

### The customer who exists five times

One human being exists five times in a mid-sized South African bank: in the current-account system, in the card system, in the home-loan book, in the vehicle-finance book — often a separate legal entity — and on the wealth platform. Each instance has its own key, its own spelling of the name, its own

address, its own know-your-customer refresh date and its own view of arrears. None of them is wrong. Each is a *record*. The bank has no *identity*.

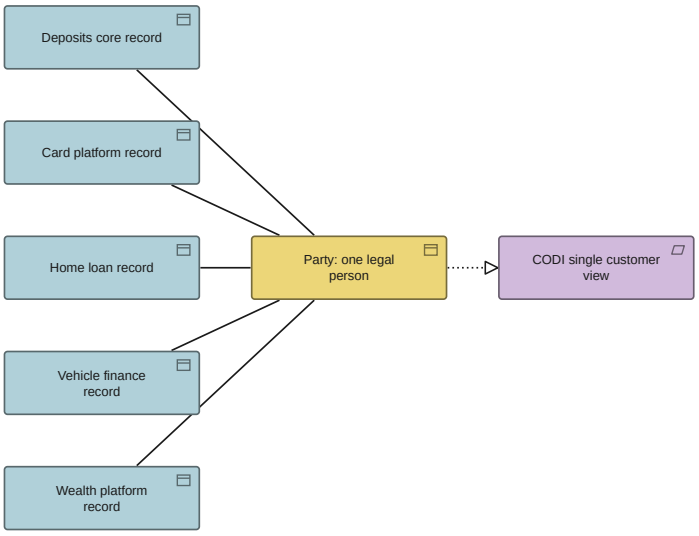



**Figure 1.4** One human being, five records, no identity. The resolution is not a sixth record: it is to make Party the thing and customer a dated role. The CODI single customer view is the first requirement in South African banking to put a legal definition and a monthly deadline on that model [20].

The usual response is to build a customer master, and the usual result is a sixth record. What actually resolves it is a different conceptual move: `Party` is the thing — a legal person, natural or juristic, that the bank can transact with or owe — and *customer* is a dated role a party plays against an account, an agreement or a payment. So are depositor, borrower, guarantor, signatory, beneficial owner and counterparty. Modelled that way, a beneficial owner behind a corporate client who holds no product can exist, which matters because anti-money-laundering law requires the bank to identify and verify them.

And there is now a test with a legal definition attached. The Corporation for Deposit Insurance single customer view must consolidate each qualifying depositor's balances across every qualifying product into one record, monthly, attested by the chief executive and the chief financial officer, executable at short notice under resolution conditions [20]. It is not a principle. It is a file with a field list, and it is the cleanest business case an architect will ever get for fixing party identity.

### The batch window

The overnight cycle is the bank's operational heartbeat and it is still the single largest constraint on architecture. Roughly in order: cut-offs close; inbound clearing files are received and applied; outbound collection and payment files are generated; interest is accrued and capitalised; fees are assessed; the arrears state machine advances days past due; statements and statutory notices are composed; the general ledger interface runs; impairment and risk extracts run; the warehouse loads; and everything downstream loads from the warehouse.

Four reasons it is still batch, in descending order of how often it is the real one. The clearing rails are batch — EFT credit and EFT debit are file-based, with action dates, cut-offs and deferred settlement, and a creditor cannot be real-time against a rail that is not. Periodic computation is genuinely periodic: interest accrual, fee assessment, amortisation, impairment staging and regulatory aggregation are defined on a period rather than an event, and making them event-driven does not make them cheaper, it makes them harder to prove. The core was designed around an end of day, so batch posting, value dating and the day-end position are the ledger's native model. And nobody has been able to afford to change it.

What makes a late batch serious is not the outage but the tail. The collection or salary file misses its cut-off, so a day's money does not move and the impact is immediate and public. Statutory notices are not produced, which is a compliance event rather than an inconvenience. Downstream systems load incomplete data, so every report built on the warehouse is silently wrong for a day. The arrears engine does not advance, so collections works a stale book. The regulatory extract misses its window. A batch failure is a one-day outage plus a reconciliation tail.

Which means the cut-off times are business facts an architect has to know. A sample of published South African ones, enough to show the shape:

| Clock | What it governs |
|---|---|
| 19:00 SAST | Registered Mandates collections submit in the late window, after DebiCheck and before EFT debits |
| 23:00 SAST | Registered Mandates response deadline on the action date; settlement 05:00 the next day |
| 20:00 same day | TT1 DebiCheck mandate authentication window; miss it and the request restarts |
| Two days | TT2 DebiCheck batch mandate authentication window, then restart |
| Early morning | DebiCheck collections process ahead of other collection methods |
| 24×7 | SAMOS settlement, with a dedicated night window for clearing-house batches |
| 15 business days | BA 900 monthly return, after month-end |
| 10 working days | Deposit insurance monthly submission, after month-end |
| Two business days | Cross-border payment reporting to Financial Surveillance, from the value date |

> The authoritative EFT credit and EFT debit submission cut-off table is not public. The clearing operator's published rules state that file submission times live in product and service manuals and in a participant fee book available under non-disclosure. This is a genuine gap: if your design depends on an EFT cut-off, get it in writing from the bank's payments operations team and date the answer.

### The core nobody can replace

Ask five people in a bank what "core banking" means and you will get five answers, because three genuinely different things have historically lived in the same box: the double-entry account ledger; the product engine that defines what a product *is*, with its rate structure, fee schedule, limits and lifecycle; and the customer master that accounts hang off. In most legacy South African estates it is all three, which is exactly why replacement is so hard.

Treat "core" as a question rather than a system. Establish separately where the authoritative balance is for each product family; whether product definition is configuration or code — and ask for the elapsed time of the last product launch, not the methodology; where authoritative customer identity lives; and where end of day actually runs. Four answers, often four systems, often not the one labelled core.

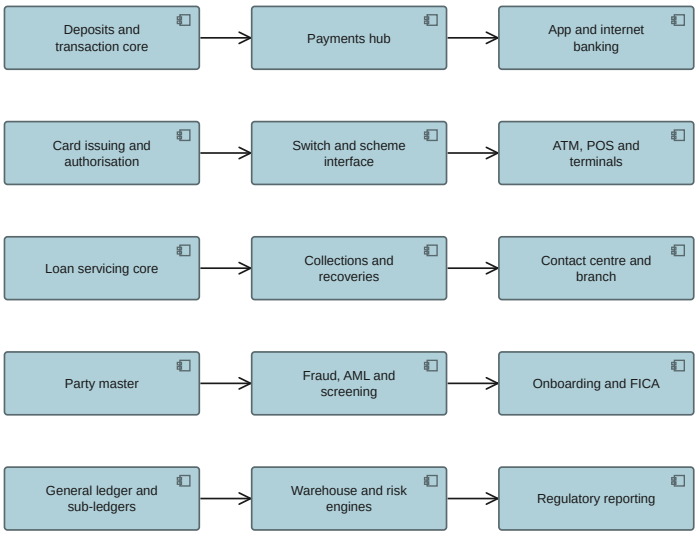



**Figure 1.5** The estate at a glance, in three bands: the systems of record on the left, the middle in the centre, and what people and other parties touch on the right. Expect more than one instance of the core, the loan servicing platform and the payments hub in any bank with an acquisition history. Chapter 12 is the real map, and it is longer.

On the record, core modernisation in South Africa is a multi-year business transformation rather than a technology upgrade. The best-documented local programme is Nedbank's Managed Evolution: it began in 2016, the reported cost was R11.7 billion, and it concluded with the delivery of the core banking platform — the last element modernised — at the end of 2024 [25]. Trade press is the source for the figure, so attribute it to the outlet rather than the bank. The counterexample is Barko, live on a vendor's software-as-a-service core in under six months; the honest comparison is one of scope, not of competence, because the two programmes were not attempting the same thing.

Two further cautions. No South African bank's core platform is publicly sourceable — not Nedbank's, Capitec's, Absa's, Investec's, African Bank's, Bidvest's, Postbank's or Sasfin's. A good deal of confident market folklore circulates on this and an architect who repeats it in a steering committee will be wrong in front of people who know. And mainframes and COBOL in the largest South African banks are real, load-bearing and widely attested by practitioners, but they are not something I can source from a published disclosure. Write it as practitioner consensus.

What you will actually find on arrival, in the composite: a hollowed-out core you were told was a monolith; more cores than the application inventory shows; a data warehouse doing operational work; three different computations of available balance, in the app, at the ATM and in the authorisation host;

a regulatory reporting estate that is the largest single consumer of change capacity; a payments hub with three bypasses; and a capable, under-documented mainframe team who are the only people who know what end of day really does.

### The payment rail that is its own country

By 2024 settlement value, South African retail payments were still overwhelmingly a batch EFT credit system: EFT credit 70.3%, real-time clearing 11.0%, cards 10.7%, EFT debit 5.8%, authenticated collections 0.8%, and rapid payments — PayShap — 0.7% [19]. Anyone arriving with the assumption that instant payments are the default rail will design for the wrong seven-tenths.

PayShap launched in March 2023, is ISO 20022 native, runs twenty-four hours a day including weekends and public holidays, supports payment to an account or to a proxy, and carries a R50,000 transaction cap [18]. It is growing fast and the published volume counts disagree with each other badly enough that the disagreement is itself the finding; use the Reserve Bank's regulatory report series and PayInc's own statements and treat secondary aggregations as unreliable. Note that the cap started at R3,000 and a design that still assumes that is wrong by a factor of about seventeen.

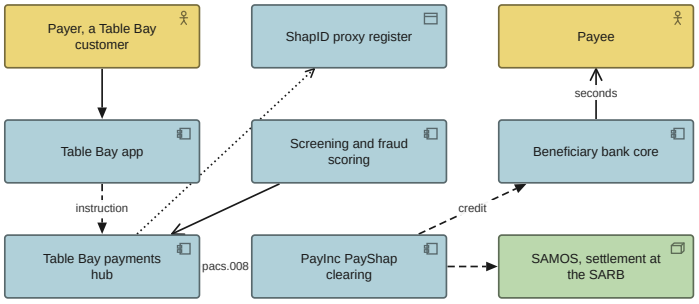



**Figure 1.6** One PayShap payment at the invented Table Bay Bank, from the app to settlement in central-bank money [18][19]. The Access edge to the proxy register is the part with no United Kingdom or United States equivalent in this shape. Screening is inline, not a queue.

And one genuinely surprising fact about the domestic rails: SAMOS completed its migration to ISO 20022 on 17 September 2022, the first real-time gross settlement system on the African continent to do so, and PayShap was ISO 20022 from the start [26]. South Africa's rails are ahead of most global bank cores. The integration problem in a South African bank is usually not that the rail is primitive. It is that the rail is structured and the core is not.

## The book from here

**The twenty-two items, and what each one is for**


**PART I — THE INDUSTRY**

1 The industry in one chapter
2 The lines of business
3 The vocabulary
4 How the business model works
5 Who does what

**PART II — HOW A BANK RUNS**

6 How banks make money
7 The operating model
8 The process model

**PART III — DATA AND SYSTEMS**

9 The conceptual data model
10 The information model
11 The conceptual application model
12 The application landscape
13 The reference architecture

**PART IV — CONSTRAINTS**

14 Technology providers
15 Regulation, risk and what the regulator inspects

**PART V — MODELLING IT**

16 Modelling banking in ArchiMate
17 Table Bay Bank, end to end
18 Your first ninety days

**APPENDICES**

19 A — The cross-maps
20 B — Glossary
21 C — Standards, bodies and where to find them
22 D — References


**Figure 1.7** The book's structure. Part I is the business, Part II is how a bank runs, Part III is the data and the systems, Part IV is the constraints, and Part V is how to model it. If you read only one more chapter, read the one for the question you were hired to answer.

**Part I — The industry.** Chapter 2 takes the twelve lines of business one at a time, with the capital and liquidity each consumes. Chapter 3 is the vocabulary, built in themed runs with the arguments attached, because the payments words in particular are a local dialect. Chapter 4 is how the business model works — the canvases, and the five things the canvas leaves out. Chapter 5 is who does what: around forty-five roles in nine groups, with what each person does all day and what they complain about, because that last line is where architecture work comes from.

**Part II — How a bank runs.** Chapter 6 does the money properly, with the arithmetic shown. Chapter 7 is the operating model. Chapter 8 is the process model, and it is the one chapter drawn in BPMN rather than ArchiMate, because in a process model the hand-off between roles is the whole point and ArchiMate has no swimlanes.

**Part III — Data and systems.** Chapter 9 is the conceptual data model: the twenty-odd things a bank talks about, and the argument about each. Chapter 10 is the information model, including bitemporality, reference data and the reporting layer. Chapter 11 is the conceptual application model — the applications a bank *needs*, named by what they do. Chapter 12 is the application landscape as you will find it, which is the longest chapter in the book. Chapter 13 is the reference architecture: BIAN, ISO 20022, FIBO, APQC, and what a reference model is actually for.

**Part IV — Constraints.** Chapter 14 is technology providers, with the vendor ownership corrections that matter — several widely-repeated attributions are out of date. Chapter 15 is regulation, risk and what the regulator inspects, with twenty instruments, each with its commencement date and its architectural consequence.

**Part V — Modelling it.** Chapter 16 is modelling banking in ArchiMate: which of the forty-two elements earn their place and which patterns to reuse. Chapter 17 works Table Bay Bank end to end — an invented mid-tier South African full-service bank with about 2.4 million retail customers, a business bank, a small corporate and transactional arm and a treasury, used throughout this book to make the teaching concrete. Nothing about Table Bay Bank is real. Chapter 18 is your first ninety days.

**Appendices.** The cross-maps between the frameworks; a glossary of more than two hundred terms; the standards and bodies with where to find them; and the references.

## Three things that will surprise you

If you have come from retail, telecommunications, mining or the public sector, three things about banking will take longer to accept than the rest.

**The product is a position on a balance sheet, and the regulator owns the constraint.** In most industries a good product idea is limited by cost, demand and delivery capacity. In a bank it is also limited by capital and stable funding, both of which are externally prescribed and scarce. This changes the shape of the conversation you will have about every proposal: not "can we build it" but "what does it consume, and does it clear the cost of equity on a risk-adjusted, capital-charged basis". A proposal that cannot answer that will not be funded, however good the customer experience, and an architect who cannot follow the answer will be in the room without a vote.

**A large share of the change portfolio is not yours to choose.** The rails change when the clearing operator and the central bank change them. Interchange is externally determined, on an annual revision cycle, as reference data rather than a commercial parameter. The interbank benchmark is being retired on a fixed timetable: the Market Practitioners Group has recommended a "no new JIBAR" milestone of 1 May 2026, revised from end-March, with JIBAR ceasing publication at the end of December 2026, and the replacement — ZARONIA — is a backward-looking overnight rate compounded in arrears [13]. That last one is worth dwelling on, because it is not a renaming. An interest calculation that was a single multiplication at the start of a period becomes a daily compounding calculation over the period, with observation shifts, lookbacks and lockouts, and it touches every loan servicing, bond, derivative, hedge accounting and internal-pricing system that references a floating rate. The deadline is fixed and the scope is "every contract". In many banks the mandatory, externally-scheduled programme load consumes the roadmap before any discretionary work is considered.

**Trust is the actual product, and it is destroyed by latency rather than by controls.** The National Financial Ombud Scheme recovered almost R443 million for consumers in 2025, of which R53 million was banking; fraud is the highest category of banking complaints, around 60% of that banking recovery went to fraud-related refunds, and the ombud's recurring finding was that banks responded *too slowly after becoming aware* of the fraud [22]. Read that as a systems requirement rather than a conduct finding. It sets a service-level expectation on fraud response — detect, freeze, trace, recall — in minutes to hours, imposed through a dispute-resolution body rather than a regulation. A fraud response that routes through a back-office queue now fails a test that did not exist five years ago.

If you stop reading here, keep four sentences. The balance sheet is the product, so every design decision is a balance-sheet decision. There are two revenue engines and they are computed from different data at different grain. Credit cost is an estimate that must stay reproducible for decades, which makes time a first-class dimension. And payments in South Africa is a local dialect that no offshore reference architecture will teach you.

**THE CHAPTER IN TEN LINES**

- A bank exists to perform four transformations — maturity, liquidity, credit intermediation and payments — and every part of its architecture is downstream of one of them.
- The balance sheet is the product, not a report about it: every product is a line on it, and every line carries a capital charge, a liquidity classification and a maturity profile.
- There are two revenue engines: net interest income computed from balances and rates, and non-interest revenue computed from events. Different grain, different latency, one platform for both usually produces numbers nobody trusts.
- In South Africa prime is repo plus 3.5 percentage points and almost all retail lending is prime-linked, so an MPC announcement reprices a whole book overnight and automatically.
- Credit cost is an estimate, and IFRS 9 measures it against the risk assessed at origination — so the origination-date risk grade must survive twenty years of a mortgage, which most cores do not manage.
- Capital, not demand, is the binding constraint on a new product: two products with the same margin are not worth the same if their risk weights differ.
- The five largest banks hold nearly 90% of sector assets, competition arrives at the edges rather than the core, and coexistence architecture is the permanent state.
- South African payments is a local dialect — cheques abolished, mandates held at the payer's bank, central-bank-determined interchange, no regulated open banking regime — and no offshore reference architecture will teach it.
- By settlement value, retail payments are still overwhelmingly batch EFT credit, while the domestic rails are ISO 20022 and ahead of most bank cores.
- What breaks is predictable: the customer who exists five times, the batch window, the core that is three systems in one box, and the rails nobody in the bank controls.
- Trust is the real product, and the ombud now penalises latency rather than controls: a fraud response that routes through a back-office queue fails a test that did not exist five years ago.

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CHAPTER 2

# The lines of business

*A bank is a dozen businesses sharing a licence, a ledger and a customer master. This chapter is the taxonomy: what each line sells, what it consumes, what it needs, and where it will not behave the way you expect.*

## Where the money is, and where the customers are

Start with the single most useful orientation fact in South African retail banking, because it is the one newcomers get backwards and it determines what the architecture should be sized for.

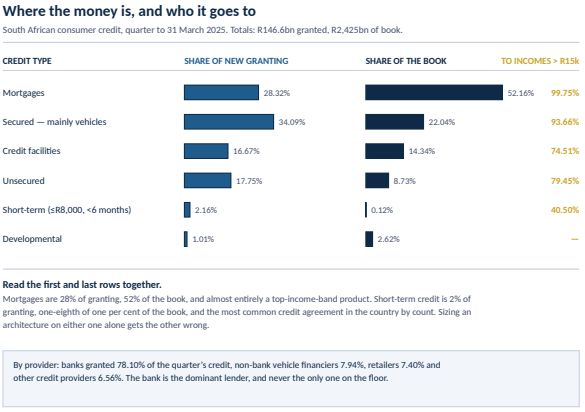



**Figure 2.1** South African consumer credit for the quarter to 31 March 2025 [1]. Three readings of the same market: what was granted, what is on the books, and who it went to. The orientation fact of the whole chapter is that the first two columns disagree and the third explains why.

Mortgages are 52.16% of the consumer credit book and 28.32% of new granting in the quarter. Short-term credit — the National Credit Act category of loans of R8,000 or less, under six months — is 2.16% of granting and **0.12%** of the book [1]. Now overlay the income distribution. Of the rand value of mortgage credit granted, 99.75% went to consumers earning above R15,000 a month. Of short-term credit, 40.50% did, with 44.84% going to people earning R10,000 or less [1].

Two businesses, then, in opposite corners of the same bank. One is a small number of very large, very long, very secured contracts held by the top income band and worth half the balance sheet. The other is an enormous number of tiny, short, unsecured contracts held at the bottom of the market and worth almost nothing on the balance sheet, which nevertheless generate most of the events, most of the collections activity and most of the regulatory attention.

> Architecture sized on customer count will be wrong for mortgages. Architecture sized on book value will be wrong for the entry-level business. These are different businesses sharing a customer master, and the usual mistake is to build one system for both and discover it serves neither.

Three further numbers from the same quarter sharpen the picture. Within secured credit, vehicles were R46.2bn, 92.40% of secured granting. Within credit facilities, credit and garage cards were R11.0bn, store cards R5.9bn, other facilities R5.1bn and bank overdrafts only R2.2bn. And by provider, banks granted R114.5bn of the R146.6bn total — 78.10% — with non-bank vehicle financiers 7.94%, retailers 7.40% and other credit providers 6.56% [1]. A bank is the dominant lender but not the only one on any dealer floor or in any shopping centre, and a competitor with a lighter regulatory load and no deposit base is a different kind of competitor.

Behind the money sits the population. There were 28.90 million credit-active consumers at March 2025, of whom 18.49 million were in good standing and **10.41 million — 36.04% — had impaired records**; 101.26 million accounts of which 20.68 million were impaired; and 434.66 million credit bureau enquiries in the quarter, of which only 14.50 million were consumers seeking credit and 62.94 million were tracing and debt collection [22]. Hold on to that last ratio. More than four times as many bureau enquiries are made chasing people as are made lending to them. The collections industry is not an afterthought in this market; in volume terms it is larger than origination.

A second orientation fact, about who is allowed to be in this market at all.

## The four statutory categories of deposit-taker

South Africa has four statutory species of deposit-taker, which is a difference from the United Kingdom and the United States where a single bank charter dominates. Membership of the Corporation for Deposit Insurance is compulsory for all of them, which gives the cleanest public list [2][3].

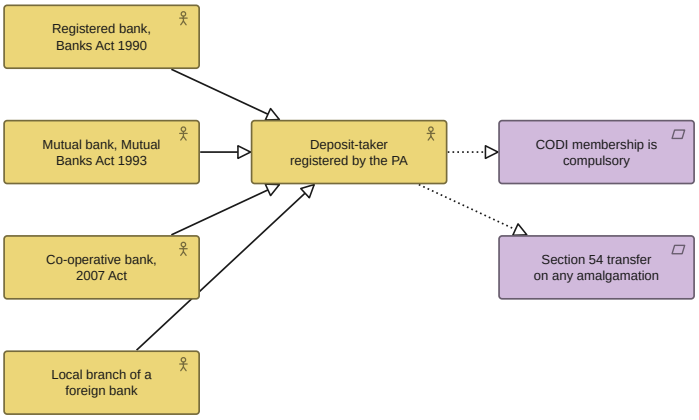



**Figure 2.2** Four statutory species of deposit-taker, not one charter [2][3]. The specialisation matters: a mutual bank need not be a company, a foreign branch runs on its parent's capital and has no local retail franchise, and all of them must belong to the deposit insurance scheme.

**Registered banks** — fourteen of them, under the Banks Act 94 of 1990: Absa, Access Bank, African Bank, Albaraka, Bidvest, Capitec, Discovery Bank, FirstRand, Investec, Nedbank, OM Bank, Sasfin, Standard Bank and TymeBank [2]. Only a public company registered under the Companies Act may carry on the business of a bank, and section 11 of the Banks Act makes it an offence for anyone else to do so.

**Mutual banks** — three: Bank Zero, Finbond and GBS [2]. Governed by the Mutual Banks Act, 1993; depositor-owned rather than shareholder-owned; and they need not be companies [3]. Note for anyone who assumes that "digital bank" means commercial bank: Bank Zero holds a mutual banking licence, not a commercial one [2].

**Co-operative banks** — four listed on the deposit insurance membership page, though the page itself states five [2]. Member-owned, under the Co-operative Banks Act, 2007. Below them sit co-operative financial institutions, supervised by the Prudential Authority but not currently inside the deposit insurance scheme [25].

**Local branches of foreign institutions** — twelve, including Bank of China, BNP Paribas, Citibank N.A., Deutsche Bank, Goldman Sachs, HSBC, JP Morgan Chase, Standard Chartered and State Bank of India [2]. These are *branches*, not subsidiaries. They operate on the parent's capital, hold no local retail franchise, and exist for corporate lending, markets and correspondent banking. If you are asked to model one, most of the retail reference architecture is irrelevant and the interesting content is limits, settlement and reporting.

> The counts do not reconcile across official sources. The Prudential Authority's operational-risk discussion for the first half of 2025 refers to twenty-eight registered banks without stating which categories are counted [4]; the deposit insurer reported thirty-three member banks in the year to 31 March 2025 — fourteen commercial, five co-operative, three mutual and eleven foreign branches [25]. A fifth category, representative offices of foreign banks, exists but is not in the deposit insurance list and I could not source a count. Do not publish a single tidy number; cite the register you used and date it.

Licensing itself is worth one paragraph because it bears on every acquisition an architect will be asked to integrate. An applicant applies to the Prudential Authority for authorisation under section 12 of the Banks Act, then must apply for registration within twelve months under section 16, and the process runs roughly ten months to a year [3]. More to the point: **every bank acquisition in South Africa runs through a section 54 transfer of business**, which is a legal migration of contracts with a gazetted effective date. That date is not a programme milestone that can slip. It is a legal event, and the migration either lands on it or the bank is in breach.

## The twelve lines

What follows is the taxonomy. For each line: what it sells, who buys it, how it makes money, the capital and liquidity it consumes, the systems it needs, and the thing about it that will not match your assumptions.

### 1. Retail transactional banking

**Sells** a current account — still called a cheque account in South Africa, nine years after the last cheque — plus savings and call accounts, a debit card, an app, cash access and bundled extras. **Buys** everyone with an income. **Money** comes from monthly account fees, per-transaction fees, cash-handling fees, card interchange on the issuing side, and — the larger half, and the part outsiders miss — the endowment earned on the credit balance.

**Capital and liquidity.** Almost no capital: this is a liability business, not an asset business. On liquidity it is the opposite of a consumer: current and savings balances are the cheapest, stickiest funding a bank has, and the liquidity ratios reward them heavily. A rand of retail current-account balance is worth considerably more to a bank than a rand of wholesale deposit of the same size, which is why deposit market share is fought over at a price that looks irrational from outside.

**Systems.** The deposit and current-account core, the card issuing platform, the app and internet channel, the ATM and branch channels, the fee engine, and gateways to every payment rail.

**Where it is different from what you would assume.** South African retail pricing has moved from pay-as-you-transact to a bundled monthly fee with rewards and non-banking extras. The 2026 Banking Charges Report, as reported in March 2026, found entry-level administration fees of R7.50 a month at both Absa Transact — with R5,000 of life cover included — and FNB Easy PAYU, which comes with four loaves of bread and a supermarket voucher a

month [5]. A bundle is not a price. It is a product with entitlements, counters and non-banking components: per-customer counters that reset monthly, a rewards accrual ledger, partner settlement, and a fee engine whose main job is to *waive* rather than to charge. Fee engines built for per-transaction pricing cannot express this without a parallel entitlement service, and that gap is the origin of most South African retail fee defects.

The second surprise is the economics at the bottom. Capitec reports 26.6 million active clients of whom only 10.4 million are "fully banked" [6]. The marginal entry-level client transacts in small amounts, often in cash, and is unprofitable on fees alone. The business works through three things and only three: a near-zero marginal cost to serve, attachment of credit, insurance, airtime and value-added services, and the endowment on a great many small balances. Which is why "fully banked" conversion, not accounts opened, is the metric that matters in this line, and why an onboarding funnel that produces an account nobody funds is producing cost.

The third surprise is that the channel estate is shrinking faster than cash usage is. Absa's outlets went from 611 in 2021 to 574 in 2026 while its ATM and cash-device fleet fell 41%, from 8,435 to 4,976, and digitally active customers rose 10% to 3.8 million with digital revenues up 21% [23]. Cash handling is being pushed to retail partners and onto a smaller, more expensive own estate. For an architect this means two things at once: the branch and ATM channels are not going away on the timescale of your roadmap, and the cash reconciliation problem between device, switch and ledger gets *harder* as the estate shrinks, because each remaining device carries more of the volume.

#### 2. Savings and deposits

**Sells** savings, call, notice — seven, thirty-two, sixty and ninety-day — fixed deposits, tax-free savings accounts, and group or stokvel accounts. **Buys** savers, from a R50 deposit to a corporate treasury placing R500 million. **Money** is not a fee. The product *is* the funding, and its value to the bank is the difference between what the bank pays for it and what equivalent funding would cost in the market.

**Capital and liquidity.** This is where the liquidity ratios live. The Liquidity Coverage Ratio tests high-quality liquid assets against net outflows over a thirty-day stress; the Net Stable Funding Ratio tests available stable funding against required stable funding over a year. South African banks ran roughly 140–165% on the first and 115–119% on the second across 2023 to 2025 [4]. One change already in bank funding plans: the available-stable-funding factor for short-term rand funding from non-bank financial corporates is being cut from 20% to 10% between January 2025 and December 2027 [4], which mechanically devalues that funding source relative to retail deposits without anyone making a commercial decision.

The competitive consequence is visible in the disclosures. Standard Bank's current and savings account balances were R515bn, up 12%, with call deposits of R630bn, up 13%, at H1 2026 [16]. Those are endowment plays as much as liquidity plays, and they are why a transactional-account price war is rational behaviour rather than a loss leader.

**Systems.** The deposit core, a maturity and rollover engine, notice-period management with early-withdrawal penalties, dormancy and unclaimed-balance treatment, and the deposit insurance single customer view feed.

**Where it is different.** A deposit product's value to the bank is a function of its behavioural stickiness under stress, not its contractual maturity. That is exactly what the liquidity ratios encode and exactly what most deposit cores do not record. The behavioural model — core balance against volatile balance, with a decay curve per product and segment — therefore lives in the asset and liability management system, fed by extracts, and the quality of the bank's liquidity position is bounded by the quality of those extracts.

Two local features have no offshore analogue. **Stokvels** are rotating savings and credit associations, estimated at around R50 billion across 800,000 groups and 11 million members [7]. Banks serve them with multi-signatory group accounts needing three signatories and no single beneficial owner in the conventional sense, which breaks naive customer-account data models and complicates beneficial-ownership logic. And **Islamic banking** is a structurally distinct segment rather than a product variant: a profit-sharing or cost-plus structure has no interest accrual at all, so it cannot be expressed in a conventional loan schema without either abusing the fields or running a parallel product engine. Albaraka is a fully fledged Islamic bank and FNB, HBZ, Absa and Standard Bank operate Islamic windows [8].

#### 3. Personal lending

**Sells** unsecured personal loans, typically over one to eighty-four months, revolving credit facilities, and overdrafts on a current account. **Buys** salaried individuals: the middle market for consolidation, the lower-middle for income smoothing. **Money** from interest at or close to the National Credit Act maximum, initiation and service fees, and attached credit life insurance.

**Capital and liquidity.** The heaviest risk weights in retail, because there is no collateral, and the highest expected loss. Unsecured is R211.6bn of book, 8.73% of the total, on R26.0bn of quarterly granting; credit facilities — cards, store cards, overdrafts — are R347.7bn and 14.34% [1].

**Systems.** Origination and decisioning: scorecard, bureau call, affordability engine, policy rules. A loan servicing system that handles rescheduling and arrears states. And — the dominant system by headcount — **collections**, with a dialler, treatment strategies, pre-legal, legal and sale of book.

**Where it is different.** This is the highest-margin and highest-loss line in retail banking, and the pricing only works if it covers the loss. African Bank's personal banking credit loss ratio was 11.5% in the six months to March 2026 [9]. The consequence for an architect is blunt: in a South African unsecured lender the collections estate is larger and more consequential than the origination estate, and it is usually the part nobody modelled.

#### 4. Home loans

**Sells** a twenty- to thirty-year advance secured by a mortgage bond registered over residential property, plus access bonds that allow redraw, further advances, switches from another lender, and building loans drawn in stages. **Buys** the top income band, almost exclusively [1]. **Money** from a thin spread over the cost of long funding — mortgages averaged prime less 0.67% in the first quarter of 2026 [10] — plus initiation fees and attached insurance.

**Capital and liquidity.** Low risk weight because of the collateral, but the revised standardised approach that took effect in South Africa on 1 July 2025 makes the residential mortgage risk weight depend on **loan to value** [11]. That is a quiet but significant change: a dated collateral valuation and a current exposure become a bucketed capital input, per exposure, per period. On liquidity, a twenty-year asset is the single heaviest consumer of required stable funding on the balance sheet.

**Systems.** Mortgage origination with valuation, conveyancing and attorney workflow, Deeds Office lodgement and bond registration. A servicing system that can reprice an entire book overnight. Redraw and access-facility management. Arrears, legal process, property in possession. And the securitisation feed where the bank uses one.

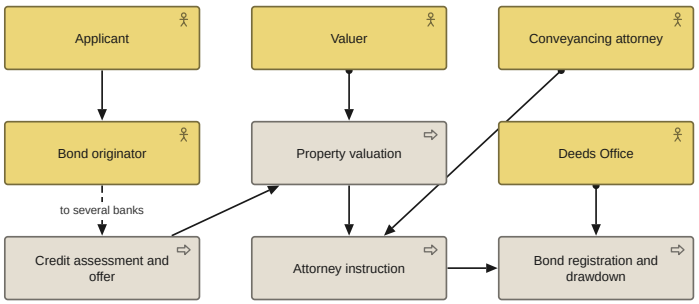



**Figure 2.3** Home loan origination, with the external parties drawn in. Three of the nine boxes are outside the bank and two of them set the elapsed time. The originator's Flow edge is the one that costs money: the same application goes to several banks, and 45.5% of declines are approved somewhere else [10].

**Where it is different.** Origination runs substantially through **bond originators**, who submit one application to several banks at once. The first-quarter 2026 data: average deposit 12.8% of purchase price, first-time buyers 8.2%, approval rate 84%, first-time buyers 48% of applications — and **45.5% of applications declined by one bank were approved by another** [10].

Read that last number slowly. Nearly half of all declines are a *policy* disagreement rather than a creditworthiness fact. The bank's decline is a competitor's booking; the originator sees both outcomes and the bank sees one; and credit policy is being arbitraged in real time by an intermediary with better information than the lender. Decision latency and policy transparency are revenue levers here, not compliance artefacts.

The second difference is that origination is a long-running, externally blocked process rather than a transaction. Attorneys and the Deeds Office take weeks and are outside the bank's control. The case state has to survive that, the external parties have to be first-class integrations with their own service levels, and the credit decision has to be versioned and replayable because it will be re-made when the valuation comes back.

#### 5. Vehicle and asset finance

**Sells** instalment sale agreements — where the financier retains ownership until the final payment — finance leases, and operating leases, usually with a residual or **balloon** payment. For commercial clients: plant, equipment and fleet finance. **Buys** individuals buying vehicles and businesses buying productive assets. **Money** from interest spread, initiation fees, and insurance and value-added product attachment.

**Capital and liquidity.** Secured, so lighter than unsecured lending and heavier than a mortgage; medium-tenor, so a moderate stable-funding consumer. Instalment debtors, suspensive sales and leases were R664.9bn on bank balance sheets at December 2025 [12]; vehicles were R46.2bn, 92.40% of secured credit granted in the quarter [1].

**Systems.** Dealer-floor origination at the point of sale, with sub-minute decisioning because the customer is standing at the dealership. Asset and vehicle identification number registration. Insurance verification as a condition of the facility. Residual and balloon management. Settlement quotes. Repossession and auction.

**Where it is different.** Because the financier retains ownership under an instalment sale, repossession is a contractual process rather than a foreclosure, and the asset register is a legal record rather than an inventory convenience. Note also that non-bank vehicle financiers were 7.94% of all credit granted in the quarter [1] — this is the line where the bank most often is not the only lender on the dealer floor, and decision latency loses the deal on the spot.

#### 6. Cards

**Sells** a revolving credit facility with a payment card attached, an interest-free period on purchases settled in full, rewards, and travel and lifestyle benefits at the premium end. **Buys** the middle and upper income bands. **Money** from three streams that behave quite differently: interest on revolved balances, annual or monthly card fees, and **interchange** received by the issuer on every purchase.

**Capital and liquidity.** Unsecured and revolving, so a heavy risk weight — and note that under IFRS 9 a revolving facility is measured over the period of credit exposure, which may extend beyond the contractual period, so the expected loss calculation needs a modelled behavioural life rather than a contract term.

**Systems.** Card issuing and authorisation, which is sub-second, twenty-four hours a day, with no acceptable downtime window. The scheme interface for clearing and settlement files. Statement and minimum-payment cycles. Disputes and chargebacks. Rewards accrual and redemption. Card-not- present authentication and fraud scoring. Tokenisation for digital wallets.

**Where it is different.** In South Africa **interchange is determined by the central bank**, not set by the schemes. The Reserve Bank determines rates for ATM, cash-back at point of sale, debit and credit cards, and authenticated collections including the Registered Mandate Service, using a cost-based methodology with rates reviewed annually and the methodology every five years [13]. Market-determined, bilateral rates apply to EFT credit, EFT debit and real-time clearing [13]. For a reader from the European Union, the nearest equivalent is the Interchange Fee Regulation caps; for the United States there is no analogue outside debit. **The architectural implication is that interchange is externally set regulated reference data on an annual revision cycle, not a commercial parameter, and it belongs in a versioned, effective-dated reference-data store rather than in the acquiring platform's pricing module.**

The second difference is the fraud shape: card fraud losses in South Africa were R1.7bn in 2025, up 18%, with card-not-present the largest contributor [14].

### 7. Business banking

**Sells** business transactional accounts, merchant acquiring, overdrafts and term loans, asset finance, commercial property finance, trade and foreign exchange for smaller importers and exporters, payroll and collections services, and increasingly scored rather than relationship- assessed credit. **Buys** everything from a one-person enterprise to a mid-corporate with revenue up to R1.5 billion. **Money** from transaction and account fees, merchant service fees, interest spread, and the operating-account deposit float, which is the real prize.

**Capital and liquidity.** Mixed on capital. On liquidity it is the best line in the bank: Standard Bank's business and commercial arm held R190bn of deposits from enterprises with revenue up to R100m and R330bn from around 100,000 mid-corporates, targeting R725bn by 2028 [15], and posted a 36.3% return on equity at H1 2026 — the highest of any segment in that group [16].

**Systems.** Business onboarding, which is far heavier than retail. Multi-user mandates and approval hierarchies. Host-to-host and API payment file ingestion. Merchant acquiring and settlement. Collections as a service, including authenticated-collection mandate origination on behalf of clients. And a credit process that has to carry both scored and judgemental decisioning in one pipeline.

**Where it is different.** Business onboarding is not a bigger version of retail onboarding. It is N retail onboardings plus an entity onboarding: every director and signatory needs full individual due diligence, so a six-director company is seven onboardings and one unverifiable director blocks the whole thing. Beneficial ownership has to be resolved through trusts, nominee structures and layered holdings to the ultimate natural persons, and that work is largely manual. Mandates are a permissions model, not a form — "any two of these four, up to R500,000, except payroll which needs the finance director" is an authorisation policy that most business banking channels express badly and that the customer changes constantly.

The open question in the segment is scoring. Capitec Business grew to 686,000 clients, up 123%, with 141,000 merchant accounts and R35.5bn of credit — of which only 12% was scored lending [6]. Small-business credit is where statistical scoring has not yet displaced relationship judgement, and the bank that resolves it changes the competitive structure of the segment.

### 8. Corporate and transactional banking

This is the sticky, high-return half of what most banks label corporate and investment banking, and it deserves separating from the trading half because the systems have nothing in common.

**Sells** cash management, accounts in several countries and currencies, liquidity structures — sweeps, physical pooling, notional pooling — payments and collections at volume, host-to-host integration into the client's enterprise resource planning system, and trade finance: letters of credit, documentary collections, guarantees and standby credits, supply-chain and receivables finance. **Buys** large corporates, multinationals, the public sector and financial institutions. **Money** from fees plus the deposit float. Standard Bank's corporate and investment bank earned R74.4bn of revenue in FY2025 on R1.5 trillion of deposits and R700bn of advances, split global markets 43%, transaction banking 35% and investment banking 22% [15].

**Capital and liquidity.** Lending consumes capital against an obligor group rather than an account. Trade finance generates off-balance-sheet exposure through guarantees and letters of credit that still carry a capital charge. Corporate deposits are less stable than retail ones under the liquidity ratios, so a rand of corporate float is worth less than a rand of retail current account even when it is larger.

**Systems.** A corporate channel. Enterprise resource planning integration. Mandate and entitlement management across legal entities. A liquidity management engine. Document-checking workflow and SWIFT MT7xx messaging for trade. Sanctions screening on every party, and for trade on every vessel. Limits and exposure management across a group of connected counterparties, covenant monitoring, syndication tracking, and collateral management.

**Where it is different.** Three things. The unit of credit decision is an *obligor group*, not an account, so the data model needs a party-to-party exposure graph rather than a customer hierarchy. The decision is a paper approved by a committee under delegated authority, so the system models authority limits and quorum rather than a score cut-off. And **covenants are testable conditions with their own monitoring cycle** — nothing in a retail lending stack has that concept, and in most banks the covenants live in a word-processing document.

### 9. Investment banking and markets

**Sells** advisory, underwriting, structured and project finance, syndication and distribution on the investment banking side; and on the markets side foreign exchange spot, forward and swap, interest rate derivatives, bonds, commodities, equities and prime services. **Buys** corporates, institutions, governments and other banks. **Money** from arrangement and underwriting fees, structuring margin, bid-offer spread and some proprietary risk. FirstRand's trading and fair value income was R7.1bn, up 42%, in FY2026 [17].

**Capital and liquidity.** The market risk framework, the credit valuation adjustment framework for counterparty risk on derivatives, and — since 1 July 2025 in South Africa — the revised versions of both as part of the final Basel III reforms [11]. As at March 2026, 58% of South African banks were on the standardised approach for market risk and 42% on the simplified standardised approach [4].

**Systems.** Order management, trade capture, pricing and curve infrastructure, real-time risk including value at risk and sensitivities, collateral and margin, confirmations and settlement, and trade and transaction reporting.

**Where it is different.** Real-time here is a *risk* requirement, not a customer-experience requirement. The architecture is dominated by limits checked before the trade rather than reported after it, independent price verification, confirmation before settlement, and nostro reconciliation. And the market-risk estate is the one place in a bank where the reference-data problem is bigger than the transaction-data problem: most unexplained profit and loss is a reference-data break.

#### 10. Wealth and private banking

**Sells** a relationship banker, credit on favourable terms assessed against a view of total assets, investment management, fiduciary services — trusts, wills, estates — offshore accounts and foreign exchange, and lifestyle benefits. **Buys** high earners and high-net-worth individuals. **Money** from lending margin at the top of the mortgage market, fees on assets under management, and transaction and foreign exchange margin.

**Capital and liquidity.** Well-secured lending, so light on capital; and private clients hold large, relatively stable deposits, so the line is a net contributor on liquidity.

**Systems.** A consolidated client, household and entity view spanning accounts, investments, trusts and companies — the hardest data problem anywhere near retail banking. Portfolio reporting. A credit process that underwrites on net worth and liquidity rather than payslip affordability. And a relationship-manager desktop that is a workflow tool, not a sales pipeline.

**Where it is different.** Value per private banking client is reported at roughly ten times that of a personal banking client [18], and the tooling is routinely worse. Investec holds around 128,000 South African private clients, a 41% share of clients earning above R5 million a year and 35% of the mortgage market above R4 million, and is moving down-market into the R800,000 to R1.5 million band where it holds about 7% [18]. The consolidated household view that the whole proposition depends on usually does not exist and is assembled by hand, every time.

#### 11. Treasury

**Does**, rather than sells: manages the bank's own balance sheet. Funding, liquidity, interest rate risk in the banking book, the foreign exchange position, the liquid asset portfolio, and the internal price of money between business units. **Money**, as an accounting matter, is whatever the funds transfer pricing framework leaves in the treasury book, plus the carry on the liquid asset portfolio.

**Capital and liquidity.** Treasury *is* the liquidity function. Both regulatory ratios are computed over the entire balance sheet, with every exposure classified by counterparty type, product, maturity bucket and behavioural assumption under prescribed run-off factors — daily for the coverage ratio, monthly for the funding ratio, with full lineage and an audit trail per classification judgement. No core banking system holds that classification. The treasury and asset-liability system exists to apply it.

**Systems.** The treasury and asset-liability management system, fed by extracts from every core; a funds transfer pricing engine; money-market dealing and the settlement account position.

**Where it is different.** Two South African specifics. First, monetary policy implementation changed in 2022 from a shortage system to a **tiered floor system** with a controlled surplus: each bank has a quota based on balance-sheet size, surplus up to quota earns the repo rate, and above quota earns repo less 100 basis points [19]. The marginal return on an idle rand is therefore a step function with a bank-specific threshold — a direct input to any treasury or liquidity system, and one that offshore treasury packages do not model. Second, the benchmark transition: JIBAR ceases publication at the end of December 2026 and the recommended "no new JIBAR" milestone is 1 May 2026 [20]. Every floating-rate contract has to be found, classified and repapered. Treasury owns that programme, and it reaches every lending system in the bank.

And one sentence on **funds transfer pricing**, because it is simultaneously the most consequential and the least well-engineered calculation in most banks. It charges asset-generating units for funds and credits liability-generating units for funds raised, with treasury acting as the internal bank. It determines the profit and loss of every product and every business unit. If the curve, the liquidity premium and the behavioural assumptions are not reproducible and versioned, no product profitability number in the bank is defensible — and product profitability drives the whole investment portfolio.

#### 12. Bancassurance

**Sells** insurance into the bank's own customer base, under the bank group's own insurance licences. **Buys** existing customers, usually at the moment of another transaction. **Money** from premium and underwriting result, plus commission. Capitec Insurance was 27% of group earnings at R2.5bn, with 2.2 million credit life policies and 3.8 million funeral policies covering 17 million lives [6]. FirstRand's insurance new business annual premium equivalent was R4.2bn, up 22% [17].

**Capital and liquidity.** Insurance sits under a different prudential regime — the Insurance Act and its own capital standards — and a banking group holding an insurer consolidates it differently. Note Prudential Authority Directive 2 of 2025, which stops banks inflating capital by recognising post-acquisition reserves from significant insurance investments [11].

**Systems.** Policy administration, which is a different animal from a loan ledger; claims; and the attachment point in the bank's own origination flow.

**Where it is different. Credit life is a mechanically attached product and is regulated as such.** The cost is capped at R4.50 per R1,000 of the deferred amount for credit facilities, unsecured loans and developmental credit, and R2 per R1,000 for mortgages; the cover must settle the debt on death or permanent disability and pay instalments for up to twelve months on retrenchment; and an unemployed consumer may not be sold retrenchment cover [21]. The systems implication is sharp: the insurance premium is a line on the credit agreement, priced off the deferred amount, under a statutory ceiling. **The loan pricing engine and the insurance pricing engine are therefore the same engine**, and a change to either is a change to a regulated disclosure.

## Three things you will be told are lines, and are not

**Securitisation** is a funding and capital tool, not a customer-facing line. Loans are pooled and ring-fenced in a special purpose vehicle, notes are issued in rated tranches to institutional investors, the proceeds repay the warehouse facility that originally funded the loans, and losses hit the originator's retained profit first, then a first-loss loan from the originator, and only then investors. The originator usually keeps servicing the pool [24]. South Africa requires Reserve Bank approval for each new securitisation under the Securitisation Regulations issued under the Banks Act, with independent trustees and an annual audit of the securitisation entity [24]. Architecturally it is a feed and a flag: every exposure needs to know whether it is in a pool, which pool, and on what date it entered, and the servicing reporting to investors is a monthly obligation with its own deadline.

**"Digital"** is a channel, not a line. A bank that reports digital revenue is reporting the same products sold through a different route. The distinction matters because treating digital as a line produces a parallel product catalogue, a parallel pricing engine and eventually a parallel customer record —

which is how banks end up with a sixth party master.

**The Rest-of-Africa estate** is not a line either, though it is frequently presented as one on an organisation chart. A group operating in a dozen African markets typically runs a different, often newer, often packaged core in each, with its own regulator and its own clearing system. Group reporting is where those meet, and it is usually the only place they meet. This is practitioner observation rather than a sourced claim — no South African bank publishes an application-portfolio count — but it is reliable enough to plan for.

## The comparison, on one page

| Line | Revenue model | Balance-sheet weight | System intensity | Regulatory load |
|---|---|---|---|---|
| Retail transactional | Fees plus endowment on balances | Liability side; large and cheap | High: channels, fee engine, every rail | Conduct Standard, deposit insurance, FICA |
| Savings and deposits | The spread against market funding | Liability side; the funding base | Medium: core, maturity engine, SCV feed | Deposit insurance, liquidity ratios, BA 900 |
| Personal lending | Interest at or near the NCA cap, fees, credit life | Small book, heavy risk weight | High: decisioning plus a large collections estate | NCA affordability, in duplum, debt review, IFRS 9 |
| Home loans | Thin spread over long funding, held for years | Half the consumer book | High: external parties, Deeds Office, repricing | NCA, LTV-driven risk weights, IFRS 9 |
| Vehicle and asset finance | Interest spread, fees, insurance attach | Second-largest secured book | High: dealer-floor decisioning, asset register | NCA, instalment sale law, IFRS 9 |
| Cards | Interest, card fees, interchange | Small book, heavy risk weight | Very high: 24×7 authorisation, schemes, disputes | SARB interchange determination, PCI DSS, scheme rules |
| Business banking | Fees, merchant service fees, spread, float | Mixed; the float is the prize | High: onboarding, mandates, acquiring, host-to-host | FICA beneficial ownership, NCA below thresholds |
| Corporate and transactional | Fees plus the deposit float | Large deposits, moderate advances | High: limits, covenants, multi-entity mandates | Large exposures, sanctions, FinSurv on cross-border |
| Investment banking and markets | Fees, structuring and bid-offer margin | Trading book and large exposures | Very high: pricing, real-time risk, settlement | Market risk, CVA, trade reporting, central clearing |
| Wealth and private banking | Lending margin, fees on assets, FX margin | Well-secured lending, stable deposits | Medium, but the hardest data problem | FAIS advice records, fiduciary duties, POPIA |
| Treasury | FTP residual and portfolio carry | Owns the whole balance sheet | High: ALM, FTP, benchmark transition | LCR, NSFR, IRRBB, BA 300/330 returns |
| Bancassurance | Premium and commission | Off the banking balance sheet | Medium: policy administration and claims | Insurance Act, credit life caps, FAIS |




**Figure 2.4** The twelve lines ranked by customer count, by balance-sheet weight and by system intensity. The orderings are the author's judgement from the figures in this chapter rather than a measured dataset; the disagreement between the columns is the point, not the exact positions.

Two readings of those three columns are worth making explicit. The lines with the most customers are not the lines with the most balance sheet, which is the orientation fact this chapter opened on. And system intensity does not track either one: cards consume a disproportionate share of engineering effort relative to their book, because the authorisation path is sub-second and always on and the dispute process has externally enforced deadlines. If you allocate architecture capacity by book size you will under-serve cards and payments every time.

## What the line structure does to the application estate

Three patterns, and all three are visible in almost every South African bank.

**The line owns the system, so the party is duplicated.** Lines were built, bought or acquired one at a time, and each came with a platform that had its own idea of a customer. Vehicle and asset finance is frequently a separate legal entity with a separate core. The result is the five records of Chapter 1, and

the cost is not the duplication itself but that nothing can be answered across lines: total exposure to one group of connected companies, total relationship value, total deposit cover.

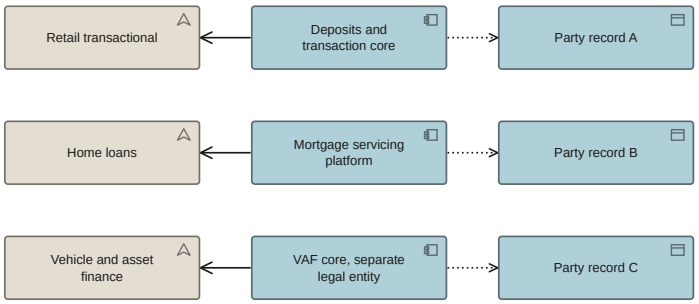



**Figure 2.5** What the line structure does to the estate. Each line arrived with its own platform and its own idea of a customer, and vehicle finance is often a separate legal entity as well. Nothing here is wrong; it is just that no question can be answered across the three rows.

**Retail and corporate share a ledger and a customer master and almost nothing else.** Retail is high-volume, low-value, standardised and straight-through. Corporate is low-volume, high-value, bespoke, and dominated by limits, collateral, confirmations and settlement finality. Attempts to unify them beyond the general ledger and the party master are a recurring and expensive failure pattern. The useful target is a shared party and product spine with deliberately separate transaction estates, not one platform.

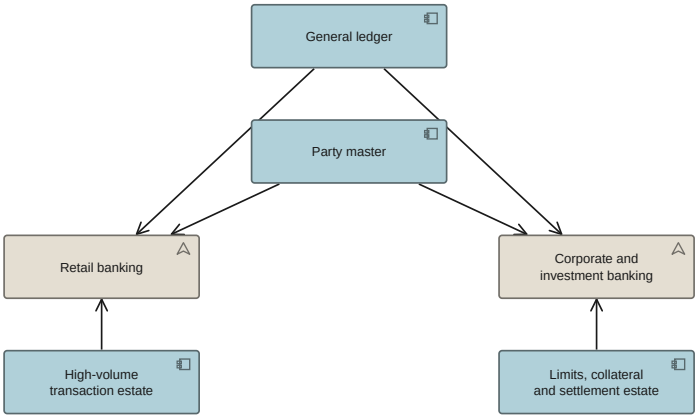



**Figure 2.6** Retail and corporate banking share a general ledger and a party master and very little else. Unifying them beyond that spine is a recurring and expensive failure pattern; the useful target is a shared spine with deliberately separate transaction estates.

**The lending conveyor is the pattern that repeats, with different parts per line.** Origination, decisioning, fulfilment, servicing, collections, recoveries. Personal lending is a decision pipeline with a huge collections tail. Home loans are a case-managed, externally blocked process with a thin collections tail and a heavy legal one. Vehicle finance is sub-minute decisioning at a dealer's desk with an asset register attached. Business lending is a credit-paper workflow with a committee at the end. Four conveyors, the same six stages, and systems that barely overlap. An architect who draws one lending capability and one lending application has drawn something that cannot be built.

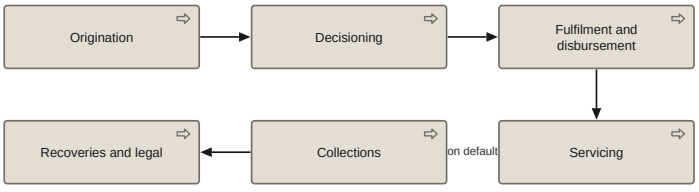



**Figure 2.7** The lending conveyor: six stages that repeat across every credit line, with entirely different systems at each stage per line. Drawing one lending capability served by one lending application describes something that cannot be built.

> The practical move at the start of an engagement is to get the line-of-business list and the application inventory onto the same page and draw the lines between them. The question to ask is not "which applications do we have" but "which line owns this, who else uses it, and what happens to the other lines when this one changes". In most banks nobody has that picture, and producing it in the first month buys you more credibility than any target-state diagram.

One closing warning about the numbers in this chapter. Bank results figures are reporting-period-specific and go stale within six months; sector aggregates within a quarter. FirstRand and Capitec have June and February year-ends, so "latest" is not a common date across the five largest banks, and comparing them on one slide without stating the periods is a reconciliation defect waiting to be found by somebody senior. Every figure above carries its as-at date for that reason.

**THE CHAPTER IN TEN LINES**

- Mortgages are 52% of the consumer credit book and 28% of new granting; short-term credit is 2% of granting and 0.12% of the book — where the money is and where the customers are is not the same place.
- South Africa has four statutory deposit-taker categories, not one charter, and every acquisition runs through a section 54 transfer with a gazetted effective date that cannot slip.
- Retail transactional banking earns more from the endowment on balances than from fees, so a bundled account is a product with entitlements and counters rather than a price.
- A deposit's value to the bank is its behavioural stickiness under stress, not its contractual maturity — which is what the liquidity ratios encode and what deposit cores do not record.
- In unsecured lending the collections estate is larger and more consequential than the origination estate, and it is usually the part nobody modelled.
- Home loan origination is a long-running, externally blocked process: attorneys and the Deeds Office set the clock, and nearly half of all declines are approved by another bank.
- Card interchange in South Africa is determined by the central bank on an annual cycle, so it is versioned regulated reference data, not a pricing parameter.
- Business onboarding is N retail onboardings plus an entity onboarding, and mandates are an authorisation policy rather than a form.
- Corporate credit decisions are made against an obligor group by a committee under delegated authority, and covenants are testable conditions with a monitoring cycle that no retail stack has.
- Credit life is priced off the deferred amount under a statutory rand cap, which makes the loan pricing engine and the insurance pricing engine the same engine.
- The same six-stage lending conveyor repeats across four lines with systems that barely overlap — and the first useful deliverable in any engagement is the map of which line owns which application.

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CHAPTER 3

# The vocabulary

*Banking runs on a few hundred ordinary words used with precise and conflicting meanings. This chapter defines them in themed runs, with the arguments attached — and ends with the five words that mean different things in different parts of the same bank.*

## Three sentences you cannot yet read

> *"NII was up on volume but the endowment unwound, so NIM is down twelve, and the CLR's ninety-five against a sixty-to-a-hundred through-the-cycle."*

> *"The RM mandates migrated in March but the TT1 fall-out is still thirty-odd per cent, so we're sitting with unpaids we can't retry inside the window."*

> *"Output floor bites on the mortgage book once we re-run standardised, and the CET1 impact lands before the Pillar 2B review."*

None of those is jargon for its own sake. Each is compressed and carries real content, and until you can decompress it you will nod along and miss the thing you were brought in to catch.

This chapter is the decompression, built in seven themed runs rather than alphabetically, because the words only make sense in clusters; the alphabetical list is Appendix B. Each run has a table of terms and then prose on the few that are routinely misused, and the chapter ends with the five ordinary English words that mean different things in different parts of the same bank — an architecture problem rather than a language problem. Where South African usage differs from the global usage the local one is flagged, because payments in particular is a dialect and offshore material will mislead you expensively.

## Run one: the balance sheet words

| Term | What it means |
|---|---|
| **Asset** | What the bank is owed or owns: loans, securities, cash at the central bank |
| **Liability** | What the bank owes. Your deposit is the bank's liability |
| **Advance** | A loan, in South African regulatory usage. "Gross loans and advances" is the book before impairments |
| **Deposit** | Money placed with the bank, repayable. Split by type: current, savings, call, notice, fixed, wholesale |
| **Impairment** | The allowance held against expected credit losses. A negative asset |
| **Impaired advances** | The South African regulatory term for non-performing loans |
| **Gross / net** | Before or after impairments. Never assume which |
| **Group / bank** | A listed group includes insurance, asset management and offshore operations; the registered bank does not |
| **Headline earnings** | The JSE's mandated primary earnings measure, excluding specified capital items |
| **Normalised earnings** | A management measure adjusting headline earnings for consolidation artefacts |
| **Point-in-time / average** | Margins and ratios are usually struck on average balances; capital is period-end |
| **BA 900** | The monthly statutory balance-sheet return; the authoritative market-share source |

**Advance** is the first trap, because it is not quite "loan", and the denominators move. The December 2025 sector figures are total assets of R8.97 trillion, loans and advances of R6.08 trillion and deposits of R6.67 trillion [1]. The same period's analysis of the six largest *groups* gives gross loans and advances of R6.8 trillion and deposits of R8.3 trillion [2]. Both are right: the first is the registered banks as the central bank measures them, the second the listed groups including everything offshore. **A management-information layer that does not carry group-versus-bank as an explicit dimension will produce two numbers for the same thing and be asked to explain the difference in front of the audit committee.**

**Impaired advances** against **impairments** is the second. Sector impaired advances were R307.6bn, 5.20% of gross loans, against specific impairments of R147.2bn, 2.50% [3] — a coverage of 47.85%, which is not under-provisioning. The gap is collateral the bank expects to realise, so a secured book carries far lower coverage than an unsecured one at identical arrears, and comparing coverage across banks without comparing book mix produces a conclusion that is plainly false.

**Headline earnings** is the third. It is a JSE-mandated measure, not an IFRS one, and it is the primary number in every South African bank results announcement; several banks also publish a "normalised" view on top. One announcement therefore carries at least three earnings figures, and a dashboard showing one of them without saying which has clarified nothing.

## Run two: the interest and margin words

| Term | What it means |
|---|---|
| **NII** | Net interest income: interest earned on assets less interest paid on liabilities |
| **NIM** | Net interest margin: NII over average interest-earning assets, in basis points or per cent |
| **NIR** | Non-interest revenue: fees, commissions, trading, insurance and investment income |
| **Endowment** | The margin benefit from non- and low-rate liabilities when the policy rate rises |
| **CASA** | Current and savings accounts: the cheapest, stickiest deposits |
| **Repo rate** | The SARB policy rate. 7.25% from 25 September 2026 |
| **Prime** | The published reference lending rate: repo plus 3.5 percentage points. 10.75% from 25 September 2026 |
| **JIBAR** | Johannesburg Interbank Average Rate. A forward-looking term rate. Ceases publication end-December 2026 |
| **ZARONIA** | South African Overnight Index Average. The backward-looking replacement |
| **Credit adjustment spread** | The spread added on conversion, to compensate for JIBAR's credit component |
| **Fallback** | Contract language naming the replacement rate if the benchmark ceases |
| **FTP** | Funds transfer pricing: the internal price of money between business units |

**Endowment** is the word a newcomer most often hears and least often understands. A portion of the liability base pays zero or near-zero interest: current accounts, low-balance savings, money in transit. When the policy rate rises, the asset side reprices to prime almost immediately, because almost all South African retail lending is prime-linked, while that liability base does not reprice at all. The spread widens with no management action, and narrows the same way when rates fall. South African banks disclose it as an explicit line in their net interest income attribution, a convention far less common in United States reporting.

So deposit market share is margin share, which is why a transactional-account price war is rational rather than self-harming. And the inflation target change of 12 November 2025 — from a 3–6% range to a 3% point target with a one percentage point band, implemented over two years [4] — lowers the steady-state policy rate and structurally compresses the endowment benefit over the next cycle.

**Prime** is the second: a real, single, national reference rate pegged to the repo rate, against which almost all retail lending is quoted [5]. There is no local equivalent at scale of the United States thirty-year fixed mortgage, so repricing is automatic rather than contractual-event-driven — an MPC announcement changes the rate on the whole book overnight, including loans mid-registration. **Repo** is the third and is a genuine homonym: it means the policy rate, and it means an actual repurchase agreement, which is secured borrowing. Context disambiguates in conversation; a data dictionary and a code list must not rely on context.

**ZARONIA and the transition** is the fourth, and it will appear on your roadmap whether you want it or not. JIBAR is forward-looking — the rate is known at the start of the period and the interest calculation is a single multiplication. ZARONIA is an overnight rate compounded in arrears, known only at the end, calculated daily with observation shifts, lookbacks and lockouts.

> This is not a renaming. It changes the shape of an interest calculation, and it reaches every loan servicing, bond, derivative, hedge accounting and funds transfer pricing system that touches a floating rate. The Market Practitioners Group has recommended a "no new JIBAR" milestone of 1 May 2026, revised from end-March, with "new" defined to include renewals, refinancings, rollovers and material amendments; JIBAR ceases publication at the end of December 2026 [6]. For legacy contracts without adequate fallbacks the SARB is amending the Financial Sector Regulation Act to designate replacement benchmarks with statutory safe harbours, and **different replacement rates may apply to different product classes** [6]. That last clause is the one to put in front of your data governance forum.

**FTP** is the fifth, and it is the most consequential calculation in a bank that almost nobody outside treasury can explain. It charges asset-generating units for the funds they use and credits liability-generating units for the funds they raise, with treasury acting as an internal bank; a deposit's value is the cost of equivalent-term borrowing less the rate paid on it, and a loan's value is its interest income less the rate paid to treasury [7]. The calculation carries a liquidity term premium and adjustments for rate and liquidity risk, and customer behaviour — prepayment, early withdrawal, uncertain timing — complicates all of it [7]. Under centralised control the transfer price does not change total profit, only its split between units [7].

Which is why it matters to you. FTP determines the profit and loss of every product and business unit, and product profitability drives the whole investment portfolio. If the curve, the liquidity premium and the behavioural assumptions are not reproducible and versioned, no product profitability number in the bank is defensible — and every unit keeps its own shadow profit and loss, which is what you will actually find.

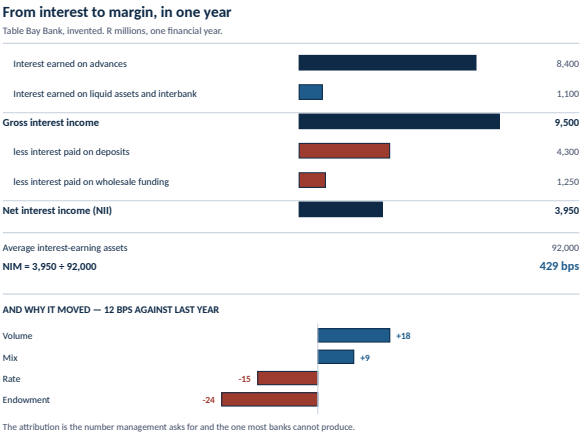



**Figure 3.1** The interest words as arithmetic, on an invented year at the invented Table Bay Bank. Gross interest income, net interest income and net interest margin are three different numbers, and the attribution underneath is what a management committee actually argues about.

## Run three: the credit words

| Term | What it means |
|---|---|
| **PD** | Probability of default |
| **LGD** | Loss given default: the share of exposure not recovered after collateral |
| **EAD** | Exposure at default |
| **ECL** | Expected credit loss: the IFRS 9 impairment measure. PD × LGD × EAD, discounted and probability-weighted |
| **Stage 1** | Performing. Twelve-month ECL |
| **Stage 2** | Significant increase in credit risk since origination. Lifetime ECL |
| **Stage 3** | Credit-impaired. Lifetime ECL, interest on the net amortised cost |
| **SICR** | Significant increase in credit risk: the stage 1 to 2 trigger. Not defined by IFRS 9 |
| **Default** | Presumed no later than 90 days past due unless reliable evidence supports a later criterion. Also not defined by IFRS 9 |
| **NPL** | Non-performing loan. "Impaired advances" in South African regulatory usage |
| **CLR** | Credit loss ratio: the impairment charge over average gross advances, in basis points |
| **Coverage** | Loss allowance over gross exposure, or over NPLs. Mix-dependent and not comparable across banks |
| **Management overlay** | A judgemental addition to modelled ECL |
| **Write-off** | Balance-sheet removal when no reasonable expectation of recovery remains. The legal claim usually survives |
| **Reckless credit** | Credit granted without a proper affordability assessment. Prohibited under NCA sections 80–81 |
| **Affordability assessment** | The prescribed Regulation 23A calculation of discretionary income |
| **Minimum Expense Norms** | The prescribed table of minimum assumed living expenses |
| **In duplum** | The statutory cap on what may accrue against a defaulting consumer |
| **Debt review** | The NCA statutory over-indebtedness restructuring process |
| **Section 129 notice** | The statutory notice a credit provider must deliver before enforcement |

**The three stages** are where the data requirement lives.

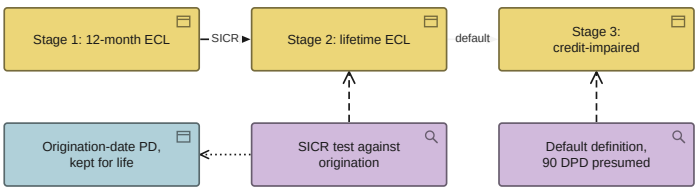



**Figure 3.2** The three IFRS 9 stages and the two tests that move an exposure between them [8]. The Access edge is the expensive one: the SICR test compares today's risk with the risk assessed at origination, so the origination-date figure must survive the life of the asset. Cure, back towards stage 1, needs its own dated event and is not drawn here.

Stage 1 is performing and carries a twelve-month expected credit loss, recognised even when the probability of default is tiny. Stage 2 is a significant increase in credit risk *since initial recognition*, and carries a lifetime loss. Stage 3 is credit-impaired [8].

Three things routinely get this wrong. SICR tests the *change in the probability of default* against origination, not the change in the loss amount. IFRS 9 defines neither "significant" nor "default", though 30 days past due is a rebuttable presumption of the first and 90 days of the second [8]. And revolving facilities are measured over the period of credit exposure even beyond the contractual period [8], so a credit card's expected loss needs a modelled behavioural life.

The architectural consequence is worth repeating in these words: **the origination-date risk grade or probability of default has to be stored, immutably, per contract, for the life of the asset** — twenty years in a mortgage, where legacy systems store current risk and overwrite it monthly. Every stage transition and cure must be a dated event, not a status field, and the macroeconomic scenarios and their weights must be versioned with the forecast vintage. Management overlays sit on top: FirstRand added R1.1bn of forward-looking provisions for geopolitical risk in FY2026 [9], and an overlay of that size stays separately identifiable in the ledger for the rest of its life.

**CLR** is the most misused ratio in South African banking commentary. The FY2025 aggregate across the six majors was 182 basis points [2]; African Bank's personal banking book ran at 11.5% in the six months to March 2026 [10]; Standard Bank reported 73 basis points [11]. Not in conflict — different book mixes. A single credit loss indicator across a group holding both mortgages and unsecured lending is meaningless, and putting one on a dashboard is a reliable way to have the dashboard disbelieved.

**In duplum** is the term most often got wrong by people who have heard of it. Under section 103(5) of the National Credit Act, while a consumer is in default the total of interest, initiation fees, service fees, credit insurance, collection costs and default administration charges **may not exceed the unpaid balance of the principal debt at the time of default** [12]. The cap is derived from a *historical event* — the principal outstanding at the moment default occurred — so the bank must capture that figure as a durable fact and cap accruals against it for as long as default persists. A system holding only the current balance cannot compute it, and a collections system that cannot compute it is over-charging consumers in a way the regulator will find. I would verify the exact enumerated charge categories against the Act's text before relying on a list.

**Reckless credit and debt review** both create states a system has to honour. Credit granted without a proper assessment, or that leaves the consumer over-indebted, is reckless, and a court may set aside rights or suspend obligations [12]; the defence is the retained assessment — income evidence, expense inputs, the version of the norms table used, the bureau enquiry and the calculation, per application and *including declines*. Debt review is a statutory restructuring through a debt counsellor, after which the provider is constrained in enforcement [12], so the party and each agreement need a debt-review state with dates, counsellor, proposal, court order and restructured schedule, driving the arrears engine. The classic failure is one system honouring that state and another not, so the bank issues a default notice to a consumer under review.

## Run four: the capital and liquidity words

| Term | What it means |
|---|---|
| **RWA** | Risk-weighted assets: exposures multiplied by prescribed or modelled risk weights |
| **CET1** | Common equity tier 1: the highest-quality capital |
| **CCyB** | Countercyclical buffer. South Africa set a positive 1% from 1 January 2026 |
| **Pillar 2A** | The South African systemic risk add-on, 0.5–2.0% of total capital |
| **Pillar 2B / ICR** | The bank-specific individual capital requirement. Must *not* be publicly disclosed |
| **D-SIB / HLA** | Domestic systemically important bank, and its higher loss absorbency add-on |
| **Output floor** | The floor on internally modelled RWA, expressed against standardised RWA |
| **LCR** | Liquidity coverage ratio: high-quality liquid assets over 30-day stressed net outflows. Minimum 100% |
| **NSFR** | Net stable funding ratio: available over required stable funding across a year. Floor 100% |
| **IRRBB / EVE** | Interest rate risk in the banking book, and the economic value of equity measure of it |

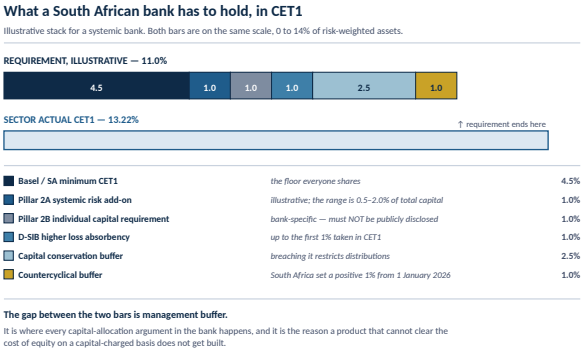



**Figure 3.3** The South African CET1 stack, illustrative for a systemic bank [15]. Note the third layer: a bank must disclose its systemic add-on and must not disclose its bank-specific Pillar 2B requirement, which is a constraint on how a capital report may be composed.

**RWA** is the word that connects architecture to the profit and loss. Each exposure is multiplied by a risk weight — low for government bonds and well-secured mortgages, high for unsecured lending and equity — and every capital ratio divides by the total. Gross credit exposure in the South African system was R9.9 trillion at December 2025 [13]; sector capital on 2025 averages was 17.22% total, 14.96% tier 1, 13.22% CET1 [3].

**The output floor** is the newest and has the biggest architectural bill attached. A bank using internal models must compute RWA **twice**, once internally and once on the full standardised approach, and take the higher of its internal number and the floored amount. That is a parallel calculation estate needing standardised inputs — external ratings, loan-to-value buckets, exposure classes, credit conversion factors — on *every* exposure, including those previously only modelled internally. The remaining Basel III post-crisis reforms took effect here on 1 July 2025 [14]. Two notes: what the market calls "Basel IV" the Prudential Authority calls Basel III, so use the local name in a local room; and the South African floor percentage and phase-in are worth verifying against the amended Regulations before quoting, because the Basel standard of 72.5% phased from 50% is not something I could confirm as the local calibration.

**Pillar 2A and Pillar 2B** carry a disclosure asymmetry that catches anyone designing external reporting. A bank must disclose its systemic add-on and must **not** publicly disclose its bank-specific Pillar 2B requirement [15]. So the capital-reporting model needs a per-component disclosure classification, and a Pillar 3 pack that derives a total from components without respecting it leaks a confidential number by arithmetic.

**LCR and NSFR** are where treasury's data problem lives. Both are computed over the *entire* balance sheet, every exposure classified by counterparty type, product, maturity bucket and behavioural assumption under prescribed run-off factors, daily and monthly respectively, with lineage per classification judgement. No core banking system holds that classification; the treasury and asset-liability system exists to apply it, and its quality is bounded by the weakest source feeding it. South African banks ran roughly 140–165% and 115–119% across 2023 to 2025 [13].

## Run five: the payments words

This is the run that needs the most care, because South African payments vocabulary is its own dialect and the words that look familiar are the dangerous ones.

| Term | What it means |
|---|---|
| **Clearing** | Exchanging and reconciling payment instructions between participants, before value moves |
| **Settlement** | The final, irrevocable transfer of value between settlement participants, in central-bank money |
| **Netting** | Offsetting obligations so that only the net amount settles |
| **SAMOS** | The South African Multiple Option Settlement system: the SARB's RTGS. Live 9 March 1998 |
| **PayInc** | The national clearing operator. Renamed from BankservAfrica in August 2025; the SARB holds 50% |
| **PCH** | Payment clearing house: one per stream, each with its own rulebook, cut-offs and participants |
| **EFT credit** | The batch push rail. Salaries and creditor payments. Deferred clearance |
| **EFT debit** | The batch pull rail. The plain, non-authenticated debit order |
| **Sort-at-source** | Submitting files directly to paying banks, bypassing central clearing |
| **RTC** | Real-Time Clearing: the older immediate-credit rail, launched 2006, for amounts up to R5m |
| **PayShap** | The instant payment service, live March 2023. ISO 20022 native, 24/7, R50,000 cap |
| **ShapID** | A proxy — usually a mobile number — that resolves to a bank account |
| **DebiCheck** | Authenticated collections: the mandate is confirmed to, held by and validated by the *payer's* bank |
| **TT1 / TT2 / TT3** | The three DebiCheck mandate authentication types |
| **Registered Mandates (RM)** | A separate rail from 10 March 2025, ISO 20022, late window, supporting business accounts |
| **RMS** | The Registered Mandate Service: the earlier non-authenticated fallback inside DebiCheck |
| **AEDO / NAEDO** | Authenticated and non-authenticated early debit orders. Switched off 31 October 2021 |
| **Mandate** | The payer's authority for a collection, with parameters the paying bank enforces |
| **Interchange** | The issuer-to-acquirer transfer per transaction. In South Africa determined by the SARB |
| **Issuing / acquiring** | Providing cards to cardholders; contracting merchants to accept them |
| **BIN / IIN** | Bank identification number: the leading card digits identifying issuer and product |
| **Chargeback** | The scheme process reversing a transaction against the acquirer and merchant |
| **Nostro / vostro** | "Our account with them" and "their account with us", in correspondent banking |
| **ISO 20022** | The structured financial messaging standard behind SAMOS, PayShap and Registered Mandates |
| **FinSurv / BOP category** | The SARB's cross-border reporting regime, and the purpose code every such payment carries |

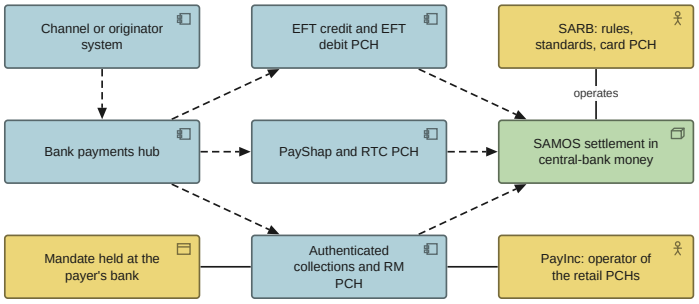



**Figure 3.4** The payments words laid out as a map: instruction on the left, clearing in the middle, settlement on the right [16][17][18]. Since September 2026 the card, high-value and several other clearing houses sit with the SARB while the retail ones named here sit with PayInc.

**Clearing, settlement and netting** are three different things and the words are not interchangeable. Clearing is the exchange and reconciliation of instructions. Netting offsets obligations so that only a balance moves. Settlement is the final, irrevocable discharge in central-bank money. All South African payments net-settle in SAMOS, live since 9 March 1998, running seven days a week and round the clock with a dedicated night window for clearing-house batches, where settlement is final and irrevocable under section 5(2) of the National Payment System Act — and the SARB does *not* guarantee it [16]. In 2024 SAMOS settled R153.7 trillion across 16.0 million transactions, 88% of that value wholesale [17].

Your data model therefore needs three separate concepts — instruction, payment, settlement — many-to-many in both directions. One instruction may produce no payment, one, or thousands over twenty years of a standing arrangement; one settlement discharges thousands of payments in a net batch.

**The clearing stream is a first-class attribute, not a channel detail.** Rulebook, cut-off, return window, dispute process and — since 2026 — the governing body all differ by stream [18]. By 2024 settlement value the split was EFT credit 70.3%, real-time clearing 11.0%, cards 10.7%, EFT debit 5.8%, authenticated collections 0.8% and rapid payments 0.7% [17]. A design that assumes instant payments are the default rail has designed for the smallest share.

> Get the institutional names right: three are in circulation and two of them mean different things. **PayInc** is the renamed BankservAfrica — a clearing and switching utility, not an industry body — founded 1972, renamed BankservAfrica 2010 and PayInc in August 2025, with the SARB's 50% stake completed in late 2025 [19][20]. **PASA** was the recognised payment system management body; its recognition has been withdrawn, functions transferring to the SARB on 11 August 2026 and to PayInc on 2 September 2026, and the SARB's own page disagrees with itself on whether completion was the 1st or the 2nd [18]. The **Payments Industry Body** is a *different thing*: a designed successor to PASA set out in a November 2022 report, which is not what happened. Material saying "the PIB replaces PASA" is describing a design, not the outcome — the commonest factual error in current South African payments commentary.

**EFT credit and EFT debit** are the batch rails. EFT credit volumes rose from 406.9 million in 2010 to 1.13 billion in 2024 [17], and clearance is deferred — the SARB's own framing is that customers wait "a day or two" [21]. EFT debit is the pull rail and is dispute-prone by design, because the *business* holds the mandate and must produce evidence within a timeframe or the collection reverses [22]. **Sort-at-source** is the local pathology worth knowing by name: users bypassing central clearing by submitting files straight to paying banks, processed as on-us transactions [21], so a corporate may hold direct file arrangements with several banks in parallel to the clearing system — which multiplies the integration surface and defeats a single-rail design.

**DebiCheck** inverts the control point, and this is the structural fact to carry out of the run. Under an ordinary EFT debit the originator collects on the strength of a mandate it holds, which the bank never sees. Under DebiCheck the debtor confirms the arrangement electronically with *their own* bank once, at contract start, and from then on that bank validates every collection against the agreed terms — amount, frequency, date, tolerance, maximum collection amount — before processing it [22][23]. The mandate is a bank-held, long-lived, first-class object with its own lifecycle: register, approve, collect, manage [23].

There is no European or United Kingdom equivalent. SEPA mandates are held creditor-side; United Kingdom Direct Debit mandates sit with the paying bank but without per-collection validation against an authenticated mandate. **A design imported from either will not model DebiCheck.** Note too that a processed DebiCheck collection cannot be reversed: a customer who disputes the mandate raises a dispute rather than getting an automatic reversal, and the burden of mandate evidence sits with the bank rather than the business [22].

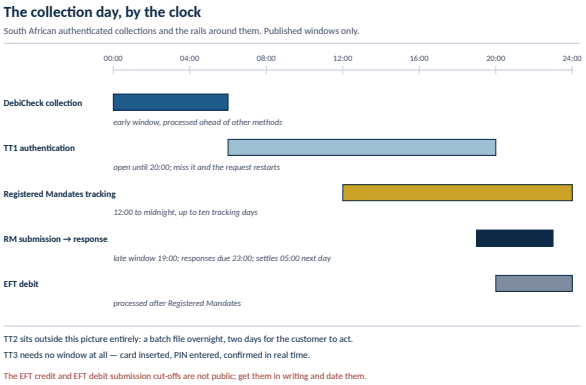



**Figure 3.5** The collection day by the clock, from published window definitions [22][25]. The move of Registered Mandates from the early to the 19:00 late window on 10 March 2025 changed the cash-flow timing, and therefore the success-rate model, of every collections business that migrated.

**TT1, TT2 and TT3** are the three authentication types and the differences are operational. TT1 is a real-time request authenticated in the app, at a point of sale or at certain ATMs, possible until 20:00 the same day; miss it and the process restarts. TT2 goes overnight in a batch file and gives the customer two days. TT3 is card-present, confirmed in real time, with no separate window at all [22][24]. A vendor reports 35–40% of TT1 attempts failing monthly against better than 99.5% for TT3 — vendor-published figures, to be treated as such [24], though the direction follows from the window mechanics.

**Registered Mandates is a new rail, not a rename**, and this is the second most common error in circulation. RMS, the Registered Mandate Service, was a mechanism inside DebiCheck for collecting where the consumer had not authenticated, running in the early window. Registered Mandates is a distinct rail meant to replace the legacy non-authenticated EFT debit order, and the shift was **mandated from 10 March 2025**, after which RMS mandates cannot be collected on DebiCheck [25]. What changed: the window moved from early to a **late window at 19:00 SAST**, with tracking from 12:00 to midnight for up to ten tracking days, responses due 23:00 on the action date and settlement at 05:00 the next day; priority sits after DebiCheck and before EFT debits; the rail is **ISO 20022**; and it supports business and **dual-signatory accounts, which DebiCheck cannot** [25]. An RM mandate can be upgraded to DebiCheck by an amendment authenticated by both banks — a state transition, not a re-registration.

**AEDO and NAEDO were switched off on 31 October 2021**, and from 1 November those instructions were not permitted, though tracking ran out and disputes, stop payments and recalls stayed permitted [26]. It still matters twice: both remain named clearing houses in the published rules, so a pre-2021 estate carries their mandate records and reason codes; and any collections trend spanning 2021 compares two incompatible instruments across a regulatory cliff.

**Interchange** here is determined by the central bank, the opposite of the European position where a regulation caps a market-set fee. The SARB determines rates for ATM and cash-back at point of sale, where the issuer pays the acquirer; for debit and credit cards, where the acquirer pays the issuer; and for authenticated collections including the Registered Mandate Service [27]. EFT credit, EFT debit and real-time clearing are market-determined and bilateral [27]. The methodology is cost-based and reviewed every five years, rates are reviewed annually, and both are enforced by directive [27]. **So interchange is externally sourced, effective-dated, versioned reference data on an annual cycle — and in a surprising number of banks it is hard-coded in the acquiring platform's pricing module.**

**Issuing and acquiring** are the two sides of the card business: the issuer carries the cardholder's credit risk, the acquirer the merchant's. The **BIN** is the leading six or eight digits of the card number, identifying issuer and product — reference data that changes. **Chargeback** reverses a transaction against the acquirer and merchant, and its governing constraint is the clock: scheme deadlines are hard, and a dispute that expires in a queue is a loss the bank absorbs. Authorised-push-payment fraud is not a chargeback at all, because the customer authorised the payment — yet the ombud has awarded refunds in those cases and criticised banks for responding too slowly after becoming aware [28]. **Nostro** is "ours with them" and **vostro** "theirs with us", reconciled daily against camt.053 statements.

One last fact, and it is the one that surprises people. **SAMOS completed its migration to ISO 20022 on 17 September 2022, the first real-time gross settlement system on the African continent to do so** [29], and PayShap and Registered Mandates are ISO 20022 native. The rails are ahead of most bank cores. The integration problem in a South African bank is usually not that the rail is primitive; it is that the rail is structured and the core is not.

## Run six: the deposit and account words

| Term | What it means |
|---|---|
| **Current account** | A transactional account. Still widely called a cheque account, nine years after cheques were withdrawn |
| **Call account** | A deposit repayable on demand, usually interest-bearing |
| **Notice deposit** | A deposit repayable after a notice period: seven, thirty-two, sixty or ninety days |
| **Fixed deposit** | A deposit for a fixed term, with a penalty for early withdrawal |
| **Stokvel account** | A group savings account, typically with three mandated signatories |
| **Covered deposit** | A qualifying deposit within the CODI limit of R100,000 per depositor per bank |
| **Qualifying product** | A product type CODI's rules include in deposit cover |
| **SCV** | Single customer view: the regulator-specified consolidation of a depositor's qualifying balances |

**Covered deposit and the single customer view** are a data specification wearing the clothes of a policy. Cover is R100,000 per qualifying depositor per bank, on the total of that depositor's qualifying balances, and it is **gross** — negative balances are not netted [30]. The bank must consolidate each qualifying depositor into one SCV record, and the rules then specify: joint accounts split by the sharing ratio *recorded by the bank*, defaulting to equal shares; formal beneficiary accounts looked through where the bank can identify beneficiaries monthly; informal beneficiary accounts such as stokvels covered per holder, split pro rata where there are several sets of signatories; sole proprietors covered separately for personal and business accounts; and a prescribed field set including a reliability-of-identification flag and not-readily-payable indicators with reasons [30].

> A sharing ratio per joint account holder is a field most banks have never captured. A reliability-of-identification flag is a concept most party masters do not have. Neither is optional now, and both are needed monthly, within ten working days of month-end. This is the cleanest example in South African banking of a regulator specifying a data structure rather than a principle, and it is the strongest business case an architect will get for resolving party identity.

**Dormancy** is a conduct obligation, not a housekeeping rule: the Conduct Standard for Banks requires a bank to identify dormant products and notify customers of them [31]. That makes it a dated rule and a notification event per product, not a flag somebody sets.

## Run seven: the regulatory acronym soup

| Acronym | Who or what it is |
|---|---|
| **PA** | Prudential Authority: safety and soundness, housed inside the SARB. Registers banks |
| **FSCA** | Financial Sector Conduct Authority: market conduct and market integrity |
| **NCR** | **National Credit Regulator.** In a South African banking sentence, never anything else |
| **NCA** | National Credit Act 34 of 2005: the consumer credit regime |
| **FIC / FICA** | Financial Intelligence Centre and its Act: the AML and counter-terrorist-financing regime |
| **CDD / EDD** | Customer due diligence, and enhanced due diligence where risk is higher |
| **PEP** | Politically exposed person: triggers enhanced due diligence |
| **STR / CTR** | Suspicious transaction report, and cash threshold report, both filed through goAML |
| **CODI** | Corporation for Deposit Insurance. Operational 1 April 2024 |
| **NFO** | National Financial Ombud Scheme. Operational 1 March 2024, consolidating four schemes |
| **POPIA** | Protection of Personal Information Act 4 of 2013. Enforceable from 1 July 2021 |
| **COFI** | Conduct of Financial Institutions Bill: the planned consolidated conduct statute |
| **FAIS** | Financial Advisory and Intermediary Services Act 37 of 2002: advice and intermediary services |
| **Conduct Standard 3 of 2020** | The FSCA's conduct standard for banks |
| **Joint Standard 1 of 2023** | IT governance and risk management. Commenced 15 November 2024 |
| **Joint Standard 2 of 2024** | Cybersecurity and cyber resilience. Commenced 1 June 2025 |
| **BA returns** | The statutory return suite. Forms removed from the Regulations by GoN 5802, 1 February 2025 |
| **FinSurv** | The SARB's Financial Surveillance department: exchange control and cross-border reporting |
| **FATF** | Financial Action Task Force. South Africa was grey-listed February 2023 and exited 24 October 2025 |




**Figure 3.6** The acronyms sorted by who is asking. Hearing "output floor" and "BA 700" means a prudential conversation; hearing "target market" and "record of advice" means a conduct one. Knowing which room you are in is half of understanding the question.

Three of these are routinely got wrong and all three matter.

**"NCR" means the National Credit Regulator** — it registers credit providers, debt counsellors and credit bureaux under the National Credit Act, with enforcement shared with the National Consumer Tribunal [32]. If the acronym means a hardware vendor to you, retrain before your first workshop.

**The Joint Standards are numbered confusingly and three of them are "number 1".** Joint Standard 1 of 2023 is IT governance and risk management, commenced 15 November 2024. Joint Standard 2 of 2024 is cybersecurity and cyber resilience, commenced 1 June 2025 with twelve months to full compliance. **Joint Standard 1 of 2024 is outsourcing by *insurers* and does not apply to banks** [33]. There is also a Joint Standard 1 of 2025 on central counterparty and trade repository exemptions. Compliance commentary miscites the insurance standard as a banking obligation often enough that you should check the instrument title yourself before accepting it in a requirements document.

**South Africa does not have a regulated open banking regime, and never write that it does.** The SARB's 2020 consultation paper states flatly that there is no policy or regulatory framework for open banking and that the National Payment System Act has no provisions on open banking, screen scraping or APIs [34]; a SARB working paper of March 2026 describes the position as market-driven, with third-party providers and APIs outside the current regulatory purview [35]; and the FSCA's Open Finance position paper explores a possible regime without dates. What exists is bilateral commercial APIs on each bank's own terms, plus **SARB Directive No. 2 of 2024**, gazetted 15 November 2024 and effective about 13 February 2025, which regulates rather than bans the issuing of EFT credit payment instructions on a payer's behalf, screen scraping included, and requires registration with the SARB, informed consent per payment, credentials never stored, five-year audit trails, monthly volume and value submissions by the 15th and compliance attestations twice a year [36]. Importing a United Kingdom Open Banking or PSD2 reference architecture designs for obligations that do not exist here and omits the one that does.

## The five words that mean different things in the same bank

Everything above is vocabulary. This last section is an architecture problem presenting as a vocabulary problem, and it is the most useful part of the chapter.

### Account

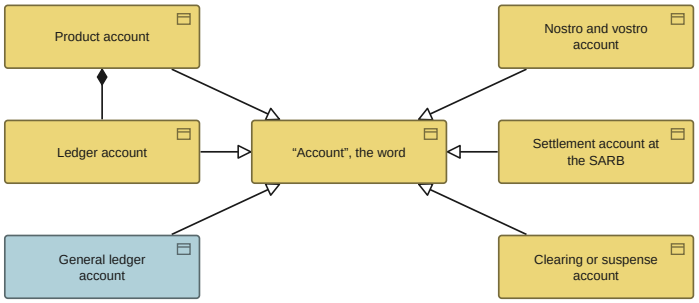



**Figure 3.7** Six things called "account", and the one real structural relationship among them: a product account is composed of several ledger accounts. One table with a type column spanning all six is how a bank ends up joining the general ledger to the customer estate by accident.

**Account** names at least four distinct things, and in most banks they share a table. A **product account** is the customer-facing instance of a product — the cheque account, the twenty-year bond, the credit card — with parties in roles, a mandate, limits, pricing, statements and a lifecycle. A **ledger account** is the balance-bearing unit entries post against, and one product account usually has several: capital, accrued interest, fees, arrears, insurance, redraw. A **general ledger account** is a line in the chart of accounts, an accounting classification rather than a customer thing, and it owns the bank's double-entry integrity. A **nostro or vostro account** is a correspondent account reconciled daily against another bank's statements. Two more sit in the corners: the **settlement account** at the central bank in SAMOS, and the **clearing or suspense account** holding in-flight value — which the deposit insurance rules expressly exclude from qualifying products, an exclusion only expressible if suspense accounts are a distinguishable kind of thing [30].

Three failures follow. One `ACCOUNT` table with a type column spanning every sense, so the general ledger and the customer estate join by accident. Interest and fees hung off the capital balance, after which arrears cannot be aged and in duplum cannot be computed. And an assumption of one owner, which joint, partnership, trust and minors' accounts all break. The settling question: *does it carry a statement-visible balance and close independently?* If it carries a balance but is never independently closed or shown, it is a ledger account inside a product account.

### Customer

**Customer** is a role, not a thing, and it is usually a bad root entity: a derived predicate — "a party with at least one live product relationship" — that goes stale the moment it is stored, and that cannot represent the prospect, the ex-customer, the guarantor holding no product, the third-party payer, the beneficial owner behind a company or the deceased estate. It is also not consistently defined inside one bank. Retail counts per natural person, business banking per enterprise group, the card issuer per plastic, regulatory reporting per single-customer-view record: four counts that never reconcile and all four defensible in context.

`Party` is the thing — a legal person, natural or juristic, that the bank can transact with or owe — and customer, depositor, borrower, guarantor, signatory, beneficial owner, counterparty, employee and supplier are all *dated roles* a party plays against an account, an agreement, a facility, a payment or another party. The settling question: *would the bank need to know this exists even with no product?* A beneficial owner behind a corporate client holds no product and FICA still requires identification and verification, so they are a party. A plastic card is not.

### Balance

**Balance** is a derivation, not a stored fact: the sum of signed entries against a ledger account to a point in time. Banks hold at least five that users assume are one number.

| Balance | Meaning |
|---|---|
| **Ledger or book** | The sum of all posted entries. The accounting truth |
| **Available** | Ledger, less holds and uncleared items, plus unused overdraft limit. The customer-facing number |
| **Cleared or value-dated** | Entries whose value date has arrived. The interest-bearing number |
| **Statement** | The ledger balance frozen at the cut-off. A historical artefact |
| **Arrears base** | Capital plus accrued interest plus fees, as aged. What collections acts on |

The deposit insurance rules show why one column cannot survive regulation: gross balances, accrued interest to the reporting date included, outstanding fees excluded, unsettled payments added or deducted, foreign currency at the close-of-business rate, two decimals, and negative balances treated as zero [30]. That is a sixth derivation from the same entry set, and none of the six is "the balance". The classic symptom is three implementations of *available* balance — app, ATM and card authorisation host — which is how a customer sees three numbers in an afternoon. The settling question: *if two reports differ on this, is one wrong?* If not, you have two named derivations, not one balance.

#### Product

**Product** is a definition; a product account is an instance. The product is the manufactured template — eligibility, pricing, fee schedule, interest basis, limits, term, currency, channel availability, required documents, regulatory classification — held in a catalogue, versioned, with effective dates. In most banks that catalogue does not exist as one thing: there is one per core system, plus a marketing catalogue, plus a pricing sheet, plus a regulatory classification maintained in finance.

The product carries classifications nothing else can: whether it is deposit-taking, credit or investment business, which decides the statute; whether it is a qualifying product for deposit insurance [30]; its BA 900 instrument and maturity classification; and its National Credit Act subtype — mortgage, credit facility, unsecured, developmental, short-term, incidental — each with different fee caps, interest caps and disclosure duties. Put qualifying status on the account instead and it will be wrong on every account opened before somebody remembered to set it.

The settling question: *would changing this change an existing customer's contract?* If yes it is a new product version with an effective date, and existing accounts stay on the old one. A marketing name, a channel or a campaign is not a product.

#### Limit

**Limit** means at least four things, enforced by different systems. A **credit limit** is granted capacity on a facility: what the customer may draw. It cannot be exceeded; it is a control. An **exposure** is the amount at risk — drawn plus undrawn committed plus contingent plus derivative add-ons, netted only where netting is legally enforceable. It can be exceeded; it is a measurement, and headroom is the derivation between the two. A **transaction or channel limit** is an operational cap: a daily app payment limit, PayShap's R50,000 scheme cap, which banks may set lower. A **mandate limit** is an authority: what a named person may approve, release or write off.

The test is the one from the credit model: *can this number be exceeded?* A limit cannot; an exposure can. Systems that store all four in a field called `limit` produce a bank that cannot answer "how much can this customer draw" or "how much are we exposed to this group", and both questions have a regulator attached.

> These five words are the reason a glossary is an architecture deliverable rather than a courtesy. The practical move is not to pick one meaning and enforce it — you will lose — but to name each sense separately in the conceptual model, map each system's usage onto the named senses, and make the mapping the thing that is governed. Chapter 9 builds that model properly.

## What to do with this chapter

Three things, in the first fortnight.

Collect the bank's own definitions before writing yours. Every bank has a data dictionary, a finance glossary and a product catalogue, and they disagree. The disagreements are the map of where the architecture work is.

Date everything. Rates, interchange, the capital stack, the payments governance arrangements and the benchmark transition all changed within the last two years and several will change again. Any design document saying "the current rate" or "the current scheme rules" needs a verification date on the page.

And when somebody uses one of the five ambiguous words in a workshop, ask which sense they mean — once, politely, and in front of everyone. It is the cheapest architecture work available and it usually surfaces a defect in the room.

**THE CHAPTER IN TEN LINES**

- Banking vocabulary is clusters, not an alphabet: the words only make sense against the mechanism they describe, which is why this chapter runs in themed groups.
- Group or bank, gross or net, headline or normalised, point-in-time or average — four qualifiers that belong on every banking number, and a reporting layer that drops them ships reconciliation defects.
- The endowment effect is why deposit market share is margin share, and the move to a 3% inflation target structurally compresses it over the next cycle.
- ZARONIA is not a renaming of JIBAR: an interest calculation moves from one multiplication at period start to daily compounding in arrears, across every system touching a floating rate, against a fixed deadline.
- IFRS 9 measures credit risk against the risk assessed at origination, so the origination-date grade must be stored immutably for the life of the asset and every stage change must be a dated event.
- In duplum caps accruals against the principal outstanding at the moment of default — a historical fact a system holding only the current balance cannot compute.
- The output floor makes a bank on internal models compute risk-weighted assets twice, which means standardised inputs on every exposure and a parallel calculation estate.
- Clearing, netting and settlement are three different things; the clearing stream is a first-class attribute because rulebook, cut-off, return window and governing body all differ by stream.
- DebiCheck inverts the control point — the mandate is held and validated at the payer's bank — and no SEPA or UK Direct Debit design will model it.
- Registered Mandates is a new rail, not a rename: mandated from 10 March 2025, ISO 20022, a 19:00 late window, and the only authenticated answer for business and dual-signatory accounts.
- Interchange here is determined by the central bank on an annual cycle, which makes it versioned regulated reference data rather than a pricing parameter.
- Account, customer, balance, product and limit each mean several different things inside one bank, and naming the senses separately in the conceptual model is the architecture work that the ambiguity is hiding.

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- **35.** Nanziri, Gbahabo and Ofori-Sasu — open banking and financial inclusion in South Africa, SARB Working Paper WP/26/07 (16 March 2026) https://www.resbank.co.za/content/dam/sarb/publications/working-papers/2026/26-07/financial-inclusion-sa.pdf
- **36.** SARB — Directive No. 2 of 2024, issuing of electronic funds transfer credit payment instructions on behalf of the payer, Notice 5550 in Government Gazette 51556 (15 November 2024) https://www.gov.za/sites/default/files/gcis_document/202411/51556gon5550.pdf

CHAPTER 4

# How the business model works

*An architect who cannot draw the business model on one page has no way to tell a system that is merely old from a system that is load-bearing. Two instruments fix that: the nine-block canvas, and the customer journey with the systems named and the failures marked.*

## The question you will be asked in week three

Somebody will put a system in front of you and ask whether it matters. It usually arrives as a budget question. Do we re-platform the deposit core. Do we keep paying for the business banking portal. Do we let the card team buy their own fraud engine.

You cannot answer that from a technical assessment. A system matters in proportion to the part of the business model it carries and the part of the customer's experience it decides. The deposit core is twenty-five years old and holds every current account: old, and load-bearing. The reporting database three analysts query is also old. One of those is an architecture problem and the other is a licence renewal, and nothing in the code tells you which is which.

This chapter gives you two one-page instruments. The first is the business model on nine blocks. The second is the customer journey, done properly, with the systems named and the failures marked. Between them you can say, in a sentence a chief financial officer will accept: this system sits on the revenue stream for 70% of our transaction fees and it is the only place the available balance is computed, so it is not a licence renewal.

Neither instrument is banking-specific. The nine-block canvas comes from Osterwalder and Pigneur [1] and journey mapping comes from service design. What makes them useful here is that banking punishes the generic version of both. A canvas for a bank whose customer segment block says "customers" and whose value proposition block says "financial wellbeing" is worse than no canvas, because somebody will believe it. A South African full-service bank has at least six segments that need different operating models, and the value proposition is not wellbeing. It is: my money is safe, it moves when I tell it to, and when I need credit the answer comes today.

There is a second reason to do this early, and it is specific to banks. In most industries the business model and the balance sheet are different documents. In a bank they are the same document. Deposits are a liability, loans are an asset, and revenue is the difference between two prices rather than a margin on one. Sector deposits were R6.67tn against loans and advances of R6.08tn at the end of December 2025, on total assets of R8.97tn [2]. Read that once more: the bank owes more than it is owed, and that is the normal, healthy state. Every instrument in this chapter has to survive contact with that fact, and the plain canvas does not — which is why the section on what the canvas leaves out is as long as the walk through the blocks themselves.

> Fill the canvas with invented names, named systems and real numbers. A canvas you cannot argue with is a canvas nobody read.

A note on the worked example. **Table Bay Bank** is invented. It is a mid-tier South African full-service bank with roughly 2.4 million retail customers, a business bank, a small corporate and transactional banking arm and a treasury. Nothing in it is a case study of any real bank, and where I give a number for the market rather than for Table Bay, it is sourced. I also name Table Bay's systems by what they do rather than by product name, because the function is the part you need to recognise in a bank that is not Table Bay.

## The canvas for a full-service bank

**Business model canvas — a South African full-service bank**

*Table Bay Bank (invented). Retail, business, a small corporate book and a treasury.*

**KEY PARTNERS**
- The SARB for settlement and PayInc for clearing — not partners, the ground you stand on
- Visa and Mastercard
- Credit bureaux
- Home Affairs, for identity
- Retailers, for cash and reach
- Bond originators and motor dealers
- Insurers, for credit life and attached cover
- Correspondent banks
- Core, card and cloud vendors

**KEY ACTIVITIES**
- Gathering and keeping deposits
- Underwriting, pricing and collecting credit
- Clearing and settling payments
- Managing capital, funding and liquidity
- Regulatory reporting and financial crime control

**KEY RESOURCES**
- The licence, and the capital behind it
- The deposit franchise
- The balance sheet itself
- Customer data and the credit models built on it
- Core ledgers and rail connectivity
- Branches, ATMs and agents
- Trust

**VALUE PROPOSITIONS**
- Money is safe, and R100,000 of it is guaranteed by statute
- An account that reaches every rail in the country
- Credit at the moment it is needed, priced today
- One provider for transacting, borrowing, insuring, investing
- For a business: collections that work and payroll that lands
- Presence where digital is not trusted

**CUSTOMER RELATIONSHIPS**
- Self-service at the bottom, relationship-managed at the top
- Primacy: whose account the salary lands in
- Depth of use, not accounts opened

**CHANNELS**
- App and internet banking
- Branches and service points
- ATMs and cash devices, in managed decline
- USSD
- Contact centre
- Relationship managers
- Originators, dealer floors, retail partners
- Host-to-host files and APIs

**CUSTOMER SEGMENTS**
- Entry-level and mass retail
- Middle market
- Affluent and private clients
- Small business and stokvels
- Commercial and mid-corporate
- A small corporate and transactional book
- Other banks, as counterparties

**COST STRUCTURE**
- Interest paid to depositors — the largest outflow of all, and not shown as a cost
- Credit losses: 182 bps of advances across the sector in FY2025
- People, and the branch and cash estate
- IT run cost, plus mandatory externally-scheduled change
- Compliance, regulatory reporting and financial crime control
- Fraud losses, and the cost of handling them

**REVENUE STREAMS**
- Net interest income: the endowment benefit plus the lending spread
- Monthly account and transaction fees
- Card interchange — a rate set by the SARB, not by the schemes
- Merchant service fees on acquiring
- Forex margin and cross-border fees
- Insurance commission and asset management fees
- Arrangement, guarantee and advisory fees in the corporate book

**Figure 4.1** The nine-block business model canvas for Table Bay Bank, an invented mid-tier South African full-service bank. Not a case study of any real bank; the shape is typical, and the shape is the point.

Now walk the blocks. Each one implies a capability, and most of them imply a system you will have to find, name and form a view about.

#### Customer segments

A bank that serves entry-level retail, the middle market, affluent and private clients, small business, commercial and mid-corporate, and a handful of large corporates is running six businesses, not one market. They differ in balance size by four orders of magnitude, in the advice they need, in how they are priced, and in what good service means. One bank's affluent segment starts above R750,000 of annual income and its private segment above R5m [3]; its commercial and mid-corporate segment runs to R1.5bn of client revenue [4]. The entry-level customer at the other end may hold a few hundred rand and transact twenty times a month.

Two segments outsiders always miss. The first is **other banks**: the interbank desk, correspondent relationships and the counterparty book are a segment with their own credit limits and their own systems. The second is **the stokvel and the informal savings group** — a real South African segment with three signatories, no single beneficial owner in the conventional sense, and a product nobody designed for it. At least one large bank has digitised remote stokvel onboarding with selfie verification and Home Affairs checks while most of the market has not [5].

*Architecture implication.* Segments this far apart cannot share one cost-to-serve. An entry-level account has to be administered for a few rand a month; a mid-corporate client can afford a relationship manager, a credit analyst and six weeks of structuring. One account platform configured to serve both ends up too expensive for one and too rigid for the other. This is the most common honest reason for the multi-core estate you will walk into. Decades of mergers are the other, and the two get blamed for each other.

#### Value propositions

Write these as the customer would say them afterwards, not as marketing says them beforehand. For a salaried retail customer: *my salary landed, the debit orders went off, and nothing surprised me.* For somebody who needs a loan: *they said yes on Tuesday and the money was there on Thursday.* For a small business: *the collections run works, and when it did not, somebody picked up.* For a corporate treasurer: *they have the balance sheet, and their settlement does not fail.*

Underneath all of them is the one proposition a bank sells and nothing else does: **safety, with a legal floor under it**. South Africa's deposit insurer has been operational since 1 April 2024 and covers R100,000 per depositor per bank, principal and interest; the Deposit Insurance Fund reached its R20bn target in its first year, and 95% of depositors were fully covered at 31 March 2025 [6].

*Architecture implication.* That guarantee is not a brochure line, it is a data requirement. Paying out within days means aggregating every balance per depositor across every product and every legal identity, computable at a point in time, to a specified field set and a monthly submission. Most legacy deposit estates cannot do this without a dedicated build, and the bank that discovers this during a resolution has discovered it too late.

#### Channels

Table Bay sells and serves through an app, internet banking, branches, ATMs and cash devices, USSD, a contact centre, relationship managers, bond originators, dealer floors, retail partners and file or API connections for corporates. Eleven channels is normal and none of them is retiring soon. The physical estate is in managed decline rather than collapse: one large bank went from its 2021 position to 574 outlets and 4,976 cash devices by mid-2026, a 41% reduction in devices [7]. Reduction is not removal, and cash demand has not fallen as fast as the estate.

*Architecture implication.* Channels set the latency budget and they decide where data is created. A card authorisation has a few hundred milliseconds. A PayShap payment carries a customer expectation of ten seconds and a R50,000 cap [8]. A corporate payroll file arrives once a month and must not fail. A mid-corporate credit application takes six weeks. The honest conclusion is one decision service, one payments hub and one mandate service behind many front ends, with what each channel may do held as enforced configuration. The conclusion you will find already built is one stack per channel, each with its own copy of the rules.

#### Customer relationships

The metric that matters here is not accounts opened. It is **primacy**: whose account the salary lands in, which card is in the wallet's default slot, which bank the debit orders point at. One listed bank reports "fully banked" penetration of 41% of its retail base as its central relationship measure, alongside 16.5 million monthly app users [9]. Depth, not breadth.

*Architecture implication.* Primacy is a derived, cross-product, behavioural measure. It needs transaction-level history, a party model that knows the salary credit from any other credit, and a definition everybody agrees on. It cannot be computed inside any single core, which is why it ends up in the warehouse and why the number in the board pack and the number in the campaign tool differ.

#### Revenue streams

A bank has two revenue engines and they behave nothing alike. **Net interest income** is the spread between what assets earn and what liabilities cost, computed from the balance sheet and the rate environment. **Non-interest revenue** is fees, interchange, forex margin, trading and commission, generated by events. In South Africa interchange is unusual: the SARB determines, approves and supervises interchange rates on a cost-based methodology reviewed every five years, with rates reviewed annually and binding on participants [10]. It is regulated reference data on an annual cycle, not a commercial parameter.

*Architecture implication.* The two engines need different data models, different grain and different latency. Net interest income is a balance-and-rate calculation over positions; non-interest revenue is an event count times a price. One management information platform built to serve both, without being honest about the difference, produces a product profitability layer that nobody trusts. Chapter 6 takes the arithmetic apart properly.

#### Key activities

Gathering deposits. Underwriting, pricing and collecting credit. Clearing and settling payments. Managing capital, funding and liquidity. Regulatory reporting and financial crime control. Changing products, prices and channels.

Note what is on that list that would not be on a retailer's: two of the six activities exist because a regulator requires them, and they consume a large share of the change budget. Note also what is missing — manufacturing. A bank does not make anything. It writes contracts and administers them.

*Architecture implication.* If the activity is administering contracts under rules that change, then the rules are data. Fee schedules, rate formulas, product terms, affordability parameters, risk weights and interchange rates are all effective-dated reference data. A bank that implements them as code has chosen a six-month lead time for a price change, and it will not be able to reconstruct what the rules were on the date of a decision somebody is now disputing.

### Key resources

The licence and the capital behind it. The deposit franchise. The balance sheet. The customer base and its data. The credit models. The core ledgers and rail connectivity. The estate. And trust, which is the only resource that can be destroyed by one visible failure.

The least intuitive of these to an architect is the **deposit franchise**. A rand of current-account balance is worth more to a bank than a rand of wholesale funding, because it is cheaper, stickier and counts better for liquidity. Deposit market share is margin share. That single sentence explains a large part of what a retail bank spends money on.

*Architecture implication.* If deposits are a resource rather than a product, the bank must classify every liability by repricing *behaviour* rather than by contractual terms, and hold a behavioural model — core versus volatile balance, a decay curve — per product and segment. Almost no deposit core does this, so the model lives in the asset and liability management system, fed by extracts, and its quality is bounded by the worst feed in the set.

### Key partners

Three kinds, and conflating them is a classic error.

**Infrastructure you cannot negotiate with.** Settlement runs through SAMOS, operated by the SARB. Clearing runs through PayInc, which is the renamed BankservAfrica and is now 50% owned by the SARB [11][12]. These are not partners. They are the ground you stand on, and when they change, you change.

**Commercial partners with their own release calendars.** Card schemes, credit bureaux, correspondent banks, insurers, bond originators, motor dealers, retail chains carrying your cash-out footprint, and your core and cloud vendors.

**The state, as a data source.** Home Affairs for identity — HANIS, ABIS and the National Population Register, including fingerprint biometrics [13]. CIPC for company registration. The deeds office for property. None of them has a service level with you.

*Architecture implication.* A large share of a bank's change portfolio is mandatory and externally scheduled. Scheme mandate releases land twice a year and are not negotiable. The domestic rails move on the SARB's programme. The shift from the old registered mandate service to the Registered Mandates rail was mandated and took effect on 10 March 2025 — a distinct ISO 20022 rail with different windows, not a rename of what came before [14]. You do not get to sequence this work. You get to absorb it, and what is left over is your roadmap.

### Cost structure

Interest paid to depositors is the largest outflow and is not presented as a cost, which catches out every architect reading a bank income statement for the first time. Then credit losses, which ran at 182 basis points of advances across the South African majors in FY2025 [15]. Then people, the branch and cash estate, IT run cost, compliance and reporting, and fraud — R2.4bn of digital banking crime and R1.7bn of card fraud reported for 2025 [16].

The cost-to-income ratio across the majors was 51% in FY2025 [15]. Within the same market, one bank runs at 36% and another at 70.7% [9][17]. That spread is not mostly about staff discipline. It is about how many cores, how many customer masters and how much coexistence machinery each one carries.

*Architecture implication.* Fraud is the clearest case of a cost whose handling exceeds its value. The loss is the smaller number; investigation, dispute handling, card reissue and customer recovery are the larger one. Architecture that shortens the path from detection to freeze pays for itself twice, and it is almost always blocked by the same thing: the freeze capability sits behind a back-office queue.

### What the canvas leaves out

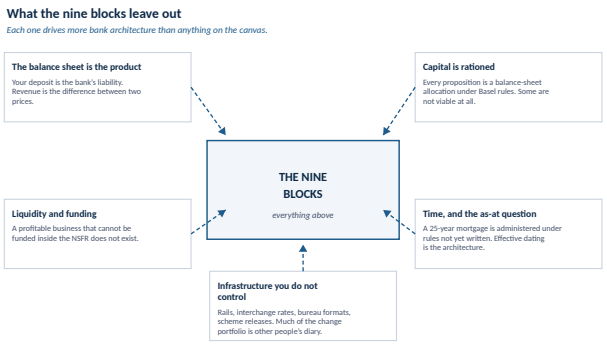



**Figure 4.2** The five things the canvas has no block for. Each drives more bank architecture than anything that is on it.

Five things, and each of them drives more architecture than anything on the canvas.

**The balance sheet is the product.** There is no block for "the inventory is other people's money, repayable on demand". A shop that loses stock loses a quarter's profit. A bank that loses depositor confidence ceases to exist, because liabilities run faster than assets liquidate. This is why "who are our largest

depositors, and are they correlated?" is a reportable daily question in a bank and a strange question anywhere else.

**Capital is rationed.** Every exposure carries a risk weight; risk weights sum to risk-weighted assets; capital ratios are capital over risk-weighted assets. The South African minimum is 7.0% common equity tier one before buffers, with Pillar 2 add-ons and a domestic systemically important bank surcharge layered on top [18]. The practical consequence is that two products with identical margins are not equally valuable, because one consumes more equity per rand of balance. A proposition that cannot clear the cost of equity on a capital-charged basis should not be built, however good the customer experience. There is no canvas block for that sentence and it kills more product ideas than any other.

**Liquidity and funding.** A profitable business that cannot be funded within the net stable funding ratio does not exist. Both the liquidity coverage ratio and the net stable funding ratio are computed over the entire balance sheet, with every exposure classified by counterparty type, product, maturity bucket and behavioural assumption, daily and monthly respectively [19]. No core banking system holds that classification. A treasury system exists to apply it.

**Time.** A 25-year mortgage commits the bank to administering a contract under rules not yet written. Every question a regulator, an auditor or an ombud asks is an as-at question: what did we know, what did the rules say, what did we tell the customer, on the date. Effective dating and as-at reconstruction are not features of a bank's architecture. They are the architecture.

**Shared infrastructure.** Covered above under partners, and worth repeating here because it belongs on the canvas and is not on it. The bank cannot change the rails, the interchange rates, the bureau data model or the scheme message formats.

> A canvas that leaves these five out will mislead you in a specific direction: it makes the bank look like a service business with unusually heavy compliance. It is not. It is a geared balance sheet with a service business attached, and when the two conflict the balance sheet wins.

## The same canvas, two other banks

The nine blocks are not wrong for other kinds of bank. They are differently weighted, and the differences are where the architecture lives.

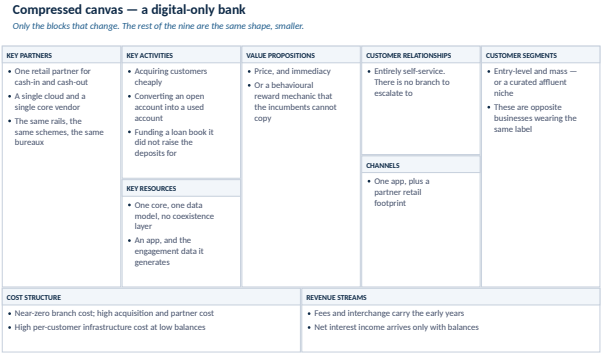



**Figure 4.3** The same nine blocks for a digital-only bank, compressed to what changes. The risk moves from the asset side to the liability side.

A digital-only bank inverts the cost structure and moves the risk. Near-zero branch cost, high customer acquisition and partner cost, high per-customer infrastructure cost at low balances. The segments skew either to entry-level and mass or to a curated affluent niche, and those are opposite businesses: one South African digital bank reported deposits of roughly R645 per customer, another roughly R17,300 per client [20][21]. The first of those figures is as at December 2024 and is materially stale; treat it as an order of magnitude, not a current fact.

The architecturally important point is that **the key risk is on the liability side, not the asset side**. Ten million customers producing a few billion rand of deposits cannot fund a meaningful loan book [20], so lending has to be funded deliberately or kept small. That is why these banks go to merchant and payroll lending rather than mortgages. And the single core, API-first, no-coexistence estate is both the reason they can run at a fraction of incumbent cost-to-income and the reason they cannot absorb an acquisition without losing that advantage.

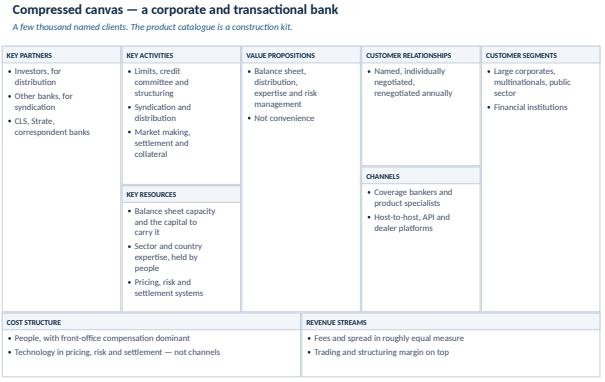



**Figure 4.4** The same nine blocks for a corporate and transactional bank. A few thousand named clients, and real time as a risk requirement rather than a customer-experience one.

A corporate and transactional bank changes the canvas more fundamentally. A few thousand named clients instead of millions. Relationships individually negotiated, so the product catalogue is a construction kit rather than a shelf. Revenue split across markets, transaction banking and investment banking in roughly those proportions — one large South African corporate and investment bank reported markets at 43% of revenue, transaction banking at 35% and investment banking at 22% [4]. People-heavy cost structure. Technology concentrated in pricing, risk and settlement rather than channels.

And one inversion worth holding on to: **real time here is a risk requirement, not a customer-experience requirement**. The corporate client does not need a sub-second response for its own satisfaction. The bank needs one because a limit has to be checked before a dealer quotes, and a settlement has to be final before an exposure can be released.

| Block | Full-service | Digital-only | Corporate and transactional |
|---|---|---|---|
| Segments | Six, spanning four orders of magnitude | One, either mass or a curated niche | A few thousand named clients |
| Value proposition | Safety, reach, credit, presence | Price and immediacy, or a reward mechanic | Balance sheet, distribution, expertise |
| Channels | Eleven, none retiring | One app plus a retail partner | Coverage bankers, host-to-host, dealer platforms |
| Relationships | Primacy and depth of use | Engagement and activation | Named, negotiated, renegotiated annually |
| Revenue | Net interest income plus fees | Fees first, interest later | Fees and spread, plus trading margin |
| Key resources | Licence, deposits, estate, data | One core, one app, one data model | Balance sheet, people, pricing and risk systems |
| Key activities | Deposits, credit, payments, capital, reporting | Acquisition and activation | Limits, structuring, distribution, settlement |
| Key partners | Rails, schemes, bureaux, the state | A retail partner and a cloud | Investors, other banks, market infrastructure |
| Cost structure | People, estate, IT run, losses | Acquisition and infrastructure | Front-office compensation |
| Dominant risk | Credit, then operational | Funding the asset side at all | Counterparty and settlement |
| What breaks first | Coexistence between cores | Unit economics at low balances | Limits, collateral and confirmations |

> Do this exercise for the bank you are in, on one page, before you touch a system inventory. If your canvas cannot distinguish your employer from the bank across the road, you have described banking rather than this bank, and your target state will be generic in exactly the same way.

## Four customer journeys, done honestly

A journey map that stops at the happy path is decoration. The version that earns its place has a column for where it breaks, and the breaks are specific enough that somebody in the room says "that is not quite how ours fails" — which is the most useful sentence a workshop can produce.

Four journeys cover most of what a retail and business bank does to people. Opening an account. Borrowing against a house. Arguing about a card transaction. Getting a company onto the bank's collections and payment rails. Each one is drawn with the systems named and the failure points marked, and then the failures worth dwelling on are expanded in prose.

### Opening a transactional account

**Opening a transactional account**

*Twelve steps, and the identity problem is the whole of it.*

| STEP | WHO AND WHAT SYSTEM | WHERE IT BREAKS |
|---|---|---|
| 1 **Intent and product choice** | App, branch consultant or agent; product catalogue | A product chosen before identity or affordability is known, so the journey has no legitimate place to pause. |
| 2 **Identity capture** | Onboarding front end: ID scan, selfie, fingerprint | ▲ Fingerprint capture fails more often for older people and manual workers. A failed capture is not a failed identity, but systems treat it as one. |
| 3 **Identity verification** | Home Affairs HANIS, ABIS and the National Population Register, through a verification service | ▲ A Home Affairs data mismatch stalls the account, and neither institution owns the fix. |
| 4 **Address verification** | Document ingestion and classification; the FICA rule set | ▲ No verified address, no onboarding at all. The accepted-document list excludes exactly the population that inclusion targets. |
| 5 **Sanctions and PEP screening** | Screening engine; externally sourced list data | False positives on common surnames land in a manual clearance queue with no service level. |
| 6 **Risk rating and due diligence** | Customer risk model; CDD workflow | ▲ The rating is set once at onboarding and never recomputed, so "expected behaviour" is a guess for the life of the relationship. |
| 7 **Terms, disclosures, acceptance** | Document generation, signature capture, document of record | The document promises what the account cannot do. Retrieval is untested until a dispute. |
| 8 **Account opened in the core** | Deposit core; party master | ▲ A second customer record is created because nobody searched the party master first. This is where the duplicate comes from. |
| 9 **Card issue, activation, PIN** | Card issuing platform; personalisation bureau | A card posted to an address that failed verification; activation stalls and nobody owns the queue. |
| 10 **Digital enrolment and device binding** | Digital identity; device registry; step-up authentication | ▲ This is the attack surface. Most reported digital banking crime starts at enrolment, not at the payment. |
| 11 **First funding** | Payments hub; PayShap or EFT credit in | ▲ An account opened and never funded is pure cost. Depth of use, not accounts opened, is the number that matters. |
| 12 **Transactionally live** | Core, switch, fraud scoring, channels | "Live" means three different things to the core, the switch and the fraud engine, and they disagree for hours. |

**Figure 4.5** Opening a transactional account at Table Bay Bank, with the systems named and the failure points marked. The triangles are the breaks that cost the bank a customer or a regulator's attention.

The South African identity problem is not a step in this journey. It is the journey.

Under section 21E of the Financial Intelligence Centre Act, an institution that cannot verify a client's residential address **may not onboard them at all** [22]. Accepted documents must generally be no older than three months, and the accepted list is long but closed: utility bills, bank statements, signed leases, municipal accounts, sworn affidavits, insurance policies, telecommunications contracts and several more, while post box addresses, handwritten notes and prepaid accounts with no registered address are not accepted [22]. For an informal settlement, a utility bill addressed to an erf and stand number with the township name is valid without a street address, a ward councillor's letter is widely accepted, and a police affidavit is the fallback [22]. The Financial Intelligence Centre's own guidance notes that excessively restrictive document lists create access barriers — the regulator saying out loud that its rule excludes people.

Identity itself is verified against Home Affairs, including fingerprint biometrics [13]. Two failure modes follow, and they are different. **A failed fingerprint capture is not a failed identity.** Capture fails more often for older people and for those who do manual work, and the system that treats a capture failure as an identity failure has turned a sensor problem into a financial exclusion [13]. And a **Home Affairs data mismatch** stalls the account with the customer caught between two institutions, neither of which owns the fix.

Then three failures that are entirely the bank's own. A **duplicate party record**, created because nobody searched the party master before opening the account, which poisons the single customer view, the deposit insurance submission and every financial crime alert for years. **Digital enrolment and device binding**, which creates the attack surface where most reported digital banking crime begins — the 2025 statistics attribute the great majority of digital banking crime cases to this area, though the precise split comes from a secondary aggregation rather than the primary report [16]. And the quiet one: **an account opened and never funded**, which is a cost with no revenue, and the reason depth of use rather than accounts opened is the number to put on the canvas [9].

*What this journey asks of the architecture.* Identity verification as a shared service with a result cache and an explicit record of verified-by-what-method, when, against what source. A document ingestion and classification pipeline. An onboarding state machine that can **hold a partially verified customer in a restricted state** rather than failing them — the capability almost nobody has built, and the one that decides whether the FICA rule excludes a customer or merely delays them. And a customer risk rating that is recomputed rather than set once.

### Applying for a home loan

**Applying for a home loan**

*A long-running case, blocked for weeks by parties the bank does not employ.*

| STEP | WHO AND WHAT SYSTEM | WHERE IT BREAKS |
|---|---|---|
| 1 **Pre-qualification** | Originator or bank calculator; soft bureau enquiry | A pre-qualification the credit system will not honour, given to a customer who now believes it. |
| 2 **Offer to purchase signed** | Outside the bank entirely | The clock starts outside your systems. The customer thinks it started with you. |
| 3 **Application, often to several banks at once** | Bond originator platform into bank origination | ▲ Nearly half of applications go to more than one bank, and a large share declined by one are approved by another. You lose on elapsed time and never learn why. |
| 4 **Income and affordability, Regulation 23A** | Document ingestion, income extraction, decision service | ▲ Three payslips or three months of statements, by regulation. This is the single largest source of origination latency in South African retail lending. |
| 5 **Bureau enquiry** | Bureau interfaces; enquiry record | ▲ Repayment history must be assessed within fourteen business days for a mortgage. A hard pipeline timer, not a target. |
| 6 **Credit decision and rate offer** | Decision engine; risk-based pricing | ▲ The decision is not final — valuation will re-trigger it. If it is not versioned you cannot explain either version. |
| 7 **Property valuation** | Valuer panel; collateral register | ▲ A valuation below the purchase price changes the loan-to-value and can flip an approval. Deposits are thin, so small movements matter. |
| 8 **Formal grant, quotation, NCA disclosures** | Document generation; pre-agreement statement | Statutory artefacts that must be reproducible, with their inputs, years later. |
| 9 **Acceptance and attorney instruction** | Attorney panel integration; case state | Three sets of attorneys — transferring, bond registering, cancelling — none of them yours. |
| 10 **Deeds office lodgement and registration** | External. The bank holds case state and a priced commitment | ▲ Weeks of external blocking while the bank holds a rate. Rate changes in the window are the architecture problem. |
| 11 **Bond registration and first drawdown** | Loan servicing; payments; collateral register | ▲ A policy rate change must reprice the whole book overnight, including loans mid-registration that are not yet on book. |
| 12 **On book, and servicing begins** | Loan servicing, collections, impairment engine | ▲ Origination data is not carried forward, so the servicing system does not know what the customer was told. |

**Figure 4.6** A home loan application: a long-running case, externally blocked for weeks, re-decided at least once, holding a priced commitment throughout.

This one is not a transaction. It is a long-running case, externally blocked for weeks, with a priced commitment held open across the blockage, and it will be re-decided at least once before it completes.

Four failure points deserve expansion.

**Parallel applications.** Bond originators submit the same application to several banks simultaneously, and the published market data shows a large share of applications declined by one bank being approved by another [23]. A slow or conservative bank loses the deal to a faster one and never learns why, because nothing comes back. Time-to-yes is a competitive variable with a measurable price, which is the argument that funds origination work when a customer-experience argument will not.

**Affordability as a regulated calculation.** Regulation 23A has been in force since 13 March 2015. It prescribes the income evidence — the latest three payslips or three months of bank statements for a salaried applicant, three months of statements or financial statements for the self-employed — and prescribes the calculation: gross income, less statutory deductions, less minimum living expenses from a prescribed table, less other disclosed commitments, with the bureau record considered [24]. Repayment history must be assessed within seven business days of approval, fourteen for a mortgage [24]. Three consequences follow. Document ingestion and income extraction become a core origination capability rather than a convenience. The bureau freshness windows are hard pipeline timers. And affordability belongs in a decision service with effective-dated rule versions and a stored, replayable decision record, because a reckless-lending challenge years later must be answerable against the rules as they stood on the day.

**The external block.** Attorney instruction, deeds office lodgement and registration happen outside the bank and take weeks. The bank's systems must hold case state across that, model the external parties as first-class integrations with their own service levels, and survive a rate change in the window. Which leads to the last one.

**Repricing.** When the policy rate moves, every prime-linked loan reprices — the whole book, overnight, including loans that are mid-registration and not yet on book. Retail lending in South Africa is quoted as prime plus or minus a margin, with mortgages averaging prime less 0.67% in one 2026 dataset [23], so the repricing event is mechanical and total. An architecture that treats rate change as a maintenance task rather than a scheduled mass event will discover the difference on a Thursday afternoon.

### Disputing a card transaction

**Disputing a card transaction**

*Ten steps, three different liability regimes, and clocks inside clocks.*

| STEP | WHO AND WHAT SYSTEM | WHERE IT BREAKS |
|---|---|---|
| 1 **The customer notices the transaction** | App, contact centre or branch | The customer cannot tell fraud from a merchant dispute. Neither can the agent who takes the call. |
| 2 **Intake and classification** | Contact centre front end; dispute case object | ▲ Fraud, merchant dispute and processing error have different rules, time limits and liability. Misclassification at the first touch decides the outcome. |
| 3 **Containment: block the card** | Card issuing; switch stand-in; token vault | ▲ A block applied in one place and not the others. Stand-in keeps approving. |
| 4 **Provisional credit decision** | Dispute workflow; deposit core | Provisional credit is its own fraud vector, and the rule for granting it is usually undocumented. |
| 5 **Chargeback raised against the acquirer** | Scheme interface; case evidence | ▲ Scheme time limits are hard deadlines. A dispute that sits in a queue expires and the bank absorbs the loss. |
| 6 **Merchant represents or accepts** | Acquiring platform; merchant | ▲ Evidence is scattered across the acquiring platform, the document store and somebody's mailbox. |
| 7 **Second chargeback, pre-arbitration, arbitration** | Scheme interface; scheme rules as reference data | ▲ Clocks inside clocks, none of them modelled as first-class state. |
| 8 **The fraud path, in parallel** | Fraud engine; payments hub; beneficiary and mule data | ▲ Latency between detection and the ability to freeze and recall is the exact failure the ombud penalises. |
| 9 **Authorised push payment loss** | None of the above applies | ▲ The customer authorised it, so no chargeback exists — and the ombud has still awarded refunds, criticising slow response rather than weak controls. |
| 10 **Outcome, letter, ombud file** | Customer communication; document of record | The complete file must be assemblable on demand, because the ombud will ask for it. |

**Figure 4.7** A card dispute. Three liability regimes meet in one conversation, and the classification made in the first thirty seconds decides the outcome.

The dispute journey is where three liability regimes collide in one conversation, and the conversation happens in the least-trained channel.

**Classification at intake decides the outcome.** Fraud, merchant dispute and processing error have different rules, different time limits and different liability outcomes. The customer cannot tell them apart. The first agent often cannot either. Everything downstream inherits that first classification, and correcting it later usually means a clock has already run.

**Scheme clocks are hard.** A chargeback has to be raised against the acquirer within the scheme's time limit. A dispute that sits in a queue expires, and the bank absorbs the loss with no recourse. These clocks have to be modelled as first-class, escalating state on the case, not as a report somebody reads on Monday.

**Authorised push payment fraud is not a chargeback.** The customer authorised the payment. Scheme rules offer no recourse. And yet the National Financial Ombud has awarded refunds in these cases and criticised banks for responding too slowly after becoming aware of the loss; of the ombud's banking recoveries, around 60% were fraud-related, and the determinative factor cited was latency rather than weak controls [25]. Read that as a design requirement: the path from detection to freeze to attempted recall must not route through a back-office queue.

*What this journey asks of the architecture.* A dispute case object that spans the card platform, the scheme interface, the fraud system and the customer channel, with scheme clocks explicit. A real-time containment path. And the ability to assemble the complete file — including the ombud response — on demand, because the ombud will ask for it.

### Onboarding a business customer for transactional banking

**Onboarding a business customer for transactional banking**

*Not one onboarding. One entity onboarding plus N individual ones, plus a permissions model.*

| STEP | WHO AND WHAT SYSTEM | WHERE IT BREAKS |
|---|---|---|
| 1 **Enquiry and needs assessment** | Relationship manager; CRM | Commitments made in a sales meeting that the provisioning process cannot keep. |
| 2 **Entity identification** | CIPC verification; party master | ▲ The entity already exists in four systems under three spellings of its name. |
| 3 **Beneficial ownership, to the natural persons** | CDD workflow; an ownership graph, if one exists | ▲ Trusts, nominees and layered holdings resolved by hand. A permanent obligation, unchanged by the grey-list exit. |
| 4 **Individual due diligence on every director and signatory** | The same stack as retail onboarding, run N times | ▲ A six-director company is seven onboardings. One unverifiable director blocks the whole thing. |
| 5 **Entity risk rating** | Financial crime risk model | ▲ Industry, cash intensity and cross-border exposure scored from what the banker typed into a free-text field. |
| 6 **Mandate and resolution capture** | A mandate service, if the bank has one | ▲ "Any two of these four up to R500,000, except payroll, which needs the finance director" is an authorisation policy, not a form. |
| 7 **Product selection and price negotiation** | Pricing tool, and a spreadsheet | ▲ A negotiated price that no downstream billing engine can hold, so it is applied by hand every month. |
| 8 **Account opening** | Business core; party master; limits | ▲ Entity, accounts and mandates created in three systems with no single owner of the truth. |
| 9 **Channel provisioning** | Online banking administration; token issue; API credentials | ▲ Entitlements configured per channel, so the mandate is implemented three times and diverges within a month. |
| 10 **Host-to-host or API integration** | File gateway; API gateway; test harness | ▲ Format agreement, test cycles and certification. A project inside a sales cycle, and the sales cycle |
| 11 **Collections setup** | Mandate rails; collections platform | ▲ Registered Mandates is a distinct rail, not a rename, and it — not DebiCheck — is the authenticated answer for business and dual-signatory accounts. |
| 12 **First live payment run** | Payments hub; the client's payroll deadline | ▲ The first payroll run is the real go-live, and it happens on the customer's date, not yours. |

**Figure 4.8** Business customer onboarding: one entity onboarding, N individual ones, a permissions model and an integration project, all inside a sales cycle.

This is the journey most banks have never properly designed, and the one where a self-service answer changes the economics most. One listed bank grew its business client count by 123% to 686,000 [9], which is only achievable by making this journey self-service, and that number is the strategic question

sitting in front of every incumbent.

**It is N retail onboardings plus an entity onboarding.** Every director and signatory needs full individual due diligence. A six-director company is seven onboardings, and one unverifiable director blocks all of them.

**Beneficial ownership chains.** Trusts, nominee structures and layered holdings are common in South Africa, and resolving to the ultimate natural persons is manual work. South Africa exited the Financial Action Task Force grey list on 24 October 2025, having been listed in February 2023 — and the exit changed no domestic obligation [26]. Beneficial ownership verification is permanent. If somebody tells you the grey-list exit reduces the due diligence requirement, they are wrong.

**Mandates are a permissions model, not a form.** "Any two of these four up to R500,000, except payroll, which requires the finance director" is an authorisation policy. The customer changes it constantly. Most business banking channels express it poorly, and because entitlements are configured per channel, the same mandate is implemented three times and diverges within a month. A mandate and entitlement service that is the single authority for who may do what, across every channel, is the highest-value unglamorous thing in a business bank.

**Integration is a project inside a sales cycle.** Host-to-host file exchange and API connections need format agreement, test cycles and certification. That is weeks of elapsed time sitting inside a sales process measured in days, and the first payroll run is the real go-live, on the customer's date.

## The make-or-break moments

**The moments that decide the relationship**

*Where the customer forms a view, and the system that decides it.*

| THE MOMENT | WHAT THE CUSTOMER CONCLUDES | THE SYSTEM THAT DECIDES IT |
|---|---|---|
| **The account goes live, or does not** | Either this bank works, or I will go to the one my cousin uses | Identity verification, document ingestion, the onboarding state machine |
| **The credit answer, and how long it took** | They understood my situation, or they ran a rule over me | Decision engine, affordability service, bureau interfaces |
| **The first payment that fails** | My money is not safe here | Payments hub, rail connectivity, the exception queue |
| **The dispute outcome** | They took my side, or they took the merchant's | Dispute case, scheme interface, fraud engine |
| **The arrears conversation** | They worked with me, or they sent a lawyer | Collections strategy, arrangement capture, the section 129 notice |
| **The fee that was not explained** | They are taking money quietly | Pricing and billing engine, customer communication |
| **The payroll run that lands** | I can keep banking here as my business grows | File gateway, payments hub, mandate service |

**Figure 4.9** The moments at which a banking relationship is decided, and the system that decides each one. The two shaded rows are the ones a customer does not forgive.

Across all four journeys the same pattern holds: the relationship is decided at a small number of moments, and at each of them a specific system makes the decision. The table above is the version to take into a steering committee, because it converts "the onboarding system is old" into "the onboarding system decides whether a customer ever becomes a customer".

Two of those moments are worth more than the others. The **first failed payment** and the **dispute outcome** are where a customer reconsiders whether their money is safe, and safety is the one proposition a bank cannot substitute. Everything else can be recovered from.

## The same five failures, every time

Read the four journeys together and the failures sort into five classes. You will meet all five in every bank you walk into, whatever its size and whatever it has spent.

1. **Identity and party.** The bank does not reliably know whether two records are the same person, or who ultimately owns an entity. Everything downstream — single customer view, deposit insurance aggregation, financial crime monitoring, credit exposure to a connected group — inherits the defect.
2. **State that has to survive.** Origination, disputes and business onboarding are long-running, externally blocked processes. Built as transactions, they lose case state, time out, or hold a commitment nobody can reconstruct.
3. **Authority expressed in the wrong place.** Who may approve what, to what limit, with what evidence — in credit, in payments, in mandates — written in a policy document and implemented separately in each channel and system.
4. **Decisions that cannot be replayed.** The affordability assessment, the credit score, the fraud decline, the provisional credit grant: made, acted on, and not stored with their inputs and rule versions. Then somebody asks why, years later.
5. **Clocks nobody models.** Scheme time limits, bureau freshness windows, regulatory submission deadlines, prescription periods, deeds office elapsed time. Each one is a hard deadline with a financial consequence, and most of them live in a spreadsheet or in somebody's head.

> Those five are the backbone of a target state for a retail and business bank. If your roadmap addresses all five you will be arguing about sequencing. If it addresses none of them, you are modernising the parts that photograph well.

## Capability first, or system first

There are two ways to turn these two pages into architecture, and only one of them survives the first budget cycle.

**System first** starts from the inventory. List the applications, assess them, rate them red, amber and green, and propose replacements for the red ones. It is quick, it is legible, and it produces a programme that cannot be defended, because nobody outside technology can tell from a red rating whether the system matters. It also produces the specific failure where a bank replaces a visible front end and leaves the capability gap behind it untouched.

**Capability first** starts from the value stream. What has to happen for the customer to get what they came for; what capability does each stage of that need; what realises each capability today. The gaps fall out, and some of them are not systems that need replacing but capabilities nobody has built at all.

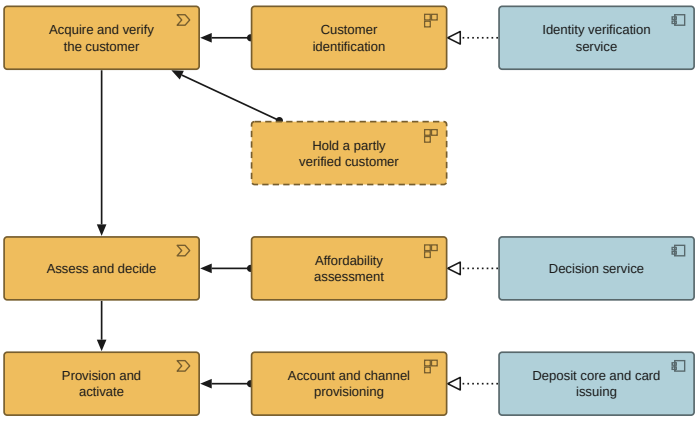



**Figure 4.10** Account opening as a value stream, the capabilities each stage needs, and what realises them. The dashed capability is real, needed, and realised by nothing — the gap an application inventory cannot show you.

The figure draws the account-opening value stream against the capabilities it needs and the applications that realise them. The dashed box is the point. **Hold a partly verified customer** is a capability the value stream needs, nothing realises, and no application inventory will ever show you — because it is not a system that is old or failing. It is a capability that does not exist, and the symptom is a drop-off at verification that gets written up as customer apathy.

That is the whole argument for doing the canvas and the journeys before the inventory. An application assessment tells you which systems are in poor health. A capability map against the value stream tells you which of them matter, and what is missing entirely. You need both, and in that order.

One honest caution to close on. Both instruments are conversations, not deliverables. The canvas is right when the chief financial officer argues with the revenue block. The journey map is right when the head of operations corrects the failure column. If you produce either one without that argument happening, you have produced a diagram of what you were told, and you will build against it.

#### THE CHAPTER IN TEN LINES

- A system matters in proportion to the part of the business model it carries and the part of the customer experience it decides — not in proportion to its age.
- The nine-block canvas works for a bank only if the segments, channels and revenue streams are specific enough to argue with.
- The canvas has no block for the balance sheet, capital, liquidity, time or the shared infrastructure, and those five drive more architecture than anything on it.
- Deposit market share is margin share, which means liabilities must be classified by repricing behaviour rather than contractual terms.
- A digital-only bank's hard problem is the liability side; a corporate bank's real-time requirement comes from risk, not from customer experience.
- Under FICA section 21E an unverifiable residential address blocks onboarding outright, which makes document ingestion and a restricted-state onboarding machine core capabilities.
- Home loan origination is a long-running, externally blocked case that will be re-decided, so decisions must be versioned and replayable.
- A card dispute is decided by the classification made in its first thirty seconds, and authorised push payment loss has no chargeback at all.
- Business onboarding is one entity onboarding plus N individual ones plus a permissions model — and the mandate service is the highest-value unglamorous thing in a business bank.
- Five failure classes recur everywhere: identity and party, long-lived state, authority in the wrong place, decisions that cannot be replayed, and clocks nobody models.

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CHAPTER 5

# Who does what

*Every role in a bank: what the person does all day, the decision they own, the data they need, the system they live in and what they complain about. This is the chapter that lets you run a discovery session without being found out.*

## The org chart will not help you

You will be given one in your first week. It will have the chief executive at the top and six or seven boxes underneath — retail, business banking, corporate and investment banking, risk, finance, technology, maybe operations as its own box. It will be accurate. It will also tell you almost nothing you can use, for three reasons.

The first is that work in a bank crosses the boxes constantly and the org chart does not draw the crossing. A business credit application touches a relationship manager, a credit analyst, a sanctioner, a committee, second-line risk, legal, operations and finance before anyone is paid, and the org chart puts those people in four different reporting lines on purpose.

The second is that some of those separations are **controls, not preferences**. The reason the person who prices a trade does not also settle it is not that treasury likes departments. It is that a bank where the same person can capture, value and settle a position has no defence against the oldest fraud in finance. The same logic puts credit sanctioning outside the business that originates the deal, and internal audit outside everything. An architect who proposes a tidier structure without knowing which lines are control lines will propose removing a control, and will be told so in a room they did not expect.

The third is that the org chart names units and you need **roles**. A role is a job with a decision attached. "Credit" is a unit. "The person who may approve R10m personally and must refer R12m upward" is a role, and it is the one that changes what you build.

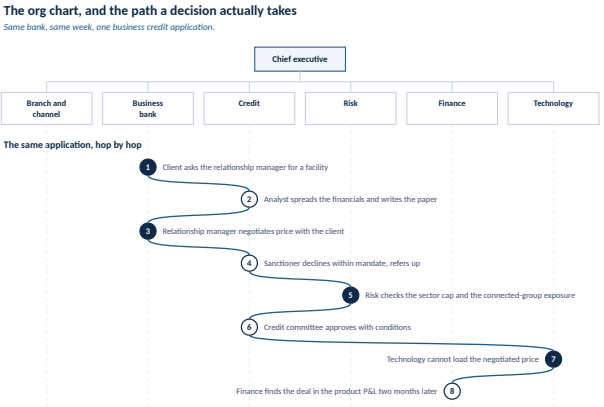


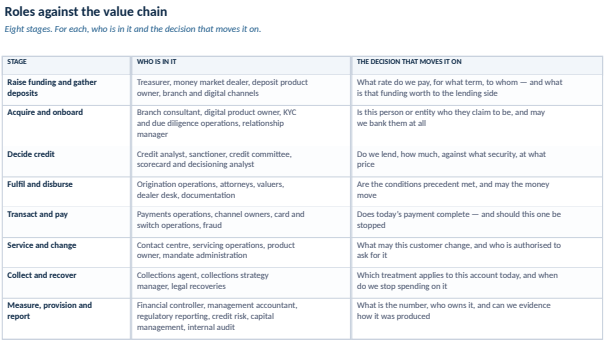

**Figure 5.1** The reporting lines, and the path one business credit application actually takes through them. The org chart is accurate and the path is the work.

So the instrument for this chapter is not an org chart. It is a list of roles, and for each role five things: what they do all day, the decision they own, the data they need, the system they live in, and what they complain about. The last of those is where your work comes from. Nobody funds architecture because the model is untidy. They fund it because a credit analyst spends 70% of their time gathering data, or because the regulatory reporting analyst takes two weeks to answer a question the regulator expects in two days.

This chapter uses **Table Bay Bank**, an invented mid-tier South African full-service bank, to keep the examples concrete. The roles are real; the bank is not.

## Roles against the value chain

Before the detail, a map. A bank's value chain has eight stages, and every role sits in one or two of them. Put a role in the wrong stage and you will interview the wrong person about the wrong system.



**Roles against the value chain**

*Eight stages. For each, who is in it and the decision that moves it on.*

| STAGE | WHO IS IN IT | THE DECISION THAT MOVES IT ON |
|---|---|---|
| **Raise funding and gather deposits** | Treasurer, money market dealer, deposit product owner, branch and digital channels | What rate do we pay, for what term, to whom — and what is that funding worth to the lending side |
| **Acquire and onboard** | Branch consultant, digital product owner, KYC and due diligence operations, relationship manager | Is this person or entity who they claim to be, and may we bank them at all |
| **Decide credit** | Credit analyst, sanctioner, credit committee, scorecard and decisioning analyst | Do we lend, how much, against what security, at what price |
| **Fulfil and disburse** | Origination operations, attorneys, valuers, dealer desk, documentation | Are the conditions precedent met, and may the money move |
| **Transact and pay** | Payments operations, channel owners, card and switch operations, fraud | Does today's payment complete — and should this one be stopped |
| **Service and change** | Contact centre, servicing operations, product owner, mandate administration | What may this customer change, and who is authorised to ask for it |
| **Collect and recover** | Collections agent, collections strategy manager, legal recoveries | Which treatment applies to this account today, and when do we stop spending on it |
| **Measure, provision and report** | Financial controller, management accountant, regulatory reporting, credit risk, capital management, internal audit | What is the number, who owns it, and can we evidence how it was produced |

**Figure 5.2** The eight stages of a bank's value chain, who sits in each, and the decision that moves work to the next stage.

Three things to notice about that map. **Funding comes first**, before any customer-facing activity, which is the reverse of how the bank's own organisation is usually drawn. **The same stage contains first-line and second-line roles** who disagree with each other by design. And **the last stage is where every earlier stage's data defects surface**, which is why the regulatory reporting analyst and the credit risk modeller are the two roles most likely to be able to tell you, precisely and bitterly, what is wrong with the architecture.

> How to read the tables that follow. Each row is one role. "The system they live in" is deliberately named by function rather than by product, because the product differs between banks and the function does not. "What they complain about" is the hook: it is the sentence you repeat back to a steering committee when you need a business sponsor for something that looks like plumbing.

## Branch and channel

The front line, and the part of the bank with the highest staff turnover and the lowest tolerance for a slow screen. The physical estate is in managed decline but not retreat: one large bank closed its way to 574 outlets and 4,976 cash devices by mid-2026, a 41% reduction in devices from 2021, while cash demand fell more slowly [1]. At the other end of the same market, one bank reports 16.5 million monthly app users [2].

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Branch consultant / teller | Opens accounts, takes deposits, captures documents, resolves queries, sells simple products | Whether this person is who they claim and whether the documents pass | Customer view, product catalogue, acceptable-document rules, cash position | Teller and servicing front end plus the onboarding workflow | Four systems per task, re-keying, the rule that blocks an obviously legitimate customer |
| Branch manager | Runs the branch result, staffing, queues, cash, audit findings | Local discretion and exception approval | Branch performance, queues, cash forecast, audit findings | Branch management reporting | Targets set centrally; reporting that arrives too late to act on |
| Self-service and cash operations | Device uptime, cash levels, replenishment, cash-in-transit routes | Cash forecast per device, and whether a device stays live | Device telemetry, cash balances, dispute volumes, route plans | Device management and cash forecasting | A shrinking estate serving undiminished cash demand; reconciliation breaks between device, switch and ledger |
| Contact centre agent | Queries, limits, card blocks, disputes, arrears calls | First-contact classification — including fraud versus dispute | Customer view, transaction history, product terms, dispute rules | Contact centre desktop over several back ends | Handling-time targets that conflict with doing it properly; no view of what the customer did in the app first |
| Digital channel product owner | Owns the app and internet banking, adoption, releases | What gets built next and what the journey does | Funnel analytics, adoption, defects, feedback | Product analytics and the delivery backlog | The back end cannot support the journey; releases gated by core change windows |

The architecturally interesting role here is the **contact centre agent**, because they own the classification that decides the outcome of a card dispute, and they make it in under a minute with the least training in the bank. Chapter 4 traced what that costs. Any design that moves classification earlier, or makes it reversible without losing a scheme clock, is worth more than it looks.

## Relationship and business banking

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Business banker / relationship manager | Manages named clients, prepares credit applications, negotiates price, handles escalations | The credit recommendation and the price proposal, within mandate | Client financials, account behaviour, exposure and limits, pricing and profitability tools | CRM plus a credit application workbench plus spreadsheets | Preparing the same credit paper three times in three systems; no reliable client profitability number; losing deals on elapsed time |
| Corporate coverage banker | Owns the institutional relationship, coordinates product specialists, originates transactions | The relationship strategy and the return-on-capital case for using balance sheet | Group-wide exposure and revenue by client, capital consumed, wallet share | CRM plus deal pipeline plus the product specialists' systems | No single client view across products, countries and legal entities; no credible return on capital per client |
| Private banker / wealth manager | Serves high-net-worth clients across banking, credit, investment and fiduciary needs | The client plan and the lending recommendation against net worth | A consolidated household and entity view across accounts, portfolios, trusts and companies | A private banking desktop, and a spreadsheet | The consolidated view does not exist and is assembled by hand every time |
| Dealer-floor finance consultant | Structures and books vehicle finance while the customer waits | Deposit, term and balloon structure to fit affordability | Instant decision, affordability, settlement quotes, insurance verification | A point-of-sale finance front end into the decision service | Decision latency losing the deal to the financier at the next desk on the same floor |

Two patterns repeat across all four. **Nobody has a trustworthy client profitability number**, because it depends on funds transfer pricing and cost allocation, and chapter 6 explains why those are the least well-engineered calculations in a bank. And **elapsed time is the competitive variable**. The dealer-floor consultant loses to the desk across the floor; the business banker loses to the bank that answered first. Both of those are architecture problems wearing a sales hat.

## Credit

Credit is the function with the most roles, the sharpest internal disagreements and the clearest decision rights. It is also where an architect can do the most damage by assuming that retail and business credit are the same thing with different numbers. They are different classes of work. Retail credit is a decision pipeline: high volume, standardised data, thin human touch, minutes to days, with a regulated affordability test as the gate. Business credit is a case-management and credit-paper problem: financial spreading, covenants, security registration, facility structures with sub-limits, committee workflow and an annual review cycle.

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Credit analyst | Spreads financials, models scenarios, writes the credit paper, proposes terms | The risk opinion and the proposed structure | Financial statements, management accounts, bureau data, account behaviour, security and valuations | Credit application system, a spreading tool, spreadsheets | Data gathering is most of the job; every annual review starts from scratch |
| Credit manager / sanctioner | Approves or declines within a personal mandate; refers upward beyond it | The decision, personally and by name | The credit paper, the applicant's full group exposure, policy and mandate rules, concentration | The credit workbench and the mandate grid | Incomplete papers; no view of aggregate exposure to a connected group; mandates written in a policy document rather than enforced |
| Credit committee | Collective approval above individual mandate; sets appetite and policy | Large exposures, policy exceptions, risk appetite | Portfolio concentration, stress results, exception trends | A paper pack, and whatever the secretariat assembled | Papers arriving late and inconsistent; no ability to ask a portfolio question in the meeting |
| Scorecard / decisioning analyst | Builds and monitors application and behavioural scorecards, cut-offs, policy rules, champion–challenger tests | The cut-off and the rule set — that is, who gets credit | Application and performance data at full history, bureau data, reject-inference samples, vintage outcomes | A decision engine plus an analytics environment | No clean link between what was decided, on what data, and what happened; models cannot be deployed without a release cycle |
| Credit risk modeller | Builds and maintains probability of default, loss given default and exposure at default models, impairment staging criteria, macro overlays | The model and its assumptions | Origination-date risk grades kept for life, stage transition history, collateral valuation history, macro scenarios, recovery outcomes | A modelling environment fed from the risk data mart | Legacy systems hold current risk, not risk at origination — the one thing the accounting standard requires |

### The credit mandate ladder

If you take one structure out of this chapter, take this one. The mandate ladder is the single clearest example in a bank of **authority expressed as data**, and the single most common example of authority expressed badly.

**The credit mandate ladder at Table Bay Bank**

*Illustrative, and invented. The shape is standard; every bank's thresholds differ.*

| TIER | WHO HOLDS IT | UP TO | WHAT MUST BE ATTACHED | WHAT FORCES IT UPWARD |
|---|---|---|---|---|
| **0 · Automated** | The decision service, under a delegated policy | R350,000 unsecured; R3m mortgage | Scored input vector, bureau record, affordability calculation, strategy and model version | Any policy knock-out, any override, any adverse bureau flag |
| **1 · Front line** | Business banker, personally | R1m group exposure | Short credit paper, twelve months of conduct, security valuation | New to bank, unsecured above R250,000, any arrears in twelve months |
| **2 · Sanctioner** | Regional credit manager, personally, second signature required | R10m | Full credit paper, spread financials, connected-group exposure, covenant set | Connected-group aggregate above the tier, any policy exception |
| **3 · Credit committee** | Head of credit and two members | R100m | Everything above, plus concentration view and a stress case | Sector or single-name cap, related-party exposure, country risk |
| **4 · Group credit committee** | Executives, chaired outside the business, chief risk officer present | R500m | Everything above, plus capital and risk-weighted asset impact and the funding view | Large-exposure threshold, appetite breach, new product or new sector |
| **5 · Board credit committee** | Non-executive directors | Above R500m, and all policy | Everything above, plus an independent second-line risk opinion | Nothing — this is the top of the ladder, and it sets the rest of it |

**Figure 5.3** A credit mandate ladder. The thresholds are invented for Table Bay Bank; the structure — personal mandates, a second signature, an evidence ladder and six escalation triggers — is standard.

The thresholds above are invented for Table Bay. The structure is not. Six things about it matter to an architect, and five of them are usually got wrong.

**One. The ladder is measured on aggregate connected exposure, not on the facility in front of you.** A R4m request from a client who already has R7m across three entities under common control is an R11m decision. Deciding it at tier two is a mandate breach, and the person who signed it is personally in breach, not the system. Almost every bank's credit workbench shows the facility clearly and the connected group poorly, which is why the sanctioner's complaint in the table above is the one to write down verbatim.

**Two. The mandate is personal.** It attaches to the individual, by name, with a recorded date and a recorded approver, not to the chair they sit in. Someone who moves from one region to another does not take the mandate with them; it is re-granted or it lapses. That makes the mandate register a joiner-mover-leaver problem as much as a credit policy problem, and it is precisely the kind of data that rots when it lives in a document.

**Three. Above tier zero, the second signature is the control.** The sanctioner is independent of the business that originated the deal. Two signatures where the second comes from the same reporting line is not a control, it is a formality with extra steps. When you model this, model the independence, not the count.

**Four. Escalation is triggered by six different things, only one of which is money.** Exposure above the tier. Product outside the holder's scope. A policy exception of any size. Connected-group aggregate. Sector, single-name or country concentration. And a request that is within every limit but that the holder does not want to own. The last one is real, it is healthy, and no system models it, which is why referrals arrive with no reason code and committees spend their first ten minutes working out why they are looking at a small deal.

**Five. The evidence ladder is as important as the money ladder.** Each tier requires more to be attached, and the attachments are the audit trail. A referral that arrives without the connected-group view does not get decided; it gets bounced, and the elapsed time that loses the deal is spent bouncing.

**Six, and this is the architectural point: the ladder is data, not policy.**

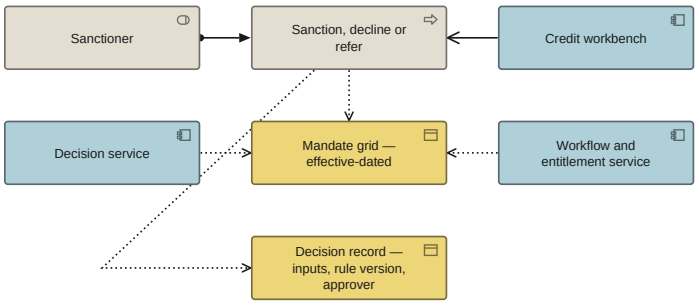



**Figure 5.4** The mandate grid as data: one effective-dated object, consulted by the decision service, the workbench and the entitlement service before the decision, with the decision record written after it.

Consider what has to change it. A repo rate move changes what R1m of instalment means. A sector going on watch moves a whole industry up a tier. A new product needs a scope entry. An appetite change at the board credit committee reprices every threshold. People join, move and leave every week. That is a dataset with a weekly change rate and an annual audit, and if it lives in a policy document it will be implemented separately in the origination workbench, the decision service, the workflow engine and the entitlement system — four implementations that diverge within one cycle, which is the finding internal audit raises every year in every bank, usually phrased as "delegated authority limits were not consistently applied".

Three design consequences follow.

- **One authority service, consulted before the decision, not after.** Enforcement in the user interface is not enforcement; a bank with a mandate rule in a screen has a mandate rule that an interface change can remove.
- **Effective dating, with history.** A decision taken in 2022 must be testable against the ladder as it stood in 2022, because that is the question an investigation asks. This is the same as-at requirement that runs through the whole of banking architecture, applied to authority.
- **The same structure recurs everywhere.** Payment release mandates on a corporate account, dealing limits in treasury, expense authority, fee waiver limits in a branch. They are all the same object: a person, a scope, a limit, an effective period, an evidence requirement and an escalation path. Banks

build them four times. You can argue for building them once, and the credit mandate ladder is the example that wins the argument, because it is the one with personal liability attached.

> When you are told "the mandate matrix is in the credit policy document", you have found a programme. Ask to see the document, then ask how many systems implement it, then ask what it said two years ago. The third question is usually the one nobody can answer.

## Collections and recoveries

Two different businesses that are routinely built as one system. Collections works a customer who still has a relationship: contact, promise to pay, arrangement, cure. Recoveries works an exposure where the relationship is over: legal process, security realisation, write-off, sale of book.

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Collections agent | Contacts customers in arrears, negotiates arrangements, records promises to pay | The treatment applied to this account today | Arrears age, balance, payment history, prior arrangements and whether they held, contact history, affordability | A collections system plus a dialler | No view of the customer's other products in arrears; arrangements agreed verbally and not enforced by the system; stale contact data |
| Collections strategy manager | Designs and tests treatment paths — early, late, pre-legal, legal, write-off, sale | Which customer gets which treatment, and when | Roll rates and cure rates by segment and treatment, cost per treatment, recovery rates | A strategy engine and an analytics environment | Cannot run controlled experiments; no attribution of recovery to treatment |
| Legal recoveries | Summons, judgment, emoluments attachment orders, repossession, debt review, sale of written-off books | The legal route, and when to abandon it | Legal case state, costs incurred, security position, recovery forecast | The attorney's system, which is not the bank's | Legal status held outside the bank; prescription computed by hand |

Two South African specifics that an architect must know about before designing anything here. **Debt review** under the National Credit Act creates a parallel, externally driven arrangement state that both servicing and collections must honour. And **prescription** is a clock per debt — three years for ordinary unsecured debt, thirty for mortgage-secured and judgment debt — running from when the debt is due, interrupted by acknowledgment of liability or by service of process, not by sending a letter [3]. Section 126B of the Act prohibits selling a prescribed debt or continuing to collect one [3]. A collections system that drives everything off days past due, with no prescription clock, either writes off recoverable debt or illegally collects prescribed debt. There is no third outcome.

## Treasury and markets

A separate landscape inside the bank, with its own vocabulary, its own regulator facing it and, in most banks, its own technology team. The decomposition into front, middle and back office is the first thing to understand, and it is the thing most often mistaken for an organisational preference.

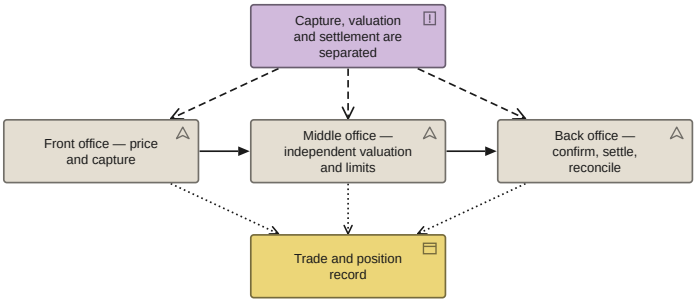



**Figure 5.5** Front, middle and back office in treasury. The separation is a control, not an organisational preference, and the shared trade record is where it is usually undone.

It is a control. The front office prices and captures. The middle office values independently, monitors limits and explains the profit and loss. The back office confirms, settles, reconciles and accounts [4]. The separation exists so that no single person or system can capture a trade, decide what it is worth and move the cash. Every large trading loss in banking history has involved that separation being incomplete, usually because one of the three functions was reading the other's data rather than sourcing its own.

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Treasurer / head of ALM | Funding, liquidity, interest rate risk in the banking book, the liquid asset portfolio, the transfer pricing framework | The funding plan and the structural rate position | The whole balance sheet by maturity, repricing and behavioural bucket; liquidity ratios; the transfer pricing curve; deposit behavioural models | A treasury and asset-liability system fed by extracts from every core | Data quality and timeliness from source systems; being unable to answer a committee question between monthly cycles |
| Money market dealer | Manages the bank's daily cash position, places and takes interbank funds | Today's cash | Intraday cash position, central bank account balance and quota use, money market rates | A dealing system and the central bank interface | Intraday position visibility arriving too late to act on |
| FX and sales dealer | Quotes and executes spot, forward and swap for clients and the book | The price quoted and the position taken | Live rates, client limits, position and profit and loss, credit lines | A dealing and pricing platform | Client credit limit checks slowing down quoting |
| Rates and derivatives trader | Makes markets in interest rate products, manages curve and basis risk | The book's risk position | Curves, position and sensitivities, value at risk, collateral and margin | A trading platform plus a risk engine | Curve infrastructure, and a benchmark transition that touches every valuation |
| Middle office / product control | Validates trades and prices, computes and explains daily profit and loss, reconciles front to back | The official profit and loss | Trade data, independent prices, reference data, position reconciliations | An independent valuation and control platform | Manual price sourcing; unexplained profit and loss caused by reference data breaks |
| Treasury operations and settlements | Confirms, settles, manages nostro accounts and collateral calls, investigates fails | Whether the cash and the securities actually move | Settlement instructions, nostro balances, confirmation status, standing instructions | A settlement and confirmations platform | Standing settlement instruction data quality; manual confirmation matching |

The benchmark point in the rates trader's row deserves a sentence of its own, because it is the largest scheduled piece of work in the South African market at the time of writing. JIBAR is being retired; the recommended milestone for no new JIBAR contracts is 1 May 2026 and publication ceases at the end of December 2026, with ZARONIA as the replacement [5]. ZARONIA is an overnight rate compounded in arrears, so an interest calculation that was a single multiplication known at the start of a period becomes a daily compounding calculation known only at the end, with observation shifts, lookbacks and lockouts [5]. Every loan servicing, bond, derivative, hedge accounting and transfer pricing system that touches a floating rate is in scope. The deadline is fixed and it is not yours.

## Product, pricing and operations

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Product owner (deposits, lending, cards, vehicle finance) | Owns the product's result, features, pricing, lifecycle and regulatory compliance | The product's definition and economics | Volumes, margins, fee income, loss rates, capital and liquidity consumption, competitor pricing | A product catalogue, a management information layer and a spreadsheet | No trustworthy product profitability number; six-month lead times to change a fee because pricing is in code |
| Pricing manager | Sets rates, fees, discounts, bundles and rewards; models elasticity and impact | The price | Elasticity and take-up data, competitor fee schedules, regulated maxima, cost to serve, transfer pricing rates | A pricing tool, and a great deal of Excel | Price changes requiring a release; no way to simulate a change before committing to it |
| Payments operations | Runs the daily payment cycles, monitors files and rails, investigates failures and returns, manages recalls and mandate exceptions | Whether today's payments complete | File and batch status, rail availability, exception queues, cut-off times | A payments operations console over the hub and the rails | Investigating one exception across five systems; no end-to-end payment trace |
| Reconciliations and back office | Matches ledgers, suspense accounts, nostro and scheme settlement files | Whether the books balance | Sub-ledger and ledger feeds, scheme settlement files, nostro statements | A reconciliation platform, and spreadsheets | Unexplained suspense balances; manual matching at month end |

The pricing manager's complaint is the one to act on first, because it is cheap to fix and expensive to leave. In South Africa a significant part of the catalogue is regulated: maximum interest rates and fee caps under the National Credit Act move with the repo rate, and card interchange is determined by the central bank rather than by the schemes [6]. That means every policy rate change is a catalogue-wide repricing event with a legal maximum attached. A bank that holds prices in code has chosen to do a release every time the monetary policy committee meets.

## Risk

The second line. Independent of the business, reporting to the chief risk officer, and structurally required to disagree with the first line.

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Credit risk (portfolio) | Monitors concentration, vintage performance, early warning indicators; stress tests the book | The portfolio view and the early warning call | Exposure by segment, sector, geography and vintage; collateral; macro scenarios | The risk data mart and a stress testing engine | Aggregating exposure across systems and legal entities |
| Market risk | Independent measurement of trading risk — value at risk, stressed measures, sensitivities, limit breaches | Whether a limit has been breached and what happens next | Positions, market data history, limit structures | A market risk engine fed from the trading platforms | Position completeness, and gaps in market data history |
| Liquidity risk | Independent oversight of the liquidity ratios, survival horizons, funding concentration, the contingency funding plan | Whether the bank's funding profile is acceptable | The same data as treasury, sourced independently | A liquidity risk platform | Having to build a duplicate pipeline because treasury's cannot be used for independent challenge |
| Operational risk | Risk and control self-assessments, loss event capture, key risk indicators, scenario analysis, capital | What counts as a loss event and what it costs | Loss event data, control test results, incident data | A governance, risk and compliance platform | Loss events reported late and inconsistently; controls documented but not evidenced |
| Model risk | Independent validation of every material model, plus the model inventory and lifecycle | Whether a model may be used at all | Model documentation, development data, validation results, performance monitoring, lineage | A model inventory and a validation environment | No complete model inventory; models living in spreadsheets outside governance; no reproducible development environment |

Model risk is the row an architect should read twice. Every number in chapter 6 — the impairment charge, the capital requirement, the value at risk, the transfer price — comes out of a model, and every model needs an inventory entry, a validation, a monitoring record and a lineage trail. The honest state of most banks is that the inventory is incomplete and a material number somewhere is produced by a spreadsheet nobody has validated. Finding that spreadsheet is a legitimate first deliverable.

## Compliance, financial crime and legal

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Compliance officer | Maps obligations to controls, monitors, advises, manages regulatory findings | Whether the bank can do the thing it wants to do | An obligations register mapped to products, processes and controls; monitoring results | A compliance management platform, or a spreadsheet pretending to be one | Obligations tracked in spreadsheets; no traceability from a regulation to the control that satisfies it |
| Financial crime analyst | Investigates alerts, files reports, conducts enhanced due diligence, screens sanctions and politically exposed persons | Whether an alert becomes a report | Customer profile and expected behaviour, transaction history across products, network data, sanctions lists, beneficial ownership | Transaction monitoring plus a case manager | Alert volumes with high false positive rates; a fragmented customer view that makes "expected behaviour" unknowable; ownership chains resolved by hand |
| Due diligence operations | Onboarding and periodic refresh due diligence; remediates incomplete files | Whether a file is complete | Identity and address evidence, entity and ownership records, screening results | The onboarding and refresh workflow | Remediation campaigns, because the original data was never captured in a re-usable form |
| Fraud analyst | Monitors real-time alerts, blocks and traces, recovers funds, builds detection rules | Whether a transaction is stopped | Real-time transaction streams, device and behavioural signals, beneficiary data, known-mule lists, case history | A fraud engine plus a case manager | Latency between detection and the ability to freeze and recall |
| Legal counsel | Drafts and reviews contracts and terms, advises on regulation, manages litigation and statutory transfers | What the bank may promise in writing | Product terms, regulatory text, case state | A matter management system, and the documents | Product terms and system behaviour diverging — the document promising what the system cannot administer |
| Information officer (POPIA) | Manages consent, data subject rights, breach notification and retention | Whether personal data may be used for this | Data inventory, consent records, retention rules | A privacy management tool over an incomplete inventory | Not knowing where personal data lives; retention rules that cannot be implemented across the estate |

The scale here surprises people. South Africa's Financial Intelligence Centre received 10,863,741 reports in 2025/26, of which 5,841,505 were international funds transfer reports and 718,464 were suspicious and unusual transaction reports [7]. Those are not compliance submissions. They are production data pipelines with a statutory deadline, and they should be designed as such.

> South Africa exited the Financial Action Task Force grey list on 24 October 2025, having been listed in February 2023 [8]. The exit eases correspondent banking friction and changes no domestic obligation whatsoever. Beneficial ownership verification is permanent. If a programme business case rests on grey-list exit reducing the due diligence burden, the business case is wrong.

## Finance and regulatory reporting

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Financial controller | Produces statutory and management accounts, runs the close, maintains the chart of accounts | The numbers | Complete sub-ledger feeds, reconciliations, accounting rules | The general ledger and the close platform | Manual journals at close; sub-ledger to ledger reconciliation breaks |
| Management accountant | Produces segment and product results, budgets, forecasts and variance analysis | The allocated view of profitability | Transfer pricing, cost allocation, capital allocation, volume and margin data | A management information layer and a planning tool | Nobody agrees with the allocated numbers, so every business unit keeps a shadow result |
| Regulatory reporting analyst | Produces the full prudential return suite, disclosures, liquidity returns and ad hoc regulator requests | The accuracy of what the regulator receives | Granular, classified, reconciled balance sheet and exposure data, with lineage | A regulatory reporting platform over a mart | Returns assembled from manual mappings; a regulator question requiring a two-week manual exercise |
| Capital management | Forecasts risk-weighted assets and capital, manages issuance and the internal capital adequacy process | The capital plan | Exposure data with risk weights at deal level, forecast scenarios | A capital planning model, usually in a spreadsheet | Risk-weighted assets not attributable at deal level, which makes optimisation a portfolio guess |

The regulatory reporting analyst deserves special attention because their work is a **production function with statutory deadlines and personal accountability**, not a reporting function. The anchor prudential return in South Africa is the BA 900 balance sheet return, due monthly within fifteen business days of the month end it relates to, requiring every balance classified by institutional sector, instrument, currency and maturity [9]. That counterparty-and-instrument matrix is effectively a mandated conformed dimension set for the bank's warehouse. A bank that cannot classify every balance to it every month does not have a reporting problem. It has a data architecture problem that happens to surface in reporting.

## Internal audit, and the control functions the law requires

The roles above are organisational choices. Some of what sits above them is not.

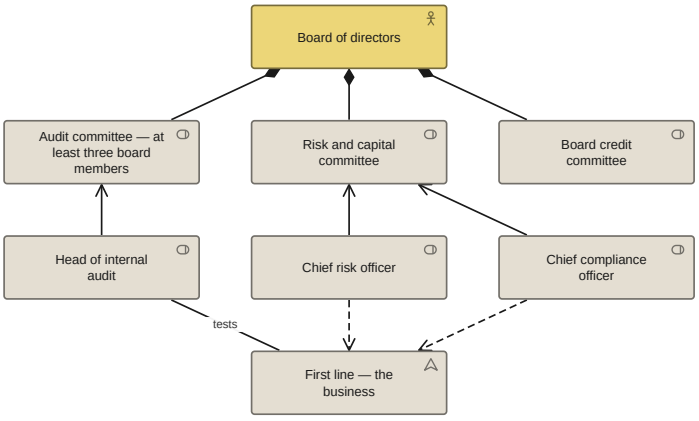



**Figure 5.6** The board-level control structure of a South African bank. The reporting lines that bypass the executive are the control; drawing them as ordinary lines is both wrong and, in an entitlement model, a finding.

In a South African bank the board must have an **audit committee of at least three board members**, under section 64 of the Banks Act [10]. The board also carries risk, capital and credit committees, and the chief risk officer, chief compliance officer and head of internal audit have reporting lines into them that bypass the executives they are reviewing. That bypass is the control. An architect who draws these functions as peers of the business, reporting the same way, has drawn something that is both wrong and, if implemented in an entitlement model, a finding.

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Internal audit | Independently tests controls and processes; reports to the audit committee | Whether a control operated, and whether to report it | Evidence that the control operated — logs, approvals, exceptions, system records | An audit management platform, and whatever evidence it can extract | Evidence not retained or not retrievable; "the control exists but cannot be evidenced" as a recurring finding |

That last complaint is the one to design against. A control that cannot be evidenced is, for audit and for the regulator, a control that did not happen. The cheapest architecture decision in a bank is to make approvals, overrides, exceptions and rule versions first-class, queryable records at the moment they occur, instead of reconstructing them from logs eighteen months later.

> A naming trap worth memorising. **Joint Standard 1 of 2024 is about outsourcing by insurers and does not apply to banks.** The instruments that do are Joint Standard 1 of 2023 on information technology governance and risk, which commenced on 15 November 2024, and Joint Standard 2 of 2024 on cybersecurity and cyber resilience, which commenced on 1 June 2025 with twelve months to full compliance [11]. Three instruments numbered "1" circulate in the same conversation. Get the year right before you quote one in a steering committee.

Those two standards matter to you more than to most people in this chapter, because they put technology governance and cyber resilience at board level, with named accountability. The practical consequence is that your architecture governance artefacts — the standards, the exception register, the technology risk view — now have a board committee as an audience, and that audience reads them.

### Technology

| Role | All day | Decision owned | Data needed | Lives in | Complains about |
|---|---|---|---|---|---|
| Enterprise or domain architect | Target state, standards, roadmaps, the coherence of change | What the target state is, and what counts as an exception | An application and integration inventory, business capability map, change portfolio | A repository, and far too many slide decks | Decisions made in delivery that contradict the target state; no reliable inventory |
| Core banking engineer | Maintains and extends the deposit, loan and card platforms | What can safely be changed, and when | Batch schedules, dependency maps, release windows | The core platform and its tooling | Batch windows constraining everything; change lead times; scarce skills on old platforms |
| Integration and API engineer | Builds and runs the layer between cores, channels and external parties | The contract between two systems | Interface catalogue, message schemas, volumes, error rates | An integration platform and an API gateway | Point-to-point sprawl; no canonical model; external parties changing formats on their own schedule |
| Payments engineer | Rail connectivity, ISO 20022 messaging, scheme and clearing interfaces | How a message is built and routed | Scheme specifications, rail calendars, certification requirements | The payments hub and the rail adapters | Mandatory externally scheduled change consuming the entire roadmap |
| Data engineer | Pipelines and the platform feeding risk, finance, regulatory reporting and analytics | What lands in the warehouse and when | Source schemas, change data capture, data contracts, quality rules | The data platform | Source systems without change data capture; no data contracts; being blamed for upstream quality |
| Security and identity engineer | Protects the estate; manages identity, entitlements and privileged access | Who may do what, technically | Entitlement models, access reviews, privileged access records | An identity and access platform | Entitlement sprawl; mandate logic duplicated per channel |

Two of those roles are your closest allies and you should find them in week one. The **data engineer** knows which source systems lie. The **payments engineer** knows exactly how much of next year is already spoken for by scheme and rail change, which is the single most useful number for roadmap planning and the one least often asked for.

### The architect's real counterparties

Put the org chart away and draw this instead: the eight people whose problems you can actually solve, and who can fund the solving.

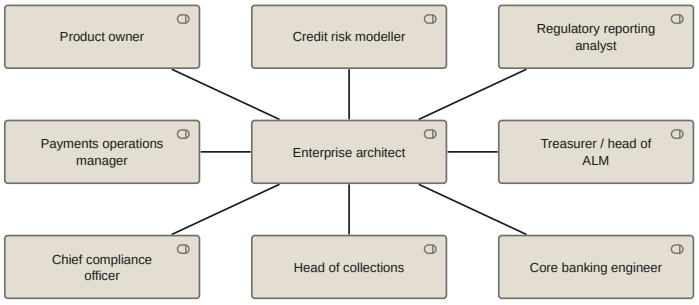



**Figure 5.7** The architect's real counterparties. Eight roles, each owning a decision and a complaint with an architectural answer.

The selection is deliberate. None of them is a chief executive, and only one of them reports through technology. Each one owns a decision, holds a budget or has a regulator's attention, and each one has a complaint from the tables above that has an architectural answer:

- The **product owner** cannot change a price without a release.
- The **credit risk modeller** cannot get origination-date risk grades.
- The **regulatory reporting analyst** cannot demonstrate lineage.
- The **payments operations manager** cannot trace a payment end to end.
- The **treasurer** cannot answer a committee question between monthly cycles.
- The **chief compliance officer** cannot trace an obligation to a control.
- The **head of collections** cannot see the customer's other arrears.
- The **core banking engineer** cannot change anything inside the batch window.

Every one of those is a sentence you can put in a business case, and none of them is "our architecture is inconsistent".

## One process, every role

The last instrument is a responsibility map over a single process. A business credit application is the right one to use, because it crosses more of the bank than anything else a mid-tier bank does routinely.

**One process, every role**

*A business credit application, end to end. R responsible, A accountable, C consulted, I informed.*

| STEP | Rel. mgr | Analyst | Sanctioner | Committee | Risk | Legal | Ops | Finance |
|---|---|---|---|---|---|---|---|---|
| Client asks for a facility | A | C | I | | | | | |
| Information gathering and spreading | R | A | I | | | | C | |
| Serviceability and affordability assessment | C | A | I | | C | | | |
| Risk grading and pricing proposal | C | R | A | | C | | | I |
| Credit paper written | C | A | R | I | | | | |
| Sanction within mandate | I | C | A | | I | | | |
| Referral to committee | I | C | R | A | C | | | |
| Conditions precedent set | C | C | C | I | | C | R | |
| Security and legal documentation | I | I | C | | | A | R | |
| Limit loaded and funds disbursed | I | I | C | | | C | A | I |
| Annual review and covenant monitoring | R | C | A | I | C | | | I |

*Read the gaps, not the letters: no column owns step 8, and finance appears three times and signs nothing.*

**Figure 5.8** A business credit application, eleven steps against eight roles. Draw it badly in the room and let them correct it — the corrections are the content.

Read it for the gaps rather than the letters. In most banks that grid, drawn honestly, shows two things. First, **a step with no accountable owner** — usually conditions precedent, or the hand-off between sanction and documentation, which is exactly where applications sit for a week with nobody chasing. Second, **a function consulted everywhere and accountable nowhere**, which in a real workshop is usually finance or risk, and which tells you that their data requirements are being treated as opinions.

Do not draw this grid yourself and present it. Draw it badly, in the room, and let them correct it. The corrections are the content.

## Four questions

You now have enough to run a discovery session without being found out. Four questions, asked of any role in this chapter, will produce more architecture than a week of document review.

1. **"Walk me through yesterday."** Not the process, the day. Which screens, in what order, with what re-keying, and where did you wait. People describe processes as they are documented and days as they are lived.
2. **"What decision is yours to make, and what happens if you are wrong?"** This separates a role with authority from a role with a workflow task, and it finds personal accountability — which is where mandates, evidence and audit trails live.
3. **"Where does the number come from, and who disagrees with it?"** Every contested number in a bank marks a data boundary: product profitability, transfer pricing, impairment, exposure, cost allocation. The disagreement is the architecture finding.
4. **"What do you keep in a spreadsheet, and why?"** The most honest question in the set. Every spreadsheet is a capability the systems do not provide, maintained by somebody who needed it badly enough to build it. Catalogue them and you have a requirements backlog that is already prioritised by someone's own effort.

> A fifth question if the room is willing: "what would you stop doing tomorrow if a system did it properly?" The answers are usually re-keying, reconciliation and assembling the same pack from the same four sources. All three are architecture problems, and all three are measurable in people.

**THE CHAPTER IN TEN LINES**

- The org chart is accurate and nearly useless: it hides the crossings, and some of the separations it shows are controls rather than preferences.
- A role is a job with a decision attached; name the decision and you have found the system that matters.
- "What they complain about" is where architecture funding comes from, because it is already a business problem in business language.
- Retail credit is a decision pipeline and business credit is a case and paper problem — different software, different data, different measures.
- The credit mandate ladder is measured on aggregate connected exposure, is personal to the holder, and escalates on six triggers of which only one is money.
- A mandate ladder held in a policy document gets implemented four times and diverges within a cycle; it belongs in one effective-dated authority service consulted before the decision.
- Front, middle and back office in treasury are separated so that no one person can capture, value and settle a position.
- A South African bank's board must have an audit committee of at least three directors, and the control functions report past the executives they review.
- A control that cannot be evidenced did not happen, so approvals, overrides and rule versions should be first-class records at the moment they occur.
- Four discovery questions — walk me through yesterday, what decision is yours, where does the number come from, what is in your spreadsheet — produce more than a week of document review.

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CHAPTER 6

# How banks make money

*A bank earns the difference between two prices, provides for losses it has not yet suffered, and is rationed by capital it must hold against both. An architect who cannot read that arithmetic will fund the wrong programme.*

## Why an architect reads the income statement

An architect who cannot read a bank's income statement will build the wrong thing. Not a slightly suboptimal thing: the wrong thing. They will fund a channel programme in a year when the problem is the cost of deposits. They will promise a product profitability dashboard without discovering that the number it would show depends on an internal price nobody has agreed. They will model impairment as a batch job rather than as the most heavily governed dataset in the building. All three happen, in that order, in real programmes.

So this chapter does the arithmetic properly. Not as finance training — you will not be asked to sign anything — but to the level where you can read a results booklet, follow an executive committee conversation, and recognise which of the things being argued about is actually a data problem.

Four qualifiers travel with every number in this chapter, and dropping one of them ships reconciliation defects into every management information layer you ever build:

- **Gross or net.** Before or after impairments and provisions.
- **Group or bank.** A listed group includes insurance, asset management and offshore operations. The registered bank does not.
- **Headline, normalised or IFRS.** South African listed companies report headline earnings as the primary measure, and several banks also publish a normalised view that strips consolidation artefacts.
- **Point-in-time or average.** Margins and ratios are usually on average balances. Capital is period-end. The two never tie exactly and somebody will ask why.

One more framing point before the numbers. In most industries the business model and the balance sheet are separate documents. In a bank they are the same document. Deposits are a liability, loans are an asset, and the income statement is what happens when you charge more for one than you pay for the other. South African banks held R6.67tn of deposits against R6.08tn of loans and advances, on total assets of R8.97tn, at the end of December 2025 [21]. The system owes more than it is owed, it is funded by its own customers rather than by wholesale markets, and that is the healthy state. Everything below follows from it.

The worked bank is **Table Bay Bank**, invented: a mid-tier South African full-service bank with roughly 2.4 million retail customers. Its figures are illustrative and internally consistent. Every market figure is sourced.

## The shape of the income statement

**The income statement of a bank, walked**

*Table Bay Bank, one illustrative year. Invented figures, real shape.*

| LINE | R MILLION | WHERE THE NUMBER IS PRODUCED |
|---|---|---|
| Interest income on advances and liquid assets | 9,800 | Loan and deposit cores, accrued daily, posted to the ledger |
| Interest expense on deposits and funding | (5,200) | Deposit core, wholesale funding register, treasury |
| Net interest income | 4,600 | Derived. The asset and liability system owns the attribution |
| Non-interest revenue | 3,100 | Billing engine, card platforms, trading systems, commission records |
| Total income | 7,700 | General ledger |
| Credit impairment charge | (760) | Impairment engine over the credit risk data mart |
| Operating expenses | (4,000) | General ledger, with an allocation layer nobody agrees with |
| Profit before tax | 2,940 | General ledger |
| Tax | (794) | Tax computation, usually outside the ledger |
| Headline earnings | 2,146 | The ledger, adjusted — and the adjustment is a judgement |
| Memo: average interest-earning assets | 104,000 | Average balances, not period-end. The two differ and both get quoted |
| Memo: average gross advances | 82,000 | Risk data mart |
| Memo: average equity | 10,500 | Capital model |
| Memo: risk-weighted assets | 72,000 | Capital engine, computed per exposure |

**Figure 6.1** A bank's income statement, with the system that produces each line. Table Bay Bank is invented and its figures are illustrative; the right-hand column is the part to read twice.

Read that table twice. The first reading is the arithmetic. The second is the right-hand column, which is the architecture.

Three observations an architect should take from it immediately.

**There are two revenue engines and they are computed from different things.** Net interest income is a balance-and-rate calculation over positions. Non-interest revenue is an event count times a price. Different grain, different latency, different reconciliation. One management information platform asked to serve both without acknowledging the difference produces a product profitability layer nobody trusts.

**The largest outflow is not presented as a cost.** Interest paid to depositors sits above the total income line, netted off. A reader used to a trading business will look for cost of goods sold and not find it, and will mis-read the cost-to-income ratio as a result.

**Two of the big lines are estimates produced by models.** The impairment charge is a statistical estimate posted as an accounting entry. Risk-weighted assets are the output of a capital engine. Both carry full audit obligations. Both are, in most banks, the numbers with the weakest lineage.

South African sector numbers for scale: the major banks ran a net interest margin of 443 basis points in FY2025, down from 451; non-interest revenue grew 9.3%, faster than net interest income; the credit loss ratio was 182 basis points; cost-to-income was 51%; and return on equity was 20% [1].

## Net interest income, decomposed

The formula is trivial and the decomposition is not.

> *Net interest income* = (*volume of assets* × *asset yield*) − (*volume of liabilities* × *funding cost*)

Management will decompose that four ways and expect your systems to support all four: **volume** (more book, same spread), **rate or margin** (same book, different spread), **mix** (reallocation between differently-spread products) and **endowment** (the policy rate flowing through the assets faster than the liabilities).

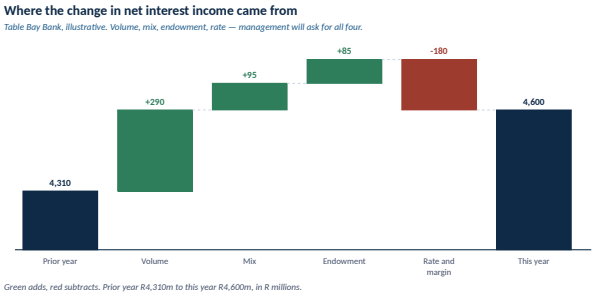



**Figure 6.2** The four-way attribution of a change in net interest income. Illustrative figures for Table Bay Bank.

Each of those four is a different question with a different owner, and each needs data the others do not. **Volume** is a balance question, answerable from the cores at daily grain. **Rate** needs the contractual and effective rate per account, which is harder than it sounds once you include promotional rates, tiered pricing and negotiated exceptions. **Mix** needs a product dimension that is stable over time, which rules out a product code that was reused when a product was withdrawn and relaunched. **Endowment** needs a split of the liability base into what reprices and what does not, which is a behavioural judgement rather than a contractual fact.

Build the attribution and you can answer the question an executive committee actually asks, which is never "what is our margin" but "why is it lower than last year and whose fault is that". Fail to build it and the conversation defaults to volume, because volume is the only part anybody can evidence — which is how a bank ends up growing a book that is shrinking its earnings. Any analytics layer that cannot produce a volume, rate, mix and endowment attribution gets rebuilt within two years. It is the most requested and least often delivered capability in bank management information, and it is worth putting on your roadmap on the first day, because by the time it is requested formally it will be requested urgently.

### The deposit franchise

A rand of current-account balance is worth more to a bank than a rand of wholesale funding. It is cheaper, it is stickier, and it counts better under the liquidity rules. That is why deposit gathering is a strategic activity rather than an administrative one, and why one large South African bank reports its low-cost current and savings balances as a headline number — R515bn, up 12%, with call deposits of R630bn, up 13%, at its 2026 interim [2].

The architectural demand this creates is specific and almost always unmet. The bank must classify every liability by its **repricing behaviour**, not by its contractual terms. A notice deposit with a 32-day contractual notice period may behave like a two-year deposit in practice; a current account with no contractual term may have a core balance that never leaves. So the bank needs a behavioural model per deposit product and segment — a core-versus-volatile split, and a decay curve. No deposit core holds that. It lives in the asset and liability system, fed by extracts, and its quality is bounded by the worst feed in the set.

The franchise also costs money in a way an architect should know about, because it generates a monthly data obligation. South Africa's deposit insurer has been operational since 1 April 2024, covering R100,000 per depositor per bank, funded by monthly premiums currently running at 0.2% a year of covered deposits, charged monthly and indexed annually [22]. To pay that premium the bank must compute its covered deposits, which means aggregating every balance to a single depositor across products and legal identities, applying joint-account shares and beneficiary look-through, every month. The deposit franchise is therefore not only a funding strategy; it is a recurring single-customer-view computation with a statutory deadline, and a bank that cannot do it has a party-data problem that will surface first as a premium submission and later as a resolution failure.

### The lending book

On the asset side, South African pricing is unusually mechanical and you should know the four reference rates.

**Prime** is the repo rate plus 3.5 percentage points [3], and retail lending is quoted as "prime plus or minus x". That makes every monetary policy committee decision a catalogue-wide repricing event, applied overnight to the whole book. **JIBAR**, a forward-looking term rate, prices corporate and wholesale lending plus a credit spread. **ZARONIA** is the overnight rate replacing it. And the **repo rate** itself drives the regulated maxima under the National Credit Act, which are expressed as formulas on repo.

The JIBAR transition is the largest scheduled piece of rate work in the market. The recommended milestone for no new JIBAR-referencing contracts is 1 May 2026, with publication ceasing at the end of December 2026 [4]. The architectural crux is not the name change. It is that ZARONIA is compounded in arrears, so an interest calculation that used to be a single multiplication performed at the start of a period becomes a daily compounding calculation known only at the end, with observation shifts, lookbacks and lockouts [4]. Every system that touches a floating rate is in scope, including the transfer pricing engine discussed below.

Mix matters more than skill in explaining margin differences between banks. A rand of unsecured lending earns multiples of a rand of mortgage. When one large bank reports a 5.29% margin and another 3.75% [5][6], the gap is mostly what they lend to whom, not how well they negotiate.

### The endowment effect

This is the one piece of bank arithmetic that is both simple and routinely misunderstood by people outside banking, and it is disclosed as an explicit line in South African net interest income attributions in a way that is far less common in United States reporting.




**Figure 6.3** The endowment effect. Free funds — non-interest deposits plus equity — earn the policy rate and cost nothing, so a rate rise widens the spread with no management action.

A portion of the funding base pays no interest at all: current accounts, low-balance savings, float in transit, and the shareholders' equity. Call it free funds. When the policy rate rises, the asset side reprices to prime almost immediately while the free funds do not reprice, because there is nothing to reprice. The spread widens with no management action, no new customers and no better pricing. When the rate falls, it narrows exactly the same way.

Two consequences follow, one commercial and one structural.

The commercial one: **deposit market share is margin share**. A bank competing hard for current-account primacy is buying endowment as much as it is buying liquidity.

The structural one is the more interesting for a planner. South Africa announced a new inflation target of 3% with a one percentage point band in November 2025 [7]. A lower steady-state inflation target implies a lower steady-state policy rate, which structurally compresses the endowment benefit over the coming cycle, and pushes more of the earnings burden onto cost-to-income and fee income. If you are building a five-year roadmap for a South African retail bank, that single sentence should change its shape: the work that pays is the work that lowers unit cost and raises fee revenue per customer, not the work that grows balances.

### Funds transfer pricing

Now the number everybody argues about.

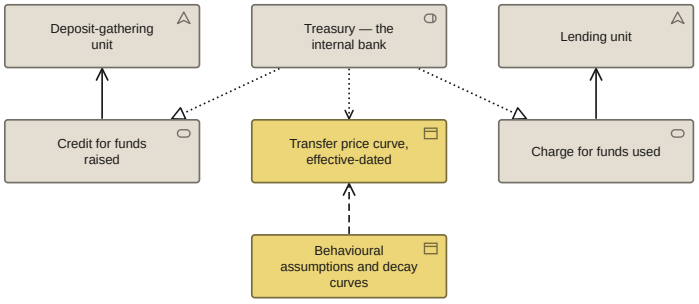



**Figure 6.4** Treasury as the internal bank. The transfer price does not change total profit; it only changes whose profit it is, which is why the curve is the most carefully attended number in the building.

Funds transfer pricing is the internal mechanism by which the bank charges asset-generating units for the funds they use and credits liability-generating units for the funds they raise, with treasury sitting in the middle as an internal bank: buying from the deposit side and selling to the lending side [8]. A deposit's value is the cost of equivalent-term borrowing in the market minus the rate paid to the depositor. A loan's value is its interest income minus the rate it is charged by treasury [8]. The calculation necessarily includes a liquidity term premium, plus adjustments for interest rate and liquidity risk, with the treatment of foreign exchange and credit risk varying between banks; customer behaviour — uncertain repayment timing, early withdrawal, prepayment — makes it harder still [8].

Here is why it is the most politically charged number in a bank. **Under centralised control, the transfer price does not change total profit at all. It only changes how profit is split between business units** [8]. It is a zero-sum allocation between executives who are paid on their unit's result. Move the curve five basis points and you have moved money from retail to corporate, or from deposits to lending, without anything happening in the outside world. Everybody knows this. That is why the ALCO meeting at which the curve is approved is the most carefully attended meeting in the bank, and why the OECD's guidance is that every institution should have an explicit transfer pricing policy, centrally managed by treasury with independent risk and finance oversight [8].

The forum where it is settled is **ALCO**, the asset and liability committee [8]. An architect should ask to attend one, once. It is the meeting at which the funding plan, the structural rate position, the liquidity ratios and the transfer price are decided together, and watching it is the fastest way to understand why the treasury system's data requirements are not negotiable. It is also where you will hear the same complaint from the other side of the table: the committee cannot ask a question between monthly cycles, because the data arrives monthly.

The architecture consequences are sharper than the politics.

- **The transfer price must be stamped on the contract when it is written, and re-struck only on a defined repricing event.** If it is recomputed retrospectively whenever the curve changes, last year's product profitability changes too, and nobody will ever trust any of it again.
- **The curve, the liquidity premium and the behavioural assumptions must be versioned and reproducible.** "What was the transfer price on this mortgage on the day it was written, and which curve version produced it" is a question that gets asked, and the inability to answer it is why product profitability discussions stall.
- **It touches everything.** Transfer pricing feeds product profitability, which feeds the investment portfolio, which funds you. It is simultaneously the most consequential and the least well-engineered calculation in most banks, and that is not a throwaway line — it is the single highest-value unglamorous fix available to a bank's data architecture.

> When somebody says "we do not have a reliable product profitability number", they have almost always said one of two things: our transfer pricing is not reproducible, or our cost allocation is not agreed. Find out which before you propose a dashboard.

## Non-interest revenue

The other engine. Fees, commissions, interchange, trading income, insurance and investment results. It grew faster than net interest income across the South African majors in FY2025 [1], and at one listed bank net non-interest income of R16.1bn exceeded net interest income of R12.7bn in the six months to August 2026, making up 70% of income from operations after credit impairments [9]. That bank processed 662 million digital transactions in the period, up 26% [9]. For an architect, the relationship between those two sentences is the whole point: non-interest revenue is a **priced event stream**, and its growth is a volume question before it is a pricing question.

Four components worth distinguishing, because they behave differently.

**Account and transaction fees.** Monthly fees, per-transaction fees, cash handling, penalty fees. Published, compared in the press annually, and regulated at the edges. The architectural requirement is a fee catalogue as effective-dated reference data and a billing engine that can be re-run for a prior period. A bank that holds fees in code has chosen a release cycle for every price change.

**Card interchange.** In South Africa this is not a commercial parameter. The central bank is the responsible authority to determine, approve, regulate and supervise interchange rates, using a cost-based methodology reviewed every five years with rates reviewed annually, binding on participants, who must submit interchange data and returns [10]. The determined streams include ATM, cash-back at point of sale, debit and credit cards, and authenticated collections [10]. For a reader from the European Union the nearest analogue is the interchange caps; for a reader from the United States there is no general analogue. The architectural implication: interchange rates are **externally set regulated reference data on an annual revision cycle**, and a bank that treats them as configuration owned by the card team will eventually be out of compliance by accident.

**Foreign exchange and cross-border.** Margin on conversion plus explicit fees. In South Africa it carries an extra obligation that reference architectures from other markets omit entirely: cross-border payments and foreign card transactions must be reported to the central bank's financial surveillance function under a reporting category that a payment system cannot infer. The category is a business classification and must be captured or derived by rule, then validated and reported.

A South African wrinkle on fee revenue that reference material from other markets will not prepare you for: a large part of the retail lending fee catalogue is **capped by statute, with the caps expressed as formulas on the repo rate**. The National Credit Act prescribes maximum interest rates by credit type, a maximum monthly service fee, and an initiation fee calculated as a percentage of the principal subject to a ceiling, all adjusted by the regulator [23]. Credit life insurance premiums are capped in rands per R1,000 of the deferred amount, with a lower cap for mortgages [24]. Two things follow. First, every monetary policy decision is a catalogue-wide repricing event *against a legal maximum*, not merely a commercial choice. Second, the rand-per-R1,000 credit life cap couples the insurance pricing engine to the lending pricing engine, which in most banks are different systems owned by different businesses. Treat the published numbers with care: the figures in circulation are computed at a repo rate that has since moved, and come from secondary sources rather than the regulator's gazette [23]. Check the gazette before you put a cap in a design.

**Knowledge-based fees.** Advisory, structuring, arrangement, guarantee and commitment fees in the corporate and investment bank. These are negotiated per deal, not catalogued, and they are the reason corporate billing is a different problem from retail billing. One South African corporate and investment bank reported its revenue mix as markets 43%, transaction banking 35% and investment banking 22% [11]. Trading income is volatile by nature: one group reported trading and fair value income of R7.1bn, up 42% in a single year [5]. Do not build a forecast model that assumes it repeats.

## Impairments and the credit cycle

The credit loss ratio is the impairment charge divided by average gross loans, quoted in basis points. Across the South African majors it was 182 basis points for FY2025 [1]. Individual banks in their 2026 reporting ranged from 73 basis points at one large group [2] to 94 and 95 at two others [12][6], 105 at a fourth [5], and 7.7% at a group whose personal banking book ran at 11.5% [13]. That spread is book mix, not management quality, and an architect who does not know that will draw the wrong conclusion from a benchmarking slide.

One number that is routinely misread. Sector impaired advances were 5.20% of gross loans on 2025 averages, against specific credit impairments of 2.50% — a coverage of 47.85% [14]. That gap is not under-provisioning. It is collateral the bank expects to realise. A secured book carries far lower coverage than an unsecured one at the same level of non-performing loans, which is why comparing coverage ratios between banks without comparing book mix is meaningless.

**The cycle is the thing to plan for.** Banks quote a through-the-cycle range for the credit loss ratio — one large group gives 60 to 100 basis points and reports 95 against it [6] — because the charge is not stable and is not meant to be. When the cycle turns, four things happen at once and each is a load on a different part of the estate. Accounts migrate from stage 1 to stage 2 in volume, which multiplies the provision without a single default occurring. Collections volumes rise, and the treatment engine that was tuned for a benign book starts queueing. The forward- looking scenarios are revised, so last quarter's number is restated for reasons that have nothing to do with any customer. And model performance drifts, which triggers monitoring breaches, which triggers revalidation, which takes months the cycle does not give you. An architect planning capacity for a credit system should plan for the turn, not the average, because the turn is when every one of those four lands in the same quarter.

### IFRS 9 in outline

**IFRS 9 expected credit loss, in outline**

*Three stages. The stage is an accounting state, and it is also a data requirement.*

| STAGE | HOW IT IS ENTERED | WHAT IS MEASURED | INTEREST COMPUTED ON | WHAT THE SYSTEMS MUST HOLD |
|---|---|---|---|---|
| **1 · Performing** | Initial recognition | Twelve-month expected credit loss, probability-weighted | Gross carrying amount | An expected loss even where the probability of default is very small |
| **2 · Significant increase in credit risk** | A significant increase in the "probability of default" since origination. 30 days past due is a rebuttable presumption | Lifetime expected credit loss | Gross carrying amount | The risk grade "at origination", kept for the life of the asset, and every stage movement as a dated event |
| **3 · Credit-impaired** | Default. Presumed no later than 90 days past due unless reliable evidence supports otherwise | Lifetime expected credit loss | Amortised cost net of the allowance | The definition of default the bank actually uses — the standard does not give one |
| **Across all three** | Measurement rules that apply everywhere | Unbiased, probability-weighted across a range of outcomes, with forward-looking scenarios | Discounted at the original effective rate for fixed, current for variable | A versioned scenario set with its weights, and revolving facilities measured beyond their contractual period |

**Figure 6.5** The three expected-credit-loss stages, and what each one obliges the systems to hold. The standard defines neither "significant" nor "default" — the bank does.

The accounting standard that governs this is IFRS 9, and the thing to understand is that it replaced an *incurred* loss model with an *expected* loss model. Under the old approach a bank provided when something had gone wrong. Under the new one it provides from the moment the asset exists [15].

Three stages. **Stage 1** is performing, and carries a twelve-month expected credit loss — the losses from defaults occurring within twelve months of the reporting date, probability-weighted, recognised even when the probability of default is very small. **Stage 2** is entered on a **significant increase in credit risk** since initial recognition, and carries a lifetime expected credit loss. **Stage 3** is credit-impaired, carries lifetime loss, and computes interest on the amortised cost net of the allowance [15].

Four details that matter more than they look.

- The significant-increase test compares the **change in the probability of default** against origination, not the change in the loss amount [15].
- **The standard does not define "significant".** Each entity sets its own criteria [15].
- **The standard does not define default either.** The entity uses a definition consistent with its internal credit risk management, with 90 days past due as a rebuttable presumption; 30 days past due is a rebuttable presumption of significant increase [15].
- Measurement must be **unbiased and probability-weighted across a range of outcomes**, with at least two scenarios where scenarios are needed, and revolving facilities are measured over the period of credit exposure even beyond the contractual period [15].

### Precisely what this forces the systems to hold

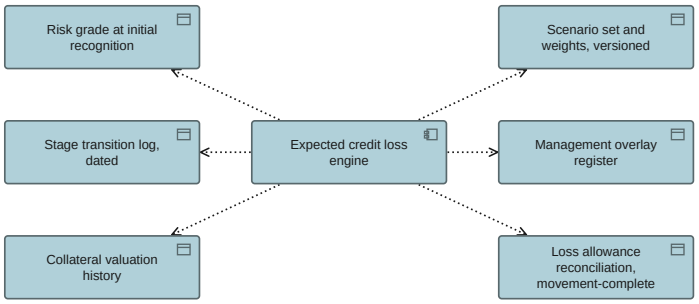



**Figure 6.6** What an expected credit loss calculation needs from the estate. Every one of these six is a data retention, lineage or event-history requirement rather than a modelling one.

Five requirements, each of which breaks a conventional loan system. This is the section to quote when somebody asks why the risk data layer costs what it costs.

**One. Credit risk at initial recognition, retained for the life of the asset.** The significant-increase test is a comparison against origination, so the origination risk grade or probability of default must be stored, immutably, per contract, for twenty years in a mortgage. Legacy systems store *current* risk and overwrite it monthly. This single requirement is the most common reason a bank's impairment calculation cannot be reproduced.

**Two. Stage, and every stage transition, as a dated, auditable event.** Movements from stage 1 to 2 to 3, and the cure path back, are the primary explanation of the impairment charge. They must be a transaction log, not a status field.

**Three. Forward-looking macroeconomic scenarios, weighted, with the weights and the scenario set versioned.** The impairment number depends on a growth and rate forecast. The auditor will ask which forecast, approved by whom, when. Management overlays — one group added R1.1bn of additional forward-looking provision for geopolitical risk in a single year [5] — must be separately identifiable in the ledger, not blended in.

**Four. Collateral valuations, dated, revalued, and linked to exposure.** Loss given default on a mortgage is a statement about a property value. The valuation date and source are part of the audit trail.

**Five. A movement-complete reconciliation of the loss allowance.** New originations, derecognitions, stage transfers, remeasurement, write-offs, recoveries, model and assumption changes — each separately derivable [15]. This is the requirement that forces an impairment data mart into existence and makes the credit risk data layer the most-governed dataset in the bank.

> Note what is *not* on that list: a better model. Every one of the five is a data retention, lineage or event-history requirement. A bank that responds to an impairment problem by buying modelling software, without fixing origination-date retention and stage history, has bought a faster way to produce a number it still cannot explain.

## Cost-to-income and jaws

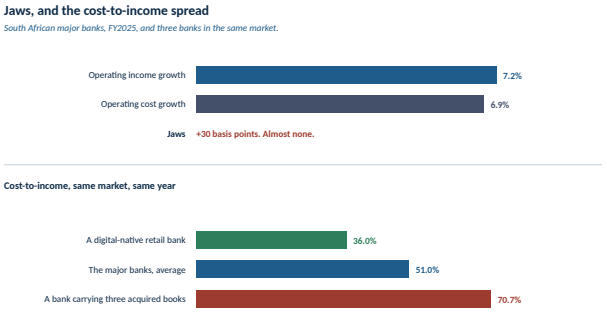



**Figure 6.7** Jaws across the South African majors in FY2025, and the cost-to-income spread within the same market in the same year.

Cost-to-income is operating expenses over total income. Across the South African majors it was 51% in FY2025, up from 50.4% [1]. **Jaws** is the gap between income growth and cost growth: income grew 7.2% against costs of 6.9%, giving positive jaws of 30 basis points, which is to say almost none [1].

Within the same market and the same year, one bank runs at 36% and another at 70.7% [9][13]. That is the clearest numerical argument for architectural simplicity anybody has ever handed an enterprise architect, and you should use it.

What drives the ratio in a South African bank specifically:

- **The branch and cash estate.** In managed decline, but still large relative to digital-only competitors, and cash handling, transport, insurance and reconciliation is a real cost that northern European banks have largely shed.
- **Core estate age and count.** Decades of mergers leave multiple cores, multiple customer masters and a permanent integration run cost. This is the line item your work actually moves.
- **Regulatory programme load.** Benchmark transition, impairment maintenance, the Basel reforms effective 1 July 2025, a 1% countercyclical buffer from 1 January 2026, and a stable-funding factor change phasing to December 2027 [16]. None of it optional, none of it scheduled by you.
- **Headcount.** Roughly 185,000 people across the six majors [17].
- **Fraud and dispute handling**, whose operational cost is a multiple of the loss.

## Return on equity, and the capital behind it

Return on equity is earnings divided by average equity. It is the measure management is paid on and the measure that connects every product decision back to the balance sheet. South African majors averaged 20% in FY2025 [1]. Individual results in the 2026 reporting season ranged from 15.0% at two large groups [12][6] through 19.8% [2] and 24.9% [5] to 31% at one [9] and negative at another [13]. Within a single bank, segment returns diverge enormously: one group reported business and commercial banking at 36.3%, corporate and investment banking at 24.8% and personal and private banking at 18.6% in the same half-year [2].

**Cost of equity is the hurdle, and it is published.** One large group states its cost of equity explicitly at 14.0%, against a return of 15.0% and a 2028 target of about 17% [6]. A business line earning 12% is destroying value while growing. Here is the sentence that connects architecture to the profit and loss, and it is worth memorising:

> *A product that cannot clear the cost of equity on a risk-adjusted, capital-charged basis should not be built, however good the customer experience.*

It helps to decompose the ratio, because each part lands on a different system. Return on equity is the product of three things: **margin** (earnings as a share of income), **asset productivity** (income as a share of assets) and **gearing** (assets as a multiple of equity). The first is the income statement and the cost base. The second is pricing and mix. The third is not a management choice at all — it is set by the capital rules, and it is the one an architect most often misses. Two banks with identical margins and identical pricing will report different returns if one holds more capital against the same book, and how much capital it holds against the same book is a function of risk weights, which is a function of data. Collateral recorded properly lowers a risk weight. Collateral recorded as a scanned document in a folder does not.

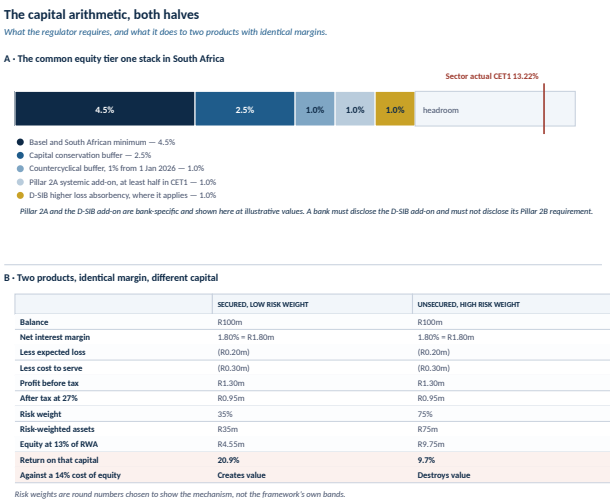



**Figure 6.8** The capital arithmetic. Above, what the regulator requires of common equity tier one. Below, two products with identical margins and different risk weights — one creates value, the other destroys it.

To see why, you need the denominator. Every exposure is multiplied by a **risk weight** — low for government bonds and well-secured mortgages, high for unsecured lending and equity — producing **risk-weighted assets**. Capital ratios are capital divided by risk-weighted assets. Gross credit exposure in the South African system was R9.9tn at December 2025, up 6.3% on the year [16].

The South African minimum common equity tier one requirement is 7.0% before buffers, with a floor of 6.5% including buffers, 8% tier one and 10% total [18]. On top sit the capital conservation buffer, a countercyclical buffer — set at 1% on a positive-cycle basis from 1 January 2026 [16] — a Pillar 2A systemic add-on of at least half in common equity, a bank-specific Pillar 2B requirement, and a higher-loss-absorbency add-on for domestic systemically important banks [18]. One disclosure quirk matters if you are building external capital reporting: **a bank must disclose its D-SIB add-on and must not disclose its Pillar 2B requirement** [18]. The sector actually ran total capital adequacy of 17.22%, tier one of 14.96% and common equity tier one of 13.22% on 2025 averages [14]. There are also two absolute floors that catch people out: a minimum capital requirement of R250m under section 70A of the Banks Act, and a legal reserve of 2.5% of deposit liabilities held with the central bank and maintained daily [19].

**Two approaches, and the floor between them.** A bank may compute credit risk weights on the **standardised** approach, where the weights are prescribed, or on an **internal ratings-based** approach, where it uses its own models subject to supervisory approval. Internal models generally produce lower risk-weighted assets, which is the point of building them. The remaining Basel III post-crisis reforms took effect in South Africa on **1 July 2025**, covering credit risk on both approaches, operational risk, credit valuation adjustment, market risk and the **output floor** [20].

The output floor is the piece with the largest architectural consequence and the least discussion. It places a floor under internally modelled risk-weighted assets, expressed as a percentage of what the standardised approach would have produced. To apply it, **every exposure must be calculated twice, every period: once under the internal model and once under the standardised approach.** That means standardised inputs — collateral classifications, loan-to-value bands, counterparty types, external ratings — must be maintained to production quality for a calculation whose only purpose is to constrain another calculation. Banks that built their data model around their internal models discover that they have one well-fed pipeline and one starved one, and the starved one is now binding.

> The output floor's calibration is set in the Basel text and phased. I have not sourced the South African phase-in percentages, so take the mechanism here and check the number against the Prudential Authority's own directive before you quote one in a design.

Approach usage in South Africa is more conservative than reputation suggests. As at March 2026, 58% of banks were on the standardised approach for market risk and 42% on the simplified standardised approach; for credit valuation adjustment, 52.63% were on the reduced basic approach, 31.58% on the simplified approach and 15.79% on the full basic approach [16].

**What "capital-efficient" means to a product owner.** Two products with identical margins differ in value if their risk weights differ, because the higher-weighted one needs more equity per rand of balance. "A capital-efficient version" means one of five things: more or better collateral, a shorter maturity, a better-rated counterparty, a risk-transfer structure such as securitisation or a guarantee, or distribution off the balance sheet. Panel B of the figure above shows the mechanism with round numbers.

The architectural conclusion is blunt. **Product and pricing systems must carry risk weight and capital charge as first-class, deal-level attributes, and the pricing calculator must produce a risk-adjusted return on capital per deal rather than a margin.** A bank whose capital team cannot attribute risk-weighted assets to the deal level is optimising its balance sheet by portfolio guesswork, which is the complaint in the capital management row of chapter 5 and one of the better-paying problems an architect can solve.

## Liquidity, and why a treasury system exists at all

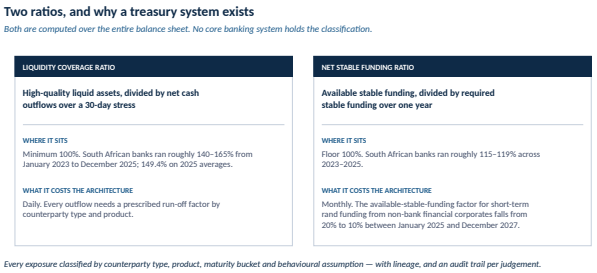



**Figure 6.9** The liquidity coverage ratio and the net stable funding ratio, and the classification obligation that is the reason a treasury system exists.

Two ratios, both with a floor of 100%, both computed over the entire balance sheet.

The **liquidity coverage ratio** is high-quality liquid assets divided by net cash outflows over a thirty-day stress scenario, and South African banks ran roughly 140–165% from January 2023 to December 2025 [16], with the financial stability review giving 149.4% on 2025 averages [14]. The **net stable funding ratio** is available stable funding divided by required stable funding over one year, and South African banks ran roughly 115–119% across 2023 to 2025 [16].

Two South African specifics. The available-stable-funding factor for short-term rand funding from non-bank financial corporates is being cut from 20% to 10% between January 2025 and December 2027 [16] — a mechanical tightening that changes the relative value of that funding source and is already in bank funding plans. Separately, interest rate risk in the banking book, measured as the sensitivity of economic value of equity, ran at about 4.4% of tier one capital against a 15% ceiling at the end of 2025 [16].

Now the architectural answer to the question in the heading. Both ratios require **every exposure on the balance sheet classified by counterparty type, product, maturity bucket and behavioural assumption, under prescribed run-off factors, daily for one ratio and monthly for the other, with full lineage and an audit trail per classification judgement**. No core banking system holds that classification. It is not what a core is for. The treasury and asset-liability system exists to apply it, and its data quality is bounded by the weakest source feeding it — which is why the treasurer's complaint in chapter 5 is about source data rather than about their own tooling.

## Reading a results booklet in ten minutes

A practical interlude, because you will be handed one.

Start at the back, with the segment analysis, not the front. The group number tells you how the bank did; the segment numbers tell you which business is paying for which, and that is the map of where your programme money is going to come from. A group averaging 20% can contain a business banking arm at 36% and a personal banking arm at 19% [2], and those two will have completely different appetites for a shared platform.

Then four readings, in this order.

1. **Net interest margin, and the attribution beneath it.** If the margin fell, was it volume, mix, rate or endowment? A booklet that shows the attribution is a bank whose asset and liability system works. A booklet that gives one number and a sentence is telling you something.
2. **Credit loss ratio against the stated through-the-cycle range.** Inside the range is a cycle. Outside it, twice running, is a book or a model.
3. **Cost-to-income, and jaws.** Negative jaws for two periods means the cost base is growing into a flat income line, which is the condition in which simplification programmes get funded and discretionary ones get cut.
4. **Common equity tier one, against the requirement rather than against last year.** The gap between the two is the bank's room to grow, and it is the constraint that decides whether a new lending proposition is even possible.

Then one question that is not in the booklet: **what is in non-interest revenue that will not repeat?** Trading income and one-off fees flatter a year and are usually disclosed in a note rather than on the face of the statement.

## The four numbers an executive watches

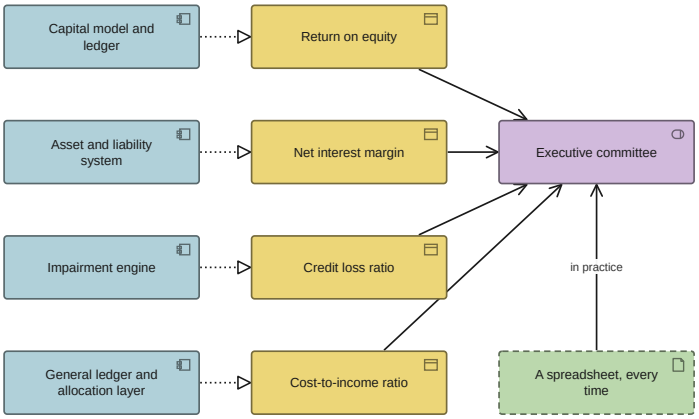



**Figure 6.10** The four numbers an executive committee watches, the system that produces each — and the dashed artifact that, in most banks, does the last mile.

Strip everything in this chapter down and four numbers survive. An executive committee will discuss all four every month, in roughly this order, and each one is produced by a different system.

| The number | What it answers | Produced by | Where it actually comes from |
|---|---|---|---|
| Return on equity | Are we worth owning | Capital model plus the ledger | A planning spreadsheet, reconciled to both |
| Net interest margin | Is the core business earning | The asset and liability system | An attribution pack built monthly |
| Credit loss ratio | Is the book turning | The impairment engine over the risk mart | A model run, with overlays applied by hand |
| Cost-to-income | Are we running efficiently | The ledger plus the allocation layer | The ledger, plus allocations nobody agrees with |

The fifth, watched by a different audience, is the common equity tier one ratio. Treat it not as a performance measure but as a constraint: it decides what the other four are allowed to be.

And now the honest point, which belongs at the end of a chapter about arithmetic. **In most banks, the last mile of each of those four numbers is a spreadsheet.** Not the whole calculation — the source systems do the work — but the final assembly, the adjustment, the overlay and the presentation. The figure above draws it, and it is drawn deliberately, because pretending otherwise produces target-state diagrams that nobody in finance recognises.

The right response to that is not to be scandalised and propose eliminating spreadsheets, which has been attempted in every bank and succeeded in none. It is to ask three questions about each one:

1. **Are its inputs reproducible?** Can you re-create the input extract for a prior period, exactly, from a governed source?
2. **Is its logic visible?** Is there a documented, reviewed statement of what it does between input and output?
3. **Is its output reconciled?** Does the number that leaves it tie back to the ledger, with the difference explained?

A spreadsheet that answers yes to all three is a presentation layer and is fine. A spreadsheet that answers no to any of them is an unmanaged model producing a number the board relies on — and that is a finding waiting to be written, a regulatory question waiting to be asked, and the most defensible business case you will find in your first ninety days.

### THE CHAPTER IN TEN LINES

- Four qualifiers travel with every bank number: gross or net, group or bank, headline or normalised or IFRS, point-in-time or average.
- Net interest income is a balance-and-rate calculation and non-interest revenue is a priced event stream — different grain, latency and reconciliation, and one platform serving both without saying so is not trusted.
- Free funds — non-interest deposits plus equity — make the endowment effect mechanical, which is why deposit market share is margin share.
- A lower steady-state policy rate structurally compresses the endowment benefit and pushes the earnings burden onto cost and fees.
- Funds transfer pricing does not change total profit, only whose profit it is, which is exactly why it is political and why it must be stamped at origination and versioned.
- IFRS 9 forces five things onto the systems: origination-date risk retained for life, stage transitions as dated events, versioned scenarios and overlays, dated collateral valuations, and a movement-complete allowance reconciliation.
- Cost-to-income differences within one market are mostly core count, customer masters and coexistence machinery — which is the clearest numerical argument for architectural simplicity available.
- A product that cannot clear the cost of equity on a capital-charged basis should not be built, so risk weight and capital charge belong in pricing as deal-level attributes.
- The output floor forces every exposure to be calculated under both the internal and the standardised approach, which makes the standardised data pipeline production-critical.
- The last mile of all four executive numbers is usually a spreadsheet; the useful response is to ask whether its inputs are reproducible, its logic visible and its output reconciled.

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CHAPTER 7

# The operating model

*A bank is two businesses welded together, running on a clock it does not own. This chapter is how one actually runs, end to end — the value chain, the channels, the business day, the regulatory calendar, and the handful of boundaries that are real.*

## Two businesses, one licence

Table Bay Bank is invented. It is a mid-tier South African full-service bank with about 2.4 million retail customers, a business bank, a small corporate and transactional arm and a treasury, and nothing about it describes a real institution. It is here because an operating model is impossible to explain in the abstract.

Ask Table Bay Bank's chief operating officer what the bank does and you will get an org chart. Ask what the bank *runs* and you get something more useful, and stranger: two businesses that share a licence, a brand and a general ledger, and almost nothing else.

The first is the **balance-sheet business**. It takes deposits and makes loans. Its unit of work is an account, its unit of record is an entry, and its clock is the day. It is regulated prudentially — capital, liquidity, provisions — and its systems are slow-moving on purpose, because a twenty-five-year mortgage is a promise to administer a contract under rules that have not been written yet.

The second is the **payments business**. It moves other people's money. Its unit of work is an instruction, its clock is somewhere between sixty seconds and ten milliseconds depending on the rail, and it is regulated as national infrastructure. In South Africa the central bank operates the settlement system, sets interchange for several card and collection streams, owns half the clearing utility and, since September 2026, holds the payment scheme rule-making function itself [1][2][3].

| | Balance-sheet business | Payments business |
|---|---|---|
| Unit of work | The account | The instruction |
| Clock | End of day | Seconds, and falling |
| What failure looks like | A wrong balance, found at close | A salary that did not arrive, found by the customer |
| Regulated as | A prudential institution | National infrastructure |
| Change driver | Product, pricing, credit appetite | Scheme mandates with external dates |
| Who decides the roadmap | The bank | Largely, somebody else |

> These two businesses have different clock speeds, different failure modes and usually different organisational homes. Treating "the bank" as one landscape mis-scopes everything that follows — the integration pattern, the availability target, the release cadence, the shape of the data model. The first question to ask about any system in a bank is which of the two businesses it serves.

There is a third thing, which is neither a business nor a function, and which an architect will underestimate exactly once: the **reporting and regulatory production line**. It is not a reporting team. It is a factory with statutory deadlines, personally attested output and, in most South African banks, the largest single claim on change capacity in the portfolio. It gets its own section.

## The value chain, end to end

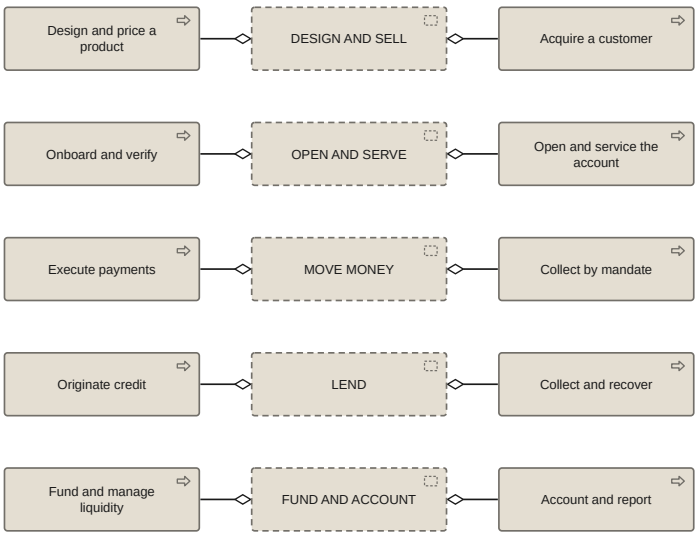



**Figure 7.1** The bank's value chain: ten end-to-end processes in five groupings. Read it as what the bank runs, not as who runs it — the org chart cuts across every one of these.

Ten steps, in five pairs. Walk them once properly and you will know where every system in the bank came from.

**Design and price a product.** Someone decides that a savings account pays 4.1% above R10 000, that a personal loan is priced at prime plus six, that the bundled monthly fee is R115 and includes five free electronic payments. Three things make this harder in a bank than it looks. Product definition in most legacy cores is code rather than configuration, so a new product is a release and a release train. Prices are partly not the bank's to set: National Credit Act maxima are formulas on the repo rate, so every monetary policy decision is a catalogue-wide repricing event [4]. And a price has a capital and liquidity consequence, which means the product owner's P&L case depends on an allocation nobody fully trusts.

**Acquire a customer.** Branch, app, retail partner, broker, bond originator, dealer floor, payroll scheme. The acquisition channel determines the data the bank starts with, and the bank will live with that difference for the life of the relationship. A customer acquired through a bond originator arrives with a full income file and no transactional history; one acquired at a retail till arrives with an identity document and nothing else.

**Onboard.** Identity verification, address verification, screening, risk rating, due diligence. In South Africa this step is the single hardest part of retail acquisition, and the reason is statutory: under section 21E of the Financial Intelligence Centre Act an institution that cannot verify a client's residential address may not establish the relationship at all [5]. The architecture consequence is a state machine, not a form: a partially verified customer must be holdable in a restricted state rather than failed.

**Open the account.** A product instance in the core, a mandate saying who may instruct what, limits, channel entitlements, and often a card. This is where a bank's data model is made or ruined, because the mandate is routinely recorded at account level instead of party level, and a customer who changes a signatory then has to change it once per account.

**Service it.** Address changes, limit changes, statements, notices, disputes about fees, debit-order stops, dormancy, death and estates. The least glamorous step and the largest consumer of operational cost. It is also where the bank's promises are tested, because a service request is the only interaction most customers initiate.

**Transact.** Payments out, payments in, card authorisations, cash in and out, collections by mandate. This is the payments business, and it is the step with the hard external clock. Its defining property is that the bank is one participant in a shared system it does not control.

**Lend.** Application, affordability, decision, agreement, disbursement, then servicing for years. Retail lending is a decision pipeline: high volume, standard data, thin human touch, minutes to days, with the regulated affordability test as the gate. Business lending is a case-management and credit-paper problem with a committee at the centre, financial statements to spread, security to register and an annual review that comes round whether anyone has time for it or not. The two are different classes of software with barely overlapping data models, they are measured differently — retail on conversion, time-to-yes and straight-through rate; business on time-to-credit-paper, limit utilisation and covenant breaches — and conflating them is the classic scoping error. One further South African specific: the National Credit Act makes affordability a *prescribed calculation* with prescribed inputs, including a minimum-expense-norms table, so it belongs in a decision service with effective-dated rule versions and a stored, replayable decision record [4].

**Collect and recover.** Two businesses again. Collections works a customer who still has a relationship. Recoveries works an exposure where the relationship is over — legal process, security realisation, sale of book, write-off, and recovery after write-off. South African practice adds a statutory step before enforcement: the section 129(1)(a) notice, with proof of delivery [6].

**Fund and manage liquidity.** The deposit franchise is the cheapest funding a bank has, and treasury's job is to make the asset book and the liability book agree about maturity, rate and currency. It is the step most invisible to the rest of the bank and the one with the shortest decision cycle: the money market dealer manages today's cash against the central bank account, and a settlement obligation that cannot be funded is not a reporting problem, it is a rejected instruction. Real-time rails change this materially, because a gross, prefunded rail means liquidity has to be managed intraday and continuously rather than netted at the close [7].

**Account and report.** Close the ledger, compute the provision, produce the returns, sign the attestations. Every other step in the chain feeds this one, and this one is where a defect in any of them becomes visible to a regulator.

> Notice what the chain does not contain. There is no "data" step, no "risk" step and no "compliance" step, because those are not steps — they are properties of every step. An operating model that puts them in the chain as boxes produces an architecture in which they are somebody else's problem nine times out of ten.

## Channels, and what each one can actually do

Eight channels, each with its own session model, authentication, latency budget and failure mode, all needing the same answer to one question: what is my balance.

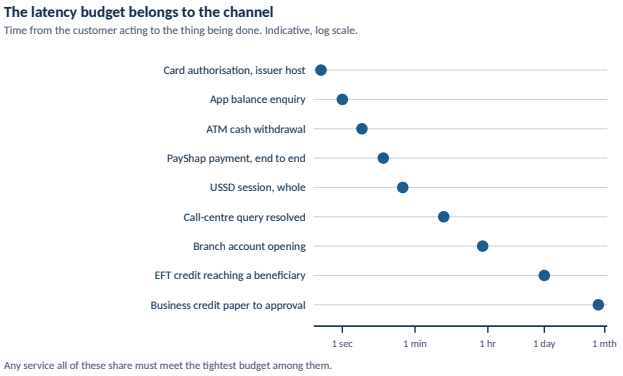



**Figure 7.2** Time from the customer acting to the thing being done, by channel. Indicative durations on a log scale; the point is the spread, not the precise values. Any shared service must meet the tightest budget among the channels that call it.

| Channel | What it can genuinely do | Latency the design must assume | What it cannot do |
|---|---|---|---|
| Branch | Identity, documents, cash, complex origination, remediation | Seconds per screen, hours per case | Scale; serve a customer 200km away |
| ATM / self-service | Cash, balance, limited transfers, card services | Under two seconds, offline-tolerant | Anything needing a document or a signature |
| Call centre | Explain, authorise, resolve, retain | Seconds, with a human waiting | Produce evidence it did not capture |
| Internet banking | Bulk files, multi-user authorisation, host-to-host | Seconds; minutes for a file | Serve a feature phone |
| Mobile app | Everything retail, well, for the device-capable | Under a second, or it is abandoned | Multi-signatory business authorisation |
| USSD | Balance, transfer, prepaid, mandate authentication | Under 30 seconds per session, stateful | Carry a payload or survive a timeout |
| WhatsApp / chat | Enquiry, notification, light servicing | Conversational, no session guarantee | Be the authoritative record of consent |
| API / host-to-host | Corporate payment and collection volume, contractual | Milliseconds to seconds, under a contract | Be assumed to pass through the user interface |

**Branch.** Teller and consultant applications over the core, plus cash handling, vault and till reconciliation and physical document capture. South African branch networks are large and shrinking: for the FY2024/FY2025 reporting period branch counts were Capitec 880, FNB 630, Standard Bank South Africa 626, Nedbank 623 and Absa 616 [8]. The branch remains the origination channel for complex products and the remediation channel of last resort for incomplete customer files.

**ATM and self-service.** Driven by the switch, not by the core — which surprises architects who assume the core answers everything. The estates are big: Capitec 8 798 devices, Absa 5 138, FNB 4 771, Nedbank 4 297, Standard Bank South Africa 3 450 for the same period, with Capitec's figure including a large retail-partner footprint [8]. Volume is falling fast. The Reserve Bank records ATM transactions declining from 509.4 million in 2023 to 289.4 million in 2024 [9]. Architecturally this matters because the estate-management, key-management and reconciliation systems behind an ATM network are expensive per surviving transaction, and the cost does not fall as fast as the volume.

**Call centre.** Computer-telephony integration, a screen-pop from the party master, a knowledge base, call recording with a retention obligation, and an agent desktop that aggregates eight systems behind one screen. The agent desktop is usually the worst piece of integration in the bank and the place where screen-scraping is most often found and least often admitted.

**Internet banking.** The 2000s channel, now frequently the *business* channel while retail moves to the app. It carries bulk payment file upload, multi-user authorisation hierarchies and host-to-host integration for corporates — capabilities the app does not have and cannot easily acquire. Do not let a digital programme retire it without first finding out what the corporates are doing with it.

**Mobile app.** The primary retail channel, and the primary risk concentration. SABRIC's 2025 statistics put banking-app-related crime at roughly 89% of reported digital banking crime cases and 70.5% of the total amount claimed, with reported digital banking losses rising from about R1.9bn in 2024 to R2.4bn in 2025 across 110 074 recorded incidents [10].

**USSD.** Session-based, menu-driven, stateful, short timeout, tiny payload, and a per-session cost the bank carries. It is load-bearing in South Africa rather than a legacy curiosity, and it is the exact opposite of what an API-first design assumes: synchronous, stateful and latency-bound. I could not source

current USSD transaction shares at South African banks; individual results presentations occasionally disclose them.

**WhatsApp and chat.** A mainstream South African channel with three awkward properties: no screen, no session guarantee, and a third-party platform in the trust path. Authentication, consent capture and audit have to be designed explicitly rather than inherited from the web channel. Per-bank adoption figures are not consolidated anywhere I could find.

**API.** Two different things wearing one name. Bank-to-corporate host-to-host and API integration is mature and contractual. Open banking in the United Kingdom or European Union sense **does not exist in South Africa as a regulated regime**. What exists is bilateral commercial APIs plus Directive No. 2 of 2024, which regulates the initiation of EFT credit payment instructions on a payer's behalf — the screen-scraping case — by requiring registration, per-payment informed consent, no credential storage, five-year audit trails, fraud insurance, monthly volume and value reporting by the fifteenth and biannual compliance attestations [11].

> **COST PER CHANNEL IS THE NUMBER EVERYBODY QUOTES AND NOBODY SOURCES**
>
> You will be shown a slide with a cost-to-serve per channel: R40 in branch, R8 at an ATM, R2 in the app. I could not source such a table for South African banks, and the ones in circulation are either foreign, a decade old, or an internal allocation presented as a fact. Use the figure the bank's own finance team will defend in writing, insist on knowing whether it carries a share of the estate's fixed cost, and never put an unsourced number in an architecture document where it will be quoted back to you for three years.

Three things follow for architecture, and they are the whole point of the section.

**The latency budget belongs to the channel, not to IT.** A USSD session will time out in under a minute. A card authorisation has to answer in well under a second. A branch consultant will wait four seconds for a screen and complain. A commercial credit paper takes three weeks. Any service that eight channels share has to meet the tightest budget among them, which in practice means the authorisation host sets the bank's non-functional requirements.

**Every channel accrues business logic and becomes an undeclared system of record.** Consent captured in the app that never reaches the party master. A limit held in the app's own store. A balance derivation implemented differently in the app, at the ATM and in the authorisation host, so three channels show three balances and all three are defensible.

**The weakest channel sets the bank's actual fraud exposure.** Authentication strength varies by channel, and an attacker picks. Modelling authentication per channel, rather than as one capability consumed at different strengths, is how a bank ends up with a 2010 control on a 2026 threat.

### The physical estate

The branches, the ATMs and the cash. Four points an architect needs and rarely gets told.

The estate is a property and logistics business inside the bank, with leases, device life cycles, cash-in-transit contracts, key management and physical security. It has its own systems and its own reconciliation chain — device to switch to ledger — and the breaks in that chain are a standing operational loss.

It is shrinking, and shrinkage is harder than growth. Closing a branch moves its customers onto channels that may not be able to serve them, re-points the FICA remediation workload, and strands the documents. SABRIC reported ATM attacks down 46% in 2025 with associated cash losses down 63%, and bank robberies down from eight incidents in 2024 to two in 2025 [10] — the risk profile of the physical estate is changing as fast as its size.

It is increasingly not the bank's. Retail partners run tills that take deposits; a device in a supermarket is on the bank's balance sheet of accountability and somebody else's floor. That is a delegated arrangement, and it belongs in the section on those.

And it is where the identity problem is solved for the population that digital cannot reach. Fingerprint capture against Home Affairs fails disproportionately for older people and for those doing manual work, and a failed capture is not a failed identity even though systems routinely treat it as one [12]. The branch is where that gets fixed by a human being.

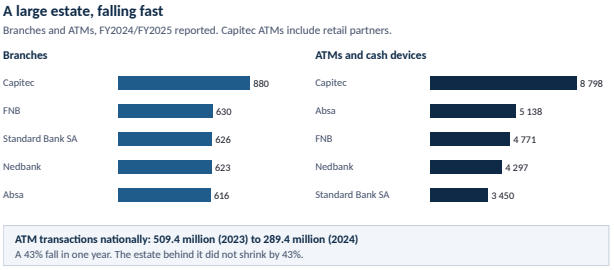



**Figure 7.3** Branch and ATM estates for the five largest South African banks, FY2024/FY2025 reported [8], against the national fall in ATM transactions from 509.4 million in 2023 to 289.4 million in 2024 [9]. Capitec's device count includes a large retail-partner footprint.

### The same chain, three operating models

One more thing before the clock. The ten steps are the same across the bank; how they run is not, and a single operating model diagram that averages them is worse than no diagram.

**Retail** is a volume machine. The acquisition channel is digital or a retail partner, the decision is a scorecard, the servicing is self-service, and the economics are per-unit: cost per account opened, cost per transaction, loss rate by vintage. The architecture question is throughput and straight-through processing, and the enemy is any step that requires a person.

**Business banking** is a hybrid, and the hardest of the three to model honestly. The customer has a named relationship manager, but the products are standardised and the volumes are real. Credit is a paper approved under a delegated authority rather than a score, the unit of decision is an obligor group rather than an account, and the mandate — who may instruct, to what limit, with whose countersignature — is a first-class object with its own lifecycle. Nothing in a retail stack has the concept of a covenant with a monitoring cycle, and business banking has thousands of them.

**Corporate and transactional banking** is a few hundred named clients, bespoke pricing, host-to-host integration and a service-level agreement per client. Real time here is a risk requirement, not a customer-experience feature: a corporate treasurer needs to know a position, not to enjoy an interface. The operating model is account management plus operations, and the systems are dominated by limits, collateral, confirmations and settlement finality.

> The practical test of whether a bank has one operating model or three: ask how a limit increase is approved in each segment, and how it reaches the system that enforces it. In retail it is a strategy in a decision engine. In business banking it is a sanctioner with a personal mandate. In corporate it is a credit committee and a facility structure with sub-limits. If one system is asked to serve all three, find out which segment it was built for — because the other two are being served by spreadsheets.

## The business day

This section matters more in banking than in any other industry, and it is the one most often missing from an architecture document.

A bank's day is not a day. It is a sequence of windows, each with a cut-off, each cut-off owned by somebody outside the bank, and between them an overnight cycle that has to complete before the next day can open.

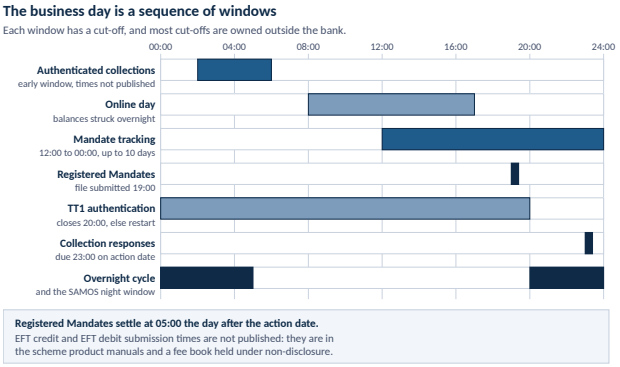



**Figure 7.4** The business day as a sequence of windows. The Registered Mandates and DebiCheck times are published [14][15]; the early-collection window and the overnight cycle are drawn indicatively because the authoritative EFT cut-off schedule is not public [16].

**The settlement layer runs around the clock, but not uniformly.** SAMOS, the domestic real-time gross settlement system, has operated since 9 March 1998 and runs seven days a week, twenty-four hours a day, with different settlement windows for weekdays and weekends which the operator may change, and a dedicated **night window** for clearing-house batch processing [7]. Two settlement models coexist: a real-time line that settles gross and prefunded, rejecting instructions where liquidity is insufficient, and a continuous processing line that queues instructions and funds them as liquidity arrives [7]. Intraday liquidity is free but bounded — a participant may use up to 70% of its cash reserve account balance and pledge up to 50% of its liquid asset requirement as collateral for intraday loans repayable by end of day — and, critically, **the night batches draw only on the cash reserve account balance** [7]. The batch window is therefore not an internal convenience. It is a settlement-infrastructure fact with a liquidity constraint attached. In October 2025 night settlement was R1.52 trillion against R17.22 trillion in day settlement [13].

**The collection windows are the sharpest clock in the retail bank.** Authenticated collections under DebiCheck process in the early-morning window, ahead of other methods, and a processed DebiCheck collection cannot be reversed [14]. Registered Mandates — the rail that replaced the Registered Mandate Service from 10 March 2025, and which is a new ISO 20022 rail rather than a rename — submits in a **late window at 19:00 SAST**, tracks between 12:00 and midnight for up to ten tracking days, requires responses by **23:00** on the action date and settles at **05:00** the following day, processing after DebiCheck and before plain EFT debits [15]. Mandate authentication has its own clocks: a TT1 request can be authenticated until 20:00 the same day and otherwise restarts; a TT2 batch request allows the customer two days [14].

**The overnight cycle, roughly in order.** Cut-offs close. Inbound clearing files are received and applied. Outbound collection and payment files are generated. Interest is accrued and capitalised. Fees are assessed. The arrears state machine advances days-past-due. Statements and statutory notices are composed. The general ledger interface runs. Expected-credit-loss and risk extracts run on their own cycles. The warehouse loads. Then everything downstream of the warehouse loads. The online day opens.

> **THE AUTHORITATIVE CUT-OFF TABLE IS NOT PUBLIC**
>
> I could not source the full EFT credit and EFT debit file submission schedule. BankservAfrica's published rules state explicitly that submission times live in the Product and Service Manuals supplied by its customer account management team and in a participant pricing and fee book available under non-disclosure [16]. This is a genuine unsourceable gap, not an oversight. If you are designing against those cut-offs, get the manuals from the bank's payments operations team or from PayInc, and put the actual times in your own design document, because no reference architecture you can download will carry them.

**One day at Table Bay Bank**, to make the sequence concrete. Again: invented. At 05:00 the previous night's Registered Mandates settle and the collection outcomes land, so the collections team opens on yesterday's unpaid reason codes. From about 06:00 the authenticated collections have already processed

in the early window, which is why a customer's salary-day balance looks different at seven in the morning than it did at midnight. Through the morning the online day runs against balances struck overnight. At 12:00 mandate tracking opens for the day's unpaid items. The afternoon is cut-off country: each rail closes its submission window, and payments operations works the queue backwards from the last one. At 19:00 the Registered Mandates file goes out. By 20:00 any TT1 mandate authentication the bank is waiting on has either happened or restarted. At 23:00 responses to collections are due. Then the overnight cycle runs, and the next morning somebody in regulatory reporting finds out whether it completed.

Notice what this does to an apparently simple requirement. "Show the customer their balance in real time" crosses four of those windows and three systems with different ideas of what time it is. That is not a user interface problem.

**Why so much of this is still batch.** Four reasons, in descending order of how often they are the real one. First, the clearing rails are batch: EFT credit and EFT debit are file-based with action dates, cut-offs and deferred settlement [27], and a bank cannot be real-time against a rail that is not. Second, periodic computation is genuinely periodic — interest accrual, fee assessment, amortisation, provision staging and regulatory aggregation are defined on a period rather than an event, and making them event-driven does not make them cheaper, it makes them harder to prove. Third, the core was designed around an end-of-day, and batch posting with value dating is the ledger's native model. Fourth, nobody has been able to afford to change it.

**What breaks when a batch runs late**, in the order it appears:

1. A collection or payment file misses its cut-off. A day's collections or a day's salary payments do not go out. The impact is immediate, public and in the press by lunchtime.
2. Statements and notices are not produced. For a statutory notice — a section 129 letter, a notice of a rate change — that is a compliance event rather than an inconvenience.
3. Downstream systems load incomplete data. The warehouse and every mart built on it are wrong for a day, and every report drawn from them is silently wrong, which is worse than being visibly wrong.
4. The arrears state machine does not advance. Collections works a stale book, and a customer who went into default is not treated.
5. The regulatory extract misses its window, and a monthly return is late. Late returns are a supervisory conversation.
6. The online day opens late, or opens on an inconsistent position.

> A batch failure is not a one-day outage. It is a one-day outage plus a reconciliation tail. The compounding is what makes it serious, and it is why the overnight cycle's critical path is a first-class architectural object that belongs in your model with its dependencies drawn.

**Where real time is going, and what it costs.** The direction is not in doubt. PayShap is real-time and ISO 20022 native [18]. Registered Mandates is ISO 20022 [15]. SAMOS migrated to ISO 20022 on 17 September 2022, the first real-time gross settlement system in Africa to do so [19]. PayInc has stated an intention to migrate Real-Time Clearing traffic to PayShap and eventually retire it, which is a roadmap rather than a dated commitment [17]. The honest costs are these. Twenty-four-by-seven availability stops being a target and becomes a precondition, because there is no batch window to hide maintenance in — which means zero-downtime deployment and online schema change, not a faster batch. Liquidity becomes an intraday discipline. Fraud and sanctions screening move into the payment path under a latency budget, deciding in milliseconds on data that used to be available overnight. Irrevocability changes the error model from reversal to recovery. Reconciliation becomes continuous. And the periodic computations do not go away.

The conclusion is unglamorous and worth stating plainly: **real time does not replace the overnight cycle. It is added alongside it, and the transitional state is permanent.** Architect for both, indefinitely, and be suspicious of any roadmap whose end state has one.

## The reporting and regulatory calendar

In a bank, reporting is a production function. It has statutory deadlines, attested accuracy, named personal accountability and a change queue that outranks yours.

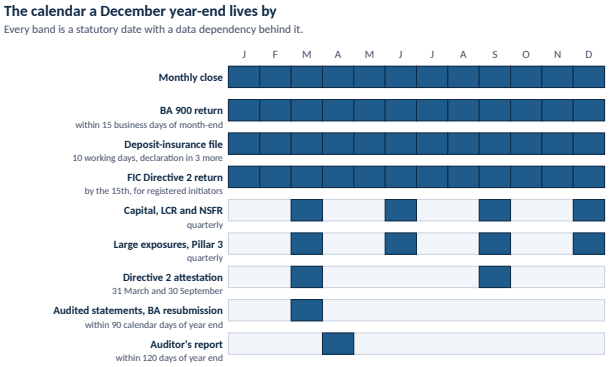



**Figure 7.5** The reporting calendar for a December year-end. Every band is a statutory date with a data dependency behind it, and each one is a claim on the same change capacity your programme wants.

**Daily.** Settlement and clearing reconciliation. Nostro and suspense. The intraday liquidity position. And a South African specific that a reader from elsewhere will not expect: cross-border card transactions by non-residents at South African merchants and ATMs are reported to the Reserve Bank's Financial Surveillance Department **daily, per authorised dealer**, under the FinSurv reporting categories [20]. Every cross-border payment is a balance-of-payments reporting event with a mandatory category code, and the specification references the ISO 20022 messages pacs.008 and pacs.009cov for interbank flows [20].

**Monthly.** The financial close. And the two returns that set the rhythm of the whole year:

- **BA 900**, the balance sheet of deposit-taking institutions, is due **within 15 business days after the month-end to which it relates**, with granular data by institutional sector, instrument, currency and maturity [21]. The suite of BA returns is specified by directive — currently D6/2025, effective 1 July 2025 — and each submission carries a signed, certified form BA 099, or BA 099A for foreign operations [22].
- **The deposit-insurance submission.** The Corporation for Deposit Insurance requires a monthly file **within 10 working days of month-end**, with a chief executive and chief financial officer declaration following **within three working days** of submission [23]. The file itself is four fields in a CSV. The calculation behind it is a per-depositor aggregation across every deposit product with prescribed treatment of joint accounts, sole proprietors, negative balances and pledges [24].

Also monthly: the Financial Intelligence Centre volume and value submission by the fifteenth for providers registered under Directive 2 of 2024 [11].

> The BA 900 counterparty-and-instrument matrix is effectively a mandated conformed dimension set for the bank's warehouse. A bank that cannot classify every balance to that matrix every month does not have a reporting problem. It has a data-architecture problem that happens to be visible monthly.

**Quarterly.** Capital and liquidity. Regulatory capital and risk-weighted assets, the leverage ratio, the liquidity coverage ratio and net stable funding ratio, large exposures, and Pillar 3 disclosures aligned to the latest Basel framework. The Prudential Authority implemented the remaining Basel III post-crisis reforms — credit risk under both standardised and internal ratings-based approaches, operational risk, the leverage ratio, credit valuation adjustment, market risk and the output floor — on **1 July 2025** [25].

**Annually.** The audited financial statements and the resubmission of annual BA returns after they are finalised, no later than 90 calendar days after year end, with the auditor's report due within 120 days [22]. The internal capital adequacy assessment. And, for providers registered under Directive 2 of 2024, biannual compliance attestations on 31 March and 30 September [11].

**Two traps in this calendar.**

The first is that the dates are not the hard part. The hard part is lineage. A supervisor's question is rarely "what is the number"; it is "show me how you got it". Returns assembled from point-in-time extracts that do not tie to the general ledger will produce a number that is right and indefensible, and the second condition is the one that fails an inspection.

The second is that a product change breaks returns. Launch a new deposit variant and four returns need a new classification, the warehouse needs a new dimension member, and the deposit-insurance aggregation needs to know whether the product qualifies. Banks that treat regulatory reporting as a downstream consumer discover this at submission time. Banks that treat it as a consumer with a veto discover it at design time, which is cheaper by an order of magnitude.

## Delegated and third-party arrangements

A modern bank does a great deal of its operating model through other people, and it remains accountable for all of it.

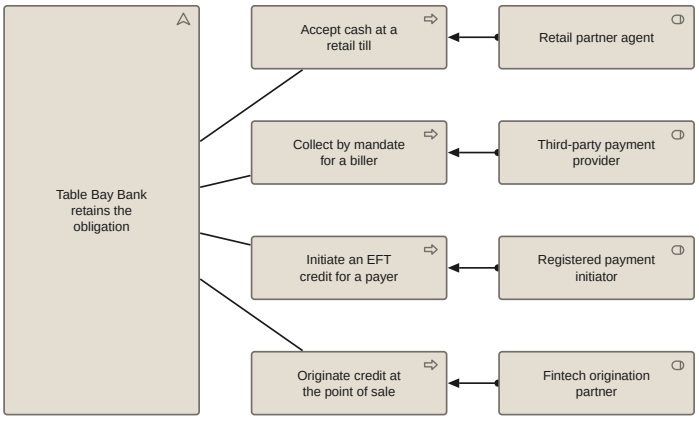



**Figure 7.6** Four delegated arrangements, each with a third party assigned to the work and the bank retaining the obligation. The association from the bank is the part that has to be evidenced: oversight, and the records the bank holds itself.

**Agents and retail partners.** Tills in supermarkets that take deposits and give cash back, kiosks that open accounts, retailers whose footprint substitutes for branches. The bank keeps the customer identification obligation, the cash controls and the reconciliation.

**Third-party payment providers.** A registered intermediary that collects from or pays to the bank's customers on a biller's behalf. The provider holds the commercial relationship with the biller; the bank holds the account, the mandate and the settlement. Existing third-party payment provider registrations granted under the old industry body remain valid until the Reserve Bank publishes its own authorisation framework, and pending applications are assessed on the previous criteria in the meantime [1].

**Payment initiators.** Directive No. 2 of 2024 created a registered category for parties that initiate EFT credit payment instructions on a payer's behalf, which is the regulated form of what the market calls screen-scraped payment initiation [11]. The architectural consequence is specific and has no British or European analogue, because there the same need is met by a mandated open-banking API: **traffic from a registered initiator is a known, named,**

**regulated flow against the bank's own internet banking channel, and the bank must be able to identify it, risk-manage it and tell it apart from fraud.** If your channel architecture cannot distinguish a registered initiator's session from an attacker's, that is a finding.

**Aggregators and fintech partnerships.** Origination partners who bring applications, lenders-as-a-service arrangements where the bank provides the balance sheet and the licence, card programme partners, and embedded finance. The split that matters is between who performs the work and who carries the obligation. The affordability assessment under the National Credit Act is the bank's if the credit is the bank's, regardless of whose screen the customer filled in.

**What governs it.** Here is a correction worth carrying, because the wrong instrument is widely cited. **Joint Standard 1 of 2024 is outsourcing by insurers and does not apply to banks.** For banks, the instruments are **Joint Standard 1 of 2023** on information technology governance and risk management, which commenced 15 November 2024, and **Joint Standard 2 of 2024** on cybersecurity and cyber resilience, which commenced 1 June 2025 with twelve months to full compliance [25]. Cloud and data offshoring are addressed in a Joint Communication issued in July 2025 [25]. There is also a Joint Standard 1 of 2025 on central counterparty and trade repository exemptions. Four instruments, three of them numbered "1", and getting the citation wrong in front of a compliance officer costs you the room.

**Four things the bank cannot delegate**, and they should appear as retained obligations in your model rather than as assumptions in a contract: the customer due diligence outcome; the regulatory report; the settlement obligation; and the complaint. On the last one, external escalation now runs to the National Financial Ombud Scheme, operational since 1 March 2024, which consolidated the banking, credit, long-term insurance and short-term insurance schemes into divisions [26]. An ombud referral arrives with a demand for the complaint file, and a bank whose complaint file is distributed across a partner's systems and three email accounts discovers the retained obligation the hard way.

## What the operating model means for architecture

Some boundaries in a bank are real. They are consequences of law, settlement mechanics, double-entry arithmetic or a clock the bank does not own. You can move work across them only by changing something external.

Some boundaries are accidents. They are consequences of an acquisition, a programme that ended, a vendor selection made in 2004, or an executive reorganisation. They look identical on an application landscape diagram, and telling them apart is most of the value an architect adds in the first ninety days.

| Real boundaries | Accidents of history |
|---|---|
| The legal entity — capital, licence, resolution, returns | One core per acquired book, nobody decommissioned |
| Settlement finality: a settled instruction cannot be recalled [7] | A separate card platform per scheme relationship |
| The mandate: who may instruct, on whose behalf, to what value | Three customer masters, no owner |
| The rail: its format, its cut-off, its clearing cycle [16][17] | A payments hub with three channel bypasses |
| Where the authoritative balance is, per product family | Available balance computed in three places |
| The period: a close, an accrual, a return date | Product logic in the channel because the core had a release freeze |
| The segment where the operating model genuinely differs — retail decision pipeline against business credit committee | The Rest-of-Africa estate as a separate landscape wearing the brand |

Four questions to ask about any proposed change. They are deliberately blunt, and between them they will catch most of what goes wrong.

**1. Which of the two businesses does this touch, and does it change a clock?** A change that moves work across the batch boundary, adds a dependency into the overnight critical path, or needs a decision inside a payment's latency budget is a different and larger change than the one on the slide. Ask for the cut-off it has to meet and who owns that cut-off.

**2. Where does the authority live, and is it enforced in the service or in the screen?** Mandates, limits, approval thresholds and segregation of duties are the controls a bank is inspected on. A control enforced in a user interface is not a control, because the host-to-host integration does not use the user interface. This single question finds more real defects than any other.

**3. What does it do to the returns?** Name the returns affected, the classification that changes and the dimension that needs a new member. If nobody in the room can answer, the regulatory reporting team will answer it later, at a worse time, with less budget.

**4. Who is accountable when it is somebody else's system?** For every delegated step, name the retained obligation, the evidence the bank holds itself, and how the bank would answer an ombud or a supervisor using only what it holds. If the answer involves asking the partner, the arrangement is not architected, it is hoped.

And four artefacts worth producing, because the four questions above are only answerable if somebody has written these down. None of them is a large document.

**A clock map.** One page listing every cut-off the bank's operations depend on, who owns it, what happens if it is missed and which systems are on the path to it. In most banks this exists only in the heads of the payments operations team, and it is the single most useful thing a new architect can assemble in week two.

**A mandate and authority register.** For each product and channel: who may instruct, up to what value, with what approval, and *where that is enforced*. Expect to find enforcement in three places with three different answers, and expect the gap between the authority people have and the authority the systems enforce to be a standing audit finding.

**A retained-obligation list for every delegated arrangement.** One row per arrangement: the partner, the work they perform, the obligation the bank keeps, and the evidence the bank holds itself. Short, politically awkward, and the thing a supervisor will ask for.

**A returns-to-systems map.** Which return draws on which system, at what granularity, with what classification. This is the artefact that turns "what does it do to the returns" from a rhetorical question into a five-minute check.

One closing note on sequencing, because operating-model work has a characteristic failure. The temptation is to draw the target operating model first, as an organisational picture, and derive the architecture from it. That produces a model of a reorganisation. Draw the current operating model instead — the clocks, the cut-offs, the mandates, the retained obligations, the returns — and the target emerges from the places where it demonstrably does not work. Those places are specific, they are few, and they are arguable with evidence, which is the only kind of target worth publishing.

#### THE CHAPTER IN TEN LINES

- A bank is two businesses sharing a licence: a slow, ledger-centric balance-sheet business and a fast, latency-bound payments business. Decide which one a system serves before anything else.
- The value chain is ten processes in five pairs: design and sell, open and serve, move money, lend, fund and account. Risk, data and compliance are not steps in it; they are properties of every step.
- Each channel brings its own latency budget, and a service shared by eight channels must meet the tightest one. The authorisation host usually sets the bank's non-functional requirements.
- Cost-to-serve per channel is quoted constantly and sourced almost never. Use the number the finance team will defend in writing, or use none.
- The business day is a sequence of windows with cut-offs owned outside the bank: DebiCheck in the early window, Registered Mandates at 19:00 with responses by 23:00 and settlement at 05:00, SAMOS with a night batch window.
- The authoritative EFT cut-off schedule is not published. Get it from payments operations or PayInc, and put the real times in your own design document.
- A late batch is not a one-day outage. It is a one-day outage plus a reconciliation tail, and the tail reaches collections, statements, the warehouse and the returns.
- Real time does not replace the overnight cycle. It runs alongside it, and the transitional state is permanent.
- Regulatory reporting is a production line with attested output: BA 900 within 15 business days, the deposit-insurance file within 10 working days, FinSurv daily for non-resident card activity. Lineage, not the number, is what fails an inspection.
- Delegation moves the work, never the obligation. For every arrangement, name the retained obligation and the evidence the bank holds itself.
- Some boundaries are real — legal entity, settlement finality, the mandate, the rail, the period. Some are accidents of acquisition history. Telling them apart is most of the value an architect adds in the first ninety days.

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CHAPTER 8

# The process model

*A process architecture is the inventory of what a bank does; a process model is the inside of one of them. Here are both — twelve end-to-end processes, and thirteen of them drawn in BPMN with the roles in lanes, because the question that matters is always who is holding the work right now.*

## The folder with 600 pages in it

Ask a bank for its process model and you will be sent a folder. In the invented Table Bay Bank the folder runs to about 600 pages, was last edited for a core upgrade three years ago, and is mostly screenshots of the collections system with the fields annotated. It describes, in forensic detail, how to key a promise-to-pay. It does not describe how an unpaid debit order becomes a collections call, which is the step that takes two days and loses the money.

That is the normal starting position, and it is a particular kind of failure rather than laziness. The bank documented its procedures and called the result a process architecture. Nobody ever wrote down the list of its processes.

> A **process architecture** is the inventory: which end-to-end processes exist, what triggers each one, what outcome ends it, who owns it, and how they fit together. A **process model** is the inside of one of them: steps, roles, decisions, data, hand-offs. The first is a map of the country. The second is a street plan of one town. A thousand street plans still will not tell you how many towns there are.

An architect needs both, and for reasons that have nothing to do with documentation.

**A process is where a capability turns into work with a cost and a clock.** "Payments" is a capability. It has no cycle time. "Execute a payment by EFT credit" is a process: it has a cut-off, a repair rate, a suspense balance and a cost per item, and every one of those numbers is negotiable. Capabilities are for deciding what matters. Processes are for deciding what to change.

**Processes are the only artefact the business will argue with.** No head of operations has ever disputed a capability map, because nothing is at stake in one. Put a swimlane diagram of the collection run on the wall and the collections manager will dispute it inside ninety seconds, which is the point of drawing it. A model that provokes no argument has told you nothing.

**A process model is the one architecture artefact that forces you to name who decides.** Application landscapes hide authority. Data models hide it completely. The moment you draw lanes you have to answer: who may release a repaired payment, who may sanction R20 million, who may write off, who may grant provisional credit on a disputed card transaction. In a bank those answers matter more than in most industries, because the gap between the authority a person has and the authority the system enforces is the standing audit finding of the sector.

**Systems are bought and replaced along process boundaries.** When a bank buys a collections platform it is buying a span of process. If the inventory says collections ends at write-off, then post-write-off recovery, agency placement and prescription monitoring fall outside the procurement, and they will be in a spreadsheet for the next decade. A wrong boundary at L1 becomes a wrong contract at signature.

### Why this chapter leaves ArchiMate

Everything else in this book is drawn in ArchiMate 4, and that is the right choice for relating a process to a capability, an actor, a business object and an application service. It is a poor notation for *sequence*. ArchiMate has no gateway. Its triggering relationship carries no condition. It cannot show a hand-over between two performers without a convention you have to teach the room before you can use it. And it has no swimlanes, which is fatal here, because in a bank the hand-over is usually the defect.

BPMN 2.0 [1] is a sequence notation with a lane per performer. It answers the operations manager's question — who is holding this work right now, and what are they waiting for — and it answers the architect's question — where does the work cross an organisational or system boundary, because that is where it stops. This chapter is drawn in BPMN, and it is the only chapter in the book that is.

Being plain about why matters, because the two notations have to stay reconcilable. ArchiMate models the process as an element *in an architecture*: one box, named for what it achieves, with a role assigned to it, a capability it realises, the objects it accesses and the applications that serve it. BPMN models what happens *inside* that box.

> THE MAPPING BACK, AND IT IS MECHANICAL
>
> Each **pool** in this chapter is one **Process** element in the ArchiMate model, and it is one of the twelve in the L1 map below. Each **lane that is a person** is a **Role**, related to that Process by **Assignment**. Each **lane that is a system** is an **Application Component**, related to the Process by **Serving**. Each **data store or data object** in a figure is a **Business Object**, related by **Access**. Keep the names identical in both notations and the models stay reconcilable; let them drift and you have two process architectures and a meeting about which one is true.

Two practical consequences. First, do not model in BPMN what you can state in ArchiMate: if the question is "which applications serve this process", draw it once in ArchiMate and stop. Second, do not let the BPMN become the system of record for structure. The pool name, the lane list and the data stores belong in the repository as ArchiMate elements; the BPMN is the sequence view of one of them.

The lanes in this chapter use the job titles from Chapter 5. A reminder of the ones that recur, because an architect who cannot name the jobs cannot run the workshop:

| Role in the lanes | What the person actually does all day |
|---|---|
| **Onboarding or KYC officer** | Works a queue of incomplete customer files, decides whether a document is acceptable, and chases the ones that are not |
| **Payments operations** | Watches files and rails all day, works the exception queue, and decides whether today's payments go out |
| **Collections agent** | Phones people in arrears, negotiates arrangements, records promises, and works a dialler list somebody else designed |
| **Credit analyst** | Spreads financials, writes the credit paper, and spends most of the day gathering data |
| **Sanctioner** | Approves or declines credit within a personal mandate, and is personally accountable for it |
| **Disputes clerk** | Classifies a complaint as fraud, merchant dispute or processing error, and starts a scheme clock they cannot stop |
| **Treasury operations** | Confirms, settles, chases nostro breaks, and maintains settlement instructions |
| **Regulatory reporting analyst** | Produces returns from data they do not own, and signs for their accuracy |

## L0, L1, L2, L3 — and the warning

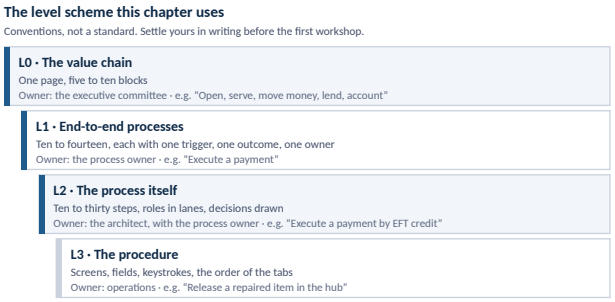



**Figure 8.1** The level scheme this chapter uses. They are conventions, not a standard — APQC numbers its levels differently and BIAN does not use process levels at all. Settle yours in writing before the first workshop.

Those are the definitions this chapter uses. They are conventions. They are not authoritative, and nobody else's are either.

APQC's cross-industry process classification numbers its levels category, process group, process, activity and task, which makes "level 2" a process *group* and shifts everything in this chapter down a number [2]. BIAN does not use process levels at all: its Service Landscape organises 322 Service Domains into 38 Business Domains, and BIAN's own description of the Landscape is "a reference structure that categorizes and organizes BIAN Service Domains for ease of access", adding that different classification criteria produce different layouts of the same set [3]. That is a service decomposition, not a process decomposition, and mapping one to the other is a real piece of work rather than a lookup. In many banks "L1" means the one-page value chain, which this chapter calls L0.

> **SETTLE THE LEVELS IN WRITING BEFORE THE FIRST WORKSHOP**
>
> This sounds like pedantry. It is the most expensive piece of pedantry you will skip. A programme with two definitions of "level 2" produces two populations of models at different grain, discovers the mismatch at integration, and reconciles by hand at the worst possible moment. Write the definitions on one page, get the process owners to initial it, and put it in the modelling conventions. Half an hour, week one.

Two more things about levels before the map.

**The levels are not a decomposition of equal depth, and should not be.** It is normal and correct for a bank to have twelve L1 processes, eight modelled at L2 because they carry the volume or the pain, two at L3 because they are being automated this year, and two left at L1 because nothing is changing there. A programme that insists on uniform depth is producing documents, not architecture.

**Below L2, stop.** L3 and L4 belong to operations. They change weekly, they are tied to a screen layout, and an architecture repository that holds them is a repository that is wrong within a month. Model L1 and L2, point at the operational procedures, and do not take ownership of them.

## The L1 process map

Here is a defensible L1 inventory for a full-service bank. Twelve processes in four groups.

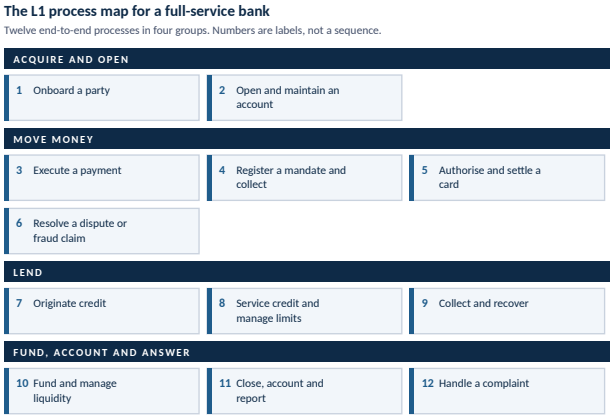



**Figure 8.2** An L1 process inventory for a full-service bank: twelve end-to-end processes grouped by what they are for rather than by which department performs them. Each one maps to a single ArchiMate Process element.

And here is what makes it an architecture rather than a list: each one has a trigger, an outcome, an owner, and an honest note.

| # | Process | Trigger | Outcome | Honest about it |
|---|---|---|---|---|
| 1 | Onboard a party | An application, or a refresh falling due | A party record that is identified, screened, risk-rated and usable | Half triggered process, half **cycle** — periodic refresh is the half that is never modelled |
| 2 | Open and maintain an account | An approved party wants a product | An account with a product instance, a mandate, limits and entitlements | The mandate is the part that matters and the part held at the wrong level |
| 3 | Execute a payment | A customer or system instructs | Funds with the beneficiary, ledger posted, status answerable | One process, six rails, and the rail determines almost everything |
| 4 | Register a mandate and run a collection | A contract, then a billing date | A mandate at the debtor's bank, and a collection that succeeds or fails with a reason | Two processes sharing a name. Split them, as this chapter does |
| 5 | Authorise, clear and settle a card transaction | A card is presented or a card-not-present transaction submitted | Approve or decline in under a second; money days later | The only process in the bank measured at the 99th percentile in milliseconds |
| 6 | Resolve a dispute, chargeback or fraud claim | A customer says a transaction is wrong | Liability allocated and the ledger reflecting it | Runs on somebody else's clock: the scheme's |
| 7 | Originate credit | An application, or a facility request | Money disbursed under a compliant agreement, or a reasoned decline | Two different processes — retail and business — behind one name |
| 8 | Service credit and manage limits | A scheduled review, a behavioural trigger, a customer request | A limit confirmed, raised, cut or withdrawn, with a reason | A **cycle**. Annual review in business banking; continuous in retail |
| 9 | Collect and recover | A missed payment; later, a cure failure | Cure, arrangement, legal process, or write-off and post-write-off recovery | Also two processes. Recovery outlives the relationship by years |
| 10 | Fund the balance sheet and manage liquidity | Continuous | A funded, hedged balance sheet inside appetite | A **cycle** with a committee, not a triggered flow |
| 11 | Close, account and report | A date | A closed ledger, submitted returns, signed attestations | A **production schedule**. Drawing it as a swimlane gives you a Gantt chart with lanes |
| 12 | Handle a complaint | An expression of dissatisfaction, anywhere | A resolution, a decision letter, a recorded root cause | Regulated, externally escalable, and owned by nobody senior enough [4] |

Four rules make an inventory like this usable.

**One trigger, one outcome, one owner.** A candidate with two triggers is two processes. Items 4, 7 and 9 above fail that test, and the table says so out loud rather than hiding it. Most banks keep each pair together because one department does both, which is an organisational fact rather than a process fact.

**Nothing between the trigger and the outcome goes on the list.** "Screen for sanctions" is a step inside several processes, not a thirteenth process. This rule is what keeps an inventory at twelve items instead of forty.

**Cross-cutting work is a service, not a process.** Sanctions and fraud screening, document production, party matching, rating, notification and payment posting are consumed by several processes and owned by none of them. Put them on the application and service side of the model. A process architecture that lists "customer communication" as an L1 process has confused a shared capability with an end-to-end flow, and will then be unable to say which process owns the outcome.

**Be honest about which items are not processes.** Four of the twelve are cycles or production schedules rather than triggered flows: the refresh half of 1, the review cycle in 8, treasury's continuous cycle in 10, and the calendar-driven close in 11. Leave them in the inventory, because removing them makes the business think you have missed the work. Mark them, because trying to draw a swimlane model of "close, account and report" produces something that is neither a process model nor a plan.

THE FOUR PROCESSES ALMOST ALWAYS MISSING FROM A BANK'S OWN INVENTORY

Ask for the bank's own list and compare. Four things are nearly always absent, and each is absent for the same reason — the owner is too junior to have a seat at the architecture table. **Mandate management.** The DebiCheck or Registered Mandates mandate is a long-lived object with a lifecycle, held at the *debtor's* bank, and in most inventories it appears only as a step inside collections. **Dispute and chargeback.** Usually filed under "card operations", which is a department, not a process. **Recoveries after write-off.** The exposure leaves the book, so it leaves the process architecture, and then runs for five more years in a spreadsheet with an attorney. **Complaints.** Treated as a service-quality activity rather than a regulated process with an external escalation path to the National Financial Ombud Scheme, operational since 1 March 2024 and consolidating four previous schemes into divisions including Banking and Credit [4]. Add a fifth if the bank has partners: **third-party and agent onboarding and oversight**, which is somebody's contract folder and nobody's process.

What follows models nine of the twelve at L2, in thirteen figures. Three processes are left at L1 deliberately: service credit and manage limits, fund the balance sheet, and handle a complaint. The first two are cycles, and the third would be identical in any industry. That is the honest shape of a real process architecture, and it is a better use of two months than thirteen uniform models.

## 1. Onboard a party and complete FICA

A customer walks into a branch or opens the app and wants an account. In South Africa this is the hardest step in retail acquisition, and it is hard for a statutory reason rather than a technical one.

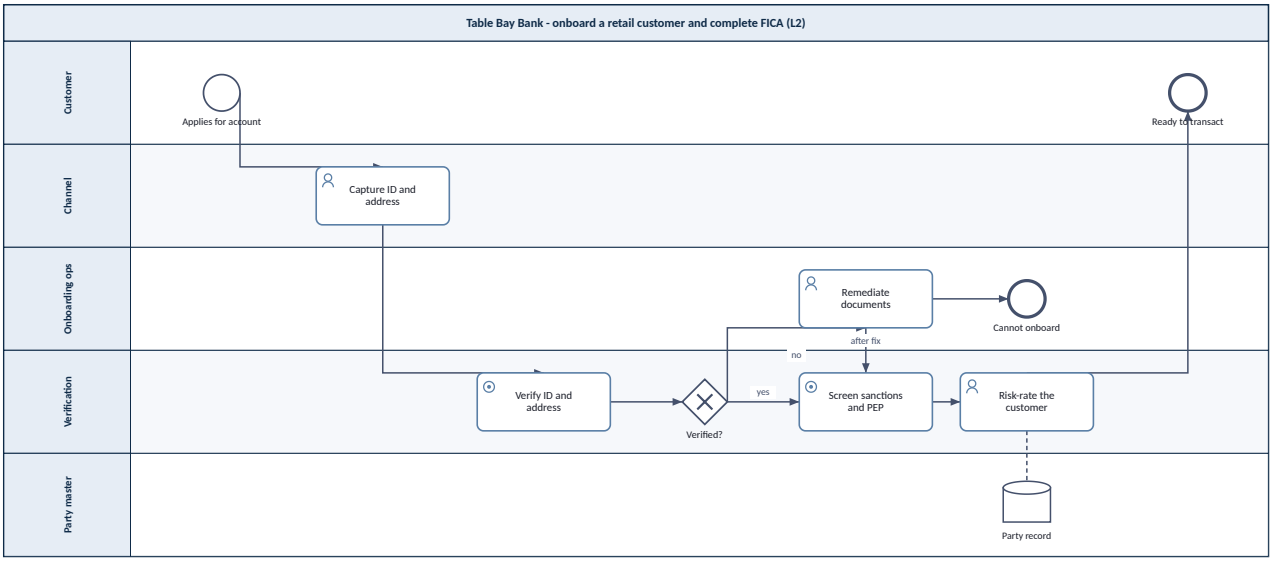



**Figure 8.3** Onboard a retail customer and complete FICA. The load-bearing element is the verification gateway: under section 21E a customer whose address cannot be verified may not be onboarded at all, so the "no" branch is a legal stop and the bank needs a third state for "not yet".

**The decision points.** Only one gateway is drawn, and it is the one that matters: *is the customer verified?* Under section 21E of the Financial Intelligence Centre Act an institution that cannot verify a client's residential address may not establish the relationship at all [5]. So the "no" branch is not an inconvenience, it is a legal stop, and the honest version of the gateway has three exits rather than two — verified, not verified, and *not verified yet*, which is where the whole population of partially onboarded customers lives. Ask what the system does with the third case. If the answer is that there is no third state, the bank is either turning away customers it could keep or opening accounts it should not.

**Data created or changed.** Party and identifiers; addresses and the evidence for them; for a juristic person the beneficial-ownership chain to the ultimate natural persons; customer risk rating; consent records; and the due-diligence evidence documents themselves. The evidence is the part that goes wrong: it is captured, and then it is not indexed to the party, so it exists and cannot be found.

**Systems and measures.** Channel, party master, content management, sanctions and politically-exposed-person screening, application fraud, the bureau and the core. Measure time-to-onboard, straight-through rate, drop-off by step, the count of accounts with deficient customer files, and the screening false-positive rate. The last one is a cost, not a quality metric: every false positive is an analyst-hour.

**Where it goes wrong.** Each channel captures a different data set, so the party record's completeness depends on where the customer walked in. Risk rating is set once at onboarding and never recomputed, which is the opposite of what a risk rating is for. Juristic-person onboarding is built as a bigger version of retail onboarding when it is a different process with an ownership graph at its centre. And the remediation backlog outlives three programmes, because nobody owns the refresh cycle — which is the half of process 1 that the inventory above flagged.

## 2. Open and maintain an account

Onboarding produced a party. This process produces something the customer can transact on.

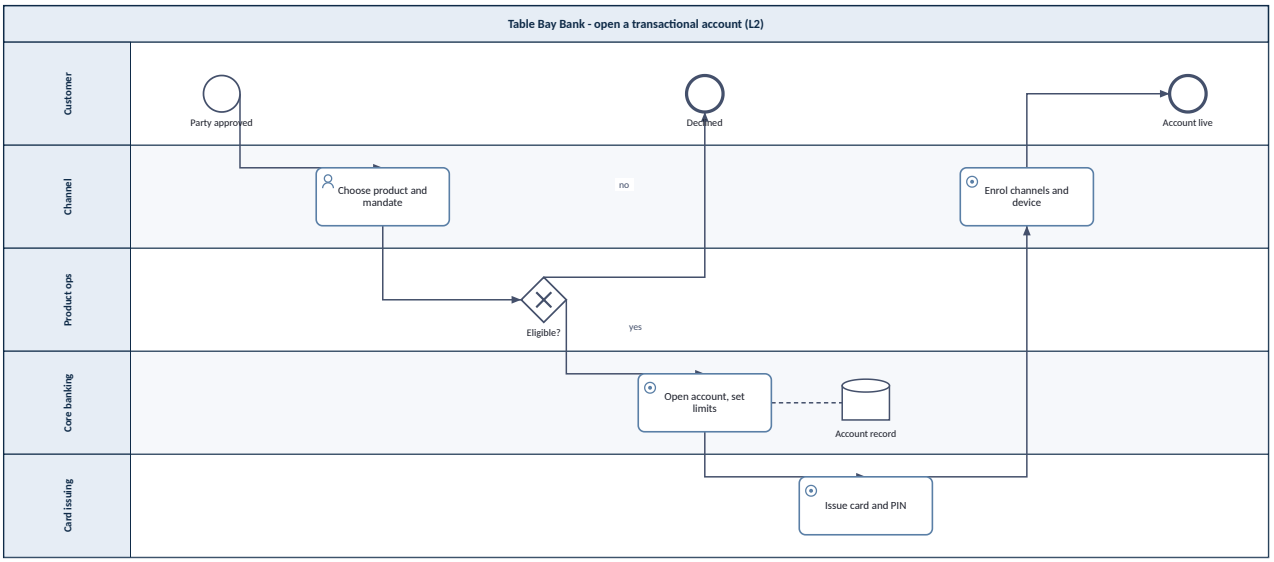



**Figure 8.4** Open a transactional account. The mandate captured in the first task is the object most often held at the wrong level; the deposit-insurance single customer view is updated here or it is not reliable at all.

**The decision points.** *Is the customer eligible for this product?* — a question about product rules, regulatory status and sometimes sanctions. In most banks it is also, silently, a question about which system the product lives in.

**Data created or changed.** The account and its product instance; the signing mandate; limits; fee structure; channel entitlements; and the card record if one is bundled. One omission from the figure is deliberate and worth naming: the deposit-insurance single customer view, which must aggregate each qualifying depositor's balances across every qualifying product under prescribed rules for joint accounts, sole proprietors, pledges and negative balances [6][7]. If that view is built as a month-end query rather than updated here, it cannot be produced at short notice, which is its entire purpose.

**Systems and measures.** Channel, core banking, card issuing, customer communication, identity and access, and the deposit-insurance feed. Measure accounts opened per channel, time to first transaction, abandoned applications, and the zero-balance rate at ninety days — which is the only honest measure of whether an account opening was worth anything.

**Where it goes wrong.** Onboarding and account opening are collapsed into one digital flow that cannot be resumed, so a customer interrupted at step six starts at step one. The mandate is recorded at account level rather than party level, so changing a signatory means repeating the change once per account and getting it wrong on the fourth. The bundled card's lifecycle is disconnected from the account's, so a closed account keeps a live card. And fee structure is captured here but computed somewhere else, which is how two systems come to disagree about what the customer pays.

## 3. Execute a payment by EFT credit

The rail determines the process, not the other way round. This figure is the batch push rail, which by value is still the dominant retail payment instrument in South Africa — 70.3% of retail settlement value in 2024, against 0.7% for rapid payments [8].

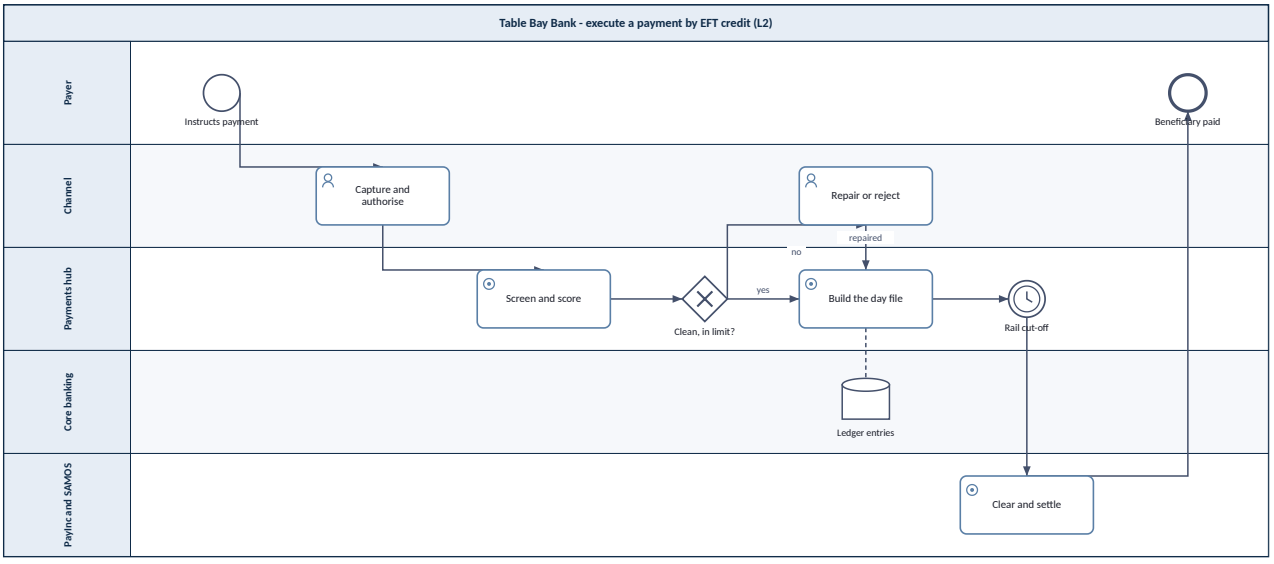



**Figure 8.5** Execute a payment on the batch EFT credit rail. One gateway stands for three separate checks with three owners and three queues — undo that simplification in the bank's own model.

**The decision points.** *Is it clean and within limit?* — one gateway standing for three checks that are often three systems: sanctions screening, fraud scoring, and the payer's mandate and limit. Each has its own answer, its own false-positive rate and its own queue. Drawing them as one gateway is a simplification you should undo in the bank's own model, because the three queues have different owners and the hand-over between them is where payments age.

**Data created or changed.** The payment instruction and its full status history; the approval trail; the beneficiary record and its verification state; the rail reference that lets a customer query be answered; and the ledger entries. For a cross-border payment, add the balance-of-payments category code, because every cross-border payment is a reporting event to the Reserve Bank's Financial Surveillance Department [9].

**Systems and measures.** Channel, payments hub, screening, fraud, core, the rail interface, and the general ledger. Measure straight-through rate, repair and reject rate, time to beneficiary by rail, cut-off breaches, suspense age and value, and screening hit rate. Suspense age is the one nobody volunteers: ask for the value and the age of the oldest item, and you will learn more about the payments estate in five minutes than from a week of interviews.

**Where it goes wrong.** Channels that bypass the hub, so three paths reach a rail and only one is controlled. Approval enforced in the user interface rather than the service, so the host-to-host integration walks straight past segregation of duties. Duplicate detection scoped per channel instead of globally. Status inferred from file acknowledgements rather than modelled, so "where is my payment" cannot be answered from one place. And the cut-off enforced by a person watching a clock.

## 4. Register a mandate: the DebiCheck cycle

This is the most South-African process in the book, and the one that imported reference material will model wrongly. There is no European or British equivalent. Under a SEPA direct debit the mandate is held creditor-side; under a British direct debit the paying bank holds it but does not validate each collection against electronically authenticated terms. **DebiCheck mandates are held at the debtor's bank, and that bank validates every collection against the agreed terms before processing it** [10][11].

Two figures, because the mandate and the collection are two processes that share a name. This one is the mandate.

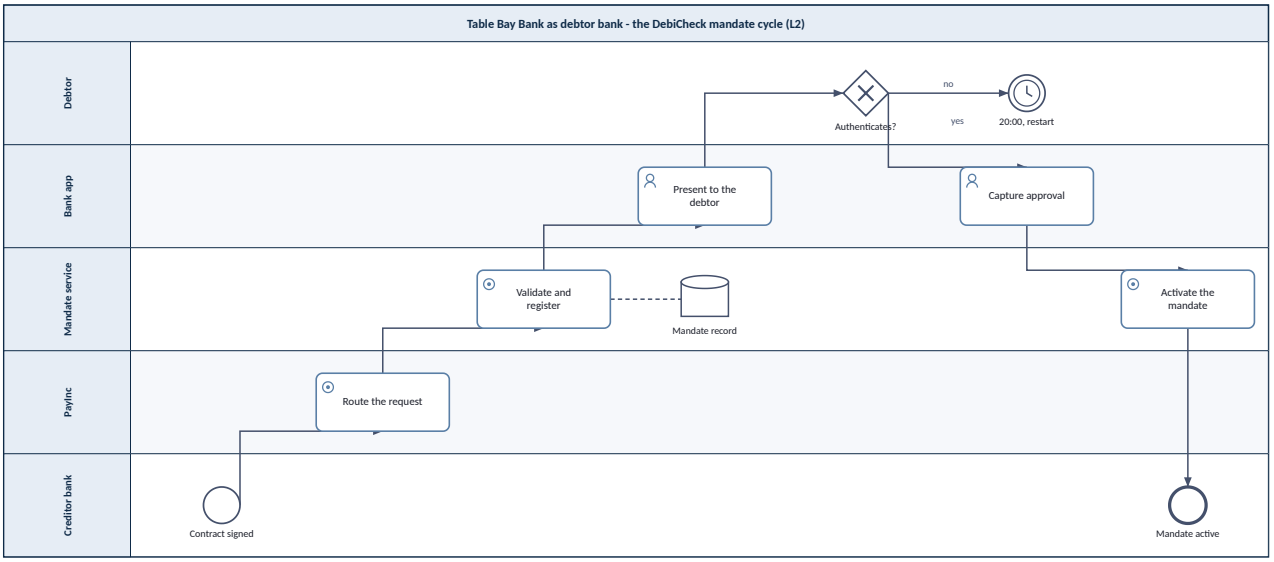



**Figure 8.6** The DebiCheck mandate cycle, seen from the debtor's bank, which is where the mandate lives. The authentication window is the architecture: TT1 closes at 20:00 the same day and the request restarts.

**The decision points.** *Does the debtor authenticate, in time?* The time is the architecture. A TT1 request is a real-time request to the debtor's bank, authenticated in the banking app, at a point of sale or at certain ATMs, and authentication is possible until 20:00 the same day; miss it and the whole request restarts. A TT2 request goes overnight in a batch file and gives the customer two days. A TT3 is authenticated card-present, with the card inserted and the PIN entered on an enabled device, and is authorised during the original transaction with no window at all [10][11].

**Data created or changed.** The mandate, with its type, its agreed terms — amount, frequency, first and last collection date, adjustment rules — and its status. Then the amendments, suspensions and cancellations, each of which is a state transition rather than a new mandate. A Registered Mandate can be upgraded to a DebiCheck mandate by an amendment authenticated by both participating banks [12]: model that as a transition, not a re-registration, or the history breaks.

**Systems and measures.** Billing or servicing, mandate management, the bank and PayInc interfaces, the core, and disputes. Measure authentication success rate by transaction type, the authorisation rate, and mandate-to-billing divergence. Across DebiCheck and the old Registered Mandate Service, mandate initiations in 2024 were 58.8 million and 9.5 million respectively, technical success was 98% and the average authorisation rate 62% [8] — a 38-point gap between technically working and actually authorised, which is the gap the architecture has to explain.

**Where it goes wrong.** Mandate terms diverge from billing after a rate change, so collections start failing validation silently and nobody connects the two. TT1 is used where TT3 was available: one vendor claims 35–40% of TT1 attempts fail monthly because customers miss the notification or cannot reach the app in time, against better than 99.5% for TT3 — a vendor figure, published by a participant, and worth testing rather than repeating [13]. And the creditor side holds no mastered view of its own mandates, because the mandate lives at the other bank.

## 5. Run a Registered Mandates collection

Now the collection. **Registered Mandates is a new rail, not a renaming.** The shift from the old Registered Mandate Service was mandated and took effect on **10 March 2025**; from that date, collections against those mandates can no longer be submitted on DebiCheck and must go through RM [12]. Four things changed that a 2024 design will have wrong.

| | Registered Mandate Service, before | Registered Mandates, from 10 March 2025 [12] |
|---|---|---|
| Collection window | Early | **Late window, 19:00 SAST** |
| Tracking | — | 12:00 to 00:00, up to **ten tracking days** |
| Response deadline | — | **23:00** on the action date |
| Settlement | — | **05:00** the day after the action date |
| Priority | Within DebiCheck | After DebiCheck, **before EFT debits** |
| Accounts | — | Business and **dual-signatory**, which DebiCheck cannot support |
| Data standard | — | **ISO 20022** |

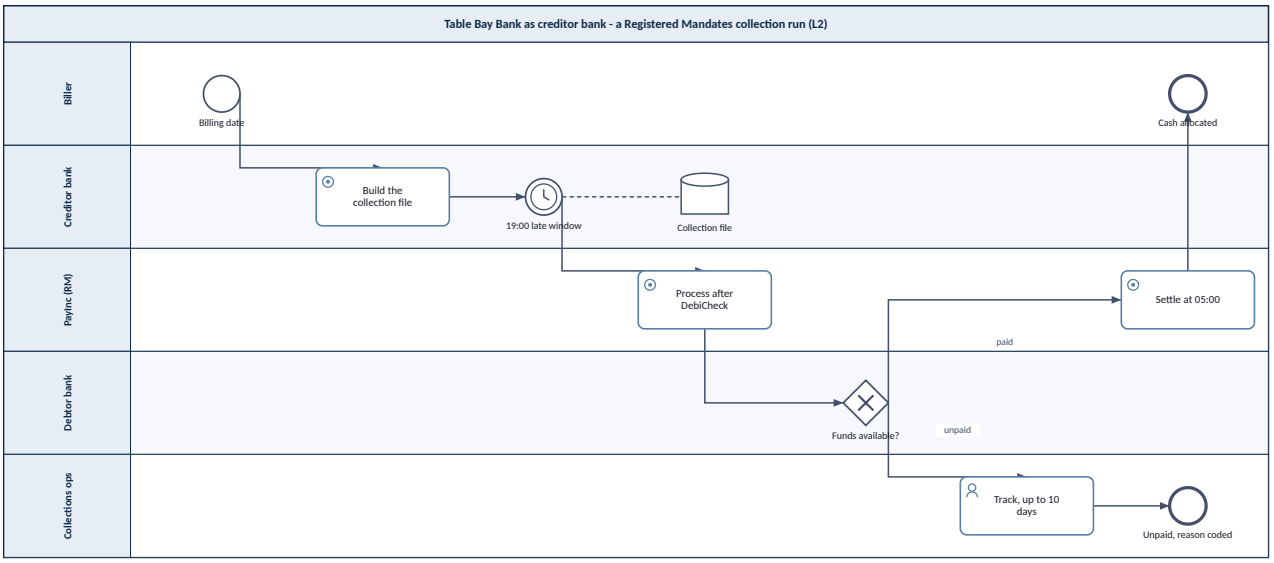



**Figure 8.7** A Registered Mandates collection run. RM is a new ISO 20022 rail mandated from 10 March 2025, not a renaming: the late 19:00 window, the tracking window and the 05:00 settlement are its own.

**The decision points.** *Are the funds there on the action date?* That is the only real gateway, and everything interesting is what happens on the "no" branch — tracking, which re-presents the collection for up to ten days inside a daily window.

**Data created or changed.** The collection instruction, its outcome and reason code, the tracking state and its remaining days, the settlement record, and any dispute. Note that RM collections are fully disputable [12], so the outcome is not final when the money arrives.

**Systems and measures.** The biller's billing system, the creditor bank's collection build, PayInc, the debtor bank and the collections operation. Measure collection success rate — 79% across DebiCheck and the old service in each of 2022, 2023 and 2024 [8] — the unpaid reason-code distribution, tracking-day yield, and the dispute rate.

**Where it goes wrong.** Three failures, all expensive. The unpaid reason codes are not mapped to differentiated retry action, so one strategy is applied to twelve different causes and the tracking window is wasted on accounts that were never going to pay. The move from an early to a 19:00 late window changed cash-flow timing and therefore the success-rate model of every collections business that migrated, and some of them are still running pre-2025 assumptions. And a creditor system built around fixed-width Bankserv record layouts needs a genuine remodelling rather than a mapping to carry ISO 20022, because the old format cannot hold the structured originator and recipient identification the new rail expects.

## 6. Authorise a card transaction

The fastest process in the bank, and the only one whose service level is quoted at the 99th percentile in milliseconds.

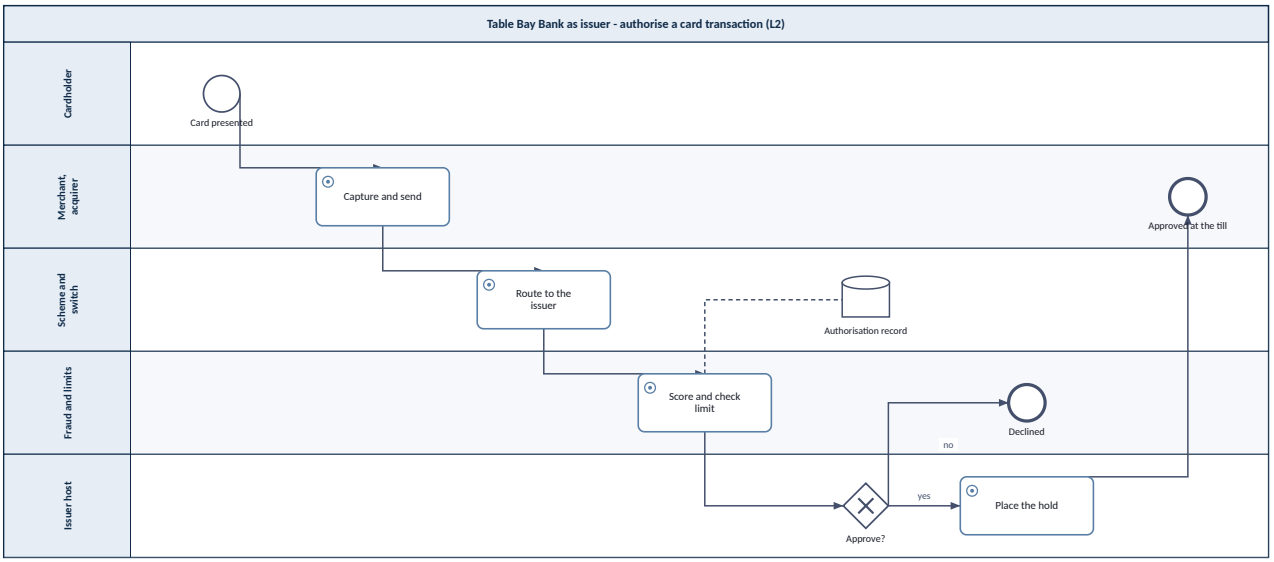



**Figure 8.8** Authorise a card transaction. Everything to the right of the gateway happens in under a second; clearing and settlement happen days later and are a separate matching problem.

**The decision points.** *Approve or decline?* — a composite of the cryptogram or PIN check, the three-domain-secure authentication result for card-not-present, velocity and fraud rules, and the available balance. The architecturally interesting part is what happens when the issuer host is unreachable: the switch stands in and decides on its behalf, against limits that somebody set and frequently nobody has reviewed since.

**Data created or changed.** The authorisation record; a hold on the available balance; the token-to-primary-account-number mapping; the authentication result; and, days later, a clearing record and a settlement entry with interchange and scheme fees. Authorisation and clearing are different events about the same purchase, and matching them is a whole function.

**Systems and measures.** Terminal or gateway, acquirer, switch, scheme, issuer authorisation host, fraud engine, core and general ledger. Measure approval rate, decline reasons, latency at the 99th percentile, stand-in volume, unmatched authorisation-to-clearing count, fraud in basis points, and the chargeback ratio against the schemes' monitoring thresholds. The card-not-present channel carries its own compliance clock: the PCI DSS v4 e-commerce script-integrity requirements became effective on 31 March 2025 [14].

**Where it goes wrong.** Expired authorisations are never released, so the available balance understates for days and the customer is declined on their own money. Authorisation-to-clearing mismatches are cleared by hand. Fraud rules and authorisation rules are owned by different teams with no combined view of why a customer was declined. And the three implementations of available balance from the core chapter meet here, where the difference between them is visible at a till.

## 7. Resolve a dispute and run a chargeback

The clock in this process belongs to the card scheme, and it does not stop for the bank's queue.

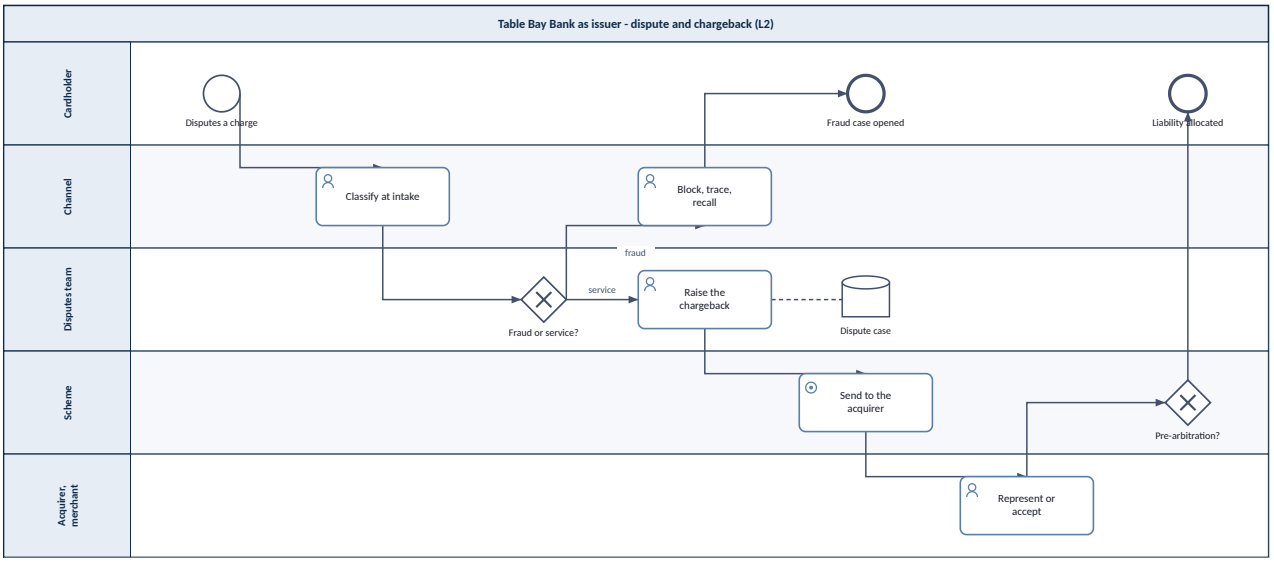



**Figure 8.9** Dispute and chargeback at the issuer. The classification at intake decides the outcome, and it is made by the least-trained channel in the bank against a clock the scheme owns.

**The decision points.** Two, and the first one decides the outcome. *Fraud, merchant dispute or processing error?* — three different rule sets, three different time limits, three different liability outcomes, and the classification is made at first contact by the least-trained channel in the bank. Then, much later, *pre-arbitration or accept?*

**The clock, because it is the process.** Vendor-published guidance on Visa's limits puts cardholder filing at generally 120 days from the transaction or expected delivery date, with 75 days for some reason codes and a 540-day outer limit for non-receipt; 30 days per phase for a merchant response, which acquirer and processor cut-offs often compress to five or ten days in practice; pre-arbitration within 30 days; arbitration filing within ten days of the pre-arbitration response; and a full dispute typically 45 to 90 days [15]. That is a vendor source and should be checked against the scheme's own dispute management guidelines before anything is built on it — but the shape is right, and the shape is what the architecture has to model.

**Data created or changed.** The dispute case, reason code, provisional credit, evidence documents, the representment, pre-arbitration and arbitration records, the final liability allocation and the ledger entries.

**Where it goes wrong.** Deadlines tracked in a spreadsheet, so an expired dispute becomes an absorbed loss. Evidence scattered across three stores and an inbox. Provisional credit granted and never recovered. No feedback from confirmed fraud outcomes back into the fraud engine, so the bank pays twice for the same lesson. And the one that is specifically South African: authorised push-payment and social-engineering fraud is not a chargeback at all, because the customer authorised it, yet the ombud has awarded refunds in such cases and criticised banks for responding slowly once aware [4]. Build the fraud path to move at the speed of a freeze, not the speed of a back-office queue.

## 8. Retail credit, application to decision

Retail lending is a decision pipeline. It does not fit one figure, and the split is instructive rather than a drafting compromise: **the decision and the disbursement are different processes with different participants, different clocks and different regulators.** Everything up to the decision happens inside the bank in minutes. Everything after it involves the customer, statutory documents and, for secured lending, attorneys and a deeds office, and takes weeks. Splitting at the credit decision is the natural seam, and if your model does not split there it is probably hiding the second half.

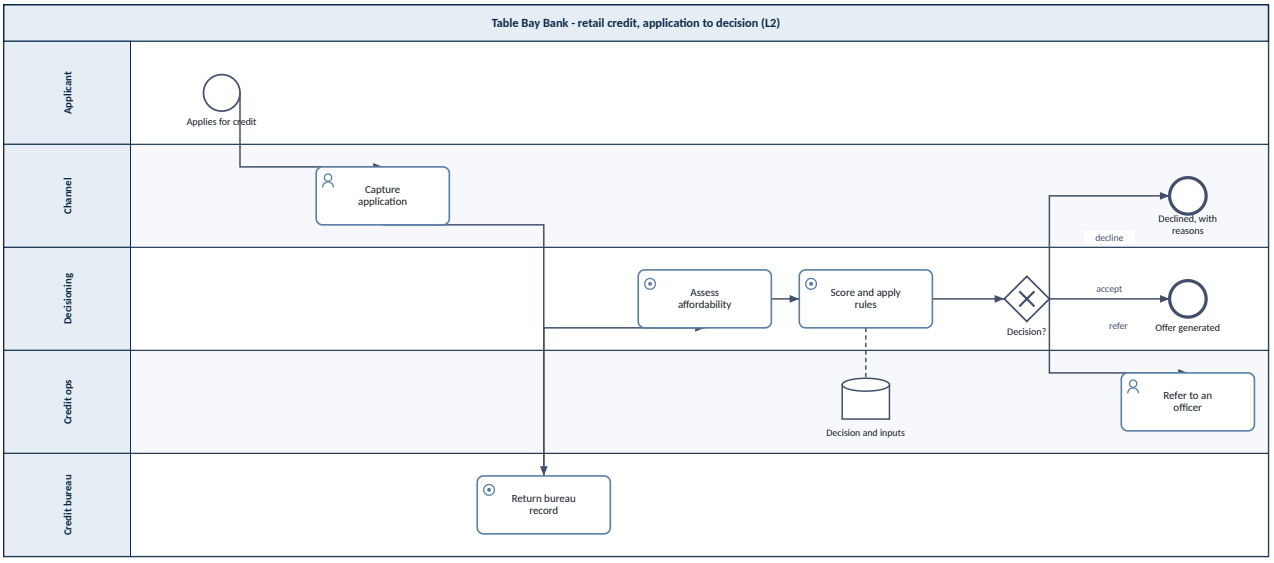



**Figure 8.10** Retail credit, application to decision. Split from the next figure at the credit decision because everything before it happens inside the bank in minutes and everything after it involves external parties and weeks.

**The decision points.** One gateway with three exits — accept, decline, refer — and the referral exit is the one architects skip. Ask what proportion of applications it takes and why, because referral rate and referral reason distribution are the only honest measure of whether the policy rules express the bank's actual appetite.

**Data created or changed.** The application; the bureau record; the affordability assessment and every input to it; the score, the strategy and model versions that produced the decision; and the decision reasons. Affordability is a *prescribed calculation* under Regulation 23A: gross income, less statutory deductions, less a prescribed minimum-expense-norms table, less disclosed commitments, with income validated by three payslips or three months of bank statements, and the bureau record considered within seven business days of approval — fourteen for a mortgage [16]. Those windows are pipeline timers, and a design that treats them as guidance will build a process that quietly breaks them.

**Where it goes wrong.** The affordability calculation lives in a spreadsheet owned by a credit analyst. The exact input vector is not retained, so the decision cannot be reproduced — and a reckless-lending challenge years later has to be answered against the rules as they stood on the decision date. Three channels run three scorecards. Policy rules are duplicated between the decision engine, the origination front end and the core. And declined-then-approved-elsewhere is never measured, so the bank never learns what its cut-off costs it.

## 9. Retail credit, offer to disbursement

The second half, and the half that is externally blocked.

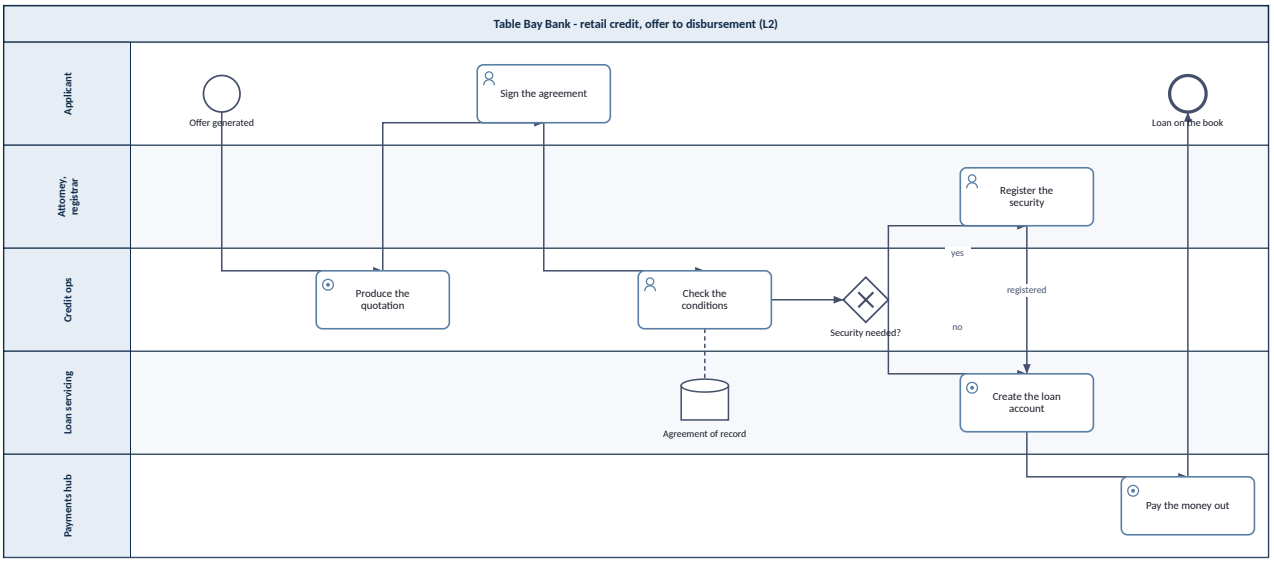



**Figure 8.11** Retail credit, offer to disbursement. The second half is externally blocked: for secured lending the bank holds a priced commitment open while attorneys and a deeds office run a process it does not control.

**The decision points.** *Are the conditions precedent met, and is security needed?* For an unsecured personal loan the answer is no and the money moves the same day. For a mortgage the answer sends the case to attorneys and a deeds office, and the bank must hold a priced commitment open while an external process it does not control runs for weeks — including across a repo rate change.

**Data created or changed.** The pre-agreement statement and quotation, which are statutory artefacts under the National Credit Act and must be reproducible years later; the signed agreement; the conditions-precedent register; the security registration; the loan account; and the disbursement instruction.

**Systems and measures.** Origination, document generation, collateral and security management, the loan servicing platform, and the payments hub. Measure time from offer to disbursement, the not-taken-up rate and its reasons, conditions-precedent ageing, and the proportion of cases where the rate at disbursement differs from the rate quoted.

**Where it goes wrong.** Origination is built as a transaction rather than a long-running case, so state does not survive weeks and external parties are not first-class integrations with their own service levels. Not-taken-up cases are never tracked, so conversion leakage is invisible. Origination data is not carried into servicing, so the loan arrives on the book without the risk grade it was priced on — which is precisely the data an impairment model needs for the life of the loan.

## 10. Business credit, and the mandate ladder

Business lending is not a bigger version of retail lending. It is a different process, and six differences matter enough to change the data model.

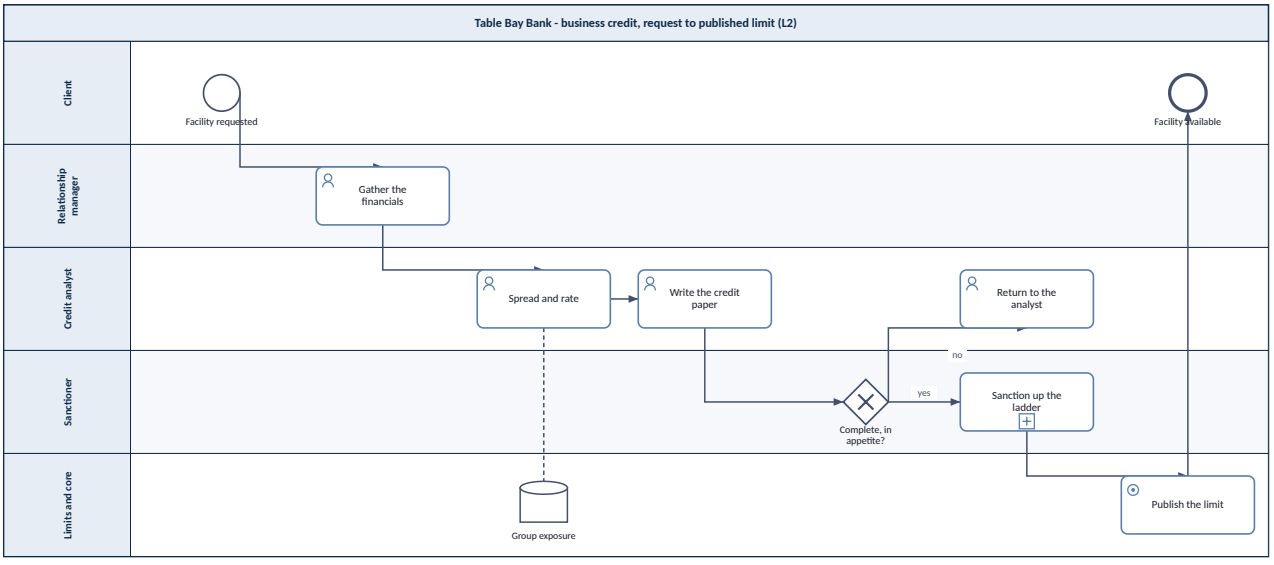



**Figure 8.12** Business credit, request to published limit. The sanction step is drawn as a sub-process because the mandate ladder is an escalation loop that will not fit in a lane.

The unit of decision is an **obligor group**, not an account, so the model needs a party-to-party exposure graph. The decision is a *paper approved by a person or a committee under delegated authority*, so the system must model authority limits and quorum rather than a score cut-off. The product is a **facility structure** — multiple sub-limits, currencies and drawdown types over shared security — not a loan. **Covenants are testable conditions with their own monitoring cycle**, and nothing in a retail stack has that concept. Review is annual and mandatory, so the process is a recurring cycle. And limits must be published in real time to the payments and treasury systems that consume them.

**The mandate ladder** is the sub-process in the figure, and it is drawn as a sub-process because it is a loop that will not fit in a lane. A credit paper goes to the lowest authority that can approve it; if the exposure, the risk grade or a policy exception exceeds that rung's mandate, it goes up one rung, and the rungs are personal, not positional. Model it as an authority structure plus an escalation rule, not as a sequence of named individuals, because the individuals change and the structure does not.

**Where it goes wrong.** Covenants in a Word document attached to an email. Group structure maintained by hand in a spreadsheet, so aggregate exposure to a connected group cannot be produced on demand. Limits held in three systems that diverge. Conditions precedent waived informally. Annual reviews overdue by years, with the exposure still live. And the mandate structure written in a policy document rather than enforced by the system, so what actually constrains a sanctioner is their own memory.

## 11. Collections and recoveries

Two businesses again. Collections works a customer who still has a relationship. Recoveries works an exposure where the relationship is over.

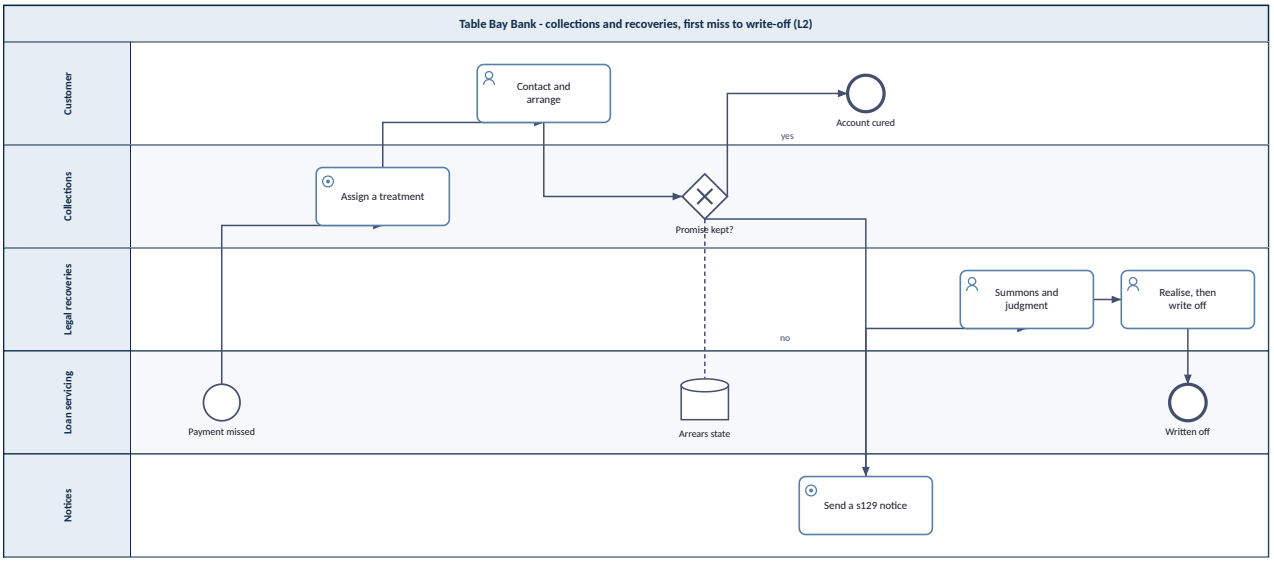



**Figure 8.13** Collections and recoveries. The section 129(1)(a) notice is a statutory gate before enforcement, and its proof of delivery is a data requirement before it is a legal one.

**The decision points.** *Was the promise kept?* — which sounds trivial and is the whole treatment engine. Behind it sits the arrears state machine, and the architecturally important fact about it is that the same account carries **three overlapping definitions of being in trouble at once**: the operational state (days past due and treatment bucket), the accounting state (the impairment stage), and the regulatory state (the Prudential Authority's distressed-restructured classification, under Directive 7 of 2015 until the end of 2026 and then its replacement) [17]. Three definitions, one account, three audiences, and they must tie.

**The statutory step.** Before enforcement, a section 129(1)(a) notice, with proof of delivery handled according to the Act's address rules; prescription runs on a clock driven by acknowledgment, payment and service of process; and prescribed debt may not be sold or reactivated [18]. Each of those is a data requirement before it is a legal one.

**Where it goes wrong.** Treatment assigned on days past due alone, ignoring everything the bank knows about the customer. Prescription managed by manual review, which is how a bank ends up enforcing a debt it may not enforce. The section 129 proof of delivery captured but not indexed, so it cannot be produced when it is needed. Debt-review status honoured by one system and ignored by another. And recovery outcomes never returned to the risk data mart, which makes loss-given-default an assumption rather than a measurement.

## 12. Treasury, deal capture to settlement

A separate landscape inside the bank, with its own vocabulary and usually its own IT team.

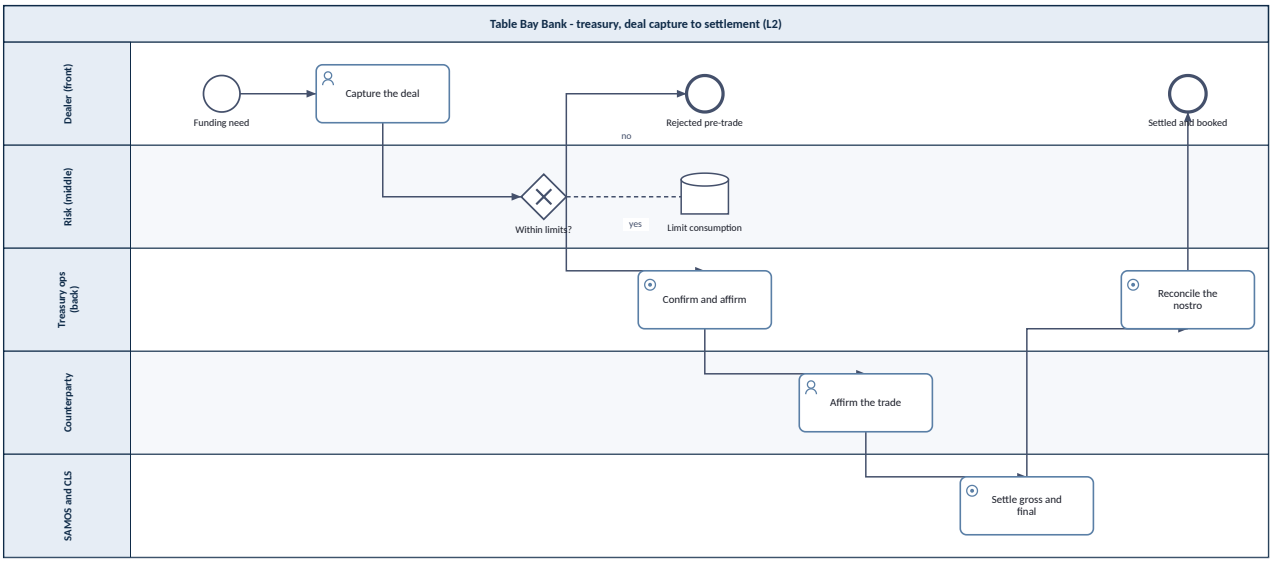



**Figure 8.14** Treasury, deal capture to settlement. The only question that matters about the limit gateway is whether it runs pre-trade in the capture system or post-trade in a report.

**The decision points.** *Is the deal within limits?* — and the only question worth asking about it is whether that check happens **pre-trade in the capture system or post-trade in a report**. The answer tells you whether the limit is a control or a commentary.

**The chain.** Capture, pre-trade control, confirmation and affirmation, settlement processing and cash management, accounting entries, reconciliation and reporting [19]. Settlement itself is unforgiving: an instruction on the real-time line of the domestic settlement system is gross and prefunded and is rejected outright where liquidity is insufficient, and settlement is final and irrevocable, with no guarantee from the central bank [20].

**Where it goes wrong.** Limits checked after the fact. Standing settlement instructions maintained in the back office with no change control, which is the classic internal-fraud vector in any bank. Nostro reconciliation manual and lagging. Independent valuation performed using the front office's own market data, which makes it neither independent nor a valuation. And a chain from deal to general ledger that crosses five systems with four reconciliations, each of which is somebody's month-end.

## 13. Close, account and report

The one in the inventory that is a production schedule rather than a process. It is drawn anyway, because the hand-overs are where it fails.

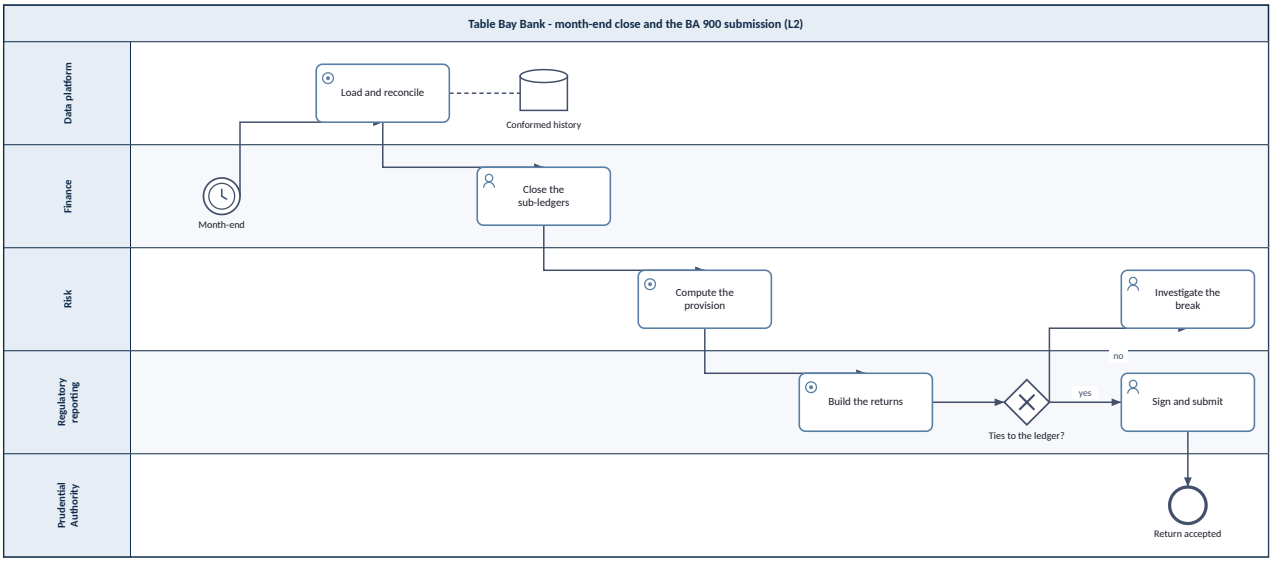



**Figure 8.15** Month-end close and the BA 900 submission. It is a production schedule rather than a process, drawn anyway because the hand-overs between finance, risk and regulatory reporting are where it fails.

**The decision points.** *Does it tie to the ledger?* A return that is correct but cannot be reconciled to the general ledger will not survive a supervisory question, and the break investigation is on the critical path to a statutory deadline.

**The deadlines.** BA 900 within 15 business days of the month-end it relates to [21], each BA submission accompanied by a signed, certified form BA 099 or BA 099A [22], the deposit-insurance file within ten working days of month-end with a chief executive and chief financial officer declaration within

three working days of submission [6], and annual BA returns resubmitted after the audited statements, no later than 90 calendar days after year end [22].

**Where it goes wrong.** The impairment computation sits on the close critical path and over-runs. Returns are sourced from point extracts that do not tie to the ledger. Lineage cannot be demonstrated, which fails an inspection even when the number is right. One product change breaks four returns. And the whole close depends on one named person who has not taken leave in December since 2019.

## From here to a target process architecture

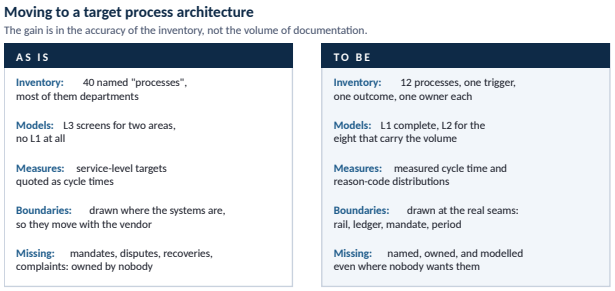



**Figure 8.16** What moving to a target process architecture actually changes. The gain is in the accuracy of the inventory and the honesty of the measures, not the volume of documentation.

Four moves, in order. They are unglamorous and they are the work.

**Fix the inventory before modelling anything.** Get to twelve processes with one trigger, one outcome and one named owner each. This takes two workshops and produces more argument than any other artefact in the programme, because naming an owner for mandates, disputes or recoveries is a political act. Do it anyway: everything downstream is unsound without it.

**Measure the current process before redesigning it.** Not with a survey. From the systems: event timestamps rather than status flags, reason-code distributions rather than averages, and a random sample of thirty closed cases traced end to end by hand. The measurement is what makes the redesign arguable. Without it, the target process is a preference.

**Decide which variants you keep.** A bank does not have one payment execution process; it has one per rail, and in several cases one per channel per rail. The useful question is not how to unify them but which forks are justified. An EFT credit file and a real-time proxy payment genuinely need different processes, because one has a cut-off and a day of deferred clearance and the other has ten seconds. A branch-initiated transfer and an app-initiated transfer probably do not. Declare the variants, cost them, and retire the ones nobody can justify.

**Attach each process to a capability and each step to a service.** This is where BPMN hands back to ArchiMate. The L2 model says who does what in what order; the ArchiMate model says which capability the process realises, which application service each step consumes, and which business object each step touches. Do that and the target becomes testable: you can ask whether the application estate can actually support the process you have drawn, and you will usually find two steps where it cannot. Those two steps are the programme.

One note on sequencing. Do not model all twelve at L2. Model the ones that carry the volume, the regulatory exposure or the pain — in a South African retail bank that is onboarding, payments, the mandate and collection pair, cards and disputes, credit origination, collections, and the close — and leave the rest at L1 until something changes.

### The trap: modelling the process you were told about

Every process model you are handed describes an idealisation, and the idealisation is not a lie. It is what the process owner believes, and they believe it because they designed it and nobody has told them otherwise.

The signs are consistent and quick to spot.

- **No rework loops.** Real processes are mostly rework. A model with no loop back for incomplete information is describing a world where forms arrive complete and debit orders are always paid.
- **Every gateway resolves one way.** A "funds available?" gateway with a "no" branch that leads to a box called "unpaid" and no further detail. In a South African collections business the "no" branch is the business.
- **Cycle times that are service levels.** Two days in the model, nine days in the data. The model is quoting the target.
- **Two exits where the system records forty reason codes.** Count the exits from the gateways. Count the distinct reason codes the system records at that decision. If the model has two and the system has forty, the model is a summary of somebody's intention.
- **A step with no owner in any lane.** Usually the hand-over. In the 600-page folder at Table Bay Bank, the unpaid-to-collections step was not missing because somebody forgot. It was missing because nobody owns it, and therefore nobody described it.

Four things defeat this, in increasing order of cost.

**Sit with the work for a day.** Not an interview — a day at the desk, watching, counting the times the clerk opens a second system or picks up the phone. One day produces more correction than a week of workshops.

**Pull thirty closed cases at random and trace each one.** Not thirty cases the operation selects; thirty from a random sample, including the ones that went wrong. The variance will be larger than anyone told you, and the distribution of paths through the process *is* the process.

**Count the reason codes, and read the free text.** The configured reason-code list is the organisation's considered view of what can happen. The free-text notes are what actually happens. Where a code called "other" accounts for a third of cases, there is a branch nobody has modelled.

**Find the spreadsheet.** There is always a spreadsheet, and it sits exactly where the process crosses a boundary that no system spans: between collections and legal recoveries, between the dispute case and the scheme's clock, between the creditor's mandate view and the debtor bank's, between the sub-ledger and the return. The spreadsheet is not a workaround to be eliminated. It is a precise, free, continuously maintained map of the missing part of the architecture, and it is the first thing to ask for.

Table Bay Bank is invented and nothing in this chapter describes a real institution. The pattern is not invented. Every figure above has a gateway whose unglamorous branch is where the money goes, and in most banks that is the branch nobody has drawn.

#### THE CHAPTER IN TEN LINES

- A process architecture is the inventory — trigger, outcome, owner. A process model is the inside of one. Most banks have hundreds of the second and none of the first.
- This is the one chapter that leaves ArchiMate, because ArchiMate has no gateway and no swimlane, and in a bank the hand-over is the defect. The mapping back is mechanical: pool to Process, human lane to Role by Assignment, system lane to Application Component by Serving, store to Business Object by Access.
- Settle the meaning of L0, L1, L2 and L3 in writing before the first workshop. APQC numbers them differently and BIAN does not use levels at all.
- Twelve L1 processes in four groups is a defensible inventory. Four of them are cycles or production schedules rather than triggered flows — say so rather than forcing a swimlane onto them.
- Four processes are almost always missing from a bank's own list: mandate management, disputes and chargebacks, recoveries after write-off, and complaints. Each is missing because its owner is too junior to be in the room.
- The mandate cycle and the collection run are two processes sharing a name, and the DebiCheck mandate lives at the debtor's bank, which no imported reference model assumes.
- Registered Mandates is a new ISO 20022 rail mandated from 10 March 2025, with a 19:00 late window, tracking to ten days, responses by 23:00 and settlement at 05:00 — not a rename of the old service.
- Split a process at a natural seam rather than cramming it: retail credit splits at the decision, because the second half runs on attorneys and a deeds office rather than on the bank.
- Measure before redesigning: event timestamps, reason-code distributions, and thirty randomly chosen closed cases traced by hand. Without that, a target process is a preference.
- Every handed-over model is an idealisation. No rework loops, gateways that resolve one way, cycle times that are service levels, and a step with no owner are the tells.
- Find the spreadsheet. It sits exactly where the process crosses a boundary no system spans, and it is a free, accurate map of the missing part of the architecture.

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CHAPTER 9

# The conceptual data model

*Fifteen boxes, no attributes, no keys and no system names. It takes an afternoon to draw, it outlives two core replacements, and the argument it settles about the word "account" is worth more than the rest of the data programme.*

## The model you draw in the first week

Table Bay Bank's lead architect drew the bank's conceptual data model on a whiteboard in her second week. It took two hours. In the room: the head of retail deposits, a product manager from transactional banking, somebody from payments operations, a credit manager and the financial controller. Fifteen boxes went up. Somebody photographed it on the way out.

Table Bay Bank is invented — a mid-tier South African full-service bank with about 2.4 million retail customers, a business bank, a small corporate and transactional arm and a treasury. Nothing about it is a case study. It is here so that the abstractions have somewhere to land.

Three years on, that photograph is the first thing a new architect at Table Bay is shown. Underneath it, the information model has been rewritten once and the physical schema twice. The photograph has not changed, because the bank has not changed. A bank in 1975 had parties, agreements, accounts, entries and balances. So does a bank in 2026, and so will one in 2050, whatever is in the database by then.

That is the whole case for the artefact. It is the cheapest thing in the data architecture to produce and the most expensive to buy back once a core replacement has started without one.

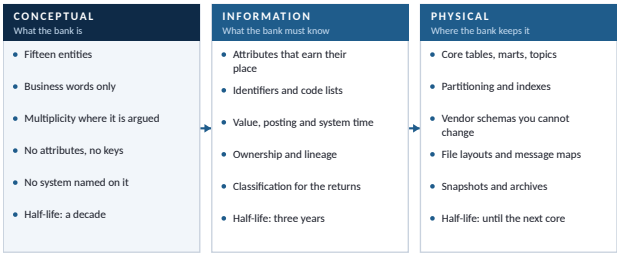



**Figure 9.1** Three layers, with the prohibitions that give the first one its long life. The half-lives in the bottom row are the argument for doing the leftmost column first and cheaply.

### What it is not

Five prohibitions. Each of them is load-bearing, and each of them will be argued about in the first hour.

**No attributes.** Not even the obvious ones. No `opened_date` on Account, no `amount` on Transaction. The moment an attribute appears the meeting becomes a negotiation about attributes.

**No keys.** Not natural, not surrogate. "Account number" is not on the model, which is just as well: chapter 10 spends a page on why a South African account number does not identify an account.

**No tables.** Many-to-many relationships stay many-to-many here. Resolving one is a design decision and this is not a design.

**No system names.** The core, the card platform and the payments hub do not appear. If the model changes when a system is replaced, it was not a conceptual model.

**No regulatory or temporal structure.** No IFRS 9 staging columns, no POPIA consent flags, no valid-from and valid-to. All three are real, all three are mandatory, and all three belong one layer down.

The point of the prohibitions is not minimalism. It is audience. Watch the rule break once and you will enforce it for ever. Somebody writes `balance` on Account, because obviously an account has a balance. Within ten minutes the treasurer is explaining that the available balance is not the ledger balance, the card person is explaining that the authorisation host holds a third number, and somebody from finance asks which one goes on the return. All three are right. None of it belongs in this meeting. The model that was going to take two hours now takes four sessions and comes out as an information model nobody owns.

### What it is actually used for

Three things. A model not used for at least one of them is decoration.

**Scoping.** Draw a boundary round the boxes a core replacement touches and you have the most honest one-page scope statement anybody will produce. It survives procurement because it is in business language. "This programme touches Product, Account, Entry, Balance and Statement; it does not touch Facility, Collateral or Impairment State" is a sentence an executive can hold a supplier to.

**Settling vocabulary.** Most of the arguments in a bank's data programme are definitional, not technical. Is a dormant account an open account? Is a reversed fee a fee? Is a recovery after write-off income or a negative provision? Those get answered on this page, or they get answered inconsistently in nine systems and then reconciled by hand every month.

**Testing a package.** Take the fifteen boxes into a vendor demonstration and ask where each one lives. The silences are informative. In my experience the longest pauses come on Facility, Mandate and Impairment State, in that order — though that is a pattern I have noticed rather than one I can source.

### Three tests for whether something is an entity

1. **Can you define it in one sentence in words the business already uses**, without saying "a record of" or "a table holding"?
2. **Does the business count it, name it, or argue about it?** "How many accounts are dormant" is a real question people fight over. "How many account-party junction rows" is not.
3. **Would it survive replacing every system tomorrow?** An Entry would. An "Account Header" would not.

The third test is the sharpest and it kills most candidates.

### How many

Twelve to twenty. Below twelve you have a diagram rather than a model and you have certainly collapsed something that matters — usually Account. Above twenty-five you have started the information model and lost the room. Fifteen is a good number for a bank and it is the number this chapter builds to.

> The conceptual model is the only artefact in the data architecture that the business owns rather than tolerates. Draw it in their words, keep it on one page, and let them change it. Everything below it is yours.

## The spine: a party, an agreement, an account

Six entities carry most of the weight. Get these right and the rest follows.

> **WHY THESE FIGURES ARE IN CROW'S FOOT AND NOT IN ARCHIMATE**
>
> The rest of this book models in ArchiMate 4, and parts of this chapter draw the same entities as Business Objects. The figures that carry multiplicity are drawn in crow's-foot notation on purpose. A conceptual data model argues about multiplicity — one party or several, one account per agreement or many — and crow's foot states that on the line, where a business reader can read it off the page and disagree with it. **ArchiMate has no notation for multiplicity on a relationship.** The argument would have to go in a note, and a note is not a model. So: draw the argument in crow's foot, hold the model in ArchiMate, where it connects to the processes, the applications and the data objects that realise it.

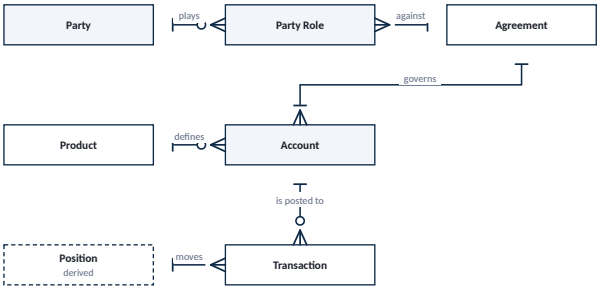



**Figure 9.2** The spine, in crow's-foot notation: a party, in a role, under an agreement, on an account defined by a product, with transactions against it. Position is drawn dashed because, like Balance, it is a derivation — and it only belongs on the model of a bank with a trading book.

### Party, and why Customer is usually a bad entity

One human being exists five times in a mid-sized South African bank: in the cheque-account system, in the card system, in the home-loan book, in the vehicle-finance book — often a separate legal entity — and on the wealth platform. Each instance has its own key, its own spelling, its own address and its own view of arrears. None of them is wrong. Each is a *record*. The bank has no *identity*.

**Party** is a legal person, natural or juristic, that the bank can transact with or owe. That is the whole definition. Everything else — depositor, borrower, guarantor, signatory, beneficial owner, counterparty, employee, supplier — is a **dated role** that a party plays *against something*: an account, an agreement, a facility, a payment, another party.

So why not Customer? Four reasons, and the fourth is the one that ends the argument.

- **It is a role, not a thing.** "A party with at least one live product relationship" is a derived predicate over agreements. Store it and it is stale by the next business day.
- **It cannot hold the people you still have to know about.** The prospect, the ex-customer, the guarantor with no product of their own, the third-party payer, the beneficial owner behind a company, the deceased estate. All of these are parties the bank must identify. None of them is a customer.
- **It is not consistently defined.** Retail counts per natural person, business banking per enterprise group, the card issuer per plastic and the resolution authority per single-customer-view record. In most banks those four counts never reconcile — a symptom I have seen often enough to expect it, though not one anybody publishes.
- **As a root entity it has nowhere to put control.** Make Customer the root and everything hangs off it. A party who is only a beneficial owner then cannot exist, and the financial-crime model — which is a model of *parties and control*, not of customers — has no home. FICA, as amended, requires identification and verification of beneficial owners behind a corporate client [1]. Those people hold no product at all.

The settling question is simple: *would the bank need to know this exists even if it held no product?* A beneficial owner, yes. A plastic card, no.

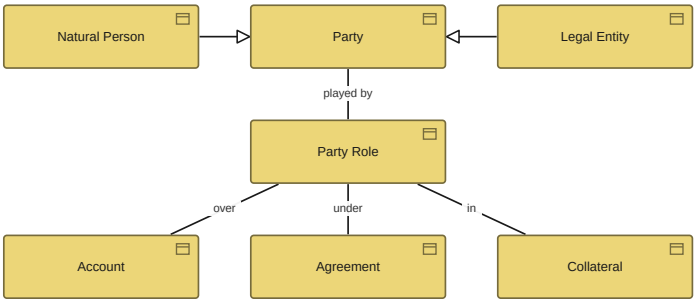



**Figure 9.3** The party pattern as ArchiMate Business Objects. Party holds identity and nothing else; every relationship in the bank lands on a dated Party Role. Natural person and legal entity differ enough to specialise, mostly because a legal entity has no capacity of its own.

**Party Role** is the second box and the one that does the work: a party, in a named capacity, against something, between two dates. Signatory, borrower, guarantor, beneficiary, cardholder, instructing party, complainant. Drawing it separately is what lets the model say that the same human is a customer of the retail bank, a director of a business client and an employee — three facts a bank is required to know about simultaneously, because that combination is a conflict-of-interest and a financial-crime signal before it is anything else.

One South African consequence worth stating early: the deposit insurer excludes the failing bank's own executive management, board, their related parties and its lead external audit partners from cover [2]. That is a rule you can only compute if the depositor data knows which depositors are those people — a party-to-party relationship, not a flag.

Natural persons and legal entities differ enough to be worth a specialisation on the page. A natural person has a thirteen-digit South African identity number or a passport and a country; a birth, a death, a marriage with a matrimonial regime, an insolvency. A legal entity has a CIPC registration number, a trust number or a non-profit number; a registration, a name change, business rescue, liquidation. More to the point, a legal entity has **no capacity of its own**. It can act only through authorised natural persons, which is why Mandate is a first-class entity in a bank and an afterthought almost everywhere else. Trusts sit awkwardly between the two: since 1 April 2023 trustees must record and lodge beneficial-ownership information with the Master [1], so a trust needs founder, trustee and beneficiary roles, not an owner field.

And POPIA protects **juristic persons** as data subjects, which GDPR does not. A corporate client in South Africa has access and correction rights. That is an SA divergence worth knowing before someone designs a data-subject-request process that only handles people.

#### Agreement, Product and the thing called Account

**Agreement** is the legal instrument: terms accepted, when, through which channel, with which disclosures, at which wording version. **Product** is the manufactured definition — eligibility, pricing, fee schedule, interest basis, term, currency, channel availability, regulatory classification — held in a catalogue, versioned, with effective dates.

A product is a definition; an account is an instance. The test for whether a change is a new product version is whether it would change an existing customer's contract. If it would, existing accounts stay on the old version. A marketing name, a campaign or a channel is not a product.

Product carries classifications nothing else can carry: whether the statute that applies is deposit-taking, credit or investment; whether the product is a **qualifying product** for deposit insurance, which the published rules define by listing current, term, notice, savings, transactional, tax-free savings and the Islamic Wadi'ah, Qard and Murabaha structures, and excluding shares, commodity holdings, electronic money and anything not repayable at par [3]; the instrument and maturity classification the monthly BA 900 return needs [4]; and the National Credit Act subtype, each with its own fee caps and disclosure duties. Put qualifying status on the account instead of the product and it will be wrong on every account opened before somebody remembered to set it.

#### The several things the word "Account" means

This is the commonest modelling failure in a bank, and it is worth slowing down for. Six distinct things share the word.

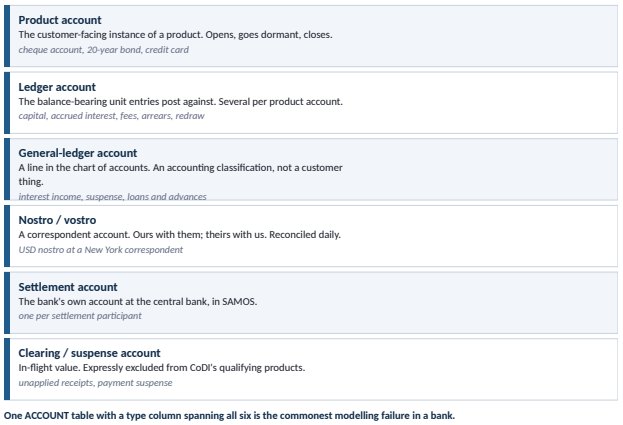



**Figure 9.4** Six things called account. The regulator can tell them apart: the deposit insurer's rules exclude suspense accounts from qualifying products, an exclusion that can only be expressed if a suspense account is a distinguishable kind of thing.

The damage from collapsing them into one `ACCOUNT` table with a type column is specific and predictable:

- The general ledger and the customer estate join by accident, and nobody notices until a reconciliation is out by one account.
- Interest and fees hang off the capital balance, so arrears cannot be aged and the *in duplum* cap — which limits what may accrue while a consumer is in default — cannot be computed.
- The model assumes one owner. Joint accounts, partnerships, trusts and minors' accounts all break it. The deposit insurer requires a **sharing ratio per joint holder**, defaulting to equal shares where none is recorded [3] — a field most core systems have never captured.
- The model assumes account numbers are unique for ever. They are recycled, migrated and inherited through acquisitions.

The settling question: *does it carry a statement-visible balance and close independently?* If it carries a balance but is never independently closed or shown to anybody, it is a ledger account inside a product account.

### Transaction and Position

**Transaction** is a business event with economic meaning: a card purchase, an electronic funds transfer, a fee levy, an interest run. The test is whether the customer recognises it as one thing that happened. One foreign-currency card purchase is one transaction and five or more entries — customer debit, settlement credit, foreign-exchange margin, interchange, cross-border fee, value-added tax on the fee.

**Position** is Balance's wholesale twin, and it is the sixth spine entity in any bank with a treasury: the net holding in an instrument, by book, at a point in time, marked to a price. Retail bankers never use the word; treasury uses nothing else. A bank with a trading book needs it on the model, and the honest way to draw it is as a derivation over deals in the same way Balance is a derivation over entries. If your bank has no trading book, leave it off and say so.

## The money and the risk

Seven entities, and the argument about each one is about whether it is a fact or a derivation.

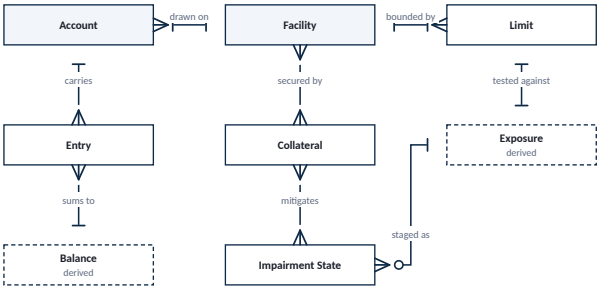



**Figure 9.5** Money and risk. The two dashed boxes are derivations that banks store anyway, for performance, for period close and for reproducibility. Everything solid is a fact somebody signs for.

**Posting and Entry.** A posting is the act of recording a transaction, at a point in time, as a set of balanced entries. An **entry** — journal line, leg — is one signed amount against one ledger account. That is the atom, and it is immutable. The invariant is double entry: per posting, debits equal credits.

The invariant is what makes a bank auditable, and it is destroyed by modelling a transaction as a row with a from-account and a to-account. That shape cannot represent a three-legged posting — debit the customer, credit income, credit VAT — nor the suspense leg of an in-flight payment, nor a posting that crosses legal entities or currencies and therefore needs a foreign-exchange position leg. Two columns are not a ledger.

**Entries are events, not versioned states.** Append only. No updates, no deletes. A mistake is corrected by a new entry, never by editing the old one. Chapter 10 takes that apart properly; here it is enough to know that it is the structural difference between a bank's data and almost everything else an architect has modelled.

**Balance is a derivation.** It is the sum of signed entries against a ledger account to a point in time. A bank nevertheless holds at least five numbers that users assume are one: the ledger or book balance, the available balance, the cleared or value-dated balance, the statement balance and the arrears base. The settling question is *if two reports differ on this, is one of them wrong?* If the answer is no — both right, different meanings — then you have two named derivations, not one balance.

Banks store balances anyway, for three honest reasons: performance, because a forty-year mortgage has hundreds of thousands of entries; closure, because period-end balances are signed off and must not move afterwards; and reporting, because a submitted figure must be reproducible years later. The right pattern is a running balance per entry plus immutable period-end snapshots, with the current balance derived from the last snapshot and the entries since. The wrong pattern is one mutable balance column updated in place. Draw Balance on the model, draw it dashed, and let the room argue about it.

**Facility and Limit.** A facility is a granted capacity to draw — an amount the customer may use, repay and re-use. The cardinality to the account is the point: a home loan is one agreement, one facility and one or more accounts, loan plus redraw; a credit card is one agreement, one facility, one account and many cards; a R50 million combined facility across an operating company and two subsidiaries is one facility, three borrowers, several accounts, a facility limit *and* sub-limits; and a revolving loan's facility survives the account reaching zero.

So Facility is not an account attribute. Exposure, limit usage, large-exposure reporting and concentration risk all operate at facility and group-of-connected-clients level, which is the level the Prudential Authority tightened in Directive 14 of 2025 [5]. The settling question: *can the customer draw, repay and redraw without a new credit decision?* If yes, there is a facility.

A **limit** is a control and cannot be exceeded. An **exposure** is a measurement and can be. Headroom is the derivation between them, and exposure is not balance: it is drawn plus undrawn committed plus contingent — guarantees, letters of credit — plus derivative add-ons, netted only where netting is legally enforceable.

**Collateral and security.** Collateral is an asset; security is the legal right over it. Conflating them is why banks cannot answer "what is our exposure to Sandton office property". The relationships are the hard part: one property secures three loans with ranked bonds; one general notarial bond secures a whole facility; one director's suretyship secures all present and future obligations. What the model needs is an **allocation** between the security instrument and the exposures it secures — many to many, with priority and share — plus a dated **valuation** with its method and valuer, not a collateral-value column. Since the revised standardised credit-risk approach took effect in South Africa on 1 July 2025, residential mortgage risk weights depend on loan-to-value [5], which turns a dated valuation into a capital input. The settling question: *does the right survive the asset's sale?* If it follows the asset, the registered instrument is its own entity.

**Impairment State.** Arrears and days past due are observations, reconstructible from the entries and the expected-payment schedule. Impairment stage and provision are *estimates*, valid only as at a date, with a model version and a scenario set attached. Storing both with the same update semantics is the commonest reason a bank cannot reproduce last quarter's expected credit loss. Stage is simultaneously an accounting attribute, a capital input and a leverage input — the Prudential Authority treats stage 1 and 2 provisions as general and stage 3 as specific, and deducts provisions for all three stages from the leverage-ratio exposure measure [6]. Make it a dated, governed fact on the exposure, not a derived reporting column.

## Instruction, payment, settlement — three entities, not one

Almost every first draft has one box called Payment. It is wrong in a way that costs a programme a year.

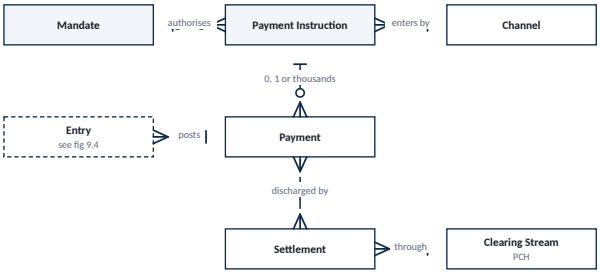



**Figure 9.6** Three entities, not one. The multiplicities are many-to-many in both directions: one instruction can raise thousands of payments, and one settlement can discharge thousands. Clearing stream is on the model because in South Africa the rulebook, cut-off and return window all differ by stream.

A **payment instruction** is the request to move value: the mandate-authorised order, with payer, payee, amount, currency, requested date, reference, channel and authorisation. Its lifecycle is received, validated, authorised, screened, then accepted, rejected or held. A **payment** is the resulting movement of value, usually spanning several banks and systems, carrying an end-to-end reference, with a lifecycle of initiated, cleared, settled, completed, returned or recalled. **Settlement** is the discharge of the interbank obligation in central-bank money — in South Africa at the Reserve Bank, in SAMOS [7] — and it goes pending, then settled, which is final and irrevocable.

The multiplicities are many-to-many in both directions, which is exactly why one box cannot hold them. One instruction produces no payment when it is rejected, one when it succeeds, or thousands when it is a batch file or a standing instruction that runs for twenty years. One payment can produce several settlements along a correspondent chain. One settlement discharges thousands of payments in a net clearing batch.

The South African addition is that the model must carry **which clearing stream** a payment went through, because the rulebook, the cut-off, the return window and — since the 2026 governance change — the body that owns the rules all differ by stream [7]. A payment model with no clearing-stream concept cannot answer a return query.

The settling question: *if this fails, who has to be told?* Instruction failure is rejected to the customer. Payment failure after clearing is returned or recalled through the scheme. Settlement failure is systemic.

## Paper, channel and the things customers say

Five more entities, all of which outsiders leave off and all of which a conduct regulator asks about.

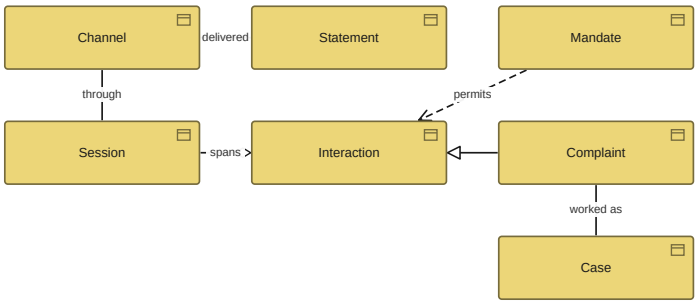



**Figure 9.7** Paper and channel as Business Objects. A complaint is a specialised interaction with a statutory shape; a case is the work done about it. Mandate influences what may be done in an interaction, which is why it is an entity rather than a permission.

**Mandate** is an authority: who may do what, on which account, up to what amount, from when to when. Three different things share the word, and all three are mandates. The **account operating mandate** says who may transact and in what combination — "any two to sign" — and is compulsory for any legal entity. The **collection mandate** is the payer's authority for a third party to collect; in South Africa it is a first-class, system-held object, because the Registered Mandates stream and Authenticated Collections both exist to make it electronically verifiable and separately held [7]. A debit order without a valid mandate is unauthorised, and the mandate is the bank's defence. The **internal mandate** is an employee's delegated authority to approve, release, override or write off, up to a limit.

All three are dated authorisations by a party, over a scope, with a limit. Model them as flags on an account and you lose the limit, the dates, the evidence and the revocation — the four things an auditor asks about. The settling question: *is there an amount or a scope beyond which this person may not act?* Then it is a mandate, not a permission.

**Statement** is a dated, frozen rendering of a period's entries and balances for one account, delivered through a channel. Three facts make it architecturally interesting. It is immutable: once issued it is evidence, so re-rendering it and getting a different answer is a control failure. It is a disclosure instrument under both the Conduct Standard and the National Credit Act. And its **delivery is an event** — recipient, channel, timestamp, outcome — which the ombud will ask for.

**Complaint** is a first-class entity with a statutory shape, not a case type in the CRM. Section 8 of the Conduct Standard for Banks requires an effective complaints framework with **prescribed categories** for retail complaints, acknowledgement and progress updates, free access, ombud information and complaints-data monitoring, with reporting to the FSCA including public reporting [8]. The regulator reviewed banks' complaints handling in March 2025 and told the industry to do better [9]. The minimum model is: complainant — a party, possibly not a customer — the product complained about, the categorised cause, dates received, acknowledged and resolved, the outcome, redress paid, ombud escalation and determination, and root cause. If the categories do not match the prescribed list exactly, the return gets built in a spreadsheet.

**Case** is the unit of work: an investigation, a dispute, an alert disposition, a debt-review file. It is not the same thing as the complaint, the alert or the claim it concerns, and drawing it separately is what lets one case span three complaints or one complaint spawn two cases.

**Channel** is the means of interaction — branch, ATM, app, internet banking, USSD, WhatsApp, call centre, point of sale, API. It is not a code on a transaction. Channels have owners, rulebooks, cut-offs, fee structures, service levels and regulatory treatment of their own.

## The whole thing on one page

Fifteen boxes. This is the figure to photograph.

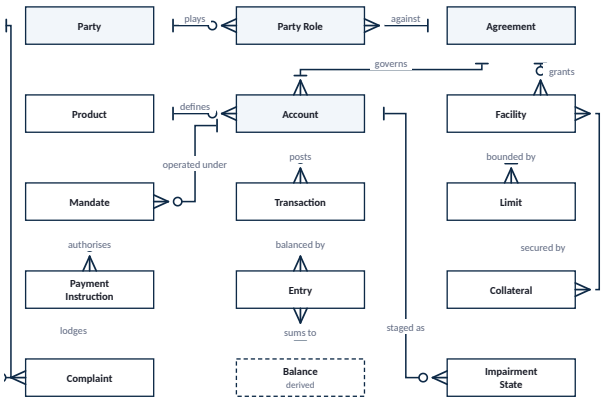



**Figure 9.8** The fifteen. Photograph this one. Shaded boxes are the identity and contract spine; the dashed box is the derivation everybody wants to store. Multiplicity is shown only where a room has argued about it.

| Entity | The question it answers | What breaks without it |
|---|---|---|
| Party | Who are we dealing with, and have we dealt with them before? | Five customer masters and no identity |
| Party Role | In what capacity, over what, between which dates? | Signatories, guarantors and beneficial owners have nowhere to live |
| Agreement | What did we actually contract to, and on which wording? | Disclosure cannot be evidenced |
| Product | What is this thing we sell, and which version? | Classification for the returns is set per account and is wrong |
| Account | What is the customer-facing instance that opens and closes? | The ledger and the customer estate join by accident |
| Facility | What may be drawn, repaid and redrawn without a new decision? | Large exposures cannot be computed |
| Limit | What may not be exceeded, by whom, to what value? | Controls become code in several places |
| Transaction | What happened, that the customer would recognise as one thing? | Multi-leg economics cannot be represented |
| Entry | What is the immutable signed amount against one ledger account? | No double entry, therefore no audit |
| Balance | What is the position at a point in time, on which definition? | Three channels show three numbers |
| Mandate | Who may act, over what, up to how much, from when? | Unauthorised debits and an undefendable dispute |
| Collateral | What asset stands behind this, worth what, as at when? | Credit-risk mitigation and loss given default are guesses |
| Impairment State | How bad is this exposure, on which date, by which model? | Expected credit loss cannot be reproduced |
| Payment Instruction | What were we asked to do, by whom, through which rail? | "Where is my payment" has no answer |
| Complaint | What did the customer say was wrong, in which category? | The conduct return is assembled by hand |

Three deliberate omissions. **Position** belongs on the model of any bank with a trading book and is left off here because Table Bay's treasury is small; where it exists it sits beside Balance as the other derivation. **Exposure** is a derivation over Facility, Limit and Balance and stays off for the same reason Balance nearly did. **Settlement** is on the payments model but not on the fifteen, because for a retail-dominated bank it is one step inside the payment lifecycle rather than a thing the business counts.

## The variants

Three shapes of bank, three different-looking models from the same fifteen boxes.

### Retail

Party is a natural person. Party Role is thin — holder, joint holder, occasionally a guardian. Agreement and Account are close to one-to-one, which tempts everybody to merge them; resist, because the credit card breaks it on day one. Mandate matters mostly as a collection mandate. Volume is the architectural problem, not structure.

### Corporate, with its hierarchy and its mandate problem

Everything that was thin in retail becomes thick.

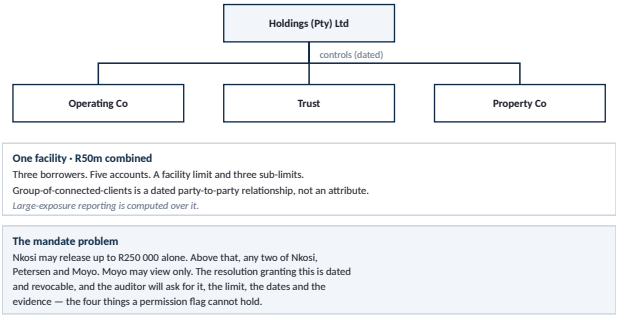



**Figure 9.9** The corporate variant. The party becomes a dated graph, the facility becomes structured, and the mandate becomes the hardest object in the bank. Names are invented.

Three things change. **The party is a graph**, not a record: a holding company, operating subsidiaries, a trust, directors, shareholders, beneficial owners resolved to natural persons, all dated, all revisable. Model the group of connected companies as an attribute rather than a dated party-to-party relationship and large-exposure reporting becomes a manual exercise. **The facility is structured**: one combined limit with sub-limits per borrower, per currency, per product, and the arithmetic of headroom is not a subtraction. **And the mandate is the hard part.** Who may instruct, who may release, in what combination, up to what value, on which account, from when, revoked by what. The resolution granting that authority is a document with a date, and the auditor will ask for the document, the limit, the dates and the revocation.

### The delegated or agent model

The third shape is the one that catches architects out, because the bank holds the account but somebody else holds the relationship. An attorney's trust account, an estate agent's rental account, a stokvel, a payroll bureau, an employer running a group savings scheme, a fintech operating accounts for its own users through a sponsoring bank.

Structurally it is the same pattern every time: a party holds the account, a second party is authorised over it by a mandate, and *behind* both there is a set of beneficial interests the bank may or may not be able to see. South African deposit-insurance rules make the distinction legally sharp. A **formal beneficiary account** — an attorney's trust account is the example — is looked through to each beneficiary, with a record per beneficiary *if the bank can identify all beneficiaries and their balances monthly*; where it cannot, there is one record for the account with no split [3]. An **informal beneficiary account** — a stokvel — is covered per holder, and is upgraded to the formal treatment if the bank holds the constitution and member details monthly [3].

Read that twice. **The cover an ordinary member of a stokvel gets depends on whether the bank's data model can see them.** That is as direct a line from a modelling decision to a customer outcome as this book contains, and it is why Party Role and Mandate earn their places on a fifteen-box model.

## The entities that should not be there

Five turn up on every first draft.

| ON THE FIRST DRAFT | WHAT IT ACTUALLY IS | WHAT TO DO |
|---|---|---|
| **Customer** | A dated role over an agreement, not a thing | Party + Party Role; count it, do not store it |
| **Transaction (as catch-all)** | Three things: the business event, the posting, the entry | Transaction, Posting, Entry — kept apart |
| **Record** | A storage artefact that has escaped into the model | Delete. Name the thing the record is about |
| **Status** | A dated attribute of something else, hoisted to a box | Lifecycle state on the entity, with its dates |
| **Address** | An attribute of a party role, with a purpose and a date | Keep it down a layer; FinSurv needs several kinds |

**Figure 9.10** The five that turn up on every first draft, what each one actually is, and what to put in its place.

Customer is the one worth fighting over and has already had its page. Transaction-as-catch-all is the second: one box labelled Transaction holding the business event, the posting and the entry, which guarantees that nobody can say whether a reversal is one transaction or two. Record is a storage artefact that has escaped into the model. Status is a dated attribute hoisted into a box, and the giveaway is that nobody can say what a Status *is* without naming what it is the status of. Address looks harmless and is not: it is an attribute of a party in a role, with a purpose and a date, and a bank needs several kinds at once — residential, postal, registered office, and the structured address ISO 20022 messages carry.

### Three that are borderline, and when to let them in

Honesty is better than rules here. Three entities sit on the line.

**Statement.** It is arguably a rendering of entries and balances, which is to say a derivation, and a purist leaves it off. Let it in, because it is immutable evidence with a delivery event attached, and because the business counts statements and argues about them.

**Regulatory Return.** A return is not a report. It is a submission, for a reporting entity, for a period, in a version, with a preparer, a reviewer, an approver, a submission timestamp, a regulator acknowledgement and — critically — resubmissions. The deposit insurer builds resubmission into its process: it notifies banks of discrepancies and may request a resubmission, and banks may request one themselves [10]. If the bank's reporting problem is on the agenda, put Return on the model. If not, leave it to chapter 10, which owns it.

**Session.** A bounded authenticated interaction: who, from what device, through which channel, when, authenticated how, doing what. It feels technical and it is not. Joint Standard 1 of 2023 requires logical access control, data-loss prevention and logging and monitoring, and Joint Standard 2 of 2024 requires multi-factor authentication for access to critical functions and sensitive information [11][12]. Neither is expressible without session, device and channel as retained things. Let Session in if the bank's problem is digital fraud, and leave it off otherwise.

## From here to the information model

A conceptual box is not a table and it is not one table. The way down is a ladder, and the rungs are worth naming before chapter 10 starts climbing them.

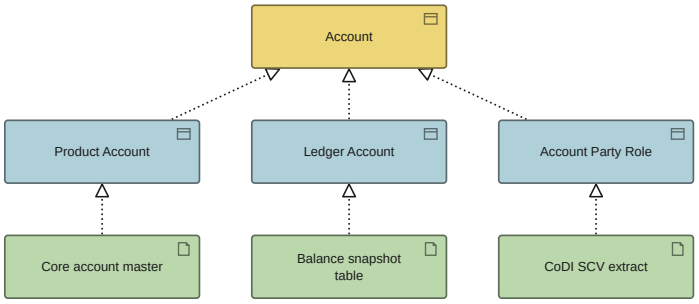



**Figure 9.11** The realisation ladder for one box. Account becomes three logical entities and at least three physical artefacts, and the mapping is many-to-many in both directions — which is why mapping a vendor core to a conceptual model is a week of work, not an afternoon.

One conceptual entity becomes several logical entities, each with attributes, identifiers, resolved many-to-many relationships and a temporal shape. Those become physical artefacts: core tables you cannot change, a snapshot table, a mart, a monthly extract file. The relationship is realisation, not decomposition, and it is many-to-many in both directions — one physical table may realise parts of three conceptual entities, which is exactly what makes a vendor core hard to map.

Chapter 10 picks the ladder up from the second rung: attributes where they earn their place, keys, classification, ownership, lineage and — at length, because a bank's version of the problem is not an insurer's — time.

## The afternoon that produces one

You do not need a project to get a conceptual model. You need one afternoon, six people and a wall.

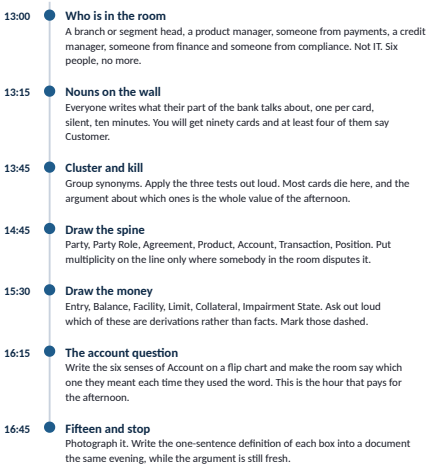



**Figure 9.12** An afternoon that produces a conceptual model. The 13:45 hour is the one that cannot be skipped and the one everybody tries to skip.

Two rules make the difference between an afternoon that works and one that produces a hundred sticky notes and no model.

**Nobody from IT writes on the wall.** Architects facilitate, hold the pen when asked, and enforce the prohibitions. The moment an architect starts proposing entities, the business stops owning the model.

**The deletion hour is the one that cannot be skipped.** Ninety cards become fifteen boxes by argument, and the argument is the deliverable. Teams that skip it get a model with forty boxes that nobody can hold in their head, and within a month it is a diagram on a wiki that no one opens.

The week after, do three things. Write the one-sentence definition of every box into a document and circulate it for correction — definitions, not descriptions. Walk the model against two real cases, one retail and one corporate, and find what it cannot hold. Then take it into the next vendor conversation and watch where the pauses come.

### THE CHAPTER IN TEN LINES

- A conceptual data model is twelve to twenty business entities with no attributes, no keys, no tables and no system names, and its value comes from the prohibitions.
- Use it for three things only: scoping a programme, settling vocabulary, and testing a package. Anything else is decoration.
- Customer is a role, not an entity. Party plus dated Party Role is the pattern, and it is the only shape that can hold a guarantor, a beneficial owner or a deceased estate.
- The word account means at least six different things in a bank, and collapsing them into one table with a type column is the commonest modelling failure in the industry.
- Balance is a derivation; banks store it anyway for performance, period close and reproducibility, so draw it dashed and keep immutable period-end snapshots.
- Payment instruction, payment and settlement are three entities with many-to-many relationships in both directions, and a South African model must also carry the clearing stream.
- Mandate is one word for three dated authorisations — account operating, collection and internal delegation — and all three lose their limit, dates, evidence and revocation if modelled as flags.
- Entries are immutable events, not versioned states; a mistake is corrected by a contra entry, never by an edit.
- The corporate variant turns the party into a dated graph and the mandate into the hardest object in the bank; the delegated model decides, through beneficiary look-through, how much deposit cover a stokvel member actually gets.
- Draw it in one afternoon with six business people and no IT writing on the wall, and do not skip the hour spent deleting cards.

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CHAPTER 10

# The information model

*The layer below the fifteen boxes: attributes that earn their place, keys that are not keys, classifications that decide what the return says, and time — which in a bank runs on three clocks and is the subject that separates people who have worked in one from people who have not.*

## The layer where the arguments get expensive

Chapter 9 ended with fifteen boxes and a photograph. This chapter is what happens when somebody asks what is *in* the boxes.

The conceptual model survives a core replacement. The information model does not, quite: it has a half-life of about three years, because attributes get added whenever a regulator asks a new question and because classification schemes change under you. It is still worth doing properly, because almost everything that goes wrong in a bank's data estate goes wrong at this layer. Not in the physical schema, which is usually competent, and not in the conceptual model, which is usually absent but cheap to recover. Here: in the attributes nobody owns, the keys that are not keys, the classifications that disagree, and above all in time.

Six things make up an information model. Attributes, where they earn their place. Keys and identifiers. Classification. Ownership. Lineage. And time — which in a bank is a bigger subject than the other five together, and gets the long treatment in this chapter.

Table Bay Bank, still invented, is the worked example throughout.

## Attributes, where they earn their place

The conceptual model banned attributes. This layer admits them, and the discipline that replaces the ban is a test. An attribute earns its place by having an answer to four questions.

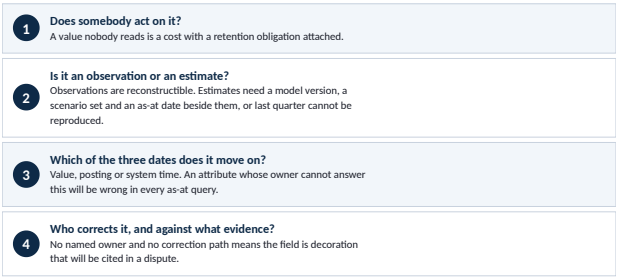



**Figure 10.1** Four questions an attribute has to answer before it goes on the model. The second one is the bank-specific one: observations and estimates cannot share update semantics.

The second question is the one that separates a bank from most enterprises. **Observations and estimates are different kinds of fact and must not share update semantics.** Days past due is an observation: it is reconstructible from the entries and the expected-payment schedule, it has one right value for any date, and a disagreement about it is a bug. An IFRS 9 impairment stage is an estimate: it is valid only as at a date, it depends on a model version and a scenario set, and two different values for the same date can both be correct if they were produced under different methodology. Store them in the same shape, with the same "latest value wins" behaviour, and the bank loses the ability to reproduce last quarter's expected credit loss. That is the commonest reason an audit of an expected-credit-loss number takes six weeks.

A short list of attributes a bank cannot do without, each of which is load-bearing somewhere later in this chapter:

- On an **entry**: posting identifier, journal reference, ledger account, general-ledger account, signed amount, currency, value date, posting date, system timestamp, transaction type, narrative, channel, originating reference, the operator or system that posted it, and a reversal pointer.
- On an **account**: product version, opening and closing dates, status with its history, the institutional sector and industrial classification of the counterparty, residual maturity bucket, currency, and the deposit-insurance qualifying flag — which, as chapter 9 argued, is derived from the product and not set on the account.
- On a **party**: the surrogate identifier, the legal identifier and its type, country of residence and of tax residence, a dated risk rating with the factors that produced it, and a reliability-of-identification measure.

That last one surprises people. The deposit insurer's prescribed field set includes a **reliability-of-identification flag** per depositor [1]. A bank is being asked, formally, to say how sure it is that it knows who somebody is. There is no equivalent in most industries, and there is nowhere obvious to put it: it is not an attribute of a name or of a document, it is an attribute of the bank's *confidence in the match*, which means the matching process has to produce a retained, graded output rather than a yes or no.

The fourth question — who corrects it — is the one that quietly kills attributes. Every bank's information model has fields that were added for a project that ended, that nobody has owned since, that no process writes to and that three downstream consumers still read. They are not harmless. A dormant attribute carries a retention obligation, appears in a subject-access response, and will eventually be cited by somebody in a dispute as evidence of what the bank knew.

## Keys, and why a South African account number is not one

Every architect arriving from a market with IBANs makes the same mistake in the same week.

South Africa routes domestic payments on a **six-digit branch code plus an account number**. There is no domestic IBAN. Account numbers are bank-specific: different lengths, different internal structures, different check-digit algorithms, and some with no check digit at all. Three consequences follow, and all three are architectural.

**An account number alone does not identify an account nationally.** It needs the branch code, and implicitly the bank. Any model that treats account number as a global key will collide the first time two banks' data meet in the same table — in a payments warehouse, a collections bureau file or a group reporting layer.

**Validation is a service, not a computation.** Because there is no universal check digit, confirming that an account exists and belongs to the named party is an interbank enquiry run through the national utility, not something a local routine can do. I could not source the current service names, coverage or rulebooks; the clearing-house rules and the utility's participant specifications would settle it. The architectural point does not depend on the detail: **payee verification is an integration, with a latency and a cost per call, and it belongs in the payment path.**

**Identifiers are recycled.** Account numbers are reissued, migrated and inherited through acquisitions. A surrogate key with no business meaning, plus the business identifier as a dated attribute, is the only shape that survives a merger.

The same logic applies to the card number. Under PCI DSS the primary account number may be stored only if rendered unreadable, and sensitive authentication data — full track data, the card verification values, the PIN block — must not be stored after authorisation at all, even encrypted. The current version is v4.0.1, and the future-dated v4 requirements became effective after 31 March 2025 [2]. So the PAN is not available as a key, a join column or a search field. Card identity needs a surrogate plus a truncated PAN for human recognition, and tokens need their own entity with their own scope and lifecycle — because a reissued card with a new PAN must keep its device tokens working, which is impossible if the token is derived from the PAN.

> In a South African bank, assume no identifier is globally unique, self-validating or permanent until somebody shows you the standard that makes it so. Surrogate key inside, business identifier as a dated attribute, verification as a service.

## Classification: the codes that decide what the return says

Classification is where an information model stops being a modelling exercise and becomes a regulatory one.

The monthly BA 900 return requires every balance to be classifiable, simultaneously, by **counterparty institutional sector**, **instrument type**, **currency** and **residual maturity**, at nominal or market value on an accrual basis, feeding the national monetary and credit aggregates [3]. Four governed dimensions, monthly, reconciled to the general ledger. If they are not governed attributes in the product and party masters, the return gets assembled in spreadsheets — and because it feeds published national statistics, the errors are visible outside the bank.

Cross-border reporting wants a different cut of the same thing: the Financial Surveillance department's reporting system requires an institutional sector code and an industrial classification on the resident customer, a country code on the non-resident, and a balance-of-payments category on the transaction [4]. The category is a business classification that no payment system can infer. It has to be captured at origination or derived by rule, and then validated.

Three further classification sets a South African bank must carry, each owned somewhere different:

| Scheme | What it classifies | Who needs it | Where it goes wrong |
|---|---|---|---|
| Institutional sector and industrial classification | The counterparty | BA 900, FinSurv | Captured at onboarding and never refreshed |
| Instrument, currency, residual maturity | The balance | BA 900 | Maturity recomputed differently by two systems |
| Deposit-insurance qualifying status | The product | CoDI | Set on the account, so wrong on the back book |
| National Credit Act product subtype | The credit agreement | NCR, fee and interest caps | Marketing names mistaken for subtypes |
| Prescribed retail complaint category | The complaint | FSCA conduct reporting | A free-text "other" bucket that swallows a third of the volume |

The architectural rule is the same in every case. **A classification is a governed code list with an owner, a version and an effective date, not a dropdown.** When the scheme changes — and the Banks Act Regulations were amended by Government Notice 5802 of 31 January 2025 so that the return forms are no longer in the Regulations but are determined by the Authority in writing [5] — the bank has to be able to reclassify a back book and keep both the old and the new classification of a historical period. That is a versioning requirement before it is a reporting one.

## Ownership and lineage

Two words that get written on slides and not implemented.

**Ownership** means a named person who decides what an attribute means and who may change it, and who is asked when two systems disagree. Not a committee, not a function, a person. The test of whether ownership is real is whether the owner has ever refused a change.

**Lineage** means that every number on a return can be traced to the entries and reference data behind it, through a derivation whose version is recorded. Not a diagram of system-to-system arrows, which is the thing most banks have. A lineage that cannot answer "this figure, on this line, of this return, for March, came from these rows through this version of this calculation" is documentation, not lineage.

Two South African obligations make attribute-level lineage non-optional. POPIA gives a data subject a right to correction, and a correction has to reach every copy — which means the party master has to know which source contributed which value. And the deposit insurer's reporting process builds resubmission in: it notifies banks of discrepancies and may request a resubmission, and a bank may request one itself [6]. You cannot correct a submission you cannot trace.

There is a practical test for both, and it takes an afternoon. Pick one line on last month's BA 900. Ask who owns the definition of that line. Ask which systems contributed to it, at what grain, through which transformation, and in which version of that transformation. Then ask to see the reconciliation between it and the general ledger, and who signed it. Four questions. In most banks the first two have answers, the third produces a spreadsheet and the fourth produces a name and an email thread. The gap between the second and the third question is the whole of the risk-data remediation backlog.

The reason it is hard is not technical. It is that lineage is captured as a by-product of building pipelines, and pipelines are built by projects, and projects end. Lineage that is not generated by the thing that moves the data will be out of date within two releases. An architect's practical contribution here is to insist that every derivation used in a return is a named, versioned artefact under change control — a view, a job, a rule set, with an identifier that the submitted figure carries with it. That single rule converts lineage from documentation into data.

#### Retention: the longest clock wins

A bank's retention schedule is not one period. It is the maximum of several statutory clocks that run on the same record, and the longest of them is longer than most architects expect.

Financial Surveillance requires integrated forms and related communications for five years [4]. The Conduct Standard requires advertising records for at least five years [8]. FICA requires customer due diligence and transaction records producible on request [7]. And **prescription** — the period after which a debt can no longer be enforced — runs three years for ordinary unsecured debt, six years for a bill of exchange or notarial contract, and **thirty years for mortgage-secured debt and for judgment debt**, running from when the debt is due, interrupted by acknowledgment of liability or by service of process [23].

Thirty years is the number that should change a design. It means the signed home-loan agreement, the security documents, the section 129 notice and its proof of delivery, and the initial-recognition credit state of the exposure all have to be retrievable under evidential conditions three decades after origination — across at least two core replacements and probably three document stores. Applying a uniform retention period at the shortest statutory clock, which is the common shortcut, destroys evidence the bank will need. Applying the longest to everything is lawful and expensive and collides with POPIA's requirement not to keep personal information longer than necessary. The only correct answer is a retention schedule keyed to record class, with the class on the record.

### Time: three dates, not two

An insurer needs two time axes. A bank needs three, because the economic date and the accounting date are themselves different facts.

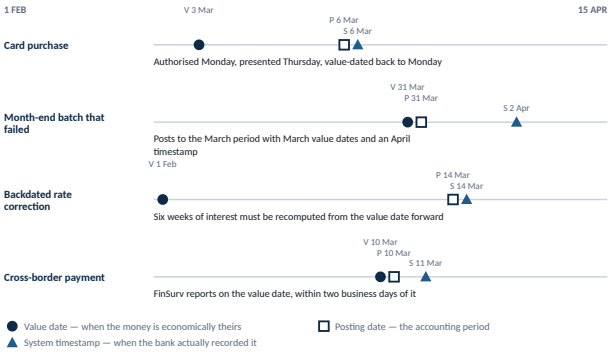



**Figure 10.2** Three dates on every entry, diverging legitimately. The backdated correction is the hard case: six weeks of interest has to be recomputed from the value date forward, which is only possible if daily accruals were retained.

**Value date** is when the money is economically the customer's or the bank's. It drives interest accrual, the available balance, the exchange rate applied and cross-border reporting. **Posting date** is when the entry is recognised in the books, in an accounting period; it drives the general ledger, the income statement and period close. **System timestamp** is when the bank actually recorded it; it drives audit, disputes and operational service levels.

They diverge routinely and legitimately, which is the part that surprises people. A card transaction authorises on Monday, presents on Thursday and values back to Monday. A month-end batch that fails and runs on 2 April posts into the March period with 31 March value dates and 2 April timestamps. The Financial Surveillance reporting rules make the distinction operational: the reportable field is the **value date**, submission is due within two business days of the value date, and value dates more than fifteen days old raise warnings [4].

Every entry carries all three. A model with one `date` column cannot compute interest, cannot close a period and cannot answer a dispute. This is not a nicety; it is the single most common structural defect in a data warehouse built by people who have not worked in a bank.

#### Backdating

Posting today with a value date in the past. It is normal — a late rate change, a fee waiver, a corrected deposit, a late-presented instrument — and it cascades.

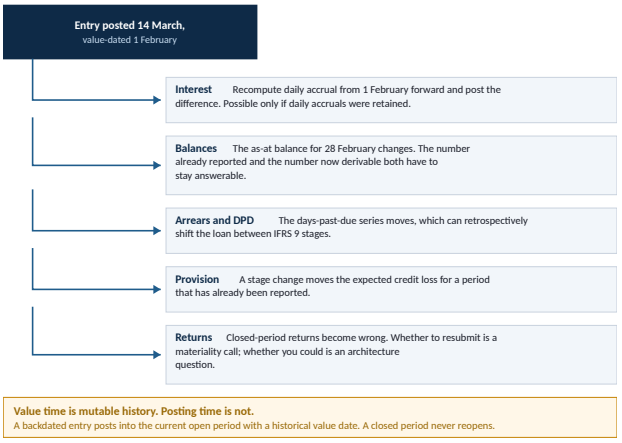



**Figure 10.3** What one backdated entry sets off. Each consequence is survivable; the combination is what makes an unbounded backdating window dangerous.

Banks bound the damage in two ways. A **backdating window** per product, with a mandate level required to exceed it. And a hard rule that a closed accounting period never reopens: a backdated entry posts into the current open period carrying a historical value date.

That asymmetry is the sentence to remember. **Value time is mutable history. Posting time is not.** Everything else about a bank's temporal design follows from it.

### Reversals, contra entries and corrections

An entry is never edited and never deleted. A mistake is undone by a **contra entry**: equal and opposite, referencing the original, posted on the date of correction, with the original left visible.

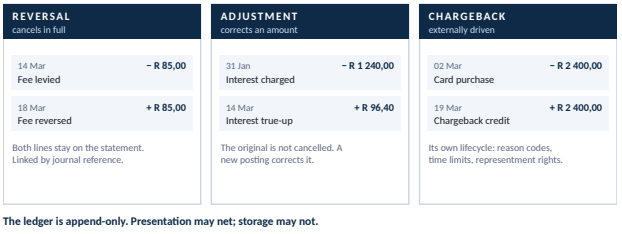



**Figure 10.4** Three correction mechanisms that get conflated. Both lines stay visible in all three, because the statement is evidence.

Three mechanisms get conflated, and the conflation shows up as a complaint. A **reversal** cancels an erroneous posting in full and both lines stay on the statement. An **adjustment** corrects an amount without cancelling the original — an interest true-up after a backdated rate change is the standard case. A **scheme reversal, chargeback or return** is externally driven, with its own reason codes, time limits and representment rights; it is a dispute object with a lifecycle, not an internal correction.

Customers read statements and so do ombuds. Suppressing the original and its reversal destroys the evidence that the bank behaved correctly. Net in the presentation if you must. Never net in the store.

### The restatement problem

Restatements arise from backdating, from late data, from model changes — a recalibrated probability-of-default model changes last year's expected credit loss — from classification corrections, and from accounting policy changes. The difficulty is that two entirely legitimate questions have different right answers.

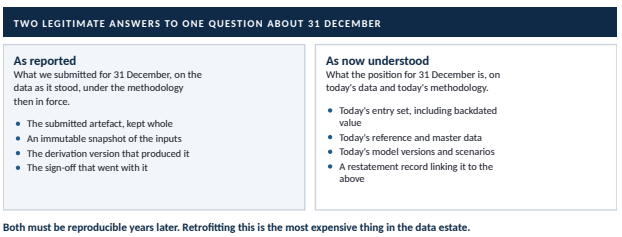



**Figure 10.5** The restatement problem. Two legitimate answers, both reproducible years later, and a record that links them.

Supporting both requires four things that are cheap to build at the start and brutal to retrofit: immutable submission artefacts; immutable period-end snapshots of the underlying data; **versioned derivation logic**, meaning the code and the parameters that turned data into a return, kept with the return; and a restatement record that links the two answers and says why they differ.

In my experience this is the most commonly missing capability in a bank's data estate and the most expensive to add late — a judgement rather than a sourced claim, though the scope of South African banks' risk-data remediation programmes would settle it either way.

#### Why history must be reconstructible

Not as a principle. As a list of specific obligations.

- **BA returns** are submitted per period and are subject to the Authority's requirements on accuracy and revision [3][5].
- **FICA** requires customer due diligence and transaction records producible to the Financial Intelligence Centre on request [7].
- **Financial Surveillance** requires integrated forms and related communications to be retained for five years [4].
- **The Conduct Standard** requires advertising records for at least five years [8].
- **IFRS 9** requires an unbiased, probability-weighted expected credit loss as at the reporting date, reflecting reasonable and supportable forward-looking information [9] — evidenced only if the inputs as at that date survive. That means the **initial-recognition credit state** of every exposure has to survive the life of the exposure, because a significant increase in credit risk is measured relative to origination. For a twenty-year mortgage that is a twenty-year retention requirement on a field most core systems overwrite.
- **Resolution.** If the bank fails, the deposit insurer must compute every depositor's cover from the data as at a point in time [1].

#### How this differs from an insurer

An architect who has done insurance will reach for the bitemporal pattern — valid-from and valid-to plus recorded-from and recorded-to on every versioned entity — and it is the wrong first move here.

| | Insurer | Bank |
|---|---|---|
| Dominant correction mode | Retroactive amendment of a *state*: an endorsement changes the policy as it validly stood | Contra posting against an immutable *event* stream |
| What is versioned | Policy, cover, sum insured, party roles — long-lived states | Rate, limit, mandate, product version, party attributes. Entries are not versioned at all |
| Finest temporal grain | Monthly, in policy periods and accident years | Daily, and intraday for payments and treasury |
| Estimate-heaviness | Extreme: reserves dominate the balance sheet | Moderate: most of the balance sheet is contractual, with estimates layered on |
| Who asks "as at" | Actuaries and reinsurers, quarterly | Finance daily, treasury intraday, risk daily, the regulator monthly, the resolution authority potentially overnight |

So the insurer's problem is solved with bitemporal versioning on entities. A bank's is solved with three things working together: an **immutable, triple-dated entry ledger**; **versioned reference and master data** around it; and **immutable period-end snapshots** over the top. Effective-dating the transaction table doubles the model for nothing and slows the daily close. Use bitemporal versioning where a bank genuinely has long-lived states — pricing, limits, mandates, product versions, party attributes — and leave the entries alone.

Four implementation traps are worth naming, because each of them is discovered late.

**The snapshot that is not immutable.** A period-end snapshot stored in a table that a reload job can rewrite is not a snapshot. It needs to be write-once, with the job that produced it recorded, and a control that proves it has not changed — a hash is enough and is rarely done.

**The reference data that is not versioned with the entry.** An entry posted in March, classified under the sector code list as it stood in March, re-reported in July under the July code list, will produce a different BA 900 line. Either the entry carries its classification as at posting, or the code list is versioned and the derivation picks the version in force — but something has to hold the date.

**The daily accrual nobody kept.** Interest is computed daily and posted periodically, and a backdated entry can only be interest-corrected if the daily accrual record survives. Banks that store only posted interest discover this during the first large backdating incident, and the reconstruction is manual.

**The derivation that lives in a report.** A figure computed inside a reporting tool, by a formula in the tool, cannot be versioned, cannot be tested and cannot be shown to a supervisor. If a number appears on a return, the arithmetic that produced it belongs in a governed artefact upstream of the tool, and the tool should only render it.

### Reference and master data

The quiet layer. Nobody funds it, everything depends on it.

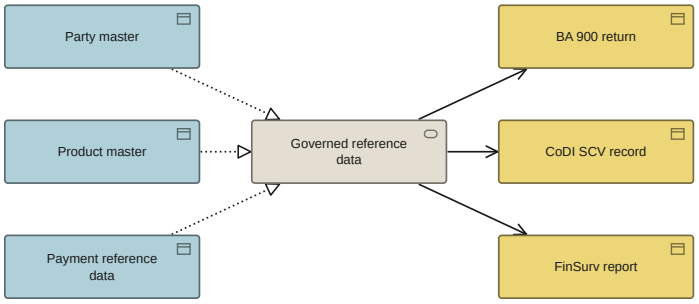



**Figure 10.6** Reference and master data as a governed service rather than three tables. The three returns on the right are different cuts of the same classification, which is why the masters cannot be owned independently.

**The customer master** holds identity, not relationships. Party type, legal name and known-as names, the thirteen-digit identity number or a passport and country or a CIPC registration or trust number, dates of birth or registration, country of residence and of tax residence, and the bank's own surrogate identifier — plus match keys, survivorship rules, source-record links and attribute-level lineage. It must not hold product relationships, balances or "customer status". Those are derived, and the moment they are stored in the master, the master becomes a reporting artefact with a nightly latency.

The South African hard parts are specific. Name order and multiple surnames. The identity number's embedded birth date, gender and citizenship digits, which tempt a modeller into deriving facts from a field that is sometimes wrong. Non-residents with no South African identity number. And the Common Monetary Area — Lesotho, Eswatini, Namibia — which is neither domestic nor fully foreign for reporting purposes [4].

Matching deserves a paragraph of its own, because this is where a party master is won or lost. Matching on identity number alone looks attractive and fails on three populations at once: foreign nationals, records created before the current identity regime, and anybody whose number was mistyped at a branch in 1998 and never corrected. Matching on name and date of birth over-merges. The working pattern is a scored match over several keys with three outcomes rather than two — automatic merge, automatic reject, and a referred queue worked by people — plus **survivorship rules** that say which source wins per attribute rather than per record, and a **reversible merge**, because the one thing worse than two records for one person is one record for two people. Reversibility is the requirement nobody writes down and everybody needs: a merged party must be separable, with the history going back where it came from, which means the merge is itself a dated event and not a deletion.

Two obligations make the whole thing non-optional rather than merely desirable: FICA's ongoing due diligence, which needs one view of a client's risk across every product they hold [7]; and the deposit insurer's single customer view, discussed below [1].

**The product master** holds versioned product definitions plus the reference sets a product must map to: institutional sector and industrial classification, instrument and maturity breakdown, deposit-insurance qualifying status and credit product subtype. Note the subtlety: sector and industry are *counterparty* classifications applied through the product's reporting line, so the product master and the party master jointly determine the return. Neither owner can fix a BA 900 problem alone.

**Branch codes.** Historically every physical branch had its own six-digit code, and an electronic transfer needed the code of the branch where the account was opened. That was a data-quality disaster, because customers move and branches close. Banks therefore issued **universal branch codes**: one code per bank that works for every account at that bank. FNB's is 250655, and each major bank publishes its own [10]. I could not source when each universal code was introduced, whether physical branch codes remain valid on all streams, or whether any stream still requires the specific branch code; the clearing-house rules and the utility's participant specifications would settle it.

Two things follow for the model. A branch is an organisational unit with a location; a branch code is a routing identifier; and they are two entities, because a branch can close while its code survives and a universal code belongs to no physical branch at all. And a bank maintains several branch identifier schemes at once — the six-digit domestic routing code is not the **eight-digit** branch code the Financial Surveillance reporting system requires on cross-border transactions, along with branch name, hub code and hub name [4].

**BIN and IIN tables.** The issuer identification number is the leading digits of a card number, historically six and migrated to eight. A bank maintains the ranges it issues on and tables for every range it acquires on, with scheme, country, card type, product, debit or credit or prepaid, regulated status and funding source. That data drives routing, interchange, fee calculation, fraud rules and decline reasons. I could not source current migration dates or status.

**Country and currency.** ISO 3166 and ISO 4217 sound trivial and are not, because a bank needs several *country* roles for the same party and transaction: residence, citizenship, incorporation, tax residence, the counterparty's bank, the beneficiary. Cross-border reporting requires a country code on the non-resident and prohibits using the South African code there except in certain card cases — and, counter-intuitively, rand values are **not** calculated by the reporting entity: the bank reports the foreign amount and the Reserve Bank applies the mid-rate [4]. A system that helpfully pre-converts has broken the return.

**SWIFT BIC and ISO 20022.** The business identifier code identifies a financial institution in eight or eleven characters and is the correspondent-banking routing key [4]. ISO 20022 has replaced the MT message set for cross-border payments, with the coexistence period ending in November 2025 [11][12]. The architectural consequence is not the format. It is the content: ISO 20022 carries *structured* party data — name, structured address, identifiers, purpose codes, remittance information, ultimate debtor and ultimate creditor. A payment store that flattens the counterparty to a free-text name line can neither populate an outbound message nor use an inbound one. That is a data-model problem wearing a messaging-standard costume, and it is the single most common reason a payments modernisation runs late.

**The proxy registry.** PayShap, South Africa's rapid payments service, launched in March 2023 [13]. Its architecturally significant feature is the **proxy**: a mobile number or alias that resolves to an account, held in a central registry, so a payer can pay to a phone number without an account or branch code. A

proxy registry introduces entities a bank does not otherwise have — the proxy itself, with a value, type and status; its dated **binding** to an account with evidence of ownership; a **resolution event**, which is a privacy-sensitive disclosure of the payee's name and therefore has to be logged; and **portability** between banks. Governance of the relevant clearing houses moved to the renamed national utility during 2026 [14]. I could not source the supported proxy types, the registry's governance or whether portability is live.

## Classification, consent and POPIA

POPIA is usually introduced as a privacy statute with a compliance checklist. For an architect it is better read as a set of data requirements, three of which change the model.

### Section 71: a fully automated decline is a regulated act

Section 71 says a data subject may not be subject to a decision with legal consequences, or affecting them substantially, based **solely** on automated processing intended to create a profile — and it names profiling of **creditworthiness** explicitly. The exceptions apply where the decision is made in the course of concluding a contract with appropriate measures protecting the data subject's legitimate interests, including an opportunity to make representations, and where sufficient information about the underlying logic is given [15].

Read that as an architect rather than a lawyer and it says four concrete things.

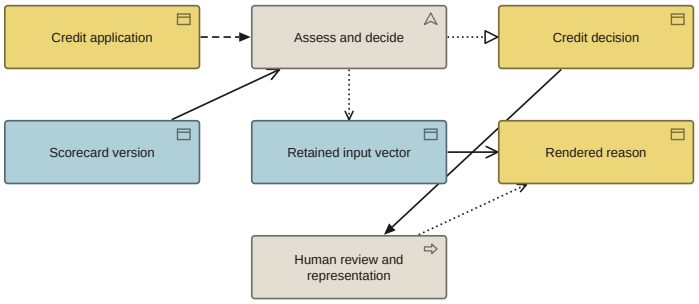



**Figure 10.7** POPIA section 71 as an architecture. The retained input vector is the load-bearing element: without it the reason given to the applicant cannot be shown to be true.

There must be a **human review path** that is real rather than nominal. There must be a **representation mechanism** the applicant can use. There must be a **renderable reason**, in language a person can act on. And the reason has to be *true*, which means the inputs that produced the decision must have been retained — the scorecard version, the strategy version and the exact input vector, not just the score.

Machine-learning scorecards are not prohibited by section 71. Opacity is the problem, not the technique. A gradient-boosted model whose inputs and version are retained, whose reasons are derived and recorded at decision time, and which has a human review path behind it, is in a better position than a logistic scorecard whose input vector was discarded after the run.

The National Credit Act arrives at a similar place from a different direction: it requires reasons for a decline, and it requires the affordability assessment — the income evidence, the expense inputs, the prescribed norm-table version, the bureau enquiry and the calculation — to be retained per application, **including declines** [16]. Between the two statutes, the origination chain becomes a regulated record in which nothing may be overwritten.

### Transfers, consent and marketing

Section 72 allows transfer to a foreign recipient only on listed bases: a law, binding corporate rules or a binding agreement providing adequate protection; the data subject's consent; necessity for a contract with them or in their interest; or their benefit where consent is not practicable [15]. The practical consequence is a **data inventory keyed to purpose, category, legal basis, retention and location** — and the prudential and conduct regulators issued a joint communication on cloud computing and data offshoring in July 2025 [17], so the inventory now has two audiences.

Section 69 restricts unsolicited electronic direct marketing, and the Conduct Standard adds that opt-outs must be honoured at no charge [8]. Together they make marketing preference **party-level master data**, dated, per channel, per purpose — not a flag on a campaign list. A bank with consent captured in four channel systems cannot evidence either obligation.

And one divergence worth repeating from chapter 9: POPIA protects **juristic persons** as data subjects. A corporate client has access and correction rights that a GDPR-shaped design will not have planned for.

## CoDI's single customer view: a regulation that is really a data model

If you take one worked example from this chapter, take this one. It is the clearest case in South African financial regulation of a regulator specifying a data structure rather than a principle — and in my view it is the most under-appreciated data requirement in South African retail banking.

The Corporation for Deposit Insurance became operational on 1 April 2024. Cover is **R100 000 per qualifying depositor per bank**, computed on the total of that depositor's qualifying balances at that bank, gross, with negative balances not netted, and separate per bank [18]. All registered banks are members.

The obligation that matters to an architect is not the cover. It is the **single customer view**: the bank must consolidate each qualifying depositor's accounts and balances in qualifying products into one SCV record, with the covered deposit being the qualifying balance capped at R100 000. The published rules

then specify, in detail that reads like a data dictionary [1]:

- **Joint accounts** split by the **sharing ratio recorded by the bank**, defaulting to equal shares where none is recorded.
- **Formal beneficiary accounts** — an attorney's trust account — looked through to each beneficiary, with a record per beneficiary *if the bank can identify all beneficiaries and their balances monthly*, otherwise one record for the account with no split. Beneficiaries that are themselves financial businesses are not reported.
- **Informal beneficiary accounts** — a stokvel — covered per holder, upgraded to the formal treatment if the bank holds the constitution and member details monthly; and where such an account has several sets of signatories, the R100 000 is split pro rata by each account's share.
- **Sole proprietors** covered separately for personal and business accounts, each to R100 000, as separate SCV records.
- **Balance rules**: gross; interest accrued to the reporting date included; outstanding account fees excluded; unsettled payments added or deducted; foreign currency at the close-of-business rate on the reporting date; two decimals; a negative balance treated as a zero qualifying balance.
- **Flags**: pledged accounts reported at full balance with the unpledged portion as the qualifying balance and marked not readily payable; deceased depositors likewise flagged.
- **A mandated field set**: holder identifier, depositor type, product name and type, account number, account balance, qualifying balance, joint and pledged flags, contact details, a reliability-of-identification flag, signatory details for informal beneficiary accounts, and the not-readily-payable indicators with their reasons.

Here is what that arithmetic does to one depositor.

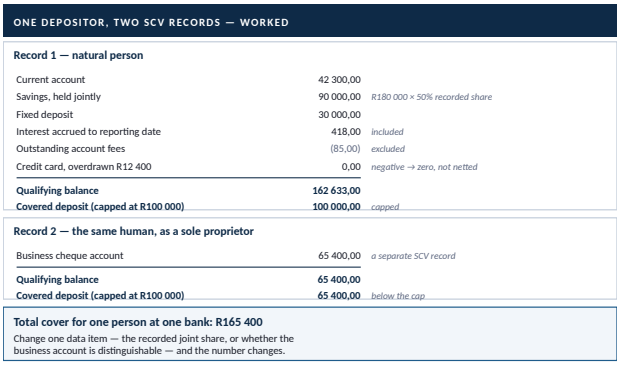



**Figure 10.8** The deposit insurer's aggregation rules applied to one invented depositor. Two SCV records, R165 400 of cover, and every line of it determined by a data item the bank may or may not have captured.

Excluded depositors are specified too, and one exclusion is a pure architecture requirement: the failing bank's own **executive management, board, their related parties and its lead external audit partners** are not covered [18]. To compute that, depositor data must be able to identify which depositors are those people — which is a dated party-to-party relationship, not a flag.

The submission looks trivial and is not. Monthly totals go in as a four-field CSV — member bank code, reporting date, total qualifying deposits, total covered deposits — due within ten working days after month-end, with compliance declarations in prescribed windows [6]. Quarterly SCV calculation submissions were exempted from 1 April 2024 to 30 September 2025, during which banks still had to produce the calculations internally because they generate the monthly totals [6]. **What regime applies after 30 September 2025 — whether full SCV record files are now submitted periodically, and in what format — I could not source**, and it is the open question most worth checking before you scope a programme around it. The deposit insurer's current reporting requirements document would settle it.

> The single customer view is a single customer view with a **legal definition and a pass/fail test**: computed monthly, auditable, and executable under resolution conditions at short notice. It forces a resolved party identity across every deposit-taking system, a product-master qualifying flag, a joint-account sharing ratio, beneficiary data for trust and stokvel accounts, contactability, an identification-reliability measure per party, and reproducible as-at calculation. A bank with five customer masters cannot produce it, and the regime has already been used in anger — a co-operative bank was placed in resolution and eligible depositors were paid within the prescribed timeframe [19].

Treat it, therefore, as the acceptance test for the whole information model. If the bank can produce the SCV on demand, from mastered data, with lineage and attestation, then party identity, product classification, balance definition and as-at discipline are all working. If it is produced as a reporting-layer query each month-end, none of them is.

## The reporting and analytics layer

Three layers get built as one, and then one of them fails.

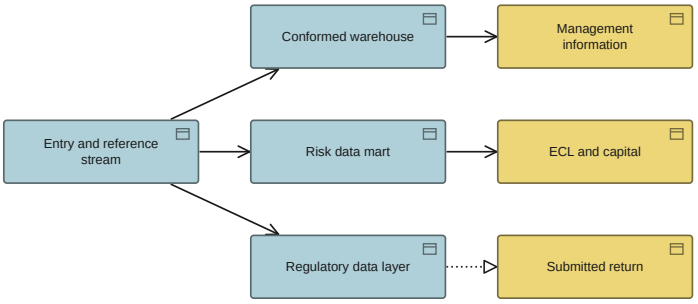



**Figure 10.9** Three layers with three different integrity requirements. The warehouse may be rebuilt; the regulatory layer may not. One store cannot be both.

The **data warehouse** holds integrated history for management information, finance and analytics, on conformed dimensions — party, product, account, general ledger, date, channel, branch — with facts at entry grain or daily-balance grain. The **risk data mart** holds exposure-grain data for credit, market, liquidity and operational risk: probability of default, loss given default, exposure at default, ratings history, collateral, stage, and the daily days-past-due series. The **regulatory data layer** holds return-line grain, plus the immutable submission and the derivation version.

They differ because their *integrity requirements* differ, and that is the whole point. **The warehouse may be rebuilt. The regulatory layer may not.** A warehouse is rebuilt whenever a dimension changes, which is normal and healthy. Run the regulatory layer out of the same store and the rebuild destroys the evidence of what was submitted. One physical store cannot be both rebuildable and immutable.

The other recurring failure is subtler: letting the warehouse master something operational — a customer segment, a risk grade, a limit, a collections treatment. The moment it does, the overnight batch latency becomes an operational latency, and a customer is treated today on the basis of what was true yesterday.

### BCBS 239

The Basel Committee's *Principles for effective risk data aggregation and risk reporting* was published in January 2013 and sets fourteen principles in four groups, with global systemically important banks expected to comply by January 2016 [20][21]. Successive progress reports have found adoption incomplete [22].

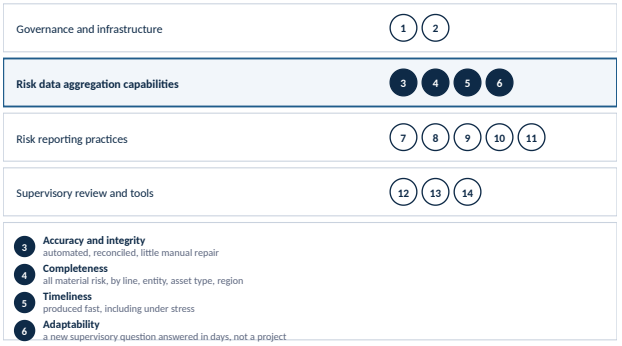



**Figure 10.10** BCBS 239's fourteen principles. The four highlighted are the aggregation capabilities, and adaptability is the one banks fail: the monthly pack is produced, the unexpected question is not answered.

The four aggregation principles are the architect's accountability, and the fourth is the one that fails. **Accuracy and integrity** means largely automated aggregation, reconciled to source, with minimal manual intervention. **Completeness** means all material risk across the group, broken down by business line, legal entity, asset type, industry and region. **Timeliness** means rapid production including under stress. **Adaptability** means an ad hoc supervisory request can be answered without a project.

Most large banks can produce their monthly pack. Very few can answer a question they have not been asked before within days. That gap is why BCBS 239 remediation sits in every large bank's change portfolio, usually under a different name, usually for longer than planned.

Whether BCBS 239 is formally adopted into South African law for all banks, or applied supervisorily to domestic systemically important banks, I could not source; the Regulations relating to Banks and the Prudential Authority's directives would settle it. The obligation is real either way, because the Authority's expectations around returns accuracy and model governance land in the same place.

## What "single version of the truth" has to mean

The phrase is on the first slide of every data strategy and it is almost always used to mean one database, which is wrong, or one number, which is worse. In a bank that reports to a regulator it has to mean four things.

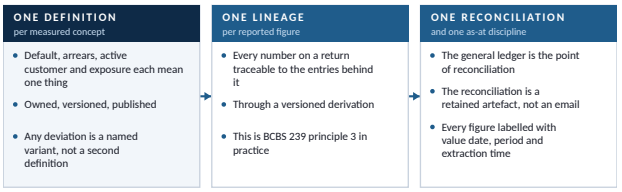



**Figure 10.11** What the phrase has to mean in a bank. Note that none of the four is a statement about where data is stored.

**One definition per measured concept.** Default, arrears, active customer and exposure each mean one thing, owned, versioned and published. Any deviation is a named variant, not a second definition. The regulatory and the accounting definitions of default are genuinely different, and both must be derivable from the same base — which is a variant, declared.

**One lineage per reported figure**, through a versioned derivation. That is BCBS 239's accuracy principle expressed as a build requirement.

**One point of reconciliation**, which is the general ledger, with the reconciliation itself a retained artefact rather than an email.

**One as-at discipline.** Every figure labelled with its value date, its posting period and its extraction time, so that two differing numbers can be shown to differ *because they are as at different times* — which is the single most useful sentence an architect can put into a disagreement between finance and risk.

What it cannot mean is one physical store. The regulatory layer's immutability and the warehouse's rebuildability are incompatible in one place, and a strategy that promises both is promising to fail at the first restatement.

Chapter 11 moves up a layer again, from what the bank must know to what the bank's applications must do about it.

### THE CHAPTER IN TEN LINES

- The information model is the layer where attributes, keys, classification, ownership, lineage and time are settled, and it is where most of a bank's data defects actually live.
- An attribute earns its place by having an owner, a correction path, and a clear answer to whether it is an observation or a dated estimate.
- There is no South African domestic IBAN: account numbers are bank-specific, check digits are not universal, and payee verification is an interbank service, not a local computation.
- A bank needs three time axes, not two — value date, posting date and system timestamp — and a model with one date column can neither compute interest nor close a period.
- Value time is mutable history; posting time is not. A backdated entry posts into the current open period carrying a historical value date, and a closed period never reopens.
- Supporting both 'as reported' and 'as now understood' needs immutable submissions, immutable period snapshots, versioned derivation logic and a restatement record.
- A bank's temporal answer is an immutable triple-dated entry ledger plus versioned reference data plus period snapshots — not the insurer's bitemporal pattern applied to transactions.
- POPIA section 71 makes a fully automated credit decline a regulated act, which is a data requirement — retained inputs, a renderable reason, a human review path — before it is a legal one.
- CoDI's single customer view is a regulator-specified data structure with aggregation rules, joint-account shares, beneficiary look-through, a mandated field set and a monthly submission; treat it as the acceptance test for the whole information model.
- Single version of the truth means one definition, one lineage, one reconciliation point and one as-at discipline — never one physical store, because immutability and rebuildability cannot share a database.

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CHAPTER 11

# The conceptual application model

*Chapter 12 names the systems a bank runs. This is the layer above: the twenty-five things a bank's applications have to do, named by what they do, with nobody's product name anywhere on the page — and the five that are missing almost everywhere, for the same reason.*

## The model you draw before anyone has chosen a vendor

Chapter 12 names the systems a bank runs. This chapter is the layer above it: the applications a bank *needs*, named by what they do, with nobody's product name anywhere on the page.

It is the artefact you draw when you are designing a target rather than describing an estate, and it is the one most often skipped — because an estate is visible and a target is an argument. The cost of skipping it shows up later, in a procurement where three vendors answer three different questions, or in a core programme whose scope is a list of systems rather than a list of capabilities.

### Three artefacts that get confused

A great deal of wasted meeting time comes from mixing these up.

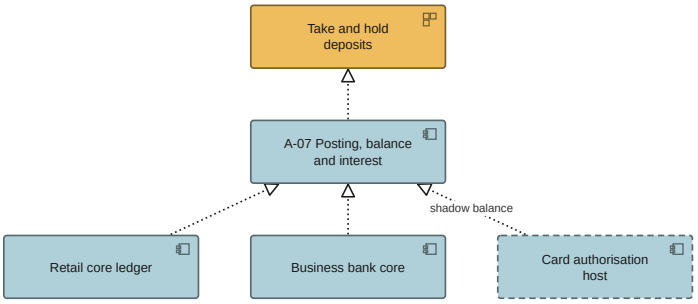



**Figure 11.1** Three layers, with one capability threaded through them. Realisation runs upward and is many-to-many: the card authorisation host realises part of A-07 with a shadow balance that will never equal the ledger's.

A **business capability** is something the bank can do: take and hold deposits, lend to a consumer, move money for a client, report to the Prudential Authority. It is independent of how, and it changes about once a decade.

A **conceptual application capability** is a coherent lump of application function that realises part of one or more business capabilities. "Posting, balance and interest" is one. It is independent of *who supplies it*, and it changes when the industry changes — ISO 20022 added one and the deposit insurance regime arguably added another.

A **system** is a thing with a licence, a version, a support contract and somebody's logo on it. It changes when a contract ends.

The relationship between the layers is realisation, and it is many-to-many in both directions. One conceptual capability is realised by three systems; one system realises eleven conceptual capabilities, which is precisely what a monolithic core is. Draw the three layers on separate pages and the conversation about replacement becomes tractable. Draw them on one page and you will produce something nobody can use.

Table Bay Bank — invented, mid-tier, South African, about 2.4 million retail customers, a business bank, a small corporate and transactional arm and a treasury — is the worked example at the end.

## What goes in a box, and what does not

Four tests, and a rule about size.

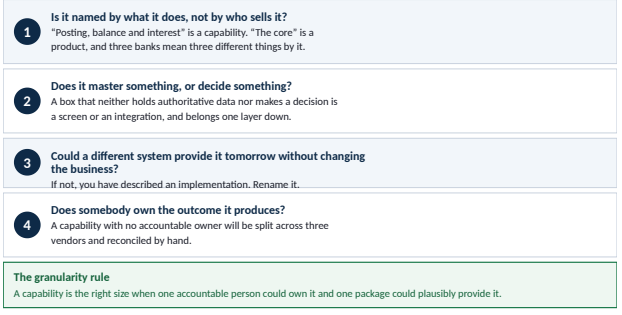



**Figure 11.2** Four tests for what goes in a box, and the rule that settles how big a box should be.

The granularity rule in the figure is doing more work than it looks. Too coarse and the model says nothing: a box called "Core banking" is a vendor category, not a capability, and it will hide the three things inside it that are actually the problem. Too fine and you have drawn a service catalogue: "beneficiary maintenance" is a feature of payment orchestration, not a capability, and a model with four hundred boxes cannot be used to scope anything.

Twenty-five is the right order of magnitude for a full-service bank. A niche lender needs about nine of them. A tier-one universal bank needs all twenty-five and several instances of some.

#### Three things that are not application capabilities

These three appear on first drafts, and each of them makes the model worse.

**"Digital."** It is a set of channels, a delivery style and sometimes a department. It is not something the bank does that another box does not. A model with a "Digital" box has a box that overlaps fifteen others, and the first time you use the model for scoping, somebody will claim that everything is in it.

**"Integration."** Integration is how capabilities reach each other. It is real, it is expensive, and it belongs in the technology architecture, not here. A box called Integration on a conceptual application model is an admission that the boundaries between the other boxes were not thought through — because the integration requirement *is* the boundary.

**"AI."** The same objection, with more funding behind it. Machine learning is a technique used inside A-13, A-19, A-21, A-22 and A-23. It masters nothing of its own and it decides nothing the hosting capability was not already accountable for. A box called AI produces the worst kind of target architecture: one where the governance question — which decision, whose model, what reason given to the customer — has no owner, which is exactly the question POPIA section 71 asks [1].

> If a vendor's architecture diagram has boxes called Digital, Integration and AI, you are looking at a marketing artefact. Ask which of the twenty-five each product actually masters, and watch the answer change shape.

## The twenty-five

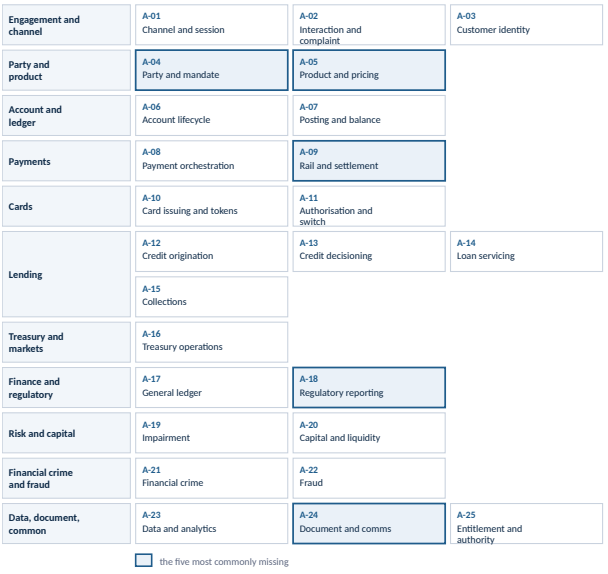



**Figure 11.3** Twenty-five conceptual application capabilities in eleven domains. The grouping is for reading; several capabilities serve more than one domain. The five outlined are the ones most often missing.

#### Engagement and channel

**A-01 Channel experience and session** presents the bank through one channel and holds the authenticated session. Seven channels in South Africa, each with its own session model, latency budget and failure mode: branch, ATM and self-service, internet banking, the app, USSD, chat, and the corporate API. USSD is not a legacy curiosity here — it is synchronous, stateful, low-latency and carries a tiny payload, which is the opposite of what an API-first design assumes.

**A-02 Interaction, service and complaint management** records what was said, works the service request and categorises the complaint. The complaint part is not a CRM feature. The Conduct Standard for Banks requires prescribed categories for retail complaints, acknowledgement and progress updates, free access and complaints-data monitoring, with reporting to the FSCA [2]. Model it as a capability and the categories become a governed code list. Model it as a CRM case type and the return gets built in a spreadsheet.

**A-03 Customer identity and authentication** registers a customer, issues and verifies credentials, binds devices and steps up when risk demands. It is separate from A-04 because identity-as-credential and identity-as-party are different problems with different owners, and conflating them is why the party master ends up holding passwords.

There is an asymmetry in this group worth stating once and remembering. A bank can make A-03 arbitrarily strong and still lose a great deal of money, because in the dominant South African fraud pattern the customer authenticates correctly and approves the payment themselves, on the criminal's behalf

[3]. Strengthening A-03 is necessary and is not sufficient, and a target architecture that puts all of its fraud response there has misread the problem.

### Party and product

**A-04 Party master, relationship and mandate** holds one identity per legal person, plus the dated relationships and authorities over accounts. This is the capability with the widest fan-out in the bank.

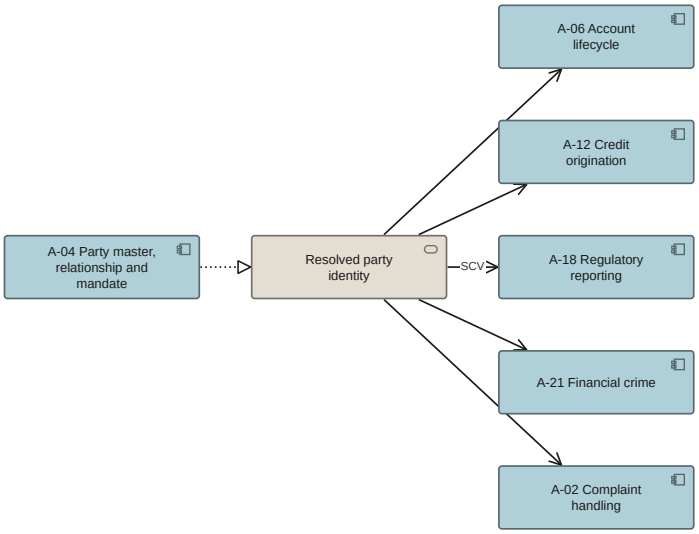



**Figure 11.4** Why A-04 is the capability with the widest fan-out. Five others consume a resolved party identity, and one of them — the deposit insurer's single customer view — has a legal pass/fail test attached.

**A-05 Product definition, catalogue and pricing** defines what a product is, versions it, prices it and publishes the definition everywhere else. Almost no bank has this as a single capability, and the consequences are chapter-12-length.

### Account and ledger

**A-06 Account lifecycle and servicing** opens, maintains, blocks, makes dormant and closes the customer-facing account. Dormancy is a regulated behaviour, not a housekeeping routine: the Conduct Standard requires a bank to identify dormant products and notify the customer [2].

**A-07 Posting, balance and interest** posts balanced entries carrying all three dates, accrues and capitalises interest by product rule, and holds the balance. This is the irreducible centre of a bank. Everything chapter 10 said about value date, posting date and immutability lands here.

Note what A-07 is not. It is not "the core". That word, in most banks, names a system that happens to contain A-05, A-06, A-07 and a good deal of A-04, which is why core replacement is so hard and why the honest first question in any core conversation is really four questions: where is the authoritative balance, where is product definition, where is the authoritative party identity, and where does end-of-day run. Four answers, often four systems, often not the one with core in its name.

Splitting A-06 from A-07 matters for the same reason. Account lifecycle is a business behaviour with regulated edges — opening, dormancy, closure and refusal, the last two of which the Conduct Standard requires to come with reasonable prior notice and reasons [2]. Posting is an accounting invariant. A target architecture that can replace one without the other has a route out of the monolith; one that cannot, has not.

### Payments

**A-08 Payment initiation and orchestration** takes an instruction from any channel in one canonical internal form, validates and repairs it, detects duplicates, checks mandates and limits, screens and scores it, selects a rail and tracks it to completion. **A-09 Rail adaptation, clearing and settlement** speaks each rail's rulebook: message or file format, cut-off, return window, mandate regime, reconciliation to the settlement report.

Splitting them is deliberate and it is the single most useful boundary on this model.

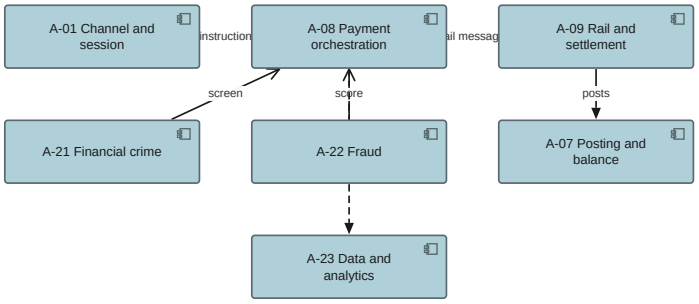



**Figure 11.5** The payment chain, and the boundary that matters. Screening and fraud serve the orchestration capability synchronously, inside the path; the rail capability owns the rulebook, the cut-off and the return window.

Two things follow from the picture. Screening and fraud are **synchronous participants** in the payment path, not downstream consumers — which matters because South African digital banking losses are dominated by social engineering, where the customer authenticates correctly and authorises the payment themselves [3]. Every control built around authentication passes; only behaviour is left. And the rail boundary is where South African specifics live: the clearing streams, their rulebooks and now their governing body all differ by stream, and governance of several clearing houses moved during 2026 [4].

### Cards

**A-10 Card issuing and token management** manufactures and manages the instrument and its digital tokens — and the token-to-PAN mapping is a mastered relationship, not a technical detail, because a reissued card with a new number must keep its wallet tokens working.

**A-11 Authorisation and switching** decides in a few hundred milliseconds, routes to the right host or scheme, and stands in when the host is unavailable. Its view of the available balance is a shadow of the ledger's and will never equal it. **That gap is a permanent architectural feature, not a defect**, and it is where the "my app showed a different number" complaint lives. Note also that there is no such thing as an "ISO 8583 interface": the schemes base their messaging on the standard and add their own fields and usages, so a design that names the standard without naming the scheme and version has specified nothing [5].

### Lending

Lending is not one capability. It is a conveyor, and the split differs by product.

**A-12 Credit application and origination** captures the application, collects evidence, orchestrates bureau and verification calls, and produces the statutory pre-agreement statement, quotation and agreement. **A-13 Credit decisioning and affordability** turns application, bureau and internal data into a decision, a price and a limit — and retains the decision, the reasons, the model and strategy versions and the exact input vector, because an applicant may ask why and a regulator may ask on what basis [1][6]. **A-14 Loan servicing, collateral and arrears** runs the loan: amortisation, rate changes, payment allocation, redraw, restructure, collateral and security administration, and the arrears state machine. **A-15 Collections and recoveries** is two businesses — working a customer who still has a relationship, and working an exposure where the relationship is over.

The arrears state machine is the architecturally interesting object, because three definitions of "in trouble" must coexist and tie: the operational state, the IFRS 9 stage, and the Prudential Authority's distressed-restructured classification, governed by Directive 7 of 2015 and replaced by Directive 11 of 2025, with banks applying Directive 7 until 31 December 2026 [7]. Three classifications of the same account, derived from the same facts, reported to three audiences, and they have to reconcile.

The retail and business conveyors are different classes of software and their data models barely overlap. Retail origination is a decision pipeline: high volume, standardised data, thin human touch, minutes to days, with the affordability test as the regulated gate. Business and commercial origination is a case-management and credit-paper problem: financial-statement spreading, covenant definition, security registration, facility structures with sub-limits and currencies, credit-committee workflow with delegated authority, and an annual review cycle. The measures differ too — conversion rate and straight-through-processing rate on one side, time-to-credit-paper and covenant-breach rate on the other. That contrast is practitioner consensus rather than something I can source, but it is reliable enough to design around: a target architecture with one A-12 has decided, probably without noticing, that business lending will be squeezed into a retail pipeline.

A-15 carries a South African obligation that catches out anybody arriving from another market. Prescription — the point at which a debt stops being enforceable — runs on events, not on days past due: it is interrupted by acknowledgment of liability, which a payment generally is, and judicially by *service* of process rather than by issuing a summons [12]. So A-15 has to compute and carry a prescription clock per debt, driven by those events. Wrong in one direction and the bank writes off recoverable debt; wrong in the other and it collects a prescribed debt, which the National Credit Act prohibits.

### Treasury, finance and regulatory

**A-16 Deal capture, position and treasury operations** covers front, middle and back office: capture, limits, independent valuation, confirmation, settlement, nostro reconciliation and accounting. In most banks it is a separate landscape with its own team, and it is the box most often forgotten when somebody says "the bank". Its recurring failure is fragmentation along the trade lifecycle — breaks between the treasury record, the risk view, the accounting entries and the settlement instruction, with rekeying between them, which shows up as delayed visibility into cash and funding and a reconciliation workload that never shrinks. Whether it is one front-to-back platform or a chain of four is the single question that determines its cost.

**A-17 General ledger and financial control** holds the chart of accounts, accepts journals, controls period open and close, and produces statutory accounts. The general ledger does not know what a mortgage is. Everything product-specific happens in sub-ledgers that interface to it — and the interface, not the

ledger, is the architectural object.

**A-18 Regulatory reporting and submission** produces, reviews, approves, submits and resubmits each return with lineage. This is not a reporting function. It is a production function with statutory deadlines, attested accuracy and personal accountability, and it consumes a large share of a bank's change capacity. The specification now moves faster than it used to: Government Notice 5802 removed the return forms from the Banks Act Regulations so that returns are determined by the Authority in writing [8], and the BA 900 alone requires every balance classified by counterparty sector, instrument, currency and maturity, monthly [9].

#### Risk and capital

**A-19 Credit risk measurement and impairment** measures exposure, stages it, computes expected credit loss and back-tests against recoveries. It is a production accounting engine, not an analytics exercise, and it must reconcile to both the general ledger and the returns.

**A-20 Capital, liquidity and balance-sheet risk** computes risk-weighted assets and capital ratios, the liquidity ratios, and interest-rate risk in the banking book. South Africa implemented the remaining Basel III post-crisis reforms from 1 July 2025, including the output floor [7] — which means a bank on internal models must compute risk-weighted assets twice, by two approaches, on every exposure. That is a parallel calculation estate, and it is a capability question before it is a modelling one.

The unifying point across A-19 and A-20 is that both need the same exposure data at different granularities and frequencies, and in most banks each gets its own extract. That is why a bank can report three different exposure numbers in the same month to three different audiences.

#### Financial crime and fraud

**A-21 Screening, monitoring and financial-crime case management** is two computational problems usually bought as one product. Screening is fuzzy name-and-attribute matching against sanctions, politically-exposed-person and adverse-media lists, run at onboarding, on a periodic re-screen of the whole base, and inline on payments. Monitoring is behavioural: rules and models over transaction history producing alerts and cases. The capability must retain **list versions**, because a historical match has to be reproducible against the list in force at the time.

**A-22 Fraud detection and intervention** needs its own box because its intervention options are different: hold, delay, challenge, warn, as well as decline. A fraud capability that can only decline is architecturally unable to address the dominant South African loss category [3].

#### Data, document and the common layer

**A-23 Data integration, history and analytics** moves and conforms data, holds history and serves marts, models and features. Chapter 10 drew the line this capability must respect: the warehouse may be rebuilt, the regulatory layer may not, and nothing operational may be mastered here.

**A-24 Document, evidence and communication** keeps the document of record under retention rules driven by several statutes at once, and produces outbound communication with proof of delivery. The test is whether the bank can reproduce the exact rendered statement a customer received five years ago — not the data behind it, the artefact. Several of its outputs are statutory instruments with prescribed content and timing, and at least one — the section 129 notice that must precede enforcement — has to have its delivery proved, sometimes years later and in court.

**A-25 Workforce identity, entitlement and authority limits** covers joiner-mover-leaver, entitlements, segregation of duties and the authority limits that are a business control implemented as an access control, usually in several places at once.

### The table

| ID | Capability | Purpose | Primary user roles | Data mastered | Typical system in chapter 12 |
|---|---|---|---|---|---|
| A-01 | Channel experience and session | Present the bank through one channel and hold the authenticated session | Digital product owner, branch consultant, contact-centre agent | Session, device binding, channel preference | App, internet banking, USSD gateway, teller front end |
| A-02 | Interaction, service and complaint | Record what was said, work the request, categorise and resolve the complaint | Contact-centre agent, service consultant, complaints officer | Interaction, case, complaint with prescribed category | Sales and service CRM |
| A-03 | Customer identity and authentication | Register, issue and verify credentials, bind devices, step up on risk | Digital fraud manager, identity engineer | Credential, device registration, step-up outcome | Customer identity and access management |
| A-04 | Party master, relationship and mandate | Hold one identity per legal person with dated roles and authorities | Party data steward, KYC analyst, business banking service manager | Party, party role, party-to-party relationship, mandate, consent | Party master or MDM platform |
| A-05 | Product definition, catalogue and pricing | Define, version, price and publish what a product is | Product manager, pricing analyst | Product version, fee schedule, rate structure, eligibility, classification | Product catalogue or the core's product engine |
| A-06 | Account lifecycle and servicing | Open, maintain, block, dormant, close the customer-facing account | Branch consultant, deposits operations | Account, status history, standing instruction, holds | Core banking, deposits side |
| A-07 | Posting, balance and interest | Post balanced entries with three dates; accrue, capitalise and hold the balance | Financial controller, core operations | Entry, ledger account, accrual, period-end snapshot | Core banking ledger |
| A-08 | Payment initiation and orchestration | Take, validate, screen, route and track a payment instruction | Payments operations manager, corporate service manager | Payment instruction and status history, beneficiary, approval trail | Payments hub |
| A-09 | Rail adaptation, clearing and settlement | Speak each rail's rulebook; manage mandates, returns and reconciliation | Payments operations, settlement clerk | Rail message, collection mandate, clearing reference, settlement position | Rail adapters, mandate gateway, RTGS interface |
| A-10 | Card issuing and token management | Manufacture and manage the instrument and its digital tokens | Card product manager, card operations | Card, truncated PAN reference, token, scope, lifecycle | Card management system or processor |
| A-11 | Authorisation and switching | Decide in milliseconds, route, stand in when the host is down | Card operations, switch engineer | Authorisation, shadow balance, stand-in decision | Authorisation host and switch |
| A-12 | Credit application and origination | Capture the application and produce the statutory quotation and agreement | Credit consultant, business credit analyst | Application, documents, offer, agreement artefact | Loan origination system |
| A-13 | Credit decisioning and affordability | Turn application, bureau and internal data into a decision, price and limit | Credit risk manager, scorecard analyst | Decision, reasons, model and strategy version, input vector, affordability calculation | Decision engine |
| A-14 | Loan servicing, collateral and arrears | Run the loan and the security behind it, and hold the arrears state | Loan servicing clerk, security administrator | Loan account, schedule, collateral item, security instrument, valuation, daily DPD | Lending servicing platform |
| A-15 | Collections and recoveries | Work the customer, then work the exposure | Collections agent, recoveries manager, legal administrator | Treatment, promise to pay, arrangement, legal state, prescription clock, write-off | Collections and recoveries system |
| A-16 | Deal capture, position and treasury operations | Capture the deal, hold the position, value, confirm, settle | Dealer, treasury middle office, back-office settlements | Deal, position, limit usage, confirmation, nostro | Treasury front-to-back platform |
| A-17 | General ledger and financial control | Hold the chart of accounts, accept journals, close the period | Financial controller, finance business partner | Journal, chart of accounts, period state, reconciliation certificate | General ledger or finance ERP |
| A-18 | Regulatory reporting and submission | Produce, review, approve, submit and resubmit each return with lineage | Regulatory reporting manager, head of finance | Return submission, version, sign-off, acknowledgement, restatement | Regulatory reporting platform |
| A-19 | Credit risk measurement and impairment | Measure exposure, stage it, compute expected credit loss, back-test | Credit risk modeller, head of credit risk | Exposure base, PD, LGD, EAD, stage, provision, scenario weights | ECL and credit risk engines |
| A-20 | Capital, liquidity and balance-sheet risk | Compute RWA and capital, liquidity ratios, banking-book interest-rate risk | Head of capital management, ALM analyst, treasurer | RWA components, capital ratios, behavioural cash flows, liquidity measures | Capital calculation and ALM systems |
| A-21 | Screening, monitoring and financial-crime cases | Screen against lists, monitor behaviour, work alerts, file reports | Financial crime analyst, money laundering reporting officer | Risk rating, list version, alert, case, disposition, filed report | Screening and transaction-monitoring platforms |
| A-22 | Fraud detection and intervention | Score inside the path and intervene: hold, delay, challenge, warn, decline | Fraud analyst, head of digital fraud | Rule set, score, intervention, confirmed-loss outcome | Card, application and transaction fraud engines |
| A-23 | Data integration, history and analytics | Move and conform data, hold history, serve marts, models and features | Data engineer, data steward, head of analytics | Conformed dimensions, history, derived measures, feature definitions | Warehouse and lakehouse stack |
| A-24 | Document, evidence and communication | Keep the document of record; produce and prove outbound communication | Records manager, communications specialist | Document of record, retention class, rendered artefact, proof of delivery | ECM and customer communication management |
| A-25 | Workforce identity, entitlement and authority | Joiner-mover-leaver, entitlements, segregation of duties, authority limits | Identity administrator, operational risk manager | Identity, entitlement, authority limit, recertification | Workforce IAM and governance tooling |

## The patterns

Five patterns run through the twenty-five, and knowing which one a capability follows tells you more about the programme risk than anything else on the page.

**Usually one.** A-04, A-05, A-07, A-17, A-23. These should exist once. When they exist more than once, the duplication is the bank's central data problem, not an inconvenience. A-17 is the only one that reliably *is* singular, because finance will not tolerate two ledgers.

**Usually several.** A-06, A-07 in practice, A-12, A-14 and A-08 all come in multiples — by segment, by acquired book, by legal entity, by country. The question to ask is not "how many" but "which one is authoritative for what", and the answer should be written down somewhere other than in people's heads.

**Usually bought.** A-11, A-16, A-19, A-20, A-21 and A-22 are bought almost everywhere, because the standards, the models or the list feeds make building uneconomic. Buying them is correct. The architectural work is the boundary, not the build.

**Usually built.** A-01 and A-02 drift towards build in every bank that competes on its app, and A-08 is increasingly built because the canonical internal payment format is the bank's own design problem. A-18 is built far more often than anyone plans, because no package arrives knowing the local return set.

The build-and-buy line is usually drawn in the wrong place. The question is not which capabilities are differentiating; individually, almost none of them are. It is which capabilities carry a *local* specification that changes faster than a vendor's release train. A-09 and A-18 both do: the mandated collection rail changed in March 2025 and the return specification moved out of the Banks Act Regulations and into a directive in the same window [8]. A capability whose specification is set by a domestic authority, and can now change in writing, needs either a vendor with a local product team or an internal build. There is no third answer, and a roadmap commitment is not one.

**Usually missing.** Five, and they are the same five in most banks.




**Figure 11.6** The five most commonly missing capabilities and why each one is absent. The common factor at the bottom is the real finding.

The common factor is worth stating plainly, because it is the most useful sentence in this chapter. **All five missing capabilities are shared.** No single product line will fund a capability it has to share with four other product lines, and no product line's executive is measured on it. That is a funding-model problem presented as an architecture problem, and it will not be solved by a better diagram. It is solved by putting the shared capabilities in one programme with one owner and one budget — which is what "core modernisation" usually turns out to mean once the scope settles.

## How the model changes with the line of business

The twenty-five boxes do not change. Their weight does.

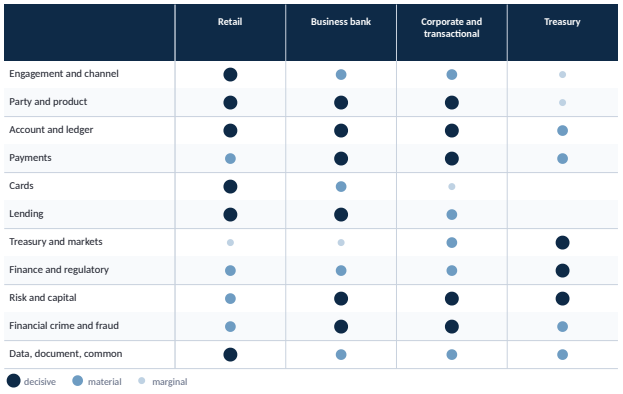



**Figure 11.7** The same twenty-five boxes, weighted four ways. An architect who carries one column's weighting into another column's conversation will be wrong in a way that is hard to recover from.

Three readings worth taking from the matrix. **Retail is a volume problem** — the engagement, card, lending and data boxes carry the load, and the architectural difficulty is throughput and consistency across channels. **Corporate and transactional banking is a mandate and payments problem** — multi-user authorisation hierarchies, host-to-host integration, bulk file upload, sub-limits, and capabilities the retail app does not have and cannot easily acquire. **Treasury is a different bank** with its own vocabulary, its own regulator-facing reporting and usually its own IT function; A-16 and A-20 dominate and much of the rest is irrelevant to it.

An architect who carries one column's weighting into a conversation about another column will be wrong in a way that is hard to recover from, because the other side will conclude you do not know their business.

## How the model changes with the size of the bank

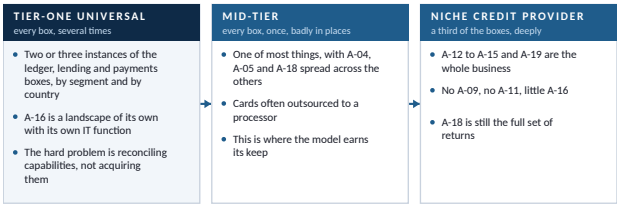



**Figure 11.8** How the model changes with the size of the bank. The mid-tier column is where the model earns its keep, because that is where capabilities exist once and are spread across systems that were never meant to carry them.

A fourth shape is worth a sentence, because South Africa has them: the **mutual or co-operative bank**. It needs A-06, A-07, A-17, A-18 and A-21 in full, because the prudential and financial-crime obligations do not scale down, and almost nothing else. The disproportion between its regulatory capability requirement and its commercial capability requirement is the defining fact of its architecture, and it is why those banks buy almost everything.

## How to use it

Three uses. The model is worth the afternoon if it is used for one of them.

### As a target architecture skeleton

Draw the twenty-five. Colour each one by whether it exists, exists in several places, or does not exist. Then draw the target: which boxes stay, which consolidate, which are new. That picture is a programme scope in business language, and because no product name appears on it, it survives a change of vendor shortlist.

The discipline that makes it useful is to resist adding the integration arrows. The arrows are the next picture, and if you draw them now the conversation will be about them.

The second discipline is sequencing, and the capability map gives it to you almost for free. Order the target by fan-out: the capabilities that the most other capabilities consume go first, because every month they stay duplicated is a month of reconciliation nobody budgeted for. On this model that ordering puts A-04 and A-05 at the front, A-07 next, and the consuming capabilities after — which is, uncomfortably, the opposite of the order a business case usually prefers, because the shared capabilities deliver nothing visible to a customer on their own. That tension is the real content of a core modernisation debate, and having it in front of a capability map is better than having it in front of a vendor's roadmap.

### As a package-fit assessment

This is where the model pays for itself fastest.

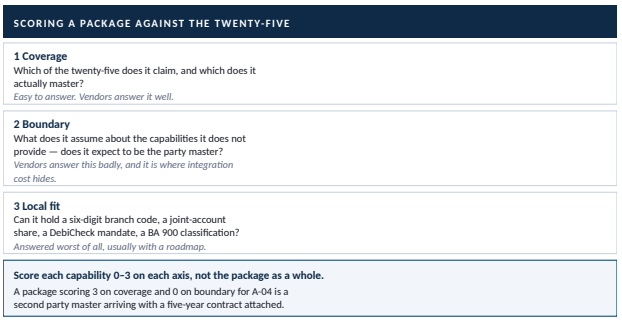



**Figure 11.9** The package-fit scorecard. Axis two is the one vendors answer badly and axis three is the one they answer worst, usually with a roadmap.

Score each capability, not the package. A suite that covers eighteen of the twenty-five and quietly assumes it is the party master has not covered eighteen capabilities: it has covered seventeen and started a war over A-04. Axis three is the one that matters most locally and gets the thinnest answers — a six-digit branch code with a universal variant, a joint-account sharing ratio, an authenticated collection mandate, a BA 900 classification set. None of those is exotic in South Africa and none of them is in a package built for a market with IBANs.

Chapter 13 covers the reference models an architect can borrow for this — and one caution belongs here rather than there: BIAN's Service Landscape is a useful dictionary for arbitrating capability boundaries, but BIAN publishes no open licence on its deliverables, so availability is not permission. Get terms in writing before embedding its content in a commercial deliverable [10].

#### As the basis for the application portfolio

Every application in the estate maps to one or more of the twenty-five. That mapping, maintained, answers questions the estate list cannot: where is capability duplicated, where is it absent, where does one system carry so many capabilities that it cannot be replaced incrementally, and what would actually have to move to retire a given platform.

Keep the mapping at capability level and refuse to let it become a function-level matrix. A function-level matrix has thousands of cells, is never maintained, and is used once.

Two derived views repay the effort of maintaining it. A **duplication view** — capabilities realised more than twice — is the shortest honest statement of a bank's integration cost, because every duplicate is a reconciliation somebody runs every day. And a **concentration view** — systems realising more than four capabilities — names the platforms that cannot be replaced incrementally, which is the single fact a board most needs and least often gets. A system carrying eleven of the twenty-five is not a system to be migrated. It is a programme.

#### What the model deliberately does not contain

Three omissions get queried every time. **No processes**: a process crosses capabilities, and it belongs in the process models of chapter 8. **No organisational units**: the bank will reorganise, and a capability map that moves when it does is worthless as a target. **No data stores**: chapter 10 says what is held, this model says who masters it, and drawing both on one page produces an artefact that is wrong at both layers.

## Table Bay Bank against the model

Table Bay is invented, and its estate is the ordinary mid-tier shape: a retail core from 1996, a second core inherited with a business bank acquired in 2014, cards outsourced to a processor, a payments hub bought in 2019 that three channels still bypass, a warehouse, and a regulatory reporting layer that is a reporting tool over a mart.

Scored against the twenty-five, Table Bay has twenty-one of them. The four it does not have are A-05, A-09 as a single owned capability, A-18 in the sense this chapter means, and A-24's rendered-artefact archive. A-04 exists twice — once operationally in the retail core and once completely in the warehouse, which makes the complete view a day old. A-07 exists three times.

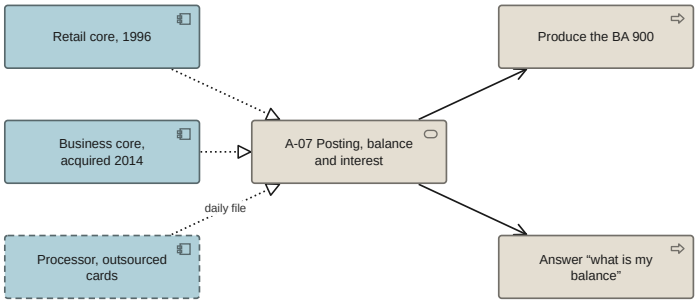



**Figure 11.10** Table Bay Bank, invented, against one capability. Three implementations of A-07, one of them somebody else's and a day behind — which is why the bank can produce its single customer view monthly and not on demand.

One consequence of that picture is worth following through, because it is the kind of thing that only becomes visible at this layer. Table Bay's deposit-insurance single customer view has to aggregate every qualifying depositor's balances across all three implementations of A-07, resolved to one party identity, monthly, within ten working days of month-end [11]. With A-04 complete only in the warehouse, the single customer view can be produced monthly and cannot be produced on demand — which is the only circumstance in which it is actually needed. The gap is not a reporting defect. It is A-04 in the wrong layer, and no amount of work on the submission will fix it.

Two further things fall out of the same picture, which is a fair return on one page of drawing. Table Bay's payments hub does not own A-09, so three channels reach rails directly; screening therefore happens inline on some payments and after the fact on others, and nobody can say which without tracing a path. And because A-05 does not exist, the deposit-insurance qualifying flag is set per account rather than derived from the product — so every account opened before the flag was introduced carries whatever default was in place that day.

That is the whole argument for drawing the conceptual application model. Each of those defects is invisible on an estate list, obvious on a capability map, and expensive to discover during a resolution weekend.

## Where this goes wrong

Four failure modes, in the order you will meet them.

**The boxes get vendor names.** Somebody renames A-07 after the core product, and within a month the model has stopped being a target and become a description. The test is whether the model would change if the contract changed. If it would, start again.

**It becomes a wiring diagram.** Arrows get added, then protocols, then queue names. The result is a solution architecture drawn at the wrong altitude, unreadable by the people whose agreement the model exists to get. Keep the arrows for a different page, and keep that page to one flow at a time — the payments picture earlier in this chapter is about as busy as a conceptual model should get.

**It mirrors the organisation chart.** One box per department, which guarantees that shared capabilities have no home and that the five missing ones stay missing. The organisation will reorganise; the capabilities will not.

**It is drawn once and never compared to the estate.** A target with no current state beside it is an aspiration. The model earns its place the first time somebody uses it to say "we have four of these and we need one", and loses it the moment the mapping to the real estate goes stale.

A closing discipline, then, for anybody drawing one of these in their first month. Draw the twenty-five from the list in this chapter, change the five or six that do not fit the bank you are actually in, and stop. Do not add boxes merely to be complete; a conceptual application model is a tool for having an argument, and a model with sixty boxes cannot host one. Then take it to three people — somebody from payments, somebody from credit, somebody from finance — and ask each of them which box owns the thing they complain about most. The boxes they cannot find are the missing capabilities, and after this chapter they will not be a surprise.

Chapter 12 is where the comparison with the real estate gets made, system by system.

#### THE CHAPTER IN TEN LINES

- Business capability, conceptual application capability and real system are three layers, related by many-to-many realisation, and most wasted meetings come from mixing them.
- A conceptual application capability is named by what it does, masters something or decides something, could be supplied by a different system tomorrow, and has an accountable owner.
- Digital, integration and AI are not application capabilities: the first overlaps everything, the second is the boundary between the others, and the third is a technique used inside five of them.
- Twenty-five capabilities, A-01 to A-25, cover a full-service bank; a niche lender needs about nine and a tier-one universal bank needs several instances of some.
- Payment orchestration and rail adaptation are separate capabilities, and screening and fraud sit inside the payment path synchronously rather than downstream of it.
- The five most commonly missing capabilities are product definition, an operational party master, rail adaptation, regulatory submission with lineage, and the rendered-artefact archive.
- All five are shared, and no product line will fund a capability it must share with four others — which is a funding problem wearing an architecture costume.
- Use the model three ways: as a target skeleton, as a per-capability package-fit score, and as the basis for the application portfolio mapping.
- Score packages per capability on coverage, boundary assumptions and local fit; the local-fit axis is answered worst and is where South African specifics bite.
- The four failure modes are vendor names on the boxes, arrows turning it into a wiring diagram, boxes mirroring the organisation chart, and never comparing it to the real estate.

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CHAPTER 12

# The application landscape

*This is the chapter you will come back to. It names the systems a bank actually runs, says what each is authoritative for, where each one hurts, and what it is called in the trade — then the wiring, the batch clock and the legacy question, with the sourced claims and the practitioner consensus kept apart.*

## The shape of the estate

Come back to this chapter. It is the one that names the systems, and the names will attach to people by your third week.

The teaching example is **Table Bay Bank** — an invented South African full-service bank, mid-tier, about 2.4 million retail customers, a business bank, a small corporate and transactional arm, and a treasury. Table Bay is a fiction and nothing about it is a case study. Wherever a number matters it is sourced to something real.

Start with the most useful thing to know about a bank's estate: it is not one estate. It is two businesses welded together, with a third layer laid across both. The **balance-sheet business** takes deposits and makes loans; its systems are ledger-centric and slow-moving, measured on correctness to the cent, and they change on a release train somebody scheduled a quarter ago. The **payments business** moves other people's money; its systems are high-volume and latency-bound, measured on availability, and they change when a card scheme or the central bank says so, on dates nobody inside the bank chose. The **control layer** — risk engines, financial crime, fraud, regulatory reporting, the data platform — reads from both and reports to people who can close the bank.

Three clock speeds, three failure modes, usually three organisational homes. Draw one landscape across all three and you mis-scope everything downstream, because "can we change this in six weeks" has a different answer in each.

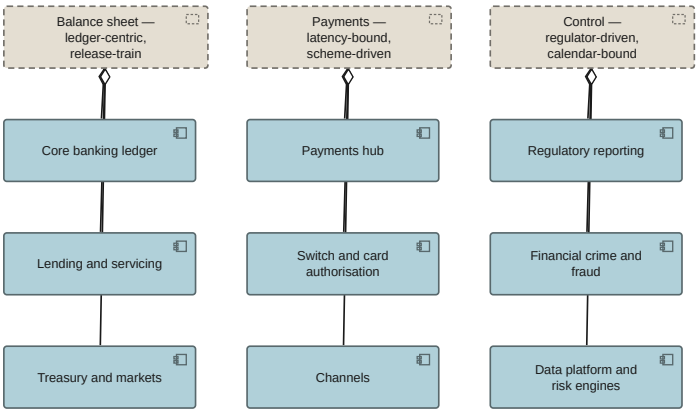



**Figure 12.1** Three clock speeds in one estate. The groupings are not org units — they are change cadences. Ask of any component which group it sits in, and you have predicted how fast it can move and who decides.

Here is the map. If you learn one table from this book, learn this one, because the right-hand column settles most boundary arguments.

| # | Class | System of record for |
|---|---|---|
| 1 | Core banking / account ledger | Account, balance, entry, product instance |
| 2 | Deposits and transactional accounts | Current, savings, notice and fixed accounts; interest; fees |
| 3 | Lending — origination | Application, offer, affordability assessment |
| 4 | Lending — decisioning | Score, strategy, decision reason |
| 5 | Lending — servicing | Loan account, amortisation, arrears state |
| 6 | Collections and recoveries | Treatment, promise-to-pay, legal state, write-off |
| 7 | Card issuing and authorisation | Card, PAN and token, authorisation, available balance |
| 8 | Card acquiring and merchant | Merchant, terminal, settlement, merchant discount rate |
| 9 | Switch | Routing, stand-in, scheme interface |
| 10 | Payments hub | Payment instruction, rail selection, status |
| 11 | Channels | Session, interaction, device, consent |
| 12 | Treasury and markets | Deal, position, limit, confirmation, settlement |
| 13 | General ledger and finance | Journals, chart of accounts, statutory accounts |
| 14 | Regulatory reporting | BA returns, FinSurv, CODI and FIC submissions |
| 15 | Data platform | Conformed history, dimensions, marts |
| 16 | Risk engines | Models, parameters, provisions, risk-weighted assets |
| 17 | AML, CFT and sanctions screening | Customer risk rating, alert, case, regulatory report |
| 18 | Fraud | Rules, scores, cases, confirmed-loss outcomes |
| 19 | CRM and party master | Party, relationship, mandate-to-act, consent |
| 20 | Documents and customer communication | Documents of record, retention, evidence; statements and notices |
| 21 | Identity and access | Credential, entitlement, authority limit |

Expect **more than one instance** of classes 1, 5, 7 and 10 — by legal entity, by acquired book, by country, by segment. Expect exactly one of class 13, because a bank can have only one set of statutory accounts. In a large South African group, add a separate retail and corporate core, and a rest-of-Africa estate that is a different landscape wearing the same brand.

> **THE HONESTY RULE FOR THIS CHAPTER**
>
> Some of what follows is sourced to a regulator, a scheme or a published disclosure, and carries a citation. Some is practitioner consensus — the thing every architect finds and nobody publishes — and is marked as such. No South African bank publishes an application-portfolio count, so any claim about the *shape* of a typical estate, including mine, is consensus rather than fact. Say "this is the practitioner view" in your own documents and you are defensible.

## 1. Core banking, and the honest argument about "core"

**What it is.** The most abused term in the industry. Ask five people and get five answers, because three genuinely different things have historically lived in one box.

The public definition is thinner than you would expect. Gartner's, as quoted in general reference material, is "a back-end system that processes daily banking transactions", posting updates to accounts and other financial records [23]. The same source frames core banking historically as the *branch-networking* achievement: before the 1970s branches ran local servers and batched to a data centre at close of business, so a transaction could take a day to appear [23]. Hold on to that. **"Core banking" originally named the solution to a batch problem, and the batch it solved is still visible in the architecture.**

Here is the argument, laid out, because you will have it in your first month.

| If "core" means... | Then it is... | And believing only this costs you... |
|---|---|---|
| The ledger | Double-entry accounts: balances, entries, accrual, value dating, end-of-day | Under-scoping. Product behaviour and customer data live elsewhere and nobody owns them |
| The product engine | The configuration that defines what a product *is* — rates, fees, limits, eligibility, lifecycle | A discovery that the ledger cannot support a product the engine can express |
| The customer master | The customer record that accounts hang off | Three customer masters and no owner |
| All three | The monolith | Nothing — the right reading for most legacy South African cores, and the reason replacement is hard |

**So treat "core" as a question, not a noun.** Establish four things separately and write the answers down: where is the authoritative balance, per product family; where is product *definition*, configuration or code; where is the authoritative customer identity; and where does end-of-day run. Four answers, often four systems, often not the one the organisation chart labels "core".

**FOUR QUESTIONS, FOUR POSSIBLE SYSTEMS**

| The question | What you may actually find |
|---|---|
| Where is the authoritative balance, per product family? | A retail core, a corporate core, a card authorisation host with a shadow balance |
| Where is product definition — configuration or code? | A product engine, a pricing engine, or COBOL nobody has read since 1998 |
| Where is the authoritative customer identity? | The core, the CRM, or the data platform — and the last one is usually complete |
| Where does end-of-day run, and what is on its critical path? | One scheduler, several, or a runbook held by one team on a rota |

**Figure 12.2** "Core" is a question with four answers, and in a legacy estate they are frequently four different systems. Write the four answers down in your first week; the rest of the landscape makes sense afterwards.

**What it masters.** The account identifier, the balance, every entry with its value date and posting date, the product instance and its parameters, the account-to-customer linkage, status history and accrual state. Scoped properly it also owns the balance types that matter more than they sound — ledger, available, cleared and uncleared, hold and earmark — plus accrual by product rule, fees and limits, the end-of-day cycles and the ledger interface.

**Where it hurts.** Customer identity and relationships, product *definition* versioning, anything event-driven, anything a product manager wants changed this quarter. The failures, in the order you meet them: product logic hard-coded, so each new product is a code release; one core per acquisition with nothing decommissioned; available balance reimplemented separately in the app, the ATM and the authorisation host, so three channels show three balances; and no event stream, so every downstream system either reads the core's tables directly or waits for the nightly extract.

## 2. Deposits and transactional accounts

**What it is.** Economically the cheapest funding a bank has and operationally the highest-volume thing it does. Current accounts — still called cheque accounts, which tells you something — savings, notice deposits, fixed deposits, call accounts, and the bundled-fee transactional accounts that dominate retail.

**What it masters.** The product catalogue and its pricing, interest by tier and balance band, fee assessment and waiver, notice periods and early-withdrawal penalties, dormancy and unclaimed balances, attachment holds, set-off against indebtedness.

There is a free taxonomy sitting in the regulatory returns and you should borrow it. The BA 900 structure splits **demand deposits** into transferable and other demand up to one day; **fixed-term deposits** by original term, up to and over one year; **notice deposits** by notice period, up to and over one year; and **other deposits** [15]. If your data model cannot classify every deposit balance into that grid you have a data-architecture problem, not a reporting problem.

**The local thing you will not find in offshore material.** Deposit insurance is new, and it carries the most under-appreciated data requirement in South African retail banking. The Corporation for Deposit Insurance became a legal entity on 24 March 2023, with member banks beginning data submission on 1 April 2024 [17]. Cover is **R100 000 per qualifying depositor**, with sole proprietors receiving R100 000 for personal accounts and a separate R100 000 for business accounts [18].

The obligation is a **single customer view**: every qualifying depositor's accounts and balances in qualifying products consolidated into one record, under aggregation rules that are specified rather than left to the bank [18]. Joint accounts split by the recorded sharing ratio, or equally where none is recorded. Sole proprietors and partnerships get a record separate from the individual's. Negative balances are reported as a zero qualifying balance. Balances are gross, in rand at close-of-business rates [18]. The required field list — identifiers, classification, product type, currency, three different balances, joint-account flag and share, pledged flag, beneficiary fields — is in effect a mandated party-and-account data model [18].

> **THE TRAP IN THE DEPOSIT INSURANCE OBLIGATION**
>
> The submission is trivial and the calculation is not. The monthly file is four fields in CSV: member bank code, reporting date, total qualifying deposits, total covered deposits [17]. Behind it sits a legally attested, chief-executive- and chief-financial-officer-signed aggregation across every deposit product, per natural person, monthly, within ten working days of month-end [17].
>
> Built as a reporting-layer query it cannot be produced at short notice in a resolution event — which is the only reason it exists.

**Where it hurts.** Interest and fee logic duplicated between core and pricing engine. Bundled-fee calculation in a monthly batch nobody can explain. Dormancy rules applied inconsistently, so unclaimed balances are understated. Holds implemented per channel.

## 3–6. The lending stack, by product and by stage

The biggest structural difference from an insurance landscape: lending is not one system. It is a **conveyor of four or five**, and the conveyor differs by product. The stages are origination, decisioning, fulfilment and disbursement, servicing, collections, recoveries.

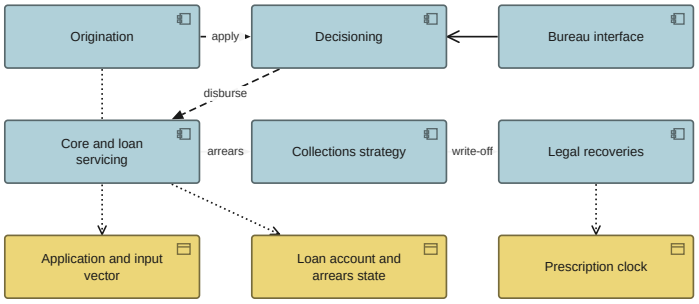



**Figure 12.3** The lending conveyor and the three objects that must survive every hand-off. The application with its exact input vector, the loan account with its arrears state, and the prescription clock. Lose any one at a boundary and the bank cannot reproduce a decision, reconcile three definitions of default, or lawfully collect.

**Origination** carries the application to an offer: capture, document and income-proof collection, bureau and employer verification, affordability assessment, product and pricing configuration, offer and acceptance, the pre-agreement statement and quotation — a statutory artefact — condition-precedent tracking, and the disbursement instruction. It masters the application, the offer and its terms, and the evidence set, which must be reproducible years later.

Retail and business origination are different classes of software. Retail is a *decision pipeline*: high volume, standardised data, thin human touch, minutes to days, with the regulated affordability test as the gate. Business is a *case-management and credit-paper problem*: financial-statement spreading, covenant definition, security registration, facility structures with sub-limits and currencies, credit-committee workflow under delegated authority, an annual review cycle. The data models barely overlap. That contrast is practitioner consensus, not a sourced market finding.

**The regulatory shape.** Lending to natural persons and small juristic persons falls under the National Credit Act 34 of 2005, enforced by the National Credit Regulator. The architecturally material obligations: a mandatory affordability assessment before credit is extended; a pre-agreement statement and quotation; statutory disclosure in the agreement; limits on the cost of credit; and, before enforcement, a **section 129(1)(a) notice** [27]. An architect arriving from the United Kingdom will look for a conduct-principles analogue and find a prescriptive, document-centric statute instead.

**Decisioning** turns application plus bureau plus internal data into an accept, decline or refer, and a price and a limit. It masters the decision, its reasons, the model version that produced it, and **the exact input vector** — a regulatory artefact, not an engineering nicety, because a declined applicant may ask why and a regulator may ask on what basis. It fails when three channels run three scorecards, when policy rules are duplicated between engine, front end and core, when the input vector is not retained so a decision cannot be reproduced, and when affordability lives in a spreadsheet owned by one credit analyst.

**Servicing** runs the loan account's life: amortisation and rate changes, payment allocation — the order in which a receipt clears fees, interest and capital, a rich source of disputes — redraw and further advance, restructures and payment holidays, early settlement quotation, and the arrears state machine. **That state machine is the architecturally important object**, and it carries three overlapping definitions that must coexist and must tie.

| Definition | What it is | Driven by |
|---|---|---|
| Operational | Days past due, treatment bucket | The collections strategy |
| Accounting | IFRS 9 stage 1, 2 or 3 | Significant-increase-in-credit-risk tests |
| Regulatory | Distressed restructured credit exposure | Directive 7 of 2015, replaced by **Directive 11 of 2025**, applying Directive 7 until 31 December 2026 then transitioning [16] |

Three definitions of "in trouble", from one account, to different audiences. When they disagree, somebody must be able to say which is right for which purpose — and that explanation is an architecture artefact.

**Collections and recoveries** are two businesses, routinely built as one system and found wanting. Collections works a customer who still has a relationship: contact strategy, promise-to-pay, arrangement, cure. Recoveries works an exposure where the relationship is over: legal process, security realisation, sale of book, write-off, post-write-off recovery. Between them they master the treatment path, contact attempts and outcomes, promises and their performance, debt-review state, the section 129 notice and its proof of delivery, legal state, write-off — and the prescription clock.

Prescription is three years for ordinary unsecured debt, thirty years for mortgage-secured and judgment debt, six years for a bill of exchange or notarial contract [27]. It runs from when the debt is due, and is interrupted by acknowledgment of liability — a payment is generally a tacit acknowledgment — or judicially by *service* of process; issuing a summons or sending a section 129 notice is not enough [27]. Section 126B, in force 13 March 2015, prohibits selling a prescribed debt under the Act, and prohibits continuing to collect one where the consumer raises prescription [27].

> **THE PRESCRIPTION CLOCK IS NOT DAYS PAST DUE**
>
> A collections system must compute and carry a prescription clock **per debt**, driven by acknowledgment, payment and service of process — not by arrears ageing. Get it wrong one way and the bank writes off recoverable debt. Get it wrong the other way and the bank illegally collects prescribed debt.
>
> Whether most systems model this explicitly rather than by manual review is not something anyone publishes. Assume manual until shown otherwise, and make it your first question of a collections platform.

Two more local facts. Debt review under section 86 creates a parallel, externally driven arrangement state that servicing and collections must both honour. And complaint escalation goes to the **National Financial Ombud Scheme**, operational from 1 March 2024, which folded the Banking Ombud

and the Credit Ombud into one body with Banking and Credit as a division [26] — so an architect looking for a standalone banking ombudsman will not find one.

**How the stack fails as a whole.** The hand-over points leak. Origination data is not carried into servicing, so the servicing system does not know what the customer was told. Write-off is executed in the core with no corresponding event in the risk mart, so loss back-testing is impossible. And the section 129 proof of delivery is a scanned file with no index, discovered at trial.

## 7–9. Cards, end to end

The card estate is where a bank's architecture is most standards-driven and least negotiable, because the schemes impose message formats, timeframes and certification. It is also where you will most often find a competent, well-run, utterly opaque platform that nobody inside the bank fully understands.

**Issuing** manufactures and manages the instrument: card record, account number, expiry, verification values, PIN, the instruction to a personalisation bureau, chip personalisation data, activation, block and replacement — and the digital equivalents, wallet provisioning and network tokenisation.

**Authorisation** is the real-time decision at the moment of purchase. Is the card valid, is the cardholder authenticated, are funds or credit available, does this look like fraud, is it within scheme and bank rules — answered in a few hundred milliseconds or the terminal times out. The authorisation host keeps its own view of available balance, an *authorisation* or shadow balance, decremented by approved authorisations and reconciled to the core when clearing arrives.

> **THE BALANCE GAP IS A FEATURE, NOT A DEFECT**
>
> The gap between the authorisation balance and the ledger balance is a permanent property of card processing. It is where the "my app showed a different balance" complaint lives. You cannot design it away. You can make sure there are two implementations of available balance rather than four, and that the complaints team knows which is which.

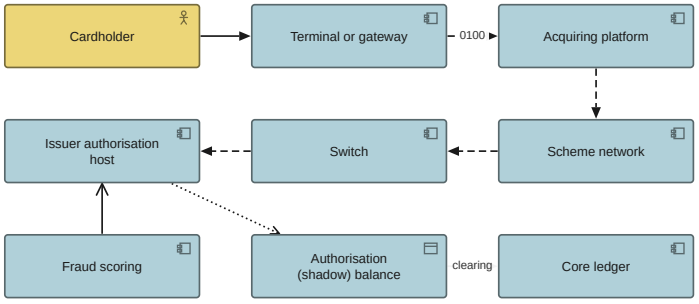



**Figure 12.4** One card transaction, end to end. Read the top row left to right and the second row right to left: the message goes out through the acquirer and the scheme and comes back to the issuer. The authorisation balance it decrements is not the ledger balance, and the ledger only learns about the transaction when clearing arrives, days later.

**The message standard is ISO 8583**, "Financial transaction card originated messages", with editions in 1987, 1993 and 2003; the 2003 edition had not gained wide acceptance as at 2017 [12]. The Message Type Indicator is four digits — version, class, function, origin — where class 1 is authorisation, 2 financial, 4 reversal and chargeback, 5 reconciliation, 8 network management [12]. Field presence is signalled by **bitmaps**: a primary bitmap covers fields 1 to 64 and its first bit flags a secondary for 65 to 128 [12]. The elements you will meet by number are DE2 the account number, DE3 the processing code, DE4 the amount, DE39 the response code and DE55 the chip data; the pairs are 0100/0110 authorisation, 0200/0210 financial, 0400 reversal, 0800/0810 network management [12].

> **THERE IS NO SUCH THING AS AN ISO 8583 INTERFACE**
>
> The standard's own reference literature says it plainly: ISO 8583 "is not typically used directly by systems or networks" — the schemes base their authorisation messaging on it but add network-specific fields and usages, and implementations do not always interpret the message-type meanings identically [12].
>
> So there is a Visa interface, a Mastercard interface, a domestic switch interface, and three mapping layers. An architect who writes "ISO 8583" in a solution design without naming the scheme and the version has specified nothing.

**3-D Secure and tokenisation** add two things the diagram usually omits. EMV 3-D Secure, the cardholder-authentication protocol for card-not-present transactions, introduces a *separate authentication flow and separate infrastructure* — an access control server on the issuing side, a 3-D Secure server on the acquiring side — which must then correlate to the authorisation. Network tokenisation replaces the account number with a scheme-issued token, and the token-to-account mapping becomes **a new master relationship** somebody must own. Current version numbers for both specifications could not be sourced and are not given. One material difference is worth carrying: network tokens have no expiry constraint, whereas pre-authorisations expire after up to thirty days [8].

**PCI DSS is a hard dated constraint** and it moved recently: version 3.2.1 was retired on 31 March 2024, and the version 4.x future-dated requirements became effective on 31 March 2025 — among them two that address e-skimming by requiring integrity verification of payment-page scripts [28]. Any design that puts a payment page inside the bank's own document object model inherits those.

**Acquiring** is the merchant side, and it is a credit business pretending to be an administrative one: merchant onboarding and underwriting — the acquirer carries chargeback risk if the merchant fails — terminal estate and key injection, capture, clearing file submission, merchant settlement, pricing,

chargeback representment. It masters the merchant and outlet hierarchy, terminal records, the captured transaction and the payout. It is bad at joining the merchant-side and issuer-side views of one transaction — which is exactly what fraud and dispute analysis needs.

**The switch** is the router: it decides whether each message goes to the bank's own authorisation host, because the card is on-us, or out to the domestic switch or scheme network. It also provides **stand-in processing**, answering on the host's behalf within limits when the host is unavailable — a secondary set of authorisation rules and a secondary balance view that must later reconcile. Domestic inter-bank switching here is historically **SASWITCH**: R5.26 billion of settlement in October 2025, against R82.8 billion of Visa Debit and R76.2 billion of Mastercard in the same month [2].

**Interchange here is set by the central bank, which is the opposite of the European picture.** The Reserve Bank determines interchange for ATM and cash-back at point of sale, for debit and credit cards, and for authenticated collections; it is market-determined and bilateral for EFT credit push, EFT debit pull and real-time clearing [36]. The Interchange Determination Project, initiated in 2011, uses a cost-based methodology, reviews rates annually and the methodology every five years, and enforces determined rates by directive [36]. Current rates could not be sourced.

> **INTERCHANGE IS REFERENCE DATA WITH A REGULATOR ATTACHED**
>
> A rate with a regulatory source, an annual review cycle and a determination date belongs in a versioned, effective-dated reference-data store. It does not belong hard-coded in the acquiring platform's pricing module, where every one of those properties disappears.

**Where the card estate hurts.** Three places compute available balance. The authorisation host's rules and the fraud engine's rules are maintained by different teams with no combined view of what gets declined. The token-to-account mapping is treated as a technical detail, so the bank cannot answer "which cards are provisioned to which wallets". And scheme mandate releases — twice a year, non-negotiable — consume the entire card change capacity.

## 10. The payments hub and the rail interfaces

**What it is.** The bank's exposure to national infrastructure. The hub pattern answers a specific problem: without it, every channel implements every rail.

A hub accepts a *payment instruction* from any channel in a canonical internal form, enriches and validates it, screens it, selects a rail, formats the rail-specific message or file, submits it, tracks status, handles returns, and posts to the ledger. It owns the instruction lifecycle and it is the single place where "where is my payment" can be answered. The pattern is practitioner and vendor vocabulary rather than a published standard; treat it as a shape, not a specification.

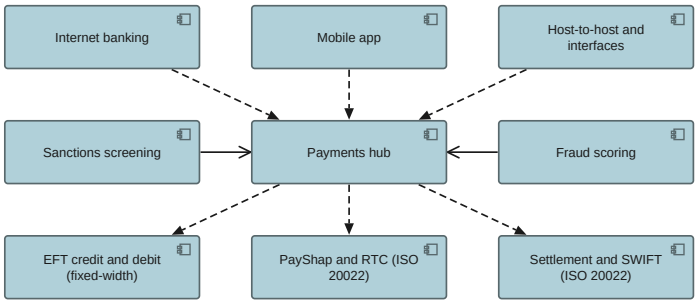



**Figure 12.5** The payments hub pattern. One canonical instruction in from any channel, screening and fraud as synchronous participants rather than downstream consumers, and one rail interface per rail. The failure to look for is a channel with a line straight past the hub to a rail — and there is usually more than one.

**What it masters.** The payment instruction and its full status history, the approval trail, beneficiary records and their verification state, and the rail-level reference that lets a client query be answered. Most of the capability list is control: validation and repair, beneficiary management with account verification, **duplicate detection** — the highest-value single control in the component — limit and mandate checking, screening and fraud scoring inline, rail selection and formatting, status tracking, returns and unapplied funds, and ledger posting with nostro and suspense management. An insurer's outbound payment is a controls problem on a modest volume; a bank's is a national-infrastructure obligation at scale, with hard cut-offs, a regulator that inspects it, and a direct line from a defect to a customer's salary not arriving.

**The rails, and what each does to the process.** The rail determines the process, not the other way round.

| Rail | Timing | Scale and limits | Data standard |
|---|---|---|---|
| EFT credit (batch push) | Deferred — customers wait "a day or two for funds to clear" [5] | 70.3% of 2024 retail settlement value [3] | Fixed-width clearing-house formats |
| EFT debit (batch pull) | Deferred; the business holds the mandate and must produce evidence or the collection reverses [7] | 5.8% of 2024 retail settlement value [3] | Fixed-width |
| DebiCheck | Early-morning window, ahead of other methods; a processed collection cannot be reversed [7] | 58.8 million mandate initiations in 2024 [3] | Clearing-house formats |
| Registered Mandates | Late window at 19:00; responses by 23:00 on the action date; settlement 05:00 next day [6] | Supports business and dual-signatory accounts, which DebiCheck cannot [6] | **ISO 20022** [6] |
| Real-Time Clearing | Credit within 60 seconds [5]; up to R5 million [8] | 11.0% of 2024 retail settlement value [3]; premium-priced, and the Reserve Bank said so in 2020 [5] | — |
| PayShap | Real-time, 24 hours a day | Cap raised to R50 000; 80% of transactions under R500 in late 2025 [9] | **ISO 20022 native** [9] |
| Cards | Authorisation now, clearing and settlement later | 10.7% of 2024 retail settlement value [3] | ISO 8583 scheme dialects [12] |
| High-value domestic | Gross, prefunded or collateral-backed, final and irrevocable, no recall [1] | R153.7 trillion across 16.0 million transactions in 2024 [3] | **ISO 20022** since 17 September 2022 [10] |
| Cross-border | Correspondent chain over SWIFT | Every one is also an exchange-control reporting event [13] | **ISO 20022** under the CBPR+ timetable [11] |

> **THE HEADLINE FACT ABOUT SOUTH AFRICAN RETAIL PAYMENTS**
>
> By value this is still overwhelmingly a batch EFT credit system: 70.3% of 2024 retail settlement value, against 0.7% for rapid payments [3]. The modern rails are growing fast and they are where the roadmaps point. They are not yet where the money is. Design for both, and do not let a conference deck tell you the batch is over.

**Where it hurts.** Each channel built its own rail integration before the hub existed, and three of them still bypass it. Approval is enforced in the user interface rather than the service, so a host-to-host integration bypasses segregation of duties. Duplicate detection is scoped per channel. And status is derived from file acknowledgements rather than modelled.

One specifically local failure. A canonical internal format designed around the fixed-width clearing-house world cannot carry ISO 20022 structured remittance data or structured addresses. PayShap, Registered Mandates and the next generation of the high-value system all assume ISO 20022. That is a remodelling, not a mapping.

## 11. Channels

Seven channels, each with its own session model, authentication, latency budget and failure mode, all needing the same answer to "what is my balance".

**Branch.** Teller and consultant applications over the core, plus cash, vault and till reconciliation and physical document capture. Networks are large and shrinking: for the 2024 and 2025 reporting years, branch counts were Capitec 880, FNB 630, Standard Bank South Africa 626, Nedbank 623 and Absa 616 [22]. Still the origination channel for complex products, and the due-diligence remediation channel of last resort.

**ATM and self-service.** Driven by the switch, not the core. Estates for the same period: Capitec 8 798, Absa 5 138, FNB 4 771, Nedbank 4 297, Standard Bank South Africa 3 450 [22]. Volume is falling fast, from 509.4 million ATM transactions in 2023 to 289.4 million in 2024 [3], so estate, key and reconciliation management cost more per surviving transaction every year.

**Call centre.** Telephony integration, screen-pop from the party master, call recording as an evidential record, and the agent desktop that aggregates eight systems behind one screen. That desktop is usually the worst piece of integration in the bank and the place screen-scraping is most often found.

**Internet banking.** The 2000s channel, now often the *business* channel while retail moves to the app. It carries bulk payment file upload, multi-user authorisation hierarchies and host-to-host integration for corporates — capabilities the app does not have.

**Mobile app.** The primary retail channel and the concentration of risk. Banking-app-related crime was about 89% of reported digital banking crime cases in 2025 and 70.5% of the amount claimed, with total reported losses rising from approximately R1.9 billion in 2024 to **R2.4 billion in 2025** across 110 074 incidents [20].

**USSD.** Genuinely load-bearing here rather than a legacy curiosity, and architecturally it is *synchronous, stateful and low-latency with a tiny payload* — the opposite of what an interface-first design assumes. Transaction shares per bank are not published.

**Chat banking.** Now mainstream, with no reliable adoption figures. The architectural point stands without them: a conversational channel has no screen, no session guarantee and a third-party platform in the trust path, so authentication, consent and audit must be designed explicitly rather than inherited from the web channel.

**Interfaces to third parties.** Bank-to-corporate host-to-host integration is mature and contractual. Open banking in the European sense **does not exist here as a regulated regime** — the Reserve Bank's own 2020 consultation says it has no policy or regulatory framework for open banking and that the governing Act has no provisions on open banking, screen scraping or interfaces [31], and a 2026 working paper puts third-party providers outside the current regulatory purview [32].

**Where channels hurt.** Each accrues its own business logic and becomes an undeclared system of record. The consent model differs per channel, so consent captured in the app never reaches the party master. And authentication strength varies, so the weakest channel sets the bank's fraud exposure.

## 12. Treasury and markets: front, middle, back

A separate landscape inside the bank, with its own vocabulary, its own regulator-facing reporting and usually its own technology team. Learn the three-office decomposition, because that is how treasury argues. The **front office** owns funding, investing, hedging, pricing and valuation inputs, cash and liquidity positioning, asset and liability management and trade capture. The **middle office** owns market and credit risk monitoring, pre- and post-trade controls, limit monitoring, stress and scenario, and independent valuation. The **back office** owns confirmation and affirmation, settlement processing, cash management, accounting entries, reconciliation and transaction reporting [25].

**Where it hurts** is well documented and you will recognise all of it: data breaks between treasury records, risk views, accounting and settlement instructions; manual re-keying; delayed visibility into cash, liquidity and exposure; and reduced auditability when approvals are dispersed [25]. The back office also connects to local market infrastructure — foreign-exchange settlement through CLS accounted for R922.8 billion of domestic settlement in October 2025, and dematerialised money-market instruments R154.3 billion [2].

## 13. General ledger and finance

The general ledger does not know what a mortgage is. It knows debits, credits, accounts and dimensions. Everything product-specific happens in sub-ledgers that *interface* to it — and the interface, not the ledger, is the architectural object.

**Three decisions determine a decade of reconciliation cost. Posting granularity**: per transaction, per account per day, or per product per day? Fine granularity gives analysis and kills performance; coarse granularity gives performance and forces an allocation layer that becomes permanent. **Chart of accounts design**: do the dimensions carry enough to serve statutory *and* management *and* regulatory reporting without a mapping layer? And **where the sub-ledger lives** — in the core, in a separate accounting engine, or in the data platform. The last is the common answer and the worst one, because it makes the authoritative accounting record a derived artefact.

**Where it hurts.** Twenty systems post with no common control framework, so month-end is a hunt for the one that did not balance. Suspense accounts become permanent. And the sub-ledger reconciliation is performed in a spreadsheet by one person who is about to go on leave.

## 14. Regulatory reporting

In a bank this is not a reporting function. It is a *production* function with statutory deadlines, attested accuracy and personal accountability, and it consumes an enormous share of change capacity. Four distinct obligations, in a combination a reader from another market will not expect.

**Prudential — the BA returns.** Submitted to the Reserve Bank's Prudential Authority under the Banks Act 94 of 1990 and its Regulations, specified by directive — currently Directive D6/2025, effective 1 July 2025, with each submission accompanied by a signed, certified form BA 099, or BA 099A for foreign operations [14]. The individual returns are listed on the Authority's website rather than in the directive text [14]. The anchor return is **BA 900, the balance sheet of deposit-taking institutions**: monthly, due within fifteen business days of the month-end it relates to, requiring granular data by institutional sector, instrument, currency and maturity, with counterparty columns running from deposit-taking institutions and the monetary authority through households and corporations to general government and non-residents [15].

> **BA 900 IS YOUR CONFORMED DIMENSION SET, WHETHER YOU CHOSE IT OR NOT**
>
> The counterparty-and-instrument matrix in BA 900 is effectively a mandated conformed dimension set for the bank's data platform. A bank that cannot classify every balance to it, monthly, has a data-architecture problem rather than a reporting problem — and the problem will present itself as a late return.

**Prudential — capital and risk.** The Authority implemented the remaining Basel III post-crisis reforms on 1 July 2025, covering standardised and internal-ratings-based credit risk, operational risk, the leverage ratio, credit valuation adjustment, market risk and the output floor [16]. Also on the backlog: Directive 14 of 2025 on large exposures, Joint Standard 2 of 2024 on cybersecurity and cyber resilience effective 1 June 2025, and a July 2025 joint communication on cloud computing and data offshoring [16].

**Exchange control and balance of payments — FinSurv.** Uniquely South African, and missing from every imported reference architecture. The banks, as authorised dealers, must report cross-border transactions to the Reserve Bank's Financial Surveillance Department in categories covering resident-to-non-resident payments, direct investment, non-resident rand flows and card transactions on both sides — the last **daily per authorised dealer** [13]. The specification references the ISO 20022 interbank payment messages [13].

The consequence is sharp. **Every cross-border payment and every foreign card transaction must carry a balance-of-payments category code, and that code is a business classification a payment system cannot infer.** It must be captured at origination or derived by rules, validated, and reported — which makes exchange-control reporting a mandatory consumer of the payments hub and the card platforms rather than an afterthought. Thresholds and the non-compliance regime are not in the material retrieved for this book.

**Financial crime.** For 2025/26 the Financial Intelligence Centre received 4 303 762 cash threshold reports, 718 464 suspicious and unusual transaction reports, 5 841 505 international funds transfer reports and 10 terrorist property reports — 10 863 741 in all, down from 13.5 million [19]. **Read the scale properly: international funds transfer reports alone are 5.8 million machine-generated reports a year. That is a data pipeline, not a compliance submission.**

**Deposit insurance.** Monthly, within ten working days of month-end, with chief-executive and chief-financial-officer attestation within three working days of submission [17].

**Where it hurts.** Each return is built as a point solution against whichever source was closest, so three returns disagree. The data platform is not the source, so returns come from extracts of extracts. Lineage cannot be demonstrated to the regulator. And a product change breaks four returns, and nobody knows which four.

## 15. The data platform

In a bank this layer carries more weight than in most industries, because it is the de facto integration layer, the regulatory reporting source, the risk-model feedstock and the only place the customer appears whole.

The canonical stack is source extracts, landing and staging, an integrated and conformed layer, marts by consumer, a semantic layer, and a feature store with model serving. Here it is commonly a hybrid: an on-premises warehouse holding the regulatory record and a cloud lakehouse holding the analytical workloads, with the migration incomplete and both authoritative for different things. Prevalence is not published; treat it as the shape to expect rather than a measured fact. It masters conformed history, the dimensions and the derived measures the business argues about. Nothing operational.

> **THE SINGLE MOST DAMAGING MISTAKE IN THIS LAYER**
>
> Letting the data platform master something operational: a customer segment, a risk grade, a limit, a collections treatment. The moment it does, the overnight-batch latency becomes an operational latency, and a customer is treated today on the basis of what was true yesterday. The failure is invisible until it is a conduct finding.

**Where it hurts.** Three warehouses from three eras, none decommissioned. A conformed layer modelled around the source systems rather than the business, so a new source means a new silo. And a feature store and a warehouse computing the same feature differently, so the model in production sees different data from the model in development.

## 16. Risk engines

Five distinct computational engines that an architect new to banking tends to collapse into one "risk system". They are not one system and they are not one team.

**Credit scoring.** Application and behavioural scorecards plus bureau scores, feeding decisioning and collections strategy. Needs model governance, version control and effective dating.

**IFRS 9 expected credit loss.** Computes provisions, from staging per exposure, probability of default, loss given default, exposure at default and forward-looking macroeconomic scenarios with weights. Runs monthly, feeds the general ledger and the BA returns, and must reconcile to both. **It is a production accounting engine, not an analytics exercise** — a distinction many banks learned late. Its staging must reconcile with the operational arrears state and the distressed-restructured classification under Directive 11 of 2025 [16].

**Regulatory capital.** Risk-weighted assets and capital ratios under the Basel framework as implemented locally, with the remaining post-crisis reforms effective 1 July 2025 [16].

**Market risk.** Value at risk and expected shortfall, sensitivities, stress and scenario on the trading book.

**Asset and liability management, and liquidity.** Interest-rate risk in the banking book, repricing gap, net interest income sensitivity, behavioural assumptions on non-maturity deposits, and the liquidity and funding ratios. This one needs the balance sheet at contractual *and* behavioural cash-flow granularity — usually the first time anyone has asked the core for cash flows rather than balances, and the answer is usually no.

**The unifying point.** All five need the same exposure data at different granularities and frequencies, and in most banks each gets its own extract. That is why a bank can report three different exposure numbers in one month, and why the first useful thing an architect can do here is propose one exposure base rather than a sixth engine.

## 17. AML, sanctions and transaction monitoring

Two different computational problems, routinely bought as one product. **Screening** is name-and-attribute matching against sanctions, politically-exposed-person and adverse-media lists, run at onboarding, on a periodic re-screen of the whole base, and inline on payments; it is fuzzy matching at scale and its false-positive rate is the dominant operational cost of the function. **Transaction monitoring** is behavioural: rules and models over transaction history producing alerts, cases and, where justified, regulatory reports.

Between them they master customer risk rating and the review schedule, due-diligence workflow, beneficial ownership for juristic persons, list ingestion and version control, alert triage and case management, regulatory report generation, and **targeted financial sanctions freezing**, which the Financial Intelligence Centre must require of accountable institutions under United Nations Security Council resolutions [19].

> **A LIST IS AN EFFECTIVE-DATED OBJECT**
>
> List versions must be retained and match results must be reproducible against the version in force at the time. A bank that cannot reproduce a historical match cannot defend a historical decision — and screening decisions are exactly the decisions that get re-examined years later.

The governing statute is the Financial Intelligence Centre Act 38 of 2001 as amended, with schedule changes effective 19 December 2022 extending supervision to designated non-financial businesses and professions [19]. The colloquial verb here is **"FICA"** — "the customer has not FICA'd" means due diligence is incomplete — used as a noun, a verb and an adjective. South Africa was removed from the Financial Action Task Force list of jurisdictions under increased monitoring on 24 October 2025, having largely addressed 22 action items [19].

> **A STALENESS WARNING YOU WILL NEED**
>
> Grey-listing between 2023 and 2025 drove a wave of remediation programmes, so material published in that window over-states the urgency and may describe controls since re-scoped. Check the date on anything you are handed about South African financial crime controls.

**Where it hurts.** Screening and monitoring from different vendors over different party views, so one knows about a customer the other does not. Alert volumes managed by raising thresholds rather than improving matching. Beneficial ownership captured as free text. A review backlog measured in years. And a payment rail that does not go through the hub, screened after the fact or not at all.

## 18. Fraud

Three distinct fraud problems with three distinct system shapes. A bank that treats them as one will be weakest at the one it most needs.

**Card fraud.** Real-time scoring inside the authorisation path under a hard latency budget, plus post-authorisation rules, plus chargeback and confirmed-fraud feedback. Card fraud losses were R1.7 billion in 2025, up 18%, with card-not-present the largest contributor [20].

**Application fraud.** Identity and document fraud at origination: synthetic identities, impersonation, falsified income. Needs identity verification, document authentication, device and velocity signals, and cross-application link analysis. Fraudulent home-loan applications fell 3.4% to R595.9 million in 2025 [20].

**Transaction and scam monitoring.** The dominant loss category and the one most often mis-designed. Digital banking crime losses were R2.4 billion in 2025 across 110 074 incidents [20]. The mechanism matters more than the number. Most incidents involve **social engineering**: impersonation of bank staff, police, government officials, fraud investigators, delivery companies and mobile-network representatives, creating urgency, and persuading victims to disclose credentials or one-time passwords — or **to approve transactions in their own banking app without realising what they are authorising** [20]. Generated voice, text, email and images have removed the old tells; poor spelling and unnatural wording are no longer reliable signals [20].

> **THE MOST IMPORTANT ARCHITECTURAL CONSEQUENCE IN THIS CHAPTER**
>
> If the customer authenticates correctly and authorises the payment themselves, every control built around authentication and authorisation passes.
>
> The bank's only remaining defence is behavioural: does this payment look like this customer, to this beneficiary, at this time, in this session, on this device. That needs beneficiary history, device and session signals, session-level telemetry, and an **intervention capability — hold, delay, challenge, warn — inside the payment path**. Which means the fraud engine must be a synchronous participant in the payments hub, not a downstream consumer.
>
> A design that places fraud monitoring after submission is architecturally unable to address roughly 70% of South African digital banking loss [20].

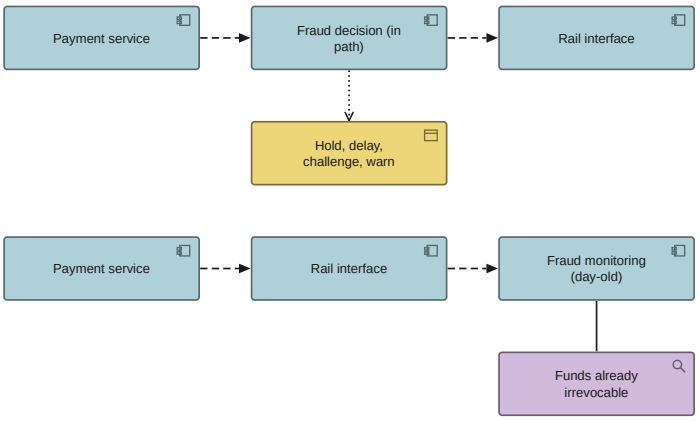



**Figure 12.6** The same fraud engine, two topologies. Above: a synchronous participant that can hold, delay or challenge before the instruction reaches the rail. Below: a downstream consumer reading the rail's output. On an irrevocable real-time rail the lower design can only produce a report, which is why it cannot address the dominant loss category.

**Where it hurts.** Card, application and transaction fraud in three systems with no shared entity resolution, so a mule account seen by one is invisible to the others. The fraud engine reading yesterday's data from the warehouse. And an intervention capability that exists only as "decline", with no delay-and-contact option.

## 19–21. Party, documents and identity

Three classes that an architect meets as one problem: who is this person, what did we send them, and what are they allowed to do.

**CRM and the party master** are two different things, both called CRM, and banks buy the second while assuming they have the first. The *operational party master* holds the authoritative party record — identity, identifiers, addresses, relationships including mandate-to-act and signing authority, risk rating, consent, and the link to every account. The *sales and service CRM* holds interactions, leads, campaigns, service requests and complaints. The aggravating difference from an insurer is that a bank has a *legal obligation* to produce an aggregated per-person view, and it is the deposit-insurance single customer view described earlier [18]. The argument about whether a single view of customer is worth building has already been settled by statute; what remains is whether it is a mastered capability or a month-end query.

**Document and content management** holds the documents of record: signed agreements, due-diligence evidence, security documents, mandates, proof of delivery, call recordings. The material properties are retention, immutability, indexing and retrieval under evidential conditions — and retention is driven by several statutes at once, including the prescription periods, the longest of which for a mortgage-secured debt is thirty years [27]. **Customer communication management** generates the outbound, with an archive of exactly what was sent and proof of delivery, several items being statutory artefacts including the section 129(1)(a) notice [27].

**Identity and access** splits by population. *Customer identity* covers registration, credentials, authentication, step-up, session management and the **transaction-authorisation** flow that is the last line of defence against the social-engineering losses above. State the asymmetry plainly: a bank can make authentication arbitrarily strong and still lose R2.4 billion a year, because the customer is authenticating correctly on the criminal's behalf [20]. *Workforce identity* covers joiner, mover and leaver, the entitlement model, privileged access, segregation of duties, **authority limits** — a business control implemented as an access control, usually in several places — and recertification.

**Where these hurt.** Three party masters, with the data platform's version the only complete one, which makes the complete view a day-old artefact. Identity matched on identity number alone, which fails for foreign nationals and pre-1994 records. Relationships modelled as account-level attributes, so a signing-authority change is made account by account. Consent held as a marketing flag rather than an effective-dated object with a lawful basis. Documents indexed by account number only, so party-level retrieval is impossible. Authority limits implemented per application, so changing one is a project. And step-up authentication triggered by amount alone, which the playbook routes around by keeping each transfer under the threshold.

## How these systems actually talk

Everything above is a box. What follows is the wiring, and the wiring is where a bank differs most from what you are used to. Five patterns, in rough order of volume of business value moved, not of architectural respectability.

**ALL FOUR ARE LIVE IN THE SAME BANK**

| Mechanism | Era | Data standard it can carry |
|---|---|---|
| File drops<br>dominant by value moved | 1970s onward | Fixed-width clearing-house formats.<br>Cannot carry ISO 20022 structures |
| Message bus / ESB<br>nobody will decommission it | 2005–2015 | A canonical model shaped by the rails it supported at the time |
| Point-to-point and direct database reads | always | The core's physical schema.<br>Every reader votes against moving |
| Interface gateway and event backbone | 2015 onward | ISO 20022 end to end — which the new domestic rails already assume |

Cards sit outside all four: ISO 8583 scheme dialects, one mapping layer per scheme.

**Figure 12.7** Four generations of integration, all live at once, with the data standard each one can actually carry. The right-hand column is the whole problem: the rails that matter most by value cannot carry the standard the new rails require.

**File drops.** Still the dominant mechanism. Fixed-width or delimited files on a schedule, landed on a managed file transfer platform, with a header, detail records, a trailer and a control total; an acknowledgement file back; and a reconciliation of counts and values. The clearing rails are file-based at the retail end [4]. **The clearing-house fixed-width format family is the lingua franca** — the record layouts for EFT credit payments, EFT debit collections and mandate files that every bank's bulk channel and every third-party collections bureau implements.

The exact current layouts are not public: the utility's rules place file submission times in product and service manuals supplied by its customer account management team, and in a participant fee book available under non-disclosure [4]. That is a genuine gap, not an oversight. Obtain the specifications from a participant bank before writing format detail into a design. **And the consequence that matters: fixed-width legacy formats cannot carry ISO 20022 structured data.** A hub whose canonical model was shaped by those formats needs a remodelling, not a mapping.

**Message buses.** The 2005 to 2015 investment: a central bus with canonical message formats, orchestration and transformation in the middle. It failed for the usual reason, which was centralisation rather than technology — one team, one release schedule, one point of contention, and orchestration logic accumulating in middleware with no business owner. What replaced it is a layering rather than a replacement: an interface gateway for synchronous request and response, an event backbone for asynchronous propagation, and the file estate left in place because it works. Expect all three generations live at once.

**Point-to-point.** The honest majority of integrations in any bank of age: a direct connection built by a project, owned by nobody, documented in that project's closed wiki. The telltale is an interface inventory shorter than the number of systems. Its specific banking form is **direct database reads** — a downstream system querying the core's tables because the core has no service. That couples the consumer to the core's physical schema, and it is the main mechanism by which a core becomes unreplaceable. Every direct reader is a vote against ever moving.

**Screen-scraping, which here is regulated rather than deniable.** Two forms. *Internal:* robotic process automation and agent-desktop integration driving a terminal emulator or a web session on the user's behalf, because the underlying system has no interface. Nobody owns it, it breaks on every screen change, it cannot be regression-tested — and it is the reason the agent desktop works at all. *External:* third parties using a customer's online banking credentials to initiate payments.

**South Africa regulates the external form rather than banning it**, which has no European analogue. Reserve Bank Directive No. 2 of 2024, published 15 November 2024 and effective about 13 February 2025, applies to any person issuing EFT credit payment instructions on behalf of a payer, by screen scraping or a similar tool, and requires registration with the Reserve Bank or an exemption [30]. The obligations include informed consent before credentials are used and again for each payment, with the request explaining that payments are irrevocable and that the payer is sharing credentials rather than logging into their bank; credentials encrypted and never stored; no use of a bank's branding without written permission and no false "instant payment" claims; audit trails including consent for five years; insurance against fraud losses; monthly volume and value submissions; and compliance attestations twice a year [30].

The consequence is specific. **Scraping traffic from registered providers is a known, named, regulated flow against the bank's own internet banking channel, and the bank must be able to identify it, risk-manage it, and distinguish it from fraud.** No imported reference architecture contains that requirement, because elsewhere the need is met by a mandated interface.

**ISO 20022, placed properly.** The one standard that genuinely unifies, and the one with hard external dates. The high-value settlement system completed its migration on 17 September 2022, the first real-time gross settlement system in Africa to do so [10]. PayShap is ISO 20022 native [9]; Registered Mandates is ISO 20022 with enhanced originator and recipient identification [6]; exchange-control reporting references the ISO 20022 interbank messages [13]. Cross-border, the correspondent-banking timetable applies: instruction migration completed in November 2025 with coexistence ended, mandatory receipt of certain exceptions-and-investigations messages from November 2026, mandatory send and receive from November 2027, and the end of coexistence for statements in 2028 — with the roadmap warning its own dates may change [11]. What is *not* ISO 20022: the legacy EFT rails and the clearing-house file family, and the card rails, which are ISO 8583 scheme dialects [12].

> **THE CORRECT SENTENCE ABOUT SOUTH AFRICA AND ISO 20022**
>
> South Africa is an early ISO 20022 adopter at the high-value and new-rail end, and a legacy fixed-width estate at the retail-batch end, **simultaneously**. The domestic rails are ahead of most global core banking systems. That is why a single canonical payment model is hard here, and why the migration is a decade-long overlap rather than a cutover.

## The batch clock and the business day

**The overnight cycle is the bank's operational heartbeat, and it is still the largest single constraint on architecture.** Roughly in order: cut-offs close; inbound clearing files are applied; outbound collection and payment files are generated; interest is accrued and capitalised; fees are assessed; the arrears state machine advances days past due; statements and notices are composed; the ledger interface runs; the risk extracts run; and the data platform loads, after which downstream systems load from it.

This is not merely an internal convenience. The domestic settlement system has a dedicated **night window for clearing-house batch processing**, and those night batches draw only on the participant's cash reserve account balance [1]. October 2025 night settlement was R1.52 trillion against R17.22 trillion in day settlement [2]. The batch window is a settlement-infrastructure fact.

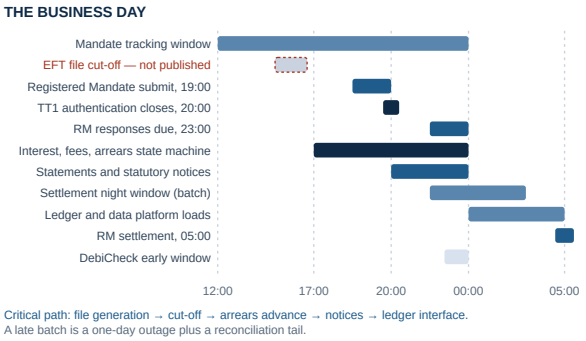



**Figure 12.8** The business day, with the published times marked and the unpublished one marked as unpublished. The EFT credit and EFT debit cut-off schedule is genuinely not public: the clearing utility's rules place file submission times in product manuals and a fee book under non-disclosure. Get it from a participant bank, and do not design a customer journey around a guess.

**Why so much is still batch.** Three reasons, in descending order of how often they are the real one. **The clearing rails are batch**: EFT credit and debit are file-based with action dates, cut-offs and deferred settlement [5], and a creditor cannot be real-time against a rail that is not. **Periodic computation is genuinely periodic**: accrual, fee assessment, amortisation, loss staging and regulatory aggregation are defined on a period rather than an event, and making them event-driven does not make them cheaper, it makes them harder to prove. And **the core was designed around an end-of-day** [23].

**The cut-off times that govern the business day** are what the business actually runs on, and an architect who does not know them will design a customer journey that silently misses one. The published ones: Registered Mandate collections submit in the late window at 19:00, with tracking from 12:00 to midnight for up to ten tracking days, responses due by 23:00 on the action date, and settlement by 05:00 the following day [6]. DebiCheck collections process in the early-morning window ahead of other methods, and a real-time mandate authentication request can be answered until 20:00 the same day, failing which it restarts [7]. The settlement system runs 24 hours a day, plus the night window for batch [1]. The EFT credit and EFT debit cut-off schedule is **not public** [4]. Get it from a participant bank, and do not let anybody tell you it is on a website.

**What breaks when a batch runs late**, in order of appearance. The collection or payment file misses its cut-off, so a day's collections or a day's salary payments do not go out, and the impact is immediate and public. Statements and statutory notices are not produced, which is a compliance event. Downstream systems load incomplete data, so every mart is wrong for a day and every report on them is silently wrong. And the arrears state machine does not advance, so collections works a stale book. **A batch failure is not a one-day outage. It is a one-day outage plus a reconciliation tail.**

**What real time costs, honestly.** The direction is unambiguous — the clearing utility intends to migrate real-time clearing flows to PayShap and eventually retire the older rail [8]. Price four things. Twenty-four-hour availability becomes mandatory, so there is no window to hide maintenance in: that means zero-downtime deployment and online schema change, not a faster batch. Liquidity management becomes intraday and continuous, because real-time gross rails are prefunded or collateral-backed [1]. Irrevocability changes the error-correction model from reversal to recovery [1]. And **the periodic computations do not go away**. So real time does not replace the overnight cycle. It is added alongside it, and the transitional state is permanent.

## The legacy question

The most consequential thing about a South African bank's architecture, and the subject on which published evidence is thinnest. This section is deliberately explicit about what is sourced and what is practitioner consensus, because the gap is wide and somebody will ask you.

**What is sourced.** Core modernisation is a multi-year business transformation rather than a technology upgrade, with phased migrations at large institutions running several years and a single cutover reserved for simpler environments [24]. The **hollowed core** is a documented target state, described as keeping only essentials such as the general ledger and transaction posting, with other services in satellite applications [23][24].

And the one well-documented local programme is Nedbank's Managed Evolution. As reported by the trade press quoting the bank's chief executive and chief operating officer at results, it began in 2016, cost R11.7 billion, and concluded nine years later with the delivery of the core banking platform — the last element modernised — at the end of 2024; the chief operating officer said "we concluded that project with the delivery of the core banking platform", and named securing qualified technology skills as a key challenge [21]. The architecture is described as modular, cloud-based and interface-driven, and **the core was delivered last** [21]. Attribute that to the outlet rather than to the bank's own disclosure, and note what the report does not say: it names no vendor, and an earlier public figure for the same programme was R9 billion, which is itself a datum about how these numbers travel [21].

Set against it, the honest counterexample. Barko Financial Services went live on a vendor-operated software-as-a-service core and digital front end **in under six months**, with a new lending service designed and built in twelve weeks and launched about three months later [33].




**Figure 12.9** The two sourced South African anchors, with the scope difference stated. Both figures are reported by the trade press and the vendor respectively rather than disclosed in an annual report, and are attributed accordingly. They are calibration, not benchmarks.

> **BOTH NUMBERS ARE TRUE, AND THEY ARE NOT COMPARABLE**
>
> Nine years and six months are both real because they are different problems. Barko was a single-product lender on vendor-operated software-as-a-service with no legacy book to migrate [33]. Nedbank was an incumbent universal bank modernising an entire estate with the core delivered last, and every product ever sold still accruing on it [21].
>
> Anyone who quotes the six months at you as a benchmark for an incumbent core replacement is comparing a greenfield launch to a book migration. The thing that takes nine years is not the software. It is the book.

**What is practitioner consensus, flagged as such.** None of the following is sourced, and none of it could be, because no South African bank publishes an application portfolio and no analyst survey of the local core estate was available for this book. Present each as the practitioner view and you are defensible.

1. **Mainframes and COBOL are present and load-bearing** in the largest banks, running retail deposit and card ledgers, with the usual mainframe data stores and transaction processing.
2. **The real constraint is the skills pipeline, not the technology.** The systems work. The people who understand them are retiring, and the closest thing to a public acknowledgement is a chief operating officer naming technology skills as a key challenge [21].
3. **Core replacement here has a poor completion record and a good *partial* record.** Programmes that set out to replace a core and instead delivered a digital front end, a product engine and a data layer are the norm — and several are successes mis-scoped as replacements.
4. **"Hollow out the core" is what actually happens, usually without being chosen.** Product definition moves to a product engine, identity to a party master, orchestration to a service layer, analytics to a lakehouse, and the core ends up as a ledger and a posting engine — the intended target state of a deliberate hollowing programme, arrived at by accretion instead.
5. **Mergers and acquisitions, not technology choices, explain most duplication.** The multiple cores, card platforms and customer masters in a local banking group are usually artefacts of corporate history, with an owner who left.
6. **The rest-of-Africa estate is a separate landscape wearing the same brand**, typically a different, newer, packaged core per market with its own regulators and clearing systems.

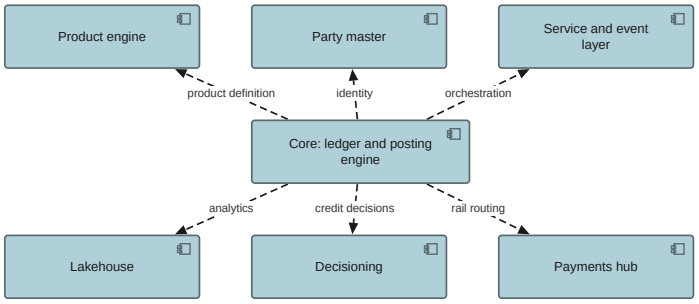



**Figure 12.10** Hollowing out the core. Each arrow is a capability that left the ledger, and every one of them is a thing somebody once called "core". Done deliberately this is a target state; done by accretion over fifteen years it is the same picture with no owner for the arrows.

**What an architect actually finds, on arrival.** A hollowed-out core they were told was a monolith. More cores than the application inventory shows. A data platform doing operational work. An agent desktop held together by screen-scraping. Three computations of available balance. A payments hub with three bypasses. A service bus nobody will decommission. A collections system that cannot compute prescription. And a capable, under-documented mainframe team who are the only people who know what end-of-day actually does.

Two pieces of local history live on in data models rather than in systems. **Cheques were withdrawn entirely, effective 31 December 2020** — issuance and acceptance both [34] — and yet "cheque account" survives as the common name for a current account, paper clearing remains a named clearing house in published rules [4], and any payment-mix analysis crossing 2020 contains an instrument that no longer exists. And **the early debit order streams were switched off on 31 October 2021** and replaced by the authenticated collection rail [35], so an older estate carries mandate records and reason codes for instruments that cannot be used.

One more change happened under everyone's feet, and it changes who you write to. The payments industry body's recognition as scheme rule-maker was withdrawn, with functions transferring during August and September 2026 — clearing-house and rule functions plus licensing and registration to the Reserve Bank itself, standards to its oversight division, and the remaining clearing houses to the renamed clearing utility [37]. That utility, formerly BankservAfrica, is now **50% owned by the Reserve Bank** [38]. So the body that writes the rules your payment systems must obey is, for most streams, now the central bank. Design decisions that would be commercial elsewhere are regulatory here.

**Ask.** What is the core replacement history here — what was attempted, what was delivered, and what was the gap? Which systems have no identified owner, and which have nobody under fifty who can change them? What is the application count, and when was it last verified by decommissioning rather than by survey? And what share of the change budget goes to regulatory and scheme mandates before any discretionary work begins?

#### THE CHAPTER IN TEN LINES

- A bank's estate is two businesses welded together with a control layer across both: a slow ledger-centric balance-sheet business, a latency-bound payments business regulated as national infrastructure, and the risk, crime and reporting systems that read from both.
- "Core" is a question, not a noun. Establish separately where the authoritative balance is, where product definition lives, where customer identity is mastered and where end-of-day runs — four answers, often four systems.
- Lending is a conveyor of four or five systems, not one, and the retail and business conveyors are different classes of software; the three objects that must survive every hand-off are the input vector, the arrears state and the prescription clock.
- There is no such thing as an ISO 8583 interface — there is a Visa interface, a Mastercard interface, a domestic switch interface and three mapping layers; and the gap between the authorisation balance and the ledger balance is permanent.
- By 2024 settlement value South African retail payments were still 70.3% batch EFT credit, while the high-value settlement system has been ISO 20022 since September 2022 and PayShap is ISO 20022 native — the domestic rails are both ahead and behind at once.
- Deposit insurance brings a mandated single customer view with specified aggregation rules, monthly within ten working days and signed by the chief executive and chief financial officer; built as a reporting query it cannot do the one job it exists for.
- If the customer authenticates correctly and authorises the payment themselves, every authentication control passes — so the fraud engine must be a synchronous participant in the payment path with hold, delay and challenge, not a downstream consumer.
- Screen-scraping here is regulated rather than deniable: Directive No. 2 of 2024 requires registration, per-payment consent, five-year audit trails, insurance and biannual attestations, so scraped traffic against your own channel is a flow you must be able to identify.
- The overnight cycle is a settlement-infrastructure fact, not an internal convenience; real time is added alongside it rather than replacing it, and the transitional state is permanent.
- Nedbank's R11.7 billion over nine years with the core delivered last, and Barko live on vendor-operated SaaS in under six months, are both true and not comparable — and the mainframe, COBOL and core-replacement claims everyone repeats are practitioner consensus, not sourced fact.

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CHAPTER 13

# The reference architecture

*Banking has several mature published answers to "what are the words" and "what is the list of things we do". BIAN, ISO 20022, FIBO, the IBM lineage, APQC and BCBS 239 all overlap, none of them is a design, and one of them has no published licence at all.*

## Four teams, four customer counts

In her second week at Table Bay Bank, the new lead enterprise architect asked how many customers the bank had. Retail said 2.4 million. The party master held 3.1 million records. The deposit-insurance return said 1.9 million qualifying depositors. The prudential balance-sheet return does not count customers at all — it counts balances by counterparty sector.

Nobody was lying. Retail counted people with an active product. The party master counted records, including duplicates, closed relationships and the deceased. The deposit-insurance figure counted *qualifying depositors* under a statutory definition with exclusions built into it. And the balance-sheet return has no concept of a customer, because prudential regulation does not care who you are, only what sector you belong to.

Table Bay Bank is an invention carried through this book, and nothing about it is a case study. The argument is not. Four weeks of meetings produced a reconciliation spreadsheet nobody trusted, which broke the following quarter.

This is the problem a reference architecture solves. Not the big problem — not *what should our architecture be*. The small, grinding, expensive one: what are the words, and what is the list of things we do. A bank that cannot answer those two questions consistently cannot be modelled, measured or modernised, and will spend the first eight months of every programme re-deriving them.

Banking is unusual in having **several mature, maintained, publicly documented answers that genuinely overlap**. So the first real architectural decision a bank makes is not which core to buy. It is which reference model governs its vocabulary.

One warning before any of the detail: **none of these is a design.** They are vocabularies, classification schemes and conformance obligations. An architect who treats one as a target architecture produces a three-hundred-component landscape diagram nobody can build or fund.

> **WHAT THIS CHAPTER DOES NOT DO**
>
> Every body described here publishes copyrighted material, and several sit behind membership or registration gates. This chapter *describes and points*. It does not reproduce service-domain lists, message definitions, process hierarchies or ontology fragments, and neither should your own documents without written permission.

## What a reference architecture is actually for

Three honest uses, in ascending order of how often they pay for themselves.

**Boundary arbitration.** Two teams both claim they own "customer onboarding". Both are right about part of it, both have a roadmap, and the argument has run for two quarters. An external decomposition settles it — not because it is better than either team's, but because it is *nobody's*. Definitional disputes are political; an industry decomposition de-politicises them. This is the highest-value use of a reference architecture in a bank, and the cheapest.

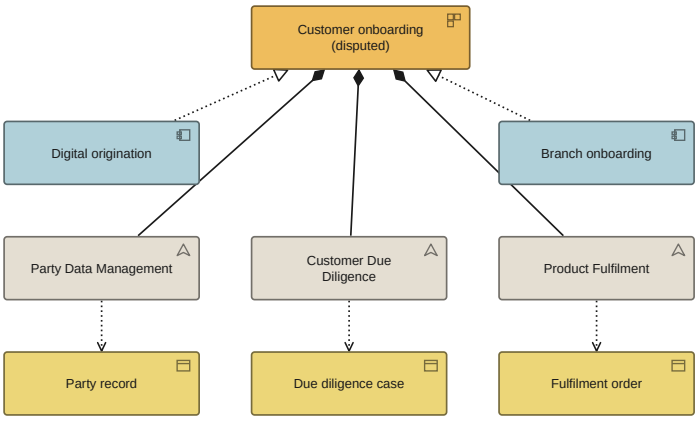



**Figure 13.1** The highest-value use of a reference architecture, and the cheapest. Two components both realise what both teams call "onboarding". Three Service Domains, each owning one control record, cut the capability where the records divide — and the argument becomes a question about which component owns which record.

**A coverage check.** Given a complete published list of what banks do, you can ask of your own estate: which of these can we actually do, where, by whom, and how well. The value is in the *absences*. Nobody writes down the capability they have never had. A published list makes the hole visible — and in chapter 12 the holes were things like a prescription clock and an exposure base, neither of which anybody had written down.

**Boundary language.** When you write a request for proposal, integrate with a scheme or negotiate with a third-party processor, a shared term for the thing being exchanged saves weeks. "BIAN-aligned" is now a testable vendor claim, which it was not ten years ago.

**The three ways architects misuse one**

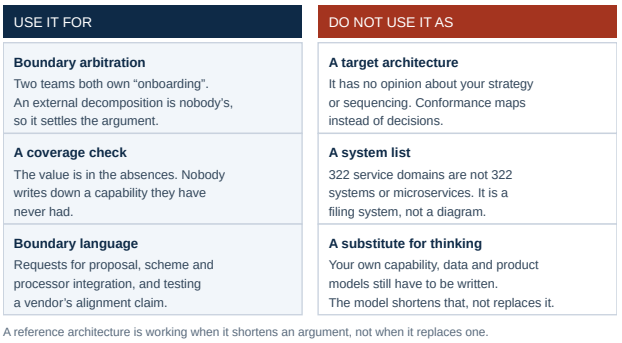



**Figure 13.2** Three honest uses and three misuses. The misuses are not careless — each is what you get by taking the published picture at face value.

**As a target architecture.** A reference model has no opinion about your strategy, your cost base, your funding mix or your sequencing. It cannot have one: a model that told every bank what to build would be wrong for all of them. The reference model describes the *problem space*; your target architecture is a set of choices inside it. Confusing the two is expensive precisely because it looks like progress — you get conformance maps instead of decisions.

**As a system list.** Treating 322 service domains as 322 systems, services or microservices is the single most costly error in this chapter, and it happens in good faith, because the published picture looks like an architecture diagram. It is not. It is a filing system.

**As a substitute for thinking.** Because reference models exist, teams conclude the bank's own capability, data and product models need not be written. They do. The reference model shortens that work and raises its quality. It does not replace it, and a programme whose deliverable is "BIAN adoption" has no deliverable.

In TOGAF terms the fit is simple, and saying it out loud ends the "where does this go" argument. An industry reference architecture is an input to the Architecture Repository, specifically the reference library. It is consumed in Phase B when you need a capability baseline and a glossary, and in Phase C when you need a conceptual information model to check your own against. It is not an Architecture Vision and it is not a Target Architecture. The deliverables remain yours. What the reference model removes is the blank page, and the three months of committee work a blank page generates.

> THE CLEAN FORMULATION
>
> **TOGAF tells you how to do architecture. BIAN tells you what the things are called. ArchiMate is how you draw either.** Anyone who asks you to "choose between TOGAF and BIAN" has been misled, and correcting that in the first meeting will save you a cycle.

## BIAN — the Banking Industry Architecture Network

**Who publishes it.** A member-owned, not-for-profit association established in 2008, a German registered association headquartered in Frankfurt [5]. Its own site reports 113 members — 40 banks and 59 software partners — and 18 active working groups, naming large banks alongside large vendors and consultancies [3]. That mix is itself an architectural fact: BIAN is a *negotiated* standard between banks and the vendors selling to them, which is why it is simultaneously very detailed and very careful not to prescribe implementation.

**What it actually is.** A semantic framework for banking services, deliberately limited to semantics — what things are called and what they are responsible for — not how they are built [5]. Six pieces you must be able to name.

A **Service Domain** is the atomic unit: a self-contained area of business capability with one responsibility boundary and one type of record that it owns. A **Control Record** is that record — the single master information object a Service Domain owns through its lifecycle. **Behaviour Qualifiers** are sub-facets of a control record that service operations act on. **Service Operations** are the verbs applied to a control record or qualifier. **Semantic APIs** are logical interface specifications, one per Service Domain that has at least one Service Operation. And **Business Domains** group Service Domains.

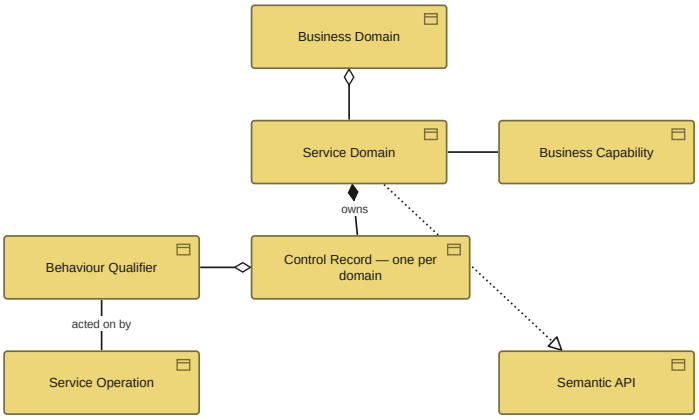



**Figure 13.3** The six pieces of BIAN you must be able to name, drawn as business objects because that is what they are — published information products, not systems. Read down the middle: a Business Domain groups Service Domains, each Service Domain owns exactly one type of control record, and the operations and the interface hang off that record.

> **ONE CONTROL RECORD PER DOMAIN IS A DATA-OWNERSHIP RULE IN DISGUISE**
>
> The control record is the most under-appreciated part of BIAN and the part that does real work. "One Service Domain owns one control-record type through its lifecycle" is not a modelling convention. It is a statement that exactly one component is authoritative for each kind of record — which is the answer to "where is the authoritative balance", "where is product definition" and "who masters the token-to-account mapping" from chapter 12.
>
> If you take one thing from BIAN and ignore the rest, take this.

**The Service Landscape, and BIAN's own modesty about it.** BIAN describes the Landscape as "a reference structure that categorizes and organizes BIAN Service Domains for ease of access", adding that different classification criteria produce different layouts of the *same* set of domains [4]. That sentence kills the commonest misuse — treating the Landscape picture as the architecture. Print it on the wall next to the poster.

**Current release, with the numbers.** Service Landscape **14.0**, release notes final version **20 February 2026**, announced 13 March 2026 [1][2]. From BIAN's own metrics: **322 Service Domains** (from 327), **38 Business Domains** (unchanged), **586 Business Capabilities** (unchanged), **4,905 Service Operations** (from 5,139), **242 Semantic APIs** (from 249), 1,585 Business Objects, 2,515 Business Object Attributes, 362 Business Scenarios, 245 control-record diagrams, 430 business-object-model diagrams — and **no metamodel change** [1].

**SERVICE LANDSCAPE 14.0 · RELEASE NOTES 20 FEBRUARY 2026**

| Metric | 13.0 | 14.0 | change |
|---|---|---|---|
| Service Domains | 327 | 322 | −5 |
| Business Domains | 38 | 38 | — |
| Business Capabilities | 586 | 586 | — |
| Service Operations | 5,139 | 4,905 | −234 |
| Semantic APIs | 249 | 242 | −7 |
| Metamodel | | unchanged | — |

18 domains added (rewards, a payments cluster, corporate card, three insurance).
7 obsoleted — **all of them payments.** That is the release, in one line.

**Figure 13.4** Service Landscape 14.0 against 13.0, from BIAN's own metrics table. A standard whose service-operation count fell by 234 in one release is being pruned, which is healthier than growth — and is still a migration event for anyone who hard-coded a name.

Two observations worth more than the counts.

**The payments re-cut is the real content of 14.0.** Eighteen Service Domains were added — a rewards and incentives cluster, a payments cluster covering payee management, customer consent, payment orchestration, payment confirmation and payment settlement, a corporate card domain, and three insurance domains — and seven were obsoleted, *all of them payments* [1]. Read that as BIAN re-cutting payments for an ISO 20022 world. The old payment domains were shaped around instruction, execution and order; the new ones are shaped around orchestration, consent and confirmation, which is how an ISO 20022 payment actually decomposes.

**And the Semantic APIs confirm it.** The 242 specifications come in four flavours: REST over BIAN's own business object model, REST over the ISO 20022 business component model, asynchronous over BIAN's model, and ISO 20022's [1]. Sixteen Service Domains have no service operations and therefore no interface [1].

> **WHAT THE DUAL FLAVOURING TELLS YOU**
>
> **BIAN has accepted that ISO 20022's data vocabulary wins inside payments**, and now offers its own service decomposition with ISO 20022 payloads. That is the single most useful fact in this chapter for a South African architect, because the domestic rails are already ISO 20022 at the high-value and instant end. You do not have to choose between the two standards. You use BIAN for *who owns what* and ISO 20022 for *what the message says*.

**A naming drift to flag rather than paper over.** The metamodel is documented as three levels — Business Area, Business Domain, Service Domain [5] — but version 14.0's release notes count Business Domains and never use "Business Area" [1]. If your repository has a Business Area layer, say where it came from and which release it tracks.

**Behaviour qualifier renames are a hidden migration cost.** Version 14.0 removed, renamed and replaced a large set of qualifiers, mostly to rationalise naming: "Account Statement" became "Financial Status Tracking", "Withdrawal" became "Debit and Credit" [1]. Harmless in a document, expensive in anything that hard-coded the old names.

**Certification lags the Landscape, and you should know by how much.** Certification is commercial and sits with a BIAN subsidiary. A third-party accredited publisher records a 2019 version one, now retired; a 2022 version two against Service Landscape 9; and a 2026 version three — Foundation, Banking Architecture Practitioner and Data Architecture Practitioner — against **Service Landscape 13**, with version two sales retiring in June 2026 and a Business Transformation Practitioner expected in the fourth quarter of 2026 [9]. Foundation is sixty questions in an hour, 70% to pass, valid two years [5]. So the current certification tests the previous Landscape. Say so when somebody waves a certificate at a design decision.

#### What BIAN costs, and the thing most people get wrong

Be precise here, because a programme plan can die on it.

BIAN's site states that the BIAN Portal is available to members and non-members [3]. Release notes for recent versions are openly linked documents [1]. The Service Landscape portals for versions 13.0 and 14.0 sit behind a registration form asking for name, email, company and country. A separate member login leads to a wiki.

And the only intellectual-property term found anywhere on the deliverables, membership and join pages is a copyright footer [6][7][8]. **No open licence. No published usage terms. No royalty-free grant.** The join page lists benefits — working-group participation, networking, custom training, governance involvement — with no tiers and no fee amounts, and the membership frequently-asked-questions page poses the question "how much is the annual membership fee" and does not answer it [7][8].

> AVAILABILITY IS NOT PERMISSION
>
> As at October 2026, BIAN's core artefacts are obtainable by a non-member through a registration gate, but **BIAN publishes no licence**. Those are different things, and a registration gate is not a grant.
>
> So: if you intend to embed BIAN content in a contractual deliverable, a product you sell, a request for proposal issued to the market, or a repository a systems integrator will hold, **get the terms in writing before you do it**. Do not infer a licence from the fact that you were able to download something. Fee amounts, licence terms for portal and repository content, and whether redistribution inside a vendor deliverable is permitted could not be established for this book, and are not stated here as fact.

#### Coreless Banking, where BIAN makes an argument

Everything above is a dictionary. **Coreless Banking** is the one place BIAN takes an architectural position: that a bank can be assembled from modular services orchestrated across domains rather than anchored on a monolith. It launched on 23 September 2019 with six large banks and five vendors, covering consumer payments, customer offers and consumer loans [10].

The participant list is itself the lesson. Two of those five vendors have since been absorbed into other companies. The current state and outputs of the initiative could not be established for this book. Treat Coreless Banking as a well-argued thesis with an unclear delivery record — which, in fairness, describes most of the "composable banking" literature.

#### How BIAN relates to TOGAF and ArchiMate

The joint work is old and real. The Open Group and BIAN launched a practical guide to enterprise architecture in banking on 14 November 2013, updated for an early Service Landscape release and framed as using BIAN content inside a TOGAF effort [11]. The Open Group also publishes ArchiMate expressions of the BIAN model as catalogue items [12].

Two cautions. Which BIAN version each Open Group artefact models, and its access terms, could not be established; the only version signal retrieved was a title referring to Service Landscape 7.0, several releases behind [12]. And BIAN's own current work centres on core systems [5], so three new insurance domains are a start, not a bancassurance reference model.

**What BIAN is genuinely good for**, in one list. Arbitrating capability boundaries. Making vendor overlap visible — map two shortlisted suites onto the same domain set and the gaps and the integration surface become undeniable. Supplying an interface and event naming convention to a bank that has none, which is most banks. Testing vendor claims: ask which domains, which version, which flavour, and whether the mapping is published. And onboarding — the 38 Business Domains are the fastest legitimate map of what a bank does.

### ISO 20022 — a reference data model wearing a message format's clothes

The commonest error in banking architecture documents is calling ISO 20022 "a message format". It is a **methodology plus a central repository of business models from which messages are generated**. The messages are an output.

**Who publishes it.** A multi-part international standard from the financial services technical committee, in eight parts covering methodology, registration and schema generation, with a **Registration Authority** maintaining the central repository that holds the models and their schema outputs [15].

**What is inside.** Two major parts: a **Data Dictionary** and a **Business Process Catalogue** [16]. The Dictionary holds *business concepts* — business components, business elements with multiplicities, business associations, business roles and constraints; *data types*, defining valid values by format or enumeration under representations such as amount, quantity, code set, date, time and text; and *message concepts* — message components, including

choice components that permit one element only, message elements and constraints [16]. Items carry a registration status: provisionally registered, registered, or obsolete with a removal date [16].

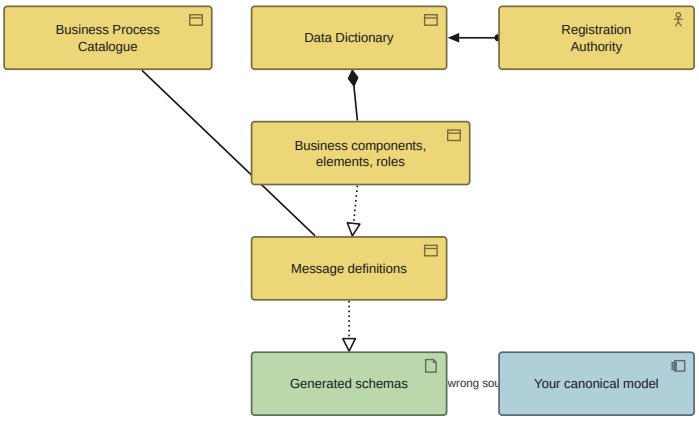



**Figure 13.5** Why calling ISO 20022 "a message format" is wrong. The business model layer is the asset; the schemas are generated from it. Most banks consume the bottom row and build a canonical internal model from it — which is building from the output rather than the model.

**The business model layer architects ignore.** Business components, associations and roles are a financial-services conceptual data model *with change control*. Model changes go to the Registration Authority on a dedicated form; new messages start from a **Business Justification**; maintenance runs through change requests; and code sets update through request forms [15].

**The governance you will meet in a steering committee.** A Registration Management Group approves business justifications. The Registration Authority validates requests. **Standards Evaluation Groups** — payments, securities, trade services, foreign exchange, cards, plus a derivatives subgroup — validate new messages. A technical support group maintains the business application header, approved by a cross-group harmonisation body [15]. If you are ever told a field cannot be added, that is the chain that said no.

**What it costs, and the detail that tells you everything.** The repository offers **free access to the Data Dictionary and the Business Process Catalogue**; the standard itself is purchased; and submitting organisations with an approved business justification can obtain the modelling editor and a version of the full repository [15]. That full repository ships in an Eclipse modelling format built on a decade-old Eclipse release, and browsing it means downloading a metamodel and generating a plugin — the page itself warns of a known performance problem [17].

> **WHY NOBODY IN YOUR BANK HAS SEEN THE MODEL**
>
> The industry's richest free financial data model ships in a decade-old Eclipse modelling format. So most banks consume generated schemas and never see the business model at all — and then build a canonical internal model from the *schemas*, which is the wrong layer. If you want the conceptual value, somebody has to go and get it.
>
> The precise intellectual-property and terms-of-use conditions on repository content could not be established for this book. The standard links separate pages for them; read those before you redistribute anything.

**Where not to use it.** Not as a service decomposition — it tells you what a payment *is*, not who owns the capability. Not as a warehouse model; deriving analytical structures from message definitions bakes transport concerns into the data layer. And it has no opinion about pricing, eligibility or the product catalogue. The pattern that survives is **canonical-internal, standard-at-the-edge**.

**Why it matters more here than almost anywhere.** The correspondent-banking migration made ISO 20022 operative for cross-border payments with a November 2025 milestone for instruction messages and the retirement of the legacy equivalents [18]. South Africa ran ahead of that: the domestic high-value settlement system completed its migration on 17 September 2022, the first real-time gross settlement system in Africa to do so [19][20]. So a core or hub that is not ISO 20022 native is a permanent translation liability in this market, not a future one.

## FIBO — the Financial Industry Business Ontology

**Who publishes it.** The EDM Council develops and maintains FIBO with industry subject-matter experts [21], with the canonical repository on a public code host [22]; a standards body also maintains a page on it [23].

**What it is.** A machine-readable **ontology** of financial terms, concepts and relations, in modular domains, with concepts, relationships and attributes, published in the semantic web's description languages and reviewed quarterly by ontologists and sector specialists [21]. Its publisher's stated uses include data discovery and classification, standardisation across firms and systems, interoperability, risk analysis for customer due diligence, integrating heterogeneous sources — and **regulatory reporting consistency, with BCBS 239 named explicitly** [21].

**Where it earns its place.** When the problem is *meaning across heterogeneous sources* — reconciling counterparty, instrument and legal-entity concepts across a markets business, a lending book and a regulatory return. It is a poor fit for transaction processing, product configuration or anything latency-sensitive.

Two honest cautions. The publisher says institutions worldwide have deployed it but **names none**, so the book says adopters are not publicly named rather than asserting a user base [21]. And the current release identifier and the exact licence could not be established from the publisher's own page; the

repository's licence file would settle it [21][22]. Pair FIBO with BIAN rather than choosing: BIAN bounds *who owns what*, FIBO defines *what the words mean*.

## IFW and BDW — the IBM lineage, and what survives

Every bank data warehouse built between roughly 1995 and 2015 either used IBM's banking models or was built by people who had. The **Information FrameWork** was developed at IBM as an alternative to Zachman and documented in the IBM Systems Journal in 1996, with content divided into information models, process models and integration models [13].

What survives does so under a different name. The lineage became part of what is generally called the Industry Models [13], and IBM's product documentation index names the current banking offering the Banking and Financial Markets Data Warehouse, explicitly labelling the older Banking Data Warehouse and Financial Markets Data Warehouse as replaced by it, their last entries from 2016 [14]. The most recent utilities release in that index is dated 28 February 2020, the newest content entry 31 January 2018, and the index page itself was last modified in March 2020 [14].

**Characterise that honestly.** The lineage is alive as a product name and visibly *quiet*. A documentation index last touched in 2020 with content dated 2018 is not evidence of discontinuation, nor of active investment. Support status, roadmap, licensing and pricing could not be established for this book.

**Where it is still useful.** Three places. **Archaeology** — if a bank's warehouse has tables named for these concepts, the models are the documentation nobody kept, and that is worth real money on a migration. **A ready-made analytical data model** with banking depth that BIAN does not attempt. And **a negotiating position** in a greenfield data programme: an existing industry model, even dated, is a baseline that "we'll model it ourselves" has to beat.

**Where not to use it.** Not as a service or interface decomposition — that is BIAN's job, and these integration models predate modern interface practice by two decades. And not as a physical design. The practitioner rule of thumb is that an industry data model delivers somewhere between sixty and eighty per cent of a logical model and none of a physical one, discovered at performance testing — **a practitioner figure, not a measurement, and you should present it as one**.

## APQC's banking process framework

**Who publishes it.** APQC maintains a cross-industry Process Classification Framework plus a set of industry-specific variants. The **Banking framework is at version 8.0, dated 26 August 2026**, published in document and spreadsheet form, with earlier editions still in the library [24].

**What it is and what it is for.** A hierarchical decomposition of business processes — categories, process groups, processes, activities, tasks — with stable identifiers, designed for benchmarking. Its value is that it is **process-shaped where BIAN is capability-shaped**. APQC answers "what are the steps, and who benchmarks them". BIAN answers "what is the responsibility boundary". A bank with both, mapped, has the beginnings of a real operating model.

**What it costs.** The framework pages show download links alongside register and sign-in prompts, and state neither whether download requires registration or membership nor under what licence [24]. Licensing and non-member access could not be established. **Do not assert it is free.**

**Where not to use it.** It is a benchmarking taxonomy with no data model, so it cannot arbitrate system boundaries. Using its identifiers as application-landscape keys produces a process catalogue that looks authoritative and maps to no system.

## BCBS 239 — the principles that force the data architecture

The others are vocabularies. This one is an obligation, and it is the only framework in this chapter a bank can be **supervised against**.

The fourteen Principles for effective risk data aggregation and risk reporting were published in January 2013, with a compliance date of 1 January 2016 for the global systemically important banks designated by November 2012, and a recommendation that national supervisors apply them to domestic systemically important banks three years after designation [26]. They fall into four groups.

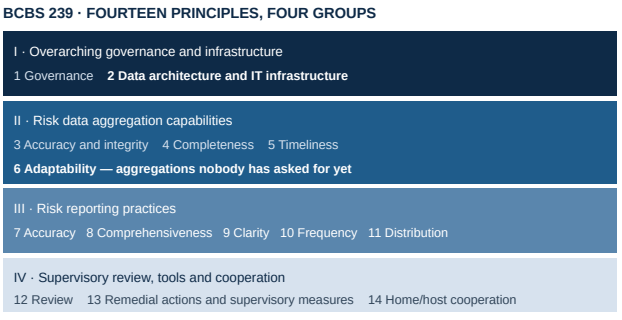



**Figure 13.6** The fourteen principles in their four groups. Principle 2 is the one that lands on an architecture backlog; Principle 6 is the one that rules out a data platform built only around the current returns.

**What they oblige, in an architect's terms.** Principle 2 is the one that lands on a backlog: design, build and maintain data architecture and infrastructure that *fully support risk data aggregation in normal and stress conditions*. In practice that means named data ownership, documented lineage from source to report, controlled reference data, aggregation on demand rather than on a monthly cycle, and fast answers to **ad hoc** questions in a crisis. Principles 3 to 6 are a data quality management system with a regulator attached.

**Principle 6 is the sleeper.** Adaptability requires producing aggregations nobody has asked for yet — which rules out a data platform designed only around the current returns. Go back to chapter 12 and read the regulatory reporting section again with that in mind: a return built as a point solution against whichever source was closest satisfies today's question and fails Principle 6 by construction.

**Why it is still on every large bank's backlog, thirteen years on.** Because compliance was not achieved, and the Committee says so in measured language. A newsletter dated 6 January 2026 states that the principles still apply but implementation has evolved, that meeting their intended outcomes "remains a continuous effort", and that a data-driven culture "remains a work in progress" at some organisations [25]. Its findings read like a list of things you will find in your first month: **data lineage remains difficult because of legacy systems and distributed data estates**; producing **timely ad hoc reports, especially during crises**, is a significant hurdle, with testing in normal times suggested as preparation; cross-border banks may standardise group-level practices to manage differing jurisdictional expectations; and automation and artificial intelligence are promising but early-stage, with outputs dependent on data quality [25]. The newsletter is explicit that it creates no new supervisory guidance [25]. Earlier evidence of the same gap: a 2018 European thematic review found that **none of the twenty-five significant institutions examined had fully implemented the principles**, attributing the weakness mainly to unclear responsibility and accountability for data quality [26].

> **THE MOST USEFUL FACT IN THIS CHAPTER**
>
> BCBS 239 is unfinished for architectural reasons, not analytical ones. It fails on lineage, ownership and ad hoc aggregation — exactly what a fragmented application estate cannot provide.
>
> That makes it **the most reliable business case an architect has** for data-platform, lineage and master-data work. It is supervisory, it is dated, it is unfinished, and the regulator has said in writing what is wrong. Use it.

One accuracy caution. Secondary summaries of these principles are unreliable on the detail — one widely-read one refers to "the eleven Principles above" when the standard lists fourteen. Quote the groupings, and cite the Committee's own publication.

## Open banking standards, and what South Africa actually has

One incautious sentence here does real damage, so teach the distinction first: **a mandate, a standard and a market practice are three different things**, and a jurisdiction can have any combination of them.

| Jurisdiction | Mandate? | Common standard? | What you design against |
|---|---|---|---|
| United Kingdom<br>v4.0.1, 1 Apr 2026 | yes | yes, central | Spec plus experience and operational guidelines |
| EEA<br>updated 18 Jun 2026 | yes | consortium<br>free, CC-BY-ND | Certificate-based identity, consent model, corporate scope |
| United States<br>recognition pending | rule in flux | market-led | Consent interface, security profile, reciprocal fraud data |
| **South Africa**<br>October 2026 | **no** | **none** | Bilateral commercial contracts, plus Directive 2 of 2024 on screen-scraped EFT credits |

**Figure 13.7** A mandate, a standard and a market practice are three different things, and a jurisdiction can have any combination. The right-hand column is the one that catches an architect importing a reference architecture.

**United Kingdom — a mandate plus a centrally governed standard.** The Open Banking Standard is published by Open Banking Limited, currently **version 4.0.1, published 1 April 2026**, covering technical specifications, Customer Experience Guidelines and Operational Guidelines [27][28]. Version 4.0.1 is described as a targeted update: additional international payments data, refinements to variable recurring payments for sweeping, field-treatment confirmations, cryptoassets added as an optional transaction risk indicator, and known-issue fixes; it shipped alongside a dynamic client registration release, with the management information standard unchanged [27]. Two features matter architecturally. The standard **bundles customer-experience and operational guidelines with the interface specification**, which is why consent flows there are consistent between banks. And **variable recurring payments** are how open banking becomes a payments rail rather than a data rail. The licence and the current scope of mandated implementers could not be established from the release page and are not stated here.

**European Economic Area — a mandate from regulation, a standard from an industry consortium.** The Berlin Group publishes the openFinance API Framework, building on its earlier work: interfaces over HTTP with transport security, third parties identified by qualified certificates, currently a second-generation suite, last updated 18 June 2026, with a two-year workplan published in January 2026 [29]. Its scope reaches beyond the payments directive into broader data sources and account types, commercial premium services, a privacy-regulation-oriented consent model, corporate banking support and multilevel strong authentication [29].

> **THE CLEANEST LICENSING IN THIS CHAPTER**
>
> The Berlin Group's specifications are published **free under a Creative Commons attribution, no-derivatives licence**, the interface definition files are on a public code host, and participation in its taskforce and advisory group **carries no fees** and does not require membership [29].
>
> Say that out loud when somebody argues that industry standards must be gated. One of them is not.

**United States — a market-led standard seeking regulatory recognition.** The FDX interface specification reached a fifth major release, its first since early 2020, covering consumer control dashboard guidance for data providers, recipients and aggregators; standardised consent including a consent interface and grant notifications; alignment with an established financial-grade security profile and, interestingly, with an insurance body's annuity data definitions; and reciprocal data sharing that lets recipients inform providers of suspected fraud [30]. Access is described as free subject to a licence agreement characterised as royalty-free, with documentation on a member wiki [30]. The body applied to the United States consumer regulator to be

recognised as a standard-setting body [31]. Its current version and date, member count, and the status of that recognition could not be established and are not stated here.

#### South Africa — write this carefully

**South Africa has no regulated open banking regime as at October 2026.** Never write that it does. What exists is consent-based data sharing by commercial arrangement, plus an unfinished policy process, plus one directive that is genuinely binding.

The conduct regulator published a draft Open Finance position paper, with comments due in August 2023. It explores a **phased, possibly mandatory** regime requiring institutions to build infrastructure to share data with third parties on customer consent, and proposes proportionate oversight of four participant types, unbundled informed consent, risk-management and disclosure frameworks, data standards and a statutory complaints framework. Critically, it states that institutions already participating are **not governed by a regulatory framework**, and that third-party providers and interfaces are not licensed and fall outside the regulator's ambit. The levers named are the data-protection Act and a conduct code. No implementation dates are given [32].

The central bank's own research calls the market **market-led**. A working paper dated 16 March 2026 gives the status as banks and fintechs collaborating voluntarily with an evolving regulatory environment, and states that the current lack of regulatory policy creates market uncertainty, making incumbents reluctant to share data [33].

And the one hard obligation is not about interfaces at all. **Directive No. 2 of 2024** regulates *screen-scraped* electronic funds transfer credit initiation — anyone issuing a credit payment instruction on behalf of a payer, by screen scraping or a similar tool, must register with the central bank or be exempted, with per-payment informed consent, credential handling rules, five-year audit trails, insurance against fraud losses, monthly reporting and biannual compliance attestations [34]. The earlier consultation had proposed a common interface standard that was never implemented [35].

> THE SENTENCE NOT TO WRITE, AND THE SENTENCE TO WRITE
>
> **Do not write:** "South Africa's open banking regime requires…". There is no regime.
>
> **Write:** South Africa has consent-based data sharing by commercial arrangement, no mandated interface standard, no licensing of third-party providers, no mandated data set, no central directory and no mandated service level — and a published regulatory intention to move toward a phased mandatory regime whose timing is not set.
>
> The architectural consequence deserves its own line. **In South Africa the integration contract with a third party is a commercial contract, not a standard.** Bilateral interface design, bilateral security models, bilateral liability and bilateral change control are the norm. An architect arriving from the United Kingdom or the European Union will badly under-estimate the per-partner cost, and will design for obligations that do not exist while omitting the one that does.

#### The interchange layer South Africa does have

There is no single banking equivalent of the insurance industry's ACORD. The interchange layer is **fragmented by instrument and by rail**, and that fragmentation is itself the thing to teach. ISO 20022 is the strategic destination for most of it. The preceding generation of securities and payment messages is still the frame of reference for legacy mapping [18]. Card messages are a paid standard whose current edition is dated 2023, and in practice every acquirer, switch and scheme runs a dialect [40]. Electronic trading has its own protocol [42], derivatives have a markup language and an open, machine-readable lifecycle model now hosted by a financial open-source foundation [41]. And the closest historical analogue to a retail-banking interchange body is a cautionary tale: it merged into a United States payments association, with its specifications now hosted there [38][39].

But the one that will actually bind you is local, and no global reference architecture contains it. **The South African credit bureau data hub.** Its operator describes it as the sole legally recognised central data transmission hub, routing member submissions to the six bureaux authorised by the National Credit Regulator to hold payment profile information, and reports roughly 54 million records submitted monthly across more than three thousand members in eleven sectors, with twenty account types and a rejection rate under one per cent [36]. The register of credit bureaux is much longer than six, and the distinction between *registered as a bureau* and *authorised to hold payment profile information* is the one an architect needs [37].

> THE STANDARD THAT OUTRANKS ALL OF THEM, LOCALLY
>
> For a South African bank, **the credit bureau submission layouts are a mandatory interchange standard with a monthly deadline and a statutory basis**. That makes them more operationally binding than anything BIAN publishes — and they appear in none of the reference architectures in this chapter. Chapter 14 goes into the regulatory position; the architectural point here is that your most non-negotiable data contract may not be in any model you licensed.

## How to choose, and how to layer two

You will not use all of them, and you should decide which before you talk to procurement.

| If the question is... | The body is... | And the test of success is... |
|---|---|---|
| Who owns "customer onboarding", and where does it stop? | BIAN Service Domains and control records | A boundary dispute ends in one meeting, with a written owner |
| What do we do, and what are we missing? | BIAN Business Domains | You can name an absence nobody had written down |
| What does this payment message mean? | ISO 20022 Data Dictionary | Your canonical model carries structured remittance and address data |
| What are the steps, and how do we benchmark them? | APQC Banking framework | A process catalogue that reconciles to a capability map |
| What do these words mean across risk, markets and lending? | FIBO | One definition of counterparty survives three systems |
| What should a banking warehouse contain? | The IBM lineage, as a checklist | Your own model gains entities you had forgotten |
| Why must we fund lineage and master data? | BCBS 239 | A funded programme with a supervisory sponsor |
| How do we expose data to a third party? | The applicable local answer — which here is a contract, not a standard | A per-partner cost you estimated before you signed |

**Layering two of them is the normal case, and there is one defensible way to do it.**

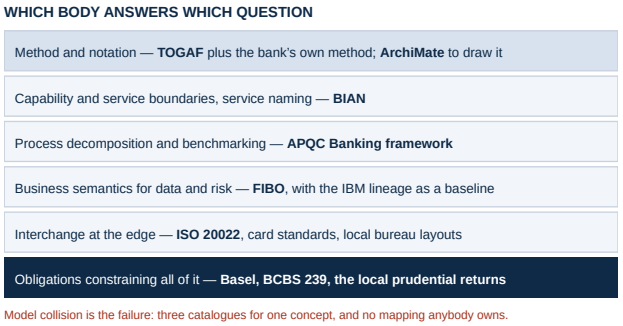



**Figure 13.8** One defensible layering, offered as one way rather than the way. The hard part is not the stack: it is the mapping between two adjacent layers, and the discipline to maintain exactly one of them.

Use **TOGAF plus the bank's own method** for how to do the work and **ArchiMate** to draw it. Use **BIAN** for capability and service boundaries and for service naming. Use **APQC** for process decomposition and benchmarking, mapped to BIAN rather than competing with it. Use **FIBO**, with the IBM lineage as a warehouse starting point, for business semantics in the data and risk estate. Use **ISO 20022** and the card standards at the edge, and the local credit bureau layouts where they apply. And treat **Basel, BCBS 239 and the local prudential returns** as obligations that constrain all of the above.

The hard part is not the stack. It is the mapping between two adjacent layers, and the discipline to maintain exactly one mapping rather than three.

### Three failure modes, and how to spot them early

**Model collision.** BIAN names, APQC identifiers and ISO 20022 element names for the same concept, with no mapping between them — three catalogues and no truth. The tell is a glossary with three entries for "payment order". The fix is cheap if done at the start and expensive later: pick one as the naming authority per layer, and record the mapping as a maintained artefact with an owner, not as a one-off spreadsheet from a workshop.

**Model worship.** A practice that produces conformance maps instead of decisions. The tell is a steering pack whose main artefact is a percentage of alignment. Ask what decision the map changed. If the answer is none, it is not an architecture artefact, it is a status report.

**Model substitution — a reference architecture used as a substitute for thinking.** This is the one the chapter opened on and the one to watch for hardest, because it is the most respectable-looking. Four tells, and any one of them should stop you.

- The bank's own capability model, data model and product model do not exist, but there is a licensed reference model and a mapping exercise.
- A question about *this* bank — which product family, which legal entity, which rail — is answered with a reference-model domain name.
- The target architecture is the reference model's picture, reorganised.
- Nobody can name a decision that would be different if the reference model said something else.

> **THE TEST**
>
> A reference architecture is doing its job when it **shortens an argument**. It is being used as a substitute for thinking when it **replaces one**.
>
> A bank that cannot say how many customers it has does not need a better reference model. It needs somebody to write down the four definitions, name an owner for each, and decide which one goes on the slide. The reference model makes that work faster and more defensible. It does not do the work, and it will never tell you which of the four numbers is the one the chief executive should say out loud.

**THE CHAPTER IN TEN LINES**

- A reference architecture earns its keep as boundary arbitration, a coverage check and boundary language — not as a target architecture, not as a system list, and never as a substitute for writing your own capability, data and product models.
- TOGAF tells you how to do architecture, BIAN tells you what the things are called, and ArchiMate is how you draw either; anyone asking you to choose between TOGAF and BIAN has been misled.
- BIAN Service Landscape 14.0 (release notes 20 February 2026) has 322 Service Domains, 38 Business Domains, 586 Business Capabilities, 4,905 Service Operations and 242 Semantic APIs, with no metamodel change — and the release is essentially a payments re-cut.
- One Service Domain owns one control record through its lifecycle: that is a data-ownership rule in disguise, and it is the most useful thing in BIAN.
- BIAN now ships its Semantic APIs over both its own business object model and ISO 20022's business component model, which is BIAN conceding that ISO 20022 wins the data vocabulary inside payments.
- BIAN certification lags the Landscape — the 2026 certification tests Service Landscape 13 while 14.0 is current — and BIAN publishes no licence: availability through a registration gate is not permission, so get terms in writing.
- ISO 20022 is a methodology plus a repository of business models from which messages are generated; the Data Dictionary and Business Process Catalogue are free, the full repository ships in a decade-old Eclipse format, and most banks therefore build canonical models from the generated schemas rather than the model.
- FIBO gives you meaning across heterogeneous sources and names BCBS 239 as a use case, but its publisher names no adopters and its current release and licence could not be established; the IBM banking warehouse lineage survives as a quiet product name and is most valuable as archaeology.
- BCBS 239 is the only framework here a bank can be supervised against, it is unfinished for architectural rather than analytical reasons, and that makes it the most reliable business case an architect has for lineage, ownership and master data.
- South Africa has no regulated open banking regime: consent-based sharing by commercial contract, no mandated interface or data set, no licensed third parties — plus Directive No. 2 of 2024 on screen-scraped EFT credit initiation, and credit bureau submission layouts that outrank every model in this chapter.

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CHAPTER 14

# Technology providers

*Somebody will ask you which core banking system to buy before you know enough to answer. This chapter will not tell you. It will teach you what kind of thing each vendor sells, which attributions the market routinely gets wrong, and why in South Africa the honest answer to "what does this bank run" is often that nobody has published it.*

## Read this before you read anything else

In your first month at a bank somebody will ask which core banking system to buy. You will not know; nobody three weeks into the industry does. What you *can* learn quickly is what **kind** of thing each vendor sells, which question it exists to answer, and what it costs architecturally to say yes.

> FRAMING, AND IT IS NOT NEGOTIABLE
>
> **Every company named in this chapter is an example of who plays in a category. Nothing here is a recommendation. Nothing is an endorsement. It is not a shortlist and it must not be used as one.**
>
> Inclusion means a vendor was recognisable enough to illustrate a category *and* had a source behind the claim made about it. Omission means nothing — some of the strongest products here are missing because no source could be found. There are no rankings, scores or quadrants, because analyst positions in this market are purchased research and age badly, and because a league table is exactly what a reader new to banking will mistake for advice.

### The research is dated, and here is what goes stale first

The research behind this chapter was done in **October 2026**. Knowing which parts to distrust first is more useful than the chapter itself.

SHELF LIFE · RESEARCH DATED OCTOBER 2026

| What | Why, and the date that moved it |
|---|---|
| Card issuing and acquiring | 12 January 2026: one group took Worldpay and sold its own issuer business the same day. Earlier text is void. |
| Core vendor ownership | Two divestments from one vendor in eighteen months. That is the tempo to expect, not the exception. |
| Reference model | BIAN Service Landscape 14.0, February 2026 — a payments re-cut that renamed behaviour qualifiers. |
| SA payments bodies | Scheme rule-making moved to the central bank and the renamed clearing utility during 2026. |
| Open finance, SA | Explored, not in force. Write so the text does not become false when it changes. |

What does not rot: the fifteen categories, and the question each exists to answer.

**Figure 14.1** What goes stale first. Distrust this chapter from the top of the list down, and put the date you re-checked next to every ownership and product name that reaches a board paper.

**Ownership rots fastest, and it rots as an architectural attribute, not a trivium.** A platform under private-equity margin pressure behaves differently in a roadmap conversation than a founder-led or listed one. A platform owned by an exchange, an index group or a card scheme has a reason to bundle data with software. Record the owner in your decision record, with the date you checked.

**Product naming rots almost as fast.** T24 became Temenos Transact [1], which matters when you read a 2015 document, and Global Processing Services became Thredd in 2023 [2]. Half the confusion in a vendor conversation is two people using names from different years.

**What does not rot is the shape of the problem.** The categories below have been recognisably the same for twenty years, and the questions each exists to answer have not changed at all.

## The corrections — get these right, because the rest of the market gets them wrong

Each of the following is wrong in current commentary, and each was checked against the vendor's or body's own material. Getting one wrong in a board paper is worse than omitting the vendor entirely.

**IN CIRCULATION, AND WRONG**

| Commonly written | Actually |
|---|---|
| Finastra Kondor / Summit / Opics | **Teciem (Apax-owned)** |
| Finastra Phoenix / MalauzAi | **CORA Group, 3 June 2026** |
| Adenza Calypso / AxiomSL | **Nasdaq's (2023, $10.5bn)** |
| Featurespace, independent | **Visa's, December 2024** |
| Pismo, independent | **Visa's, January 2024** |
| FICO Debt Manager | **C&R Software's** |
| SAP core banking | **SAP Fioneer (JV, 2021)** |
| SimCorp, independent | **Deutsche Börse's (2023)** |
| Compuscan, SA bureau | **Experian's (agreed Dec 2018)** |
| "NCR" the ATM maker | **National Credit Regulator** |

Getting one of these wrong is worse than omitting the vendor entirely.

**Figure 14.2** Attributions that are wrong in current commentary. Each was checked against the vendor's or body's own material. These are not trivia — they are wrong lines in somebody's live architecture repository.

- **Kondor, Summit, Opics, Sophis, Sophis Invest and Arc are Teciem's, not Finastra's.** Finastra sold its treasury and capital markets division to funds managed by Apax, and the business was spun out as Teciem carrying those products [4].
- **The Phoenix core and MalauzAi digital banking are CORA Group's, not Finastra's.** On 3 June 2026 Finastra sold its United States mid-market banking business — the Phoenix core, MalauzAi, Analyzer IQ and enterprise content management — to CORA Group, part of Jonas Software within Constellation Software [3].
- **Calypso and AxiomSL are Nasdaq's.** Nasdaq acquired them as Adenza in a $10.5 billion transaction announced on 12 June 2023, its largest deal [5]. **Do not write "Adenza" as a present-tense independent company.**
- **Featurespace is Visa's**, completed December 2024 [6]. **Pismo is Visa's**, completed January 2024 at around $1 billion [7]. A card scheme owning both a fraud platform and a core and issuer-processing platform is a new structural fact, and it belongs in a concentration assessment.
- **Debt Manager is C&R Software's, not FICO's.** The collections and recovery business moved to Jonas Software, and C&R publishes an explainer saying exactly that [8]. Bank documentation and job advertisements still say "FICO Debt Manager". The transaction date and consideration could not be established.
- **SAP's banking software is sold through SAP Fioneer**, a joint venture with DEDIQ carved out of SAP in 2021 [9]. Writing "SAP core banking" without naming Fioneer sends a reader looking for a product that is not there.
- **SimCorp is Deutsche Börse's**, the takeover completed in 2023 [10], with Axioma risk analytics merged in.
- **Compuscan is Experian's.** Experian agreed to acquire Compuscan and ScoreSharp in December 2018, reported locally at R3.7 billion, from Actis and management [11][12]. **Do not write "Compuscan" in the present tense as an independent bureau.**
- **On 12 January 2026 Global Payments completed its acquisition of Worldpay and simultaneously sold its own Issuer Solutions business to FIS** [13][14]. Net effect: **issuing processing consolidated into FIS, merchant acquiring consolidated into Global Payments.** Anything written before that date describes a market that no longer exists. A $24.25 billion Worldpay consideration circulates in secondary reporting; the completion release discloses no price, so the figure is not stated here as fact.
- **And the one that will embarrass you locally: "NCR" in a South African banking sentence means the National Credit Regulator** [15], not the former automated-teller-machine manufacturer. Two very different conversations, one acronym.

> **HOW TO USE A CORRECTIONS LIST**
>
> Not as trivia. Each of these is a *wrong line in somebody's current architecture repository*, and most got there from a vendor slide. When you inherit a repository, spot-check the vendor and ownership fields against the vendor's own site, with a date. You will find three or four, and finding them is cheap credibility.

## Part one: the categories, and who plays in each

Vendors sell suites, built so you cannot see where one capability stops and the next begins. Invisible boundaries are harder to argue with and harder to leave. So the categories below are organised by capability, and you should decompose any suite onto them before comparing anything.

**ONE VENDOR, HOW MANY CATEGORIES**

| | core | payments | cards | lending | collections | markets | reporting | crime/fraud | front end |
|---|---|---|---|---|---|---|---|---|---|
| Temenos | ● | ● | | | | | | | ● |
| Finastra | ● | ● | | ● | | | | | |
| Fiserv | ● | | ● | | | | | | |
| FIS | ● | | ● | | | | | | |
| Oracle FS | ● | | | | | | | ● | ● |
| SAP Fioneer | ● | ● | ● | ● | | | | | ● |
| Nasdaq | | | | | | ● | ● | | |
| Visa | ● | | ● | | | | | ● | |

Presence only, from this chapter's sourced statements. Not scope, not share, not quality.

**Figure 14.3** A readable subset of the market, by capability rather than by vendor. Read the rows: a vendor in one row is a product decision; a vendor in four or more is a concentration decision with a different approval path.

Read that figure by rows, not cells. A vendor in one row is a product decision. **A vendor in four or more rows is a concentration decision** — a different kind of risk, with a different approval path, and in a regulated institution a different conversation with the second line.

### Core platforms

**What it is for.** The system of record for the account, the balance, the posting and the product terms. It decides what the bank can sell, how fast, and at what unit cost. Separate three axes before comparing anything: retail against corporate and transaction banking against treasury, which are different data models entirely; suite against ledger-plus-product-layer; and licensed software you run against software the vendor operates, which decides who carries the regulatory outsourcing obligation.

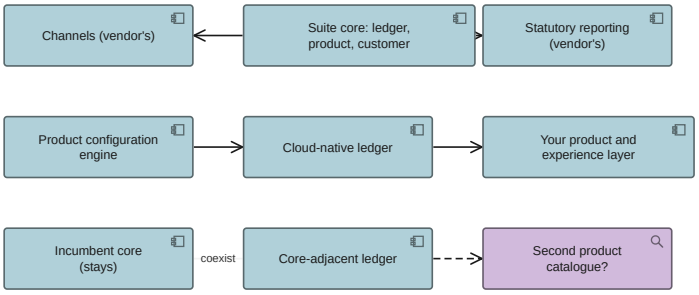



**Figure 14.4** Three shapes sold as "a core", and they are not substitutes. Top: the suite, where product definition and the customer master come with the ledger. Middle: a ledger plus a product-configuration engine, with channel and product built on interfaces. Bottom: a core-adjacent ledger beside an incumbent core — which is two estates unless somebody writes down the migration thesis.

**Core platforms — examples, not recommendations**

- **Temenos** — the broadest independent. Its site names Temenos Core with variants for retail, business, corporate, treasury, credit unions and Islamic banking, claiming 950-plus banks in 150-plus countries [1]. Those are vendor-published counts; attribute and date them. **T24 was renamed Temenos Transact** [16].
- **Finastra** — after two divestments, its own description names **Loan IQ, LaserPro, Trade Innovation, Essence, Global PAYplus, Payments To Go** and Financial Messaging [3]. Essence is the retail core.
- **Fiserv** — the clearest published segmentation anywhere: **DNA**, **Finxact** as the interface-first cloud-native option, **CoreAdvance** as a turnkey modern core, **Signature** for mid-sized to large commercial banking, plus Cleartouch, Precision and Premier [17].
- **FIS** — a large United States base; a 2019 analyst figure cited by the Federal Reserve Bank of Kansas City has FIS serving 78 banks above $10 billion in assets [18]. Current core product naming could not be established.
- **Oracle Financial Services** — **FLEXCUBE** and Oracle Banking Cloud Services, spanning retail, corporate, digital experience, compliance, risk and finance [19].
- **TCS BaNCS** — one family across banking, capital markets and insurance, with modules including a global banking platform, payments, securities processing, treasury and corporate loan origination [20].
- **Infosys Finacle** — the other large Indian-origin core, met alongside BaNCS in African tenders. Current product naming and customer counts could not be established.
- **SAP Fioneer** — banking lines covering core banking, virtual account management, digital banking, loans and collateral management, mortgage, card management and a payment hub [9].

**The regional tier matters more than an Anglophone vendor list suggests.** In surveyed United States banks in 2022, Jack Henry held 21% and CSI 7%, with COCC at 3% [18]; CORA Group now holds the former Finastra Phoenix core in that segment [3]. Beyond that sits a tier of regional-language, regional-regulation cores an Africa-facing architect will meet in tenders, which this book does not name because their current positioning could not be verified. Tell your steering committee the shortlist they were handed is not the market.

**The cloud-native cohort sells something different, and the difference is the point.** They sell a **ledger and a product-configuration engine**, not a bank in a box, and they expect the buyer to build product and channel on interfaces.

**Cloud-native cores — examples, not recommendations**

- **Mambu** — software-as-a-service core with composable positioning; it raised €235 million at a €4.9 billion valuation [21]. Later reporting gives a different valuation, so the current figure is unreconciled and not stated here.
- **Thought Machine** — **Vault Core** and **Vault Payments**, using a contract-like mechanism as the product-definition language [22], with named go-lives including a United Kingdom specialist lender starting with a single mortgage product [23].
- **10x Banking** — **SuperCore**, founded and led by a former Barclays group chief executive [24], with £40 million raised in 2026 [25]. Directly relevant to South Africa; see part two.
- **Pismo** — cloud-native core and issuer processing, **acquired by Visa** in January 2024 [7].
- **Finxact** is now inside Fiserv [17] and **Technisys** inside SoFi — a cohort consolidating into incumbents, which is itself a datum about the category's economics.

### Payments hubs and switches

**What it is for.** Two jobs that get conflated. A **switch** authorises and routes in real time against a scheme or rail. A **payments hub** centralises initiation, validation, enrichment, routing, screening, scheme formatting, clearing, settlement, exceptions and reporting across *many* rails, so the bank can add a rail without rebuilding a channel. Chapter 12 described the pattern; this is the market that sells it.

**Payments hubs and switches — examples, not recommendations**

- **ACI Worldwide** — the BASE24 lineage from 1982 across point-of-sale and branch channels, clearing, settlement and fraud detection, plus the switching heritage described below [26].
- **Finastra** — **Global PAYplus** as the enterprise hub and **Payments To Go** as its software-as-a-service form [3]. **Volante Technologies** — ISO 20022-native payments-as-a-service and a payment hub [27]. **SAP Fioneer** — a payment hub inside the banking range [9].
- **Form3** and **Icon Solutions** sit at the cloud-native and build-your-own ends. Their current product naming could not be verified, so no product names are given.
- **Electrum** and **Sybrin** are the South African entrants; see part two.

**One lineage worth knowing, because it runs through South Africa.** The switching software later sold as Postilion came from **Mosaic Software**, formed in April 1994 with South African offices and Business Connexion among its shareholders; S1 Corporation agreed to buy it for up to $52 million [28], and **ACI Worldwide acquired S1 in 2012** [26]. Whether the product carried the Postilion name at the time of the S1 transaction could not be established — the contemporaneous report describes the software without naming it. Say "the Mosaic lineage" and you are safe.

### Card issuing and acquiring

**What it is for.** Issuing covers card lifecycle, product configuration, authorisation decisioning, cardholder authentication, tokenisation, disputes and chargebacks, and scheme mandates. Acquiring covers merchant onboarding and underwriting, the terminal and gateway estate, authorisation and capture, settlement and funding, interchange and fee calculation, and representment.

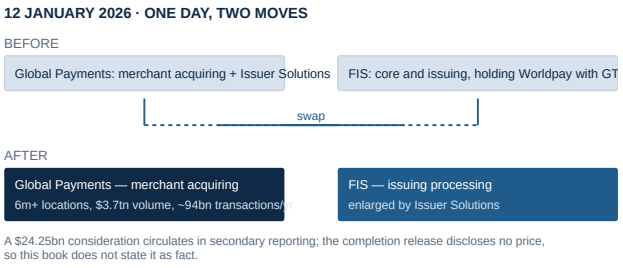



**Figure 14.5** The 12 January 2026 transaction, which dates everything written about this category before it. One group completed its acquisition of a large acquirer and sold its own issuer-processing business to another on the same day.

**The fact that dates everything in this category** is the 12 January 2026 transaction described above, which produced a merchant business with more than six million merchant locations, $3.7 trillion of payment volume and around 94 billion transactions a year across more than 175 countries [13][14].

**Card platforms — examples, not recommendations**

- **FIS** — issuing processing, enlarged by the Issuer Solutions acquisition of 12 January 2026 [13][14].
- **Global Payments** — merchant acquiring, enlarged by Worldpay on the same date [13][14].
- **Thredd**, formerly Global Processing Services, rebranded in 2023 [2] — the processor behind a large part of European fintech card issuing.
- **Marqeta** and **Episode Six** in modern issuer processing [29].
- **Pismo**, now Visa's, spans core and issuer processing [7].
- A tier of pan-European and global issuer and acquirer processors also plays here. Their current product naming could not be verified, so they are named only as a category: ask each for its own current product list rather than trusting a book.

### Lending origination and decisioning

**What it is for.** Three separable jobs, and conflating them is the commonest error in retail lending: **origination workflow**; **credit decisioning** — scorecards, policy rules, affordability, bureau data in the decision path; and **credit risk modelling with model governance**. Pricing is sometimes bundled and sometimes separate.

**Lending origination and decisioning — examples, not recommendations**

- **nCino** — Salesforce-native commercial and consumer lending and onboarding. It **acquired the South African business-onboarding specialist DocFox for around $75 million in March 2024** [30][31], since folded into the nCino brand. That it is built on Salesforce is the defining architectural fact about it: it inherits that platform's data model, extensibility and licensing economics, for better and worse.
- **Finastra** — **Loan IQ** for syndicated and commercial lending, **LaserPro** for United States loan documentation [3].
- **SAP Fioneer** — loans management, collateral management, complex loans, a credit workplace and a mortgage cloud offering [9]; **TCS BaNCS** sells corporate loan origination [20].
- The decisioning and scorecard specialists are a real and crowded tier. Their current product naming could not be verified and is deliberately omitted — ask the vendor for its own product sheet, because the names change with every release cycle.

### Collections

**What it is for.** The post-delinquency lifecycle: treatment strategy, contact orchestration, arrangements, hardship, legal process, write-off, book sale and recovery — all under conduct rules. In South Africa, add the prescription clock from chapter 12 and the authenticated-collection rails, neither of which a global product models natively.

**Collections — examples, not recommendations**

- **C&R Software — Debt Manager**, the former FICO collections and recovery product, now within Jonas Software [8]. Watch the name: bank documentation and job advertisements still say "FICO Debt Manager".
- **Amplifin** (South Africa) — published material covers its management platform, payment streams and **Registered Mandate collections** [32] [33], which is the rail chapter 12 described and which almost no global product knows about.
- **Nutun** (South Africa) — the former Transaction Capital Risk Services, now a business-process and collections operation [34].
- Several global vendors have collections lines. Their current naming could not be verified and is omitted.

### Treasury and markets

**What it is for.** Front-to-back for trading and treasury — pricing and risk across rates, foreign exchange, credit, equity and commodities; position and limit management; collateral and margin; confirmations, settlement and lifecycle events; valuation and profit and loss. And separately **banking-book treasury**: asset and liability management, liquidity, interest-rate risk in the banking book, and funds transfer pricing.

**Treasury and markets — examples, not recommendations**

- **Murex MX.3** is the long-standing front-to-back incumbent in large trading estates.
- **Nasdaq** holds **Calypso** and **AxiomSL**, acquired as Adenza in a $10.5 billion transaction announced 12 June 2023 [5].
- **Teciem** holds **Kondor, Summit, Opics, Sophis, Sophis Invest and Arc** after Finastra's divestment to Apax-managed funds [4]. A reported value of around $2 billion rests on secondary reporting and is not stated here as fact [35].
- **SimCorp**, with Axioma risk analytics merged in, is **Deutsche Börse's** [10].
- **TCS BaNCS** sells a treasury module inside the same family [20].
- Other large capital-markets vendors play here; their current product naming could not be verified and is omitted.

> **THE STRUCTURAL OBSERVATION TO RECORD AT SELECTION**
>
> In three of the four verified ownership cases above, the owner is now **an exchange, an index and data group, or a private equity fund** — not a software company.
>
> That changes roadmap behaviour, pricing behaviour and the likelihood that data and software get bundled. It is a legitimate architectural attribute, it belongs in the decision record, and it is the kind of thing a selection committee never asks about and later wishes it had.

### Regulatory reporting

**What it is for.** Producing jurisdiction-specific returns on a calendar, with controls, sign-off, an audit trail and the ability to explain a number back to source. What you are really buying is **a calculation engine plus a maintained taxonomy subscription** — the vendor's commitment to track your jurisdiction's rule changes.

**Regulatory reporting — examples, not recommendations**

- **Nasdaq AxiomSL** [5].
- **Regnology** — note that it hosts an analyst vendor spotlight on itself [36]; treat a vendor-hosted analyst document as marketing, whoever wrote it.
- **Wolters Kluwer OneSumX** and **Vermeg AgileREPORTER** are the other names a reader will meet. Their current naming and scope could not be verified here.

THE SOUTH AFRICAN POINT NO GLOBAL VENDOR LIST ANSWERS

The instrument is the **BA returns** series, revised on a rolling basis by Prudential Authority directive [37]. Whether any global reporting vendor ships maintained BA-return content **could not be established for this book**.

The practitioner expectation — and present it as expectation, not fact — is that BA returns are built or heavily configured locally, which makes the reporting layer the place a global platform selection most predictably generates custom work. Settle it by asking three vendors directly, in writing, for their South African taxonomy coverage and the service level on rule changes. The answers are usually revealing.

### Financial crime, and fraud

**Two categories, not one**, and regulators assess them separately. Financial crime bundles four controls: customer due diligence; sanctions and watchlist screening, both party and in-path; transaction monitoring; and case management with regulatory reporting. Fraud protects the bank and customer from loss in real time and is justified by loss avoidance rather than compliance.

**Financial crime and fraud — examples, not recommendations**

- **LSEG Risk Intelligence** publishes World-Check risk intelligence data and sanctions screening carried over from Refinitiv [38][39]. Older documents say "Refinitiv World-Check"; a reader should know it is LSEG's now.
- **Oracle** sells anti-money-laundering and compliance solutions inside its banking portfolio [19], and **Sybrin** (South Africa) ships the same capability alongside its payments hub [40].
- **Featurespace** is **Visa's**, completed December 2024 [6]. **Entersekt** (Stellenbosch) is the South African authentication and app-security entrant; see part two, with the caveats that belong there.
- This chapter deliberately names no other vendors in these two categories. There is a long list of well-known platforms whose current product naming could not be verified behind this book. The category description above is safe to use; the vendor-to-product mappings are not, so go to each vendor's own product page.

THE ONE ARCHITECTURAL CONSTRAINT THAT RESHAPES THIS CATEGORY

On an instant rail with a sub-second scheme response window, fraud detection must be **in the authorisation path** — which makes it a latency-critical dependency of the payment service, not an analytics capability. That single constraint reshapes topology, failure mode and the vendor conversation. In South Africa, where the instant rail runs around the clock with a R50 000 cap [41], this is live reality rather than a design exercise.

### Credit bureaux and credit data

Globally the familiar names are TransUnion, Experian and Equifax, with FICO as the dominant United States score provider; nothing beyond the South African registry facts in part two could be verified.

**The structural point holds anywhere.** A bureau call is an **external dependency inside the decision path**, with a cost per call, a latency, an availability profile and data-protection consequences. Caching, fallback and consent scope are architecture, not operations. And the richest bureau assets are country-specific, so plan for a thinner external data layer than global vendor material implies.

### Data platforms, CRM, process platforms and digital front ends

Four categories that behave alike from a selection point of view: each is a general-purpose platform with a banking framing laid over it, and in each the architectural question is *where state lives*.

**Platforms and front ends — examples, not recommendations**

- **Data platforms.** The hyperscalers address banking less through distinct banking products than through industry framing of general-purpose services plus a partner ecosystem. The verified South African engagements are in part two.
- **CRM. Salesforce** is a named BIAN member [42] and the platform nCino is built on, which is the defining architectural fact about nCino [30]. **Microsoft Dynamics 365** is the other large incumbent. Banking-accelerator naming for both could not be verified.
- **Process and low-code. Sybrin** sells a low-code platform with prebuilt digital banking, payments and automation components [40][43], recognised in analyst market guides [44]. Honestly described, this is **the layer where the process actually lives because no single system owns it** — indispensable, and a notorious source of shadow architecture.
- **Digital front ends. Backbase** reports being named a leader in an analyst wave for digital banking engagement platforms in the second quarter of 2026 [45] — present that as a vendor-reported placement, with the year, and nothing more. **Temenos Infinity** is the channel layer met alongside Transact, selected with it at Barko [46]. **Oracle** and **SAP Fioneer** complete the suite-attached set [19][9].

**The question that discriminates in each.** Data platforms: where does the core's data land, in what form, who pays for the extract, and what is your concentration position across hyperscalers? CRM: **which system is master for the customer, and what is the reconciliation obligation against the core?** Process platforms: what stops this becoming an unsupervised application estate, and what is the licence model at mass-market scale? Front ends: does the channel layer own any state, can it release independently of the core, and does it depend on the core's interface surface or a bank-owned one? Only the second survives a core replacement.

### Core-adjacent ledgers

The newest category, and the one most worth explaining to an architect new to banking. A **core-adjacent ledger** is a double-entry posting engine with product and balance primitives, deliberately *without* the surrounding suite — no branch system, no statutory reporting, no customer master, often no product catalogue beyond a configuration language. Banks buy them to launch a proposition without touching the incumbent core, or to build the target core incrementally behind an interface layer.

**Core-adjacent ledgers — examples, not recommendations**

- **Thought Machine Vault Core** [23], **10x Banking SuperCore** [24], **Mambu** [21], **SAP Fioneer virtual account management** [9] and **Pismo** [7] all sit in this shape.
- **Zafin** sits adjacent as a **product and pricing layer outside the core**, and was a named vendor in BIAN's Coreless Banking pilot [47].
- Several smaller ledger-only products exist. Their positioning could not be verified and they are not named.

THE FOUR QUESTIONS THAT DECIDE WHETHER THIS WORKS

**What does the ledger *not* do, written down** — because that list is your build backlog. **Where does the product catalogue live**, and have you now got two? **How does statutory reporting aggregate across both estates**, given chapter 12's reporting obligations? And **what is the migration thesis** — does the adjacent ledger become the core, or is it a permanent second estate?

Banks that cannot answer the last one get the second answer by default, and then discover that the second estate has its own regulatory reporting, its own reconciliation and its own skills market.

## Part two: the South African picture

This is the half you cannot get from a global analyst report. Four structural features, established before any vendor is named. **A concentrated, sophisticated market with a very large in-house build tradition** — the big banks have built far more of their estates than a global vendor list implies, and the clearest evidence is how hard it is to source a named core platform for most of them. **The payments institutional architecture was rebuilt in 2026**, ending bank self-regulation and moving scheme rule-making to the central bank and the renamed clearing utility [48]. **Collections against thin-margin accounts is a first-class architectural problem**, under an authentication regime with no global equivalent. And **credit bureaux are a licensed, regulated category** under the National Credit Act, with a legally recognised central data hub — so credit data flows here are a compliance architecture, not an integration choice.

### The credit data layer, and why it is not a data feed

Start here, because it is the most under-appreciated difference and the one an imported reference architecture will get wrong.

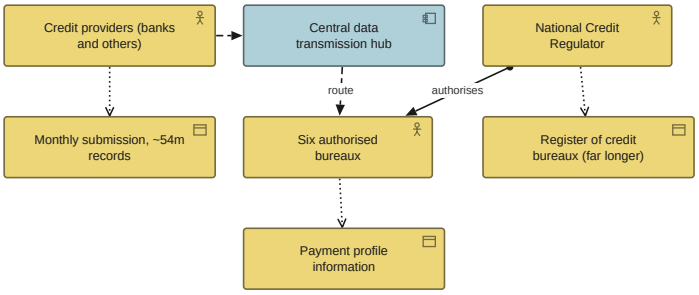



**Figure 14.6** Why the credit data layer here is a compliance architecture rather than an integration choice. Submissions route through one legally recognised hub to the six bureaux authorised to hold payment profile information. The wider register of registered bureaux is far longer — and being on it is not the same as being on the submission path.

The industry data association describes its hub as **the sole legally recognised central data transmission hub in South Africa**, routing member submissions to the **six bureaux authorised by the National Credit Regulator to hold payment profile information**: Consumer Profile Bureau, Experian, ITC Credit Bureau, TransUnion, VeriCred Credit Bureau and Xpert Decision Systems [49]. It reports roughly 54 million records submitted monthly across more than three thousand members in eleven sectors [49].

**The wider register is much longer than six** — the regulator's register runs to dozens of records, including specialist tenant and screening bureaux [15]. **The distinction between "registered as a credit bureau" and "authorised to hold payment profile information" is the one an architect needs**, because only the second set sits on the mandatory submission path.

The regulatory position is prescriptive in a way the word "bureau" does not convey. Registration is required under the National Credit Act, with competence and resourcing standards, complaints procedures and tax registration. **A controlling shareholder may not be a credit provider or a debt collection agency** — a structural separation rule with direct consequences for any bank contemplating owning a bureau. The application requires dozens of documents, including data intake, accuracy, retention and deletion procedures, information security, business continuity, exception reports and an infrastructure overview. Fees are R500 to apply and R10 000 to register, plus R5 per thousand enquiries over the prior twelve months, capped at R200 000 [50].

**ONE LINE TO CARRY INTO ANY CREDIT DESIGN**

In South Africa the credit data layer is a **regulated, licensed, centrally routed utility with a statutory submission obligation and prescribed retention and deletion rules** — the opposite of the "buy a data feed" posture a reader may import from another market. And **Compuscan is Experian's** [11][12], so a design document naming it as an independent bureau is at least seven years out of date.

### Local vendors and the local arms of global ones

**South African vendors — examples, not recommendations**

- **Electrum Software** (Cape Town) — cloud-native payments software with the most locally specific product list in this chapter: DebiCheck, PayShap, real-time clearing, electronic funds transfer debit and credit, account verification services, cross-border and settlement connectivity, cash vouchers and value-added services. Linked case studies name large banks, large retailers and a mobile network operator, and the site claims the platform is trusted by four leading banks [51]. Attribute the claim; some logos are unlabelled, and only the linked case studies are safe to cite.
- **Sybrin** — founded 1991, headquartered in Johannesburg since 2002; a low-code platform plus prebuilt payments hub, know-your-customer and anti-money-laundering, document processing, liveness detection and digital onboarding, with clients described as including **central banks and national automated clearing houses** [40][43]. That client type is the architecturally interesting part. Ownership: acquired by EOH in October 2013, then in 2021 by a consortium including One Thousand & One Voices, Crossfin and Isaac Mophatlane for R410 million [40].
- **Entersekt** (Stellenbosch) — mobile authentication and app security, founded in 2010, with Transakt as the named technology and named partners from 2016 [52]. **Treat with care:** the retrievable sources are from 2016 and 2017, and current ownership, product naming and South African bank customers could not be established. Entersekt's technology is widely believed to sit inside South African banks, and this book does not assert it without a source. Ask the bank, not the market.
- **JUMO** — South African-founded banking-as-a-service and lending platform operating with partner banks and mobile operators across emerging markets, having raised $120 million [53]. Its current partner-bank list could not be established. **DocFox** — business-onboarding specialist, acquired by nCino for around $75 million in March 2024 [30][31].
- **Payments fintechs with bank-facing roles** — Stitch, Ozow, Yoco, Lulalend, Mama Money and Mukuru appear in the central bank's own research on the local market [54].
- **PayInc**, formerly BankservAfrica — not a vendor in the normal sense but the dominant shared infrastructure, now 50% owned by the central bank, and intended to become a shared national payments utility non-banks can connect to directly [48].

**WHY THIS BELONGS IN A VENDOR CHAPTER AT ALL**

Almost no global platform models the local rails natively. **DebiCheck mandate authentication, Registered Mandate presentment windows, reason-coded retry strategies, salary-date-aligned collection cycles, proxy resolution on the instant rail, request-to-pay and an R50 000 cap** are local facts that a global core or collections product will meet with integration work [41].

This is the most reliable place a global platform selection in South Africa requires local build or a local partner. It is usually discovered late — after contract, during design — and it is the strongest argument in this chapter for component architecture rather than a single suite in this market.

### What is actually on the public record at South African banks

**The rule for this chapter: if it is not sourceable, say so rather than repeating market folklore.**

**THE PUBLIC RECORD, AND ITS LIMITS**

| Bank | What is sourceable | Source quality |
|---|---|---|
| Nedbank | R11.7bn, 2016 to end-2024; no vendor | trade press, executives |
| FirstRand | Finxact, announced 10 Mar 2025, no date | trade press, both sides |
| OM Bank | 10x; 2016 to 2025; >$135m | trade press |
| TymeBank | Mambu on AWS, Feb 2019 | dated 2019 — re-check |
| Bank Zero | Self-built on LinuxONE; 2 people | vendor case study |
| Barko | Temenos SaaS, live <6 months | vendor release |
| Capitec, Absa, Investec and others | **Core platform: not publicly sourceable** | say so; do not repeat market folklore |

No count of South African banks on any given platform exists. Do not estimate one.
Systems-integrator attributions here are effectively unsourceable: releases name the vendor and the bank only.

**Figure 14.7** What is on the South African public record, and what is not. The right-hand column is the honest answer, and the attribution column is why: three of the load-bearing facts rest on trade press quoting executives rather than on a filing.

What is on the record, with the attribution it deserves:

- **Nedbank — Managed Evolution.** R11.7 billion since 2016, concluded at the end of 2024 with the core banking platform as the last element delivered, per trade-press reporting quoting the chief operating officer and chief executive at results [55]. **The report names no vendor**, and which core platform was delivered, bought or built, **could not be established**. An earlier public figure for the same programme was R9 billion [56] — a 30% escalation between public statements is itself a datum.

- **FirstRand — Fiserv Finxact.** Announced 10 March 2025: Finxact as system of record and transaction processing engine, with FNB and RMB transitioning from their current provider over time, FNB described as having 30 million active accounts and products, and FirstRand the first institution outside the United States to commission Finxact [57]. **No timeline was given.** This is the chapter's live case study, and it rests on trade press quoting executives on both sides rather than on a filing.
- **Old Mutual — OM Bank on 10x Banking.** Reported 10 September 2025: a wholly owned cloud-native platform, the partnership dating to 2016, banking-licence approval in 2024, and more than $135 million invested [58][59]. **Nine years from technology partnership to national launch, on greenfield.**
- **TymeBank — Mambu on AWS**, confirmed at launch in February 2019, with Mambu's software-as-a-service core as the ledger and back-end microservices in Java [60]. **Date the attribution**; the source is seven years old and one corroborating survey warns its own attributions may be out of date.
- **Bank Zero — self-built on IBM LinuxONE.** IBM's own case study records an open-source banking platform with pervasive encryption and a container and Java stack, **two people managing the platform**, estimated capacity for two to three million customers, and a breakeven target of 100 000 [61]. Lesaka Technologies agreed to acquire Bank Zero for R1.1 billion, announced June 2025, with competition approval reported in November 2025 and other approvals outstanding [62].
- **Standard Bank — SAP, and Azure.** Described as one of the first banks in the world to move its core banking systems onto SAP, a process the reporting calls costly and painful, with a 2020 plan to move operational core systems to Azure under a dual-cloud strategy; its chief information officer for personal and business banking is quoted saying "it's been a long and difficult journey" [63]. Later reporting gives group technology operating expenses of R13.834 billion in 2025, 71% of migratable servers in cloud, 88% of legacy servers decommissioned, and **intangible amortisation down 16% to R2.076 billion, reflecting the retirement of older core banking platforms** [64]. Which SAP banking products, and whether the cloud migration of core completed, **could not be established**.
- **Discovery Bank — SAP, and a disputed price.** SAP's own Africa channel reports the core was built on its platform, that Discovery Bank paid more than R4 billion for its core banking system — a figure critics considered high — and quotes the chief executive on daily transaction volumes [65]. Treat the R4 billion as **reported, vendor-channel, and presented inside a rebuttal**, not as a disclosed programme cost. Earlier reporting named three bidders without saying who won, so the attribution rests on the vendor's own channel plus executive quotes.
- **Barko Financial Services — Temenos Transact and Infinity on Temenos software-as-a-service**, live in under six months, with a new lending service built in twelve weeks [46]. The clearest sourced go-live-speed data point here, and the honest counterweight to the nine-year arcs.
- **Capitec — no sourced platform attribution, and an on-record rejection of the category.** Its chief technology officer rejects the traditional core banking model, says credit and savings are leaving core banking and that "this is going to be a ledger thing", and expects competitors' core upgrades to cost R20 billion within five years [66]. That last figure is **a named executive's prediction about competitors**, not a measured number; label it as such. Capitec's own core platform could not be established.
- **Absa — in-house risk and pricing, and AWS.** Its market risk chief operating officer is quoted saying the bank felt it had the expertise internally for risk and pricing; a market-risk project ran from 2019, with a year-long parallel run from July 2024 against a 1 July 2025 compliance date; Absa also moved to AWS and works closely with Deloitte on regulatory matters [67]. Its core platform could not be established.
- **Investec, African Bank, Bidvest Bank, Postbank and Sasfin** — no credible named core platform attribution could be found for any of them.

> SAY THIS OUT LOUD, BECAUSE THE ALTERNATIVE IS FOLKLORE
>
> **No South African bank's core platform is publicly sourceable** for Nedbank, Capitec, Absa, Investec, African Bank, Bidvest Bank, Postbank or Sasfin. Three attributions that do circulate — FirstRand on Finxact, OM Bank on 10x, and the R11.7 billion figure — rest on credible **trade press** quoting executives rather than on an annual report or a filing [57][58][55]. Attribute them to the outlet, not the bank, and re-check them against the relevant integrated report before they go anywhere binding.
>
> **And no count of South African banks running a given platform exists.** Do not estimate one. A reader will quote your estimate back at you as a fact within a year.

#### Systems integrators, and honesty about them

**What is sourceable here is almost nothing.** The named facts found: Absa works closely with Deloitte on regulatory matters, per its own executive [67], and Business Connexion was a Mosaic Software shareholder in the switching lineage described earlier [28].

**The roles of the global consultancies and the large local integrators in named South African bank core implementations could not be sourced for any of them. Systems-integrator attributions in this market are effectively unsourceable**, because press releases name the software vendor and the bank and leave the integrator out.

The honest framing is more useful than a list anyway. The local banking integrator market is **thinner and more specialised than the global tier-one roster suggests**; the large banks hold substantial in-house engineering capability — the in-house risk engines, the self-build, the vendor-free public account of a nine-year programme; and for global cloud-native cores the implementation capability is often **imported or vendor-supplied**. That last point is a programme risk worth naming at selection, not at mobilisation.

### What a core replacement costs, and how long it takes

The most useful thing this chapter can do is destroy your intuition about duration.

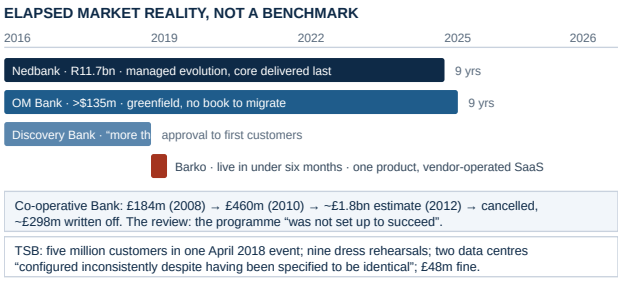



**Figure 14.8** The sourced duration anchors, with scope attached. These calibrate expectations. They are not benchmarks, because between a selection announcement and a go-live announcement sit re-scoping, phasing and things no press release mentions.

**The sourced anchors**, all from part two, with their scope attached. **Nedbank: R11.7 billion over nine years, 2016 to end-2024**, an *evolution* programme rather than a single cutover, with the core delivered last [55]. **Old Mutual / OM Bank: nine years and more than $135 million** on greenfield with no book to migrate [58]. **Discovery Bank: "more than R4 billion" reported**, vendor-channel and inside a rebuttal [65]. **Barko: live on vendor-operated software-as-a-service in under six months**, a single-product lender with no legacy book [46]. **Standard Bank: "long and difficult"** — and the cost and duration of its original programme could not be established, so do not use the figures that circulate in market conversation [63].

Six months and nine years are both true because they are different problems.

### The two fully documented failures, and what they actually teach

**The Co-operative Bank, United Kingdom.** Replacement of a system originally built in the 1970s began before a 2009 merger; the official review called the programme "ambitious", noting that no United Kingdom full-service bank had successfully done it. After the merger the scope expanded to include the acquired bank's systems. Expected cost rose from £184 million in 2008 to £460 million in 2010, and early-2012 due diligence found the whole programme would cost about £1.8 billion rather than the £0.8 billion assumed. It was paused with £349 million invested and cancelled after a related transaction collapsed, with a total write-down of about £298 million. The review's "litany of deficiencies": unstable executive sponsorship, weak capability, poor coordination, excessive complexity and frequently changing release schedules. Its conclusion was that the programme **"was not set up to succeed"**, and that blaming the collapsed transaction was a **"misconception"**. The bank's subsequent strategy favoured incremental change [68].

**TSB, United Kingdom.** Separated from its former parent in 2013 with inherited legacy systems, TSB migrated to a new platform developed by a subsidiary of its new owner, with the main event in April 2018 covering all five million customers' core banking data and payment records. An independent review published 19 November 2019 identified planning and preparation that could have been done differently, including stronger oversight of suppliers and questions about testing — this after nine dress rehearsals and a pilot involving more than 1 600 employees. TSB's board disagreed with aspects, stating that a key cause was that **the two data centres were "configured inconsistently despite having been specified to be identical"**, with coding and capacity issues and call volumes exceeding contingency [69]. TSB was subsequently fined £48 million [70]; the split between authorities and the exact date could not be established.

**Three lessons, and they are this section's spine. Scope expansion mid-programme is the common accelerant. The failure is almost never the software**: both inquiries land on governance, supplier oversight, testing and capability. And **both organisations ended up with an incremental strategy** — which is why the South African pattern of managed evolution and core-adjacent ledgers is learned behaviour rather than timidity.

### Why switching is rare, with the honest caveat

The Federal Reserve Bank of Kansas City's briefing of 27 March 2024 reports Fiserv at 42% of surveyed United States banks and 31% of credit unions; Jack Henry 21% and 12%; FIS 9% and 3%; CSI 7%; Finastra and COCC 3% each — with the largest three serving more than 70% of surveyed banks in 2022 [18]. On stickiness: 61% of banks had used their core provider for more than ten years [18]. The authors attribute this partly to long-term lock-in terms, exclusive dealing and high switching costs; over half of banks surveyed in 2020 cited implementation and upgrade fees as problematic; deconversion fees may be unclear; and contracts have been described as hundreds of pages long [18].

**Use that honestly.** The figures are United States, from 2020 to 2022 surveys, and not South African. What transfers is the **mechanism** — concentration, contract structure and deconversion economics — not the percentages.

### The cost lines that are not the licence

Licence or subscription is the smallest line. Enumerate the rest at selection, in writing.

**Duration and opportunity cost** — nine years at two of the sourced programmes above. **Data migration**, usually the largest and most under-estimated package, because legacy account data encodes decades of undocumented product variation; that characterisation is practitioner consensus, not a measurement. **Parallel run**, with both estates live and reconciled, and TSB's dual data centres as the cautionary case [69]. **Integration rebuild** — every downstream report, bureau feed, payment interface, BA return, scheme connection and reconciliation from chapter 12. **The local gap** — BA returns, authenticated collections and Registered Mandates, proxy resolution, and the size of the local implementation-skills pool. **Deconversion and exit cost**, asked at selection and written into the contract. And **scope discipline as a cost control**, with the Co-operative Bank as the case [68].

**CALIBRATE EXPECTATIONS; DO NOT BENCHMARK**

Public dates establish *elapsed market reality*, not how long any single implementation took. Between a selection announcement and a go-live announcement sit re-scoping, phasing, a second product-line selection and things no press release mentions.

So: "Nedbank took nine years" is a legitimate calibration of what an incumbent estate programme looks like. "A core replacement takes nine years" is not a finding, and "Barko did it in six months" is not a benchmark for a bank with a book.

## Build, buy or configure

**The question is not build or buy. It is where on the spectrum each capability sits**, and the answer differs per capability inside one bank.

**WHERE ON THE SPECTRUM DOES THIS CAPABILITY SIT**

| Position | Use it when | Local evidence |
|---|---|---|
| **Buy** whole product | Commodity, and the regulator effectively specifies the answer | Barko: vendor SaaS end to end, live in <6 months |
| **Buy and extend** configure hard | Standard capability, but your distribution or collections are not | Most of SA retail banking, because of the local rails |
| **Buy the ledger, build the layer** | You want product definition and experience; rent the record-keeping | TymeBank on Mambu; OM Bank on 10x |
| **Build** own it outright | It is how you compete and no product is shaped like your intent | Bank Zero's self-build; Absa's risk and pricing |

The failure mode: building commodity and buying differentiation.

**Figure 14.9** Four positions on the spectrum, with the question each answers and the local evidence for each. The failure mode is the diagonal: building commodity and buying differentiation, which is what happens when procurement rather than architecture sets the boundary.

The South African evidence is unusually good here, so lean on it rather than on principles. **Where banks here have demonstrably built**: a self-built core on open source with two people running the platform [61]; proprietary risk and pricing engines, because the bank judged it had the expertise internally [67]; and a chief technology officer rejecting the core-banking category outright and framing the future as a ledger problem [66]. **Where they have bought**: Finxact as system of record at FirstRand [57]; 10x for a new digital bank [58]; Mambu at TymeBank with everything built around it [60]; and Temenos software-as-a-service end to end at Barko [46].

**The heuristic that fits that evidence. Buy** where the capability is commodity and the regulator effectively specifies the answer — general ledger, sanctions screening, bureau submission, statutory reporting content. **Build** where it is how you compete and no product is shaped like your intent. **Buy and extend** where the capability is standard but your distribution, pricing or collections are not, which here is most of retail banking. The failure mode is **building commodity and buying differentiation** — the usual result of letting procurement, rather than architecture, set the boundary.

**The forgotten third option is now the mainstream one.** Buy the ledger; build the product and experience layer on its interfaces. That is TymeBank with Mambu, Old Mutual with 10x, and what BIAN's Coreless Banking pilot argued for in chapter 13. You own product definition and customer experience and rent the record-keeping and compliance machinery. The cost is that **you now own the integration and the product catalogue** — and you must say out loud where a second product catalogue is not allowed to appear.

### The digital-only stack against the incumbent's

They are **not the same problem**, and conflating them is the most common error a reader will import from conference material.

The digital-only stack starts with no book to migrate, one product set, one channel, one regulatory entity and a chosen operating model. So it can take a software-as-a-service ledger or self-build on open source and run on a very small team — two people on one platform, with estimated capacity for two to three million customers and a breakeven target of 100 000 [61]. Its real constraints are distribution and the cost of deposits, and even so one of them took nine years and more than $135 million to reach a national launch [58].

The incumbent's stack carries every product ever sold, including ones closed to new business and still accruing; decades of undocumented product variation in the data; statutory reporting lineage; a branch, ATM and card estate; scheme certifications; and a customer base that cannot be asked to re-onboard. It cannot choose its starting point — which is why the incumbent patterns on the record here are **managed evolution** [55] and **phased system-of-record replacement**, announced without a timeline [57].

**THE HONEST READ FOR AN INCUMBENT ARCHITECT**

"Build like a digital bank" is not advice. It is a category error.

The transferable lessons are **small operations teams as a design goal, product definition as configuration, and a single ledger with everything else on interfaces**. Not the timeline, and not the headcount. And the transferable lesson from the incumbents is that the thing which takes nine years is not the software. It is the book.

## And once more, because it matters

**This chapter is a map of what plays in each category. It is not a recommendation, a ranking, an endorsement or a shortlist.** Named vendors are examples chosen for recognisability; omission is not a judgement and inclusion is not a blessing. There are no scores and no quadrants. Ownership and product naming change faster than books print, and **the research behind this chapter is dated October 2026** — re-verify every ownership and product name before it reaches a board paper, and put the date you checked next to it.

What goes stale first, in order: **card issuing and acquiring**, where the structure changed on 12 January 2026 [13]; **core vendor ownership**, where one vendor made two divestments in eighteen months [3][4]; **BIAN release content**, which moved in February 2026; **the South African payments institutions**, rebuilt in 2026 [48]; and **open finance here**, which is explored rather than in force.

And the sentence to leave you with. **By the time three vendors are in a procurement, the architecture has been chosen** — data model, extension model, release cadence, integration style, cloud posture, the skills market you hire from, and the exit cost you will one day pay. Negotiation moves price and dates, and very little else. So write the capability model first. Fix the non-negotiable constraints before the market scan. Shortlist for the shape of the problem rather than the size of the vendor. Take references in your own jurisdiction and size band. Record the choice as an architecture decision with the alternatives and the constraints that excluded them. And name the reversibility, because an option that cannot be undone belongs at board level, not in a procurement scorecard.

**THE CHAPTER IN TEN LINES**

- Every company named here is an example of who plays in a category — not a recommendation, not an endorsement, not a shortlist — and the research is dated October 2026, so re-verify every ownership and product name before it reaches a board paper.
- Card issuing and acquiring goes stale first: on 12 January 2026 Global Payments completed its acquisition of Worldpay and sold its own Issuer Solutions business to FIS the same day, consolidating issuing into FIS and acquiring into Global Payments.
- Attributions to get right: Kondor, Summit, Opics, Sophis and Arc are Teciem's; the Phoenix core and MalauzAi are CORA Group's; Calypso and AxiomSL are Nasdaq's and “Adenza” is not a present-tense company; Featurespace and Pismo are Visa's; Debt Manager is C&R Software's; SAP's banking software is SAP Fioneer's; SimCorp is Deutsche Börse's; and Compuscan is Experian's.
- In a South African banking sentence “NCR” means the National Credit Regulator, and the credit data layer here is a licensed, centrally routed utility with a statutory submission obligation — six bureaux are authorised to hold payment profile information, and the wider register is far longer.
- A vendor appearing in four or more capability categories is a concentration decision rather than a product decision, and the owner's type — software company, exchange, index group, private equity fund or card scheme — is a legitimate architectural attribute.
- No South African bank's core platform is publicly sourceable, and the three attributions that circulate rest on trade press quoting executives: attribute them to the outlet, not the bank. Systems-integrator attributions here are effectively unsourceable.
- The sourced duration anchors are Nedbank at R11.7 billion over nine years with the core last, OM Bank at nine years and more than $135 million on greenfield, and Barko live on vendor-operated software-as-a-service in under six months — all true, none comparable.
- The two fully documented failures, the Co-operative Bank and TSB, land on governance, supplier oversight, testing and capability rather than on software, and both organisations ended up with an incremental strategy.
- Buy where the regulator effectively specifies the answer, build where it is how you compete, and buy-and-extend where the capability is standard but your collections and distribution are not — which in South Africa is most of retail banking.
- By the time three vendors are in a procurement the architecture has been chosen: data model, extension model, release cadence, integration style, cloud posture, skills market and exit cost. Negotiation moves price and dates, and very little else.

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CHAPTER 15

# Regulation, risk and what the regulator inspects

*No regulator asks how your systems are built. They ask questions only a particular kind of estate can answer — twice, on time, with evidence. This chapter is about the constraints that are not negotiable, and where each one lands in the model.*

## What inspection actually looks like

Table Bay Bank is invented for this book: a mid-tier South African full-service bank, about 2.4 million retail customers, a business bank, a small corporate and transactional arm and a treasury. Nothing about it is a case study. The pattern below is. Marieke de Villiers is its chief risk officer; in eighteen months she has sat through five supervisory engagements, and not one contained a question about technology.

The Prudential Authority asked for the credit risk-weighted assets on the BA 700 for the quarter, the exposure-level data behind them, the model version behind each probability of default, and the standardised calculation run alongside — then the same for the previous quarter, with the movement explained.

The FSCA, separately, asked for every retail complaint of the last twelve months by prescribed category, with time to resolution, outcome, redress paid and ombud referrals; then the accounts closed or refused, the reason for each, and the notice given.

The Corporation for Deposit Insurance asked the narrowest question and the hardest one: for the reporting date, total qualifying deposits and total covered deposits. Two numbers — and be able to show, on request, the record for every depositor they came from.

Three regulators, three grains, one estate. None asked how the systems were built. All three asked questions only a particular kind of estate can answer, twice, on time, with evidence. Regulation does not ask for architecture. It asks for **answers that reproduce**, and reproducible answers have four properties most banking estates lack.

- **Reproducibility.** The same question over the same period returns the same number — after a restatement, a model change, a migration or a staff departure.
- **Lineage.** You can walk from the published figure back to the entry through recorded transformations rather than remembered ones.
- **Attribution.** For any judgement inside the number — an impairment stage, a credit decline, a sanctions-alert disposition — you can name who or what decided, on what inputs, under which rule version.
- **As-at reconstruction.** You can rebuild the position as it was understood on a past date.

Every instrument below lands on one of those four, and each section closes on what it forces in the model. Nothing here is legal advice, and the dates are sourced as at October 2026; several will have moved by the time you read this. What does not move is the shape of the constraint — an entity you must hold, an attribute you must capture at the point of transacting, a decision you must retain, a report you must reproduce.

## Twin Peaks, and the seven authorities a bank answers to

South Africa supervises financial institutions through two peaks established by the **Financial Sector Regulation Act 9 of 2017**. The Act and both regulators took effect **1 April 2018**; the Financial Services Board closed on 31 March 2018 and the FSCA succeeded it; Chapter 14, establishing the Ombud Council, took effect **1 October 2018** [1].

The **Prudential Authority** sits inside the South African Reserve Bank and supervises safety and soundness: capital, liquidity, credit and model governance, large exposures, returns accuracy and, increasingly, technology and cyber risk. The **FSCA** supervises conduct: product design, disclosure, complaints, fair treatment. Where a matter touches both they issue **Joint Standards** together [1]. The split exists because the two failures pull in opposite directions — lending generously to over-indebted customers improves this year's net interest income and next year's impairment charge — and one supervisor holding both mandates trades them off invisibly. But a bank does not have two regulators. It has at least seven.

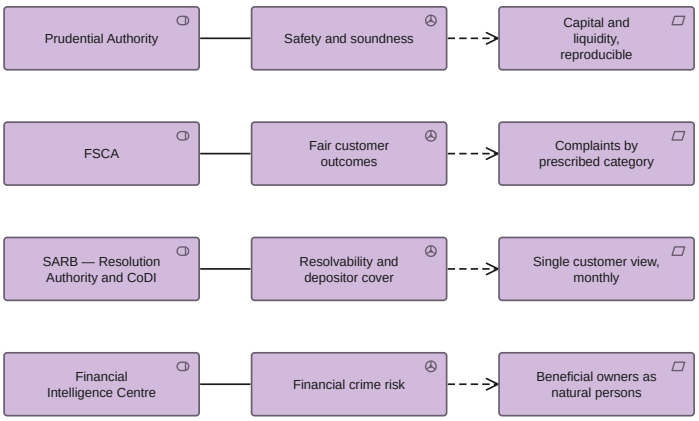



**Figure 15.1 Seven authorities, four grains of data.** The SARB appears three times under different hats. Each stakeholder carries a concern, each concern produces a duty an architecture must satisfy — and the duties want the same events described four different ways, which is the argument for one event record rather than four reporting estates.

The SARB appears three times under different hats: central bank and overseer of the payment system, **Resolution Authority** since 1 June 2023, and parent of both the Prudential Authority and the deposit insurer. Add the **Financial Intelligence Centre** for money laundering, the **National Credit Regulator** for consumer credit, the **Information Regulator** for personal information, and the **National Financial Ombud Scheme** for escalated complaints.

They want data at different grains. The Prudential Authority wants **money at a reporting date**: balances, exposures, provisions, capital, classified several ways at once. The FSCA wants **process and outcome over a period**: what was sold, to whom, after what disclosure, how long it took, who complained.

The temptation is three reporting estates. They are three derivations from the same events: a loan disbursement is an exposure to the Prudential Authority, an agreement with a disclosure record to the FSCA, and a transaction with an expected pattern to the FIC. Build the event record once at the finest grain any consumer needs, derive all three, and keep one obligations register — regulator, instrument, obligation, owner, evidence, system.

## The Banks Act, and why a licence is a data attribute

The **Banks Act 94 of 1990** and its Regulations are the licensing and prudential statute. No person may conduct the business of a bank without registration, and controlling-company approval, large-exposure limits, auditing requirements and the return regime flow from it [2]. Authorisation comes under section 12, registration within twelve months under section 16; only a public company may carry on the business of a bank, and section 11 makes it an offence for anyone else to conduct "the business of a bank", defined in section 1(1) around accepting deposits from the public.

**Section 54** governs amalgamations and transfers of business, and every bank acquisition in South Africa runs through a section 54 transfer — a legal migration of contracts with a fixed, gazetted effective date, not a milestone that can slip by a sprint.

And the licence with its conditions is an **attribute of products and accounts**, not a footnote in a compliance file. A bank that cannot state, per product, which licensed activity it performs under which authorisation cannot detect the product that has drifted outside it — and the Conduct of Financial Institutions Bill moves conduct regulation to **activity-based licensing**, with National Treasury gazetting notice of its introduction in the National Assembly on **17 April 2026**, no commencement date given, and at least one tracker still reporting the Bill as not tabled [8].

## The BA returns, and the change that moved them out of the law

The return series is how the Prudential Authority sees a bank: forms with defined content and deadlines, and the reason a bank has a regulatory reporting department.

| Return | What it carries |
|---|---|
| **BA 100 / 110 / 120** | Balance sheet; off-balance-sheet activities; income statement |
| **BA 200** | Credit risk — exposures by counterparty, approach and asset class, with impairments |
| **BA 300 / 310** | Liquidity risk; minimum reserve balance and liquid assets |
| **BA 330 / 350** | Interest rate risk in the banking book; derivative instruments |
| **BA 700** | Capital adequacy — risk-weighted assets, capital, ratios, leverage ratio |
| **BA 900** | Balance sheet of deposit-taking institutions on statistical principles — monthly |

Titles are as published on the SARB's banking-sector data pages [4]. One more, **BA 325**, appears in the regulation 46 reporting annexure with year-end and other-than-year-end variants but is untitled there and on the data pages [7], so it is out of the table rather than guessed at.

**Read BA 900 first**, which is not the obvious choice. It goes monthly to the Prudential Authority *and* the SARB's Economic Statistics Department, and feeds the monetary and credit aggregates, the flow of funds and the Integrated Economic Accounts [6]. It is the clearest single statement of what a bank's data model must carry: every balance classifiable by **counterparty institutional sector**, **instrument type**, **currency** and **residual maturity**, simultaneously, monthly, reconciled to the general ledger. If those four are not governed attributes in the product and party masters, the return is assembled in spreadsheets — and because BA 900 feeds national statistics, the errors are publicly visible.

Now the change that matters most, and that commentary routinely conflates with something else. By **Government Notice 5802, Government Gazette 52013, published 31 January 2025 and effective 1 February 2025**, the Regulations were amended to **remove the BA return forms from the Regulations**. Regulation 7 was replaced so that financial, risk-based and other returns are determined by the Authority **in writing**, and each form-specific regulation now requires completion in accordance with instructions determined or directed in writing by the Authority. Twenty-five forms, from BA 100 to BA 940, were deleted [3].

**Until 31 January 2025**

BA 100 … BA 940 **forms printed in the Regulations relating to Banks**

change path → National Treasury → Government Gazette → amended Regulations

lead time: a legislative cycle. Consultation, drafting, gazette.

↓

GoN 5802, GG 52013 · published 31 Jan 2025 · effective 1 Feb 2025

**From 1 February 2025**

Twenty-five forms **deleted** from the Regulations. Regulation 7 replaced: returns are **determined by the Authority in writing**.

change path → Prudential Authority directive

lead time: a consultation and a letter. A proposed directive of 21 April 2026 sets returns to be submitted from 1 July 2026.

**Design consequence.** Source-to-return mapping must be metadata a business analyst can change, the return must be a versioned entity, and the data layer must sit at a finer grain than any current form.

**Figure 15.2 What GoN 5802 changed.** Before 1 February 2025 a BA return form was in the Regulations, so changing it needed a gazette. After, regulation 7 puts the returns in the Authority's written determination. The specification can now move faster than any release train built around it.

TWO INSTRUMENTS, ROUTINELY CONFUSED

**GoN 5802 is not the Basel III post-crisis reform instrument.** It is dated 31 January 2025, effective 1 February 2025, and concerned with moving return forms out of the Regulations. The Basel III post-crisis reforms took effect **1 July 2025** through separate amended Regulations and prudential standards [8]. Commentary that merges the two will tell you the capital rules changed in February 2025. They did not.

The consequence is one sentence and it is expensive. **The reporting specification is now a directive, not a gazette.** It can change without a legislative cycle, and the Authority is exercising the power: a proposed directive published **21 April 2026** sets out returns to be submitted **with effect from 1 July 2026**, including new BA 900, BA 920, BA 930 and BA 940 [5].

So the source-to-return mapping has to be **metadata a business analyst can change**, the return must be a versioned entity so a figure submitted under the old form stays reproducible under it, and the data layer must sit at a grain *finer* than any current return. An estate built to the shape of today's forms is an estate that gets rebuilt.

## Basel III, the output floor, and the calculation you run twice

South Africa implements Basel through the Regulations relating to Banks and Prudential Authority directives: risk-weighted assets for credit, market and operational risk; minimum capital ratios plus the conservation, countercyclical and domestic systemically important bank buffers; the leverage ratio as a backstop; and the liquidity coverage and net stable funding ratios. South Africa adds a leverage-ratio buffer for domestic systemically important banks at 50% of the higher-loss-absorbency requirement on common equity tier 1 [8].

The Authority implemented the remaining **Basel III post-crisis reforms with effect from 1 July 2025** — revised standardised and internal-ratings-based credit risk, revised operational risk, leverage ratio, credit valuation adjustment and market risk frameworks, and the **output floor** — with Pillar 3 disclosures aligned to the latest Basel framework. The crypto-asset standard is the only component not yet implemented, with no date given [8]; commentary confirms the date and calls it a fundamental shift in how bank capital requirements are determined [9]. **Directive 12 of 2025** sets the implementation roadmap with interim reporting and phased model-change timelines, and **Directive 14 of 2025** strengthens large-exposure governance [8].

The output floor is the heavy one, and the clearest example here of a capital rule that is really a data and compute requirement.

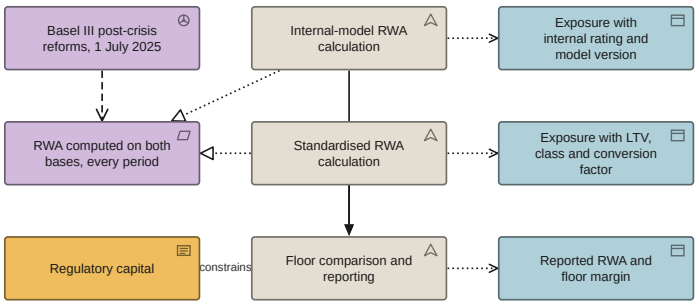



**Figure 15.3 The output floor is a parallel calculation estate.** A bank on internal models must compute risk-weighted assets twice and report the higher of the internal figure and the floored amount. That needs standardised-approach inputs — loan-to-value, exposure class, credit conversion factor — on every exposure, including ones previously only modelled internally. The arrows into the data objects are the expensive part.

A bank on internal models must compute risk-weighted assets **twice** — internal approach and full standardised — and report the higher of the internal figure and the floored amount. That needs standardised-approach inputs on *every* exposure, including exposures previously only modelled internally:

ratings or the applicable risk-weight bucket, loan-to-value for residential mortgages, exposure classes, credit conversion factors for undrawn commitments.

Two data consequences are worth stating. The revised standardised approach makes **loan-to-value a capital driver**, so a dated collateral valuation becomes a capital input and a stale valuation becomes a capital finding. The revised operational risk approach makes the business indicator and internal loss multiplier capital drivers, so a **ten-year operational loss history with defined quality standards** becomes capital data — and most banks hold operational losses in an incident register built for management reporting, with no retention policy, no completeness assertion and no reconciliation to the general ledger. That register is now in the capital calculation.

> WHAT I COULD NOT CONFIRM
>
> The **output floor percentage adopted in South Africa and its phase-in** are not in the sources I used. The Basel standard sets the floor at 72.5% of standardised risk-weighted assets, phased from 50%; whether South Africa adopted that schedule unchanged I could not verify, and the amended Regulations would settle it [8].

## IFRS 9, and the most data-hungry accounting standard a bank faces

IFRS 9 governs classification, measurement and impairment of financial instruments, effective for annual periods beginning on or after **1 January 2018** [10]. Impairment runs in three stages: stage 1 performing, twelve-month expected credit loss; stage 2, credit risk significantly increased since initial recognition, lifetime loss; stage 3, credit-impaired, lifetime loss with interest on the net carrying amount. The estimate must be unbiased and probability-weighted, reflecting the time value of money and **reasonable and supportable forward-looking information** [11].

The Prudential Authority treats **stage 1 and 2 provisions as general and stage 3 as specific**, and deducts provisions for all three stages from the leverage-ratio exposure measure on the BA 700 [10]. I could not confirm whether the proposed directive on the regulatory treatment of accounting provisions, replacing Directive 5 of 2017, is final, or under what number. Impairment stage is therefore an accounting attribute, a capital input and a leverage input at once: a governed, dated, auditable fact on the exposure, not a column computed in the reporting layer at month-end. What the standard demands of the data:

1. **The initial-recognition credit state, retained for the life of the exposure** — a significant increase in credit risk is measured *relative to origination*, so the origination-date rating or score must survive twenty years of a mortgage. Most core systems overwrite it on the first behavioural rescore.
2. **A daily days-past-due time series per exposure**, because the 30-day presumption, the 90-day default definition and every backtest need it. Monthly snapshots do not do.
3. **Dated stage classification with the trigger that caused it**, per exposure per reporting date.
4. **Behavioural lifetime for revolving facilities**, modelled and evidenced, not the contractual term, and **macroeconomic scenarios with weights**, versioned and retained with the forecast vintage.
5. **Model versions and parameter sets per reporting date**, or last quarter's charge cannot be reproduced and the inspection question above has no answer.
6. **Post-model adjustments as a separate, governed, explainable layer** — where judgement lives, and judgement is what gets asked about.

One trap. Concessions granted because of financial difficulty were governed by **Directive 7 of 2015**, replaced by **Directive 11 of 2025**, with banks applying Directive 7 until **31 December 2026** [8]. Two definitions of one event in parallel is the normal condition of regulatory data, and the reason a restructure flag needs a rule version attached.

## Resolution, and the test you take over a weekend

The **Financial Sector Laws Amendment Act 23 of 2021** made the SARB the **Resolution Authority** for banks and non-bank systemically important institutions and created the deposit insurance scheme, assented to in **January 2022** [12]. The Minister published the commencement schedule on **24 March 2023**, and the resolution framework took effect **1 June 2023**, repealing the Banks Act curatorship provisions — from that date a bank likely to fail goes into **resolution** instead [13].

Losses fall on shareholders and creditors before public funds; the SARB has stabilisation powers including **bail-in**; and systemically important banks must issue **first loss after capital** instruments able to recapitalise the institution in resolution [13]. **Prudential Standard RA04**, on valuators for resolution, was implemented in **March 2026** [8], and the regime has been used — Ditsobotla Co-operative Bank was placed in resolution and the deposit insurer paid eligible depositors within the prescribed timeframe [13]. I could not confirm the first-loss-after-capital quantum, its eligibility criteria or the deadline for holding it.

Resolvability is a data requirement before a legal one. A plan needs, at short notice: the liability structure by creditor class and legal entity with **bail-in eligibility flagged per instrument**; critical functions and the systems and third parties supporting them, across group entities; intra-group exposures; and valuation-ready data for a valuator.

The test, plainly: could the bank hand a valuator a complete, current balance sheet by legal entity over a weekend? Most cannot, and the reason is almost never the balance sheet. It is that the legal-entity dimension was never carried on the entries, so the consolidated view is the only real one and the entity views are reconstructed by allocation.

## CoDI, and a regulation that is really a data specification

The **Corporation for Deposit Insurance** is a statutory deposit insurer and a wholly owned SARB subsidiary, completed by the **Financial Sector and Deposit Insurance Levies Act 11 of 2022**, its companion **Act 12 of 2022** and the **Deposit Insurance Regulations of 2024** [14]. It became operational **1 April 2024**, and every registered bank is a member.

Cover is **R100 000 per qualifying depositor per bank**, on the total of that depositor's qualifying balances, **gross** — negative balances are not netted — and separate per bank. Qualifying products include current, term, notice, savings and transactional accounts, tax-free savings and the Islamic Wadi'ah, Qard and Murabaha structures; shares, commodity holdings, electronic money, suspense accounts and anything not repayable at par are excluded. Excluded depositors include banks and the non-bank private financial sector, government and public-sector entities, and the failing bank's own executive management, board, their related parties and lead external audit partners [15].

Stop on that last exclusion: it is a pure architecture requirement hiding in a coverage rule. The bank must identify, **in its depositor data**, which depositors are its own executives, directors, their related parties and its lead audit partners — a party-to-party relationship plus an employment role, resolvable at a reporting date, across every deposit-taking system.

Member banks submit total qualifying deposits and total covered deposits monthly, and those figures determine each bank's contributions under Prudential Standard CoDI 1 on fund liquidity [14][17]. The fund reportedly reached a R20 billion target in its first year [37]. On the premium, one source gives 0.2% divided by twelve of covered deposits monthly, indexed annually to consumer inflation, while the primary documents state no rate — treat it as reported rather than confirmed.

### The single customer view

The clearest case in South African banking regulation of a regulator specifying a data structure. Not a principle — a structure, with a pass/fail test. The bank must consolidate each qualifying depositor's accounts and balances in qualifying products into **one single customer view record**, the covered deposit being the qualifying balance capped at R100 000. The rules then specify the hard parts [16]:

- **Joint accounts** split by the **sharing ratio recorded by the bank**, defaulting to equal shares where none is recorded. Most banks have never captured a sharing ratio, so the default applies by accident rather than by decision.
- **Formal beneficiary accounts** — an attorney's trust account — looked through to each beneficiary, with a record per beneficiary **if the bank can identify all beneficiaries and their balances monthly**, else one record for the account with no split.
- **Informal beneficiary accounts** — stokvels — covered per holder to R100 000, upgraded to formal treatment if the bank holds the constitution and member details monthly; with several sets of signatories the R100 000 is split pro rata by each account's share.
- **Sole proprietors** covered separately for personal and business accounts, each to R100 000, as separate records. A model that cannot distinguish one person's personal accounts from the same person's business accounts cannot compute cover.
- **Balance rules** prescribed: gross, interest accrued to the reporting date included, outstanding fees excluded, unsettled payments added or deducted, foreign currency at the close-of-business rate, negative balances reported as zero. Pledged accounts carry the full balance with the unpledged portion as the qualifying balance, flagged not readily payable; deceased depositors likewise.
- **The field set** prescribed, down to a **reliability-of-identification flag** per depositor, contact details, joint and pledged flags, signatory details for informal beneficiary accounts, and not-readily-payable indicators with reasons.

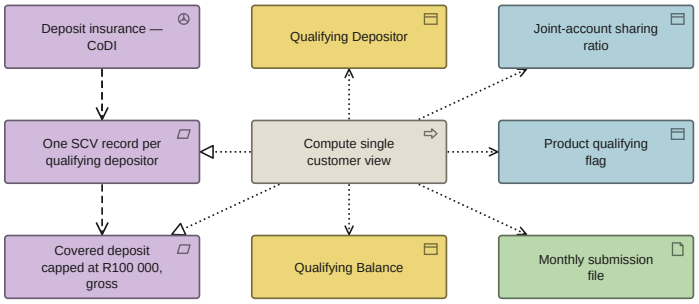



**Figure 15.4 A regulation that is a data specification.** The deposit insurer's single customer view is not a principle to aspire to. It is a computed record with a legal definition, a prescribed field set and a pass/fail test, and three of the data objects it accesses — the sharing ratio, the qualifying flag on the product rather than the account, the identification-reliability measure — are fields most banks have never captured.

The submission mechanics are modest next to the calculation: a monthly CSV with four mandatory fields — member bank code, reporting date, total qualifying deposits, total covered deposits — due **within ten working days after month-end**, the first for March 2024 balances by 5 April 2024, plus compliance declarations in prescribed windows [17]. Two numbers, derived from a legally defined aggregation across every deposit product, per natural person, executable under resolution conditions at short notice.

> **THE OPEN QUESTION I COULD NOT CLOSE**
>
> **Quarterly single-customer-view submissions were exempted from 1 April 2024 to 30 September 2025**, during which banks still had to produce the calculations internally to derive the monthly totals [17]. What applies *after* 30 September 2025 — whether full record files are now submitted periodically, and in what format — is not stated in the documents I read, and it is the most important open item in this chapter for a 2026 architecture. CoDI's current reporting requirements document would settle it. Ask before you size the work.

So: a resolved party identity across every deposit-taking system; a qualifying flag on the product master rather than the account; a joint-account sharing ratio; beneficiary data for trust and stokvel accounts; a contactability record; an identification-reliability measure per party; and a reproducible as-at calculation. A bank with five customer masters cannot produce it — the honest reason this requirement has driven more party-master investment in South Africa than any customer-experience argument did.

## FICA, the risk-based approach, and life after the grey list

The **Financial Intelligence Centre Act 38 of 2001** makes banks accountable institutions with anti-money-laundering and counter-terrorist-financing duties, enforced by the Financial Intelligence Centre and supervised for banks by the Prudential Authority.

The **General Laws (Anti-Money Laundering and Combating Terrorism Financing) Amendment Act 22 of 2022** rewrote much of it, with staggered commencement: **31 December 2022** for FICA sections 18 to 51 and related Companies Act, FSR Act and Nonprofit Organisations Act provisions, and **1 April 2023** for the Trust Property Control Act beneficial-ownership provisions. The amended FICA Schedules 1, 2 and 3 took effect **19 December 2022**, adding co-operative banks, credit providers, money or value transfer providers, high-value goods dealers and **crypto asset service providers** to the accountable institutions [18]. Substantively, the amendments expanded the **definition of beneficial owner** and due diligence for legal persons, trusts and partnerships, and made Security Council resolutions enforceable immediately on adoption [18].

Read as data, the obligations are: a documented risk methodology and a **dated risk rating per client, product, channel and jurisdiction** with its factor inputs; a board-approved compliance programme the systems can be shown to implement; identification and verification of the client, of persons acting for them and of the **beneficial owners** — a control graph resolved to natural persons, not a field; risk-driven review plus monitoring against a **stored expectation** of normal behaviour; **suspicious transaction reports** through goAML no later than **fifteen working days** after becoming aware [21]; and **cash threshold reports** for cash **above R49 999,99**, no later than **three days** excluding weekends and public holidays [20].

One subtlety catches architects importing assumptions from elsewhere: the Centre's guidance assesses **each transaction** against the cash threshold, so separate sub-threshold cash buy-ins on the same day or at different branches are not cash-threshold-reportable, though the pattern may require a suspicious transaction report [20]. Check which rule your monitoring platform implements; the two produce very different alert volumes.

South Africa was placed on the FATF list of **Jurisdictions under Increased Monitoring** in **February 2023** and **exited on 24 October 2025**, after plenary sessions in Paris from 22 to 24 October. The action plan had **twenty-two action items**, tracked over roughly thirty-two months, with an on-site assessment at the end of **July 2025**. The next **mutual evaluation is expected to begin in the first half of 2026 and to conclude in October 2027** [19].

> **EXIT CHANGES NOTHING YOU HAVE TO BUILD**
>
> Grey-listing drove the 2022 amendment Act, beneficial-ownership registers at the Companies and Intellectual Property Commission and the Master's Office, far more aggressive supervision, and enhanced due diligence applied to South African counterparties by foreign correspondent banks. **Exit reduced none of the statutory obligations** [19]. With a mutual evaluation beginning in the first half of 2026, the data has to stay *demonstrably* in order rather than merely be in order — and the difference is an evidence architecture.

What FICA forces, then: a party-and-control graph resolving beneficial ownership to natural persons; a dated risk rating per party with its factor inputs; monitoring with retained alerts, dispositions and reasons; report generation with the submitted artefact kept rather than re-derived; and verification against external registers with the match result and its provenance stored.

## Sanctions screening, and the twenty-four-hour clock

Targeted financial sanctions sit alongside FICA with a separate failure mode. FIC Act sections 26A to 26C require the Centre to administer Security Council sanctions in South Africa, the Protection of Constitutional Democracy against Terrorist and Related Activities Act prohibits dealing with property linked to terrorism or sanctioned persons, and section 49A makes failure to comply a criminal offence [21].

The duties: screen clients and prospective clients **at onboarding, on an ongoing basis, and whenever the Security Council adopts or expands measures**; **freeze** a listed person's property; do not transact with them unless the Minister has granted a permitted exception. A **terrorist property report** under section 28A — which requires freezing and ceasing business — is filed no later than **five days** after establishing that the bank holds a listed person's property; where suspicion cannot be confirmed, a section 29 report goes in within fifteen working days [21].

Two details change the architecture. The Centre's list is a **replica of the Security Council consolidated list, not a domestic list**, with permanent reference numbers identifying the regime. And **the Centre updates it within twenty-four hours of a Security Council change** [21]. So the obligation is not "screen at onboarding". It is: re-screen the entire party and counterparty estate against a new list version within a day.

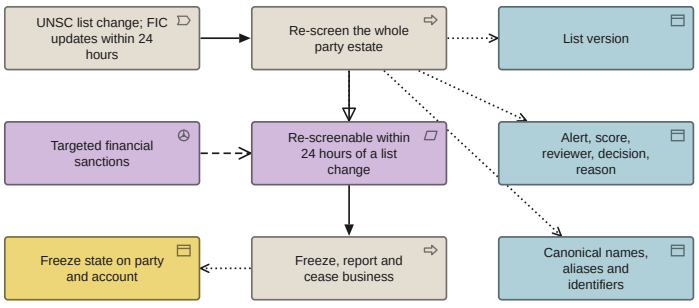



**Figure 15.5 Screening is a re-run, not a gate.** The Financial Intelligence Centre updates its list within twenty-four hours of a Security Council change, so the obligation is to re-screen the whole estate against a new version in a day. The list-version data object is what lets a bank prove what it screened against and when, and the freeze is a state on the party and the account rather than a note in a customer system.

Then the sentence from the Centre's own manual that should be read aloud in a design review: **relying on commercial screening software, or judging the risk low, is not a defence** [21]. That is aimed at banks that can do the screening and cannot show the rest — the alert, the match score, the reviewer,

the decision, the reason, and the list version the screen ran against. And a freeze must be a real state on the party and the account, surviving system boundaries, not a note in a customer system the payment hub never reads.

## The National Credit Act, which turns lending into a retained record

The **National Credit Act 34 of 2005** is more prescriptive than its United Kingdom or European equivalents, and the layer an architect new to South African banking most often underestimates. The **National Credit Regulator** regulates it; commencement was phased, with the main provisions from 1 June 2007; the **National Credit Amendment Act 19 of 2014** tightened it; and the affordability assessment regulations were gazetted as Government Notice 202, Government Gazette 38557, on **13 March 2015** [22]. The **National Credit Amendment Act 7 of 2019** on debt intervention was enacted but **not brought into operation** [23]; whether any part has commenced since I could not confirm as at October 2026.

> **A TERM THAT CATCHES OUTSIDERS**
>
> In a South African banking sentence, **"NCR" means the National Credit Regulator** — not the automated-teller-machine manufacturer, however many of its machines are in the branch network.

**Affordability assessment.** The lender must take practicable steps to validate **gross income**, apply the prescribed **minimum expense norms** table, and account for existing obligations from the bureau [22]. Income validation is specified by employment type — salaried consumers, the latest three payslips or bank statements showing three salary deposits; self-employed, three months of statements or the latest financial statements. The calculation runs gross income, less statutory deductions, less prescribed minimum living expenses, less other disclosed commitments, with the remainder available for instalments; declared expenses below the norms may be accepted only exceptionally. Repayment history must be assessed within **seven business days of initial approval or limit increase, fourteen for mortgages** [22]. That is a document-ingestion capability, a versioned rule set and two hard pipeline timers. Affordability is a **regulated calculation**: it belongs in a decision service with effective-dated rule versions and a stored, replayable decision record, because a reckless-lending challenge years later is answerable only against the rules as they stood on the decision date.

**Reckless lending.** Credit granted without a proper assessment, or that leaves the consumer over-indebted, is reckless, and a court may suspend the consumer's obligations. The defence is the retained assessment, not the policy saying one is done.

**In duplum.** Under section 103(5), while a consumer is in default, the total of interest, initiation fees, service fees, credit insurance, collection costs and default administration charges **may not exceed the unpaid balance of the principal debt at the time of default**. So the bank must capture **principal outstanding at the moment of default** as a durable fact and cap accruals against it: a per-account cap derived from a historical event, which means the arrears engine needs the default event, its date and the balance as at that date rather than the current balance. I could not confirm the precise enumerated charge categories, and they matter to the calculation.

**Debt review.** Under section 86 a consumer may apply through a debt counsellor, after which the provider is constrained in enforcement and a restructured plan may be ordered. That needs a debt-review **state** on the party and on each agreement, with dates, counsellor, proposal, court order and restructured schedule, driving the arrears engine — or one system honours debt review, another does not, and the bank issues default notices to a consumer under review.

**Credit bureau obligations.** Prescribed consumer credit information goes to registered bureaux in prescribed form at prescribed intervals, with consumer access and correction rights. A bureau submission is a regulatory return with the same lineage needs as a BA return, and corrections must flow back.

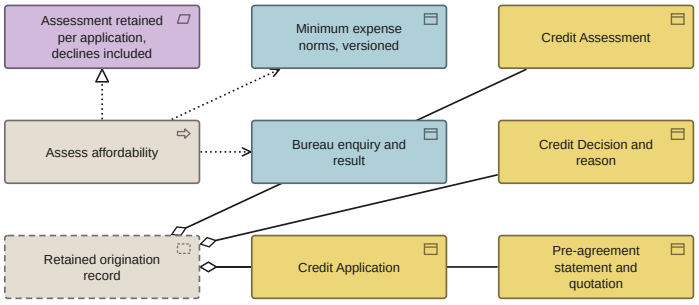



**Figure 15.6 The National Credit Act turns origination into a record that may not be overwritten.** The grouping is the point: application, assessment, decision with its reason, and the quotation as given are all retained, including for declines, because the retained assessment is the defence against a reckless-lending finding years later. The versioned expense-norm table is what makes the calculation replayable against the rules as at the decision date.

Add prescribed disclosure — a pre-agreement statement and quotation with total cost of credit, interest rate, all fees and the instalment schedule — and the shape is clear. The Act turns the origination chain into a regulated record, and **nothing in that chain may be overwritten.** A lending estate that treats an application as a transient workflow object, discarded once a loan is booked, has destroyed its own defence. One more consequence, expensive at scale: the Act's maximum rates and fees are formulas on the repo rate, so every monetary policy decision is a **catalogue-wide repricing event** the product master must absorb without a release.

## The Conduct Standard for Banks

**Conduct Standard 3 of 2020** is the FSCA's conduct standard for banks under the FSR Act, applying to banks including mutual and co-operative banks. Commencement was staggered: sections 1, 2 and 11 on **3 July 2020**; sections 3 to 6 — culture and governance, product design, retail customer

requirements, advertising — on **3 March 2021**; and sections 7 to 10 — disclosures, complaints, refusal or closure of accounts by the bank, and termination by the customer — on **3 July 2021** [24].

> **WHERE IT DOES NOT REACH**
>
> **Sections 4 to 10 do not apply to the provision of credit under a credit agreement** [24]. The National Credit Act governs there. That boundary is why a conduct obligation can be real for a transactional account and inapplicable to a personal loan, and why conduct requirements cannot be implemented once at customer level and assumed to cover everything.

The obligations that leave marks in a model: products designed with customer needs in mind and monitored on an ongoing basis, with **target-market assessment** for retail products and a prohibition on **unfair contract terms**; advertising clear, fair and not misleading, with **records kept at least five years** and marketing opt-outs honoured **at no charge**; disclosure of costs, risks and recourse including the **effective annual interest rate** where interest is paid; a complaints framework with **prescribed retail complaint categories**; **reasonable prior notice and reasons before the bank refuses or closes an account**, except where law requires otherwise or illegal use is suspected and reported; and a duty to **identify and notify customers of dormant products** [24]. The FSCA inspects complaints categorisation above all; its review found critical gaps and low recorded filing rates [25].

As entities: a **target market** on the product with monitoring evidence; a **complaint** with the prescribed categories as a governed code list; a **marketing preference** per party, dated, honoured across every channel and free to exercise, which makes preference party-level master data rather than a campaign flag; a **dormancy** rule per product; and a **closure or refusal decision** with reason and evidence of notice.

The closure duty is the conduct obligation that most often collides with a financial-crime one. Notice and reasons are required *except* where illegal use is suspected and reported — and the exception is the whole design: the closure decision must record which basis was used, by whom, on what evidence, because a regulator will ask which path you were on.

## POPIA, with the banking bite

The **Protection of Personal Information Act 4 of 2013** commenced substantially on **1 July 2020** with a one-year transition and became enforceable on **1 July 2021** [27]. Three provisions bite harder in a bank than almost anywhere else.

**Section 71 — automated decision-making.** A data subject may not be subject to a decision with legal consequences, or that affects them substantially, based **solely** on automated processing intended to create a profile — including profiling of **creditworthiness**, reliability, conduct or performance at work. Exceptions apply where the decision is made in concluding a contract with appropriate measures protecting the data subject's legitimate interests, including an opportunity to make representations, and where sufficient information about the underlying logic is given [26]. A fully automated credit decline is therefore a regulated act: the bank needs a human-review path, a representation mechanism, a renderable reason and the retained inputs that make the reason true. Machine-learned scorecards are not prohibited; **opacity** is the problem. A digital lending business case that promises ninety-five per cent automated decisioning has just acquired a designed fifth path, with a service level and an audit trail.

**Section 72 — cross-border transfer.** Transfer to a foreign recipient requires one of the listed conditions: the recipient is subject to a law, binding corporate rules or binding agreement providing adequate protection with substantially similar principles and onward-transfer restrictions; the data subject consents; the transfer is necessary for a contract with or in the interest of the data subject; or it is for their benefit and consent is not reasonably practicable. So cloud and offshore processing need a **mapped transfer basis per data category per destination**, and the Prudential Authority and FSCA issued **Joint Communication 2 of 2025 on cloud computing and the offshoring of data** in July 2025 [8]. Retrofitting data residency is a migration, not a configuration change.

**Section 69 — direct marketing.** Unsolicited electronic direct marketing is restricted; for non-customers consent is required and may be requested only once. That needs a dated consent and opt-out record per party, per channel, per purpose, plus proof the single-request rule was respected. The Conduct Standard's free-opt-out duty layers on top [24].

One last thing. Breach notification runs to the Regulator and to affected data subjects *alongside* the Joint Standard duty to notify the financial-sector authority of a material incident: **two notifications, two formats, one incident**. And in South Africa, unlike Europe, **juristic persons are data subjects**.

## The payment system, and who writes the rules now

The **National Payment System Act 78 of 1998** is the statute under which the SARB oversees the national payment system and regulates participation in clearing and settlement. Clearing happens in **Payment Clearing Houses** per stream, each with a participant group, clearing rules, a system operator and service levels; settlement happens at the SARB in **SAMOS** [28]. National Treasury published a review policy paper in September 2018; whether a replacement Bill has been introduced I could not confirm.

The governance changed in 2026, which is why an obligations register needs repointing. The Payments Association of South Africa, recognised as the payment system management body in 1999, has had its **recognition withdrawn**, and its functions were split. The SARB took over rules and standards, the licensing and registration of payment institutions, and the card, immediate-settlement, cash-settlement, equities, bonds, derivatives and money-market clearing houses on **11 August 2026**. The electronic funds transfer credit and debit, Authenticated Collections, PayShap, Registered Mandate and real-time clearing houses moved to **PayInc** on **2 September 2026**. High-value arrangements transferred to the SARB under a directive issued **2 June 2026**, and the standards structures became a standalone SARB function from **1 September 2026**. Existing authorisations remain valid, with third-party payment provider registrations valid until the SARB publishes an authorisation framework [29].

DATES THAT DISAGREE WITH EACH OTHER

**The SARB's own page is internally inconsistent on the completion date**, giving both "fully completed by 1 September 2026" and 2 September 2026 in different sections [29]. Where a primary source disagrees with itself, say so rather than picking. Two further corrections: **PayInc is the renamed BankservAfrica**, now half-owned by the SARB — a utility, not an industry body; and the "Payments Industry Body" of commentary was a November 2022 design proposal, not the outcome.

For the model this lands in one attribute. Carry the **clearing stream** on every payment, because the rulebook, the cut-off, the return window, the dispute process and now the governing body all differ by stream. A payment model with a single "EFT" type cannot express that a Registered Mandate collection and a DebiCheck collection follow different rules in different windows under different rulebooks.

One more thing, because it is the one people get wrong. **South Africa has no regulated open banking regime** — only bilateral commercial interfaces plus **SARB Directive 2 of 2024** on screen-scraped electronic funds transfer credit initiation. The FSCA's open finance work explores a regime; it has not created one.

## Exchange control, and the field no system can infer

Exchange control derives from the Currency and Exchanges Act and the Exchange Control Regulations, administered by the SARB's **Financial Surveillance Department** through Authorised Dealers — the banks — and limited-authority dealers, under the Currency and Exchanges Manual for Authorised Dealers, its limited-authority equivalent, and exchange control circulars.

The **FinSurv Reporting System** is the most prescriptive data specification a South African bank faces, and it is worth reading as a data model rather than a compliance document. Cross-border transactions are reported through modules — BOPCUS, BOPCARD RESIDENT, BOPCARD NON RESIDENT, NON RESIDENT RAND, BOPDIR, INTERBANK — each with a long prescribed field set: flow direction, a replacement indicator and the original reference, the **value date**, an **eight-digit branch code** with hub code and name, originating, receiving and correspondent banks with SWIFT country codes, up to 999 monetary-detail sequences per transaction, rand and foreign values with the SARB applying the mid-rate so the reporter must *not* pre-convert, the **balance-of-payments category and sub-category**, and full party details including **institutional sector code** and **industrial classification** [30].

The timelines are tight: submit within **two business days of the value date**; resubmit rejected data and investigate warnings within one business day, warnings triggering on value dates older than fifteen days; an integrated-form exemption for natural-person inward payments under **R50 000** and an outward remittance dispensation at **R5 000 and less**; retain integrated forms **five years** [30].

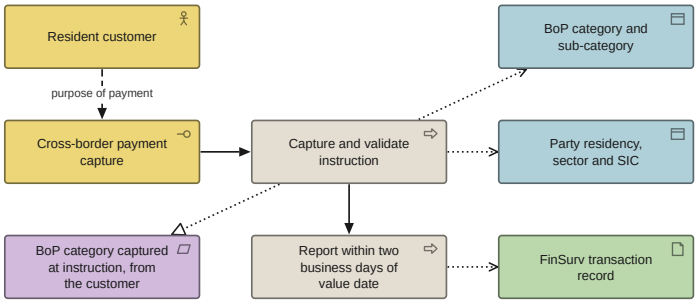



**Figure 15.7 The one field no system can infer.** The balance-of-payments category encodes the economic purpose of a cross-border payment. It cannot be derived from the amount, the beneficiary or the country, so it flows from the customer at the point of instruction — in every channel that can originate one. Residency, institutional sector and industrial classification are governed master data on the party, not free text on a form.

Here is the point that decides whether this works. **The balance-of-payments category encodes economic purpose, and no system can infer it.** It is not derivable from the amount, the beneficiary, the country or the account. It has to be captured **at the point of transacting, from the customer**, which makes it a mandatory field in every channel that can originate a cross-border payment — app, branch, host-to-host file, corporate portal, application programming interface. Architects discover this when the corporate channel goes live without the field and the reporting team starts phoning clients to ask what the payment was for. Residency, institutional sector and industrial classification are governed master data on the party for the same reason.

## The Joint Standards, with the numbering got right

Joint Standards are made jointly by the Prudential Authority and the FSCA under the FSR Act. Four matter, three of them are numbered "1", and the commentary gets them wrong often enough to warrant a table.

| Instrument | Subject | Applies to banks? | Commencement |
|---|---|---|---|
| **Joint Standard 1 of 2023** | IT governance and risk management | **Yes** — banks, branches of foreign institutions, bank controlling companies, mutual banks and others | **15 November 2024** [31] |
| **Joint Standard 2 of 2024** | Cybersecurity and cyber resilience | **Yes** — banks and mutual banks, plus insurers, market infrastructures and others | **1 June 2025**, twelve months from that date to comply fully, no transitional period in the standard [32][8] |
| **Joint Standard 1 of 2024** | **Outsourcing by insurers** | **No** — insurers under the Insurance Act 2017, except Lloyd's and branches of foreign reinsurers | Published 17 May 2024; in operation **1 December 2024** [33] |
| **Joint Standard 1 of 2025** | Central counterparty and trade repository exemption criteria | Derivatives activity rather than general banking | Effective **October 2025** [8] |

THE ERROR IN CIRCULATION

**Joint Standard 1 of 2024 is an insurance outsourcing standard and does not apply to banks** [33]. A learner guide or compliance summary listing it as a banking outsourcing obligation is wrong, and you will see it listed that way. Banks' outsourcing obligations come from the Banks Act Regulations and Prudential Authority directives and guidance notes, and from the third-party provisions inside the two technology Joint Standards — Joint Standard 1 of 2023 has no dedicated outsourcing section, but requires service providers to be fit and proper and contractually bound to protect confidential information, third parties to be considered in resilience planning, and recovery testing with linked service providers where applicable [31]. I could not confirm which current Prudential Authority instrument governs bank outsourcing, or its number. Find out before you tell a procurement committee what the rule is.

Between them the two standards oblige a specific set of **governed artefacts** rather than documents. Joint Standard 1 of 2023: a board-approved IT strategy reviewed annually with quarterly action-plan review; an IT risk framework with a **risk register**; an **IT asset inventory** and a **register of critical applications**; data-loss prevention from endpoints *and call centres*; backup with **restore testing at least annually**; **recovery time and recovery point objectives**, a geographically separate disaster recovery site and annual resilience testing including site loss; and notification of **material incidents** [31]. Joint Standard 2 of 2024 adds an **inventory of information assets**; assessment of risks to **critical operations**; least privilege and **multi-factor authentication** for users reaching critical functions and applications holding sensitive information; and notification of material cyber incidents [32]. I could not confirm the notification deadline — the source states the duty without a timeframe.

Each item in that list is a thing a regulator can ask to see, on a day of their choosing, in a form that must match what the systems actually do. So the application register, the information asset inventory, the recovery objectives, the critical-operations map and the incident classification scheme are **data products with owners**, not appendices to a policy.

## King IV, and the board's version of the same question

The **King IV Report on Corporate Governance for South Africa, 2016** is not legislation. It is given force through the JSE Listings Requirements and framed as apply-and-explain across seventeen principles [34]. **Principle 12** is the relevant one: the governing body should govern technology and information in a way that supports the organisation in setting and achieving its strategic objectives — covering oversight of information architecture, data management including privacy, technology risk including cyber, and major technology investments and third-party arrangements. I could not verify the exact wording or the recommended-practice list; the King IV Report itself would settle it.

King IV is the governance frame; **Joint Standard 1 of 2023 is the enforceable version for a financial institution**, making the governing body ultimately responsible for compliance [31]. Which produces a deliverable most architects resist and should not: a **board-consumable architecture view** — a different document from the one delivery teams use, true, short, and the same model underneath.

## The ombud landscape, after consolidation

Four industry schemes — Credit, Banking, Long-term Insurance and Short-term Insurance — voluntarily amalgamated into the **National Financial Ombud Scheme South Africa**, expected to begin operations on **1 March 2024** following recognition by the Ombud Council [35]. Its Banking and Credit division is where a bank's escalated complaints now go; the Pension Funds Adjudicator, the FAIS Ombud and the JSE Ombud remain outside the interim amalgamation [35][36].

Small change, two real consequences. The **complaint** entity needs an **external forum** attribute on a code list that has just changed, with referral and determination recorded against the original complaint. And because the Conduct Standard requires customers to be told about ombud recourse, disclosure templates are date-sensitive — every statement, decision letter and web page naming the old schemes is now wrong.

## What the regulator actually asks to see

Eight questions. If the estate answers them without a project, the architecture is sound enough for a South African bank. If it cannot answer three, you have found your first year's work.

1. Can you reproduce last quarter's published capital and impairment figures, from the data as it stood, with the model versions and parameter sets as they stood?
2. Can you walk any published number back to the entries behind it, through recorded transformations, without a spreadsheet in the middle?
3. Can you produce the single customer view for every qualifying depositor, at a chosen reporting date, and attest it?
4. For any decision affecting a customer — a decline, a closure, a freeze, an impairment stage — can you say whether it was wholly automated, on what inputs, under which rule version, approved by whom?
5. Can you re-screen the entire party estate against a new sanctions list version within twenty-four hours, and prove which version you screened against?

6. Can you classify every balance by counterparty institutional sector, instrument, currency and residual maturity at once, without a reconciliation step?
7. Can you name every third party holding customer data, in which country, on which transfer basis?
8. Can you hand an independent valuator a complete balance sheet by legal entity over a weekend?

None of those is a technology question. All eight are answered by architecture, and all eight are cheap at design time and ruinous afterwards. That asymmetry is the only reason a chapter about regulation belongs in a book written for architects rather than for compliance officers.

**THE CHAPTER IN TEN LINES**

- Regulation never asks for architecture; it asks for answers that reproduce, which means reproducibility, lineage, attribution and the ability to rebuild a past position as it was then understood.
- A South African bank answers to at least seven authorities wanting the same events at different grains, so build the event record once at the finest grain and derive every return from it.
- GoN 5802, effective 1 February 2025, took the BA return forms out of the Regulations: returns are now determined by the Authority in writing, so a reporting estate must absorb a changed return without a legislative cycle — and it is not the Basel III instrument, which took effect 1 July 2025.
- BA 900 is the clearest statement of what the data model must carry: every balance classifiable by counterparty sector, instrument, currency and residual maturity at once, monthly, tied to the general ledger.
- The Basel output floor forces risk-weighted assets to be computed twice, which makes loan-to-value, exposure class and a ten-year operational loss history into capital data.
- CoDI's single customer view is a regulator-specified data structure with a prescribed field set, a joint-account sharing ratio and a monthly submission — the most under-appreciated data requirement in South African retail banking.
- Exiting the FATF grey list on 24 October 2025 reduced no statutory obligation, and the next mutual evaluation is expected to begin in the first half of 2026.
- Sanctions screening is a re-run against a list version within twenty-four hours, not a gate at onboarding, and the Centre says software is not a defence.
- The National Credit Act makes origination a retained record — application, assessment, bureau data, calculation, decision, disclosure, agreement — where nothing may be overwritten, and makes every repo change a catalogue-wide repricing event.
- For banks the technology Joint Standards are 1 of 2023 (15 November 2024) and 2 of 2024 (1 June 2025); Joint Standard 1 of 2024 is insurance outsourcing and does not apply.
- Every cross-border payment is a central-bank reporting event carrying a balance-of-payments category that no payment system can infer, so the field belongs in every originating channel.

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CHAPTER 16

# Modelling banking in ArchiMate

*Forty-two elements, a thousand banking words, and no dictionary. This chapter is the mapping — and the reasoning behind every choice, because the reasoning is what you will need for the concept this chapter missed.*

## Forty-two elements, a thousand banking words

An architect arriving in a bank wants a dictionary. Account equals this element, payment equals that one. There is no such dictionary, and the reason is arithmetic. ArchiMate 4 gives you forty-two elements, nine of which carry no layer at all: Role, Collaboration, Process, Function, Service, Event, Path, Grouping and Location. Banking brings a thousand terms before you reach the treasury. The glossary at the back of this book runs past two hundred and twenty and is deliberately short; BIAN's service landscape, release 14.0, carries 322 service domains, 586 business capabilities and 4,905 service operations [1].

So the mapping cannot be one concept to one element, and three things happen instead.

**Many concepts collapse onto one element.** Depositor, borrower, guarantor, signatory, beneficial owner, cardholder, mandate holder and counterparty are all Role. That is the notation telling you something true: these are positions a party occupies, and the differences live in the data, not in the structure.

**Many concepts are not elements at all.** Limit, balance, days past due, interest rate, sharing ratio, loan-to-value and risk weight are attributes. Drawn as elements they produce a diagram that is unarguably complete and that nobody will read.

**A few need a relationship rather than an element.** Value moving from a payer's bank to a payee's bank is a Flow. One payment instruction producing thousands of payments is an Association whose cardinality ArchiMate cannot express.

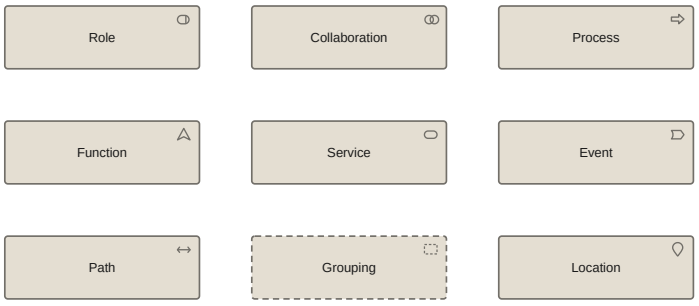



**Figure 16.1 The nine elements that carry no layer.** Role, Collaboration, Process, Function, Service, Event, Path, Grouping and Location are drawn in one neutral colour, so a Process looks identical whether it is business or application behaviour. Colour no longer tells you the layer — context does: what the element is assigned from, what it realises, what it accesses. Uncomfortable for a fortnight, then useful, because a bank is full of behaviour that sits across the seam.

> **THE ONE RULE THAT PREVENTS MOST OF THE DAMAGE**
>
> ArchiMate models **types**, not instances. There is no element for account 62004471920, no element for Mrs Dlamini, no cardinality, no effective dating, no quantity. If you need to say "a facility has one or more accounts, each with a dated limit", you need a data model, and you should build one — this book's chapter on the conceptual data model exists for that. The ArchiMate model answers a different question: what kinds of thing exist, what behaviour acts on them, and which system is responsible. An architect who tries to make ArchiMate do entity-relationship work produces something that fails as both.

One more thing before the decisions. The layer-agnostic elements are drawn in a single neutral colour, so a Process looks the same whether it is business or application behaviour. Colour no longer tells you the layer; context does — what the element is assigned from, what it realises, what it accesses. That is uncomfortable for a fortnight and then it is useful, because banking is full of behaviour that sits across the seam. A credit decision is partly a scorecard execution and partly a human sanctioner exercising a mandate, and a notation that forces you to pick one is lying for you.

## Five decisions that settle most of the model

Nearly every argument in a banking model reduces to one of five questions. Learn each test and you will answer the rest by analogy.

### Is it a product or a business object?

A **Product** is what the bank sells: a definition, priced and bundled, with an agreement attached. A **Business Object** is a thing the bank holds information about and acts on. The test: can a customer be sold it? A Gold Transactional Account is a product. The account that results is a business object. So are the agreement, the mandate, the entry, the statement and the credit application.

The trap runs the other way. Bankers habitually say "product" when they mean the account ("the product went into arrears"), and an architect obligingly draws a Product with a lifecycle. A Product has no lifecycle. It has versions, and versions are data. If you catch yourself wanting to say "this product was closed on 3 March", you are talking about an account.

### Is it a service or a process?

A **Service** is behaviour visible from outside, with a consumer who would notice its absence. A **Process** is the sequence inside that produces it. The test has two parts: would someone outside the boundary ask for this by name, and could you reorganise it completely without them caring? If both, it is a service, and the sequence behind it is a process.

Banking gets this wrong in a specific way. Departments are named after functions — Payments, Credit, Collections, Treasury — and those names arrive in the model as processes. They are not processes. "Collections" has no sequence; it is behaviour with no fixed order, which is exactly what a **Function** is for. "Recover an arrears mortgage to section 129 notice" does have a sequence, and it is a process. Draw the function for the operating model, the process for the flow of work, and connect them with Realization.

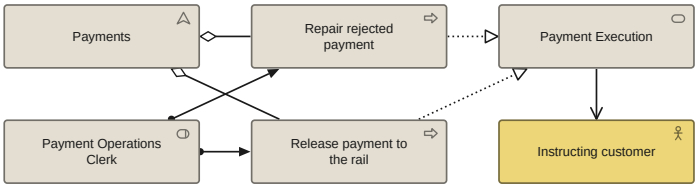



**Figure 16.2 Service or process.** Payment Execution is what the instructing customer consumes and would notice the absence of; the two processes are how the bank produces it, and either could be reorganised tomorrow without telling anyone. "Payments" is the department's name, so it is a Function — behaviour with no fixed sequence — aggregating the processes. The clerk's Role is assigned to the processes, never to the service.

### Is a channel an interface or an actor?

"Channel" conflates three different things, and the model has to separate them: the **party** (the customer, a broker, a merchant, a third-party payment provider), the **access point** (the app, a branch counter, an automated teller machine, USSD, a host-to-host link, an application programming interface), and the **entitlement** (what may be instructed through it, up to what value, by whom).

The party is a Business Actor in a Role. The access point is a Business Interface, with whatever Application Component sits behind it. The test for interface versus actor is whether the thing can decide. A merchant decides whether to accept a card; a terminal cannot decide anything.

Why it matters more here than elsewhere: the real architectural question about a banking channel is an entitlement question. May it originate a payment? Above what value? With how many approvers? May it open an account without a human seeing the documents? That question can only be asked of a party in a role under a mandate. Draw "Digital Channel" as a Business Actor and the question becomes unaskable, which is roughly how a bank ends up with three payment paths and two of them bypassing the hub.

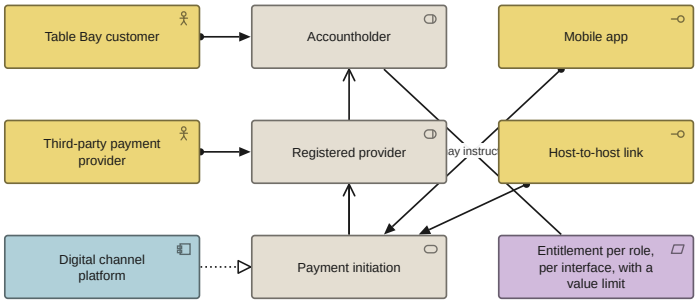



**Figure 16.3 A channel is three things.** The customer and the third-party payment provider are actors; accountholder and registered provider are the positions they hold; the app and the host-to-host link are access points. The entitlement question — may this party instruct a payment, to what value, with how many approvers — can only be asked of the role, which is why drawing "Digital Channel" as an actor makes it unaskable.

### Is it behaviour or a thing?

The hardest decision in banking, and the one the next sections return to twice. A payment is behaviour. A payment instruction is a thing. A credit decision is behaviour; the decision record is a thing. An interest accrual is behaviour; the accrual entry is a thing.

The test: can it be filed? If you can imagine retrieving it, showing it to an ombud and asking whether it was correct, it is an object. If it only happens, it is behaviour, and what you file is the record it wrote.

Most banking models fail here in the same direction — they draw the artefact and lose the behaviour, so the model has a Payment object and nothing that explains why a payment can be in flight for three days.

### Is it an element at all?

An element earns its place if behaviour acts on it, something depends on it, or someone owns it. If a concept is a property of another concept, write it down in the data model and keep it out of the picture.

Limit, balance, rate, margin, days past due, risk weight, sharing ratio, exposure at default, expected credit loss: attributes and derivations. Clearing stream: reference data, and an attribute of a payment, but a **Grouping** when you need to show that four streams have four rulebooks. Impairment stage:

an attribute, but a dated and governed one, which is why chapter 15 insists it is a fact on the exposure rather than a reporting column.

> **THE COMPLETENESS TRAP**
>
> There is a kind of banking model, produced around month four of a programme, that contains every noun in the product terms and conditions. Nine hundred elements, it validates, and it answers no question anyone asked. The test for an element is not "is it real" but "does a decision change if it is here".

## The mapping

Here is the mapping itself, in five parts. The third column is the part that matters: it is the reasoning, and the reasoning is what you will use on the concept this table does not contain.

### Parties, mandates and channels

| Banking concept | ArchiMate element | Reasoning |
|---|---|---|
| Party — a natural or juristic person | Business Actor | Identity, not behaviour. One actor, many roles, modelled once. There is no Customer element because there is no customer. |
| Customer, depositor, borrower | Role | A position a party holds against an account or agreement, for a period. "Customer" is a derived predicate over agreements, which is why it is a role and not a thing. |
| Guarantor, surety, signatory | Role | Often holds no product at all. Naming the role is what makes those parties expressible. |
| Beneficial owner | Role, on a Party-to-Party structure | FICA requires identification to natural persons. The ownership chain is Association between actors, with the percentages in the data model. |
| Trust, partnership, sole proprietor | Business Actor, with Specialization | A trust is neither an ordinary natural nor juristic person, and a sole proprietor is one person with two economic identities. Specialise the actor; do not invent a Trust element. |
| Employee, sanctioner, dealer, collections agent | Role | Never a Business Actor. People move; the process needs the position. |
| Account operating mandate | Business Object | Dated authority with signatories, combinations and limits. A flag on an account loses the dates, the limit and the revocation. |
| Collection mandate (DebiCheck, Registered Mandate) | Business Object held by an external Business Actor | The authenticated mandate lives at the payer's bank. Drawing it inside your own bank misstates who controls it. |
| Internal delegation of authority | Business Object, read by a Process | The mandate ladder below. Model it or the credit view has no authority in it. |
| Channel | Business Interface, plus a Role and a Business Service | Three things in one word. See the decision above. |
| Device, session | Not elements — data | Required by the technology Joint Standards, and they belong in the data model, with Device reserved for the physical thing when you are modelling the estate. |

### Accounts, products and agreements

| Banking concept | ArchiMate element | Reasoning |
|---|---|---|
| Product | Product | The manufactured definition: eligibility, pricing, fee schedule, interest basis, limits, regulatory classification. The one composite in the business domain. |
| Product variant | Product, with Specialization | A student version of a transactional account inherits; it does not get cloned. |
| Product version | Not an element — a Data Object in the catalogue | ArchiMate cannot date anything. One Product per rate change gives you four hundred elements and no insight. |
| Product account (arrangement) | Business Object | The customer-facing instance type: cheque account, twenty-year bond, credit card. |
| Ledger account (position) | Business Object, composed into the product account | Capital, accrued interest, fees, arrears, insurance, redraw. One product account, several balance-bearing units — which is why arrears can be aged and *in duplum* computed. |
| General-ledger account | Data Object in the general ledger | An accounting classification, not a customer thing. Model it in the application layer or the two estates join by accident. |
| Agreement | Business Object | Terms accepted, when, through which channel, at which wording version. Retained, never overwritten. |
| Facility | Business Object | A granted capacity to draw. Not an account attribute: exposure, utilisation and large-exposure reporting all operate at facility and connected-group level. |
| Balance | Not an element — a derivation | Five named derivations from one entry set. If two reports differ and neither is wrong, you have two derivations, not one balance. |

### Money, payments and the rails

| Banking concept | ArchiMate element | Reasoning |
|---|---|---|
| Payment instruction | Business Object | The request. Has its own lifecycle — received, validated, authorised, screened, accepted, held, rejected — and it is the thing a customer asks about. |
| Payment | Process, plus a Flow between actors | Behaviour spanning several banks. The Flow is what makes the inter-bank part visible; without it the model implies the bank moves money by itself. |
| Settlement | Event | Final and irrevocable, at a point in time, in central-bank money. An Event, because what matters is that it happened and when. |
| Clearing stream / payment clearing house | Grouping, with an Association to the payment process | Rulebook, cut-off, return window and governing body differ by stream. The grouping is how you show four rulebooks without drawing four payment models. |
| Entry, posting, journal | Business Object; posting is a Process | The entry is the atom and it is immutable. A correction is a new entry, which is why entries are not versioned states. |
| Transaction | Event, with Business Objects for its entries | What the customer recognises as one thing that happened. One card purchase abroad is one event and five or more entries. |
| Interest accrual | Function, writing a Data Object | No fixed sequence, runs daily, and the daily accrual record is what makes a backdated correction computable. |
| Fee | Three things: a definition in the product, an Event when levied, an entry when posted | A bank that cannot decompose a monthly charge into named components cannot evidence its disclosure. |
| Card, token | Business Object for the card; the token is a Data Object | The card is an instrument with its own lifecycle and its own cardholder. The primary account number is an attribute that may not be used as a key. |

### Credit, risk and capital

| Banking concept | ArchiMate element | Reasoning |
|---|---|---|
| Credit application, assessment, decision | Three Business Objects, written by Processes | All three retained, declines included. The chain is the defence against a reckless-lending finding. |
| Scorecard, decision engine | Application Component, plus a Service for the decision | The service is the thing four origination paths share. Draw the component only and the duplication stays invisible. |
| Mandate level | Business Object, read by the sanctioning process | A dated, amount-bounded delegation. The ladder is the model's answer to "who may approve this". |
| Collateral item | Business Object | The asset: an erf, a vehicle identification number, a share portfolio. It has a life independent of any loan. |
| Security instrument | Business Object, associated with the collateral and the exposures | The legal right over the asset. Conflating the two is why a bank cannot answer "what is our exposure to Sandton office property". |
| Valuation | Business Object, as a dated series | Since the revised standardised credit-risk approach, effective 1 July 2025, loan-to-value is a capital driver, so a dated valuation is capital data [11]. |
| Exposure, limit, utilisation | Not elements — derivations and attributes | A limit cannot be exceeded; an exposure can. Headroom is the derivation between them. |
| Impairment stage, provision | Attributes of the exposure, computed by a Function | Dated estimates with a model version attached, not current values. |
| Regulatory capital | Resource | The thing that actually constrains the balance sheet. See the capital section below. |
| Risk-weighted assets | Not an element — the output of a Function | Two functions, in fact, since the output floor. |

| Banking concept | ArchiMate element | Reasoning |
|---|---|---|
| Core banking system | Several Application Components | Never one. The argument is three sections below, and this single choice decides whether a modernisation case is arguable. |
| Payments hub | Application Component, exposing a payment-initiation Application Service | The hub earns its place in the model by being the one place a payment's status can be answered. |
| Data warehouse, reporting layer | Application Components, with Data Objects | Three layers routinely built as one. Separate them in the model and the regulatory reporting conversation becomes possible. |
| Regulatory return | Business Object with a lifecycle, realised by an Artifact | A submission for a period in a version, with preparer, approver, acknowledgement and resubmissions. Not a report. |
| Capability | Capability | An ability the bank has, with an owner and a maturity. Not an activity, and not a department. |
| Value stream | Value Stream | Stages delivering value to a named consumer, served by capabilities. The chief operating officer's view of the same facts. |
| Regulation — FICA, the NCA, the Banks Act | Driver, with the specific duty as a Requirement | A Driver cannot be realised by a system; a Requirement can. That Realization edge is the trace an examiner asks for. |
| Regulator | Stakeholder, and Business Actor where it receives something | It is both: a stakeholder with concerns, and a counterparty receiving returns on a schedule. |
| Roadmap | Work Packages, Deliverables and Plateaus | The last section. Plateaus are the only place the model says what is true at the end of each year. |

## Party and account, done properly

The costliest mistake an outsider makes in banking is building Customer as an entity. The architectural version of the same mistake is drawing Depositor as a Business Actor.

Here is what is actually true. One human being can be, at the same moment: an accountholder on a cheque account; a borrower on a home loan; a joint accountholder on a bond with a spouse; a surety for a company's overdraft; a signatory on an employer's account; a beneficial owner behind a close corporation; a cardholder on a card drawing on someone else's account; and an employee of the bank. Eight positions, one party. CoDI's deposit cover rules then ask the bank to look at that same human and say which of those positions make them a *qualifying depositor*, and whether their personal and sole-proprietor accounts are two records or one [2].

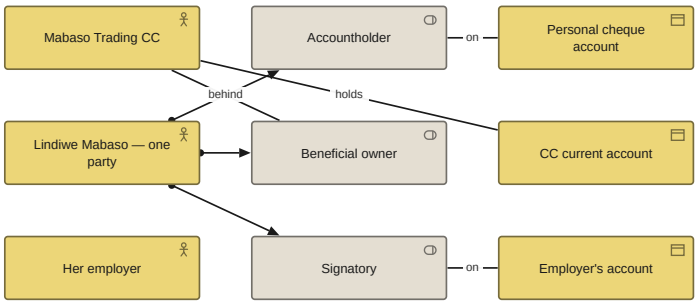



**Figure 16.4 One party, three positions.** Lindiwe Mabaso is an accountholder on her own account, the beneficial owner behind a close corporation that holds its own account, and a signatory on her employer's. Two of those three positions give her no product of her own — and the beneficial-owner position is the one FICA cares about most. Eight such positions are ordinary; the model has to make all of them sayable.

Note what that figure says that the wrong version cannot. The employer holds the account and the employee is only a signatory. The spouse appears on the bond with a share. And the beneficial owner behind the company holds no product at all, which is the case FICA cares about most.

Now the version to refuse.

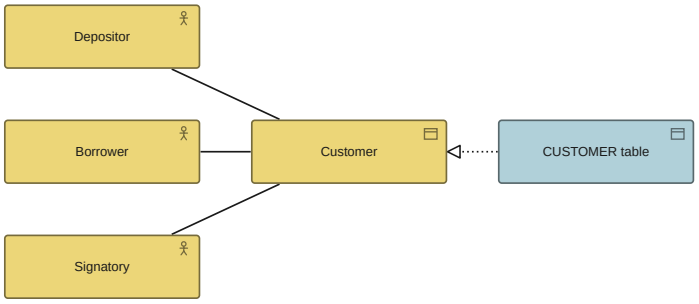



**Figure 16.5 The version to refuse.** Three actors and a Customer object. It validates, it is wrong, and it propagates downward: the Business Actor becomes an application entity, which becomes a table, which becomes the reason retail counts per natural person, business banking per enterprise group, cards per plastic and regulatory reporting per single-customer-view record — four customer counts that never reconcile.

Three things go wrong when roles are drawn as actors, and they compound.

First, you cannot say that one party holds several positions, so the model implies depositors, borrowers and signatories are different populations. They are not, and every single-view, cross-sell, financial-crime and deposit-insurance exercise depends on knowing it.

Second, the error propagates downward. A Business Actor called Customer becomes an application entity called Customer, which becomes a table, which becomes the reason retail counts per natural person, business banking counts per enterprise group, the card platform counts per plastic and regulatory reporting counts per single-customer-view record — four customer counts that never reconcile.

Third, you lose the regulatory distinction. A depositor consented to a relationship. A beneficial owner did not, is not a customer, and must still be identified and verified. A third-party payer appears in your systems because they paid your customer. Their lawful basis under POPIA, their retention and their reporting treatment are all different, and a role-based model is the only one in which that difference is even sayable.

Then the word "account", which in a bank means at least four things.

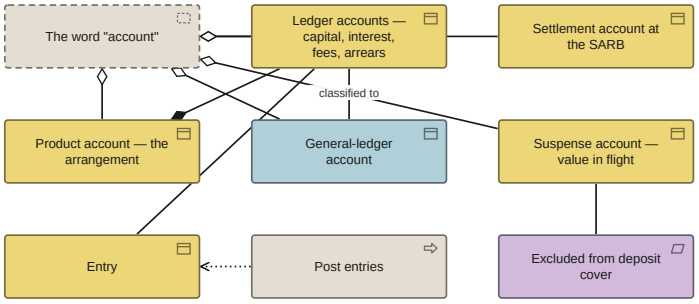



**Figure 16.6 Four things called "account".** The product account is the customer-facing arrangement; the ledger accounts inside it are the balance-bearing units entries post against, which is what lets a bank age arrears and cap accruals at the *in duplum* ceiling; the general-ledger account is an accounting classification and belongs in the application layer; and the settlement and suspense accounts are different things again — the deposit insurer's rules expressly exclude suspense balances from cover.

That figure is the single most useful thing an architect can draw in the first fortnight, because it ends an argument that otherwise runs for a year. The product account is the customer-facing arrangement. The ledger accounts inside it are the balance-bearing units that entries post against — capital, accrued interest, fees, arrears — and they are why a bank can age arrears and cap accruals at the *in duplum* ceiling. The general-ledger account is an accounting classification that belongs in the application layer. And the settlement account at the central bank is a different thing again, as is the suspense account holding value in flight, which the deposit insurer's rules expressly exclude from cover [2].

A bank with one account table and a type column spanning all four has joined its customer estate to its general ledger by accident, and nobody can safely change either.

## Product, and the catalogue underneath it

Product is the only composite element in the business domain, and banking needs it for a specific reason: the product is where the **regulatory classification** lives. Deposit-taking or credit or investment, which determines the statute. Qualifying for deposit insurance, which determines cover [2]. Instrument and maturity classification, which determines the BA 900 line. Credit subtype under the National Credit Act, which determines the fee caps and the disclosure duties.

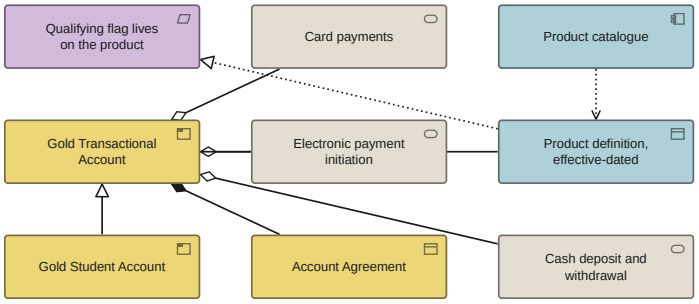



**Figure 16.7 Product is the one composite in the business domain.** It aggregates the services it promises — aggregation, because a debit card sells under the cheap bundle and the expensive one — and composes the agreement that governs them. The student variant specialises the base product rather than cloning it. And the effective-dated definition sits as a Data Object in the catalogue, because ArchiMate cannot date anything and the deposit-insurance qualifying flag has to live at definition level or it will be wrong on every older account.

Three choices in that figure are worth arguing, because each will be challenged.

**The Product aggregates the services it promises and composes the agreement that governs them.** Aggregation for the services, because the same service sells under several products — a debit card and electronic payments appear in the cheap bundle and the expensive one, and composition would force a copy. Composition for the agreement, because it exists only as part of that product.

**Qualifying status belongs on the Product, not the account.** Put it on the account and it will be wrong on every account opened before someone remembered to set it. This is the clearest example in banking of a regulatory attribute that has to live at the definition level, and the deposit insurer's product list is the reason [2].

**Versions are data, not elements.** The effective-dated definition lives as a Data Object in the product catalogue. The question people actually ask — "what were this account's terms on 14 March 2023?" — is answered by the catalogue's version store, not by a diagram. And the architectural consequence is the industry's oldest complaint: if variants are Products with inherited structure and versions are data in one place, a variant is a configuration exercise; if variants are copies and versions are code, a variant is a release.

## A payment is behaviour, not a thing

This is the section where a banking model either earns its keep or reveals that nobody checked it.

Open ten bank architecture models and nine have a box labelled Payment. It is usually an Application Component, sometimes a Business Object, and it is the reason nobody in the bank can answer "where is my salary".

Here is the shape that works.

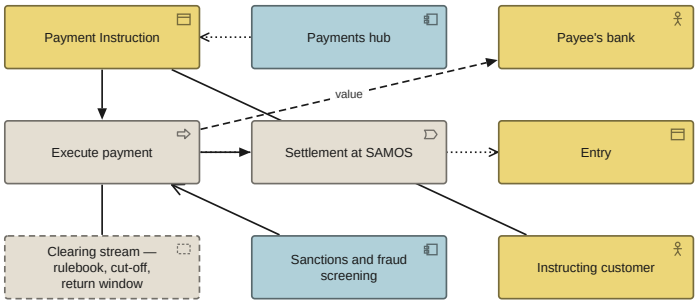



**Figure 16.8 The instruction is a thing; the payment is behaviour.** The instruction has the customer-visible lifecycle and is what an ombud asks to see. The payment is a process spanning several banks, and settlement is an Event because the only questions about it are whether and when. Screening serves the process inline rather than following it. The clearing stream is a Grouping, because four streams mean four rulebooks, four cut-offs and — since 2026 — two governing bodies.

Read it and notice five choices.

**The instruction is an object; the payment is a process.** The instruction is the thing with a customer-visible lifecycle and the thing an ombud will ask to see. The payment is behaviour that spans several banks and several days, and modelling it as an object is what makes status unanswerable.

**Settlement is an Event.** It is final and irrevocable at a point in time, in central-bank money, and the only questions about it are whether and when. In South Africa that happens at the Reserve Bank in SAMOS, which moved to ISO 20022 on 17 September 2022 — the first real-time gross settlement system in Africa to do so [3].

**The cardinalities are many-to-many in both directions, and ArchiMate cannot say so.** One instruction produces no payment when rejected, one when it succeeds, or thousands when it is a batch file or a twenty-year standing instruction. One payment can produce several settlements through a correspondent chain; one settlement discharges thousands of payments in a net clearing batch. Write the cardinalities in the view's notes and in the data model, because the picture cannot hold them.

**The clearing stream is a Grouping.** Four streams, four rulebooks, four cut-offs, four return windows, and since 2026 two different governing bodies [14]. A payment model with a single "EFT" type cannot express that a Registered Mandate collection runs in a 19:00 window the authenticated collections stream does not use [4], that real-time clearing and PayShap carry different value ceilings [13], or that a high-value instruction straight to SAMOS cannot be recalled at all.

**The screening and fraud steps are inline, not downstream.** Draw them as a Serving relationship into the payment process, from components outside it. Drawn as a subsequent process, the model implies a payment can be released and screened afterwards, which is exactly the control failure chapter 15's sanctions section is about.

> **WHAT THIS BUYS YOU IN WEEK TWO**
>
> Walk into a payments architecture review with this figure and ask one question: how many distinct paths exist from an originating system to a rail, and how many bypass the hub? In most banks the answer is six and three. The figure is what makes the question askable, because it names the hub as the place that owns the instruction lifecycle.

## The core banking system, which is three systems wearing one name

The most abused term in the industry, and the most consequential single modelling choice in this chapter.

"Core banking" has three genuinely different meanings, which have historically lived in the same box. The **ledger**: balances, entries, interest accrual, value dating, end of day. The **product engine**: the configuration that defines what a product is — rate structure, fee schedule, limits, eligibility, lifecycle. The **customer master**: the record accounts hang off. The term originally named the solution to a batch problem — before centralisation, branches ran local servers and batched to a data centre overnight, so a transaction could take a day to appear [5]. The modern vendor framing is a lean core that keeps the ledger and posting and pushes everything else into satellites [8], which is a target state and not a description of any South African estate. The batch it solved is still visible in the architecture.

So treat "core" as a question, not an answer. Establish separately: where is the authoritative balance; where is product definition, in configuration or in code; where is the authoritative party identity; and where does end of day run. Four answers, often four systems, often not the one labelled "core".

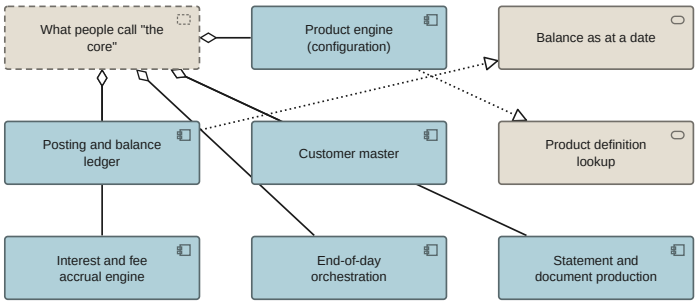



**Figure 16.9 One box on the board slide, six components here.** They have different owners, different change rates and wildly different replaceability. The two services on the right are the ones other systems actually depend on — "give me the balance as at a date" and "tell me what this product is" — and naming them is what lets you put something in front of the ledger without touching the accrual engine. This decomposition turns "replace the core" into a sequence of affordable decisions.

Why the single box costs money: every modernisation argument depends on separability. If the core is one thing, the only option is replacement — and the one well-sourced South African example runs to a decade. Nedbank's Managed Evolution started in 2016, is reported in the trade press at R11.7 billion, and delivered the core last, at the end of 2024 [6]. The counterexample is Barko Financial Services, reported live on Temenos software as a service in under six months [7]. Both numbers are true and they are not comparable: one is a full-service bank's whole estate, the other is a single-product lender's core. Quote them together or not at all.

If the core is a dozen components, there are options. Externalise the pricing. Lift collections out. Put a service in front of the balance read so four channels stop reimplementing available balance. Leave the accrual engine alone for a decade, because twenty-year amortisation schedules must reproduce to the cent and nothing else needs touching. The decomposition is not documentation. It converts an unaffordable programme into a sequence of affordable ones, and it is the highest-value hour an architect spends inside a bank.

> **DO NOT REPEAT MARKET FOLKLORE**
>
> No South African bank's core platform is publicly sourceable for Nedbank, Capitec, Absa, Investec, African Bank, Bidvest, Postbank or Sasfin. Three attributions circulate widely and all three rest on trade press rather than on the banks: FirstRand with Finxact, OM Bank with 10x, and the R11.7 billion figure above. Attribute them to the outlet, not to the bank, and do not put a vendor name on a box in a client's model because you read it somewhere.

## The credit decision, and the mandate ladder

A retail credit decision and a business credit decision are different animals, and a model that draws one process for both has already lost the argument.

Retail is a scored decision against a cut-off, with an affordability calculation whose inputs are prescribed. Business is **a paper approved by a committee under delegated authority**, where the unit of decision is an obligor group rather than an account, the outcome is a facility structure with sub-limits and

covenants rather than a loan, and review is annual and mandatory.

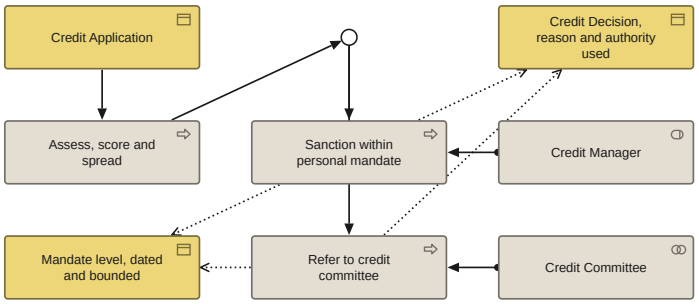



**Figure 16.10 The mandate ladder.** The or-junction routes by exposure: inside the credit manager's mandate, or up to the committee. Each path reads a dated, amount-bounded mandate-level object and writes a decision carrying the authority actually used. Three questions become answerable that are otherwise argued in workshops — who approved this, under what authority, and what happens to the queue when that person leaves. Note the Collaboration: a credit committee performs behaviour no member performs alone.

The ladder is the point. An or-junction routes the application by exposure: inside the analyst's mandate, inside the credit manager's mandate, or up to the credit committee. Each step reads a **mandate level** object — a dated, amount-bounded delegation — and writes a decision with the authority actually used. Model it this way and three questions become answerable that are otherwise asked in workshops: who approved this, under what authority, and what happens to the queue when that person leaves.

Two more things belong in the credit picture and are usually missing.

**Collateral and security are two elements, not one.** The collateral item is the asset, with a life independent of any loan. The security instrument is the legal right over it — a mortgage bond with a registered amount and a ranking, a notarial bond, a cession, a suretyship. One property secures three loans with ranked bonds; one director's suretyship secures all present and future obligations. The allocation between instrument and exposures is many-to-many with a priority and a share, and it is the thing a `collateral_value` column destroys.

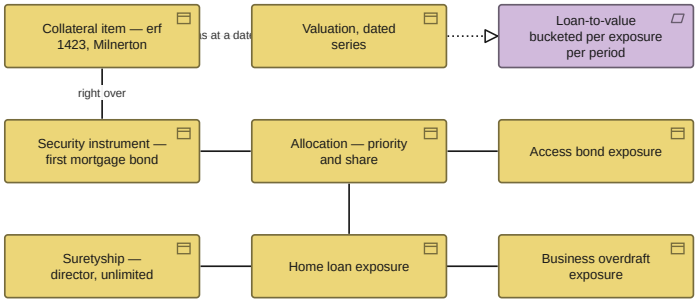



**Figure 16.11 Collateral is an asset; security is a right.** One property carries a first bond and an access-bond facility; one director's suretyship secures all present and future obligations of a business. The allocation between instrument and exposures is many-to-many with a priority and a share, which is exactly what a single `collateral_value` column destroys. Since the revised standardised credit-risk approach, the dated valuation is capital data rather than a credit-administration record.

**POPIA section 71 is in this picture, not beside it** [10]. A decline that results solely from automated processing is a regulated act, so the model needs the human-review path drawn, the decision record retained with its input vector and rule version, and a renderable reason. Chapter 15 has the law; this is where it lands in the notation.

## Capital as a constraint

Here is something banking has that most industries do not: a hard, externally audited constraint on how much business the enterprise may write. Most architects model constraints as Requirements. For capital that is wrong, and the right answer is more useful.

Regulatory capital is a **Resource**. It is a finite asset the enterprise holds, it is consumed by the business the bank writes, and the strategy layer exists precisely to model what constrains a course of action.

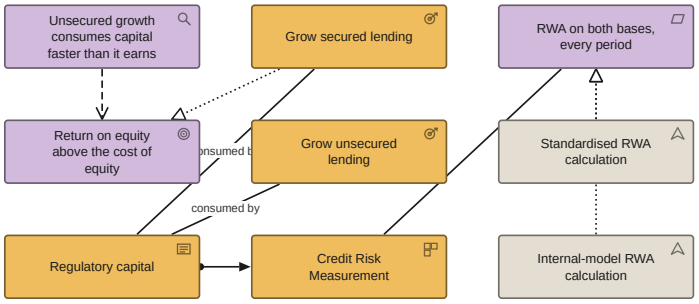



**Figure 16.12 Capital is a Resource, not a Requirement.** It is a finite asset the enterprise holds and the business consumes, which is why the strategy layer is the right place for it. An Assessment influences the Goal; two Courses of Action compete for the same Resource; the Resource is assigned to the capability that consumes it, which is the sentence a chief financial officer recognises. And the output floor's requirement is realised by two Functions rather than one.

Read the figure as an argument. An Assessment — "unsecured growth consumes capital faster than it earns it" — influences a Goal. Two Courses of Action compete for the same Resource. A Capability has a Resource assigned to it, which is the sentence a chief financial officer recognises: this capability is capital-hungry and that one is not. And the Requirement that falls out of the output floor — compute risk-weighted assets on both bases, every period — is realised by two Functions rather than one.

That last edge is worth the whole figure. It is how an architecture model carries a capital rule without pretending to be a capital model.

## A regulation, from driver to control

Chapter 15 is a list of constraints. This section is the half that usually gets skipped: the Realization edges running from a requirement down into the systems, so the model answers one question — which system satisfies which rule, and what breaks if we switch that system off.

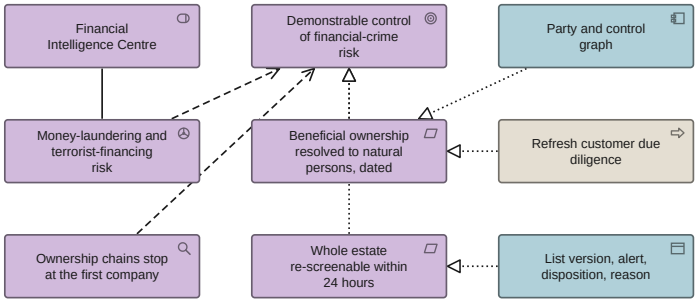



**Figure 16.13 From a statute to the thing that satisfies it.** The Driver cannot be realised by anything, which is why writing "FICA compliance" as a Requirement makes the model untestable. The two Requirements are sentences a system can satisfy or fail, and each is realised by a named component, process or data object. That is the view that answers the only question justifying a motivation layer: if the party master goes in 2029, which rules lose their realisation?

The chain is Stakeholder, Driver, Assessment, Goal, Requirement, and then the part people skip: Realization from a core element to the Requirement.

The discipline each element enforces is what matters. Writing "FICA compliance" as a Requirement is the commonest error in this layer, and it makes the model untestable — FICA is a Driver, and it cannot be realised by anything. A Goal is a direction. A Requirement is a sentence a system can satisfy or fail: "every party's beneficial ownership is resolved to natural persons and dated", or "the whole party estate is re-screenable against a new sanctions list version within twenty-four hours".

Then the realisation. An Application Component realises the requirement. A Process realises it. A Data Object realises it. Now you can answer the question that justifies the entire motivation layer: if the party master is decommissioned in 2029, which regulatory requirements lose their realisation, and what has to take them over? No other artefact in a bank answers that — not the control register, not the application inventory, not the programme plan.

> **THE MOTIVATION TEST**
>
> A motivation view with no Realization edges from core elements to requirements is decoration. A motivation view where every Requirement is realised by at least one named component, process or data object is the most valuable thing an architect can hand a chief risk officer, and the only version of it that survives an examination.

## Capabilities and value streams

Every bank has a capability model. Most of them are wallpaper. The difference is whether anything hangs off each box.

A **Capability** is an ability the enterprise possesses, with an owner. It is realised by core behaviour — functions and processes — and enabled by **Resources**. It is not an activity and not a department. "Payment Execution" is a capability; "Release an outward SWIFT payment" is a process; "Payment

Operations" is a department and belongs nowhere in the model.

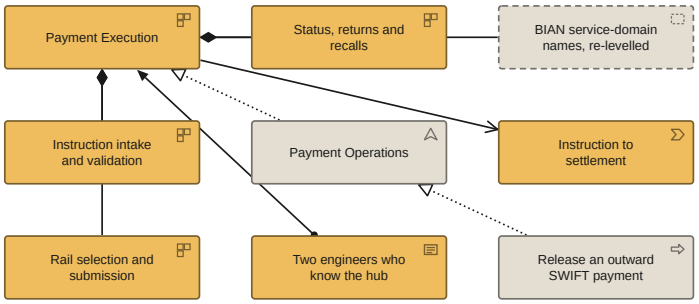



**Figure 16.14 A capability that carries something.** Three sub-capabilities, the function that realises the ability in the operating model, the process that realises the function, and the Resource that is the real constraint — two engineers who understand the hub. The grouping holds BIAN's names, re-levelled to how this bank is managed, attached as a property rather than by a Realization edge, because a BIAN service domain and an ArchiMate capability are different kinds of statement.

BIAN is the obvious source of capability names for a bank, and it is worth being precise about what it gives you. Service landscape 14.0, with release notes dated 20 February 2026, carries 322 service domains, 38 business domains, 586 business capabilities, 4,905 service operations and 242 semantic application programming interfaces, with no metamodel change from the previous release; version 14.0 is largely a payments re-cut for ISO 20022, it reduced the service-domain count rather than growing it, and certification lags the landscape [1][9].

Two warnings before you adopt it. **BIAN publishes no licence** — availability is not permission, so check before you put its content in a client deliverable [1]. And a BIAN service domain is not an ArchiMate Capability: it is a unit of service decomposition built for cross-institution comparison, while an ArchiMate Capability is an ability of *this* bank, with an owner, a maturity and an investment case. Record the mapping as a property, not a relationship. Draw Realization from your capability to a BIAN service domain and you are asserting a semantic equivalence you cannot defend.

Then value streams, which are the same facts arranged for a different audience.

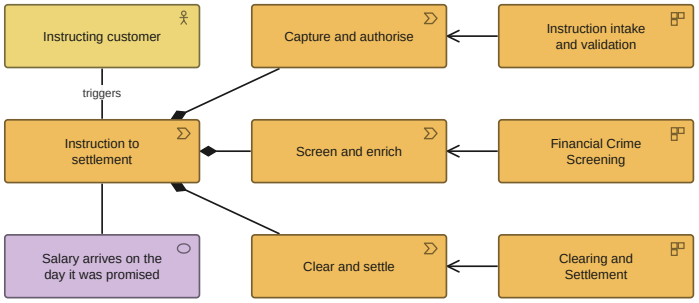



**Figure 16.15 The same facts, arranged for the chief operating officer.** Three stages from a customer trigger to a value the customer would recognise — a salary arriving on the day it was promised. Each stage is served by a capability, which is the edge that lets you argue cost to serve and turnaround on one page. "Payments" is not a value stream; it is a capability. The outcome is.

A **Value Stream** is stages from a customer trigger to the value delivered, served by capabilities, carrying a Value at the end. In a bank the useful ones are short and they cross every organisational boundary: application to disbursement, instruction to settlement, arrears to cure. The test for a value stream is whether a customer would recognise the trigger and the outcome. "Payments" is not a value stream; it is a capability. "Salary arrives in the beneficiary's account" is the outcome of one.

## The roadmap, and what is true at the end of each year

The implementation layer has three elements and they are the least abused in the notation, which is a shame, because they are where a model stops being a picture and starts being a commitment.

A **Work Package** is work with a start, an end and a budget. A **Deliverable** is what it produces. A **Plateau** is a relatively stable state of the enterprise — and the only construct in ArchiMate that lets you say what is true on 31 December.

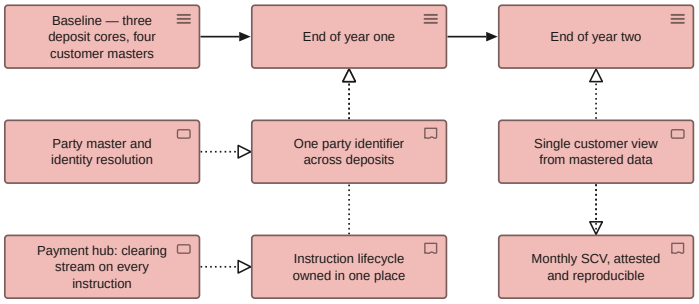



**Figure 16.16 Plateaus are the only place the model says what is true on 31 December.** Three states, two years, four work packages and the deliverables that move the bank between them. The discipline is one rule: every plateau has exit criteria you could test — one party identifier across the deposit systems, the instruction lifecycle owned in one place, the single customer view produced from mastered data. A plateau whose criteria cannot be tested is a date with ambition attached.

The discipline is one rule: every Plateau has exit criteria you could test. Not "foundation laid". Something like: one party identifier resolved across the three deposit systems, the single customer view produced monthly from mastered data rather than a query, and the payment hub carrying the clearing stream on every instruction. If a plateau's criteria cannot be tested, it is a date with ambition attached, and the programme board will discover that in month fourteen.

## Viewpoints that earn their keep in a bank

A viewpoint is a view for a named audience with a named question. These are the ones asked for repeatedly inside a bank, and the last two are missing from the standard set and should not be.

| Viewpoint | What it shows | Who asks | What it settles |
|---|---|---|---|
| Requirements realisation | A requirement traced to the component, process or data object that satisfies it | Chief risk officer, compliance, internal audit, the regulator's examiners | Which system satisfies which rule, and what loses its realisation when a system is retired |
| Capability map with maturity | Capabilities, owners, supporting applications, maturity | Executive committee, chief financial officer | Where the next two hundred million rand goes, and which capability is weakest |
| Value stream | Stages from a customer trigger to the value delivered, served by capabilities | Chief operating officer, heads of payments and credit | Cost to serve and turnaround time, argued on one page |
| Information structure and data realisation | Business object to data object to artifact, with the system holding each | Chief data officer, regulatory reporting, finance | Granularity and lineage, and whether a required number can be produced at all |
| Application co-operation | Components, services and the dependencies between them | Chief information officer, integration lead | Coupling, duplication, and which channels bypass the hub |
| Technology and deployment | Nodes, networks, facilities, what runs where | Operations, resilience, the Joint Standard 1 of 2023 evidence pack [12] | Concentration, data residency, and tested recovery |
| Implementation and migration | Plateaus, work packages, deliverables | Programme board, finance | Sequence, and what is true at the end of each year |
| **Mandate and authority view** | Every role that may approve, release, override or write off, with its limit and its system enforcement | Chief risk officer, internal audit, the first examiner through the door | Who may do what to what value, and whether the system enforces it or a policy document asserts it |
| **Payment path view** | Every route from an originating system to a rail, with the controls on each | Head of payments, operational risk | How many paths bypass the hub, and where duplicate detection is scoped |

The last two are the first questions a new chief risk officer asks in a bank, and neither is answerable from an application inventory. Build them early. A mandate and authority view in particular is the artefact that most quickly earns an architecture function the right to be consulted, because it is the one picture that nobody else in the bank is producing and several people urgently need.

## The honest limits

ArchiMate is good at what a bank is made of and bad at four things a bank runs on. Say so early, and say what you use instead, because the alternative is a reviewer discovering the gap and concluding the model is wrong rather than bounded.

**No cardinality.** The notation cannot say that one facility has one or more accounts, that one settlement discharges thousands of payments, or that a joint account has two to four holders with shares summing to one. Every one of those is load-bearing in a bank. Put them in a conceptual data model — crow's-foot notation if cardinality is the point of the picture — and reference it from the view.

**No quantity, no money, no time.** There is no element for R100 000, no way to express a 30-day stress horizon, no way to date a relationship. A limit, a rate, a cut-off and an effective date are all attributes, and a model that tries to draw them becomes unreadable before it becomes complete. This is why capital appears above as a Resource rather than as a number.

**No reporting line.** Assignment from a Business Actor to a Role is not "reports to". ArchiMate has no organisational hierarchy, deliberately, and that is correct — the org chart changes next March and the model does not describe anything that survives the change. Model Roles and the behaviour assigned

to them, and leave the hierarchy in the human resources system where it belongs.

**No swimlanes, and therefore no hand-offs.** This is the big one for banking, because in a bank the hand-off *is* the failure mode: the payment that waits for a human to release it before the cut-off, the credit paper that sits with an analyst, the arrears account that one system treats as under debt review and another does not. ArchiMate has no lane, so a process view shows the sequence and hides who was holding the work.

So this book draws level-2 process models in **BPMN 2.0**, with a pool, lanes for the roles and systems, and explicit gateways — and chapter 8 owns that decision and argues it. Use ArchiMate for the architecture and BPMN where the hand-off is the point. Two notations, each doing what it is good at, is a better answer than one notation used dishonestly.

#### FOUR TRAPS, EACH OF WHICH PRODUCES A MODEL THAT VALIDATES

**Modelling the org chart.** Business Actors named after departments, composed into a hierarchy, with processes named after the departments that run them. Quick, recognisable, worthless, and it teaches the organisation that architecture is documentation.

**Modelling the system menu.** Application functions named after screens: Account Enquiry, Payment Capture, Statement Run. That is what a model built from a user manual looks like. The test for an application service is whether something outside the component depends on it. Nobody outside depends on Account Enquiry; four systems depend on "give me the balance as at a date".

**Drawing the account as a process.** A chain of boxes labelled Open, Transact, Dormant, Arrears, Close, with arrows. Those are **events** in the life of a Business Object, each triggering different processes, in no fixed order, often concurrently and sometimes retroactively. Drawing them as a sequence encodes the false assumption that arrears follows dormancy, and that assumption shows up later as a system that cannot process a backdated reversal on a closed account.

**One box for the core.** Covered above. It is the expensive one.

#### THE CHAPTER IN TEN LINES

- ArchiMate 4 gives you forty-two elements and banking gives you a thousand words, so the mapping collapses many concepts onto one element, demotes many more to attributes, and turns a few into relationships.
- The model holds types, not instances: no cardinality, no quantity, no effective dating — so build a conceptual data model alongside it instead of asking the notation to be one.
- Five decisions settle most arguments: product or object, service or process, interface or actor, behaviour or thing, and element or attribute.
- Party is a Business Actor and depositor, borrower, signatory and beneficial owner are Roles; drawing them as actors creates the Customer entity and the four irreconcilable customer counts that follow.
- "Account" means at least four things — the arrangement, the ledger accounts inside it, the general-ledger classification and the settlement or suspense account — and one table with a type column joins the customer estate to the general ledger by accident.
- A payment instruction is an object with a lifecycle, a payment is a process spanning banks, and settlement is an event; the clearing stream is a Grouping because four streams mean four rulebooks.
- The core banking system is never one component, and the hour spent decomposing it into ledger, product engine, customer master and the rest is what turns an unaffordable replacement into a sequence of affordable moves.
- Model the mandate ladder: an or-junction by exposure, a dated amount-bounded mandate level, and a decision that records the authority actually used.
- Capital is a Resource rather than a Requirement, consumed by courses of action and assigned to the capabilities that eat it.
- A regulation is a Driver, the specific duty is a Requirement, and a motivation view with no Realization edges from components and processes to requirements is decoration.
- ArchiMate has no cardinality, no quantity, no reporting line and no swimlane — so level-2 process models go in BPMN, cardinality goes in a crow's-foot data model, and the hierarchy stays in the human resources system.

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- **10.** POPIA section 71 — automated decision making, including profiling of creditworthiness https://popia.co.za/section-71-automated-decision-making/
- **11.** Prudential Authority Annual Report 2025/26 — Basel III post-crisis reforms effective 1 July 2025, including the revised standardised credit-risk approach that makes loan-to-value a capital driver https://www.resbank.co.za/content/dam/sarb/publications/reports/pa-annual-reports/2026/chapters/governance-and-strategy/policy-regulate-25-26.pdf
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- **14.** South African Reserve Bank, payment system management body transition — functions split between the SARB (11 August 2026) and PayInc (2 September 2026) https://www.resbank.co.za/en/home/what-we-do/payments-and-settlements/psmb

CHAPTER 17

# Table Bay Bank, end to end

*One invented South African bank, modelled from motivation through to technology, a view at a time — with the argument for every choice shown rather than asserted, and the two drafts that were wrong kept in, because those are the ones that teach.*

## A warning before we start

**Table Bay Bank does not exist.** It is invented for this book, and so is everyone in it. Nothing here is a case study of any South African bank, and no system name below belongs to a real product. Where the chapter uses a real fact — a commencement date, a regulatory requirement, a published programme figure — it carries a source, and everything else is construction.

What it is, in one paragraph. A mid-tier full-service bank, roughly 2.4 million retail customers, founded in 1974, listed. A business bank serving small and medium enterprises. A small corporate and transactional banking arm that mostly does payments and trade for mid-market corporates. A treasury that funds the balance sheet and runs a modest trading book. Two data centres, in Montague Gardens and Century City. One acquisition it has never finished digesting: a vehicle and asset finance book bought in 2014, which came with its own servicing system and its own customer records.

The people you will meet: **Andile Mthembu**, chief executive. **Yvonne Lategan**, chief financial officer. **Rajesh Naidoo**, chief information officer. **Marieke de Villiers**, chief risk officer. **Zanele Mahlangu**, head of payments. **Dieter Grobler**, chief data officer, appointed fourteen months ago. **Nokuthula Dube**, head of credit. **Aisha Essop**, lead enterprise architect — whose job this chapter is.

The systems: **Helm**, the deposit and transactional core, in production since 1991. **Tiller**, the loan and vehicle-finance servicing platform that arrived with the 2014 acquisition. **Harbour**, the customer relationship platform, which everyone calls the party master and which is not one. **Quay**, the payments hub, five years old. **Signal**, the app and internet banking stack. **Tern**, card management and authorisation. **Arbiter**, the credit decision engine. **Anchor**, collections. **Meridian**, treasury. **Lookout**, financial-crime monitoring and sanctions screening. **Keel**, the general ledger. **Chartroom**, regulatory reporting. **Basin**, the data warehouse.

Twelve views follow, each building on the last. Two of them are drafts that were wrong, kept in, because the second draft is only interesting if you can see what it replaced.

## The motivation view: why this bank is changing

Aisha's first deliverable was not a landscape diagram. It was six elements and eight edges, built in two afternoons from documents that already existed: the board pack, the risk appetite statement, the last two supervisory letters and the internal audit report on regulatory reporting.

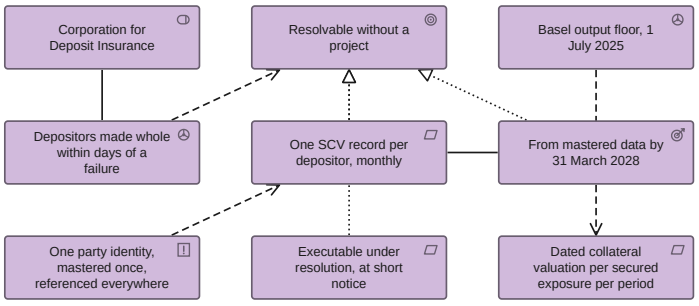



**Figure 17.1 Table Bay's motivation view, in nine elements and eight edges.** The deposit insurer is a Stakeholder and the regime is a Driver — "CoDI compliance" as a Requirement would have made the model untestable. The two Requirements are different in kind: one is a data structure, the other an operational property, and a bank can pass the first with a monthly report while failing the second completely. The Outcome carries a number and a date because the chief financial officer insisted.

Four choices in that figure are worth defending, because each was challenged in the review.

**The deposit insurer is a Stakeholder and the regime is a Driver.** Someone wanted "CoDI compliance" as a Requirement. That is the single commonest error in this layer and it makes the model untestable — you cannot write a test for compliance with an institution. The Driver is the concern: depositors must be made whole within days if the bank fails. The Requirements are sentences a system can satisfy or fail.

**There are two Requirements, not one, and they are different in kind.** The first is a data requirement: one single customer view record per qualifying depositor, computed monthly, with the joint-account sharing ratio recorded rather than defaulted [1]. The second is an operational one: the calculation must be executable at short notice under resolution conditions, which is a different test and a different system property. A bank can pass the first by building a monthly report and fail the second completely. Splitting them is what made the second visible.

**The Outcome carries a number and a date.** "Single customer view produced from mastered data by 31 March 2028, reconciling to the general ledger within R50 000." An outcome you cannot state as a number with a date is a goal in the wrong box, and Yvonne Lategan was the person who insisted on this — finance people are reliably good at this part.

**The second Driver is not a regulation.** Basel's output floor took effect in South Africa on 1 July 2025 and obliges a bank on internal models to compute risk-weighted assets twice, taking the higher of the internal figure and the floored standardised amount [8]. Table Bay runs internal models on mortgages and the standardised approach everywhere else, so the floor lands on it as a data problem rather than a modelling one: loan-to-value,

exposure class and credit conversion factors are now needed on every exposure, including exposures that were previously only ever modelled. Aisha put it in the motivation view as a Driver with one Requirement — a dated collateral valuation per secured exposure per reporting period — because that Requirement has a system consequence and "implement Basel" does not.

**The Principle is the one that will be quoted back at you.** "One party identity, mastered once, referenced everywhere." Principles are cheap to write and expensive to honour, and the test of one is whether it has ever caused a project to be told no. This one has: it is why the digital onboarding programme was not allowed to create its own customer table in month three, which is the only reason the principle is still in the model.

> **WHAT THIS BOUGHT IN WEEK THREE**
>
> Marieke de Villiers had been asking for eighteen months who owned the deposit-cover calculation. The motivation view did not answer that, but it made the question precise enough to be answerable: the Requirement had no realising component. Six weeks later it had an owner, which is what a motivation view is actually for.

## The strategy view, first draft — and why it was wrong

Aisha's first strategy view had five value streams across the top: Payments, Lending, Deposits, Cards and Treasury. It was approved in a working session and nobody objected.

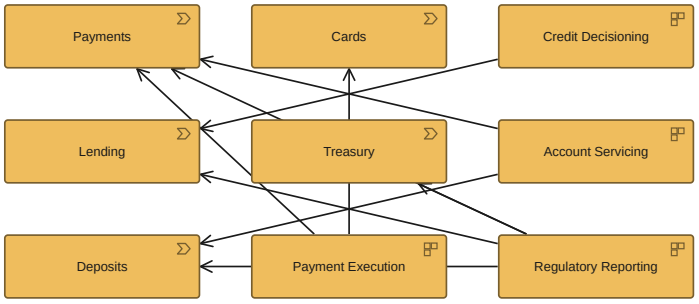



**Figure 17.2 The first strategy view, and why it was wrong.** Five "value streams" named after departments, with every capability serving every one of them. When a view's edges form a complete bipartite graph, the two sides are the same thing: these are capability groupings, not value streams. Neither has a customer trigger or a customer-recognisable outcome, so turnaround, cost to serve and failure rate cannot be measured — and the chief operating officer stops attending.

It is wrong, and the reason is worth more than the figure.

Those five are **capability groupings named after departments**. A value stream has a customer trigger and a customer-recognisable outcome, and "Payments" has neither. You cannot say what starts it or what ends it, which means you cannot measure its turnaround, cost to serve or failure rate. Draw it this way and the next question — where does this value stream break — has no answer, so the chief operating officer stops attending.

The giveaway is the Serving edges. Every capability served every value stream, because the value streams were the capability groups under another name. When a view's edges form a complete bipartite graph, the two sides are the same thing.

## The strategy view, second draft

The second draft has four value streams, each with a trigger, an outcome and a named consumer, and capabilities serving some of them rather than all of them.

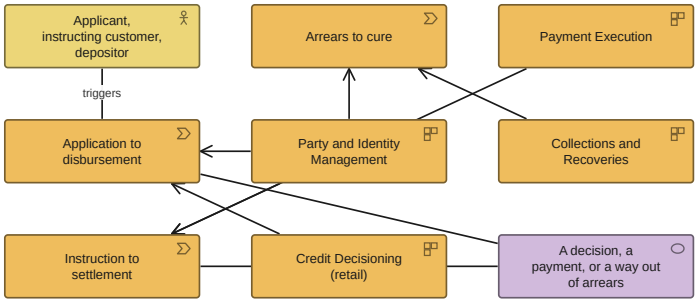



**Figure 17.3 The second draft.** Three value streams, each with a trigger and an outcome a customer would recognise, and capabilities serving some rather than all. Party and Identity Management serves every stream, which is the first time anyone at Table Bay saw that it is the constraint on all of them — and it is the capability nobody on the executive committee had an opinion about.

The maturity judgement is the part that makes it useful, and it is a judgement — Aisha's, argued with each capability owner, not a benchmark. Table Bay scored each capability one to five on two axes an executive committee will actually discuss: does it work today, and could it absorb twice the volume.

| Capability | Owner | Works today | Would scale | Judgement |
|---|---|---|---|---|
| Party and Identity Management | Dieter Grobler | 2 | 1 | The constraint on everything else. Four masters, no owner until last year |
| Payment Execution | Zanele Mahlangu | 4 | 3 | Works well; three channels still bypass the hub |
| Credit Decisioning (retail) | Nokuthula Dube | 4 | 4 | The strongest capability in the bank |
| Credit Decisioning (business) | Nokuthula Dube | 2 | 2 | Credit papers in documents, covenants in spreadsheets |
| Collections and Recoveries | Nokuthula Dube | 3 | 2 | Treats days past due as the only signal |
| Complaints and Redress | Sipho Khumalo | 2 | 2 | Categories differ by channel, so the prescribed return is built by hand [11] |
| Regulatory Reporting | Yvonne Lategan | 2 | 1 | Returns assembled from point extracts that do not tie to the ledger |
| Financial Crime Screening | Farhana Moosa | 3 | 2 | Screens at onboarding; cannot re-screen the estate in a day |
| Product Management | Sipho Khumalo | 2 | 2 | Four product catalogues; a fee change is a release |

Two things to notice about that table, because they are the general lesson rather than Table Bay's particulars.

The weakest capability is **not** the one the executive committee was arguing about. Everyone had an opinion about the core. Nobody had an opinion about party and identity management, and it is the constraint on the deposit-cover requirement, the financial-crime requirement and every customer-experience ambition in the strategy. A capability map's job is to move the argument to where the constraint actually is.

And the two columns disagree in a useful way. Regulatory reporting *works* — the returns go in on time, because eleven people make them go in on time. It does not scale, and it cannot survive the reporting specification changing by directive rather than by gazette [2]. A single maturity score would have hidden that; two columns made it the obvious first conversation.

## The business layer

One value stream, expanded. Roles assigned to processes, processes realising services, services consumed by a party in a role.

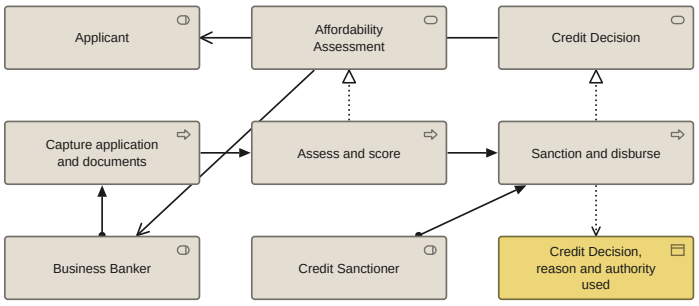



**Figure 17.4 One value stream, expanded.** Roles assigned to processes, processes realising services, the service consumed by a party in a role. Affordability Assessment is drawn as a Service because three origination paths consume it — and drawing it that way is what revealed that one of the three has its own copy of the calculation. The sanctioner is a Role, which is what survives her being on leave.

Three decisions in that view, each of which was argued.

**The sanctioner is a Role, not Nokuthula Dube.** Obvious when stated, routinely got wrong, and the test is what happens when she is on leave. A Role survives the person; a Business Actor named after an incumbent does not, and a model full of incumbents is out of date within a quarter.

**Affordability Assessment is a Service, not a step.** It is consumed by three originating paths — the branch, the app and the business banker's workstation — and the moment you draw it as a service you can see that two of them call it and one has its own copy of the calculation. That is a finding, and it came out of a naming decision rather than a system review.

**The decision record is accessed, not produced.** The Access relationship from the sanctioning process to the Credit Decision object is dotted and easy to overlook, and it is where chapter 15's National Credit Act section lands. Nothing in that chain may be overwritten, so the model's job is to show that every step writes rather than updates.

What this view cannot show is the hand-off, and in Table Bay's business-lending process the hand-off is where the time goes. An application waits an average of nine working days with a credit analyst, most of which is spent gathering documents the bank already holds. ArchiMate has no swimlane, so the view above shows the sequence and hides who was holding the work. For that conversation Aisha drew a second picture in BPMN, with a lane per role, and took both into the same meeting: the ArchiMate view to argue about which service the three origination paths share, the BPMN view to argue about the nine days. Two notations, each doing what it is good at.

One more observation from this view that an architect would miss on a first pass. The sanctioning process reads a mandate level, and Table Bay's mandate levels live in a policy document, not in Arbiter. So the system cannot enforce them: a credit manager's authority is checked by the credit manager. That finding came out of drawing an Access edge to an object that turned out not to exist in any system, which is a surprisingly common way for a model to earn its keep.

## Product and account structure

This is the view that ends the longest-running argument in the bank, and it takes four elements.

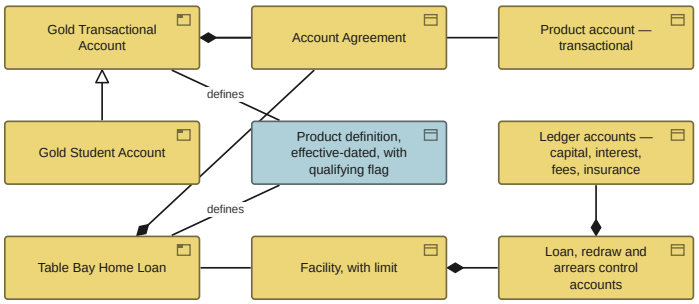



**Figure 17.5 One agreement, one facility, three accounts, four ledger accounts.** The home loan is why this view exists: an architect arriving from outside draws one box labelled Home Loan Account and loses the ability to age arrears by bucket, cap charges at the principal outstanding at default, or decompose a monthly charge for disclosure. Note where the deposit-insurance qualifying flag sits — on the Product, realised by the effective-dated definition, not on the account.

Table Bay sells a Gold Transactional Account, a Gold Student Account that specialises it, and a twenty-year home loan. The products aggregate the services they promise and compose the agreements that govern them. Under each sold account sit the ledger accounts that entries actually post against.

The home loan is the instructive one, and it is why this view exists. One agreement, one facility, and **three** accounts: the capital loan account, the redraw account and the arrears control account. Four ledger accounts under the loan account alone — capital, accrued interest, fees and insurance. An architect arriving from outside draws one box labelled Home Loan Account and loses the ability to express three things the bank is legally required to do: age arrears by bucket, cap the total of interest and charges at the principal outstanding at default, and decompose a monthly charge into named components for disclosure.

The second instructive thing is where the deposit-insurance qualifying flag sits. On the Product, realised by the effective-dated definition in the catalogue — not on the account. Helm has an account-level flag, populated by a migration script in 2024, and it is wrong on about 40,000 accounts opened before the script ran. That is an invented number for an entirely real failure mode, and it is the single most common way the deposit-cover calculation goes wrong [1].

The third thing this view settles is the joint account. Table Bay has about 180,000 of them and has never recorded a sharing ratio, so the deposit insurer's default applies — equal shares — by accident rather than by decision [1]. Drawn as a Business Object with holders in a Role, the gap is obvious: there is a role on the account and no attribute on the role. Drawn as one account box with a customer pointer, it is invisible, which is why it survived three previous attempts at a customer-data programme.

## The application landscape, first draft

Aisha's first application view had eleven boxes and one of them was labelled Core Banking.

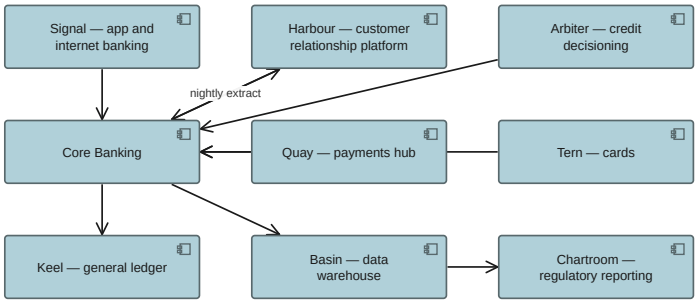



**Figure 17.6 The first application view. Eleven boxes, every arrow true, and the most expensive diagram in this chapter.** With "Core Banking" as one component there are exactly two strategies — keep it or replace it — and replacement is out of reach, so the conversation ends. The second thing it hides: nothing in this view masters a party. Harbour is drawn as the customer system, so everyone assumed it was the master. It reads from the core nightly.

Rajesh Naidoo liked it. It fitted on a slide, it was recognisable, and every arrow was true. It is also the most expensive diagram in this chapter, because of what it makes unsayable.

With Core Banking as one component, there are exactly two strategies: keep it or replace it. Replacement is a multi-year programme — the one well-sourced South African comparison is Nedbank's Managed Evolution, reported in the trade press at R11.7 billion from 2016, with the core banking platform delivered last at the end of 2024 [3]. Table Bay does not have R11.7 billion. So the board hears "keep it", the architecture function has nothing further to say, and the conversation ends.

The other thing the single box hides is that there is no component anywhere in that view that masters a party. Harbour is drawn as the customer system, so everyone assumes it is the master. It is not: it is a relationship platform that reads from Helm nightly and holds its own records for prospects. Nobody

noticed, because the view had a box labelled Customer Relationship Platform and that was close enough to a master to stop the question.

## The application landscape, second draft

The second draft decomposes Helm, names what masters what, and makes the duplicate-mastering problem impossible to look away from.

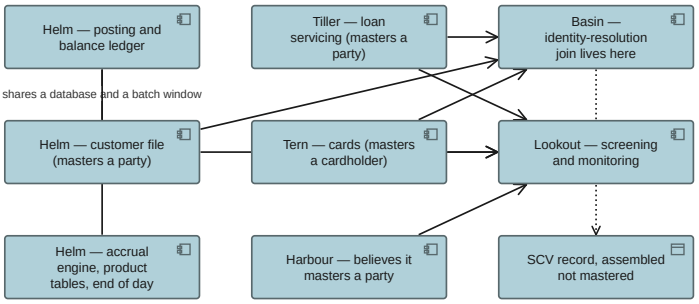



**Figure 17.7 The second draft, with the core decomposed and the mastering named.** Four components master a party: the core's customer file, Tiller's own records from the 2014 acquisition, Tern's cardholder file keyed per card, and Harbour, which believes it is the master. Four stores, four keys, four spellings of the same surname. The deposit-cover record is assembled by a join in the warehouse — and the four arrows into Lookout are the picture of a screening capability that cannot re-screen an estate in a day.

Read it for the four components that master a party. Helm's customer file, for retail deposit and transactional customers. Tiller's own customer records, which arrived with the 2014 acquisition and were never merged. Tern's cardholder file, keyed per card. And Harbour, which holds prospects and relationship data and believes it is the master. Four stores, four keys, four spellings of the same surname, four views of whether a customer is in arrears.

What that costs, in the bank's own terms:

- **The deposit-cover calculation cannot be produced from mastered data.** It is assembled in Basin by joining Helm, Tiller and Tern on identity number where present and on name and date of birth where not. That join has an error rate nobody has measured, and the output is attested monthly [1].
- **Financial-crime screening runs four times** against four name stores, with four alert queues and no single view of a party's risk rating.
- **The retail customer count, the business banking customer count and the regulatory customer-record count have never reconciled**, and each has a defender who is correct within their own definition.

Now the decomposition. Helm is not one component; it is six that happen to share a database and a batch window: the posting and balance ledger, the interest and fee accrual engine, the product tables, the customer file, end-of-day orchestration and statement production. Naming them changes the strategy from two options to six, and three of the six are affordable this year:

1. Put a balance service in front of the ledger, so Signal, Tern and the branch stop reimplementing available balance three different ways.
2. Move product definition out of Helm's tables into a catalogue that can be changed without a release, which is the only way the National Credit Act's repo-linked fee maxima can be repriced on the day the rate changes.
3. Stand up a party master as a new component, migrate identity resolution into it, and leave every source system's records where they are for now, pointing at a resolved identifier.

Leave the accrual engine alone for a decade. Twenty-year amortisation schedules have to reproduce to the cent, nothing else depends on changing them, and the risk of touching it is out of proportion to any benefit.

Two further things fall out of the second draft that the first could not express. Tiller's presence as a separate servicing platform with its own customer records is not a legacy accident to be tidied away: the 2014 acquisition came through a transfer of business with a gazetted effective date, and the book it brought has agreements whose terms are governed by the wording in force when they were signed. That is why "migrate Tiller into Helm" is not a data exercise. And Lookout appears in the second draft with four inbound arrows rather than one, which is the picture of a screening capability that cannot meet a twenty-four-hour re-screening obligation — not because the engine is slow, but because there are four name stores to re-screen and no single party estate to point it at.

> **THE PATTERN, NOT THE PARTICULARS**
>
> The first draft was not sloppy. It was a faithful picture at the wrong granularity, and the wrong granularity made the only affordable strategy invisible. That is the general case: in a bank, the decomposition of the thing everybody calls "the core" is where the architecture either creates options or removes them.

## The payments slice

Zanele Mahlangu's question was simple and nobody could answer it: how many ways can money leave this bank?

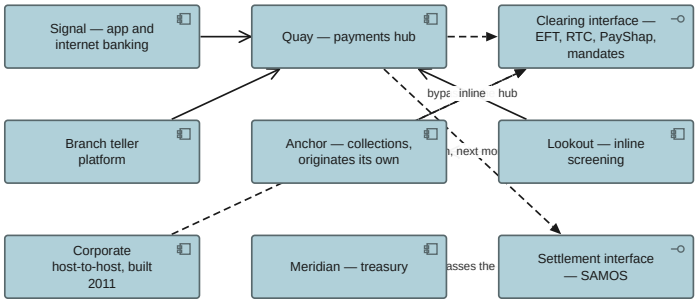



**Figure 17.8 Six ways money leaves the bank, three of which bypass the hub.** Two of the bypasses are defensible — card settlement through the scheme and high-value instructions straight to settlement. One is not: the collections system originates its own authenticated collections because a project in 2019 was three weeks from its date. Duplicate detection, inline screening and the clearing-stream attribute all cover only the hub paths.

Six paths, three of which bypass Quay. Signal goes through the hub. The branch teller platform goes through the hub. The corporate host-to-host file goes straight to the electronic funds transfer interface, because it was built in 2011 and predates Quay. Tern settles card transactions directly through the scheme interface, which is correct and should stay that way. Meridian sends high-value instructions straight to the settlement interface, which is also defensible. And Anchor, the collections system, originates its own authenticated collections, which is not defensible at all and exists because a project in 2019 was three weeks from its date.

The architectural findings fall straight out of the picture:

- **Duplicate detection is scoped per path.** A payment submitted twice through two paths is not caught. Quay's duplicate check is the highest-value control in the whole component and it covers three of six paths.
- **Sanctions screening is inline on the hub paths and batch on one of the others.** Chapter 15's twenty-four-hour re-screening obligation cannot be met for a path whose instructions the screening engine sees the following morning.
- **The clearing stream is not carried on the instruction** in two of the six paths, so the return window and the dispute process have to be inferred from the file format. Four streams mean four rulebooks and four cut-offs, and since 2026 two different governing bodies — the Reserve Bank for some clearing houses from 11 August 2026 and PayInc for the retail streams from 2 September 2026 [4].
- **Status is derived from file acknowledgements rather than modelled**, which is why "where is my payment" takes a human twenty minutes and three systems.

That last point is the one that got the work funded, and not because of the regulation. It was because the contact centre's average handling time for a payment query was the single worst number in the customer-experience pack, and nobody had connected it to an architectural cause.

There is a second dimension this view deliberately does not carry, and it is worth saying why. Payments in South Africa run on cut-offs, and a cut-off is a time of day, which ArchiMate cannot express. The electronic funds transfer credit file has a submission deadline and deferred clearing; real-time clearing and PayShap are immediate but capped; the authenticated collections stream runs in an early window and Registered Mandates in a late one at 19:00 [9]; a high-value instruction straight to settlement is final and irrevocable the moment it settles. Six paths, four clocks, and a model that cannot hold a clock.

So the cut-off calendar is a separate artefact — a one-page table, maintained by Zanele's team, that the model references and does not try to contain. Table Bay's payments incident log says the same thing in a different way: four of the last seven payment incidents were cut-off breaches, and none of them was a system failure. They were a human releasing a batch too late, which is a process control and a hand-off, and that belongs in the BPMN model with a timer event on it.

## The data view

The data realisation ladder: a business object, the data objects that realise it, the artefact that leaves the building, and the systems that hold each.

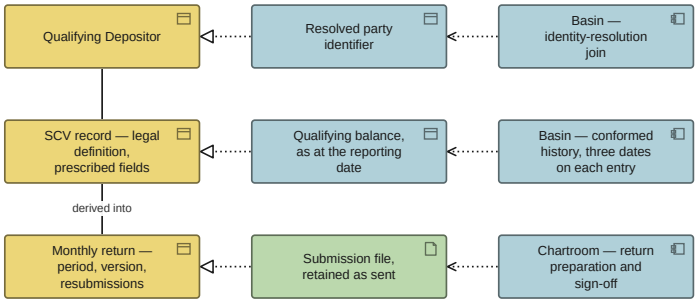



**Figure 17.9 The realisation ladder, read upward from the artefact.** A business object, the data objects that realise it, the file that leaves the building, and the system holding each. The single customer view record is a Business Object with a legal definition and a pass/fail test, not a report; the return is an object with a lifecycle including resubmissions, not a file. Basin appears twice — once as a warehouse and once as the identity-resolution system of record it should not be.

The ladder is worth drawing for exactly one reason — it is the only view in the set that shows **lineage as structure** rather than as a spreadsheet. Read it upward from the artefact and you can answer the question an examiner asks: where did this number come from, through what transformations, and can you do it again.

Three things in that view are deliberate.

**The single customer view record is a Business Object, not a report.** It has a legal definition, a prescribed field set and a pass/fail test [1]. Drawing it as a report in Basin is how a bank ends up unable to produce it under resolution conditions, which is its entire purpose.

**The regulatory return is a Business Object with a lifecycle, realised by an Artifact.** A submission, for a period, in a version, with preparer, reviewer, approver, acknowledgement and — critically — resubmissions. CoDI builds resubmission into the process explicitly [5]. A bank that models the return as a file has nowhere to put the restatement.

**The ladder has to be readable as at a date, and that is the hard part.** Three dates matter on every entry in a bank: when the thing happened in the world, the value date the interest calculation uses, and the system time the record was written. A backdated correction changes the first two and not the third, which is how a bank can restate last month's figure and still prove what it believed when it submitted. Table Bay's entries carry all three, which is the one genuinely good thing about Helm. Basin's conformed history carries two, because the migration into the warehouse in 2019 dropped the system timestamp as redundant. That single decision is why the bank cannot reproduce the impairment charge it published in March 2023, and nobody has owned up to it.

**Basin appears twice, and that is the finding.** It holds the conformed history that the return is derived from, and it also holds the identity-resolution join that should be in a party master. One of those is a warehouse's job. The other is a system of record wearing a warehouse's clothes, and every regulatory-reporting problem in the bank traces back to it.

## The technology view

Shorter than people expect, and it exists for a specific reason: two of the Joint Standard 1 of 2023 obligations are technology-view facts and nothing else can evidence them [6].

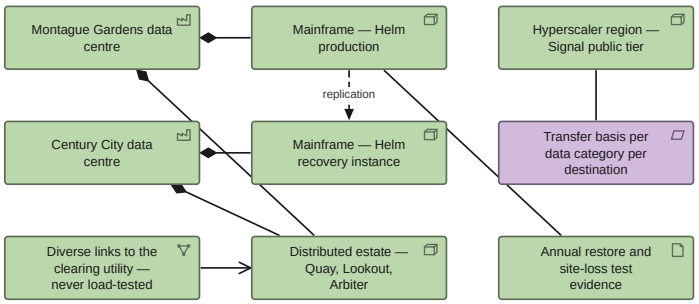



**Figure 17.10 Two facilities, geographically separate, which is what the standard requires.** The mainframe runs the core in Montague Gardens with its recovery instance in Century City; the hub is active-active across both; the public tier of the digital stack sits in a hyperscaler region, which is where the cross-border transfer question becomes concrete. What the view cannot hold is whether the recovery works — but it can make the set of things that must be tested explicit and complete.

Two facilities, Montague Gardens and Century City, geographically separate, which is what the standard requires. A mainframe node running Helm in Montague Gardens, with its disaster recovery instance in Century City. The distributed estate in both. Quay deployed active-active across the two. And Signal's public-facing tier in a hyperscaler region — which is the element that makes the POPIA section 72 question concrete, because the question POPIA section 72 actually asks is not "are we in the cloud" but "which data category, to which destination, on which transfer basis" [10] — and the Prudential Authority and FSCA issued a joint communication on cloud computing and the offshoring of data in July 2025 [7].

The second reason this view exists is concentration. Draw the nodes and the networks and one thing becomes visible that no application diagram shows: every one of the six payment paths in the previous view terminates on a single network link to the clearing utility, in one facility, with a diverse path that has never been exercised under load. That is not an application finding and no application view can produce it.

The honest limitation of this view: it says nothing about whether the recovery works. Recovery time and recovery point objectives are attributes, the standard requires annual restore and resilience testing including a site-loss scenario, and the model cannot hold a test result. What the view can do is make the set of things that must be tested explicit and complete, which is the half an auditor checks first.

## Gaps, and the roadmap in two plateaus

Gaps are the difference between the capability map's judgement and the requirements the motivation view carries. Table Bay has fourteen. Four are severe.

| Gap | Severity | What it blocks |
|---|---|---|
| No party master; four stores master a party | Severe | Deposit cover from mastered data; financial-crime screening; one customer count |
| Three payment paths bypass the hub | Severe | Global duplicate detection; inline screening; payment status |
| Product definition in Helm's tables, not a catalogue | Severe | Repricing on the day a repo change lands; qualifying flag at definition level |
| Returns assembled from point extracts that do not tie to the ledger | Severe | Reproducibility and lineage, which is what inspection asks for |
| Covenants and credit papers in documents | High | Business credit review cycle; covenant breach ageing |
| Collections treats days past due as the only signal | Medium | Differentiated treatment; debt-review state honoured consistently |

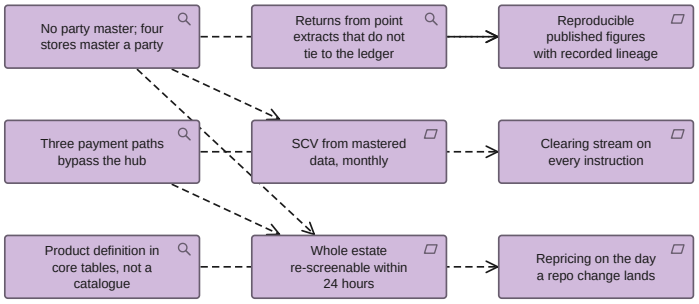



**Figure 17.11 Gaps as elements, not rows in a spreadsheet.** Three of the four severe gaps depend on the same one, which is the whole argument for sequencing and not an argument a table can make. Each gap here is drawn as an Assessment with the Requirement it blocks, so the dependency is visible and the owner is attached to something.

Drawn as elements rather than listed in a spreadsheet, a gap gains two things it otherwise lacks: an owner, and a visible dependency on the gaps that have to close first. Three of the four severe gaps above depend on the same one. That is the whole argument for sequencing, and it is not an argument you can make from a table.

Then the sequence, as two plateaus with testable exit criteria.

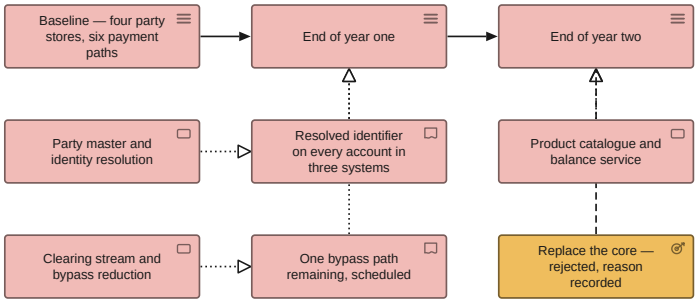



**Figure 17.12 Two plateaus with exit criteria written so they can be failed.** Nothing in year one replaces anything: the party master is additive, the source systems keep their records and gain a resolved identifier. A programme whose first year requires a decommissioning has no first-year benefit, and a programme with no first-year benefit is cancelled in month eleven. The core replacement is in the model as a rejected Course of Action, with the reason recorded.

Plateau one, end of year one. Exit criteria, written so they can be failed: a party master in production with a resolved identifier present on every account in Helm, Tiller and Tern; the deposit-cover calculation produced from that master rather than from a join in Basin, reconciling to the general ledger within a stated tolerance; the clearing stream carried on every instruction in Quay; and the three bypass paths reduced to one, with the remaining one scheduled.

Plateau two, end of year two. A product catalogue outside Helm holding definitions with effective dates, with the qualifying flag and the credit subtype as governed attributes; a balance service in front of the ledger consumed by all three channels; regulatory returns sourced from conformed history with recorded lineage; and the screening engine able to re-run against the whole resolved party estate within twenty-four hours of a list change.

Two things about this roadmap that an architect should be able to defend in a steering committee.

**Nothing in year one replaces anything.** Every work package adds a component or moves a responsibility. The party master is additive; the source systems keep their records and gain a resolved identifier. That is deliberate, because a programme whose first year requires a decommissioning has no first-year benefit, and a programme with no first-year benefit is cancelled in month eleven.

**The core replacement is not on the roadmap at all**, and saying so in writing is the point. It is in the model as a rejected Course of Action, with the reason recorded: the bank cannot fund a programme of the only comparable scale on the public record [3], the decomposition offers six cheaper moves, and the accrual engine — the genuinely hard part — does not need to change. Keeping the rejected option in the model is how an architecture decision stays decided when a new chief information officer arrives.

## What the first supervisory visit asked for

Eight months in, the Prudential Authority's supervisory team spent three days at Table Bay. Aisha was in two of the sessions, which is unusual and worth arranging if you can.

They asked for four things that the model could answer and three that it could not.

Answered. **Which systems hold customer personal information, in which country.** The technology and application views together, with the data inventory they reference. **Which applications are critical, with owners and recovery objectives.** The register that Joint Standard 1 of 2023 requires, which in Table Bay's case is a view of the application model rather than a separate spreadsheet [6]. **What the bank is doing about the four party masters, and by when.** Plateau one, with testable exit criteria. **Who owns the deposit-cover calculation.** A name, finally.

Not answered. **Reproduce the impairment charge published in March 2023.** The system timestamp is not in Basin's history, and no view can fix that; it is a remediation. **Show the mandate levels as the systems enforce them.** They are not enforced in the systems. **Show the lineage of three named BA 700 line items to the entries behind them.** Partially — two of three, with the third assembled in a spreadsheet that one person maintains.

The useful observation is not that the model answered four of seven. It is that the three it could not answer were already in the gap register, with owners and dates, which turned three findings into three in-flight remediations. Supervisors are considerably more interested in a bank that already knows what is broken than in one that produces a perfect answer to the first question and is surprised by the second.

## What the model is for

Twelve views, built over eleven weeks, mostly from documents that already existed. None of them is a system inventory and none required a tool the bank did not have.

What they produced, in order: a precise question about who owns the deposit-cover calculation; the discovery that the weakest capability was the one nobody was arguing about; a finding that one origination path has its own copy of a regulated calculation; the end of a year-long argument about what the word account means; six affordable options where there had been two unaffordable ones; a count of the ways money can leave the bank; and a roadmap whose first year replaces nothing.

Two of the twelve were wrong first. The strategy view had departments pretending to be value streams, and the application view had a single box where a decomposition was needed. Both were approved before they were corrected, which is normal and is the reason a model should be shown early and often rather than perfected privately. The second draft is where the value is, and you only get a second draft if somebody argues with the first.

THE CHAPTER IN TEN LINES

- Table Bay Bank is invented; the failure modes in this chapter are not, and every real fact in it carries a source.
- The first deliverable was a motivation view built in two afternoons from documents that already existed — not a landscape diagram.
- Write the regulator as a Stakeholder and the regime as a Driver; "compliance" as a Requirement makes the model untestable.
- The first strategy view named departments as value streams, and the giveaway was that every capability served every stream.
- Score capabilities on two axes — does it work today, would it scale — because a single score hides the function that works only because eleven people make it work.
- The weakest capability was the one nobody was arguing about: party and identity management, which constrains deposit cover, screening and every customer ambition at once.
- One agreement, one facility, three accounts and four ledger accounts is what a home loan actually is, and one box loses arrears ageing, the in duplum cap and fee disclosure.
- A single box labelled Core Banking leaves two strategies, both unaffordable; the decomposition produced six, three of them fundable this year.
- Six paths out of the bank, three bypassing the hub, is a picture that funds work — and the thing that funded it was contact-centre handling time, not the regulation.
- Gaps drawn as elements show which gap has to close first; three of four severe gaps depended on the same one.
- Nothing in plateau one replaces anything, because a programme with no first-year benefit is cancelled in month eleven.
- Keep the rejected course of action in the model with its reason, which is how an architecture decision stays decided when a new chief information officer arrives.

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CHAPTER 18

# Your first ninety days

*Ninety days is not enough to understand a bank, but it is enough to be useful — if you read the record before the strategy, ask each specialist the one question only they can answer, and build something small enough that the people who do the work will correct it.*

## Ninety days is not enough, and it is enough to be useful

You will not understand a bank in ninety days. Nobody does. People who have been inside one for twenty years do not understand the whole of it, which is why they are so generous with the parts they do.

What ninety days is enough for is this: to know what the bank is for, to know what hurts, to have drawn one true thing that the people who do the work have corrected, and to have earned the right to be in the room when the next decision is made. That is the whole ambition of this chapter, and achieving it has less to do with modelling than with the order in which you do things.

The order matters because of a specific trap. An architect arriving in a bank is handed a mandate that sounds like "give us a view of the landscape". It is the wrong first task and it is almost irresistible, because it is the thing you know how to do, it requires no political capital, and it produces something visible in a fortnight. Resist it. A landscape diagram produced before anyone has told you what hurts is a diagram nobody asked a question of, and it sets the price of architecture at the cost of documentation.

## Before week one: what is already public

Two evenings of reading before you start will put you ahead of most people who have been there a year, because almost nobody inside a bank reads what the bank publishes about itself.

**The annual report and the Pillar 3 disclosure.** Read the segment results first: which business lines make money, which lose it, and what the chief executive says has to change. Then the risk section, which tells you where the impairments are. Then the Pillar 3, which is where the capital and liquidity numbers live, now aligned to the latest Basel disclosure framework [1].

**The results presentation, and the analyst questions at the end.** The questions are the most honest document a bank produces. Analysts ask about the things management has been evasive about, and in a mid-tier South African bank they will ask about cost-to-income, impairment coverage, and whatever the technology programme is costing.

**The Prudential Authority's annual report.** Not about your bank specifically, but it tells you what the supervisor spent the year on, which is what they will ask you about. The 2025/26 edition is the one that sets out the Basel III post-crisis reforms taking effect on 1 July 2025 and the directive programme around them [1].

**The sector data.** The Reserve Bank publishes BA return data for total banks, which gives you the shape of the industry you have joined and the names of the returns you are about to hear mentioned constantly [2].

## Week one: eight documents, and where they live

Ask for these by name on day one. Asking by name is itself a signal: it says you know what a bank has, which shortens the period during which people explain banking to you.

| Document | Who has it | What it tells you |
|---|---|---|
| The current application inventory, however bad | Architecture, or the service management tool | Not the truth, but the vocabulary. Learn the system names before you meet anyone |
| The last two supervisory letters and management's responses | Compliance, or the chief risk officer's office | The most valuable documents in the bank. They say what the regulator thinks is wrong |
| The internal audit report on regulatory reporting | Internal audit | Where the manual steps and the spreadsheets are |
| The obligations or compliance register | Compliance | Every instrument, and whether anyone owns each obligation |
| The business continuity and recovery test results | Operational risk | Joint Standard 1 of 2023 requires annual restore testing and resilience testing including a site-loss scenario, so there should be evidence [3] |
| The register of critical applications | Architecture or operational risk; sometimes nowhere | Also required by that standard [3]. If it does not exist, you have found your first artefact |
| The programme portfolio with budgets and dates | The programme office | What is already funded, which constrains everything you will propose |
| Last month's end-of-day and batch failure log | Operations | The truest document about the estate. Nothing else tells you what actually breaks |

Two notes on reading them. The supervisory letters are usually treated as confidential in a way that excludes architects; push, politely, in writing, and explain why. And the batch failure log is the one people are surprised you want. In most South African banks the business day is still organised around an end-of-day cycle, and the critical path through that cycle is the single most useful thing you can know about the estate in week one.

## Weeks one to three: who to meet, in what order

Order by what each person can tell you that nobody else can, and start with the people furthest from your own discipline. If your first five meetings are with technologists you will learn the estate and not the bank, and you will spend month two unlearning.

1. **The head of regulatory reporting.** First, always. This person knows which numbers are produced by machine, which by spreadsheet, and which by one named individual who cannot take leave in the first ten working days of the month. Ask: which return costs you the most pain, and what breaks it? The answer is a map of where the data model is wrong.

1. **The financial crime or money laundering reporting officer.** Ask: can you re-screen the whole client base against a new sanctions list version within twenty-four hours, and prove which version you screened against? The Financial Intelligence Centre updates its list within twenty-four hours of a Security Council change, and relying on the software is expressly not a defence [4]. The answer tells you immediately whether there is one party estate or several.

1. **The head of payments operations.** Ask: how many distinct paths exist from an originating system to a rail, and how many bypass the hub? Then ask how long a payment query takes to answer. These two questions will get you more credibility in payments than a month of reading.

1. **The head of credit, then a credit manager who sanctions.** The head will tell you the strategy; the sanctioner will tell you what actually happens. Ask the sanctioner: where are your mandate limits enforced — in a system, or in a policy document? In most banks it is the document.

1. **A collections agent, on the floor, with a headset on.** Spend an hour. This is the single most informative hour available to a new architect in a bank, because collections sits downstream of every upstream failure. Ask what they cannot see about a customer who is in arrears. The answer is usually: the customer's other products.

1. **The chief data officer, if there is one.** Ask who owns the customer master. The quality of the pause before the answer is the finding.

1. **The treasurer.** Ask: what question can you not answer between monthly asset and liability committee cycles? Treasury is the part of the bank most dependent on data from everywhere else and least able to get it, and it is reliably ignored by architecture functions.

1. **The product owner of the busiest retail product.** Ask how long a fee change takes and why. Under the National Credit Act the maximum rates and fees are formulas on the repo rate, so every monetary policy decision is a catalogue-wide repricing event [5]. If a fee change takes a release, you have found something a board cares about.

1. **Operations: the person who runs the batch.** Ask what is on the critical path and what happens when it overruns.

1. **Only now, the technologists.** The core team, the integration lead, the infrastructure lead. By this point you can ask them questions in the bank's language rather than in yours, and the difference in what you get back is substantial.

> **THE ONE QUESTION TO ASK EVERYONE**
>
> "What do you have to do by hand every month, and what would have to be true for you to stop?" Ask it of all ten. The overlaps are your architecture backlog, and they are ranked by how many people independently named the same thing.

## What to draw first, and why it must be small

Your first model is not a landscape. It is one slice, narrow enough to be finished and checkable, chosen because somebody has already told you it hurts.

Good candidates in a bank, in order of how reliably they pay off: the payment paths out of the bank; the deposit-cover calculation from source to submission; the origination chain for the busiest credit product; and the customer counts.

Here is what the first one looks like, and notice how little is in it.

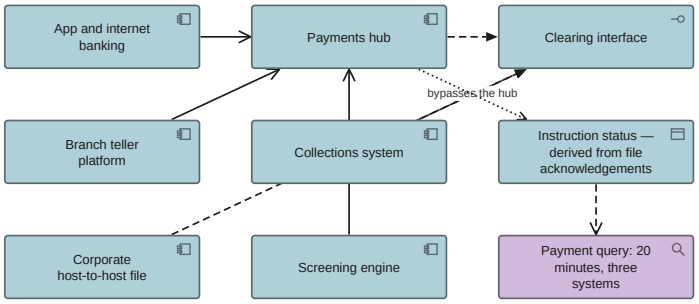



**Figure 18.1 Your first model, at the right size.** Nine elements, drawn in two days, wrong in three places. The two Flow edges that bypass the hub are the point: that is an edge somebody has to respond to, which is what separates a first slice from an inventory. Take it to the people who do the work and let them correct it — the correction is the relationship.

Nine elements. One value stream's worth of behaviour, the components involved, and the one data object that reveals the problem. Drawn in two days, wrong in three places, and that is the point — you take it to the people who do the work and let them correct it. A model that has been corrected by a payments operations manager is worth ten that have not, and the correction is the relationship.

Three rules for the first slice.

**It must have an edge that is a finding.** Not a description. In the figure above it is the two Flow edges bypassing the hub: that edge is a sentence somebody has to respond to. If your first model contains no edge that anybody would argue with, you have drawn an inventory.

**It must be finishable in a week.** Not a sprint, a week. An unfinished model is indistinguishable from no model, and the first one has to exist before you know enough to make it good.

**It must name no vendor you cannot source.** This matters more in South Africa than people realise. No South African bank's core platform is publicly sourceable for most of the major names, and a great deal of market folklore circulates as fact. Put the bank's own internal system names on your boxes, not a vendor's, unless you have seen the contract.

## What to refuse in month one

Four things, all of which will be asked of you, and all of which are better refused politely with a reason and an alternative.

**A full current-state landscape.** Offer a slice and a date for the next slice. The honest argument is that a landscape built before you know the questions will be organised around the wrong things, and will then be defended because it took four months.

**A target-state architecture.** You cannot have a target state in month one, because you do not know what constrains the bank. Offer principles instead — three or four, each with a decision it has already caused — and note that principles are the only part of a target state that is cheap to write and expensive to honour.

**A core banking recommendation.** Everyone will ask. Everyone has an opinion. Say that you will have a view in month five, and that the first thing you will do is decompose what the bank currently calls "the core", because the decomposition changes the question. If you are pushed, the honest framing is that the only well-sourced South African comparison runs long: Nedbank's Managed Evolution is reported in the trade press at R11.7 billion from 2016, with the core delivered last at end-2024 [6]; and the fastest published go-live, Barko on vendor-operated software as a service in under six months, was a single-product lender with no legacy book [7]. Both are true and they are different problems. Quoting one without the other is how a board gets misled.

**A diagram for a board pack, in week two.** This one is the hardest to refuse, because it comes from someone senior who is being helpful. The problem is not the board pack. It is that a diagram drawn to be reassuring in week two becomes the thing your function is known for, and you will spend a year being asked for more of them. Offer one page in month three, built from a model, and say that it will contain something the board does not already know.

## Six numbers that tell you whether the estate is healthy

You can form a defensible view of a banking estate in a month from six numbers, none of which is a technology metric. Ask for each by name; if nobody can produce one, that is the finding.

1. **The length of the end-of-day window, and what is on its critical path.** If the window has no headroom, every programme in the bank is constrained by it and nobody has told you.
2. **The age and value of the oldest item in payments suspense.** Suspense is where in-flight value accumulates when a process fails. An old, large suspense balance means failures are cleared by hand, at a rate slower than they arrive.
3. **The number of distinct implementations of available balance.** Three is common — the app, the automated teller machine and the card authorisation host — and three is why a customer gets three answers.
4. **Days to close, and how many manual journals it takes.** The manual journal value at close is the purest measure of how much of the bank's accounting is done outside its systems.
5. **The proportion of integrations that use an event or an interface rather than a file or a direct database read.** Nobody will have measured it. The estimate people give you is itself informative, and it is always optimistic.
6. **The number of customer masters.** Covered above, and the one to ask last, because by then you will know not to accept the first answer.

Two of those six were in the batch failure log you asked for in week one, which is why it is on the list.

## The three artefacts that buy you credibility

Credibility in a bank is bought with artefacts that other people need and nobody is producing. These three qualify, in nearly every bank, and all three are achievable in ninety days.

**One: the customer-count reconciliation.** Four functions in a bank count customers, and the numbers never agree. Retail counts natural persons. Business banking counts enterprise groups. The card platform counts cards, or cardholders, or both at different times. And regulatory reporting counts deposit-insurance records, which follow a prescribed definition with its own rules about joint accounts and sole proprietors [8].

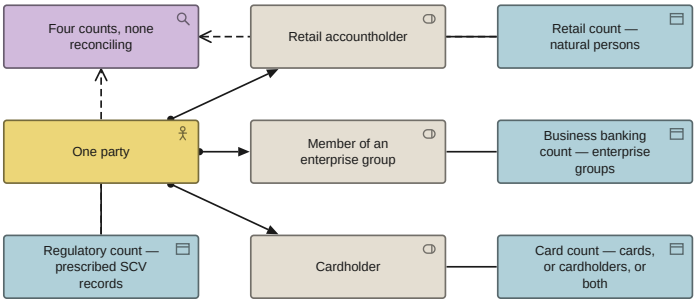



**Figure 18.2 The customer-count reconciliation, and why nobody is wrong.** Four functions count four different things and each is correct within its own definition. Put the four actual numbers on this picture with the definitions underneath and you have established, without accusing anyone, that the bank holds records rather than an identity — which is the premise of every data conversation you will have afterwards.

Produce that picture with the four actual numbers on it and the definitions underneath, and you will have done something genuinely useful on day sixty. Nobody is annoyed by it, because nobody is wrong — each count is correct within its own definition. What the artefact establishes is that the bank has records and not an identity, and that is the premise of every data conversation you will have afterwards.

**Two: the register of critical applications, with owners and recovery objectives.** Joint Standard 1 of 2023 requires it, along with an information technology asset inventory, recovery time and recovery point objectives, a geographically separate recovery site and annual resilience testing including a site-loss scenario [3]. In many banks it exists in three partial versions owned by operational risk, architecture and service management. Consolidating it is unglamorous, it is inherently collaborative, and it is a thing the chief risk officer will be asked for by a supervisor. Own it.

**Three: the obligations-to-systems map, for ten obligations.** Not the whole register. Ten obligations, each traced to the system, process or data object that satisfies it. Pick ones that matter: the deposit-cover calculation, the sanctions re-screening, the affordability assessment record, the complaint categorisation, the balance-of-payments category on cross-border payments. Then show which have no realising component at all. There will be some, and finding them is the single most valuable thing an architecture function does in a regulated institution.

## The politics, honestly

Three situations are near-universal in a bank, and none of them is a technical problem.

**The core replacement everyone has an opinion about.** There will be a faction that wants to replace it and a faction that will not let anyone near it, and both have been arguing for years. What makes this tractable is refusing the binary. The box labelled "core" is six or twelve components with different owners, change rates and failure modes, and once that is on paper the argument moves from one decision to several — most of which are small enough to be made. Do not take a side in month one. Produce the decomposition in month four and let it take the side for you.

**The data owner who does not know they are one.** Somewhere in the bank there is a manager in operations who, in practice, decides what a customer record means — because their team does the corrections, their workarounds define the actual rules, and nothing changes in that data without them. They have no data title and have never been to a governance forum. Find them, early. They will tell you more about the data model in an hour than the model documentation will, and if you try to appoint a data owner over their head, the appointment will fail quietly.

**The executive who wants a diagram for a board pack.** Not a bad person, and not a bad request. The risk is what it makes your function. The way through is to agree the picture and set a condition: it will be generated from a model rather than drawn, and it will say something the board does not already know. The second half is what turns a reassurance exercise into an architecture deliverable. If you cannot find anything the board does not know, that is also a finding, and it means you are not ready yet.

One more, smaller but corrosive. **The programme that has already decided.** There is always one funded initiative whose solution was chosen before you arrived, and which will now be routed to you for architectural approval. You cannot stop it and you should not try. What you can do is record the decision, the alternative that was not considered, and the consequence you expect — in writing, once, without drama. Six months later that record is either forgotten or it is the most credible document in the bank.

## What week thirteen should look like

| Weeks | What you do | What exists at the end |
|---|---|---|
| **0** | Annual report, Pillar 3, results Q&A, PA annual report, sector data | You know what the bank is for |
| **1** | Ask for the eight documents by name. Read the supervisory letters and the batch failure log first | The vocabulary, and a list of what breaks |
| **2–3** | Ten conversations, in order, furthest from your discipline first. One hour on the collections floor | Three things more than one person named |
| **4–5** | Draw the first slice. Finish it. Take it to be corrected | One corrected model, and the relationships that came with the correction |
| **6–8** | The six numbers. Second slice started. Refuse the landscape, in writing, with an alternative | A defensible view of estate health |
| **9–11** | Build one of the three artefacts and hand it to the person who needed it | Something other people now depend on |
| **12–13** | Ten obligations traced to realising components. Name the ones with none | The finding that gets you into the next decision |

> **Not on this plan:** a current-state landscape, a target-state architecture, a core banking recommendation, or a board diagram drawn rather than generated.

**Figure 18.3 Thirteen weeks, in the order that works.** Reading before meeting, meeting before drawing, drawing one slice before anything wider, and an artefact somebody needed before any opinion about the core. Nothing on this plan is a landscape diagram or a target state, and that is deliberate.

By the end of week thirteen, these should be true.

- You can describe, in two minutes and without notes, how the bank makes money and which business line is under the most pressure.
- You have met the ten people above and asked each the one question only they could answer, and you can name the three things that more than one person independently said hurt.
- You have one finished model slice that somebody who does the work has corrected, and a second slice in progress.
- You have produced at least one of the three artefacts, and handed it to someone who needed it.
- You know which obligations have no realising system, for ten obligations.
- You have refused at least one thing, with a reason and an alternative, and the relationship survived.
- You have not produced a target-state architecture, and you can explain why without sounding defensive.

What you should not have: a complete landscape, a strategy document, or a core banking recommendation. If you have those at week thirteen, you produced them from assumptions, and the assumptions are about to be corrected expensively.

## Then the method

Ninety days gets you oriented and trusted. It does not get you an architecture practice, and nothing in this book is a method. This volume teaches the industry: what a bank is for, how it earns, what its systems do, what the regulator demands, and how to put all of that into a model somebody on the executive committee would recognise as true.

Volume 2 is the method — the repository, the metamodel, the governance, and a worked architecture in a different bank, built the way a practice would build it. That bank is called Ardent Union, and it is deliberately not this one. Table Bay taught you to see; Ardent Union is where you do the work.

### THE CHAPTER IN TEN LINES

- Ninety days will not make you understand a bank; it is enough to know what hurts, to have drawn one true thing, and to have earned a place in the next decision.
- A landscape diagram produced before anyone has told you what hurts sets the price of architecture at the cost of documentation.
- Two evenings on the annual report, the Pillar 3 disclosure, the analyst questions and the supervisor's own annual report will put you ahead of most people who have been there a year.
- Ask for eight documents by name in week one; the supervisory letters and the batch failure log are the two that nobody expects an architect to want.
- Meet the head of regulatory reporting first and the technologists last, and ask everyone what they do by hand every month and what would have to change for them to stop.
- Spend an hour on the collections floor: collections sits downstream of every upstream failure, and the thing they cannot see is the customer's other products.
- The first model is one slice, finishable in a week, containing at least one edge somebody has to argue with.
- Refuse four things in month one — a full landscape, a target state, a core recommendation and a board diagram — each politely, with a reason and an alternative.
- Six numbers tell you whether the estate is healthy, and not one of them is a technology metric.
- Three artefacts buy credibility: the customer-count reconciliation, the register of critical applications with owners and recovery objectives, and ten obligations traced to the systems that satisfy them.
- Find the data owner who does not know they are one before you try to appoint one over their head.
- At week thirteen you should have no target-state architecture and be able to say why without sounding defensive.

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APPENDIX A

# The cross-maps

*The reference pages an architect actually uses. Five dimensions defined with IDs, then seven maps that join them — all hinged on business process, because process is the only dimension the business will argue about.*

## What these are, and what they are not

Everything here hangs off **business process**, because process is the only dimension the business will argue about and therefore the only safe hub. A capability map on its own provokes no argument; a data model on its own provokes the wrong one. Put process at the centre and every other dimension has somewhere to attach.

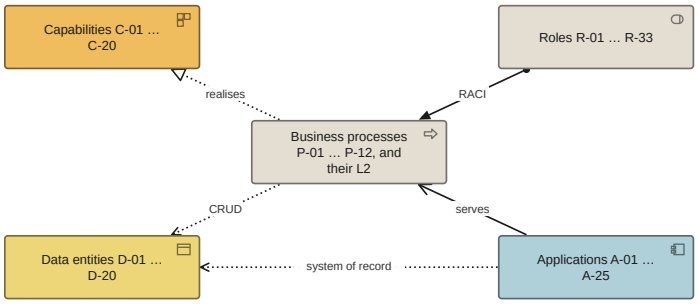



**Figure 19.1** Process as the hub, and each cross-map as the ArchiMate relationship it really is: a process realises a capability, a role is assigned to it, it accesses the data, an application serves it. The one map that does not touch process — application to data — is the Access edge along the bottom, and it is the one that settles the system of record.

Five warnings.

**A worked starting set for Table Bay Bank, not a universal reference.** Table Bay Bank is the invented South African bank carried through this book — mid-tier, 2.4 million retail customers, a business bank, a small corporate arm and a treasury. Nothing here describes a real institution; the assignments are mine.

**The IDs are local, and they are not BIAN's.** `P-04`, `C-07`, `A-10` mean something inside this book and nothing outside it. They are not BIAN Service Domain identifiers and not APQC process identifiers [1][2]. If you want BIAN's, use BIAN's — and note that BIAN publishes no licence, so get terms in writing before its names appear in work you are paid for [1].

**Illustrative, not measured.** A real cross-map comes out of workshops with the named role in the room. Where my assignment is contestable the row says so.

**The grain is L1 and L2 only.** Chapter 8 argues for stopping there, and these maps would be unmaintainable at L3.

**Expect gaps, and read them as findings.** A capability with no process, an entity with no system of record, a process with no accountable role: not errors, but the three things most worth taking to a steering committee.

Adapt it in this order. **Processes first**, because everything hangs off them. Then **roles**, the cheapest to verify — walk the list past an operations manager and they will fix a third of it in twenty minutes. Then **data**, then **applications**, and **capabilities last**, because a capability map is what people most enjoy arguing about and least need early.

## Half one — the definitions

### Business processes

Chapter 8's L1 inventory, verbatim, with its triggers and outcomes. Two notes on what has been added. **Chapter 8's inventory carries no owner column**, so the owners below come from the research behind the book and should be the first thing you reconcile with an operations leader. And chapter 8 models **thirteen** L2 processes in BPMN; two L1 processes — P-08 and P-12 — have no modelled L2, so one has been added for each at the same grain and both are marked.

| ID | L1 process | Trigger → the outcome that ends it | Owner |
|---|---|---|---|
| P-01 | Onboard a party | An application, or a refresh falling due → a party record that is identified, screened, risk-rated and usable | Client onboarding and compliance, jointly. Half flow, half cycle |
| P-02 | Open and maintain an account | An approved party wants a product → an account with a product instance, a mandate, limits and entitlements | Product operations |
| P-03 | Execute a payment | A customer or system instructs → funds with the beneficiary, ledger posted, status answerable | Payment operations. One process, six rails |
| P-04 | Register a mandate and run a collection | A contract, then a billing date → a mandate at the debtor's bank, and a collection that succeeds or fails with a reason | Collections. Two processes sharing one name |
| P-05 | Authorise, clear and settle a card transaction | A card is presented or a card-not-present transaction submitted → approve or decline in under a second; money days later | Cards |
| P-06 | Resolve a dispute, chargeback or fraud claim | A customer says a transaction is wrong → liability allocated and the ledger reflecting it | Disputes. Runs on the scheme's clock |
| P-07 | Originate credit | An application, or a facility request → money disbursed under a compliant agreement, or a reasoned decline | Retail credit and business credit. Two processes, one name |
| P-08 | Service credit and manage limits | A scheduled review, a behavioural trigger, a customer request → a limit confirmed, raised, cut or withdrawn, with a reason | Credit risk. A cycle |
| P-09 | Collect and recover | A missed payment; later, a cure failure → cure, arrangement, legal process, or write-off and post-write-off recovery | Collections; legal recoveries beyond write-off |
| P-10 | Fund the balance sheet and manage liquidity | Continuous → a funded, hedged balance sheet inside appetite | Treasurer. A cycle with a committee |
| P-11 | Close, account and report | A date → a closed ledger, submitted returns, signed attestations | Financial controller; regulatory reporting. A production schedule [5] |
| P-12 | Handle a complaint | An expression of dissatisfaction, anywhere → a resolution, a decision letter, a recorded root cause | Client experience. Regulated and externally escalable [3][4] |

| ID | L2 process, as chapter 8 models it | What happens |
|---|---|---|
| P-01.1 | Onboard a party and complete FICA | Identity, residential address, screening, risk rating, and the queue of files that fail one of them |
| P-02.1 | Open and maintain an account | Product instance, mandate at the right level, limits, entitlements, and the bundled card |
| P-03.1 | Execute a payment by EFT credit | The batch rail in full: validation, screening, cut-off, file, acknowledgement, reconciliation |
| P-04.1 | Register a mandate: the DebiCheck cycle | Register through the creditor's bank, the debtor authenticates at their own, the terms are stored there [7] |
| P-04.2 | Run a Registered Mandates collection | The 19:00 late window, up to ten tracking days, response by 23:00, settlement at 05:00 [6] |
| P-05.1 | Authorise a card transaction | Cryptogram, velocity, fraud and stand-in, inside a budget measured at the 99th percentile |
| P-06.1 | Resolve a dispute and run a chargeback | Provisional credit, evidence, representment, arbitration — on the scheme's clock, not yours [8] |
| P-07.1 | Retail credit, application to decision | Bureau, the prescribed affordability calculation, score, decision, and the retained reason |
| P-07.2 | Retail credit, offer to disbursement | Quotation, agreement, security registration, dual-control disbursement |
| P-07.3 | Business credit, and the mandate ladder | Spread, rate, paper, committee under delegated authority, conditions precedent, limits published |
| P-08.1 | Review a limit and re-decide it | *Added here.* Behavioural trigger or scheduled review, re-assessed affordability, notification and reasons |
| P-09.1 | Collections and recoveries | Treatment, arrangement, the section 129 notice, legal state, write-off, the prescription clock |
| P-10.1 | Treasury, deal capture to settlement | Pre-trade limit check, independent valuation, confirmation, settlement, nostro reconciliation |
| P-11.1 | Close, account and report | Sub-ledger close, impairment, capital, the returns and their signed certifications |
| P-12.1 | Receive, classify, resolve, record cause | *Added here.* Prescribed categories, acknowledgement, redress, root cause, and the ombud referral |

## Business capabilities

Twenty capabilities in the business's own words, each of which stays true if you switch off every computer. **Built for a bank, with local IDs — these are not BIAN's 586 Business Capabilities** [1].

| ID | Capability | What the business must be able to do |
|---|---|---|
| C-01 | Product management | Decide what to sell, define it legally, classify it, change it |
| C-02 | Pricing and rate management | Set a rate and a fee, know the margin, stay inside the statutory maxima |
| C-03 | Channel and distribution management | Reach customers through chosen channels, and know what each cannot do |
| C-04 | Party and relationship management | Know who it deals with, in what capacity, as one party |
| C-05 | Customer due diligence | Establish and maintain who a customer is, and how risky |
| C-06 | Deposit taking and account servicing | Hold other people's money, pay interest, service the account for life |
| C-07 | Payment execution and clearing | Move money on the right rail, inside the cut-off, traceably |
| C-08 | Collection by mandate | Take money from someone else's account on a verifiable authority |
| C-09 | Card issuing and acquiring | Put an instrument in a customer's hand, and take one at a merchant |
| C-10 | Credit assessment and decisioning | Decide who to lend to, on what terms, and prove it was lawful |
| C-11 | Credit administration and limit management | Hold the facility, limit and security, and publish the limit |
| C-12 | Arrears, collections and recovery | Get paid late, lawfully, and know what the book is then worth |
| C-13 | Treasury, funding and liquidity management | Fund the book, price funds internally, survive a stress |
| C-14 | Balance-sheet and capital management | Hold enough capital, know its risks, run the calculation twice |
| C-15 | Financial crime management | Detect, report and act on laundering, sanctions and terrorist financing |
| C-16 | Fraud management | Stop money leaving on an instruction the customer did not really give |
| C-17 | Financial control and accounting | Close the books and produce a number it can defend to the cent |
| C-18 | Regulatory reporting and supervisory engagement | Produce the returns on their dates, with lineage, and answer questions |
| C-19 | Customer service, conduct and complaints | Answer, resolve, record the cause, treat the customer as required |
| C-20 | Information, data and model management | Know what the data means, where it came from, who may see it |

## Roles

Chapter 5 names roughly fifty jobs in eleven groups. These thirty-three appear in the maps, under chapter 5's own names. **R-20 merges its foreign-exchange dealer and its rates trader. Its independent risk roles, management accountant, capital management, legal counsel, information officer, internal audit and technology roles are not rows here**, because in the maps they appear only as consulted or informed. And the omission that matters: **chapter 5 has no business role for data ownership** — its nearest equivalents are technology roles. That absence is the finding behind C-20's coverage row, not a gap in this table.

| ID | Role | What the person does |
|---|---|---|
| R-01 | Branch consultant / teller | Opens accounts, takes deposits, captures documents, decides who is who |
| R-02 | Branch manager | Runs the branch result, queues and cash; owns local discretion |
| R-03 | Self-service and cash operations | Device uptime, cash levels, replenishment, routes |
| R-04 | Contact centre agent | Queries, limits, card blocks; classifies fraud or dispute first |
| R-05 | Digital channel product owner | Owns the app and internet banking, and what gets built next |
| R-06 | Business banker / relationship manager | Manages named clients; owns the credit recommendation |
| R-07 | Corporate coverage banker | Owns the institutional relationship and its return |
| R-08 | Private banker / wealth manager | Serves high-net-worth clients across banking, credit and fiduciary |
| R-09 | Dealer-floor finance consultant | Books vehicle finance while the customer waits |
| R-10 | Credit analyst | Spreads financials, writes the credit paper; owns the risk opinion |
| R-11 | Credit manager / sanctioner | Approves or declines within a personal mandate, by name |
| R-12 | Credit committee | Collective approval above individual mandate; sets appetite |
| R-13 | Scorecard / decisioning analyst | Builds and monitors scorecards; owns the cut-off |
| R-14 | Credit risk modeller | Builds default, loss and exposure models and staging criteria |
| R-15 | Collections agent | Contacts customers in arrears, negotiates, records promises |
| R-16 | Collections strategy manager | Decides who gets which treatment, and when |
| R-17 | Legal recoveries | Summons, judgment, attachment, repossession, debt review, book sales |
| R-18 | Treasurer / head of ALM | Funding, liquidity, banking-book rate risk, the transfer price |
| R-19 | Money market dealer | Manages the daily cash position and interbank funding |
| R-20 | Foreign exchange, sales and derivatives dealer | Quotes spot, forward and swap; makes rate markets inside limits |
| R-21 | Middle office / product control | Validates trades and prices; owns the official profit and loss |
| R-22 | Treasury operations and settlements | Confirms, settles, manages nostros, investigates fails |
| R-23 | Product owner | Owns a product: definition, pricing, lifecycle, compliance |
| R-24 | Pricing manager | Sets rates, fees, bundles and rewards. Owns the price |
| R-25 | Payments operations | Runs the daily cycles, monitors rails, works failures and recalls |
| R-26 | Reconciliations and back office | Matches ledgers, suspense, nostro and scheme files |
| R-27 | Operational risk | Control self-assessments, loss events, indicators, capital |
| R-28 | Compliance officer | Maps obligations to controls, monitors, manages findings |
| R-29 | Financial crime analyst | Investigates alerts, files reports, screens sanctions and PEPs |
| R-30 | Due diligence operations | Onboarding and refresh due diligence; remediates files |
| R-31 | Fraud analyst | Works real-time alerts, blocks and traces, builds rules |
| R-32 | Financial controller | Runs the close; owns the accounts and the chart of accounts |
| R-33 | Regulatory reporting analyst | Produces the prudential return suite and the disclosures |

## Conceptual data entities

Chapter 9's fifteen, the three it admits as borderline and lets in, and the two more its payment section names but its summary table does not. No attributes, no keys, no system names. The five it throws out — Customer, Transaction-as-catch-all, Record, Status and Address — are deliberately absent.

| ID | Entity | What it means | What it must carry |
|---|---|---|---|
| D-01 | Party | A person or organisation we deal with | Identity, identifiers, type, verification state, risk rating |
| D-02 | Party Role | What a party is to us, over what, between two dates | Role type, scope, valid-from, valid-to |
| D-03 | Agreement | What we contracted to, on which wording | Product version, terms, disclosure evidence, signature |
| D-04 | Product | What we sell, in a version | Version, effective dates, classification, rate and fee structure |
| D-05 | Account | The customer-facing instance that opens and closes | Number, product instance, status, open and close dates |
| D-06 | Facility | What may be drawn and redrawn without a new decision | Type, currency, sub-limits, drawdown rules, expiry |
| D-07 | Limit | What may not be exceeded, by whom, to what value | Scope, value, authority, effective dates, utilisation |
| D-08 | Transaction | What happened, that the customer would call one thing | Type, date, amount, counterparty, channel, instruction |
| D-09 | Entry | The immutable signed amount against one ledger account | Amount, sign, ledger account, posting and value date, reversal link |
| D-10 | Balance | The position at a point in time, on a stated definition | Definition, as-at, currency. **A derivation, not a stored fact** |
| D-11 | Mandate | Who may act, over what, up to how much, from when | Type — account, collection, internal — scope, limit, status |
| D-12 | Collateral | What asset stands behind this, worth what, as at when | Type, identity, valuation and its date, registration |
| D-13 | Impairment State | How bad an exposure is, on which date, by which model | Days past due, stage, provision, model version. Append-only |
| D-14 | Payment Instruction | What we were asked to do, by whom, through which rail | Payer, payee, amount, date, channel, authorisation, status |
| D-15 | Payment | The resulting movement of value | End-to-end reference, clearing stream, status, return or recall |
| D-16 | Settlement | Discharge of the interbank obligation in central-bank money | Stream, date and time, finality. Final and irrevocable |
| D-17 | Complaint | What the customer said was wrong, in which prescribed category | Category, cause, dates, outcome, redress, escalation |
| D-18 | Statement | A frozen rendering of a period's entries and balances | Period, artefact, channel, delivery event. Immutable |
| D-19 | Regulatory Return | A submission, for an entity, for a period, in a version | Preparer, reviewer, approver, timestamp, resubmissions |
| D-20 | Session | A bounded authenticated interaction | Party, device, channel, authentication, start and end |

## Conceptual applications

Chapter 11's twenty-five, by ID and under its own names. That chapter carries each one's purpose, primary user roles, data mastered and the real-world category it maps to in chapter 12; this is the index.

| ID | Application capability | Domain |
|---|---|---|
| A-01 | Channel experience and session | Engagement |
| A-02 | Interaction, service and complaint | Engagement |
| A-03 | Customer identity and authentication | Engagement |
| A-04 | Party master, relationship and mandate | Party |
| A-05 | Product definition, catalogue and pricing | Product |
| A-06 | Account lifecycle and servicing | Account and ledger |
| A-07 | Posting, balance and interest | Account and ledger |
| A-08 | Payment initiation and orchestration | Payments |
| A-09 | Rail adaptation, clearing and settlement | Payments |
| A-10 | Card issuing and token management | Cards |
| A-11 | Authorisation and switching | Cards |
| A-12 | Credit application and origination | Lending |
| A-13 | Credit decisioning and affordability | Lending |
| A-14 | Loan servicing, collateral and arrears | Lending |
| A-15 | Collections and recoveries | Lending |
| A-16 | Deal capture, position and treasury operations | Treasury and finance |
| A-17 | General ledger and financial control | Treasury and finance |
| A-18 | Regulatory reporting and submission | Treasury and finance |
| A-19 | Credit risk measurement and impairment | Risk and capital |
| A-20 | Capital, liquidity and balance-sheet risk | Risk and capital |
| A-21 | Screening, monitoring and financial-crime cases | Financial crime and fraud |
| A-22 | Fraud detection and intervention | Financial crime and fraud |
| A-23 | Data integration, history and analytics | Data and common |
| A-24 | Document, evidence and communication | Data and common |
| A-25 | Workforce identity, entitlement and authority | Data and common |

## Half two — the maps

Seven maps, keyed on the IDs above. Each is a list of pairs rather than a wide matrix, because a twenty-by-twenty-five grid does not fit a page and, more usefully, because a list of pairs survives being pasted into a repository.

## Capability to process

Read it both ways: a capability exercised by no process is an aspiration, and a process exercising seven capabilities is probably two.

| Capability | Processes that exercise it |
|---|---|
| C-01 | **None.** Product design is not in chapter 8's L1 inventory |
| C-02 | **None as a decision.** P-07.1 and P-08.1 consume a price; nothing sets one |
| C-03 | P-01.1, P-02.1, P-03.1, P-05.1 |
| C-04 | P-01.1, P-02.1, P-07.3, P-12.1 |
| C-05 | P-01.1 |
| C-06 | P-02.1, P-11.1 |
| C-07 | P-03.1, P-10.1 |
| C-08 | P-04.1, P-04.2 |
| C-09 | P-05.1, P-06.1 |
| C-10 | P-07.1, P-07.3 |
| C-11 | P-07.2, P-07.3, P-08.1 |
| C-12 | P-09.1 |
| C-13 | P-10.1 |
| C-14 | P-11.1 |
| C-15 | P-01.1, P-03.1, P-11.1 |
| C-16 | P-03.1, P-05.1, P-06.1 |
| C-17 | P-11.1 |
| C-18 | P-11.1 |
| C-19 | P-02.1, P-12.1 |
| C-20 | P-11.1, and every process as a producer |

Two findings fall straight out, and both are about chapter 8's inventory rather than about Table Bay. **C-01 and C-02 are exercised by no process at all.** Nobody in the L1 map designs a product or sets a price — the work happens, but it is not in the process architecture, which is why a product change arrives as a release rather than a configuration. And **four capabilities — C-14, C-17, C-18 and C-20 — collapse onto one L2**, the close. That is what the close is, and it is why it consumes more change capacity than anything else in the bank.

## Process to role, as RACI

One accountable role per process: that is the whole discipline. Where I could not honestly name one, the row says so, because that is the finding.

| Process | Accountable | Responsible | Consulted | Informed |
|---|---|---|---|---|
| P-01 | R-30 nominally; shared with R-28, and the sharing is the problem | R-01, R-04, R-30 | R-29, R-31 | R-02, R-05 |
| P-02 | R-23 | R-01, R-04 | R-11, R-28 | R-25, R-26 |
| P-03 | R-25 | R-25, R-26 | R-29, R-31 | R-18, R-32 |
| P-04 | R-16 | R-15, R-25 | R-26, R-28 | R-23, R-33 |
| P-05 | R-23 for the card product; nobody for the end-to-end flow | R-25, R-31 | R-26, R-28 | R-32 |
| P-06 | R-26 for the chargeback; R-28 for the conduct side. **Two accountables** | R-04, R-26 | R-02, R-31 | R-23, R-32 |
| P-07 | R-11 for retail; R-12 for business, under delegated authority | R-06, R-09, R-10, R-13 | R-14, R-28 | R-16, R-23 |
| P-08 | R-11 | R-10, R-13 | R-14, R-16 | R-23, R-28 |
| P-09 | R-16 | R-15, R-17 | R-11, R-28 | R-14, R-32 |
| P-10 | R-18 | R-19, R-20, R-22 | R-21, R-32 | R-14, R-27 |
| P-11 | R-32 and R-33 jointly; contested | R-32, R-33 | R-14, R-21, R-27 | R-27, R-28 |
| P-12 | R-28 | R-02, R-04 | R-26, R-31 | R-23, R-32 |

Four rows are the honest ones. **P-01's accountability is shared between onboarding and compliance**, which is why the file-remediation backlog outlives three programmes. **P-05 has nobody accountable end to end** — the product owner owns the card, operations the file, fraud the decline, and the unreleased authorisation hold belongs to all three. **P-06 has two accountables on two external clocks.** And **P-11 has two**, which is why impairment is reconciled to the ledger by hand at close.

## Process to data, as CRUD

Which entities each process touches, and how. Which *application* masters each entity is below; the two together are what you need before a migration.

| Process | Entities touched |
|---|---|
| P-01 | D-01 **CU**, D-02 CU, D-20 C, D-18 R |
| P-02 | D-05 **CU**, D-03 C, D-11 **CU**, D-07 CU, D-01 RU, D-04 R |
| P-03 | D-14 **C**, D-15 **C**, D-16 R, D-09 C, D-08 C, D-10 R, D-11 R |
| P-04 | D-11 **CRUD**, D-14 C, D-15 C, D-09 C, D-05 R, D-17 C on dispute |
| P-05 | D-08 **C**, D-14 C, D-09 C, D-10 RU as the hold, D-05 R |
| P-06 | D-17 C, D-08 R, D-09 C as redress, D-18 R, D-20 R |
| P-07 | D-03 **C**, D-06 **C**, D-05 C, D-07 C, D-12 C, D-01 RU, D-09 C |
| P-08 | D-07 **CU**, D-06 U, D-12 U, D-18 C as the notice, D-01 R |
| P-09 | D-13 **CU**, D-12 U, D-09 C, D-03 R, D-05 U, D-18 R |
| P-10 | D-16 **C**, D-08 C, D-09 C, D-07 R, D-10 R |
| P-11 | D-19 **CU**, D-13 RU, D-09 R, D-10 R, D-04 R, D-01 R |
| P-12 | D-17 **CU**, D-18 R, D-08 R, D-05 R, D-02 R |

Bold marks the natural *creator* of an entity, and that column repays reading alone. **Two processes create a payment instruction** — execution and mandated collection — which is right, because a collection is an instruction the bank did not get from its own customer. Three rows are worth arguing about. **Nothing creates D-04, Product**, which is the hole the capability map found. **Two processes create D-17, Complaint**, which is why complaint volumes by root cause cannot be produced from one place. And **D-10, Balance, is never created**, only read and updated, because it is a derivation.

## Process to application

| Process | Applications |
|---|---|
| P-01 | A-01, A-02, A-03, A-04, A-21, A-24 |
| P-02 | A-01, A-04, A-05, A-06, A-07, A-10, A-24 |
| P-03 | A-01, A-07, A-08, A-09, A-17, A-21, A-22 |
| P-04 | A-07, A-08, A-09, A-14, A-15 |
| P-05 | A-07, A-10, A-11, A-17, A-22 |
| P-06 | A-02, A-10, A-11, A-22, A-24 |
| P-07 | A-04, A-06, A-07, A-12, A-13, A-14, A-24 |
| P-08 | A-13, A-14, A-19, A-24 |
| P-09 | A-14, A-15, A-17, A-24 |
| P-10 | A-08, A-16, A-17, A-20 |
| P-11 | A-07, A-17, A-18, A-19, A-20, A-23 |
| P-12 | A-02, A-23, A-24 |

Count the applications per process and the procurement lesson appears. **Account opening, payment execution and credit origination touch seven each.** Anyone proposing to replace one of them with one product is proposing to replace seven things, and that belongs before the shortlist. **A-07, posting and balance, appears in seven of twelve rows** — which makes a ledger replacement a programme, not a project.

## Application to data, with the system of record

The most useful table here. **Bold means system of record**: it creates the entity, owns its lifecycle, and everything else holds a copy.

| Application | System of record for | Also reads |
|---|---|---|
| A-01 | **D-20** | D-04, D-05, D-08, D-10, D-18 |
| A-02 | **D-17** | D-01, D-05, D-08, D-18 |
| A-03 | nothing conceptual; masters credential, device and step-up outcome | D-01, D-20 |
| A-04 | **D-01, D-02, D-11** | D-05, D-06, D-07 |
| A-05 | **D-04** | D-03 as wording |
| A-06 | **D-05** | D-01, D-04, D-07, D-11 |
| A-07 | **D-08, D-09**, and D-10 as its definition of it | D-04, D-05 |
| A-08 | **D-14** | D-05, D-10, D-11, D-15 |
| A-09 | **D-15, D-16**, and the rail's view of D-11 | D-14 |
| A-10 | nothing conceptual; masters card, token and lifecycle | D-02, D-05 |
| A-11 | nothing conceptual; masters authorisation, shadow balance, stand-in | D-05, D-08, D-10 |
| A-12 | **D-03** | D-01, D-04, D-12 |
| A-13 | nothing conceptual; **masters the decision, its reasons and the input vector** | D-01, D-03, D-07 |
| A-14 | **D-06, D-12, D-13** | D-03, D-05, D-09 |
| A-15 | **D-13** where arrears live in collections. Contested with A-14 | D-03, D-05, D-12 |
| A-16 | **D-16** for its own settlements; deal and position sit below this grain | D-07, D-08, D-09, D-10 |
| A-17 | nothing conceptual; postings sit below D-09 | D-04, D-09, D-10, D-13 |
| A-18 | **D-19** | all |
| A-19 | nothing conceptual; masters exposure base, parameters and scenarios | D-06, D-09, D-12, D-13 |
| A-20 | nothing conceptual; masters capital, liquidity and behavioural measures | D-05, D-06, D-09, D-10 |
| A-21 | nothing conceptual; masters risk rating, alert, case and filed report | D-01, D-02, D-14, D-15 |
| A-22 | nothing conceptual; masters rules, scores and loss outcomes | D-01, D-08, D-14, D-20 |
| A-23 | history of everything; authoritative for nothing current | all |
| A-24 | **D-18**, and the documents of record behind D-03 | D-01, D-03, D-05, D-17 |
| A-25 | nothing conceptual; masters workforce identity, entitlement and authority limit | D-01, D-02, D-11 |

Four rows deserve argument. **No application masters D-07, Limit.** Four read it and A-14 holds one view of it, which is exactly why a limit in the core and a limit in the authorisation host drift apart with nobody owning the difference — the clearest single finding here. **D-13 has two candidate masters**, servicing and collections, and in most banks the answer is "both, differently", which is the commonest reason an impairment number cannot be reproduced. **A-13 masters nothing conceptual yet masters the decision, its reasons and the input vector**, the three things a National Credit Act challenge asks for. And **A-09 holds the rail's view of a mandate whose authoritative record sits at the debtor's bank.**

## Application to role

Chapter 11 names its own primary user roles. Where one is not among the thirty-three, the cell names the job in words rather than forcing it into an ID.

| Application | Roles |
|---|---|
| A-01 | R-01, R-04, R-05 |
| A-02 | R-04, R-28; and a complaints officer, which chapter 5 does not carry |
| A-03 | R-31; and an identity engineer, a technology role |
| A-04 | R-30; and a party data steward, which chapter 5 does not carry |
| A-05 | R-23, R-24 |
| A-06 | R-01; and deposits operations |
| A-07 | R-32; and core operations |
| A-08 | R-25; and a corporate service manager |
| A-09 | R-25, R-26 |
| A-10 | R-23; and card operations |
| A-11 | card operations and a switch engineer. **No role on this list** |
| A-12 | R-06, R-09, R-10 |
| A-13 | R-11, R-13; R-14 as model owner |
| A-14 | R-26; and a loan servicing clerk and a security administrator |
| A-15 | R-15, R-16, R-17 |
| A-16 | R-19, R-20, R-21, R-22 |
| A-17 | R-26, R-32 |
| A-18 | R-33; R-32 as signatory |
| A-19 | R-14 |
| A-20 | R-18; and capital management, which this list omits |
| A-21 | R-29, R-30 |
| A-22 | R-31 |
| A-23 | R-13, R-14, R-27, R-33 |
| A-24 | R-01, R-17; and a records manager |
| A-25 | R-27; and an identity administrator |

Read the role column instead and look for roles appearing once or twice. **R-17, legal recoveries, appears in two applications and works in those two and a spreadsheet**, because no system models the prescription clock. **R-07, the coverage banker, appears in none.** And **A-11 has no business role at all**: the switch is run by engineers, which is fine until somebody must explain a decline to a customer.

## Capability to application — the coverage map

The map that goes in the board pack. Coverage is my assessment against the Table Bay estate chapter 11 describes: a retail core from 1996, a second core inherited with a business bank in 2014, cards outsourced, a payments hub three channels bypass, a warehouse, and a reporting tool over a mart. **Full** where one application serves the capability and masters its data, **partial** where it is served but not mastered or served by several, **gap** where nothing serves it.

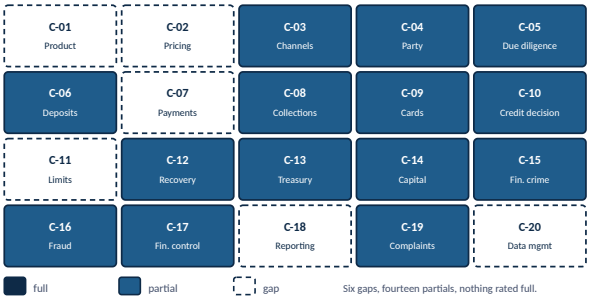



**Figure 19.2** The coverage map as one picture, against the invented Table Bay Bank estate as chapter 11 describes it. The dashed cells are the board paper; that nothing is rated full is the finding under them.

| Capability | Applications | Table Bay coverage |
|---|---|---|
| C-01 | A-05 | **Gap.** Chapter 11 finds no A-05 at Table Bay, so the deposit-insurance qualifying flag is set per account instead of derived from the product |
| C-02 | A-05 | **Gap**, the same one — and no process sets a price either |
| C-03 | A-01, A-02, A-03 | Partial. One session model, three channels, three paths to the rails |
| C-04 | A-04 | Partial. A-04 exists twice: operationally in the retail core, completely in the warehouse, so the complete view is a day old |
| C-05 | A-04, A-21, A-24 | Partial. Verification works; refresh and the deficient-file queue do not |
| C-06 | A-06, A-07 | Partial. A-07 exists three times |
| C-07 | A-08, A-09 | **Gap.** The hub does not own A-09, so screening is inline on some payments and after the fact on others, and nobody can say which |
| C-08 | A-09, A-15 | Partial. Mandates registered; reason codes not mapped to retries |
| C-09 | A-10, A-11 | Partial. Issuing sits with a processor, so the bank holds a copy of its own card data |
| C-10 | A-12, A-13 | Partial. Sound for retail; for business the credit paper is a document |
| C-11 | A-14 | **Gap.** Nothing masters D-07; limits in A-14 and A-07 diverge, covenants in a spreadsheet |
| C-12 | A-14, A-15 | Partial. Treatment on days past due alone; prescription reviewed by hand |
| C-13 | A-16, A-20 | Partial. Front, middle and back in one product, limits checked after the fact |
| C-14 | A-20 | Partial. Assembled from extracts, run once, not twice |
| C-15 | A-21 | Partial. Inline for payments, batch for onboarding |
| C-16 | A-22 | Partial. Scoring exists; the pre-release hold on a real-time rail does not |
| C-17 | A-07, A-17 | Partial. Close works; reconciliation certified by hand across three A-07s |
| C-18 | A-18, A-23 | **Gap.** No A-18 in chapter 11's sense: a reporting tool over a mart, with no submission, version or restatement record |
| C-19 | A-02, A-24 | Partial. Complaints recorded; A-24's rendered-artefact archive is missing, so proof of delivery cannot be produced |
| C-20 | A-23 | **Gap.** No lineage from a return line to a source field, and no model register |

Six gaps, fourteen partials and nothing rated full. **The nothing-rated-full is the finding**, not the gaps. A capability is full only when one application serves it and masters its data, and duplication prevents that almost everywhere at Table Bay: the party master twice, the posting engine three times, cards at a processor, limits in two places. That state reads as healthy on an application list, which is why the list is not enough.

The useful output is the ordering, in three bands.

**Fund first, where the downside is quantified and nobody will resist.** C-18 and C-20 carry a supervisory consequence, and the second is the precondition of the first. C-07 is a financial-crime exposure with a named regulator behind it and a hub already bought.

**Second, the organisational fights.** C-01 and C-02 are one build and one argument: moving product definition and pricing out of the core means somebody must own a catalogue. C-11, because the fight is about who may say no.

**Last, the one nobody will sponsor.** C-12 is the cheapest to improve and its beneficiary is the credit loss ratio rather than a department.

## Keeping these alive

**One owner per dimension, named.** Processes to an operations leader, capabilities to strategy, data to the head of data, applications to the lead architect, roles to human resources — who will be surprised to be asked and then produce a better list than yours.

**Never change the hub.** Renumbering processes invalidates every map here. Add `P-13`; do not renumber. Retire an ID; do not reuse it. Treat the series as product versions, because that is what they are.

**Review on a trigger, not a date.** Quarterly will be skipped by the second quarter. Before an architecture board, before a package evaluation, after a reorganisation, after a regulatory instrument lands — triggers survive where calendars do not.

**Publish the gaps with the maps.** A cross-map circulated without its gaps reads as a healthy estate, and the next reader will believe that. The gaps are the only part of this appendix anybody will act on.

### THE CHAPTER IN TEN LINES

- Hinge every cross-map on business process. It is the only dimension the business will contest, and therefore the only one that ever gets corrected.
- Five dimensions, five ID series — processes P, capabilities C, roles R, data entities D, applications A. Keep them local, version them, and never renumber.
- These IDs are not BIAN's, and BIAN publishes no licence for its own: if you want its identifiers, get terms in writing before they appear in a deliverable.
- Two capabilities — product management and pricing — are exercised by no process in chapter 8's inventory. Product design missing from the L1 map is the commonest omission in a bank.
- No application masters Limit. That single empty cell explains why a limit in the core and a limit in the authorisation host drift apart with nobody owning the difference.
- One accountable role per process. Where you cannot honestly name one — the end-to-end card flow — write that in the table instead of inventing an owner.
- At Table Bay nothing is rated full, and that is the finding rather than the five gaps: duplication, not absence, is what stops a capability having one owner.
- This is a worked starting set for an invented bank. Rebuild it in workshops with the named role in the room, and expect a third of the role assignments to change in the first twenty minutes.

### SOURCES FOR THIS CHAPTER


- **1.** BIAN — Service Landscape 14.0 release notes, v1.0 final version (20 February 2026): 322 Service Domains, 38 Business Domains, 586 Business Capabilities; and the deliverables pages, which carry a copyright notice and no licence https://bian.org/wp-content/uploads/2026/02/BIAN-v14.0-Release-Notes-v1.0_-Final-Version.pdf
- **2.** APQC — industry-specific process classification frameworks (Banking PCF 8.0, 26 August 2026); licensing and non-member access terms not stated https://www.apqc.org/process-frameworks/industry-specific-process-frameworks
- **3.** Moonstone — "Customer complaints: FSCA tells banks to raise their game" (March 2025), on the regulator's review of complaints handling under Conduct Standard 3 of 2020 https://www.moonstone.co.za/customer-complaints-fsca-tells-banks-to-raise-their-game/
- **4.** National Financial Ombud Scheme — About us (operations commenced 1 March 2024; four schemes consolidated into Banking & Credit, Life and Non-Life divisions) https://nfosa.co.za/about-us/
- **5.** South African Reserve Bank / CoDI — Member Bank Data Handbook (monthly submission within ten working days of month-end, with a chief executive and chief financial officer declaration) https://www.resbank.co.za/content/dam/sarb/what-we-do/deposit-insurance/CODI%20-%20Member%20Bank%20Data%20Handbook.pdf
- **6.** Precium — "RMS to Registered Mandates: beyond just a technical upgrade" (the mandated 10 March 2025 shift, the 19:00 SAST late window, tracking, and business and dual-signatory account support) https://www.precium.com/blog/rms-to-registered-mandates-beyond-just-a-technical-upgrade
- **7.** Stitch — "What is DebiCheck? South Africa's guide to authenticated debit collections" (mandate held at the debtor's bank; per-collection validation; TT1, TT2 and TT3) https://stitch.money/blog/debicheck-the-ultimate-guide
- **8.** Chargebacks911 — "Visa chargeback time limits: the 2026 guide" (vendor source; verify against Visa's own Dispute Management Guidelines before relying on a specific deadline) https://chargebacks911.com/chargeback-rules/chargeback-time-limits/visa-chargeback-time-limit/

APPENDIX B

# Glossary

*Most banking vocabulary is ordinary English used with a precise and different meaning. These are the words, in plain language, with the South African ones flagged and the dangerous ones marked.*

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Most banking vocabulary is ordinary English used with a precise and different meaning. A balance is not a number you store. An account is three different things. A mandate is three more. A bond in South Africa is not a bond anywhere else. This appendix is the list, written for somebody who has never worked in a bank, with the dangerous ones marked.

Four notes on how to read it.

**Where a word means different things in different parts of a bank, both meanings are given.** The entries for *account*, *mandate*, *balance*, *core banking*, *single customer view* and *bond* are the ones that will save you a wasted meeting.

**Where a South African term has no overseas equivalent, or an overseas term means something different here, it says so.** The entries that matter most are *DebiCheck*, *Registered Mandates*, *NCR*, *bond*, *interchange*, *EFT*, *USSD*, *stokvel*, *in duplum*, *section 129 notice* and *open banking* — the last because South Africa does not have it.

**Abbreviations are filed as they are spoken.** A banker says "LCR", "ECL" and "BA 900", not the words behind them, so that is where you will find them.

**A definition is not a specification.** Where the book's research could not settle a detail — the *in duplum* rule and *covered bonds* are the two clearest cases — the entry says so rather than guessing.

| Term | What it means |
|---|---|
| **12-month ECL** | The impairment a bank holds on a loan that is still performing: the losses it expects from defaults in the next twelve months only, not over the loan's whole life. |
| **3-D Secure** | The extra authentication step on an online card payment — a prompt from your bank in its app or by one-time code. It moves liability for a disputed transaction towards whoever did not use it. |
| **ACB file** | The fixed-width file format family used for South Africa's batch payment and collection rails, named after the Automated Clearing Bureau. Every bulk banking channel implements it, and the current layouts are not public. |
| **Access bond** | A South African home loan feature letting you redraw capital you have already repaid. Useful for the customer, awkward for a system that assumed a loan balance only goes down. |
| **Account** | The customer-facing thing that opens, holds a balance and closes. In a bank the word means at least three different things — the customer's product instance, a ledger account in the general ledger, and a nostro account at another bank — and conflating them is a classic modelling error. |
| **Acquiring** | The merchant side of card payments: signing up shops, providing terminals, and paying them for what they sell. The acquirer carries the risk that the merchant does not deliver. |
| **Action date** | The date a batch collection or payment is meant to hit the other party's account. It is not the date the file was submitted, and the gap between them is where most payment queries live. |
| **AEDO / NAEDO** | Authenticated and non-authenticated early debit orders, designed to collect from low-income debtors right after payday. Both were switched off on 31 October 2021 and replaced by DebiCheck, but their records and reason codes survive in historic data. |
| **Affordability assessment** | The calculation a South African lender must perform before granting credit, prescribed in regulation 23A of the National Credit Act, with prescribed inputs including a table of minimum assumed living expenses. It is a legal test, not a credit policy choice. |
| **Agent desktop** | The screen a contact centre or branch consultant works in, which aggregates several underlying systems. Usually the worst integration in the bank and the place screen scraping is most often found and least often admitted. |
| **ALCO** | The asset and liability committee: the forum where a bank decides how it is funded, what internal price it charges its own lending units for money, and how much interest-rate risk it will carry. |
| **ALM** | Asset and liability management — managing the mismatch between what a bank owes (mostly short) and what it is owed (mostly long), and the interest-rate and liquidity risk that mismatch creates. |
| **Amortisation** | Repaying a loan's capital gradually so the balance reaches zero at the end of the term. The opposite of a loan where the capital is all repaid at the end. |
| **APQC Banking PCF** | A published hierarchy of banking business processes with stable identifiers, designed for benchmarking against other banks. Useful for decomposition; it has no data model, so it cannot settle which system owns what. |
| **ASF / RSF** | Available and required stable funding: the two halves of the net stable funding ratio. Regulators prescribe a factor for each kind of funding and each kind of asset, and the ratio of the two must stay above 100%. |
| **ATM interchange** | The fee moving between banks when someone uses another bank's cash machine. In South Africa the central bank determines it, which is unusual — in most markets it is commercial or capped by regulation. |
| **Attach rate** | The share of customers who take a second product. In a bank it is the main lever on the profitability of a customer the bank has already paid to acquire. |
| **Authorisation** | The issuer's real-time yes or no on a card transaction, which also reserves the money. It is a promise to pay later, not a payment, and the two are days apart. |
| **Authorisation hold** | The amount reserved on a card account by an authorisation, reducing what the customer can spend. If nothing releases an expired hold, the customer's available balance is wrong and nobody notices. |
| **BA 099** | The signed certification that accompanies a bank's prudential return submission in South Africa. Somebody puts their name to the number, which is why lineage matters. |
| **BA 900** | South Africa's monthly statutory bank balance-sheet return. It is also the authoritative public source for market shares, because it is the one place every bank's numbers are compiled on the same basis. |
| **BA returns** | The series of prudential returns South African banks file with the Prudential Authority. Since GoN 5802 of 31 January 2025 the forms sit outside the Regulations and are determined by the Authority in writing, so they move without a gazette. |
| **Balance** | Not a stored fact but a derivation: an amount at a point in time, on a stated definition. A bank typically computes at least three — ledger, available and cleared — and showing the wrong one is a complaint. |
| **Balloon** | A large final instalment at the end of a vehicle-finance agreement, which keeps the monthly payment low. Common in South Africa and a frequent source of customer surprise. |
| **Bancassurance** | Insurance sold through a bank, sometimes on the bank's own licence and sometimes on a partner's. It matters architecturally because the regulatory regime, the advice record and the product model all come from the insurance world. |
| **Bank Zero** | One of South Africa's digital banks, notable for self-building its platform on open-source software rather than buying a core. Mentioned in this book only where its own published case study supports the detail. |
| **BankservAfrica** | The former name of PayInc, South Africa's clearing and switching utility. If you read "BankservAfrica" in a document, check its date: the rename was in August 2025. |
| **Basel III / Basel IV** | The international standards for how much capital and liquidity a bank must hold. South Africa's Prudential Authority calls the final 2025 package Basel III, not Basel IV, which matters when you are searching for the local instrument. |
| **Batch window** | The hours overnight in which a bank runs interest accrual, fees, arrears progression, statement production and file generation. It only ever shrinks, and a late batch is not a one-day outage but a one-day outage plus a reconciliation tail. |
| **BCBS** | The Basel Committee on Banking Supervision, which writes the international standards. It has no power of its own: national regulators transpose its standards into local law, and the local version is what binds you. |

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| **BCBS 239** | Fourteen principles on risk data aggregation and risk reporting, published in 2013. It is the only framework in banking a bank can be supervised against for its data architecture, it is still unfinished nearly everywhere, and it fails on lineage rather than on analytics. |
| **Behaviour Qualifier** | In BIAN, a named sub-part of the single record a Service Domain owns, which service operations act on. Version 14.0 renamed a large number of them, which is a hidden cost for anyone who hard-coded the old names. |
| **Behavioural maturity** | How long a deposit actually stays, as opposed to when the contract says it can be withdrawn. A current account is legally repayable on demand and behaviourally sticky for years, and the whole funding model rests on the difference. |
| **Beneficial owner** | The actual human being who ultimately owns or controls a company. Finding them is a legal obligation when onboarding a business customer, and it requires modelling an ownership graph rather than a field. |
| **BIAN** | The Banking Industry Architecture Network: banking's main reference model, a vocabulary of Service Domains and the records they own. It is a dictionary, not a design, and BIAN publishes no licence for its content. |
| **BIC** | The Business Identifier Code, the eight or eleven characters identifying a financial institution in international messaging. Often called a SWIFT code. |
| **BIN** | The leading digits of a card number, identifying the issuing bank and the product. The tables that map them are reference data a card system cannot work without. |
| **Bitemporality** | Keeping two kinds of time: when something was true in the world, and when the bank recorded it. In a bank you usually need a third — when it takes effect — and the regulator will ask you to reproduce all three. |
| **Bond** | In South Africa this means the registered mortgage over a property, not a tradeable debt security. Context tells you which, and getting it wrong in a workshop is embarrassing. |
| **Bond originator** | A South African intermediary who takes one home loan application and submits it to several banks at once. It changes the data a bank starts with and the competitive dynamics of the decision. |
| **BOP category** | The balance-of-payments reporting code South Africa requires on every cross-border payment, under exchange control. No system can infer it, so somebody has to supply it, and that makes it a hard field requirement with no overseas equivalent. |
| **Brownfield migration** | Replacing a core platform while carrying the existing book across. The opposite of greenfield, vastly harder, and the thing most "core replacement" programmes actually are. |
| **Bullet** | A loan structure where the whole capital is repaid at maturity and only interest is paid before that. |
| **Business day** | In a bank, not a calendar concept but an operational one: cut-offs, clearing windows, settlement windows and an end-of-day position. A customer journey designed without knowing the business day will silently miss a window. |
| **Business Process Catalogue** | One of the two halves of the ISO 20022 repository, alongside the Data Dictionary. Both are free to access, which makes ISO 20022 the richest free financial data model in existence. |
| **Capital conservation buffer** | An extra slice of best-quality capital, up to 2.5%, that a bank must hold above the minimum. Dipping into it does not close the bank; it restricts dividends and bonuses, which is the point. |
| **CAR** | Capital adequacy ratio: total qualifying capital divided by risk-weighted assets. A headline supervisory number and a board-level constraint on growth. |
| **Card-not-present** | A card transaction without the physical card — online, by phone, in an app. South Africa's largest card fraud category, which is why three-domain-secure exists. |
| **CASA** | Current and savings accounts. The cheapest and stickiest money a bank has, and the main reason a retail franchise is worth owning. |
| **CBPR+** | Swift's programme that moved cross-border payment messaging to ISO 20022 and retired the legacy MT equivalents. It is an external timetable no individual bank controls. |
| **CCyB** | The countercyclical capital buffer: extra capital required when credit growth looks exuberant, released when it does not. South Africa set a positive 1% buffer from 1 January 2026. |
| **CDD / EDD** | Customer due diligence, and the enhanced version for higher-risk customers: establishing who someone is, what they do, and whether that story holds together. Not a one-off at onboarding — it has to be refreshed. |
| **Central Data Transmission Hub** | SACRRA's routing service, which it describes as the only legally recognised hub for sending South African credit data to the bureaux. It is a mandatory monthly interchange obligation with no equivalent in any global reference architecture. |
| **CET1** | Common equity tier 1: the best quality capital, essentially ordinary shares and retained earnings. The ratio every bank's results presentation leads with. |
| **Champion-challenger** | Running a new credit or collections strategy on a sample of customers against the existing one, to measure whether it is actually better. It requires holding two strategies live at once, which not every system can do. |
| **Chargeback** | The scheme process that pulls a disputed card payment back from the merchant, through the acquirer. It runs on scheme deadlines that are not negotiable, which is why the clocks, not the case, drive the process. |
| **Check digit** | A digit computed from the rest of an account or card number, which lets a system reject a typo before it becomes a misdirected payment. |
| **Cheque** | Withdrawn entirely from the South African payment system at the end of December 2020. "Cheque account" survives as a product name, which confuses every newcomer and some data models. |
| **CIT** | Cash in transit: moving physical notes around in armoured vehicles. A real cost line and a real risk register entry in a country where cash demand has not fallen as fast as the branch network. |
| **Clearing** | Exchanging and reconciling payment instructions between banks so each knows what it owes. It is not settlement, which is the actual transfer of value afterwards. |
| **CLR** | Credit loss ratio: the impairment charge as a proportion of average loans, usually quoted in basis points. The single number that tells you how a lending book is behaving. |
| **CoDI** | The Corporation for Deposit Insurance, operational from 1 April 2024, which protects up to R100,000 per depositor per bank. Its real architectural weight is the data it demands, not the cover it provides. |

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| **COFI** | The Conduct of Financial Institutions Bill: South Africa's planned consolidated conduct framework for the whole financial sector. Still a bill, with its parliamentary stage moving, so check the date on anything you read about it. |
| **Collateral** | The asset a lender can take if the borrower does not pay. Its type, value, valuation date and registration state all matter, because an unregistered security is not a security. |
| **Compounded in arrears** | Working out interest by compounding an overnight rate across the period, rather than fixing a term rate up front. The convention that replaces JIBAR-style rates, and it changes when the borrower learns what they owe. |
| **Conditions precedent** | Things that must be done before a loan can be drawn — security registered, insurance in place, a guarantee signed. Waiving them informally is one of the commonest control failures in business lending. |
| **Conduct Standard for Banks** | Conduct Standard 3 of 2020, South Africa's market-conduct rules for banks. Section 8 prescribes complaint categories, acknowledgement, reporting and monitoring, which turns complaint handling into a data obligation. |
| **Control Record** | In BIAN, the single master record a Service Domain owns through its whole lifecycle. One record per domain is a data-ownership rule hiding inside a vocabulary, and it is the most useful part of BIAN. |
| **Core banking** | The most abused term in the industry. It can mean the account ledger, or the deposit and loan product engine, or everything the bank runs — and asking five people will get five answers, so ask what they mean before agreeing anything. |
| **Coreless Banking** | A BIAN initiative arguing a bank can be assembled from modular API services across its domains rather than anchored on a monolith. Launched in 2019; two of its original vendor participants have since been absorbed by others. |
| **Correspondent banking** | One bank holding an account for another so it can make payments in a currency or country where it has no presence. The chain of correspondents is why a cross-border payment can take days. |
| **Cost of deposits** | What a bank pays specifically for customer deposits, as opposed to all its funding. The cheaper this is, the more competitive the lending book can be. |
| **Cost of equity** | The return shareholders expect for carrying a bank's risk. Earning below it destroys value even while reporting a profit. |
| **Cost of funds** | The weighted average rate a bank pays on everything that funds it. Lower is better, and a retail deposit base is the cheapest source there is. |
| **Cost to serve** | The fully loaded cost of looking after one customer or processing one transaction. The number that decides whether a branch, a channel or a product survives. |
| **Cost-to-income** | Operating costs divided by total income. The headline efficiency measure for a bank; the South African majors ran around 51% in FY2025. |
| **Covenant** | A testable promise in a business loan — keep this ratio below that level, file accounts by this date. It has its own monitoring cycle, and nothing in a retail lending system has the concept at all. |
| **Coverage ratio** | Loss allowance as a proportion of gross exposure, or of non-performing loans. It depends heavily on product mix, so comparing it across banks tells you less than it appears to. |
| **Covered bond** | A bank bond secured on a ring-fenced pool of assets that stays on the balance sheet. Whether these are actually issued in South Africa could not be confirmed in the research behind this book. |
| **Covered deposit** | A deposit that qualifies for deposit-insurance protection, within the limit. Working out which deposits are covered and to what extent requires aggregating per depositor, which is a data problem before it is a policy one. |
| **CRA** | Cash reserve account: the balance a South African bank holds at the central bank. Up to 70% of it can be used for settlement during the day, and the overnight clearing batches can draw only on this balance. |
| **Credit adjustment spread** | The extra margin added when a contract converts from a term rate like JIBAR to an overnight-based rate, compensating for the credit risk the old rate contained. |
| **Credit bureau** | A registered holder of consumer credit records. In South Africa there is a long register of bureaux and a much shorter list authorised to hold payment profile information, and only the second set is on the mandatory submission path. |
| **Credit committee** | The forum that approves business facilities above an individual's mandate. Its decisions are papers approved under delegated authority with a quorum, which is a different system shape from a retail score cut-off. |
| **Credit cost multiple** | The ratio of the total cost of credit to the amount borrowed, which South African lenders must disclose. It is the figure that makes a small short-term loan look as expensive as it is. |
| **Credit enhancement** | Protection built into a securitisation so the senior notes are safe: retained profit, junior tranches, or a first-loss loan from the originator. |
| **Credit life insurance** | Cover that settles a debt if the borrower dies or is disabled. In South Africa the premium is capped by regulation at R4.50 per R1,000 of outstanding debt, and R2 for mortgages. |
| **Credit tracking** | Repeatedly re-presenting an unpaid collection over a tracking period, waiting for money to arrive in the account. Originally a feature of NAEDO; the Registered Mandate rail now runs tracking over up to ten days. |
| **Curatorship** | Court-appointed management of a failing bank under the Banks Act. One of several outcomes in a resolution, and the one most people have heard of. |
| **Cure** | An arrears account returning to good standing. The measure that tells a collections strategy whether it is working. |
| **Cut-off** | The time by which a file must be submitted for a batch rail to process it that day. South Africa's authoritative cut-off schedule is not published, which is a genuine and uncomfortable gap. |
| **CVA** | Credit valuation adjustment: capital held against the risk that a derivatives counterparty fails. A markets-business concept that lands in a bank's capital calculation. |
| **Data Dictionary** | The ISO 20022 repository's catalogue of business concepts, elements, associations and roles. A financial-services conceptual data model with change control, which almost nobody reads because most banks only ever see the generated XML. |
| **DebiCheck** | South Africa's authenticated collection rail: the customer confirms the arrangement once with their own bank, which then validates every collection against the agreed terms. There is no European or British equivalent, and a design imported from either will not model it. |
| **Debit order** | A recurring collection pulled from the payer's account by the party owed money. The opposite of a stop order, which the payer pushes. |

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| **Debt review** | South Africa's statutory process for restructuring the debts of an over-indebted consumer, run by registered debt counsellors. A customer under debt review must be treated differently by every system that touches them. |
| **Deconversion** | Leaving a core banking provider: getting your data out, in a usable form, at a known price. Ask about it at selection, not at exit, and write it into the contract. |
| **Deposit franchise** | A bank's ability to raise cheap, stable money from ordinary customers. In South Africa it is the main source of margin advantage, which is why the big banks fight over transactional accounts. |
| **DIF** | The Deposit Insurance Fund, which pays out if a South African bank fails. It reached its R20 billion target in the year to 31 March 2025. |
| **Directive 2 of 2024** | The South African Reserve Bank instruction regulating third parties that initiate electronic payments using a customer's internet banking credentials. It requires registration, per-payment consent, no credential storage, five-year audit trails and monthly reporting — regulating screen scraping rather than banning it. |
| **Drawdown** | Actually taking the money under a facility that has been approved. Approval and drawdown are different events, and the gap between them is where conditions precedent live. |
| **D-SIB** | A domestic systemically important bank: one large enough that its failure would matter nationally, which attracts extra capital and closer supervision. |
| **DTI** | Debt to income: total debt service as a share of gross income. One of the two ratios, with loan to value, that decide a mortgage. |
| **EAD** | Exposure at default: how much the bank expects to be owed at the moment a borrower defaults. Harder than it sounds for anything with an undrawn limit. |
| **EAO** | An emoluments attachment order: a court order requiring an employer to pay part of an employee's salary to a creditor. Often loosely called a garnishee, which strictly means something else. |
| **ECL** | Expected credit loss: the forward-looking impairment measure IFRS 9 introduced. It replaced waiting for a loss to happen with modelling the loss you expect, which turned impairment into a model-governance problem. |
| **Economic profit** | Earnings after charging for the capital the business consumed, at the cost of equity. A positive accounting profit can be a negative economic one. |
| **Effective dating** | Recording the date from which a change applies, separately from when it was captured. Without it a bank cannot backdate a correction, and backdating corrections is routine. |
| **EFT** | Electronic funds transfer: in South Africa, specifically the batch credit and debit rails, not a generic term for any electronic payment. EFT credit remains the overwhelming majority of retail payment value by settlement. |
| **EMV** | The chip-card standard. Its real significance is liability: the party that is not EMV-compliant carries the cost of counterfeit fraud. |
| **EMVCo** | The body, owned by the card schemes, that publishes the chip, contactless, three-domain-secure and tokenisation specifications. |
| **Endorsement** | A recorded change to a contract. More an insurance word than a banking one, but it turns up in bancassurance conversations. |
| **Endowment** | The margin a bank earns because some of its funding pays little or no interest. When the policy rate rises, endowment income rises with it for no extra effort, which flatters earnings in a rate upcycle. |
| **Entry** | The immutable, signed amount posted against one ledger account. Double entry means they come in balanced pairs, and nothing in a bank is more fundamental or less interesting to the business. |
| **ESB** | An enterprise service bus: the central integration platform most banks bought between 2005 and 2015. It usually failed for organisational rather than technical reasons, and it is usually still running. |
| **EVE** | Economic value of equity: what the bank's equity is worth in present-value terms if rates move. A measure of structural interest-rate risk, with a supervisory ceiling attached. |
| **Exchange control** | South Africa's regime for money leaving the country, administered by the Reserve Bank's Financial Surveillance Department. Every cross-border payment is a reporting event with a mandatory category code, which surprises every architect arriving from elsewhere. |
| **Facility** | An agreed availability of credit, drawn or not. A facility is a structure with sub-limits, not a loan, and business lending systems need that distinction where retail ones do not. |
| **FAIS** | The Financial Advisory and Intermediary Services Act, which governs giving financial advice in South Africa. Where a bank advises rather than only sells, FAIS brings a record-keeping and suitability regime with it. |
| **Fallback** | The contractual wording that says what rate applies if the benchmark it references stops existing. Repapering fallbacks is most of the work in a benchmark transition. |
| **FATF** | The Financial Action Task Force, which sets the international anti-money-laundering standards and publishes the lists. South Africa was grey-listed in February 2023 and exited on 24 October 2025. |
| **FATF grey list** | The FATF's list of jurisdictions under increased monitoring. Being on it raises the cost and friction of every correspondent relationship, which is why exiting it in October 2025 mattered commercially and not just reputationally. |
| **FDX** | The Financial Data Exchange and its API, the market-led data-sharing standard in the United States. Free to use under its own royalty-free licence. |
| **Fee-to-income ratio** | Non-interest revenue as a share of total income. A rough measure of how much a bank earns from doing things rather than from lending. |
| **FIBO** | The Financial Industry Business Ontology, published by the EDM Council: a machine-readable model of what financial words mean. A semantics asset for the data and risk estate, not a design for the operational one. |
| **FICA** | The Financial Intelligence Centre Act: South Africa's anti-money-laundering statute, and the source of the identity and address verification every new customer has to pass. If a residential address cannot be verified, the relationship may not be opened at all. |
| **Finance lease / operating lease** | Two kinds of lease: one transfers substantially all the risks and rewards of owning the asset to the customer, the other does not. The distinction drives the accounting and the asset model. |
| **Financial Intelligence Centre** | South Africa's financial intelligence unit, which receives suspicious transaction and cash threshold reports and publishes the sanctions guidance. The FIC in "FICA". |

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| **FinSurv** | The Reserve Bank's Financial Surveillance Department and its reporting system, which captures every cross-border payment for balance-of-payments purposes. A hard integration requirement with no overseas analogue. |
| **FIX** | The de facto messaging protocol for electronic securities trading. Relevant to a markets business, irrelevant to a retail core. |
| **FLAC** | Flow-through loss-absorbing capacity: instruments that can be written down to recapitalise a failing bank. Part of South Africa's resolution regime, with details still settling. |
| **Float** | Value in transit between leaving one account and arriving in another, which somebody economically holds in the meantime. Faster rails shrink it, which is one reason banks were slow to want them. |
| **FpML** | The XML standard for over-the-counter derivatives. Its modern successor, the Common Domain Model, models trade lifecycle events rather than documents. |
| **FRN** | A floating rate note: a bond whose coupon moves with a benchmark rate rather than being fixed. |
| **FRTB** | The revised framework for market risk capital, which South African institutions had to comply with from 1 July 2025. It is the shape of a Basel-driven architecture programme: years long, often built in-house, with a parallel run. |
| **FSCA** | The Financial Sector Conduct Authority: South Africa's market-conduct regulator under Twin Peaks, and the author of the Conduct Standard for Banks and the draft Open Finance paper. |
| **FTP** | Funds transfer pricing: the internal price a bank charges its lending units for money and pays its deposit units for raising it. It decides which products look profitable, which is why nobody fully trusts it. |
| **Fully banked** | A customer who uses this bank as their main transactional relationship, with salary in and spending out. The metric retail banks actually compete on, because it predicts everything else. |
| **Garnishee** | Strictly, a third party who owes money to a debtor. In South African conversation it is used loosely to mean an emoluments attachment order, which is a different thing. |
| **GoN 5802** | The January 2025 amendment that removed the prudential return forms from South Africa's Regulations relating to Banks, so returns are now determined by the Authority in writing. It is not the Basel III post-crisis reform instrument, which took effect separately on 1 July 2025. |
| **Greeks** | The named sensitivities of a derivative's value to the things that move it — rates, volatility, time. A markets-desk vocabulary that shows up in risk reporting. |
| **Greenfield launch** | Building a new bank with no existing book to migrate. It is a genuinely easier problem than replacing an incumbent's core, and conflating the two is the commonest error a reader imports from conference material. |
| **Haircut** | The discount applied to the value of security in a funding transaction, so the lender is covered if the collateral falls in price. |
| **Headline earnings** | The earnings measure the Johannesburg Stock Exchange requires listed South African companies to report, excluding specified capital items. The number the market quotes. |
| **HHI** | The Herfindahl-Hirschman Index of market concentration. South African banking scores around 0.180, which is concentrated by most standards. |
| **HLA** | Higher loss absorbency: the extra capital a systemically important bank must hold, up to 2.5%, because its failure would matter more. |
| **Hollowed core** | A core banking system reduced to a ledger and a posting engine, with product definition, customer identity, orchestration and analytics moved out around it. It is a deliberate target state for some banks and an accidental destination for most. |
| **HQLA** | High-quality liquid assets: the government bonds and central-bank balances that count towards the liquidity coverage ratio. Safe, liquid and low-yielding, which is the point. |
| **ICAAP** | The internal capital adequacy assessment process: the bank's own argument, in writing, that it holds enough capital for its own risks. The supervisor reads it. |
| **IFRS 9** | The accounting standard that made loan impairment forward-looking, through expected credit loss and three stages. It is the most data-hungry accounting standard a bank faces, because every number has to be reproducible years later. |
| **IFW / BDW / BFMDW** | IBM's banking data warehouse lineage — the Information FrameWork, the Banking Data Warehouse, and the Banking and Financial Markets Data Warehouse that replaced both. Visibly quiet as a product, and still the only documentation of many older bank warehouses. |
| **IFX** | The Interactive Financial eXchange, the closest historical equivalent to a retail-banking interchange standard. Its forum merged into Nacha in 2018, which is a cautionary tale about industry standards bodies. |
| **IIN** | The issuer identification number: the modern name for a card BIN, and currently migrating from six digits to eight, which is a reference-data change with a long tail. |
| **Ijarah** | A Sharia-compliant lease where the bank owns the asset and the customer pays to use it, with ownership passing at the end of the term. |
| **Impaired advances** | The South African regulatory term for non-performing loans. Around 5.2% of gross loans on 2025 sector averages. |
| **Impaired record** | A credit bureau record showing serious arrears, an adverse listing or a judgment. Around 10.4 million South African consumers had one as at March 2025, against roughly 28.9 million who were credit-active. |
| **Impairment stage** | Which of IFRS 9's three buckets a loan sits in: performing, significantly deteriorated, or credit-impaired. The stage decides how much provision is held, so the rule that moves a loan between stages is one of the most consequential pieces of logic in a bank. |
| **In duplum** | A South African common-law rule capping accrued interest at the amount of capital outstanding. Widely referred to and, in the research behind this book, not sourceable to a primary text — treat the detail with care. |
| **Inflation target** | The range the central bank aims to keep inflation within. South Africa moved to a 3% point target with a one percentage point band from 12 November 2025, which changes the expected path of the policy rate. |
| **Instalment sale agreement** | South African vehicle finance where the financier keeps ownership of the car until the last payment is made. It affects how the asset and the security are modelled. |
| **Interchange** | The fee that moves between the issuing and acquiring banks on a card transaction. In South Africa the central bank determines it for several streams, which is the reverse of the European position where regulation caps a commercially set fee. |

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| **IRRBB** | Interest rate risk in the banking book: the risk that changes in rates hurt the bank's earnings or the value of its equity, arising from ordinary lending and deposit-taking rather than trading. |
| **ISO 20022** | Not a message format but a methodology plus a repository of business models from which messages are generated. South Africa's high-value settlement system and its instant payment rail are both native to it, while the retail batch rails are not. |
| **ISO 8583** | The message standard a card authorisation travels in, from terminal to switch to scheme to issuer. Every participant runs a dialect of it, which is why switch integration is never a standards exercise alone. |
| **Issuing** | The cardholder side of cards: giving customers cards and carrying the credit risk on what they spend. |
| **Jaws** | The gap between income growth and cost growth. Positive jaws improves the cost-to-income ratio; negative jaws is what gets a chief executive asked difficult questions. |
| **JIBAR** | The Johannesburg Interbank Average Rate, South Africa's forward-looking term benchmark. It ceases at the end of December 2026 and is being replaced by ZARONIA, which means repapering contracts and changing interest calculations. |
| **Joint Standard** | A standard issued jointly by South Africa's Prudential Authority and the FSCA. Four are commonly confused and three are numbered "1": for banks the ones that apply are Joint Standard 1 of 2023 on IT governance and risk and Joint Standard 2 of 2024 on cybersecurity, while Joint Standard 1 of 2024 is outsourcing by insurers and does not apply to banks. |
| **King IV** | South Africa's corporate governance code, which puts board-level accountability on technology and information. It is the reason an architecture governance forum in a South African bank has a governance mandate and not just a technical one. |
| **KYC** | Know your customer: the industry shorthand for identifying and understanding a customer. In South Africa the statutory version of it is FICA's customer due diligence. |
| **LCR** | Liquidity coverage ratio: high-quality liquid assets divided by the net cash outflow expected in a thirty-day stress, which must stay above 100%. South African banks have typically run around 149%. |
| **Legacy** | In banking, not an insult but a description: systems that work, are load-bearing, and are understood by fewer people every year. The constraint is usually the skills pipeline rather than the technology. |
| **Leverage ratio** | Tier 1 capital divided by total exposure with no risk weighting at all. A deliberately crude backstop against clever risk modelling. |
| **LGD** | Loss given default: the share of an exposure a bank does not recover after default and after realising security. It can only be validated if recovery outcomes are fed back, which is often where the data chain breaks. |
| **Lifetime ECL** | The impairment held on a loan whose credit risk has significantly increased, or which is already impaired: expected losses over the loan's whole remaining life rather than just twelve months. |
| **Limit** | The amount that may not be exceeded, by whom, over what, from when. A limit held in three systems that disagree is one of the most reliable findings in any bank. |
| **Lineage** | Being able to trace a number on a regulatory return back to the field it came from, through every transformation. The thing BCBS 239 asks for and almost nobody can demonstrate. |
| **Liquidity premium** | The part of an internal funding charge that compensates for how long the money is tied up. Without it, long lending looks artificially profitable. |
| **Lookback / observation shift / lockout** | Conventions for when the overnight rates in a compounded-in-arrears calculation are actually observed, so an interest amount is known slightly before payment is due. Small-sounding choices with real system consequences. |
| **LTV** | Loan to value: the advance as a proportion of the property or asset value. Along with debt to income, it is what decides a mortgage and what drives its risk weight. |
| **Managed Evolution** | Nedbank's core modernisation programme: nine years from 2016, R11.7 billion, with the core delivered last at the end of 2024. The best-documented South African example, and attributed here to the trade press that reported it rather than to the bank. |
| **Management overlay** | A judgemental addition to a modelled impairment number, for something the model cannot see. Legitimate, necessary and the first thing an auditor asks about. |
| **Mandate** | Authority to act: who may do what, over what, up to how much, from when. A bank has at least three kinds — the account operating mandate, the collection mandate, and an employee's internal approval authority — and all three are dated authorities with limits, not flags. |
| **Mashonisa** | An informal, usually unregistered, South African moneylender. Part of the credit landscape a bank competes with and a reason some customers' obligations never appear on a bureau record. |
| **Minimum Expense Norms** | The table of assumed minimum living expenses prescribed under South Africa's affordability regulations. A statutory input to a credit decision, which means it belongs in versioned reference data and not in code. |
| **MIRS** | The Market Infrastructure Resilience Service: the contingency system behind South Africa's settlement system, implemented in 2023 and running ISO 20022. |
| **Model risk** | The risk that a model is wrong or wrongly used, and the independent function that validates every material model and decides whether it may be used at all. A bank runs on models, so this is a real control and not a formality. |
| **MPC** | South Africa's Monetary Policy Committee, which sets the repo rate. Its decisions reprice a bank's whole lending catalogue, because the statutory credit maxima are formulas on the repo rate. |
| **MPG** | The Market Practitioners Group, the body overseeing South Africa's move away from JIBAR. Its milestones are the dates a bank's transition plan has to hit. |
| **MSF** | Merchant service fee: the total charge an acquirer levies on a merchant, of which interchange is only a part. |
| **MTM** | Mark to market: revaluing a position at today's prices rather than what was paid for it. It makes profit and loss move daily, which is why independent price verification exists. |
| **Mudaraba** | A Sharia-compliant arrangement where the bank provides the capital, the client provides the management, and profits are shared on agreed terms. |
| **Murabaha** | Sharia-compliant cost-plus financing: the bank buys the asset and resells it to the customer at a disclosed mark-up, payable over time. Economically similar to a loan, structurally not one, and the product model has to reflect the difference. |

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| **Musharaka** | A Sharia-compliant joint venture where bank and client both contribute capital and share profits on agreed terms and losses in proportion. |
| **NCA** | The National Credit Act 34 of 2005: South Africa's consumer credit statute. It is prescriptive rather than principles-based — a mandated affordability calculation, a pre-agreement quotation, a notice before enforcement — which makes lending a retained-record problem. |
| **NCD** | A negotiable certificate of deposit: a wholesale deposit that can be traded. One of the ways a bank funds itself beyond retail money. |
| **NCR** | In a South African banking sentence this means the National Credit Regulator, which registers credit providers and bureaux. It does not mean the technology company of the same initials, and the ambiguity has wasted real meeting time. |
| **Netting** | Offsetting what parties owe each other so only the difference moves. It is what makes batch clearing affordable, and it is why the amount settled is far smaller than the amount paid. |
| **NFO** | The National Financial Ombud Scheme, operational from 1 March 2024, which merged four previous ombud schemes into divisions for banking and credit, life insurance and non-life insurance. Complaints escalate here, with their own clock. |
| **NII** | Net interest income: interest earned on loans and investments less interest paid on deposits and funding. One of a bank's two revenue engines. |
| **NIM** | Net interest margin: net interest income as a proportion of average interest-earning assets. The number that tells you whether lending is being priced well. |
| **NIR** | Non-interest revenue: fees, commissions, trading and insurance income. The other revenue engine, and computed completely differently from the first. |
| **Nostro** | "Our account with them": an account a bank holds at another bank, usually in a foreign currency. Reconciling nostros is unglamorous, lagging in most banks, and a classic internal-fraud vector. |
| **NPL** | Non-performing loan: a loan where the borrower has stopped paying as agreed. In South African regulatory language, an impaired advance. |
| **NSFR** | Net stable funding ratio: stable funding available against stable funding required over a year, which must stay above 100%. It constrains how much long-dated lending short-dated money can support. |
| **Obligations register** | A written map from each legal obligation to the control that satisfies it and the system that evidences it. Compliance officers want one; very few banks have one that is current. |
| **Obligor group** | The set of related companies and people whose credit risk must be assessed together, because they would fail together. Business lending decides at this level, which means the data model needs a party-to-party exposure graph. |
| **Ombud Council** | The South African body that recognises and oversees the financial ombud schemes. The scheme a bank actually answers to is the National Financial Ombud. |
| **Open banking** | Mandated, standardised sharing of customer banking data and payment initiation with licensed third parties. **South Africa does not have it.** What exists is bilateral commercial APIs and a regulated screen-scraping regime, and writing otherwise is the most damaging single error in this subject area. |
| **Open finance** | The broader version of open banking, covering savings, investments, insurance and credit as well as payments. South Africa's conduct regulator has published a draft paper exploring a phased mandatory regime, with no dates. |
| **openFinance API Framework** | The Berlin Group's European API standard, free under a Creative Commons licence with no fee to participate. The cleanest licensing terms of any standard in this book. |
| **Output floor** | A Basel limit on how much less capital an internally modelled approach may produce compared with the standardised one. It means a bank must run the calculation twice, which is an architectural requirement before it is a capital one. |
| **Overdraft** | A revolving facility on a current account, repayable on demand, which lets the balance go negative up to a limit. The undrawn part still consumes capital, which is a frequent modelling omission. |
| **pacs.008** | The ISO 20022 message for a customer credit transfer between banks. Worth knowing by name, because it appears in South Africa's exchange-control reporting rules. |
| **PAN** | The primary account number: the long number on a payment card. What may be stored, where, and in what form is set by the card security standard, and getting it wrong is a reportable incident. |
| **PASA** | The Payments Association of South Africa, formerly the recognised body that set scheme and clearing rules. **Its recognition was withdrawn in 2026 and its functions were split between the Reserve Bank itself and PayInc** — so material describing PASA in the present tense is out of date. |
| **PayInc** | South Africa's national clearing and switching utility, renamed from BankservAfrica in August 2025, and 50% owned by the Reserve Bank since late 2025. It is a utility, not an industry body, and the distinction matters. |
| **Payment clearing house** | A scheme arrangement for one payment stream — EFT credits, card, instant payments — with its own rulebook, participants and cut-offs. In South Africa some now sit with the Reserve Bank and some with PayInc. |
| **Payment instruction** | What the bank was asked to do. Distinct from the payment, which is the resulting movement of value, and from settlement, which is the discharge between banks — one box called "payment" costs a programme a year. |
| **Payment profile information** | The detailed record of how a consumer has actually paid their accounts, which only a subset of South African credit bureaux are authorised to hold. The distinction between being registered as a bureau and being authorised to hold this is the one an architect needs. |
| **Payments hub** | A single component that takes a payment instruction from any channel, decides which rail it should use, screens it, and tracks its status. Most banks have one and most have channels that bypass it. |
| **PayShap** | South Africa's instant payment service, live since March 2023, native to ISO 20022, capped at R50,000, and addressable by a mobile-number proxy. Published volume figures from different sources do not reconcile. |
| **PayShap Request** | The request-to-pay feature on PayShap, launched in December 2024: the payee asks and the payer approves. Per-bank limits differ widely. |
| **PCI DSS** | The card industry's data security standard, currently version 4.x, which sets what a bank or merchant may store and how it must be protected. Its requirement on stored account data is where your card data model is effectively decided. |
| **PD** | Probability of default: how likely a borrower is to stop paying, over a stated horizon. One of the three model outputs behind both impairment and capital. |

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| **PEM** | The Reserve Bank's Payments Ecosystem Modernisation programme, launched in September 2025: a new multi-currency settlement system, faster payments expansion, activity-based licensing and a national QR standard among its workstreams. |
| **PEP** | A politically exposed person, whose accounts attract enhanced due diligence. Identifying them, and their close associates, is a screening and data-matching problem rather than a policy one. |
| **Pillar 2A / Pillar 2B** | South African capital add-ons: 2A for systemic risk, applied across the sector, and 2B a bank-specific requirement which may not be publicly disclosed. The second one being secret is itself an architectural fact for a reporting system. |
| **Pillar 3** | The public disclosure part of the Basel framework. In South Africa it comes through Regulation 43, and it is the reason a bank's risk data has to be presentable as well as correct. |
| **POPIA** | The Protection of Personal Information Act, South Africa's data protection statute. Its section 71 on automated decision making is the one a credit decisioning architecture has to answer to. |
| **Posting date** | When an entry was recorded in the ledger. Different from the value date, which is when it counts for interest, and from the transaction date, which is when the thing happened. |
| **Pre-legal / legal** | The two stages of collections either side of issuing a summons. Different systems, different costs, different measures, and a customer can move backwards between them. |
| **Prescription** | The point at which a debt becomes legally unenforceable through the passage of time. South African law prohibits selling or reactivating prescribed debt, and few collections systems can actually compute the clock. |
| **Prime** | South Africa's published reference lending rate, set at the repo rate plus 3.5 percentage points. Most retail lending is quoted as "prime plus or minus something", so the whole book reprices when the policy rate moves. |
| **Provision** | Money set aside against losses the bank expects but has not yet suffered. Under IFRS 9 it is modelled rather than observed, which makes it a model-governance artefact. |
| **Prudential Authority** | South Africa's prudential regulator for banks, which sits inside the Reserve Bank under the Twin Peaks split. It publishes the directives, prudential standards and return requirements a bank lives by. |
| **QR+** | South Africa's planned interoperable national QR payment standard, supporting merchant- and payer-presented codes and both bank and non-bank participants. |
| **RAROC** | Risk-adjusted return on capital: return after expected loss, over the capital the business consumes. The measure that stops a lending team being rewarded for growing a bad book. |
| **Reckless credit** | Credit granted without a proper affordability assessment, which South African law prohibits and which a court can set aside. The reason the affordability calculation and its inputs must be retained and reproducible. |
| **Reconciliation** | Proving that two records of the same thing agree, and investigating where they do not. The hardest architecture problems in a bank are reconciliation problems, and the function that does it carries the largest hidden operational risk. |
| **Recovery** | Money that comes back on a loan already impaired or written off. If recovery outcomes are not fed back to the models, loss given default cannot be validated — which is where the data chain usually breaks. |
| **Registered Mandates** | A South African collection rail, mandated from 10 March 2025, for collections where the customer has not completed authenticated approval. It is a new rail and not a rename: ISO 20022, a 19:00 late window, up to ten tracking days, and support for business and dual-signatory accounts, which DebiCheck cannot do. |
| **Reject inference** | Estimating how applicants you declined would have behaved, so a scorecard built only on accepted customers is not biased. Unglamorous, necessary, and it needs the declined applications retained. |
| **Repo** | Selling securities with an agreement to buy them back: borrowing money against collateral. The reverse repo is the same transaction seen from the lender's side. |
| **Repo rate** | South Africa's policy interest rate, set by the Monetary Policy Committee. Everything else in the lending book is priced relative to it. |
| **Representment** | The merchant's rebuttal of a chargeback, with evidence. One stage in a dispute process whose deadlines are set by the card scheme. |
| **Request to pay** | A payment the payee asks for and the payer approves, rather than the payer initiating. In South Africa it arrived as PayShap Request in December 2024. |
| **Residual value** | The value an asset is assumed to have at the end of a finance or lease term. Guess it too high and the lender absorbs the difference. |
| **Resolution** | The regime for dealing with a failing bank without a disorderly collapse, effective in South Africa from June 2023. It implies a bank can produce particular data on a weekend, which is an architectural requirement disguised as a legal one. |
| **Restatement** | Re-issuing a published number after finding it was wrong. Being able to restate requires knowing what you knew and when you knew it, which is the practical reason a bank needs more than one kind of time. |
| **Return** | A prescribed submission to a regulator for a period, in a version, with a preparer, an approver and — critically — the possibility of resubmission. Modelling it as a report loses everything that matters about it. |
| **Reversal** | Undoing a posting by making an equal and opposite one, rather than deleting the original. Banks never delete, which is why a reversal is two entries and an audit trail. |
| **Revolving credit** | A facility that can be drawn, repaid and drawn again up to a limit, rather than reducing to zero on a schedule. |
| **RM** | See Registered Mandates. The abbreviation appears constantly in South African collections material and means nothing to an outsider. |
| **RMS** | The Registered Mandate Service: the older mechanism inside DebiCheck for collecting without completed authentication. Superseded by Registered Mandates in March 2025, and mandates not migrated are now unusable. |
| **ROA** | Return on assets: earnings over average total assets. Lower than return on equity by exactly the amount of leverage the bank runs. |
| **ROE** | Return on equity: earnings divided by average shareholders' equity. The number a bank's board is judged on. |
| **Roll rate** | The proportion of arrears accounts that move from one bucket to the next worse one. The basic measure of whether a collections treatment is working. |

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| **RoRWA** | Return on risk-weighted assets: profitability measured against the capital a business actually consumes. It explains why a product with a thin margin can still be attractive. |
| **RTC** | Real-Time Clearing: South Africa's older immediate-transfer rail, launched in 2006, priced as a premium service, used for amounts up to R5 million. PayInc intends to migrate its traffic to PayShap and eventually retire it. |
| **RTGS** | Real-time gross settlement: settling each payment individually, immediately, in central-bank money, with no netting and no reversal. South Africa's is called SAMOS. |
| **RWA** | Risk-weighted assets: exposures multiplied by a risk weight, which is the denominator of every capital ratio. "Capital-efficient" in a bank means producing less of this per rand of income. |
| **SACRRA** | The South African Credit and Risk Reporting Association, which runs the hub that routes credit data from lenders to the bureaux. Its layouts are a mandatory interchange standard with a monthly deadline, and they appear in no global reference architecture. |
| **SADC-RTGS** | The regional settlement system for southern Africa, which settles in rand and is operated by the South African Reserve Bank on behalf of the region's central banks. |
| **Sale in execution** | A court-ordered auction of attached property. The end of the road in a secured recovery, and the point at which the loss given default is finally known. |
| **Sale of book** | Selling written-off or non-performing receivables to a third party. South African law prohibits selling prescribed debt, so the book has to be filtered before it is sold. |
| **SAMOS** | The South African Multiple Option Settlement system: the Reserve Bank's settlement system, live since 1998, where every South African payment ultimately settles. It migrated to ISO 20022 on 17 September 2022, the first settlement system in Africa to do so. |
| **SASWITCH** | The domestic interbank switching arrangement for card and cash-machine traffic between South African banks. |
| **Scheme** | A card network — Visa, Mastercard and the others — which sets the operating rules, the dispute reason codes and the clocks. Its rulebook goes to participants and is not published. |
| **Scorecard** | A statistical model that ranks how likely an applicant or an existing customer is to default. Application scorecards use what you are told; behavioural ones use what the customer has since done. |
| **Screen scraping** | Using someone's login credentials to read or act on their accounts because there is no API. South Africa regulates the external version of it rather than banning it, which means a bank must be able to identify that traffic on its own channel and tell it apart from fraud. |
| **Section 129 notice** | The notice a South African credit provider must deliver before enforcing a debt, with proof of delivery. A missing or unindexed proof of delivery stops a legal process dead. |
| **Section 16** | The Banks Act provision under which a full commercial banking licence is granted in South Africa. "Section 16 approval" is how people refer to being allowed to be a bank. |
| **Section 54** | The Banks Act provision governing amalgamations and transfers of banking business. It is why moving a book between legal entities is a regulatory project, not a migration. |
| **Section 65J** | The Magistrates' Courts provision governing emoluments attachment orders in South Africa. |
| **Section 70A** | The Banks Act minimum capital requirement: R250 million or a risk-based percentage, whichever is higher. The floor under being a bank at all. |
| **Securitisation** | Funding a book of loans by selling rated notes against them through a separate legal vehicle. It moves funding off the bank's own balance sheet and brings a tranche structure with it. |
| **Semantic API** | One of BIAN's published interface specifications, one per Service Domain that has operations. They come in BIAN's own and ISO 20022 flavours, which is BIAN conceding that ISO 20022's vocabulary wins inside payments. |
| **Service Domain** | BIAN's atomic unit: one self-contained area of business capability, with one responsibility boundary and one record it owns. Version 14.0 has 322 of them, and treating them as 322 systems is the category's most expensive mistake. |
| **Service Landscape** | BIAN's own arrangement of its Service Domains into groups. Its own description calls it a structure "for ease of access", which is the cure for anyone treating the picture as an architecture. |
| **Settlement** | The final, irrevocable transfer of value between banks, in central-bank money. In South Africa it happens in SAMOS, the Reserve Bank does not guarantee it, and once settled it cannot be reversed. |
| **Settlement finality** | The legal moment after which a settled payment cannot be undone. It is why a real-time rail needs a recovery process rather than a reversal process. |
| **ShapID** | A proxy — usually a mobile number — that resolves to a bank account for PayShap payments. It has to be registered before it can receive, which makes the registry a piece of national reference data. |
| **SICR** | A significant increase in credit risk: the trigger that moves a loan from twelve-month to lifetime impairment. IFRS 9 deliberately does not define it, so every bank defines it itself and must be able to defend the definition. |
| **Single customer view** | One authoritative record of a customer across every product and channel. In South Africa it is also a specific regulator-defined data structure for deposit insurance, with mandated aggregation rules, joint-account shares, beneficiary look-through and a monthly submission — so the phrase means both an aspiration and a compliance artefact, and you must be clear which you mean. |
| **Sort-at-source** | A South African practice where a corporate bypasses central clearing by sending payment files directly to several banks. It multiplies a bank's integration surface and defeats a single-rail design. |
| **SPV** | A special purpose vehicle: the separate legal entity that holds the ring-fenced assets in a securitisation. |
| **SSI** | Standing settlement instruction: the agreed account details for settling with a counterparty. Maintained in the back office, often with weak change control, and therefore a classic internal-fraud route. |
| **Stage 1** | An IFRS 9 loan that is still performing, carrying twelve-month expected credit loss. |

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| **Stage 2** | An IFRS 9 loan whose credit risk has increased significantly since it was written, carrying lifetime expected credit loss. Movement between stage 1 and stage 2 is the single most consequential rule in impairment. |
| **Stage 3** | An IFRS 9 loan that is credit-impaired, carrying lifetime expected credit loss, with interest recognised on the net amount. |
| **Standardised / IRB approach** | Two ways to compute risk weights: prescribed numbers from the regulator, or the bank's own models. Using models requires permission, and since the output floor a bank must run both anyway. |
| **Stand-in** | A switch approving or declining card transactions on an issuer's behalf when the issuer's own system cannot be reached, within preset limits. Limits nobody has reviewed for years are a standing risk. |
| **Stokvel** | A South African rotating savings and credit group, where members contribute regularly and take turns receiving the pot. Estimated at tens of billions of rand across hundreds of thousands of groups, and banks increasingly build products for it. |
| **Stop order** | A recurring payment the payer instructs their own bank to push. The opposite of a debit order, which the beneficiary pulls. |
| **STR / CTR** | Suspicious transaction reports and cash threshold reports, filed with the Financial Intelligence Centre. The thresholds and aggregation rules are set by FIC guidance, and the aggregation rule in particular is often implemented wrongly. |
| **Subordinated debt** | Debt that ranks behind ordinary creditors if the bank fails, and which can therefore count as Tier 2 capital. |
| **Sukuk** | A Sharia-compliant instrument giving the holder a share of income from an asset, rather than interest on a loan. |
| **Suspense account** | A holding account for items that have not been matched to anything yet. Every bank has them, their age and value are a direct measure of operational health, and they are a standing audit finding. |
| **SWIFT** | The member-owned co-operative whose network carries international financial messages. It is also the only message carrier South Africa's settlement system uses, a dependency the Reserve Bank's own disclosure flags as a material risk. |
| **Switch** | The component that routes card and cash-machine traffic between devices, acquirers, schemes and issuers, and stands in when a host is unavailable. Cash machines are driven by the switch, not by the core, which surprises architects who assume the core answers everything. |
| **System of record** | The one application that is authoritative for a given thing, creates it and owns its lifecycle. Everything else holds a copy, and writing that down per entity is the single most useful table an architect can produce. |
| **Targeted financial sanctions** | Obligations to freeze assets and not deal with named individuals and entities. Compliance is a screening and matching problem against lists that change without notice, and the clock on acting is short. |
| **TCIB** | Transactions Cleared on an Immediate Basis: the low-value cross-border payment scheme for southern Africa, which mandates ISO 20022 for all participants. |
| **Through-the-cycle / point-in-time** | Two ways to estimate default probability: on a long-run average, or on conditions right now. Capital generally wants the first and impairment the second, which is why a bank runs two sets of numbers. |
| **Tier 1 / Tier 2** | The two tiers of regulatory capital: Tier 1 absorbs losses while the bank is a going concern, Tier 2 when it is not. |
| **Tiered floor system** | South Africa's framework for remunerating banks' surplus balances at the central bank: up to a quota at the policy rate, above the quota at a penalty. It changes the economics of holding cash at the Reserve Bank. |
| **Tokenisation** | Replacing a card number with a surrogate value for use in wallets and online checkouts, so the real number is never stored. It adds a token-to-card mapping that something has to master. |
| **TPPP** | A third-party payment provider, registered to collect or pay on behalf of others. The South African registration regime for them is in transition as the Reserve Bank publishes its own authorisation framework. |
| **Tranche** | A rated class of notes in a securitisation, with a defined place in the queue for losses. The junior tranches absorb losses first, which is what makes the senior ones highly rated. |
| **Transaction** | What happened, that the customer would recognise as one thing. Distinct from a posting and from a ledger entry, and one box holding all three guarantees nobody can say whether a reversal is one transaction or two. |
| **TT1 / TT2 / TT3** | The three ways a DebiCheck mandate can be authenticated: a real-time request to the customer's banking app with a same-day deadline, an overnight batch request with two days, and card-and-PIN confirmation at a point-of-sale device. The third has by far the best success rate and needs the customer present. |
| **Twin Peaks** | South Africa's split of financial regulation into prudential — the Prudential Authority, inside the Reserve Bank — and conduct, the FSCA. A bank answers to both, on different timetables, about different things. |
| **Universal branch code** | A single code per South African bank that works for all its branches, replacing the need to know the specific branch. Both forms circulate in customer data, which is a reference-data problem rather than a trivia question. |
| **Unpaid reason code** | The code returned when a collection fails — insufficient funds, no mandate, account closed, payment stopped. Treating a dozen different causes with one retry strategy is the commonest collections design failure in South Africa. |
| **USSD** | The text-menu service that works on any mobile phone without data. In South Africa it is a live, load-bearing retail banking channel, not a legacy curiosity, and it constrains any customer journey that has to work for everyone. |
| **Value date** | The date from which an entry counts for interest, as distinct from when it was posted. Backdating a value date is routine and is the reason interest has to be recomputable. |
| **VaR** | Value at risk: the loss a trading book is not expected to exceed, at a stated confidence, over a stated period. A useful summary and a famously poor guide to the worst day. |
| **Variable recurring payment** | A standing permission for a third party to initiate payments from an account within agreed limits. It is what turns open banking from a data service into a payments rail, and South Africa has no mandated equivalent. |
| **Vintage** | A group of loans written in the same period, tracked as a cohort. Comparing vintages is how a lender sees whether its recent decisions are worse than its older ones. |

| | |
|---|---|
| **Volume/rate/mix effect** | The three-way split of why net interest income moved: more lending, different rates, or a different blend of products. The decomposition every results presentation shows and every finance system struggles to produce. |
| **Vostro** | "Their account with us": the mirror image of a nostro, where this bank holds the account for a foreign correspondent. |
| **Write-off** | Removing an exposure from the balance sheet when recovery is no longer reasonably expected. It is an accounting act, not a legal one — the debt still exists and can still be recovered. |
| **Yield curve** | Interest rates plotted against how long the money is lent for. It is the input to internal funds pricing, to valuation, and to any view about the future. |
| **ZARONIA** | The South African Overnight Index Average: the backward-looking overnight benchmark replacing JIBAR. Backward-looking means the borrower does not know the rate until the period has ended, which changes systems as well as contracts. |

Three habits are worth more than this list. **Ask what somebody means by a word before agreeing anything** — especially "core", "account", "balance", "customer" and "limit", each of which has cost a programme real money. **Write the bank's own definitions down**, in the bank's own words, and get them corrected rather than approved. And **date every definition that has a regulator behind it**, because the instrument will be amended by notice and the definition will move without anybody telling you.

**THE CHAPTER IN TEN LINES**

- Banking's vocabulary problem is not jargon, it is ordinary words used precisely: account, balance, mandate, limit, customer and core each mean several things.
- Three words will cost you most: “core”, because five people give five answers; “account”, because it means a product instance, a ledger account and a nostro; and “balance”, because it is a derivation and a bank computes several.
- South Africa has terms with no overseas equivalent — DebiCheck, Registered Mandates, the universal branch code, the stokvel — and terms that mean something else here, of which “bond” and “NCR” catch the most people.
- South Africa has no regulated open banking regime. The glossary says so because the error is both common and damaging.
- Where the research could not settle a definition, the entry says so: an honest gap is more useful than a confident guess.
- Write the bank's own definitions down in the bank's own words, and date the ones with a regulator behind them, because those move by notice.

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APPENDIX C

# Standards, bodies and where to find them

*A working register of the reference models, interchange standards, obligations and institutions a banking architect has to cite: what each one is, who owns it, whether you can read it without paying, and what to read first.*

An architect in banking spends a great deal of time hunting for the authoritative text of something, and in South Africa more time still establishing which version is operative. This appendix is the register. Checked October 2026.

Three warnings. **Availability is not a licence** — the most-used reference model in banking publishes no usage terms at all, so read the access column as "can I obtain this", not "may I redistribute it in work I am paid for". **Almost everything South African is free, because it is gazetted** — the difficulty is that instruments are amended by notice rather than reissued, so the operative text is the original plus a trail. And **keep three kinds of thing apart**: *reference models* are vocabularies, not designs; *interchange standards* belong at the boundary; *obligations* set the grain and history your data must carry.

## Reference architectures and industry models

| Item | What it is | Access and cost | Read first |
|---|---|---|---|
| **BIAN** | Banking's principal reference model: **Service Domains**, each owning one **control record**, with Behaviour Qualifiers, Service Operations and Semantic APIs. Service Landscape **14.0** (20 February 2026): 322 Service Domains, 38 Business Domains, 586 Business Capabilities, 242 Semantic APIs, no metamodel change — largely a payments re-cut for ISO 20022. Published by a member-owned non-profit, Frankfurt. | Release notes are open PDFs; the Landscape portal needs a registration form. **No licence is published**, only a copyright footer. | The 14.0 release notes, then the Service Landscape page — whose own phrase "for ease of access" cures anyone treating the picture as an architecture. |
| **The Open Group** | TOGAF the method, ArchiMate the notation; also ArchiMate expressions of BIAN (G205, Y201) and a joint guide to enterprise architecture in banking. | Standards free on registration; certification paid. | The joint guide. Which BIAN version each ArchiMate artefact models is not established. |
| **FIBO** | A machine-readable ontology of financial terms and relations, in RDF/OWL, reviewed quarterly. Published by the EDM Council, which names BCBS 239 among its purposes. | Called open-source by its publisher, on GitHub; exact licence and release not stated. | The FIBO Viewer. A semantics asset for the data estate, not a design for the operational one. |
| **APQC Banking PCF** | A process hierarchy with stable identifiers, for benchmarking. **Banking PCF 8.0, 26 August 2026.** Process-shaped where BIAN is capability-shaped. | Download links sit beside sign-in prompts and **the licence is not stated. Do not assert it is free.** | The Excel edition, for the identifier scheme. It has no data model, so it cannot settle system boundaries. |
| **IBM BFMDW** | The surviving IBM banking warehouse lineage, from IFW and BDW. | Commercial. Index last modified March 2020, content dated 2018. | The documentation index, as archaeology for an older warehouse. |

## Messaging and interchange standards

| Item | What it is | Access and cost | Read first |
|---|---|---|---|
| **ISO 20022** | Not a message format. A methodology plus a repository of business models from which messages are generated: a **Data Dictionary** and a **Business Process Catalogue**. The model layer is a conceptual data model with change control. Maintained by ISO TC68 with a Registration Authority. | **Be precise.** The repository gives **free** access to the Dictionary and Catalogue; the **standard is purchased from ISO**. Different terms. The e-Repository ships in a decade-old Eclipse format, which is why most banks see only the XSDs. | The Data Dictionary explainer. SAMOS migrated 17 September 2022, first in Africa; PayShap and Registered Mandates are native to it. |
| **ISO 8583, and the rest of the interchange layer** | The card message standard — what an authorisation travels in — currently **ISO 8583:2023**. Banking has no single ACORD: alongside it sit FIX for securities trading, FpML and the FINOS Common Domain Model for derivatives, and IFX, whose forum merged into Nacha in 2018. | Paid. Every acquirer, switch and scheme runs a dialect. | Your switch vendor's dialect specification, which is what matters. |
| **Swift** | Messaging for cross-border correspondent banking, and operator of the **CBPR+** migration. Also the only carrier SAMOS uses — a reliance the SARB's own disclosure flags as a material risk. | Membership commercial; roadmaps public. | The CBPR+ roadmap, for dates no bank controls. |
| **EMVCo** | Chip, contactless, three-domain-secure and payment tokenisation specifications. | Downloadable subject to terms; certification commercial. | The specification index. Current versions were not retrievable here — check before quoting one. |
| **PCI SSC** | Payment card data security: **PCI DSS**, currently v4.x, plus the PIN and software standards. E-commerce script-integrity requirements effective 31 March 2025. | Standards **free**; assessor qualification commercial. | Requirement 3, on stored account data — where your card data model is decided. |
| **The card schemes** | Visa, Mastercard, and locally the Amex, Diners and Fleet arrangements: rules, interchange structures, dispute reason codes and clocks. | Rules go to participants, not the public. Dispute timeframes in circulation come from vendor secondary sources. | Your own rulebook. The dispute clocks *are* the process and are not negotiable. |
| **Open banking standards** | **UK**: Open Banking Standard v4.0.1, April 2026, APIs bundled with customer-experience guidelines. **EU**: the Berlin Group openFinance API Framework. **US**: the FDX API. | UK: no licence statement. EU: **free under CC-BY-ND**, no participation fee — the cleanest terms here. US: royalty-free. | The Berlin Group framework: readable without an account, and it shows what a consent model looks like. **South Africa has none of this.** |

## Capital, accounting and risk standards

| Item | What it is | Access and cost | Read first |
|---|---|---|---|
| **BCBS and the Basel framework** | International capital, liquidity and disclosure standards. Architecturally it obliges one reconcilable exposure view per counterparty, controlled reference data, reproducible historical calculations and retained granularity. | **Free**, all of it. | The consolidated framework, then your own jurisdiction's transposition — here, the Regulations relating to Banks. |
| **BCBS 239** | Fourteen principles on risk data aggregation and reporting, January 2013. Principle 2 lands on a backlog; Principle 6, adaptability, is the sleeper — it requires aggregations nobody has asked for yet. | **Free** | The standard, not a summary. Then the January 2026 newsletter, which says lineage is still difficult and crisis-time ad hoc reporting still a hurdle. |
| **IASB — IFRS 9** | The standard that made impairment forward-looking: expected credit loss, three stages, a deliberately undefined deterioration trigger. It makes a model estate part of the financial close. | Available with registration; full access commercial. Good free secondary material exists. | A Big Four in-depth guide, then the Prudential Authority's directive on provisions. |
| **FATF** | The standard-setter for anti-money-laundering and counter-terrorist financing. **South Africa was grey-listed February 2023 and exited 24 October 2025**; the next mutual evaluation is expected in the first half of 2026. | **Free** | The Recommendations, then South Africa's mutual evaluation report. |

## South African regulators and the statute book

| Item | What it is | Access and cost | Read first |
|---|---|---|---|
| **SARB** | The central bank, and an architectural actor. It **operates** the settlement system, **determines** interchange for several streams, **owns half** the clearing utility, and from 2026 **holds the scheme rule-making function**. Its Financial Surveillance Department taps every cross-border payment. | **Free** | The SAMOS PFMI disclosure, the NPS Regulatory and Oversight Report, then Directive 2 of 2024. |
| **Prudential Authority** | The prudential regulator for banks, inside the SARB: directives, prudential standards, joint standards and the **BA returns**. **GoN 5802** (31 January 2025) removed the return forms from the Regulations, so they now move without a gazette. | **Free** | The current directive on returns. GoN 5802 is **not** the Basel III post-crisis reform instrument, which took effect 1 July 2025. |
| **FSCA** | The conduct authority: conduct standards, including **Conduct Standard 3 of 2020** for banks, plus the draft Open Finance paper. **Four joint standards are confused and three are numbered "1".** For banks: **Joint Standard 1 of 2023** (IT governance, 15 November 2024) and **Joint Standard 2 of 2024** (cybersecurity, 1 June 2025). **Joint Standard 1 of 2024 is outsourcing by insurers and does not apply to banks.** | **Free** | Conduct Standard 3 of 2020, then the two joint standards that apply. Get the numbering right in writing. |
| **Financial Intelligence Centre** | The financial intelligence unit under FICA: due diligence, record-keeping, cash threshold and suspicious transaction reporting, sanctions lists. | **Free** | The sanctions manual, then Guidance Note 5C — its aggregation rule is routinely implemented wrongly. |
| **National Credit Regulator** | In a South African banking sentence, **"NCR" means the National Credit Regulator**, never a technology vendor. The National Credit Act layer is prescriptive statute: a mandated affordability assessment, a pre-agreement quotation, a section 129 notice, a bar on selling prescribed debt. | **Free** | The Credit Bureau Monitor, and the bureau registration requirements. |
| **National Treasury** | The policy department: legislation, the Twin Peaks trail, the NPS Act review paper, and the statements that date things — the FATF exit, the ombud consolidation, the November 2025 inflation target. | **Free** | Its media statements, often the only dated primary record of a change. |
| **Information Regulator** | The data protection authority under POPIA, and the access-to-information authority. Registers information officers, issues guidance and enforcement notices. | **Free** | POPIA section 71 on automated decision making — the section a credit decisioning architecture must answer to. |
| **King IV** | South Africa's corporate governance code, which places board-level accountability on technology and information. It is why an architecture governance forum in a South African bank has a governance mandate and not only a technical one. | **Free** in summary form from several published guides; the Report itself is issued by the Institute of Directors. | Principle 12 and its recommended practices. |

## South African utilities, associations and schemes

| Item | What it is | Access and cost | Read first |
|---|---|---|---|
| **PayInc** | **The renamed BankservAfrica** — a clearing and switching utility, not an industry body. Automated Clearing Bureau 1972, BankservAfrica 2010, PayInc from August 2025, with the **SARB taking 50%** in late 2025 and participating banks holding the rest. It holds the EFT Credit, EFT Debit, Authenticated Collections, PayShap, Registered Mandate and Real-Time Clearing clearing houses. | Rules published; **the ones that matter are not.** Cut-offs sit in Product and Service Manuals; the fee book is under non-disclosure. | The key rules, for the clearing-house names and the mentoring-bank requirement. Then ask for the manuals in writing. |
| **The payments association position** | The commonest error in current commentary. **PASA's recognition as payment system management body was withdrawn and its functions split between the SARB itself and PayInc**, staged across 11 August, 1 and 2 September 2026 — the SARB's own pages disagree on which date completed it. The **Payments Industry Body** was a 2022 design proposal and is **not** what happened. | **Free** | The PSMB transition page and the 11 August 2026 media release. Material saying "the PIB replaces PASA" describes a design, not an outcome. |
| **SACRRA** | The credit reporting association, operator of what it describes as the **sole legally recognised Central Data Transmission Hub**, routing member submissions to the bureaux authorised to hold payment profile information — roughly 54 million records a month across some 3,349 members. | The description is free; **the layouts are member material.** | The hub description. These layouts are a mandatory interchange standard with a monthly deadline, absent from every global reference architecture. |
| **The credit bureaux** | Two sets. **Registered** with the NCR: a long register, 55 records. **Authorised to hold payment profile information**, and so on the mandatory submission path: six — Consumer Profile Bureau, Experian, ITC, TransUnion, VeriCred and XDS. **Compuscan is Experian**, acquired 2018. | Registers free; data commercial. Section 43(2) of the NCA bars registration where a controlling shareholder is a credit provider — a structural separation rule. | The register, then the registration requirements, which list the data intake, retention and deletion procedures required. |
| **BASA** | The Banking Association South Africa: the industry representative body. Position papers, statistics releases, commentary. | **Free** | Whatever it has published on your subject. A representative body, not a rule-maker. |
| **SABRIC** | The industry's financial-crime information body, publisher of the Annual Banking Crime Statistics Report. | The report is **free**; intelligence sharing is for members. | The report itself, not the tertiary aggregations of it. |
| **National Financial Ombud Scheme** | The consolidated ombud, operational **1 March 2024**, merging four schemes into Banking & Credit, Life and Non-Life divisions. It recovered around R443 million for consumers in 2025. Recognised under the Ombud Council. | **Free** | Its case material. A referral with no retrievable complaint file is a records failure with a deadline attached. |

## What will date fastest

In order. **Payments governance**, which changed six weeks before this research was compiled and is still settling: the Payments Industry Body's status is unresolved and the National Payment System Bill was out for comment on 1 September 2026. **Standards versions** — BIAN moves roughly annually, and payments is where it moves. **Regulatory timetables** — the COFI Bill, the open-finance intention, the JIBAR cessation at end-2026, the CoDI submission regime. **Scheme rules and dispute clocks**, reissued with no public notice.

One thing will not date. **South Africa has no regulated open banking regime.** What exists is bilateral commercial APIs plus SARB Directive 2 of 2024 on screen-scraped EFT credit initiation, and an FSCA position paper that only explores a phased mandatory regime, with no dates. The FSCA's own draft says institutions already sharing data are not governed by a framework, and that third-party providers and their APIs are not licensed and fall outside its ambit. Write that sentence, not the other one.

THE CHAPTER IN TEN LINES

- Availability is not a licence. BIAN's artefacts are obtainable through a registration form and BIAN publishes no usage terms at all — get terms in writing before embedding its content in anything commercial.
- ISO 20022's repository gives the Data Dictionary and Business Process Catalogue away free; the standard itself is bought from ISO. Those are different terms and the difference matters in a contract.
- BCBS 239 and the whole Basel framework are free, and BCBS 239 is the only framework here a bank can be supervised against — which makes it the most reliable business case for data-platform work.
- Four South African joint standards are commonly confused and three of them are numbered “1”. For banks it is Joint Standard 1 of 2023 and Joint Standard 2 of 2024; Joint Standard 1 of 2024 is outsourcing by insurers.
- PayInc is the renamed BankservAfrica, now half-owned by the SARB. It is a clearing utility, not an industry body, and the documents an architect most needs from it are not public.
- PASA's recognition was withdrawn in 2026 and its functions split between the SARB itself and PayInc. The Payments Industry Body was a 2022 design proposal, not the outcome.
- The SACRRA layouts and the authorised-bureau submission path are a mandatory South African interchange standard that appears in no global reference architecture.
- South Africa has no regulated open banking regime. Bilateral commercial APIs, SARB Directive 2 of 2024 on screen scraping, and an FSCA paper that only explores a regime.

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APPENDIX D

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*Every source this book rests on, de-duplicated across four research passes and grouped by theme — then an honest list of what could not be sourced at all.*

Everything asserted in this book either carries a source or is marked as the author's judgement. This appendix collects the sources in one place, de-duplicated across the four research passes behind the book, grouped by theme and numbered continuously. Where one source supports several chapters it appears once.

**When the research was done.** The underlying research was compiled and verified on **15 October 2026**, in four passes: the business of banking and the South African market; the systems and the payment rails; the reference architectures and the technology providers; and the data model and the regulatory environment. The four passes cited 52, 57, 130 and 82 sources. De-duplicated across passes that is **298 distinct sources**, which is the list below. Every market figure, ratio, volume and version number is as published at that date.

**The source hierarchy used.** Primary sources first: statutes and gazetted regulations, prudential directives and standards, standard-setters' own publications, central-bank statistics and audited company results. Then near-primary: exchange filings, scheme and utility rule documents, vendor technical documentation and industry-body publications. Then trade press, used for facts a primary source should have carried and did not — which in South African banking is most of what you can learn about a bank's own technology, because the banks do not publish it. Vendor marketing is used only to record what a vendor claims, is attributed to the vendor, and is dated.

**Two source problems specific to banking in South Africa.** The first is that the most operationally binding specifications are not public. The authoritative cut-off times for the batch rails, and the record layouts every bulk channel implements, live in PayInc's Product and Service Manuals and in a participant pricing book held under non-disclosure. You cannot cite them, and nor can anyone else. The second is that **almost nothing about the technology inside a South African bank is on the public record.** Where a platform attribution appears in this book it is attributed to the outlet that reported it, not asserted as the bank's own disclosure, and where no source exists the book says so rather than repeating what circulates in the market.

**What will date fastest, in order.** *Payments governance.* The arrangements changed in August and September 2026, six weeks before this research was compiled, and the SARB's own pages disagree on the completion date. Anything written earlier about who sets scheme rules is wrong. *Vendor ownership and product naming.* Several entries in the vendor list are one transaction from being wrong; the January 2026 Worldpay and Issuer Solutions moves are already in it. *Regulatory timetables.* The National Payment System Bill, the COFI Bill, the FSCA's open-finance intention, the JIBAR cessation at end-December 2026 and the CoDI submission regime after September 2025 are all in motion. *Market results.* Every earnings, margin and ratio figure is a point in time and the point has passed. *Standards versions.* BIAN Service Landscape 14.0, Open Banking Standard v4.0.1, the Berlin Group V2 Suite, APQC Banking PCF 8.0 and PCI DSS v4.x will all move.

**What will not date.** The mechanisms. That a bank is a ledger with a sales channel attached, and that its hardest problems are reconciliation problems. That the rail determines the process. That a mandate held at the debtor's bank is a different object from a mandate held by the creditor. That a balance is a derivation and not a stored fact. That banking data is tri-temporal because somebody will ask what happened, when it was recorded, and when it took effect. That the overnight cycle is a settlement-infrastructure fact and not an internal convenience. Those outlast any statute, platform or version number in the list below.

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## What could not be sourced

The four research passes carried 27, 21, 37 and 31 flagged items between them. Most are details that do not reach the book. The following could not be settled and are material enough that an architect should know the gap exists, because each one is a question somebody will eventually ask you.

**The EFT cut-off schedule.** There is no public table of file submission and cut-off times for the EFT credit and EFT debit streams. BankservAfrica's own published rules say so explicitly: submission times live in the Product and Service Manuals supplied by its customer account management team, and in a participant pricing and fee book available under non-disclosure. This is the single most consequential unsourceable fact in the book, because a customer journey that misses a cut-off fails for a whole business day. Get the table from a participant bank or from PayInc directly, in writing, and treat it as an architectural constraint with an owner.

**PayInc's record layouts.** The fixed-width Bankserv/ACB formats for EFT credit payments, EFT debit collections, the sunset AEDO and NAEDO streams and the DebiCheck mandate files are implemented by every bulk channel and every collections bureau in the country, and none of the current specifications is public. Bank-published format guides exist but were not retrievable. The authoritative documents are PayInc's manuals. Do not write field-level detail from secondary material.

**Which core platform each South African bank runs.** For **Nedbank, Capitec, Absa, Investec, African Bank, Bidvest Bank, Postbank and Sasfin** no credible named attribution could be found. Three attributions that do circulate rest on trade press rather than company disclosure — FirstRand's selection of Fiserv Finxact, OM Bank on 10x Banking, and Nedbank's R11.7 billion Managed Evolution figure — and this book attributes each to the reporting outlet. Nedbank's own results reporting names no vendor at all. No count of South African banks running any given platform exists, and none should be estimated.

**The systems-integrator picture.** Almost nothing is sourceable. One named fact was found: Absa's own market risk chief says the bank works closely with Deloitte on regulatory matters. Beyond that, no source confirming a named core banking engagement at a named South African bank was found for any global or local integrator. Press releases in this market name the software vendor and the bank and leave the integrator out.

**The PayShap volume figures do not reconcile.** For overlapping periods, published figures include 136 million transactions and over R100 billion cumulative, 461 million transactions and R403 billion, 905 million transactions, and a SARB series giving 251 million transactions and R198.5 billion to March 2025. The participating-bank count is variously ten, eleven, twelve or thirteen depending on source and date. The discrepancy is unresolved; use the SARB's regulatory and oversight report series and PayInc's own statements, and treat aggregator figures as unreliable.

**BIAN's membership fees and licence terms.** BIAN's own membership FAQ poses the question "How much is the BIAN annual membership fee?" and does not answer it. No fee amounts, no tiers. More importantly, **no licence or usage terms were found on any BIAN deliverable** beyond a copyright footer. Availability through a registration form is not permission to redistribute. Get terms in writing before embedding BIAN content in a commercial deliverable.

**The CoDI single-customer-view regime after 30 September 2025.** The exemption from full SCV submission ran to that date, and what replaced it — whether full SCV record files are now submitted, and in what format — could not be established. For a 2026 South African retail banking architecture this is the most important open question in the research, because the answer determines a mandated monthly data product with aggregation, joint-account and beneficiary-look-through rules attached.

**And the rest, briefly.** A single consistent count of registered banks, mutual banks, co-operative banks, foreign branches and representative offices. Current NCA maximum prescribed rates and fee caps at the 7.25% repo. The in duplum rule, named in search results but never fetched. FICA cash-threshold and suspicious-transaction reporting thresholds. RICA's requirements and how they interact with FICA verification. Whether covered bonds are issued in this market at all. Current interchange rates and the content of Interchange Determination Project Phase V. A decommissioning date for Real-Time Clearing. The final status of the Payments Industry Body. The South African output floor percentage and its phase-in. Whether BCBS 239 is formally adopted in South African law or applied supervisorily to domestic systemically important banks only. The incident-notification deadline in Joint Standard 2 of 2024. Visa's own dispute time limits, currently cited through a vendor secondary source. Whether any global regulatory reporting product ships maintained SARB BA-return content. The support status, roadmap and licensing of IBM's Banking and Financial Markets Data Warehouse. FIBO's current release identifier and exact licence. APQC's licensing and whether non-members can download the Banking PCF. And the whole of what practitioners agree about legacy — mainframes, COBOL, the skills pipeline, the core-replacement completion record, merger-driven duplication and the separate Rest-of-Africa estate — none of which any South African bank discloses.

**Why this list is here.** An appendix of sources tells a reader what the book rests on. A list of gaps tells them where the book stops, which is more useful, because the gaps are where they will be asked for an answer and will have to go and get one. An honest list of what could not be sourced is worth more than a confident guess.

**THE CHAPTER IN TEN LINES**

- The research behind this volume was compiled and verified on 15 October 2026; four passes citing 52, 57, 130 and 82 sources de-duplicate to 298 distinct sources.
- Primary sources first — statutes, prudential directives, standard-setters, central-bank statistics and audited results — then near-primary, then trade press, with vendor material used only to record what a vendor claims.
- In South African banking the trade press is not a shortcut, it is often the only record: no bank publishes its platform estate, so every attribution is to the outlet that reported it.
- The authoritative EFT cut-off schedule and the Bankserv/ACB record layouts are not public, by PayInc's own account — obtain them in writing and treat them as architectural constraints with an owner.
- Payments governance will date fastest, then vendor ownership, then regulatory timetables, then market results, then standards versions.
- The gaps are published deliberately: an honest list of what could not be sourced is worth more to a reader than a confident guess.

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